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FM Cardio

Important basic heart failure points	
Left heart failure	- Presents as pulmonary findings (i.e., dyspnea, orthopnea, paroxysmal nocturnal dyspnea). - This is because left-heart problems cause a backup of pressure onto the pulmonary circulation, leading to increased pulmonary capillary hydrostatic pressure → transudation of fluid into the alveolar spaces (pulmonary edema). Sometimes this can also cause pleural effusion. - What USMLE will do is give you some sort of left-heart pathology + dyspnea, and then ask for the cause of the dyspnea → answer = "increased pulmonary capillary hydrostatic pressure." Another answer in this case is "increase alveolar-arteriolar (A-a) oxygen gradient." - Left atrial pressure (LAP) = pulmonary capillary wedge pressure (PCWP), which is high in left heart pathology. - For FM forms, you need to know that hyponatremia sometimes seen in left heart failure is due to "baroreceptor-mediated ADH release" – i.e., the carotid baroreceptors stretch less if the systolic impulse from the heart is reduced, so this is interpreted as fluid depletion, where more ADH facilitates retention of free water. Sounds Step 1-esque, but it's asked on an FM form.
Right heart failure	- Presents as systemic findings – i.e., jugular venous distension (JVD) and peripheral edema. - Since blood cannot enter the right heart as easily, it backs up to the neck veins (JVD) and venous circulation (increased hydrostatic pressure in veins → transudation of fluid into legs). The Q might mention that central venous pressure is high. - Hepatosplenomegaly can also be seen in RHF but is very rare on USMLE. - Normal jugular venous pressure (JVP) is 3cm above the sternal angle. JVD would be higher than this. Sometimes questions can write that jugular venous pulsations are seen 3cm above the sternal angle and the student erroneously thinks this means JVD, but this is not the case.
Congestive heart failure	- Congestive heart failure = left heart failure + right heart failure. - The most common cause of right heart failure is left heart failure. Simply adding the two together, we now call that congestive heart failure. - In congestive heart failure, we'll see both left- and right-heart failure findings – i.e., patient will have dyspnea, JVD, and peripheral edema. - PCWP is elevated in these patients, since the left heart has pathology.
Cor pulmonale	- Cor pulmonale is defined as right-heart failure due to a pulmonary cause. In other words, the left heart is completely normal in cor pulmonale and PCWP is normal. - Cor pulmonale will be a patient who has JVD and peripheral edema in the setting of obvious and overt lung disease, such as 100-pack-year smoking history, cystic fibrosis, or pulmonary fibrosis. These can present with lung findings such as wheezes, where you as the student need to say, "It just doesn't seem like they're focusing on left-heart failure as the cause of the right-heart failure here. It seems the 100-pack-year smoking Hx causing COPD is why the right heart is failing." - The patient can have a "boot-shaped" heart colloquially, which refers to right ventricular hypertrophy without left ventricular hypertrophy. - If the patient has COPD, the massively hyperinflated lungs will push the heart to the midline, causing a long, narrow cardiac silhouette, with a point of maximal impulse in the sub-xiphoid space. - You must know that *pulmonary hypertension* is the reason the right heart decompensates. In both cor pulmonale and congestive heart failure, the right heart experiences increased afterload because of *pulmonary hypertension*.

	- Bosentin is an endothelin 1 receptor antagonist that can be used for pulmonary hypertension. - A loud P2 and tricuspid regurgitation are HY findings in cor pulmonale. I discuss these in more detail in the tables below.

Hyper-quick causes of bilateral pitting peripheral edema	
Cardiac	- Right heart failure (either due to cor pulmonale or congestive) → ↓ ability to fill right heart → ↑ central venous pressure → ↑ systemic venous hydrostatic pressure → transudation of fluid from systemic veins/venules into interstitium of legs.
Hepatic	- Cirrhosis → ↓ hepatic production of albumin → ↓ intravascular oncotic pressure → transudation of fluid from systemic veins/venules into interstitium of legs.
Nephrogenic	- Proteinuria → hypoalbuminemia → ↓ intravascular oncotic pressure → transudation of fluid from systemic veins/venules into interstitium of legs.
Drugs	**- Dihydropyridine calcium channel blockers (i.e., amlodipine, nifedipine).** **- Exceedingly HY for FM.** **- Don't confuse with the non-dihydropyridine, verapamil, which causes constipation.**
Dietary	- Strict vegetarianism or veganism → ↓ dietary protein consumption → ↓ **intravascular oncotic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Pregnancy	- A little bit of peripheral edema is normal in pregnancy due to compression of IVC.

Hyper-quick causes of unilateral non-pitting edema	
Lymphatic insufficiency	- Malignancy (e.g., peau d'orange of breast), Hx of surgery (e.g., mastectomy), *Wuchereria bancrofti* (elephantiasis).
Thyroid	- Pretibial myxedema (Graves) → mucopolysaccharide deposition in skin + surrounding edema. - Myxedma (severe hypothyroidism) → despite the name, it refers to general "severe hypothyroidism," not just skin changes; can cause carpal tunnel syndrome. - "Pretibial myxedema" is only seen in Graves. Paradoxical hyperthyroidism seen in Hashimoto causing pretibial myxedema is astronomically rare and will get you questions wrong on USMLE.

HY Valvular / flow abnormalities on USMLE	
Atrial septal defect	- **Fixed splitting of S2** - Can sometimes be associated with a systolic flow murmur, since more blood L → R from the LA → RA means more blood flow across the pulmonic valve. So Q might say "fixed splitting of S2 and a systolic murmur." - Sometimes can be seen in Qs as "wide, fixed splitting." I only mention this because some students get pedantic / ask about this. "Wide splitting" just means right ventricular hypertrophy. So if the Q says "wide, fixed splitting," they're saying the patient has RVH due to an ASD. - Patent foramen ovale = ASD on USMLE. Don't confuse with patent ductus arteriosus (discussed below).
Ventricular septal defect	- Holosystolic (aka pan-systolic) murmur at lower left sternal border. - Can be associated with a diastolic rumble or enlarged left atrium (if more blood going L → R across VSD, then more blood is returning to the LA from the lungs → LA dilatation).

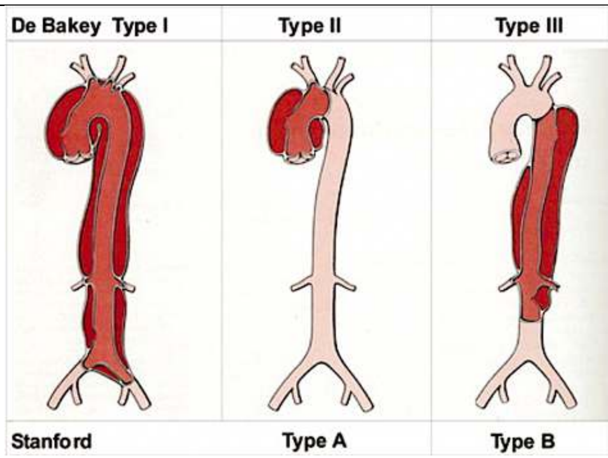
	- Seen as part of tetralogy of Fallot (VSD, RVH, overriding aorta, pulmonic stenosis). - VSD does *not* cause cyanosis at birth. Only years later after the higher blood flow to the lungs results in pulmonary hypertension, followed by right ventricular hypertrophy and reversal R → L (Eisenmenger) does the patient become cyanotic.
Mitral regurgitation	- Holosystolic (pan-systolic) or just regular "systolic," 29 times out of 30. - Most USMLE questions will not mention it radiating to the axilla. - Highest yield cause of MR on USMLE is post-MI papillary muscle rupture. USMLE is obsessed with this. They'll say hours to days after an MI, patient has new-onset systolic murmur → answer = MR. - Seen **acutely** in rheumatic heart disease (valve scars over years later and becomes mitral stenosis). - Can be caused by general ischemia / dilated cardiomyopathy. - Can cause **JVD** (i.e., back up all the way to the right heart); this is asked multiple times on the new NBMEs.
Mitral stenosis	- Described as "rumbling diastolic murmur with an opening snap"; can also be described as "decrescendo mid-late diastolic murmur" (i.e., following the opening snap). - Can cause a right-sided S4 if the pressure backs up all the way to the right heart (seen on NBMEs sometimes; this confuses students because they think S4 must be LV, but it's not the case). An S4 is a diastolic sound heard in either the LV or RV when there is diastolic stiffening due to high afterload. - 99% of mitral stenoses are due to Hx of rheumatic heart disease (i.e., the patient had rheumatic fever as a child, where at the time it was mitral regurg, but years later it has now become mitral stenosis). - One 2CK NBME Q mentions patient with history of rheumatic heart disease who, years later, now has 4/6 rumbling diastolic murmur **without** an opening snap; this is still mitral stenosis. Although opening snap is buzzy for MS, just be aware it's not mandatory and that this Q exists on NBME. - Other HY presentation on USMLE is pregnant women with new-onset dyspnea in 2nd trimester and a diastolic murmur. This is because 50% increase in plasma volume by 2nd trimester causes the underlying subclinical MS to become symptomatic. Don't confuse this with severe dyspnea and peripheral edema in late third-trimester, which is instead peripartum cardiomyopathy (antibody-mediated).
Mitral valve prolapse	- Most common murmur. - Described as mid-systolic click. - "Myxomatous degeneration" is buzzy term that refers to connective tissue degeneration causing MVP in Marfan and Ehlers-Danlos. I've seen this term on 2CK forms. - Almost always asymptomatic. On 2CK forms, they want you to know about "mitral valve prolapse syndrome," which is symptomatic MVP that presents as repeated episodes of "fleeting chest pain" on the left side in an otherwise healthy patient 20s-30s. They might say there is Hx of MI in the family, but this is MVPS, not MI. Answer on surgery form is "no treatment necessary." - USMLE loves using MVP as a distractor in panic disorder questions. They will give long paragraph about panic attack/disorder + also mention there's a mid-systolic click; they'll ask for cause of patient's presentation → answer = panic disorder, not MVP → student is confused because they say mid-systolic click, but the MVP isn't the cause of the patient's presentation; the panic disorder is; MVP's are usually incidental, benign, and asymptomatic.

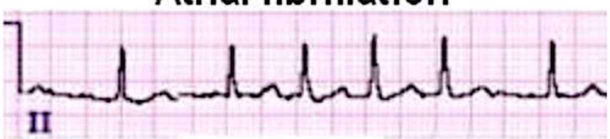
Aortic regurgitation	- Decrescendo holo-diastolic (pan-diastolic) murmur; can also be described as "early diastolic murmur," or "diastolic murmur loudest after S2." - Causes wide pulse pressure (i.e., big difference between systolic and diastolic pressures, e.g., 160/50, or 120/40) → results in head-bobbing and bounding pulses (don't confuse with slow-rising pulses of aortic stenosis). - The bounding pulses can be described on NBME as "brisk upstroke with precipitous downstroke." In turn, they can just simply say, "the pulses are brisk," meaning the systolic component is strong. - I would say 4/5 times bounding pulses means AR. The other 1/5 will be PDA and AV fistulae (discussed below). Bounding pulses occur when blood quickly leaves the arterial circulation. In AR, the blood quickly collapses out of the aorta back into the LV. In PDA, it leaves the aorta and enters the ductus arteriosus; in AV fistulae, it leaves for a vein. - Highest yield cause on USMLE is aortic dissection → can retrograde propagate toward the aortic root causing aortic root dilatation and AR. - Even though MVP is most common in Marfan and Ehlers-Danlos, AR is second most common in these patients, since if they get aortic dissection, this can lead to AR. - Can lead to volume overload on the LV and eccentric hypertrophy.
Aortic stenosis	- Mid-systolic murmur, or just "systolic" murmur; can also be described as "late-peaking systolic murmur with an ejection click." - Radiates to the carotids. This descriptor shows up quite frequently on NBME (way more than radiation to the axilla for MR). - Causes slow-rising pulses, aka "pulsus parvus et tardus" (don't confuse with bounding pulses of AR). - SAD → Syncope, Angina, Dyspnea; classic combination seen in AS, albeit not mandatory. If you get a question where they say systolic murmur but you're not sure of the diagnosis, if they say chest pain or fainting, you know it's AS. - Often caused by bicuspid aortic valve. The patient need not have Turner syndrome and often won't. Bicuspid valve is usually inherited as an autosomal dominant familial condition. - The bicuspid valve need not calcify in middle-age prior to the AS forming. Bicuspid valve can present with AS murmur in high schooler on NBME.
Tricuspid regurgitation	- Will be described on USMLE as a holosystolic murmur that increases with inspiration. - Right-sided heart murmurs get worse with inspiration → diaphragm moves down → decreased intra-thoracic pressure → increased right-heart filling. - Can cause pulsatile liver. - Highest yield cause of TR on USMLE is **pulmonary hypertension / cor pulmonale.** I see this all over the NBME exams. For whatever reason, these conditions do not cause pulmonic regurg; they cause tricuspid regurg. In other words, if you see tricuspid regurg in a Q, your first thought should be pulmonary hypertension or cor pulmonale (right heart failure due to a pulmonary cause). - IV drug user endocarditis is obvious risk factor for TR, but weirdly nonexistent on USMLE. - Carcinoid syndrome is theoretical cause but lower yield.
Functional (flow) murmur	- Systolic murmur seen in the setting of higher heart rate caused by infection, anemia, or pregnancy. Caused by increased flow across the pulmonic and/or aortic valves. - Known as a functional murmur because this means it goes away once the heart rate comes back down.

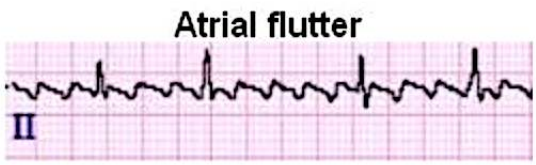
	- Seen all over 2CK forms, where they try to trick you into thinking the kid has a valvular pathology of some kind, but there isn't; there will merely be an infection or simple viral infection. - Can be seen sometimes with ASD, where the patient will have fixed splitting of S2 "plus a systolic murmur" → merely higher right-sided volume, so more flow across the pulmonic valve.
Venous hum	- On 2CK form; described as a murmur in the neck that abates when the kid is laid supine + the neck rotated. - Benign + don't treat.
"Ball-in-valve" murmur	- Associated with cardiac tumors (i.e., myxoma in adult, or rhabdomyoma in kids for tuberous sclerosis). - Described as a diastolic rumbling murmur that abates when the patient is re-positioned unconventionally (e.g., onto his or her right side).
S3 versus S4	- Both are diastolic sounds. - S3 is due to high volume/preload in the left ventricle, causing a reverberation against the wall. - S3 can sometimes be physiologic (i.e., normal / no problem) in pregnancy and high-endurance athletes. Patient will have eccentric hypertrophy (sarcomeres laid in linear sequence). If pathologic, it is due to dilated cardiomyopathy with reduced ejection fraction (<55%) or high-output cardiac failure (EF >70%). - There is one question on 2CK form where they give an S3 in diastolic dysfunction. I'm convinced this is an erratum, but I need to mention it because it exists on the NBME form. - S4 is due to high pressure/afterload on the left (but sometimes right) ventricle, causing a stiffened ventricle with diastolic dysfunction and concentric hypertrophy (sarcomeres laid in parallel). It is always pathologic. It is usually caused by systemic hypertension causing afterload on the LV, or aortic stenosis. - S4 can sometimes be right-sided on USMLE. There is a 2CK Q where they give severe mitral stenosis and say there's an S4, but it's for the RV not LV. Some weird/annoying points: - The combo of S3 and S4, seen together in the same vignette, can be seen in high-output cardiac failure. For example, they will say a patient as an AV fistula/conduit, or has Paget disease, and they will say there's S3 and S4 and ask for diagnosis → answer = high-output cardiac failure. The take-home point is that high-output failure can present with either an isolated S3 or the combo of S3 and S4 together, but never S4 alone on USMLE.
Loud P2	- One of the highest yield cardiac sounds on USMLE; almost always overlooked by students. - Means pulmonary hypertension or cor pulmonale on USMLE. - The pulmonic valve slams shut due to high pressure distal to it. - For example, they'll give a smoker who simply has a loud P2 → this just means patient has pulmonary hypertension. Not complicated. - Also recall that I said above that highest yield cause of tricuspid regurg on USMLE is pulmonary hypertension / cor pulmonale. So both what I want you to remember is both TR and loud P2 for this. - Sometimes the UMSLE will just say "loud pulmonic component of S2," or "loud S2," rather than saying "loud P2." I've never seen "loud A2" on USMLE, but in theory this means systemic hypertension. - A soft P2 refers to pulmonic stenosis, but is LY.
Wide splitting of S2	- Means right ventricular hypertrophy on USMLE. - A2 and P2 are far apart. - You don't have to worry about the mechanism. But in short, the more pressure you have in a ventricle, the more delayed the semilunar valve will close. So if we have RVH, P2 occurs later, widening the split.

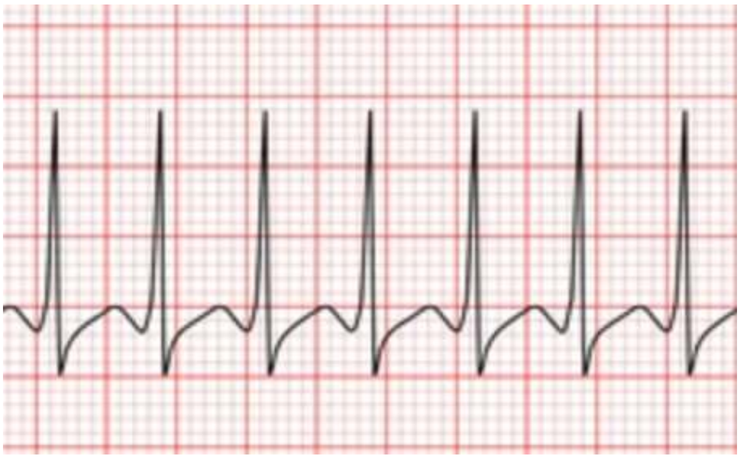
	- Wide splitting of S2, right-axis deviation on ECG, and right bundle branch block (RBBB) all = right ventricular hypertrophy on USMLE.
Paradoxical splitting of S2	- Means left ventricular hypertrophy on USMLE. - A2 occurs after P2 (normally we have A2 before P2). - Left ventricular pressure is high and A2 delayed to the point that it actually occurs on the opposite side of P2. - Paradoxical splitting of S2, left-axis deviation on ECG, and left bundle branch block (LBBB) all = left ventricular hypertrophy on USMLE.
Subclavian steal syndrome	- Confusing condition when you're first learning things but is high-yield on 2CK. - The vertebral artery (goes to brain) is the first branch of the subclavian artery (goes to arm). - If there is a narrowing/stenosis of the proximal subclavian prior to the branch point of the vertebral artery, this can lead to lower pressure in the vertebral artery. - This can cause a backflow of blood in the vertebral artery, producing miscellaneous neuro findings such as dizziness. - **Blood pressure is different between the two arms.** - USMLE will ask the Q one of two ways: 1) they'll give you dizziness in someone who has BP different between the arms and then ask for merely "subclavian steal syndrome," or "backflow in a vertebral artery" as the answer. Or 2) they'll give you BP in one of the arms + give you dizziness, then the answer will be, "Check blood pressure in other arm." - Next best step in Dx is CT or MR angiography (asked on 2CK NBME). - I should point out that probably 3/4 questions on USMLE where blood pressure is different between the arms, this refers to aortic dissection. But 1/4 is subclavian steal syndrome. As per my observation.
Vertebral artery stenosis	- Presents same as subclavian steal syndrome with otherwise unexplained dizziness, but blood pressure is not different between the arms because the subclavian is not affected. - Caused by atherosclerosis. CT or MR angiography can diagnose. - "Vertebrobasilar insufficiency" is a broader term that refers to patients who have either subclavian steal syndrome or vertebral artery stenosis.
Vertebral artery dissection	- 2CK Neuro forms assess vertebral artery dissection, where they want you to know a false lumen created by dissection in a vertebral artery can lead to stasis and clot formation, which in turn can embolize to the brain and cause stroke. - NBME can mention recent visit to a chiropractor (neck manipulation is known cause). - The answer on the NBME is **heparin** for patients who have experienced posterior stroke due to vertebral artery dissection. Sounds weird because it's arterial, but it's what USMLE wants. Take it up with them if you think it's weird.
Carotid artery dissection	- Shows up on 2CK form as patient with stroke-like presentation + who simultaneously has **ipsilateral facial/neck pain.** **-** The pain is due to stretching of nociceptors secondary to vascular dilation. - Stasis within false lumen can lead to embolus to brain/eye.
Carotid artery stenosis	- Caused by atherosclerosis. - HTN biggest risk factor for atherosclerosis specifically of the carotids (strong systolic impulse pounds the carotids → endothelial damage → atheromatous plaque formation). - Carotid bruit only seen in about 25% of Qs. Don't rely on this as crutch. - Vignette will give a stroke, TIA, or retinal artery occlusion in the setting of a patient with HTN. → You have to be able to make the association that a plaque from one of the carotids has launched off, since HTN = ↑ risk.

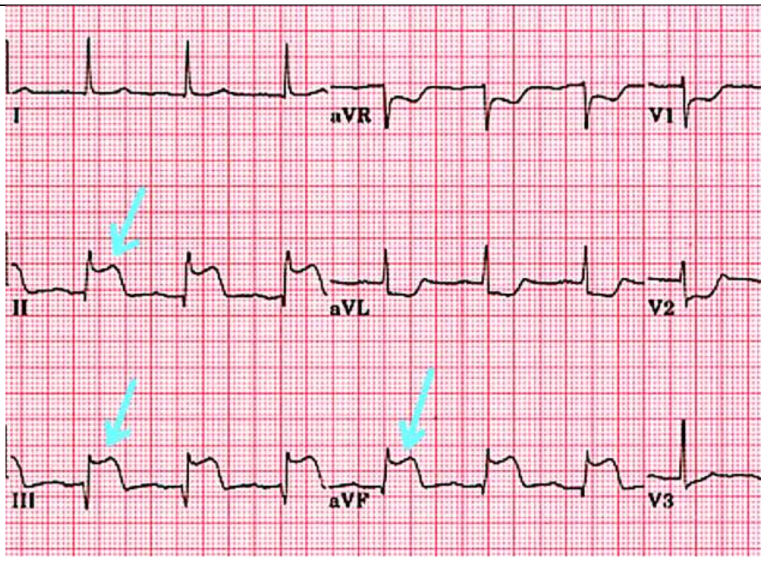
	- USMLE will then ask for management (2CK only): - Do carotid duplex ultrasonography as next best step in diagnosis to look for degree of occlusion. I've never seen carotid angiography as a correct answer on NBME exams. - If occlusion >70% symptomatic, or >80% asymptomatic, then do endarterectomy. "Symptomatic" = stroke, TIA, or retinal artery occlusion. A mere bruit is not a symptom; that is a sign. - If under these thresholds, do medical management only, which requires a triad of: 1) statin; 2) ACEi or ARB; and 3) anti-platelet therapy. - The USMLE will not force you to choose between low- and high-potency statins. - USMLE tends to list lisinopril as their favorite ACEi for HTN control. - It's to my observation aspirin alone is sufficient on NBME exams for anti-platelet therapy, even though in real life patient can receive either aspirin alone; the combo of aspirin + dipyridamole; or clopidogrel alone. - USMLE will not give borderline carotid occlusion thresholds – i.e., they'll say either 30% or 90%. If they list the % as low, look at the vignette for the drugs they list the patient on. Sometimes they'll show the patient is already on statin, lisinopril, and aspirin, and then the answer is just "continue current regimen." I have once seen "add clopidogrel" as a wrong answer in this setting, which makes sense, since the combo of aspirin + clopidogrel is never given anyway. - Sometimes they will give you a low carotid occlusion % + say the patient is on 2 of 3 drugs in the triad, and then the answer is just "add aspirin," or "add statin," or "add lisinopril." - If the vignette doesn't mention elevated BP but says you have some random dude over 50 with a stroke, TIA, or retinal artery occlusion, the next best step is carotid ultrasonography to look for carotid stenosis. In other words, it is assumed the patient has a carotid plaque in this setting. - If the vignette gives patient with episodes of unexplained *syncope* or light-headedness, but not stroke, TIA, or retinal artery occlusion, then the next best step is ECG, followed by Holter monitor, looking for atrial fibrillation (AF causes LA mural thrombus that launched off to brain/eye). - The triad of 1) statin; 2) ACEi or ARB; and 3) anti-platelet therapy is also done for general peripheral vascular disease unrelated to carotid stenosis (i.e., if a patient has intermittent claudication). - Stroke, TIA, or retinal artery occlusion, if they don't mention HTN, but they mention an *abdominal* bruit, you will still do a carotid duplex ultrasound. The implication is that the bruit in the abdomen could be a AAA or RAS, where atherosclerosis in one location means atherosclerosis everywhere, so the patient likely has carotid stenosis by extension. They once again need not mention carotid bruit; apparently it is not a sensitive finding (i.e., we cannot rule-out ↑ occlusion just because we don't hear it).
Aortic dissection	- Classically presents as severe upper chest pain radiating to the back between the scapulae. - USMLE loves dissection as the most common cause of AR due to retrograde propagation toward the aortic root. For example, patient with Hx of HTN, cocaine use, or a connective tissue disorder (i.e., Marfan, Ehlers-Danlos) who has a diastolic murmur, you should be thinking immediately that this is dissection. - As mentioned above, 3/4 Qs where BP is different between the arms refers to aortic dissection. A Q on 2CK form has "thoracic aortic dissection" where not only is the BP different between the arms, but it's also different between the L and R *legs* (i.e., L-leg BP is different from R-leg BP) → sometimes thoracic aortic dissections can anterograde propagate all the way down to the abdominal aorta.

	 - You do **not** need to memorize these aortic aneurysm types. I'm just showing you that if the common iliacs are involved (as with left image), BP can differ as well *between the legs*.
Aortic aneurysm	- Can present as "visible pulsation" on USMLE. - For aortic aneurysm, they can say "visible pulsation above the manubrium," or "pulsatile mass above the manubrium." **There can also be a tracheal shift. I've seen students select pneumothorax here.** But for whatever reason you can get tracheal shift in thoracic aortic aneurysm. For AAA, there can be "visible pulsation in the epigastrium." - Biggest risk factor for AAA is smoking. - For Family Med, do a one-off abdominal ultrasound in both men and women 65+ who are ever-smokers. This screening used to be just performed on men, but now it includes women. - For Surgery, AAA repair is indicated if the aneurysm is >5.5 cm or the rate of change of size increase is >0.5cm/month for 6 months. This is on Surgery form, where they give a patient with a 4-cm AAA and ask why serial ultrasounds are indicated → answer = "size of aneurysm." - Diabetes is protective against aneurysm. Non-enzymatic glycosylation of endothelium causes stiffening of the vascular wall. - Don't do AAA repair on USMLE in patient who has advanced comorbidities or terminal disease (e.g., stage 4 lung cancer).

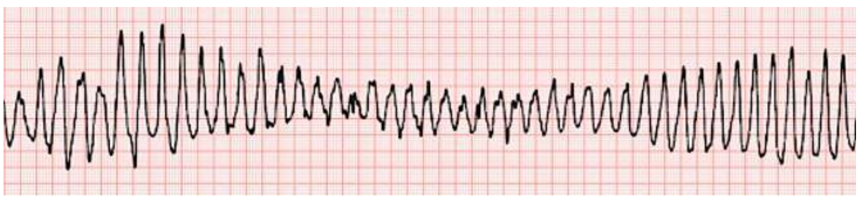
HY Murmur / ECG points for USMLE	
Atrial fibrillation (AF)	- Described as "irregularly irregular" rhythm with absent p-waves. Atrial fibrillation  - Notice how the QRS complexes are at random and irregular distances from one another. This is the "irregularly irregular" pattern. - AF is hugely important because it can cause turbulence/stasis within the left atrium that leads to a LA mural thrombus formation. This thrombus can launch off (i.e., become an embolus) and go to brain (stroke, TIA, retinal artery occlusion), SMA/IMA (acute mesenteric ischemia), and legs (acute limb ischemia). - AF HY in older patients, especially over 75. Vignette will usually be an older patient with a stroke, TIA, or retinal artery occlusion, who has *normal* blood pressure (this implies carotid stenosis is not the etiology for the embolus).

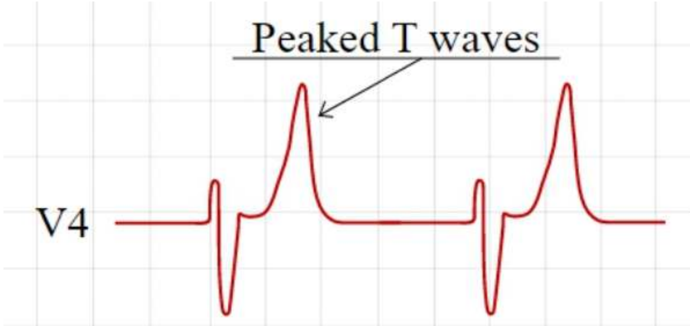
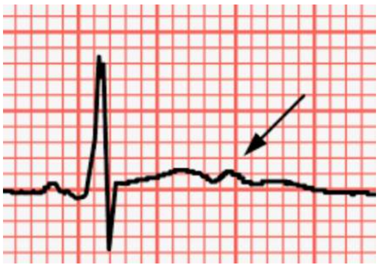
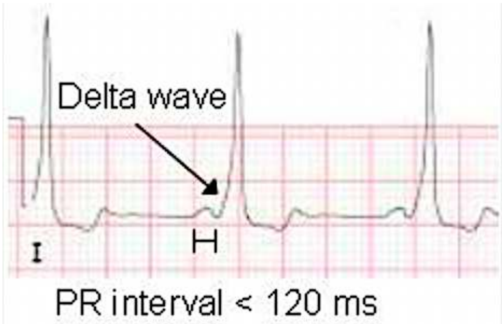
	- AF usually is paroxysmal, which means it comes and goes. The vignette might say the patient is 75 + had a TIA + BP normal + ECG shows sinus rhythm with no abnormalities → **next best step is Holter monitor (24-hour ambulatory ECG monitor)** to pick up the paroxysmal AF (e.g., when the patient goes home and has dinner). - After AF is diagnosed with regular ECG or Holter, 2CK wants **echocardiography** as the next best step to visualize the LA mural thrombus. - Patient who has severe abdominal pain in setting of AF or hyperthyroidism (which can cause AF), diagnosis is acute mesenteric ischemia; next best step is mesenteric angiography; Tx is laparotomy if unstable (answer on NBME). - Severe pain in a leg + absent pulses in patient with irregularly irregular rhythm = acute limb ischemia; USMLE wants "embolectomy" as answer. - Any structural abnormality of the heart, either due to LV hypertrophy, ischemia, growth hormone/anabolic steroid use, prior MI, etc., can lead to AF. - For 2CK, you need to know AF patient will get either NOAC or warfarin. Aspirin is no longer indicated for AF management. This is determined by the CHADS2 score. There are variations of the score, but the simple **CHADS2** suffices for USMLE → **C**HF, **H**TN, **A**ge 75+, **D**iabetes, **S**troke/TIA/emboli. Each component is 1 point, but stroke/TIA/emboli is 2 points. More points favors warfarin. - "Emboli" refers to Hx of AF leading to stroke, TIA, acute, mesenteric ischemia, or acute limb ischemia – i.e., any Hx of embolic event. 2CK IM form 7 gives short vignette of 67F with chronic AF + Hx of acute limb ischemia + no other info relating to CHADS, and answer is warfarin to prevent recurrence; aspirin is wrong. - Some students will ask about NOACs, e.g., apixaban, etc., for non-valvular AF → I've never seen NBME care about this stuff. They seem to be pretty old-school and just have warfarin as the answer, probably because there isn't debate around whether it can be used; use of NOACs is less textbook. - AF patient should also be on rate control before rhythm control. The USMLE actually doesn't give a fuck about this component of management, although in theory metoprolol or verapamil is standard. You could be aware for Step 3 that flecainide is first-line for rhythm control if patients fail rate-control and have a structurally normal heart and no coronary artery disease. - 2CK form has "electrical cardioversion" as the answer for patient with AF who has hemodynamic instability (i.e., low BP). What you need to know is: sometimes AF can trigger "rapid ventricular response," where HR goes >150 and low BP can occur.
Atrial flutter	- Has classic sawtooth appearance.  - Low yield for USMLE. I think it's asked once on a 2CK NBME. But as student you should know it exists / the basic ECG above.
Ventricular tachycardia (VT)	- Causes wide-complex QRS complexes (>120 ms; normal is 80-120 ms). 

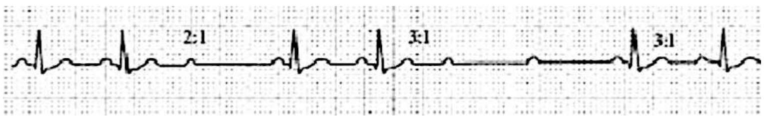
	- Exceedingly HY for 2CK that you know VT is wide-complex, whereas SVT is narrow-complex. If you look at above ECG, even if you say, “No idea what I’m looking at.” You can tell the complexes look wide like mountains in comparison to a typical ECG. - VT is treated with anti-arrhythmics – i.e., amiodarone. If patient has coma or hemodynamic instability (low BP), the NBME answer is direct current countershock or cardioversion (same thing). - Premature ventricular complex (PVC) is asked on 2CK.  - Note on the above strip, we have a wide complex (meaning ventricular in origin) that occurs earlier (hence premature). What they do on the NBME is show you this strip and ask where this abnormality originates from, then the answer is just “ventricle.” - Don’t treat PVCs on USMLE.
Supraventricular tachycardia (SVT)	- Causes narrow / needle-shaped complexes. Make sure you’re able to contrast this with VT above, which is wide-complex.  - Notice the complexes are narrow / look like needles. This means the tachy originates above the ventricles (hence SVT). - Treatment of SVT exceedingly HY on 2CK. - First step is carotid massage (aka vagal maneuvers). In kids, they can do icepack to the face. - If the above doesn’t work, the next step is give adenosine (not amiodarone). - Same as with VT, if the patient has coma or low BP, shocking the patient is the first step. In other words, for both SVT and VT, you must shock first in the setting of coma or hemodynamic instability. It’s for stable SVT and VT that the treatments differ on USMLE.
Acute MI (STEMI)	- Will present as ST-elevations in 3-4 contiguous leads.

	 - The above is an inferior MI, as evidenced by ST-elevations in leads II, III, and aVF. The answer for the affected vessel is the posterior descending artery (PDA supplies the diaphragmatic surface of the heart); since >85% of people have right-dominant circulation (meaning the PDA comes of the right main coronary), sometimes the answer for inferior MI can just be "right coronary artery." - The apex of the heart is supplied by the left anterior descending artery (LAD). If there are ST-elevations in leads V1-V3, choose LAD as the answer. - The left-lateral heart is supplied by the left circumflex artery. If there are ST-elevations in leads V4-V6 for lateral MI, choose left circumflex. - Reciprocal ST-depressions in the anterior leads V1-V3 can reflex posterior wall MI (i.e., we have "elevations" out the back of the heart, so they look like depressions on the anterior wall leads). - Post-MI papillary muscle rupture resulting in mitral regurg is exceedingly HY. As discussed earlier, if patient has MI followed by new-onset systolic murmur hours to days later, with or without dyspnea, that's mitral regurg. - Stroke-like presentation in patient who had MI weeks ago → answer = "embolus from ventricular septal aneurysm." - Most common cause of death due to MI is ventricular fibrillation (VF). - Fibrosis of myocardium in the months-years post-MI increases risk of arrhythmias such as AF, SVT, VT, etc. There's no specific arrhythmia you need to memorize. Just know the risk is there in the future. - Q waves on an ECG mean old MI / history of MI. The vignette might give you patient who has light-headedness / fainting + they say patient has Q waves in II, III, aVF, and the answer will be something like "paroxysmal supraventricular tachycardia." Student thinks this specific arrhythmia matters, but it doesn't. The point is that Hx of MI means patient is at risk for nearly any arrhythmia now. - With cardiogenic shock as a result of MI, the important arrows are: ↓ cardiac output, ↑ peripheral vascular resistance, ↑ PCWP. For Family Med forms, they won't ask arrows as answers, but they'll say in the vignette itself whether these parameters are up or down. - MI can lead to acute tubular necrosis from cardiogenic shock → acute drop in renal perfusion. This is not pre-renal. I discuss this in detail in the renal section. - First treatment for MI is aspirin. After aspirin is given, the next drug to give is clopidogrel (an ADP P2Y12 blocker) as dual anti-platelet therapy. - USMLE wants you to know anyone with acute coronary syndrome (i.e., MI or unstable angina) gets coronary catheterization. This is answer on new 2CK NBME exam. - It's to my observation that more extensive management of MI on USMLE, such as use of beta-blockers, nitrates, morphine, oxygen, statin, percutaneous

	coronary intervention, etc., isn't assessed in detail. I can comment, however, that one 2CK Q wants you to know nitrates are contraindicated in right-heart MIs, which includes inferior MI in most people due to the right coronary supplying the PDA. This is because right-sided MIs are preload-dependent, which means they need sufficient preload to maintain BP.
Pericarditis	- Shows up on ECG as diffuse ST-elevations (i.e., in all leads rather than 3-4 contiguous leads as with MI). PR depressions can also be seen, but I've never seen the USMLE give a fuck about the latter. - Patient will have pain that's worse when lying back, better when leaning forward. In turn, the patient can walk through the door into the clinic bent over at the waist. - Serous pericarditis will be post-viral, secondary to autoimmune disease, or due to cocaine use. - NBME Q gives pericarditis + a bunch of different organism types (i.e., bacterium, fungus, etc.), and answer is "virus." - Patient with rheumatoid arthritis or SLE notably at risk for pericarditis. In other words, don't get confused if they mention pericardial friction rub in vignette of RA or SLE; this is common. - For cocaine use, they'll say a 22-year-old male has chest pain after a night of heavy partying + ECG shows diffuse ST-elevations → Dx = pericarditis. - Uremic pericarditis is HY for 2CK. Q will give ultra-high creatinine and BUN and say there's a friction rub → treatment = hemodialysis. - Treatment for pericarditis is same as acute gout → NSAIDs, colchicine, steroids. - Fibrinous pericarditis is post-MI and occurs as two types: 1) literally "post-MI fibrinous pericarditis," which will simply be friction rub within days of an MI; 2) Dressler syndrome (antibody-mediated fibrinous pericarditis occurring 2-6 weeks post-MI). - ECG is first step in Dx of pericarditis, but USMLE wants echocardiography as next best step in order to visualize a concomitant effusion that can occur sometimes.
Chronic constrictive pericarditis	- I should make note that chronic constrictive pericarditis is a separate condition that doesn't present with the standard pericarditis findings as described above. - This is low-yield for USMLE, but students ask about it because it can be confused with tamponade. - There's two ways this can show up: 1) **Tuberculosis** is a classic cause; there may or may not be calcification around the heart on imaging. So if you get a Q where patient has TB + some sort of heart-filling impairment → answer = chronic constrictive pericarditis. 2) **Kussmaul sign** will be seen in the Q, where JVD occurs with *inspiration* rather than expiration. - Normally, inspiration facilitates RA filling (↓ intrathoracic pressure → ↑ pulmonary vascular compliance/stretching → ↑ high-low pressure gradient from right heart to the lungs → ↓ in afterload on RV from the lungs → blood moves easier from right heart to the lungs → blood is pulled easier from SVC/IVC to the RA). - However, if there is ↑ compressive force on the heart, the ↑ in negative intrathoracic pressure during inspiration is not transmitted to the right side of the heart, so JVP does not ↓ (and can even paradoxically can ↑). - In tamponade, however, as discussed below, the ↑ in negative intrathoracic pressure during inspiration *is able* to be transmitted to the right side of the heart, so Kussmaul sign does not occur. This is likely because in constrictive pericarditis, the rigid pericardium prevents expansion of the right heart altogether, whereas in tamponade, the pericardium isn't rigid per se, but is just

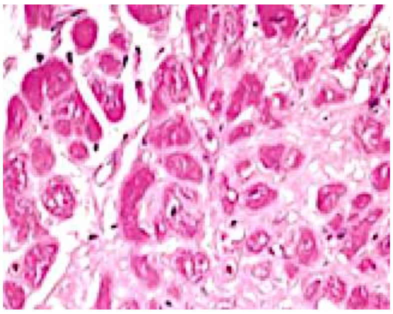
	filled with blood that can move/shift during the respiratory cycle, thereby allowing right heart expansion during inspiration.
Pericardial effusion / Cardiac tamponade	- Cardiac tamponade = pericardial effusion + low blood pressure. - What determines whether we have a tamponade or not is the *rate* of accumulation of the fluid, not the volume of the fluid – i.e., a stab wound or post-MI LV free-wall rupture resulting in fast blood accumulation, even if smaller volume, might cause tamponade, but cancer resulting in slow, but large, accumulation might not cause tamponade. - Tamponade presents as **Beck triad: 1) hypotension, 2) JVD, 3) muffled/distant heart sounds.** The question will basically always give hypotension and JVD. Occasionally they might not mention the heart sounds. But you need to memorize Beck triad as HY for tamponade. - Pulsus paradoxus (i.e., drop in systolic BP >10 mm Hg with inspiration) is classically associated with tamponade, although not frequently mentioned in vignettes. I've seen a 2CK NBME Q where they say "the pulsus paradoxus is <10 mm Hg," which is their way of saying the Dx is not tamponade. I consider that wording odd, but it's what the vignette says. - ECG will show electrical alternans / low-voltage QRS complexes.  - You can see the amplitudes (i.e., heights) of the complexes are short. This refers to "low-voltage." You can also see the heights ever so slightly oscillate up and down. This refers to electrical alternans. They show this ECG twice on 2CK NBMEs. - USMLE wants ECG → echocardiography → pericardiocentesis or pericardial window as the management sequence. Students often erroneously jump on pericardiocentesis, but NBME wants echo first to confirm diagnosis. 2CK NBME has pericardial window as answer, where pericardiocentesis isn't listed.
Torsades de pointes (TdP)	- HY type of VT that has sinusoidal pattern on ECG.  - USMLE wants you to know this can be caused by some anti-arrhythmic agents, such as the sodium- and potassium-channel blockers, such as quinidine and ibutilide, respectively. They ask this directly on the NBME exam, where Q will say patient is given ibutilide + what is he now at increased risk of → answer = torsades. - QT prolongation is risk factor for development of TdP. Agents such as anti-psychotics, macrolides, and metoclopramide prolong the QT.

	- Tx USMLE wants is magnesium (asked directly on new 2CK form), which stabilizes the myocardium in TdP.
Peaked T wave	- Seen in hyperkalemia.  - Asked once on one of the 2CK forms, where they show the ECG. - Highest yield point is that if a patient has hyperkalemia and ECG changes, the Tx USMLE wants is IV calcium gluconate or calcium chloride, which stabilizes the myocardium. Calcium gluconate is classic, but calcium chloride shows up as an answer on a 2CK NBME.
U-wave	- Means hypokalemia.  Shows up on 2CK NBME in anorexia patient. First time I've ever seen it show up anywhere on NBME material. But Q doesn't ride on you knowing it means hypokalemia to get it right. It's HY and pass-level to know that purging (anorexia or bulimia) causes hypokalemia anyway.
Delta wave	- Seen in Wolff-Parkinson-White syndrome (WPW; accessory conduction pathway in heart that bypasses the AV node, resulting in reentrant SVT). - Classically described as a "slurred upstroke" of the QRS, where the PR interval is shortened.  - Both the delta-wave and WPW have basically nonexistent yieldness on USMLE, but I mention them here so you are minimally aware.
J waves	- They mean hypothermia. You don't need to be able to identify on ECG. Just know they exist, as they show up in a 2CK vignette where patient has body temperature of 89.6 F (not 98.6).

HY Heart block points for USMLE	
First degree	- Prolonged PR interval (>200 ms). Should normally be 80-120 ms.  - Note that above on the ECG, the PR-segment in particular (just prior to the QRS complex) is extra-long. - Not really assessed on USMLE. Just know the definition. - Don't treat on USMLE.
Second degree Mobitz type I (aka Wenckebach)	- Gradually prolonging PR interval until QRS drops. Then cycle repeats.  - Don't treat on USMLE.
Second degree Mobitz type II	- *No* gradual prolongation of PR interval, followed by a random dropping of the QRS.  - Can also sometimes occur as patterns of 2:1, 3:1, etc., where there will be a P to QRS ratio of 2:1 or 3:1, etc.  - Regardless as to whether the dropped QRS is random or in a numerical pattern, there is no gradual prolongation of the QRS before the dropped complex. - More dangerous than Mobitz I. This is because Mobitz II has higher chance of progression into type III heart block. - Treatment on USMLE is insertion of pacemaker. This is asked on a new 2CK NBME exam.
Third degree	- Two things you want to look for on ECG: - 1) Ultra-slow HR (i.e., 30-40). You'll see the QRS's are super far apart. This is the ventricular escape rhythm. - 2) No relationship between the P-waves and QRS complexes. **Third degree AV block**  - Treatment on USMLE is insertion of pacemaker.

	- So what you want to remember is that Mobitz II and 3rd-degree are the ones where we insert pacemaker; 1st-degree and Mobitz I we don't.

	HY Cardiomyopathy points for USMLE
Dilated (DCM)	- Can be isolated ventricular or diffuse 4-chamber dilation. Causes are multifarious, but a key feature is **systolic dysfunction,** where ejection fraction is reduced (i.e., <55%, where normal range is 55-70). - CXR shows enlarged cardiac silhouette.  - An S3 heart sound can sometimes be heard. - Cardiac exam shows lateralized apex beat. Can be described as the point of maximal impulse being in the anterior axillary line. Should be noted that this lateralization just means an enlarged LV, so it is non-specific, and can also be seen in LV hypertrophy from any cause. But many vignettes will mention it. - The arrows USMLE wants are: ↓ EF; ↑ LVEDV; ↑ LVEDP. I discuss this stuff in more detail in my HY Arrows PDF. - Causes of DCM are **ABCD**: - **A**: Alcohol. - **B**: Wet Beriberi (thiamine deficiency). - **C**: Coxsackie B virus, Cocaine, Chagas disease. - **D**: Drugs → doxorubicin (aka Adriamycin). - Other notable causes are: - Pregnancy (peripartum cardiomyopathy); hemochromatosis; rheumatic heart disease (myocarditis leading to DCM).
Hypertrophic (HCM)	- Heart failure due to systemic hypertension. - Characterized by diastolic dysfunction (the heart can pump just fine but cannot expand as easily). - Can be associated with S4 heart sound. - The arrows USMLE wants are: ↔ EF; ↔ LVEDV; ↑ LVEDP. - Ejection fraction is normal because the heart can pump perfectly fine. - Students get confused about LVEDV, thinking it should be low, if the heart cannot expand as easily. But this is not the case for USMLE. They want you to know normal volume can be achieved; *it just merely requires more force/pressure to get there.* - As with DCM, the apex beat / point of maximal impulse can be lateralized, which merely reflects LV hypertrophy. - As described earlier, paradoxical splitting of S2, left-axis deviation on ECG, and LBBB can all be seen due to LVH. When you see these findings in vignettes, don't get confused. They just mean LVH. - Often associated with hypertensive retinopathy (fundoscopy shows narrowing of retinal vessels, flame hemorrhages, and "AV-nicking"),

	hypertensive nephropathy (hyperplastic arteriolosclerosis with increased creatinine).
Hypertrophic obstructive (HOCM)	- Caused by mutations in β-myosin heavy-chain gene; autosomal dominant; results in disordered/disarrayed myocardial fibers. - HOCM causes asymmetric septal hypertrophy that results in the anterior *mitral* valve leaflet obstructing the LV outflow tract (so it can sound similar to aortic stenosis). - Classically sudden death in young athlete; cause of death is **ventricular fibrillation** due to acute left heart strain in the setting of fast heart rate. - Presents with systolic murmur that worsens with Valsalva or standing (aortic stenosis, in contrast, gets softer or experiences no change with Valsalva) or standing. - HOCM and MVP are the only two murmurs that get worse with less volume in the heart. All other murmurs get worse with more volume. Valsalva increases intra-thoracic pressure and decreases venous return, so there's less volume in the heart. Standing simply decreases venous return. - Beta-blockers (metoprolol or propranolol are both answers on NBME) are given to slow heart rate, which maximizes diastolic filling and decreasing symptoms / risk of death.
Restrictive (RCM)	- Heart failure due to diastolic dysfunction, where HTN is not the cause. - **JVD** is HY for RCM. An S4 can also be seen. The heart will not be dilated. - HY causes are Hx of radiation (leads to fibrosis), amyloidosis, and hemochromatosis. - Student might say, "I thought you said hemochromatosis was DCM. So if we have to choose on the exam, which one is it?" The answer is, whichever the vignette gives you. If they say a large cardiac silhouette with an S3 and lateralized apex beat, that's DCM. If they say JVD + S4 + nothing about a lateralized apex beat, you know it's RCM. - Amyloidosis is protein depositing where it shouldn't be depositing. Highest yield cause of amyloidosis on USMLE is multiple myeloma, which will lead to RCM.  Cardiac amyloidosis. Myocardium is pink; amyloid is white. - Since RCM is diastolic dysfunction, the arrows are the same as HCM, which are: ↔ EF; ↔ LVEDV; ↑ LVEDP. Once again, for FM vignettes, they will incorporate these arrows within stems as the values they give you.

HY Angina points for USMLE	
Stable angina	- Chest pain that occurs predictably with exercise. - Due to atherosclerotic plaques causing >70% occlusion; can be calcific. - Classically causes ST depressions on ECG. - Nitrates (e.g., sublingual isosorbide dinitrate) used as Tx. - Nitrates are contraindicated with PDE-5 inhibitors (i.e., sildenafil, tadalafil) due to risk of low blood pressure. This is because PDE-5 inhibitors prevent breakdown of cGMP; nitrates increase production of cGMP.
Unstable angina	- Chest pain that is unpredictable and can occur at rest. - Due to partial rupture of atherosclerotic plaque leading to partial occlusion. - ST depressions on ECG. - Diltiazem is answer on new 2CK NBME for patient with unstable angina. - Patients need cardiac catheterization.
Prinzmetal angina (variant angina pectoris)	- Vasospastic angina that occurs at rest (i.e., watching TV or while sleeping) in younger adults; it is not caused by atherosclerosis. - ST elevations are seen on ECG. - You must know that Prinzmetal is also known as variant angina pectoris. There is an NBME Q that gives vignette of Prinzmetal, but answer is "variant angina pectoris." - Treatment is nitrates (can cause coronary artery dilation unrelated to the venous pooling effects) or dihydropyridine calcium channel blockers (e.g., nifedipine). Avoid α1-agonists in these patients (cause vasoconstriction), as well as non-selective β-blockers like propranolol (can cause unopposed α effects).

HY Endocarditis points for FM	
Acute endocarditis	- Bacterial infection of valve in patient with no previous heart valve problem. - Caused by *Staph aureus* on USMLE. - Left-sided valves (i.e., aortic and mitral) most commonly affected because of greater pressure changes (i.e., from high to low) within left heart, resulting in turbulence that enables seeding. - IV drug users → venous blood inoculated with *S. aureus* → travels to heart and causes vegetation of tricuspid valve.
Subacute endocarditis	- Bacterial infection of valve in patient *with history* of valve abnormality (i.e., congenital bicuspid aortic valve, Hx of rheumatic heart disease). - Caused by *Strep viridans* on USMLE. You need to know *S. viridans* is can be further broken down into: *S. sanguinis, S. mutans,* and *S. mitis*. - Hx of dental procedure is HY precipitating event, where inoculation of blood occurs via oral cavity → previously abnormal valve gets seeded.
HY points	- New-onset murmur + fever = endocarditis till proven otherwise on USMLE. - Reactive thrombocytosis (i.e., high platelets) can occur due to infection. This is not unique to endocarditis, but it is to my observation USMLE likes endocarditis as a notable etiology for it. In other words, if you get an endocarditis question and you're like, "Why the fuck are platelets 900,000?" (NR 150-450,000), don't be confused. - Hematuria can occur from vegetations that launch off to the kidney. - Endocarditis + stroke-like episode (i.e., focal neurologic signs) = septic embolus, where a vegetation has launched off to the brain. - Janeway lesions, Osler nodes, splinter hemorrhages, etc., are low-yield for USMLE and mainly just school of medicine talking points. You don't need to worry about management, treatment, or surgical prophylaxis guidelines for FM. You just need to be able to look at a vignette and merely diagnose endocarditis. Then in practice, you would send to hospital.

Rheumatic heart disease (rheumatic fever) HY points
- *Strep pyogenes* (Group A Strep) oropharyngeal infection results in production of antibodies against *S. pyogenes*' M-protein that cross-react with the mitral valve (i.e., molecular mimicry; type II hypersensitivity). - Can occur with the aortic valve in theory, but on USMLE, it is always mitral valve. - Results in mitral regurgitation acutely and mitral stenosis late, as discussed earlier. - Presents as **JONES** (J♥NES) → **J**oints (polyarthritis), ♥ **C**arditis, subcutaneous **N**odules, **E**rythema marginatum (annular, serpent-like rash), **S**ydenham chorea (autoimmune basal ganglia dysfunction that results in dance-like movements of the limbs). - Cutaneous Group A Strep infections don't cause rheumatic fever, but can still cause PSGN. - Treatment is penicillin.  Erythema marginatum

Conditions confused for cardiac path	
Panic attack	- NBME forms love trying to make you think this is an MI. - They'll give you young, healthy patient who feels doom / like he or she is going to die. - Sometimes they mention in stem Hx of MI in family as distraction. - They can say patient has mid-systolic click, as discussed earlier, and then they ask for cause of patient's symptoms → answer = panic disorder, not MVP. Student gets confused, but MVP is almost always asymptomatic, where panic attack is clearly cause of the patient effusively hyperventilating. - Treat with benzo.
Orthostasis	- Orthostatic hypotension is defined as intravascular fluid depletion causing a drop of systolic BP >20 mmHg and diastolic BP >10 mmHg when going from supine to standing. - Shows up on 2CK IM form as exactly a drop of 20 and 10, respectively, for systolic and diastolic BPs in a patient with fainting → answer = "intravascular fluid depletion." - Diuretic use is big risk factor.
Vasovagal syncope	- Fainting in response to stressor (e.g., emotional trigger). - Stress triggers an initial sympathetic response, which in turn triggers a compensatory parasympathetic response. This latter response is excessive in some people, where the peripheral arterioles dilate and the heart slows too much → decreased cerebral perfusion → lightheadedness/fainting. - 2CK wants you to know a tilt-table test can be used to diagnose, where a reproduction of symptoms can occur.
Carotid sinus hypersensitivity	- USMLE likes this for both Steps 1 and 2. - They'll say dude was shaving then got lightheadedness or fainted. Mechanism is ↑ stretch of carotid sinus baroreceptors → ↑ afferent CN IX

	firing to solitary nucleus of the medulla → ↑ efferent CN X parasympathetic firing down to cardiac nodal tissue → ↓ HR → ↓ CO → ↓ cerebral perfusion.
Costochondritis	- Inflammation of cartilage at rib joints. - Will present as chest pain that worsens with palpation or when patient reaches over the head or behind the back. These two findings are clear indicators we have an MSK condition, not cardiac. - Can be idiopathic, caused by strain (e.g., at the gym), or even post-viral.
Pleurodynia	- MSK condition asked twice on 2CK material (once on FM form; also on Free 120) that has nothing to do with the lungs, despite the name. - This is viral infection (Coxsackie B) causing sharp lateral chest pain due to intercostal muscle spasm. Sometimes students choose pericarditis, etc., even though the presentations are completely disparate. - Creatine kinase can be elevated in stem due to ↑ tone of muscle.
Viral pleurisy	- Viral infection causing inflammation of the pleura (layers covering the lungs), leading to sharp chest pain. - If this is the answer, CK will be normal (unlike pleurodynia, because it's not MSK).
Diffuse esophageal spasm	- Can cause angina-like pain in patient without cardiovascular disease. - Barium swallow shows a corkscrew appearance of the esophagus.
Gastroesophageal reflux	- Can present as chest pain confused for MI. ECG will be normal, clearly. - I discuss GERD in detail in GIT section later.

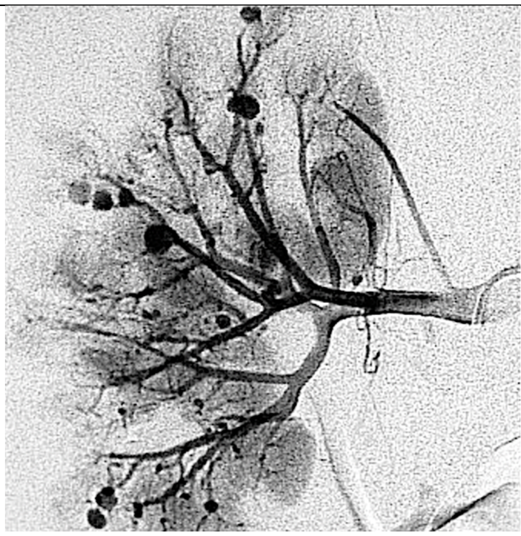
Arterial vs venous disease	
Arterial disease	- Caused by atherosclerotic disease; presents as **diminished peripheral pulses** in patient over 50 who has risk factors, e.g., diabetes, smoking, HTN. - Lower legs can be shiny and glabrous (trophic changes). - Arterial ulcers are small and punched-out; located on tops/bottoms of feet and toes.  - **Ankle-brachial indices (ABIs) are first step in diagnosis** (exceedingly HY on 2CK), which compare BP in ankle to the arm; if <0.9, this reflects ↓ peripheral blood flow due to atherosclerosis. - If ABIs are not listed as first step in diagnosis for whatever reason, choose Doppler ultrasound. There is one 2CK NBME Q where this is the case. - After ABIs, next step is exercise stress test (if listed) in order to determine exercise tolerance. If not listed, go straight to "recommend an exercise / walking program." Do not choose cilostazol first or arteriography as answers. - 2CK form has "prescription for an exercise program" as the answer. Students say, "Why does it say 'prescription'?" No fucking idea. Ok?

Venous disease	- Congestion of venous system usually from valvular incompetence; idiopathic / familial; varicose veins are one type of venous disease and are not synonymous; patients can have venous disease without varicosities. - **Peripheral pulses are normal** (those reflect arteries, not veins). - Lower legs demonstrate "brawny edema," which is a brown, hemosiderin-laden edema due to ↑ pressure / micro-extravasations; hyperpigmentatory changes resulting in brown/red skin is known as stasis dermatitis, aka post-phlebitic syndrome; the latter is a term is asked on 2CK, so know the annoying vocab. - Venous ulcers are large and sloughy, and located at the malleoli.  - Diagnose with venous duplex ultrasonography of the legs; first treatment is compression stockings. Never choose answers such as venous stripping or glue agents, etc. - Venous disease ↑ risk for DVT and superficial thrombophlebitis. If patient has active DVT or STP, answer = subcutaneous enoxaparin (heparin) over compression stockings.

Cardiac stress test points (2CK only)	
- Most 2CK Qs that ask about stress tests are in the context of evaluating patients for perioperative MI risk, however they are still asked in Family Med, where you need to know when a patient requires one. - The good news is that it is rare the Q will force you to choose between different types of stress tests. 4/5 Qs will just list one stress test, where it is simply assessing, "Do you know a stress test should be done, period, in this scenario." - Stress tests are also done for peripheral arterial disease prior to recommending an exercise/walking program (as mentioned above).	
Exercise ECG	- Most common stress test. - The answer on USMLE for patients who have stable angina, where you're looking for ST depressions (i.e., evidence of ischemia) with exertion. - **Requires a patient has a normal baseline ECG in order to perform.** - In other words, the Q will give you a big 15-line paragraph + mention in the last line that the patient's baseline ECG shows, e.g., a LBBB from a year ago that's unchanged. This means ECG stress test is wrong in this situation, since you need to have a normal ECG to do it. The 1/5 Qs that force you to choose between stress tests want you to know this detail, basically, where you just choose the non-ECG stress test instead.
Exercise echo	- Used to look for heart failure (i.e., ↓ EF) with exertion, not overt ischemia. - In other words, the answer on USMLE for patients who don't get chest pain with exertion (i.e., don't have stable angina), but who get shortness of breath with exertion. This reflects, at a minimum, left heart decompensation with possible ↓ EF.

	- Also the answer for patients who have abnormal baseline ECG.
Pharmacologic	- Refers to numerous answer choices on USMLE – i.e., dobutamine-echo, dipyridamole-thallium. - The answer on USMLE for patients who cannot exercise, such as in the setting of angina when merely walking up a single flight of steps, or in patients imminently undergoing major surgery (e.g., AAA repair), where perioperative MI risk needs to be assessed. I have seen both of these scenarios on 2CK forms. - The USMLE will typically not force you to choose between stress tests. As I mentioned at the top of this table, they will usually just have the pharmacologic stress test as the only one listed. - Dobutamine is a β1-agonist that stimulates the heart (i.e., ↑ oxygen demand). Echo can then be done to look for ↓ EF (i.e., heart failure). - Dipyridamole is a phosphodiesterase inhibitor that dilates arterioles. HR goes up to compensate, thereby ↑ myocardial oxygen demand. Thallium is then used to look at perfusion of the myocardium.
Cardiac scintigraphy	- "Cardiac scintigraphy" is a broad term that refers to any evaluation of the heart in which some form of radiotracer is used (i.e., thallium, technetium, sestamibi). - This is the same as pharmacologic stress test for all intents and purposes on USMLE, even though technically it need not require myocardium is stimulated and can just be used to look at blood flow to the heart in the resting state. - The point is: This is an answer on 2CK sometimes as just another way of them writing "pharmacologic stress test." Choose it if the patient cannot exercise.
Myocardial perfusion scan	- "Myocardial perfusion scan" is one type of cardiac scintigraphy that evaluates blood flow to myocardium. It is non-invasive, whereas coronary angiography is invasive and evaluates coronary blood flow via the use of a catheter. - This is interchangeable with cardiac scintigraphy and pharmacologic stress test on USMLE for all intents and purposes.

HY Vasculitides (fancy word that's pleural for vasculitis)	
Granulomatosis with polyangiitis	- Formerly known as Wegener granulomatosis. - Answer on USMLE for adult with triad of 1) hematuria, 2) hemoptysis, and 3) "head-itis" – i.e., any problem with the head, such as nasal septal perforation, mastoiditis, sinusitis, otitis. - Associated with cANCA and anti-proteinase 3 (anti-PR3) antibodies. - Causes "necrotizing glomerulonephritis" that can lead to rapidly progressive glomerulonephritis (RPGN).
Eosinophilic granulomatosis with polyangiitis	- Formerly known as Churg-Strauss. - Presents as combo of asthma + eosinophilia +/- head-itis. - Head-itis always seen in Wegener vignettes, but maybe only ~50% of CS Qs. - Renal involvement rare for CS. - Associated with pANCA and anti-myeloperoxidase (anti-MPO) antibodies.
Microscopic polyangiitis	- Will just present as hematuria in a patient who is pANCA / anti-MPO (+). - Similar to Wegener, can cause RPGN.
- The above three conditions can be associated with a weird neuropathy called mononeuritis multiplex, which means neuropathy of "one large nerve in many location" – e.g., wrist drop + foot drop in same patient. - You don't need to worry about the fancy term "mononeuritis multiplex," but what I do want you to be aware of is that neuropathy will sometimes show up in these vasculitis vignettes, so don't be confused about it.	
Polyarteritis nodosa (PAN)	- Medium-vessel vasculitis that causes a "string of pearls" appearance of the renal vessels.

	 - **Can be caused by hepatitis B.** - For whatever reason, USMLE wants you to know PAN spares the lungs – i.e., it does not affect the pulmonary vessels.
Takayasu arteritis	- Aka "pulseless disease." Classically affects Asian women 40s or younger. - Inflammation of large vessels, including the aorta. - Always affects the subclavian arteries (which supply the arms), which is why it can cause weakly, or non-palpable, pulse in the upper extremities.
Temporal arteritis	- Aka giant cell arteritis. - 9/10 Qs will be painful unilateral headache in patient over 50. I've seen one Q on NBME where it's bilateral. - Flares can be associated with low-grade fever and high ESR. - Patients can get proximal muscle pain and stiffness. This is polymyalgia rheumatica (PMR). The two do not always go together, but the association is HY. (Do not confuse PMR with polymyositis. The latter will present with ↑ CK and/or proximal muscle weakness *on physical exam*. PMR won't have either of these findings. I talk about this stuff in detail my MSK notes.) - Patients can get pain with chewing. This is jaw claudication (pain with chewing). - Highest yield point is we give **steroids before biopsy** in order to prevent blindness. - An NBME has "ischemic optic neuropathy" as the answer for what complication we're trying to prevent by giving steroids in temporal arteritis. - IV methylprednisolone is typically the steroid given, since it's faster than oral prednisone. - It's to my observation many 2CK NBME Qs will give the answer as something like, "Steroids now and then biopsy within 3 days," or "IV methylprednisolone and biopsy within a week." Students ask about the time frames, but for whatever reason USMLE will give scattered/varied answers like that. - Another 2CK Q gives easy vignette of temporal arteritis and then asks next best step in diagnosis → answer = biopsy. Steroids aren't part of the answer. Makes sense, since they're asking for a diagnostic step.
Thromboangiitis obliterans	- Aka Buerger disease; technically a vasculitis. - Dry gangrene of the fingers or toes seen generally in male over 30 who's a heavy smoker. - Treatment is **smoking cessation.** - Don't confuse with Berger disease, which is IgA nephropathy.
Ascending aortitis	- Tertiary syphilis can cause ascending aortitis + aortic aneurysm. - Causes "tree-barking" of the aorta.

Thrombophlebitis for FM	
Deep vein thrombosis	- DVT will be unilateral thigh or lower leg swelling in patient with risk factors such as: post-surgery, prolonged sedentation, OCP use, Hx of thrombotic disorders (e.g., Factor V Leidin, prothrombin mutation). - Virchow triad for ↑ DVT risk: 1) venous stasis (e.g., post-surgery sedentation), 2) hypercoagulable state (e.g., estrogen use, underlying malignancy), 3) endothelial damage (i.e., smoking). **- OCPs contraindicated in smokers over 35** because estrogen causes hypercoagulable state for two reasons: 1) estrogen upregulates fibrinogen; 2) estrogen upregulates factors Va and VIIIa. - USMLE loves nephrotic syndrome as cause of DVT (loss of antithrombin III in the urine → hypercoagulable state). - Antiphospholipid syndrome → DVTs despite paradoxical ↑ PTT (i.e., if PTT is high, you'd think you have bleeding diathesis, not thromboses); may or may not be due to SLE. Antibodies against phospholipids cause *in vivo* clumping of platelets + ↑ clot initiation, but disruption of *in vitro* PTT assay means ↑ PTT. - Major danger is DVT can embolize to lungs causing PE → acute-onset shortness of breath and tachycardia + death if saddle embolus. - Homan sign can mean DVT, which is pain in the calf with dorsiflexion of foot. - Diagnose DVT with duplex venous ultrasound of the leg/calf. - Treatment is heparin.
Trousseau sign of malignancy	- Migratory thrombophlebitis classically due to head of pancreas adenocarcinoma. But this can also be seen with adenocarcinomas in general, e.g., pulmonary.
Superficial thrombophlebitis	- Painful palpable cord in the ankle that may or may not track up to the knee. - Seen in patients with venous insufficiency. - Answer is "subcutaneous enoxaparin." Compression stockings are typically the answer for first step in venous insufficiency, but if you have an active ST or DVT, heparin must be given as first step.

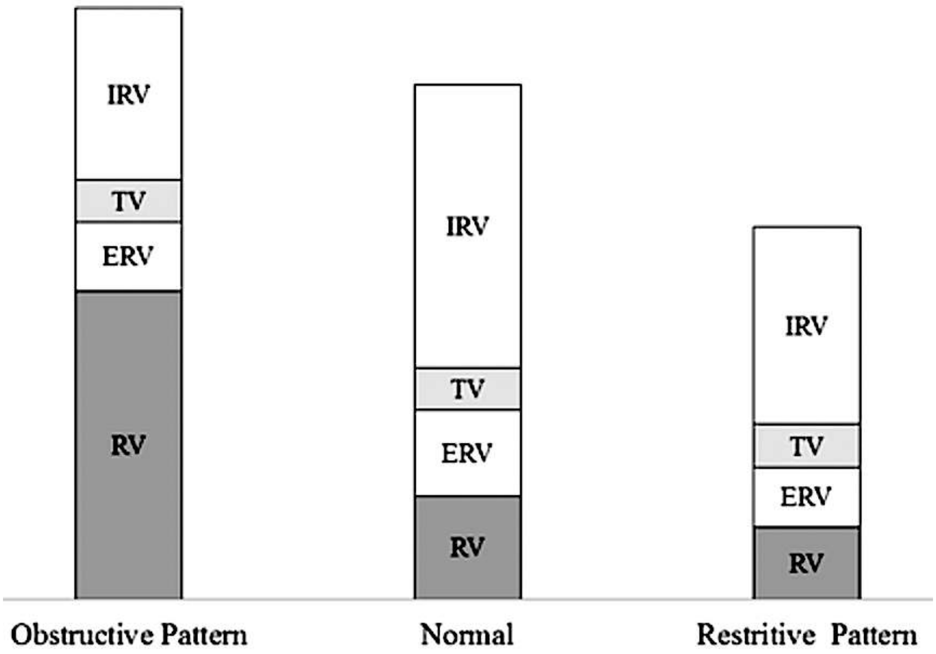
HY cardio pharm points for FM	
Verapamil	- Non-dihydropyridine calcium channel blocker (acts on nodal calcium channels). - **Causes constipation** (HY on Family Med). - Used for AF for rate control sometimes in place of metoprolol (don't worry about the use-case; USMLE doesn't give a fuck; just know MOA and side-effect).
Nifedipine	- Dihydropyridine calcium channel blocker (acts on arteriolar calcium channels → dilates arterioles → ↓ peripheral vascular resistance → ↓ BP). - **Causes peripheral edema / fluid retention** (HY on Family Med). - Used for essential HTN in patients without diabetes, atherosclerotic disease, or renal disease (if that sounds confusing, I talk about this in extensive detail later in this PDF in the Risk Factor bullet points section.
β-blockers	- Can cause depression and sexual dysfunction. - Avoid in patients with lung disease or history of severe or psychotic depression. - Metoprolol used for rate-control in AF. - Labetalol used in pregnant women for HTN (or methyldopa or nifedipine). - Timolol can be used topically for glaucoma. - Propranolol used for innumerable things → Tx of tachycardia in hyperthyroidism (because also has additional effect of ↓ peripheral conversion of T4 → T3); akathisia (restlessness as an extra-pyramidal side-effect of antipsychotics); essential, alcoholic, or idiopathic tremors; social phobia (if patient has asthma, 2CK wants benzo instead); HOCM (↓ HR ↑ LVEDV → ↓ murmur/obstruction).
Digoxin	- Directly blocks myocardial Na^+/K^+ ATPase pump. - Increases contractility + slows HR. - Toxicity presents classically as yellow/wavy "Vincent van Gogh" vision.

Lisinopril	- USMLE-favorite ACE inhibitor. - Can cause dry cough; also can ↑ serum K^+; avoid in hereditary angioedema. - Used for HTN in patients with pre-diabetes, diabetes, atherosclerotic disease, or renal disease (I talk about this in Risk Factors bullet point section in more detail).
Valsartan	- Angiotensin II receptor blocker (ARB). - Use-cases are identical on USMLE to ACEi (i.e., if you see both as answer choices to a question, they're usually both wrong because they're the "same"). - Doesn't cause dry cough the way ACEi do.
Oxymetazoline, Phenylephrine	- α1 agonists. - Highest yield use on USMLE is as intra-nasal spray for **nasal decongestion** → constrict capillaries within nasal mucosa → ↓ inflammation → relief of congestion. - Can cause rhinitis medicamentosa, which means rebound nasal congestion upon withdrawal if used non-stop for ~5 days. In other words, patients should use only as needed for a maximum of about ~3-4 days while sick.
Methyldopa, Clonidine	- α2 agonists. - Methyldopa used for HTN in pregnancy (nifedipine and labetalol also used). - Clonidine used for various psych treatments (e.g., Tourette).
Mirtazapine	- α2 antagonist. - Used to treat depression in patients who have anorexia (stimulates appetite).
Nitrates	- As discussed earlier, used for angina pectoris. - Must not take with PDE-5 inhibitors, or vice-versa.
Statins	- Statins have 2 HY MOAs on USMLE: 1) inhibit HMG-CoA reductase; 2) upregulate LDL receptors on hepatocytes. - Can cause myopathy and toxic hepatitis. An offline Step 1 NBME has myopathy as correct over toxic hepatitis, so I'd go with this as more important for FM as well. - Indications for statins on Family Med vary depending on the source (i.e., whether to use the 70 vs 100 mg/dL cutoff in certain scenarios), but for FM/USMLE, give if: - Age 20-39 if LDL > 190 mg/dL. - Age 40-75 if LDL > 100 mg/dL. - Age 20-75 in diabetic if LDL >100 mg/dL. - Some sources use 70, rather than 100, for the latter two cutoffs. What I can say is that I've seen 2CK NBME Qs where they use 100 as the cutoff (i.e., they give an LDL of 95 and statin is wrong, implying LDL is satisfactory). - Some sources incorporate a CVD risk % of >7.5%, which is more of a moot / pedantic talking point. I've seen one 2CK NBME Q where a CVD risk % shows up in the stem, but it doesn't rely on you knowing that point to get it right.
Fibrates	- Fibrates (e.g., fenofibrate) upregulate PPAR-α and lipoprotein lipase; best drugs to decrease triglycerides. - Give if triglycerides >300 mg/dL. - Can cause myopathy and hepatotoxicity, as well as cholesterol gall stones.
Orlistat	- Pancreatic lipase inhibitor used for obesity; can cause steatorrhea.
Niacin	- Vitamin B3. Two MOAs USMLE wants you to know: 1) ↓ VLDL export by the liver; 2) ↑ HDL more than any other medication. - Deficiency → pellagra → 3Ds → dementia, dermatitis, diarrhea. - Can present as delirium instead of overt dementia (biggest risk factor for delirium is underlying dementia, so old patient with delirium often has underlying cognitive decline); dermatitis will present on USMLE as either hyperpigmentation of the skin of the forearms or Casal necklace. - Administration can cause flushing (caused by prostaglandin, so mitigated by asprin), gout, and insulin resistance.
Omega/fish oils	- Decrease triglycerides.

FM Pulmonary

Obstructive vs restrictive lung disease		
	Obstructive	**Restrictive**
HY conditions	- Asthma. - COPD (chronic bronchitis + emphysema). - **Old age (i.e., healthy person age 70 vs 20)**	- Pulmonary fibrosis due to any cause (i.e., idiopathic; radiation; drugs; auto immune disease). - Pneumoconioses.
FEV1	↓	↓/↔
FVC	↓	↓
FEV1/FVC	↓ (<70%; normal is 70%)	↑/↔ (>70-80%)
TLC	↑	↓

- The above lung volumes are the HY ones you need to know for USMLE.
- FEV1 = Forced Expiratory Volume in 1 second = amount of air patient can forcefully blow out in 1 second, starting from maximal inhalation.
- FVC = Forced Vital Capacity = amount of air patient can blow out starting from maximal inspiration all the way until maximal expiration.
- The reason FEV1/FVC is ↑/↔ in restrictive, as compared to ↓ in obstructive, is because of **radial traction,** which is a "stickiness" on the outside of the airways (such as from fibrosis) that prevents them from closing as rapidly as obstructive.
- TLC = Total lung capacity → increased in obstructive due to air-trapping (hence "obstructive").

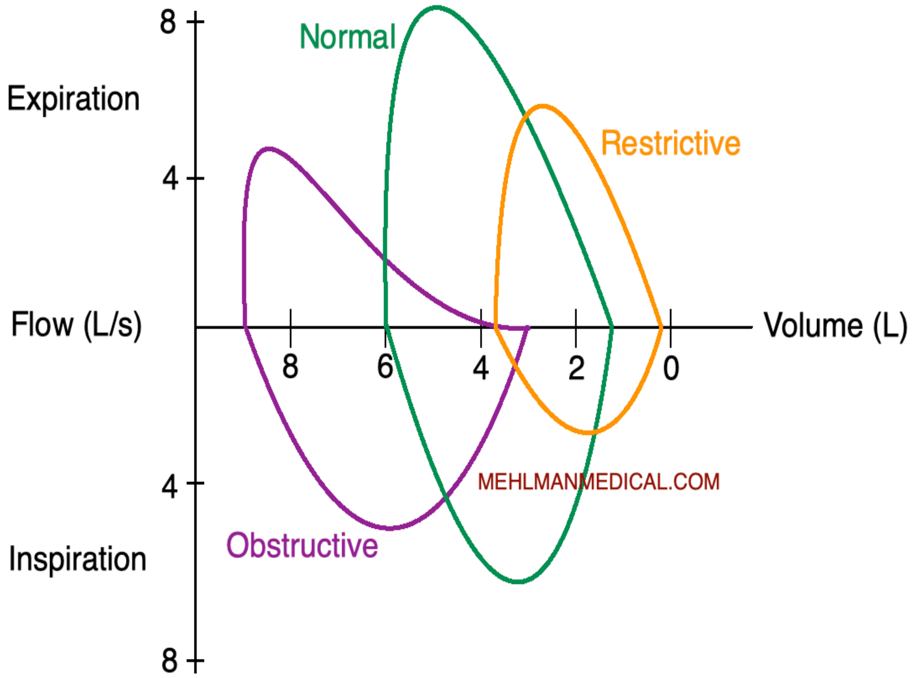


The below lung volumes are lower yield for USMLE. But I include them here for greater elucidation. For the sake of passing the Step, it is the arrows at the top of this chart that are most vital.

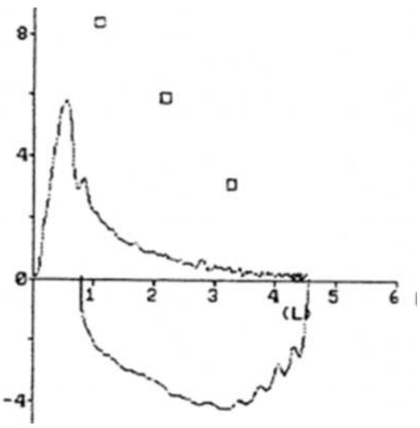
TV	↔	↔
ERV	↓	↓
IRV	↓	↓
RV	↑↑	↓
FRC (ERV + RV)	↑	↓
DLCO	↓ (but ↑ in asthma on USMLE)	↓ (↔ if neuromuscular)

- TV = Tidal Volume; generally preserved in lung diseases.

- ERV = Expiratory Reserve Volume = additional amount that can be forcibly exhaled following the end of a normal exhalation.
- IRV = Inspiratory Reserve Volume = additional amount that can be forcibly inhaled following the end of a normal inhalation.
- RV = Residual Volume = amount of air remaining in the lungs after the end of maximal expiration.
- FRC = Functional Residual Capacity = total amount of air remaining in lungs after end of normal expiration.



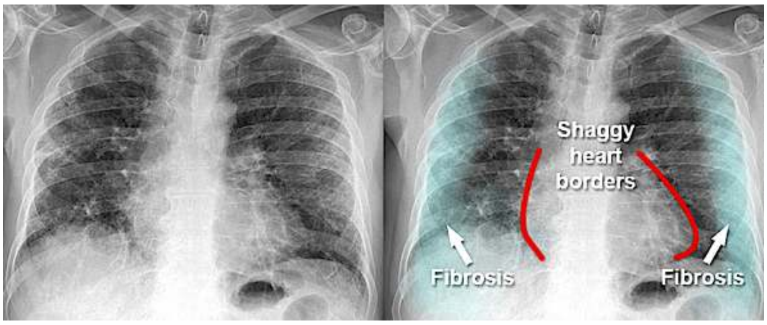
- Don't freak out about the above flow-loops. For USMLE (all Steps), you just need to know their general shape. I'd say the most important point is that the expiratory component of the obstructive curve (purple) tends to have a scooped-out, or concave, or L-shape.

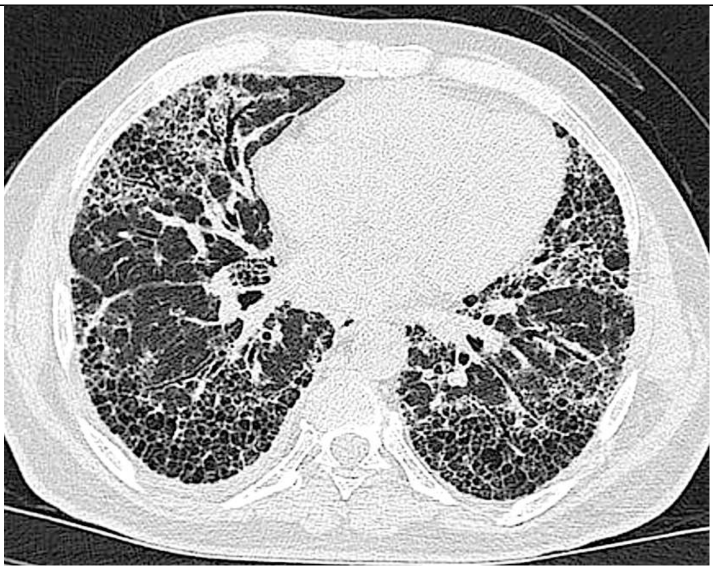


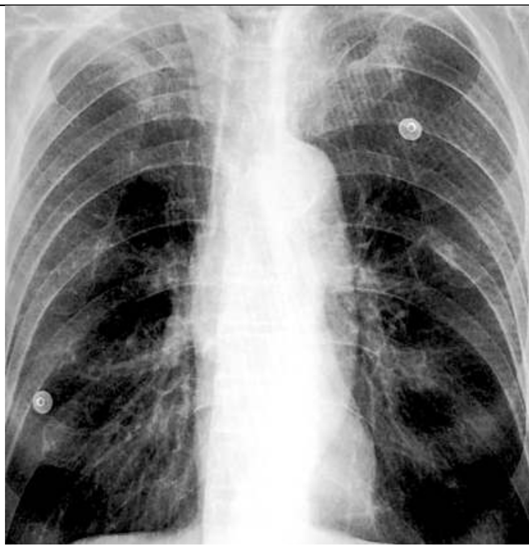
- What USMLE will do is give you a real bootleg Windows 95-type spirometry curve as per above in a 34-year-old + no family Hx + 5-year smoking Hx. You look at curve and say, "No idea what I'm looking at, but top of curve is concave, and I know that's obstructive. So between COPD or asthma here, which are both concave and hence possible answers, I'll choose asthma as most likely in this particular patient."

HY pulmonary / respiratory tract cancers for FM	
Small cell bronchogenic carcinoma	- Smoking biggest risk factor. - Occurs centrally in the lung (i.e., hilar / medially). - Paraneoplastic syndromes exceedingly HY: 1) SIADH (ADH secretion); 2) Cushing syndrome due to ACTH secretion; 3) Lambert-Eaton syndrome (production of antibodies against presynaptic voltage-gated calcium channels); 4) Cerebellar dysfunction / ataxia (anti-Hu/-Yo antibodies). - Treatment on 2CK is "chemotherapy." Do not do surgery. I believe this is the only time I've seen straight-up "chemotherapy" as correct answer on NBME.
Squamous cell carcinoma	- Smoking biggest risk factor. - Similar to small cell, occurs centrally in the lung (i.e., hilar / medially). - In other words, the two cancers that start with an "**ssss**" sound for "**c**entral" (i.e., **S**mall cell and **S**quamous cell) are **C**entral. - Highest yield point is that it secretes PTHrp (parathyroid hormone-related peptide). This acts like PTH and increases calcium / decreases phosphate, but this is not the same as PTH. Endogenous PTH is *suppressed* due to negative feedback from high calcium. There is a 2CK Q where they show low PTH in labs for squamous cell.
Adenocarcinoma of the lung	- Adenocarcinoma means cancer of glands. If the Q says something about biopsy showing glandular morphology, you know they're talking about adenocarcinoma. - The answer on USMLE for lung cancer in a non-smoker; classically "female non-smokers," but I've seen NBME Qs with this in men. - Normal ground/Earth radiation due to radon is accepted cause of lung cancer in non-smokers. There is NBME Q where they mention non-smoker living in a basement and develops lung cancer. The correlation is probably nonsense in real life, but it's in an NBME question somewhere. - Does not occur centrally on NBME exam, unlike small cell and squamous, and hence will be described as **apical or peripheral** lung lesion in non-smokers. - Apical tumors can are known as Pancoast tumors and cause Horner syndrome (miosis, ptosis, anhidrosis) due to impingement on C8 superior cervical ganglia (sympathetic nerves). They can also cause SVC syndrome (flushing of the face + congestion of neck veins) or brachiocephalic syndrome (only right side of face/neck) due to impingement on venous return. (+) Pemberton sign is worsening of flushing + neck vein congestion when raising the arms above the head. - Can be associated with migratory thrombophlebitis (Trousseau sign of malignancy). The latter is not limited to head of pancreas cancer. Adenocarcinomas in general are known to be associated with hypercoagulable state due to malignancy. USMLE won't ask specific mechanism, but liberation of tissue factor (factor III) by these cancers is a proposed etiology. - Associated with hypertrophic osteoarthropathy (clubbing + hand pain due to lung cancer); mechanism is fibrovascular proliferation. Literature says adenocarcinoma of lung is most common cause, although the association isn't 100% specific.
Bronchogenic carcinoid tumor	- Pulmonary nodule that secretes serotonin or serotonin-like derivatives, resulting in carcinoid syndrome (i.e., tachycardia, flushing, diarrhea). - A type of neuroendocrine tumor. - Carcinoid tumors are classically appendiceal and of the small bowel, but you should be aware that bronchogenic carcinoids exist. - USMLE wants urinary 5-hydroxy indole acetic acid (5-HIAA) for initial step in Dx.
Mesothelioma	- Asbestosis occurs first, which then gives rise to mesothelioma years later. - Shipyard and construction workers are buzzy for prior occupational exposure. - Asbestosis will be described as pleural or supradiaphragmatic plaques ("soft tissue plaques seen on CXR"). Pulmonary biopsy shows ferruginous bodies.
Nasopharyngeal carcinoma	- Can be caused by EBV. - A type of squamous cell carcinoma of the airway.
Laryngeal cancer	- Smoking is major risk factor. - Squamous cell carcinoma of vocal cords.

Pneumoconioses for USMLE	
Asbestosis	- As already mentioned above in the mesothelioma section, this is associated with shipyard workers, construction workers, and electricians. It can give rise to mesothelioma later in life. - Causes restrictive lung pattern.
Berylliosis	- Occupational exposure to beryllium in the aerospace / aeronautical industry. - Causes restrictive lung pattern. - Can cause granulomas.
Silicosis	- Occupational exposure to silicon (sand) in foundry or stone quarry workers. - Can cause egg-shelf calcifications in upper lobes. - Increases risk of tuberculosis infections. - Avoid anti-TNF-α agents (i.e., infliximab, adalimumab, etanercept) in these patients due to increased TB risk (TNF-α needed to suppress/fight TB).
Anthracosis	- Aka "coal miner's lung." - Black discoloration of the lung. - Can be either obstructive or restrictive.
Caplan syndrome	- Rheumatoid arthritis + any pneumonociosis, presenting as pulmonary nodules. - Clinical relevance is that patients with RA are at increased risk for developing pneumoconioses if they have a workplace exposure.

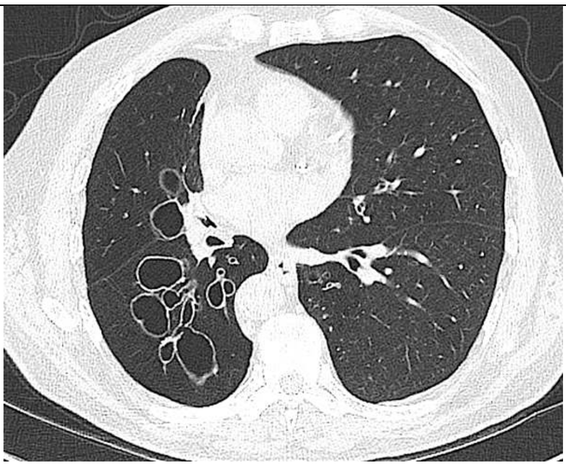
HY General lung conditions for USMLE	
Idiopathic pulmonary fibrosis (Usual interstitial pneumonitis)	- The answer on USMLE for a patient over the age of 50 who has 6-12+ months of unexplained dry cough. This is how it shows up 4/5 times. - Textbook restrictive lung disease, with $\leftrightarrow$ or $\uparrow$ FEV1/FVC. - CXR and CT scan show "reticular" or "reticulonodular" pattern. These descriptors are exceedingly HY on USMLE, where students will overlook them in the vignette, but they are hugely buzzy for restrictive lung disease. They are colloquially known as "honeycombing," but I have not seen the USMLE give a fuck about the latter colloquialism. They frequently just say "reticular" and "reticulonodular," and then you know right away, "Boom. Restrictive lung disease," i.e., fibrosis, etc. Tangentially, it's to my observation that NBME will say "reticulogranular" frequently for NRDS, but the two vignettes are clearly disparate anyway.  - After the CXR and spirometry are performed, 2CK wants "high-resolution CT of chest" as answer for next best step.

	 "Honeycombing" = **reticular / reticulonodular pattern**. - New 2CK NBME wants "lung biopsy" as answer to confirm diagnosis of interstitial lung disease (i.e., idiopathic pulmonary fibrosis) after imaging. - Vignette can also mention loud P2 (means pulmonary hypertension) with "dry inspiratory crackles heard bilaterally." - 1/10 times, the Q will be patient over 50 with increasing fatigue and shortness of breath over 6-12 months, with no mention of cough, where it initially sounds like heart failure, and they'll say CXR shows "interstitial markings" instead of reticular/reticulonodular patterning. However, they say patient has "↑ FEV1/FVC showing restrictive pattern" in the stem, which gives it away. - You need to know that **"usual interstitial pneumonitis" (UIP)** is another name for idiopathic pulmonary fibrosis. Yes, the name is weird, but it's not my opinion and it's asked twice on the NBMEs, where instead of writing "idiopathic pulmonary fibrosis" as the answer, they write "usual interstitial pneumonitis." UIP is technically a broad term that can refer to many restrictive lung conditions, but as I said, on NBME they use this synonymously with idiopathic pulmonary fibrosis. - **Tx on 2CK = pirfenidone** → anti-fibrotic agent that inhibits TGF-β-mediated synthesis of collagen.
COPD	- COPD = chronic bronchitis + emphysema. - Smokers will have combination of the two. When we say a smoker has COPD, we are saying he/she has chronic bronchitis + emphysema at the same time. - The term COPD can in theory apply to any obstructive disease of the lung that is chronic (e.g., asthma, Kartagener, etc.) but when the term is used without any specific condition attached, it refers to the combo of chronic bronchitis and emphysema.

	 - Hyperinflated lungs in COPD (due to air trapping) can push the heart to the midline. NBME will say there's a "long, narrow cardiac silhouette," or a "point of maximal impulse palpated in the sub-xiphoid space." In left ventricular hypertrophy, in contrast, there will be a lateralized apex beat, or a point of maximal impulse in the anterior axillary line. - Home oxygen is indicated on 2CK if O_2 sats are: - <88% saturation (55 mm Hg), or - <89% saturation (60 mm Hg) if the patient has cor pulmonale. - First-line Tx is now considered to be a long-acting muscarinic receptor antagonist (i.e., LAMA such as tiotropium) or a long-acting β_2 agonist (i.e., LABA such as olodaterol), either alone or in combination. - If insufficient, add an inhaled corticosteroid (e.g., fluticasone). - Should be noted that a question on an earlier Family Med form has ipratropium (SAMA) as the answer for 1st-line in COPD, but there aren't any other anti-muscarinic or β_2 agonists listed. So the point is that if you are forced to choose a SAMA or SABA (i.e., albuterol) on USMLE, these are OK, but newer guidelines technically say start with a LAMA or LABA. - "Exacerbation of COPD" as a diagnosis on USMLE will always give a patient with *high CO_2*. This is really important. In other words, if they give you a big paragraph and you're not sure of the Dx, if you see CO_2 is not elevated, the answer is not COPD exacerbation. - COPD exacerbations can be triggered by minor chest infections, so always **give antibiotics** on 2CK, even though etiology is usually viral. - Patients with COPD are chronic CO_2 retainers, so they will have a chronic respiratory acidosis (i.e., $\uparrow$ CO_2 and $\uparrow$ bicarb; pH can either be $\downarrow$ or compensated back into normal range). - For 2CK, lung cancer screening with annual low-dose chest CT is done in patients who meet all of the following: 1) age 50-80; 2) have 20-pack-year Hx of smoking; and 3) smoked within the past 15 years.
Chronic bronchitis	- Chronic bronchitis = productive cough for at least 3 months in a year, for 2+ years. - Reid index >0.5 (ratio of the thickness of bronchial mucous-secreting glands to the bronchial wall itself). <0.4 is considered normal. - Chronic bronchitis is known as "blue bloater" because the mucous sits in the alveolar spaces and impairs gas exchange. This results in a shitload of hypoxic vasoconstriction $\rightarrow$ patient can become acutely blue and hypoxic during exacerbations.

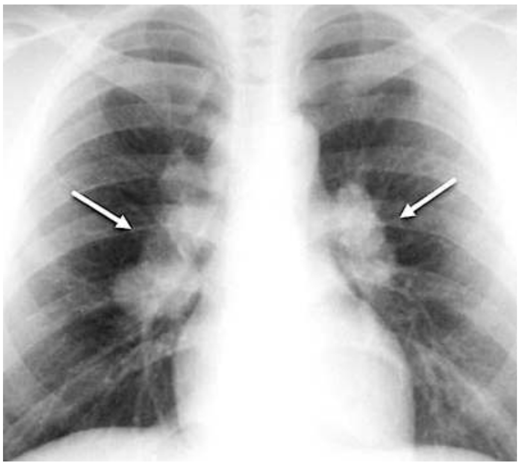
	- The hypoxic vasoconstriction of pulmonary vessels causes pulmonary hypertension (i.e., if the vessels constrict, then pressure in the more proximal pulmonary arterioles increases). This increased afterload on the right heart can lead to right ventricular hypertrophy and right heart decompensation. When right heart failure (i.e., evidence of JVD or peripheral edema) occurs due to a pulmonary cause, we now call that cor pulmonale, as discussed in the Cardio PDF. - The pulmonary hypertension can cause a loud P2 and tricuspid regurg prior to cor pulmonale occurring. - Don't confuse chronic bronchitis with acute bronchitis, which can present as a worsening cough in patient (with or without COPD) following a viral infection. This is a temporary irritation/inflammation of the airways that is self-resolving.
Emphysema	- Emphysema = loss/destruction of alveolar surface area. - When alveolar surface area is reduced, so is alveolar capillary surface area, since the capillaries are within the alveolar walls → ↓ gas exchange. - Known as "pink puffer," since although gas exchange is impaired, sudden and acute increases in hypoxic vasoconstriction, as with chronic bronchitis, are not a feature. - "Bullous changes" on CXR are synonymous with emphysema on USMLE. - Smokers can get centri-acinar emphysema (proximal alveolar structure is destroyed). - α1-antitrypsin deficiency patients get pan-acinar emphysema (entire alveolus destroyed).
α1-antitrypsin deficiency	- Codominant genetic condition resulting in emphysema and hepatic cirrhosis. - *ZZ* allele combo is worst and results in disease (asked on USMLE). - α1-antitrypsin is an enzyme produced by the liver that travels to the lungs and breaks down neutrophilic elastase. Elastase is an enzyme that normally causes damage to the alveoli. Homeostatically, elastase is required in small amounts for normal pulmonary function and cell turnover, but in high amounts it destroys the alveoli, resulting in emphysema. - Vignette will give young adult who has sibling or parent who's had early-onset emphysema or cirrhosis. - The USMLE Q can absolutely say the patient is a smoker or drinks alcohol. For instance, they might say the patient is 34 and has been smoking for 5 years and has bullous changes on CXR, or that the father died from alcoholic cirrhosis, and the student thinks, "Oh that can't be α1-antitrypsin deficiency though because they said smoking/alcohol." But the key point is that this condition *increases the risk* for early-onset emphysema and cirrhosis. Normally, COPD should take 20+ years of smoking to develop, not 5.
Asthma	- Bronchospasm that occurs either idiopathically/hereditarily, or in response to certain allergens or cold air. - One-third of patients with asthma only present with a dry cough and no problems breathing. This is called cough-variant asthma. It will often present as a patient with a dry cough that's worse in the winter. - Can also present as part of atopy constellation – i.e., dry cough in winter, seasonal allergies / rhinoconjunctivitis / urticaria in spring, and eczema in summer. - Aspirin-induced asthma mechanism HY for USMLE → inhibition of COX by aspirin → shunting of arachidonic acid down lipoxygenase pathway → increased leukotrienes → increased bronchoconstriction. - Samter triad = aspirin allergy, asthma (due to aspirin), nasal polyps.

	- "Increased expiratory phase" is a buzzy phrase that will be thrown into quite a few asthma vignettes. It is not specific for asthma and can refer to any obstructive pathology, but I just make note of it here because you'll see it quite a bit for asthma and say, "What's that mean?" → in obstructive conditions, it takes us a lot longer to exhale (FEV1 is ↓↓). - 2CK Q gives vignette of asthma and then asks for best initial step in diagnosis → answer = "spirometry." The expiratory component of the curve, as discussed earlier, will appear concave. - Should be aware methacholine challenge can also be done to diagnose. This is a muscarinic agonist that bronchoconstricts and can induce symptoms. Never give during acute episode. This can be tried between episodes. But if you're forced to choose diagnostic modality, go with simple spirometry. - USMLE cares about both outpatient management of asthma as well as acute attacks. - For outpatient management: 1) First-line Tx is β2-agonist (albuterol) for acute attacks. 2) If the Q says the patient has weekly episodes, or they ask for what could decrease risk of future episodes if patient is already on albuterol, the answer is inhaled corticosteroid (ICS; i.e., fluticasone). 3) If the combo of albuterol + ICS is insufficient, the next step is increasing the dose of the ICS. 4) If that doesn't work, adding a LABA (i.e., salmeterol) is the next step. 5) If insufficient, agents such as mast cell stabilizers, leukotriene blockers, etc., can be tried. If patient has Hx of aspirin allergy, the latter in particular can be effective. 6) Last resort for outpatient asthma is oral prednisone. It is most effective at decreasing recurrence of episodes, but we want to avoid it if at all possible because of the risk of Cushing syndrome and ↓ linear bone growth in Peds. - Acute management of severe asthma attack (i.e., not outpatient) is nebulized albuterol, oxygen, and IV methylprednisolone. Inhaled corticosteroids have no role in acute asthma management. - NBME for 2CK wants you to know that any asthma patient who requires hospital management for an acute episode must automatically be given ICS (i.e., fluticasone) on discharge. In other words, if patient isn't currently on an ICS, it must be commenced at discharge. I'm explicit here because this particular NBME Q doesn't mention albuterol anywhere in the stem, but inhaled fluticasone is the answer. - Another 2CK NBME Q gives a patient who is currently not receiving any asthma medications but who gets 2+ episodes weekly → answer = "inhaled fluticasone + albuterol." Students think this is weird because they jump straight to dual management + are even audacious enough to put the ICS before albuterol in the wording of the answer, but it's what they want. Any patient with 2+ episodes per week needs ICS in addition to the albuterol and can be commenced right away on dual therapy.
Bronchiectasis	- Dilation of the airways (ectasia means dilation) that occurs due to "loss of musculature of the airways." - Transverse CT scan will show cystic dilation of the airways.

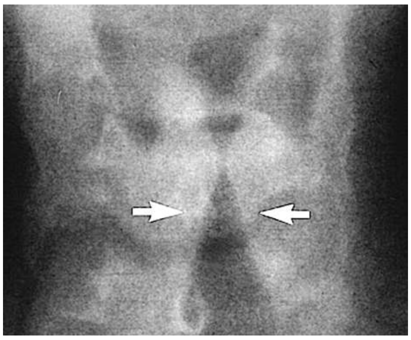
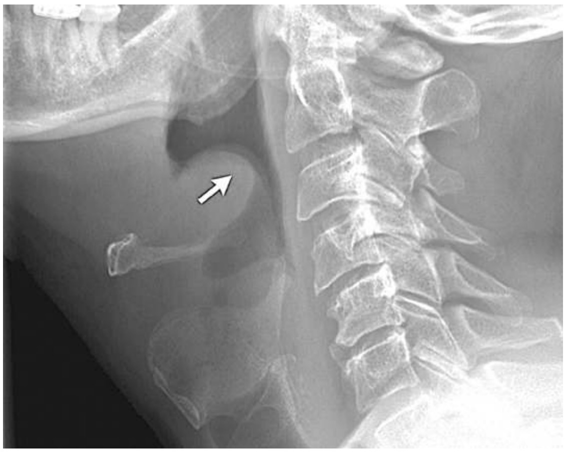
	 - Most common cause worldwide is TB. Most common cause in western countries is CF. But on USMLE, it will usually be a smoker. Asthma does not cause bronchiectasis. - Presents almost always as "cups and cups of foul-smelling sputum." - "Foul-smelling" means anaerobes, such as *Bacteroides*. - One question on 2CK Peds form gives bronchiectasis as the answer in a child who has right middle lobe syndrome, where they say there is scant white sputum (not cups and cups) and a thin, linear opacity visualized in the right middle lobe. If you think it's weird, complain to NBME not me. - **Clubbing** tends to be seen in these patients. It's not mandatory, but it's to my observation USMLE likes it for bronchiectasis.
Obstructive sleep apnea (OSA)	- Can be obstructive (i.e., usually from obesity) or central (i.e., brain-related). What USMLE wants you to know: - These patients develop chronic respiratory acidosis – i.e., ↑ CO2, ↑ bicarb, pH ↔/↓. - Cor pulmonale can occur as a result of pulmonary hypertension from hypoxic vasoconstriction. JVD or peripheral edema will be seen with cor pulmonale. Descriptors such as RBBB, right-axis deviation on ECG, and wide splitting of S2 all mean RVH. If the patient merely has pulmonary hypertension but not yet cor pulmonale, the vignette can say loud P2 or tricuspid regurg (holosystolic murmur that ↑ with inspiration). - Chronic fatigue and poor oxygenation can lead to dysthymia / depression. The answer on NBME is "mood disorder due to a medical condition." - Polysomnography (sleep study) is what USMLE wants to diagnose.
Anaphylaxis	- Acute dyspnea and bilateral wheezes in patient who was gardening (stung by a bee), who ate a particular food (e.g., peanuts), or who was commenced on a recent drug (e.g., TMP/SMX). - Vignette often gives tachycardia and low BP. - Mechanism is IgE crosslinking on surface of mast cells and basophils that leads to degranulation and histamine + prostaglandin release. Eosinophils can be recruited in response. - USMLE wants: ↓ vascular resistance, ↑ CO, ↓/↔ PCWP. - Tx = **intramuscular epinephrine** → the strong β2-agonistic effect opens the airways; the strong α1-agonistic effect constricts the arterioles and restores BP. - NBME 9 for 2CK asks, "In addition to self-injectable epinephrine therapy, what is most appropriate therapy to ↓ recurrences" → answer = **venom immunotherapy (VIT).** What this does is desensitizes the patient to the antigen by allowing him/her to develop neutralizing IgG antibodies against it.

Scombroid	- Often misdiagnosed as seafood allergy. - Bacteria in decaying meaty fish (e.g., mahimahi) can break down histidine in the fish into histamine, which causes an allergic-/asthma-like presentation. - Patients with history of asthma have greater risk of mortality.
Shellfish allergy	- Can cause allergic-/asthma-like presentation. - Occurs literally after eating shellfish (i.e., clams, mussels). - Don't confuse with scombroid. Students get trigger-happy when they learn weird conditions. If USMLE gives shellfish, the answer is shellfish allergy, not scombroid. The latter is from meaty fish and not an allergy.
Primary pulmonary hypertension (PPH)	- The answer on FM shelf for a woman 20s-30s, non-smoker, who has increased pulmonary vascularity/markings and either a loud P2 or triscupid regurg (as I talked about in cardio section, these are HY findings for pulmonary hypertension). - "Primary" means the pulmonary hypertension inherently *starts with* the lungs and is not due to a secondary (i.e., external) cause such as smoking, CF, systemic sclerosis, etc. - Due to mutations in BMPR2 gene. - Do not confuse this with cor pulmonale. Recall that the latter is right heart failure findings or RV structural changes due to a pulmonary cause. If the question gives evidence of RVH (i.e., RBBB, right-axis deviation on ECG, boot-shaped heart on CXR), or has overt right heart failure findings (i.e., JVD or peripheral edema), then we can say the patient now has cor pulmonale due to primary pulmonary hypertension. A loud P2 and tricuspid regurg, however, are HY findings for pulmonary hypertension that don't necessarily mean cor pulmonale. - Bosentan blocks endothelin-1 receptors and is HY Tx.

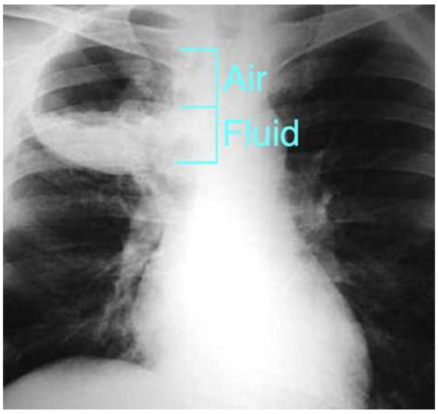
HY Autoimmune-related pulmonary conditions	
Systemic Sclerosis (aka scleroderma)	- Idiopathic autoimmune disease characterized by multi-organ system fibrosis and hardening of tissues (i.e., sclerosis). - Divided into limited and diffuse subtypes. - **Limited scleroderma** = **CREST** syndrome (**C**alcinosis, **R**aynaud's, **E**sophageal dysmotility, **S**clerodactyly, **T**elangiectasias). - USMLE can describe Raynaud as color change of the fingers with cold weather. They describe sclerodactyly as tightening of the skin of the fingers. Esophageal dysmotility presents as GERD. I haven't seen NBMEs give a fuck about calcinosis, which is abnormal Ca^{2+} deposition in tissues. - **Diffuse scleroderma** = CREST syndrome + renal involvement (presenting as ultra-HY BP, e.g., 220/120, due to renal involvement causing a surge in RAAS. If patient doesn't have high BP, it's not diffuse type on USMLE. - Apart from just being able to diagnose these conditions, the highest yield point on USMLE is that both types cause **pulmonary fibrosis,** leading to **pulmonary hypertension.** - Pericardial fibrosis can also occur. - USMLE wants "dress warmly in cold weather" as an answer for how to ↓ recurrence of Raynaud. Dihydropyridine CCBs can also be used (e.g., nifedipine). - USMLE wants to know which drugs you avoid in patients with Raynaud → answer = α1 agonists (e.g., phenylephrine, oxymetazoline), since these constrict arterioles/capillaries.
Sarcoidosis	- Idiopathic autoimmune disorder that is one of the highest yield conditions on USMLE. You must know this condition extremely well.

<table>
<tr><td></td><td>- Characterized by non-caseating granulomas within lung tissue. These consist of activated macrophages called epithelioid macrophages, aka histiocytes.
- These histiocytes in the lung secrete 1α-hydroxylase (normally produced in the PCT of the kidney in response to PTH), which will convert inactive 25-OH-D3 into active 1,25-(OH)2-D3 (i.e., causing high vitamin D; aka hypervitaminosis D), which then goes to the small bowel and ↑ absorption of calcium, causing hypercalcemia.
- Archetypal presentation is African-American woman 20s-30s with 6+ months of dry cough and red shins (erythema nodosum). Other findings like low-grade fever with flares can occasionally be seen.
- Bihilar lymphadenopathy seen on CXR or CT. They can also describe this as CXR or CT “shows hilar nodularity.”

- There is a Family Med Q that gives early-30s African-American woman with dry cough + they say CXR shows no abnormalities → answer = “activation of mast cells” (i.e., asthma is the answer); wrong answer is “non-caseating granulomatous inflammation.” So your HY point here is: if the vignette sounds like it could be sarcoidosis but they say CXR is normal or shows “mild hyperinflation” (also buzzy for asthma), choose asthma over sarcoidosis.
- A 2CK IM CMS form gives a sarcoidosis vignette where they mention bihilar lymphadenopathy on a CT scan + high hepatic AST + ALP + weight loss; they don’t mention hypercalcemia. Unusual combo of findings, but the answer is inferable based on the CT.
- Lupus pernio is an enlarged nose due to sarcoidosis (not SLE, despite the name).
- USMLE wants steroids (i.e., oral prednisone) as treatment.</td></tr>
<tr><td>Rheumatoid lung</td><td>- Rheumatoid arthritis can cause a restrictive lung disease known as rheumatoid lung.
- In addition, methotrexate, the first-line DMARD for RA, can cause pulmonary fibrosis. So patients with advanced RA can have pulmonary fibrosis often from a combo of the two.</td></tr>
</table>

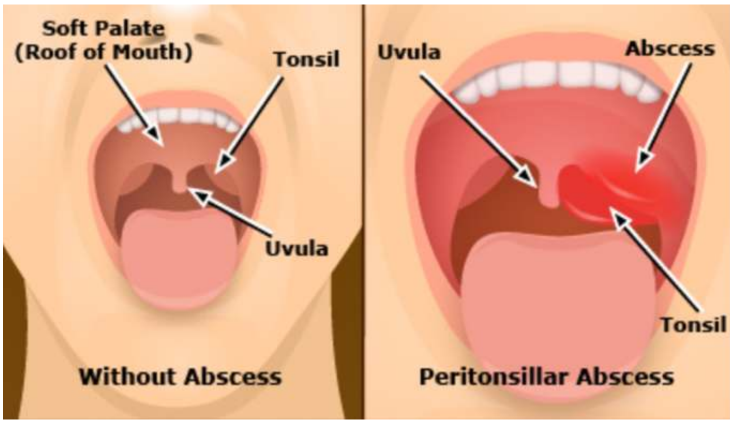
HY Infection-related stuff	
Pneumonia	- Lobar pneumonia = *Strep pneumoniae* on USMLE (right lower lobe consolidation with dullness to percussion). - Bilateral interstitial pneumonia (aka atypical pneumonia) in immunocompetent patients = *Mycoplasma* on USMLE. - Lobar pneumonia where they say "interstitial markings" and *Strep pneumo* isn't listed → answer = *Mycoplasma* (the word "interstitial" wins over location). - Bilateral interstitial / "ground-glass" pneumonia in AIDS patient → *Pneumocystis jirovecii* pneumonia (PJP). - Lobar pneumonia in AIDS patient → *Strep pneumo*, not PJP. - Bacterial pneumonia specifically post-influenza infection → *S. aureus*. - Bilateral pneumonia + low Hb or (+) Coombs test → *Mycoplasma* → can cause cold agglutinins, which means IgM against RBCs → hemolysis). - Pneumonia + hyponatremia and/or diarrhea → *Legionella*. - Pneumonia in 3-wk-old neonate who had conjunctivitis 1-2 weeks ago → *Chlamydia trachomatis* (the STI; drains through nasolacrimal duct to lungs). - Pneumonia in newborn first few days of life → Group B Strep (*Strep agalactiae*), which is gram (+) cocci. If they say gram (+) rods, that's instead *Listeria*. If they say gram (-) rods, that's *E. coli*. - Pneumonia + rabbits → *Francisella*. - Pneumonia + bird keeper → *Chlamydia psittaci.* - Pneumonia + southwest US and/or earthquake dust → *Coccidioides*. - Patients who have lung cancer are prone to obstructive pneumonias (on 2CK form). - Pneumonia in CF → *Pseudomonas* or *S. aureus*. - USMLE wants for pneumonia: adventitious/bronchial (i.e., abnormal) breath sounds + ↑ **tactile fremitus** (air vibrates due to movement through infective consolidation within alveoli). - Community-acquired pneumonia (CAP) empiric Tx = azithromycin on 2CK (on NBME). This covers the atypicals (*Mycoplasma, Legionella, Chlamydia)* as well as *S. pneumo*. - If patient has been on antibiotics in the past 3 months or has severe lung disease, levofloxacin (respiratory fluoroquinolone) can be given first-line. - CAP that results in sepsis or septic shock → give ceftriaxone (if listed, choose cefotaxime for peds). - Nosocomial pneumonia (i.e., hospital- or ventilator-acquired) requires coverage for MRSA and *Pseudomonas*. USMLE wants vancomycin PLUS either ceftazidime (a 3^{rd}-gen cephalosporin) or cefepime (a 4^{th}-gen). - For fungal pneumonia, Tx = fluconazole. - For fungal pneumonia + fungemia (high fever, chills) → Amphotericin B. - NBME for 2CK wants you to know **sputum culture, followed by blood cultures** are done in all patients with pneumonia who are septic. What they do in this patient is give you patient who has sputum culture performed, then they ask what should be done next for diagnosis? → answer = blood culture. - If you get *Pneumocystis* pneumonia, jump straight to **bronchoalveolar lavage** as the answer. - If you get a patient who has CXR or CT showing cavitary lesions in the lungs filled with a mass (likely *Aspergillus* fungus ball), they want "open lung biopsy" as the most confirmatory test.
Bronchiolitis	- Answer on USMLE for a kid <18 months old who has low-grade fever and bilateral wheezes. - Caused by respiratory syncytial virus (RSV).

	- Tx is supportive care on USMLE. Don't choose answers like ribavirin or palivizumab.
Laryngotracheal bronchitis (aka croup)	- Caused by paramyxovirus (aka parainfluenza virus). - Presents as hoarse, barking, or seal-like cough in school-age kid. The Q can say the cough gets better when his dad brings him out into the cold air. - Neck x-ray shows "steeple sign," which is sub-glottic narrowing.  - Sometimes the Q can give you easy vignette of croup, but then the answer is just "larynx" (literally inflammation of the larynx, trachea, and the bronchi). "Sub-glottic" means below the area of the vocal cords. The larynx is the area encompassing the vocal cords. - Tx is supportive. If they force you to choose an actual Tx, the answer is steroids if listed, followed by nebulized racemic epinephrine.
Epiglottitis	- Caused by *Haemophilis influenzae* ***type B***. - Seen in unvaccinated and immigrants (can be unvaccinated), as well as patients with asplenia or sickle cell (auto-splenectomy). - Can also cause meningitis. - X-ray of neck shows "thumbprint sign."  - Presents as child who has fever + difficulty breathing. They can say the kid is drooling and/or in tripod positioning (facilitates use of accessory muscles). - USMLE wants intubation as answer. Epiglottitis is a medical emergency that can lead to sudden occlusion of the airway. - Tx = 3^{rd}-gen cephalosporin (cefotaxime in peds, or ceftriaxone); give rifampin to close contacts.
Bacterial tracheitis	- Diagnosis of exclusion on USMLE (i.e., you eliminate to get there). - Classically caused by *S. aureus* following a viral URTI. - Shows up on 2CK form as patient with recent viral URTI who now has stridor (suggesting upper airway narrowing) + no drooling (suggests

	against epiglottitis) + **can fully open the mouth without difficulty** + does not improve with racemic epinephrine (suggests against croup). - Being able to open the mouth without difficulty is important because this contrasts with peritonsillar abscess, where the patient has difficulty fully opening the mouth.
Pertussis	- Classic whooping cough presents as succession of many coughs followed by an inspiratory stridor. - What you need to know for USMLE is that this can absolutely present in an adult and that they can be vague about it, just describing it as a regular cough. The way you'll know it's pertussis, however, is they will say there's either **hypoglycemia** or **post-tussive emesis,** which means vomiting after coughing episodes. - Pertussis can cause super-high WBC counts in the 30-50,000-range, where there are >80% lymphocytes. *This makes it resemble ALL.* So you should know for Peds that ALL-like laboratory findings + cough = pertussis. - Q will ask number-one way to prevent → answer = vaccination (not hard, but they ask it). Pertussis is part of TDaP. The pertussis component is killed-acellular; the tetanus and diphtheria are toxoid. - Erythromycin can be given to patients with active cough; USMLE doesn't give a fuck about pertussis stages. - Close contacts should also receive erythromycin.
Pleurodynia	- As mentioned earlier, this is an MSK condition I've seen asked twice on 2CK material that has nothing to do with the lungs, despite the name. - This is viral infection (Coxsackie B) causing sharp lateral chest pain due to intercostal muscle spasm. - Creatine kinase can be elevated in stem due to ↑ tone of muscle. - In viral pleurisy, CK will not be ↑.
Tuberculosis	- Can present similar to lung cancer, where patient can have B symptoms (i.e., fever, night sweats, weight loss) and hemoptysis. - Living in a homeless shelter or immigrant status from endemic area is buzzy. I've seen rural India and Albania as two locations in NBME Qs. - First step in diagnosis is PPD test (type IV hypersensitivity). - If PPD test is (+), the next best step is CXR. - If PPD is (+) but CXR (-), next best step is "treat for latent TB, " or "give TB prophylaxis." This is isoniazid (INH) for 9 months + vitamin B6 (since INH can cause B6 deficiency). It is exceedingly HY you know that neuropathy in a patient being treated for TB has B6 deficiency. - If PPD and CXR are both (+), the next best step is "treat for active TB," which is RIPE for 2 months + RI for 4 more months (6 months total). RIPE = rifampin, isoniazid, pyrazinamide, ethambutol. - BCG vaccine is live-attenuated. USMLE wants you to know Hx of BCG vaccine does not change management based on PPD guidelines. - If USMLE asks you how long after TB exposure will someone's sputum cultures be positive, the answer is 2-5 weeks. - Interferon-gamma release assay (Quantiferon Gold) can be used in patients who have Hx of BCG to ↓ false (+)s, but USMLE doesn't assess it. - What is considered a positive PPD test differs depending on risk factors: **- >5mm (+):** Hx of close contact to someone with active TB; immunocompromised patient (AIDS, organ transplant recipient receiving immunosuppressants, chronic corticosteroid user); calcification on CXR.

	- >10mm (+): Health care worker or prisoner/prison worker; immigrant from endemic area; TB laboratory personnel, children <4. **- >15mm (+):** everyone else. - If a PPD test is (+), never repeat it. If it is negative, it must be repeated in 1-2 weeks (i.e., sometimes false-negatives).
Pulmonary abscess	- Exceedingly HY on USMLE. - USMLE wants "aspiration of oropharyngeal normal flora," or "aspiration of oropharyngeal anaerobes" as the cause. - Q will give aspiration risk factor, such as alcoholism, dementia (can cause loss of gag reflex), Hx of stroke (leading to dysphagia), or epilepsy. - Q can also mention broken or missing teeth (hypodontia) as risk factor. - Often described on NBME as pulmonary lesion with an **air-fluid level.** This is buzzy, but not a mandatory descriptor. This refers to the top half of the circle being air, and the bottom being pus, the latter settling due to gravity.  - The stem can say the patient has "foul-smelling sputum." This descriptor is exceedingly HY and is synonymous with anaerobes on USMLE. - Oropharyngeal normal flora = *Bacteroides* (strictly anaerobic gram-negative rods); as well as *Peptostreptococcus* and *Mobiluncus*. The latter two are not HY, but *Bacteroides* is. I mention all three, however, because the Q can say sputum sample shows "gram-negative rods, gram-positive cocci, and gram-positive rods," which refers to all three. But the bigger picture concept is, this = mixed normal flora. - **Tx = clindamycin**. USMLE loves this.
Acute bronchopulmonary aspergillosis (ABPA)	- Presents as asthma-like presentation in patient with ↑ sensitivity to aspergillus skin antigen.
Otitis media (OM)	- Infection of portion of ear just deep to tympanic membrane. - Most commonly *Strep pnuemo*. - Will present as red, **immobile** tympanic membrane. Immobility of the tympanic membrane is highly sensitive for OM, meaning that if the Q says mobility is normal, we can rule out. - "Ear tugging" can be a sign in children of either otitis media or externa. - Tx is amoxicillin or penicillin. - Augmentin (amoxicillin/clavulanate) is classically given for *recurrent* OM. So if you are forced to choose between amoxicillin/penicillin alone or Augmentin, go with the former. - For 2CK FM/Peds, a tympanostomy tube (aka grommet) is used if the kid has >3 OM occurrences in 6 months, or >4 in a year.

Serous otitis media	- Aka otitis media with effusion. - Presents as fluid behind the tympanic membrane in a kid weeks after resolution of 1 or 2 otitis media infections. - Almost always benign and self-resolves in 4-8 weeks. Answer is observation.
Tympanic membrane perforation	- "Tympanic membrane perforation" is the answer on new 2CK NBME for 2-year-old who had 3-day Hx of viral infection followed by awakening with severe ear pain + has dried blood on ear lobe and pillow + otoscopy cannot visualize tympanic membrane because of seropurulent fluid draining from the ear canal. - Can occur due to otitis media, although vignette on NBME doesn't sound like classic OM and is as described above.
Mastoiditis	- Inflammation of mastoid bone caused by untreated otitis media. - The mastoid process is the posterior part of the temporal bone that is felt just behind the ear. - Can present as a painful ear pinna that is displaced (e.g., upward and outward). - Diagnosis is made by **CT or MRI.** X-ray is wrong answer. - 2CK IM Q gives a 2-year-old with mastoiditis where the answer is "CT of the temporal bone." Sounds wrong, since this is ↑ radiation for a kid, but it's what they want.
Myringitis	- Isolated inflammation of the tympanic membrane. - Can be bullous (i.e., bullous myringitis). - Caused by *Strep pneumo* or *Mycoplasma*.
Otitis externa (OE)	- Infection of ear superficial to tympanic membrane. - Classically caused by *Pseudomonas*. - Increased risk in swimmers and diabetics. - An NBME form has "necrotizing otitis externa" as answer for black skin within the ear canal in a patient. This is aka "malignant otitis externa." - USMLE wants "acetic acid-alcohol drops" as prophylaxis in college student who does crew + continues to have water exposure. - Tx (not prophylaxis) = "topical ciprofloxacin-hydrocortisone" drops.
Sinusitis	- The answer if they tell you a school-age kid has a lingering fever after an upper respiratory tract infection (URTI) for 10-14+ days. - Whenever a URTI lingers for more than ~10ish days, you want to think about sinusitis as a differential. - A 2CK vignette gives nocturnal cough (reflects aspiration; in this case, from the sinuses) and grey membranes in the oropharynx. - The grey oropharyngeal membranes detail sounds weird, since that is normally buzzy for *Diphtheria*, but it shows up on an NBME Q where the answer is sinusitis and *Diphtheria* isn't listed. - IgA deficiency Qs, which presents as recurrent sinopulmonary infections, can say patient has Hx of pneumonias + presents today with sore left cheek → reflects sinusitis. - For 2CK, **CT scan** is done if chronic sinusitis >12 weeks. After CT is performed for chronic sinusitis, nasal endoscopy can be performed. - Tx is amoxicillin/clavulanate (Augmentin). This is in contrast to OM and *Strep* pharyngitis, which are treated with just amoxicillin or penicillin alone, without the clavulanate (unless recurrent).

Other HY respiratory DDx for Family Med	
Allergic rhinitis	- May or may not be part of atopy in patients with asthma. - "Cobblestoning" of nasal mucosa is buzzy for allergy. This same word is used to describe the tarsal conjunctiva in allergic conjunctivitis. - 2CK form has "use of pillow and mattress covers" as the answer. - Otherwise, USMLE wants **intra-nasal corticosteroids** as first-line med. - USMLE does not want anti-histamines as first-line med. - 2nd-gen H1 blockers (i.e., loratadine) are the answer if *intra-nasal* corticosteroids are not listed. Oral prednisone is a **wrong** answer. - I've seen students choose prednisone and then get irascible and say, "But you said steroids were first-line." → Yeah, bro. Intra-nasal. Not fucking oral.
Adenoid hypertrophy	- Just know that enlarged tonsils (adenoids) are a cause of stridor in kids. - Answer = tonsillectomy.
Peritonsillar abscess (Quinsy)	- Pus collection causing pain and swelling at posterior oropharynx. - Can be seen as complication of tonsillitis. - Patient can have **difficulty opening his/her mouth;** this contrasts with bacterial tracheitis, where the patient can fully open his/her mouth (on 2CK NBME, where this detail is important in an otherwise vague vignette). 
Laryngomalacia	- Most common cause of stridor in infants. - Softening of airway cartilage. - Vignette will give kid under the age of 1 who has noisy breathing that is mitigated when placed prone (on stomach) or upright. - Self-resolves almost always (i.e., don't treat).
Vascular ring	- Weird cause of stridor that USMLE likes to contrast with laryngomalacia. - Vascular embryology variant can lead to airway compression. - In contrast to laryngomalacia, mitigated when neck is extended. - Tx = surgery.
Cholesteatoma	- Idiopathic squamous proliferation in middle ear behind the tympanic membrane that presents as an enlarging mass. - Can cause obstructive hearing loss + grow into inner ear. - Requires surgery. - Sounds like an obscure diagnosis, but it's not. Be aware it exists for Peds.

Differentiating viral vs bacterial URTI
- The CENTOR criteria are exceedingly HY for Family Med and Peds.
- If 0 or 1 point, the URTI is unlikely to be bacterial (i.e., it's likely to be viral). If 2-4 points, the chance is much greater that the URTI is bacterial. - **1) Absence of** cough (i.e., no cough = 1 point; if patient has cough = 0 points). - **2)** Fever >38 C. - **3)** Tonsillar exudates. - **4)** Lymphadenopathy (cervical, submandibular, etc.). - There is a version of the criteria that includes age, but on the USMLE it can cause you to get questions wrong. So just use the simplified above four points. - If 0-1 point, answer = "supportive care"; or "no treatment necessary"; or "warm saline gargle" (same as supportive care); or "acetaminophen." Latter is answer for 3M with viral URTI + fever on Peds NBME form 2. - If 2-4 points, next best step = "rapid Strep test." If rapid Strep test is negative, answer = throat culture, **NOT** sputum culture. - While waiting on the throat culture results, we send the patient home with amoxicillin or penicillin for presumptive Strep pharyngitis. - If child is, e.g., 12 years old, and develops a rash with the beta-lactam, answer = beta-lactam allergy. - If the vignette is of a 16-17 year-old who has been going on dates recently (there will be no confusion; the USMLE will make it clear) + gets a rash with the beta-lactam, the answer = EBV mononucleosis; therefore do a heterophile antibody test (Monospot test). - **EBV is the odd virus out that usually presents with all four (+) CENTOR criteria and presents like a bacterial infection.** - This is why it's frequently misdiagnosed as Strep pharyngitis. It is HY to know that beta-lactams given to patients with EBV may cause rash via a hypersensitivity response to the Abx in the setting of antibody production to the virus. EBV, in a patient who does not receive Abx, can cause a mild maculopapular rash. But the rash with beta-lactam + EBV causes a more intense pruritic response generally 7-10 days following Abx administration on the extensor surfaces + pressure points.

HY Pulmonary drug points	
Bosentan	- Endothelin-1 receptor antagonist used for pulmonary hypertension. - Endothelin is a vasoconstrictor ↑ in pulmonary hypertension.
Methotrexate	- Dihydrofolate reductase inhibitor. - 1st-line DMARD for RA + 1st-line oral agent for psoriasis that fails topicals. - Causes pulmonary fibrosis (and hepatotoxicity and neutropenia). - Toxicity can be mitigated with "leucovorin rescue," aka folinic acid (not folic acid).
Albuterol	- Short-acting β2-agonist used for asthma. - Can cause tremor.
Fluticasone, Beclomethasone	- Inhaled corticosteroids (ICS) used for asthma and COPD. - Used for prevention / to decrease recurrence of episodes. They have no role in acute exacerbations. For the latter, IV methylprednisolone is used.
Prednisone	- Most effective at ↓ recurrent asthma attacks, but used last resort because oral steroids can cause Cushing syndrome and growth stunting.
Montelukast, Zafirlukast	- Leukotriene receptor antagonists. - Leukotrienes normally cause bronchoconstriction. So these agents ↓ airway constriction. - Can be used for asthma in patients with aspirin-induced asthma (blockage of COX shunts arachidonic acid down the lipoxygenase pathway toward leukotrienes), or for asthma in general in between a LABA and prednisone.
Zileuton	- Lipoxygenase inhibitor. Blocks synthesis of leukotrienes. - - Use-case same as the -lukasts.

Nedocromil, Cromolyn sodium	- Mast cell stabilizers used for asthma. Prevent release of histamine.
Tiotropium	- Long-acting muscarinic receptor antagonist (LAMA) used first-line for COPD.
Olodaterol	- Long-acting β2-agonist (LABA) used first-line for COPD. - As discussed earlier, either a LAMA or LABA can be used first-line for COPD.
Ipratropium	- Short-acting muscarinic receptor antagonist (SAMA) used for COPD. - Not first-line anymore, but still shows up on Family Med material as an agent that's used. Some old-school practitioners still use it, so don't disregard it.
Epoprostenol	- PGI2 prostacyclin that can be used for pulmonary hypertension.
Sildenafil	- Viagra; PDE-5 inhibitor that can also be used for pulmonary hypertension.
Nifedipine	- Dihydropyridine calcium channel blocker that can be used for pulmonary HTN. - As discussed earlier, causes peripheral edema/fluid retention.
Omalizumab	- Monoclonal antibody against IgE. - Can be used in theory in severe asthmatics who have high IgE levels.
Pirfenidone	- Used for idiopathic pulmonary fibrosis. - Anti-fibrotic agent that inhibits TGF-β-mediated collagen synthesis.

FM Renal

HY Nephritic syndromes	
Post-streptococcal glomerulonephritis (PSGN)	- The answer on USMLE for red urine 1-3 **weeks** following a sore throat caused by Group A Strep (*Strep pyogenes*). This is in contrast to IgA nephropathy (which I discuss in detail below), which is red urine **1-3 days** following a sore throat. - PSGN can be caused by skin infections. This can be impetigo (school sores), cellulitis, or erysipelas. For example, the Q might say a 10-year-old has yellow crusties on his arm for the past 7 days (impetigo) + now has red urine → answer = "acute glomerulonephritis." - PSGN usually self-resolves in kids without sequela. But USMLE wants you to know that ↑ age is means worse prognosis, where chance of renal failure is ↑ it if occurs in adults. This is probably related to the more robust immune response resulting in more advanced renal damage.
IgA nephropathy	- Aka Berger disease; IgA deposition in renal mesangium. - Red urine 1-3 days after a sore throat. This is in contrast to PSGN, which is red urine 1-3 weeks after a sore throat. I just mentioned it above obviously. But students fuck this up despite the inculcation so I'm reiterating it like an asshole. - Caused by **viral infection,** not Group A Strep. - Can sometimes be caused by GI infections, but USMLE usually avoids this etiology unless including it in the Henoch-Schönlein purpura constellation. - Henoch-Schönlein purpura is HY for 2CK. It's a tetrad: 1) Palpable purpura (usually on buttocks/thighs). 2) IgA nephropathy (red urine). 3) Arthralgias. 4) Abdominal pain. - All 4 need not be present for HSP, but the abdo pain component here is presumably viral gastroenteritis leading to IgA nephropathy.
Alport syndrome	- X-linked disease; *mutation in* collagen IV gene. - Do not confuse with Goodpasture syndrome, which is antibodies against type IV collagen. FM shelf won't assess mechanisms, but you should know the difference. - The answer on USMLE for a male who has red urine + an eye or ear problem (collagen IV is present in basement membranes in the kidney, ear, and eyes). - The "eye problem" can be blurry vision/cataracts; the "ear problem" will be neurosensory hearing loss (due to organ of Corti dysfunction).
Membranoproliferative glomerulonephritis (MPGN)	- The answer on USMLE for red urine in a patient with hepatitis C, heroin use, or malignancy. - Answer is "renal biopsy" as the next best step. This guides our management. - A 2CK form has "heroin" as the answer (heroin-induced nephropathy) for patient with protein and blood in the urine. In contrast, FSGS (discussed below) from heroin has no blood in the urine.
Diffuse proliferative glomerulonephritis (DPGN)	- The answer on USMLE for red urine in a patient with SLE. - Same as with MPGN, USMLE wants "renal biopsy" as the next best step, since this guides our management.

HY Nephrotic syndromes	
Minimal change disease	- The answer on USMLE for otherwise unexplained edema in a kid who doesn't have blood in the urine. - Can present as peripheral edema, periorbital edema, and/or ascites. - Almost always this is pediatric. Very rarely it can be due to Hodgkin in adult. - Classically post-viral (i.e., URTI), but ~50% of vignettes won't mention that.

	- In other words, textbook vignette is an 8-year-old who has the sniffles for 4 days, followed by peripheral and periorbital edema a week later, without blood in the urine. Once again though, the stem need not mention the viral infection. - Called minimal change disease because light microscopy (LM) shows no abnormalities. EM, however, shows effacement of the podocytes. - Mechanism for nephrotic syndrome is “loss of size and charge barrier.” - Corticosteroids are the treatment and are highly effective.
Membranous glomerulonephritis / nephropathy	- The answer on USMLE for nephrotic syndrome in patient who has: - Exposure to drugs: dapsone, gold salts, sulfonamides. - Infection: hepatitis B (Hep C can cause it too, but rare). - Visceral cancers, e.g., breast, pancreatic. - Autoimmune (primary): antibodies against phospholipase A2 receptor.
Renal amyloidosis	- The answer for nephrotic syndrome in multiple myeloma. - Amyloidosis means protein depositing where it shouldn’t be depositing. - Multiple myeloma = cancer of plasma cells that produce too much immunoglobulins → high serum IgG kappa/lambda light chains. Immunoglobulins are proteins → these fly through the kidney and cause Bence-Jones proteinuria; they also deposit in the renal parenchyma, causing renal amyloidosis. - Biopsy shows classic apple-green birefringence with Congo red stain.
Diabetic glomerulosclerosis	- Diabetes is the most common cause of chronic renal failure. - Renal failure simply means ↓ glomerular filtration rate (GFR). **- ACEi (e.g., lisinopril) or ARBs (e.g., valsartan) are the first drugs used in diabetes if there is HTN (>130/80 in diabetes, not 140/90), proteinuria, or ↑ creatinine or renin (HY for Family Medicine!!).** - AT-II normally constricts efferent arterioles leaving the kidney. ACEi/ARBs ↓ constriction of efferent arterioles thereby ↓ filtration fraction and reducing GFR. This might sound like a bad thing – i.e., “Why would we want to ↓ GFR in diabetes if it’s the most common cause of chronic renal failure?” → The answer is because the slightly reduced GFR and filtration fraction early on ↓ the rate at which excess glucose is filtered across the glomerulus. Over time, it is inevitable that the glomerular basement membrane will non-enzymatically glycosylate and thicken, so ACEi/ARBs slow this process (i.e., slightly reduce GFR early in order to prevent a massive decrease later).
Focal segmental glomerulosclerosis (FSGS)	- The answer on USMLE for nephrotic syndrome in sickle cell. - Can also be caused by heroin and HIV. - Not as responsive to steroids as MCD.

Interstitial nephritis
- Exceedingly HY renal condition. - Aka interstitial nephropathy, or tubulointerstitial nephritis/-nephropathy. - Think of this as “an allergic reaction of the kidney.” - 4/5 Qs will be an **NSAID, β-lactam, cephalosporin** followed by getting a maculopapular rash and WBCs (eosinophils) in the urine. This presentation is textbook/pass-level. - Only ~50% of vignettes will mention the maculopapular rash. - The stem need not say “eosinophils” either. They can just say patient is on β-lactam + now has WBCs in the urine + no mention of rash → answer = interstitial nephritis. If they say the rash, it’s even easier. - 1/5 Qs will just say a patient was on an NSAID, β-lactam, or cephalosporin and now has mild proteinuria and hematuria. They don’t mention a rash or WBCs in the urine. - There is also a singular NBME Q where they just say a patient has simple peripheral edema due to an NSAID, with no mention of anything else, and the answer is interstitial nephropathy. This is more unusual, since “NSAIDs + edema” classically = pre-renal, but I’ve seen the NBME assess interstitial nephropathy for this as well. - NBME also can give you simple vignette of interstitial nephritis, and then ask for the location in the kidney that’s affected (e.g., efferent arterioles, etc.) → answer = “renal tubule.”

- For example, 40M + treated with double course of penicillin + now has maculopapular rash and eosinophils in the urine; diagnosis? → interstitial nephritis. - 60F + using naproxen (an NSAID) for 6 weeks for her osteoarthritis + has peripheral edema → answer = interstitial nephropathy. - 35F + just recently finished 10-day course of cephalexin + has mild proteinuria and hematuria → answer = interstitial nephropathy. - Shows up in one NBME Q due to allopurinol (which is a sulfa drug), but as I said, the classic drugs that cause this frequently on the exam are **NSAIDs, β-lactams,** and **cephalosporins.**

Renovascular hypertension	
Renal artery stenosis (RAS)	- Narrowing of one or both renal arteries due to atherosclerosis that causes ↑ renin-angiotensin-aldosterone system (RAAS) and ↑ BP. - Q will be patient over the age of 50 with cardiovascular disease risk factors, such as diabetes, HTN, and/or smoking. - Patients who have pre-existing HTN causing atherosclerosis leading to RAS will often have 10-20 years of background HTN that then *becomes accelerated* over a few-month to 2-year period. What this means is: the slowly developing atherosclerosis in the renal arteries finally reaches a point at which the kidney is unable to maintain autoregulation, and the RAS is now clinical (i.e., accelerated HTN of ↑↑ BP within, e.g., 3 months). - Another way USMLE will give RAS is by giving ↑ BP in patient with significant evidence of atherosclerotic disease (i.e., Hx of coronary artery bypass grafting, intermittent claudication), and then ask for the most likely cause → RAS. You have to say, "Well he clearly has atherosclerosis in his coronaries and aortoiliac vessels, so that means he'll have it in the renal arteries too." - Q can say older patient with carotid bruit has recent ↑↑ in BP and then ask for diagnosis → answer = RAS. Similar to above, if the patient has atherosclerosis in one location (i.e., the carotids), then he/she will have it elsewhere too. - HY factoid about RAS is that **ACEi or ARBs will cause renin and/or creatinine to go up.** This is a HY point that is often overlooked and is asked on NBMEs. I have not seen NBME care whether it's uni- or bilateral in this case. → Kidney can autoregulate across flux in perfusion. Patients with already-compromised renal blood flow are more sensitive to the subtle ↓ in filtration fraction that occurs secondary to ACEi/ARB use, so renin/creatinine ↑. - If USMLE gives you unilateral RAS, renin is only ↑ from that kidney. The other kidney will not produce ↑ renin. - After renin and aldosterone levels are obtained, **MR angiography** of the renal vessels is what USMLE wants for the next best step in diagnosis.
Fibromuscular dysplasia (FMD)	- The answer on USMLE for narrowing of the renal arteries in a woman 20s-40s. - **Not** the same as renal artery stenosis, and **not** caused by atherosclerosis. - If you broadly say "renal artery stenosis," that specifically refers to atherosclerosis of the renal arteries in patient >50 with CVD. - FMD is tunica media hyperplasia (not dysplasia, despite the name) that results in a "string of beads" appearance on renal angiogram. - MR angiography is answer on NBME for diagnostic modality. - Can affect the carotid vessels. A 2CK Surg Q gives FMD vignette and also says there is 25% occlusion of one of the carotids.

Urolithiasis	
- Urolithiasis is broad, umbrella term that refers to both nephrolithiasis and ureterolithiasis.	
Stone type	**HY points**
Calcium	- Most common type of stone. Can be calcium oxalate or calcium phosphate, although I've never seen USMLE once assess or give a fuck about phosphate stones. - Most young adults with idiopathic kidney stones will have "normocalcemia and hypercalciuria" – i.e., normal serum calcium but elevated urinary calcium. - Family Med will give you healthy male in his 20s with sharp pain in the flank or groin + RBCs in the urine → answer = urolithiasis. - **Crohn disease** and disorders causing fat malabsorption increase the risk for calcium oxalate stones (↑ fat retained in GI tract → ↑ chelation with calcium in GI tract → ↓ calcium available to bind oxalate → ↑ oxalate absorption by GI tract → ↑ urinary oxalate). - Primary hyperparathyroidism and malignancy causing hypercalcemia are HY etiologies. - Milk-alkali syndrome: ↑ calcium + ↑ bicarb + calcium stones. Textbook scenario is a patient taking too many antacids, but vignette will usually not mention this and it will simply be a diagnosis of exclusion (i.e., you eliminate to get there). - Ethylene glycol (Anti-Freeze) and hypervitaminosis C can cause oxalate stones. - First step in prevention and treatment of stones is "adequate hydration." - **Thiazides** can be used to prevent recurrent stones by ↓ urinary calcium. NBME has a Q that asks why → answer = ↑ reabsorption of calcium. - In theory, alkalinization of the urine can help treat oxalate stones, and acidification of the urine can help treat phosphate stones.
Struvite	- Composed of ammonium magnesium phosphate. - Formed in ↑ pH in the presence of urease (+) bacteria. - *Klebsiella, Serratia,* and *Proteus* are HY causes. - These stones are large, ram-horn like. - Apart from adequate hydration, Tx is acidification of urine with NH_4Cl + surgery.
Uric acid	- Seen in patients who have gout. Not complicated. - First-line Tx for *chronic* gout is xanthine oxidase inhibitors (i.e., allopurinol or febuxostat). They are preferred over uricosurics (meaning ↑ urinary excretion of uric acid) like probenecid and sulfinpyrazone because the latter agents ↑ risk of uric acid stones.
Cystine	- Cystine stones (hexagonal) can occur in young adult due to an inability to reabsorb the **COLA** (dibasic) amino acids → **C**ysteine, **O**rnithine, **L**ysine, **A**rginine. - Cystinuria can be diagnosed with cyanide-nitroprusside test. - Cystine is two cysteines put together (not important for the exam, but this is to prevent annoying DMs from people asking about the spelling).

ADPKD
- Autosomal dominant polycystic kidney disease. - The answer on USMLE if disease starts as an adult (i.e., 30s-40s). - Cysts are technically present early in life, but only become clinical as adult (i.e., ↑ BP and ↑ RFTs). - These patients have ↑ BP due to ↑ RAAS (compression of microvasculature of kidney due to enlarging cysts). - Can cause saccular (berry) aneurysms of the circle of Willis → risk for subarachnoid hemorrhage. - Highest yield point for Family Med is that **serial blood pressure checks** are correct over circle of Willis MR angiogram screening. Latter is wrong answer on USMLE. MR angiogram screening of circle of Willis is only done when there is (+) family Hx of SAH or saccular aneurysms. - Family Med won't ask about ARPKD, but you could be aware that it occurs in pediatrics and can be associated with hepatic fibrosis. ADPKD is more important for FM because it is an important DDx for unexplained sudden HTN in patient 30s-40s + the need for serial blood pressure checks.

Genitourinary infection-related stuff	
Acute pyelonephritis	- Infection of the kidney. "Pyelo" means kidney. - 9/10 questions will mention fever + **costovertebral angle (CVA) tenderness,** which is pain with percussion of the flank. - Major risk factors are vesicoureteral reflux (especially in pregnancy) and posterior urethral valves. - Most common organism is *E. coli* (same as for simple UTIs). - It's to my observation **bacteria can be few in the urine** in acute pyelo. This confuses some students. But it's typically what I see on NBME forms. If the infection is further down, i.e., UTI in the urethra, then bacteria are more copious. - Treatment for pyelo is **ciprofloxacin** or **ceftriaxone.** USMLE is known to ask these. - For example, old dude + high Cr (caused by post-renal from BPH) + treated for pyelo → now gets sore ankle → was treated with cipro (causes tendonopathy). - 2CK form has ceftriaxone as an answer in pyelo Q where cipro isn't listed. But cipro is classic Tx. It's to my observation that ceftriaxone is HY drug on 2CK for community-acquired "general sepsis" or "general complicated/severe infections" – i.e., it is hard-hitting and covers wide array of community organisms.
Cystitis	- Suprapubic tenderness in female. Patient need not have fever. - *E. coli* most common cause. - Will have bacteria and WBCs in the urine (i.e., bacteriuria + pyuria). - Urinary nitrites and/or leukocyte esterase can be positive. - Can be caused by **recent catheterization.** The relevance to Family Med is these patients will get discharged from hospital + become outpatient, or a patient might be on home catheterization. - **Nitrofurantoin** is HY drug to treat cystitis on USMLE.
Chronic interstitial cystitis	- Not an infection. - This is >6 weeks of suprapubic tenderness + dysuria (pain with urination) that is unexplained, where laboratory and urinary findings are negative. - They can mention **anterior vaginal wall pain** (bladder is anterior to vagina). - USMLE wants you to know you **don't treat.** Steroids are wrong answer. "Treatment" is standard placating placebo nonsense such as "education," "self-care" and "physiotherapy."
General UTI	- Classically *E. coli* infection of the urethra. - Inoculation of the urethra following sexual activity is most common mechanism. - Can be caused by catheters (for FM, if patient is on home catheterization). You need to know **dysuria + Hx of catheter = UTI.** - Trimethoprim, trimethoprim/sulfamethoxazole (TMP/SMX), or nitrofurantoin are standard treatments for UTI. - For prevention, post-coital voiding confers ↓ risk of recurrence. - If post-coital voiding doesn't work, "post-coital nitrofurantoin therapy" is answer on 2CK NBME form. - If post-coital nitrofurantoin therapy doesn't work, "daily TMP/SMX prophylaxis" is the answer. Sounds absurdly wrong / like a bad idea, but it's an answer a couple times on 2CK assessments. - There's also a Q where they say a girl was treated successfully in the past with TMP/SMX for a UTI + ask how to prevent recurrent UTIs now, and they just jump straight to "daily TMP/SMX prophylaxis." Once again, sounds absurdly wrong. - If patient is treated for a UTI and dysuria persists, the next best step is testing for *Chlamydia* and *Gonorrhea*.

HY Bladder incontinences	
Stress	- The answer for loss of urine with ↑ intra-abdominal pressure from laughing, sneezing, coughing. - Stereotypical risk factor is grand multiparity (i.e., Hx of many childbirths) leading to weakened pelvic floor muscles. - I'd say only ~50% of Qs will mention Hx of pregnancy. The other ~50% are idiopathic. - Buzzy vignette descriptors and answer choices are: "downward mobility of the vesicourethral junction," "urethral hypermobility," and "urethral atrophy with loss of urethrovesical angle." - Do **not** give medications for stress incontinence on USMLE. - If Kegel exercises fail, patients can get a mid-urethral sling (LY; asked once).
Urge	- The answer on USMLE for patient who has an "urge" (NBME will literally say that word and I've seen students get the Q wrong) to void 6-12+ times daily **unrelated** to sneezing, coughing, laughing, etc. (otherwise stress incontinence). - Ultra-HY for **multiple sclerosis.** I've had students ask whether MS is urge or overflow. It shows up repeatedly on the NBMEs as urge; I've never seen it associated with overflow. I'd say ~1/3 of urge incontinence vignettes on NBME forms are MS. - Other vignettes will be peri-menopausal women, or idiopathic in old women. - Mechanism is "detrusor hyperactivity," or "detrusor instability." - Vignette can mention woman has urge to void when stepping out of her car, or when sticking her key in the car/front door of her house. Sounds weird, but these are important Qs to ask when attempting to diagnose urge incontinence. - UTIs can present similarly to urge incontinence. Some students have asked, "Well isn't that because UTIs are a cause of urge incontinence?" Not really. It just happens to be that UTIs can sometimes cause transient urinary urgency. For example, if they give a Q where they say patient had Hx of urinary catheter + now has dysuria and urinary urgency, answer = "urinary tract infection" on NBME; "detrusor hyperactivity" is wrong answer. - Treatment is **oxybutynin** (muscarinic receptor antagonist); this ↓ activity of the detrusor muscle of the bladder. - Some students get hysterical about mirabegron (β3-agonist) because it has unusual mechanism, but I've never seen NBME forms assess this.
Overflow	- Will be due to either BPH or diabetes on USMLE. - Will have ↑ **post-void volume.** Normal is < ~50 mL. On USMLE for overflow, they'll give you 300-400 mL as post-void volume. - As I talked about earlier for BPH, they will give old dude + high creatinine (post-renal azotemia). Next best step is "insertion of catheter" to relieve the obstruction. If they don't have this listed, "measurement of post-void volume" can be an answer. We then treat the BPH with finasteride (5α-reductase inhibitor) or an α1-blocker (tamsulosin, terazosin). - For diabetes, the mechanism is neuropathy to the bladder causing "neurogenic bladder," or "hypotonic bladder," or "hypocontractile bladder/detrusor muscle." - For neurogenic bladder causing overflow incontinence +↑ post-void volume, remember that USMLE is first obsessed with "measure post-void volume" and "insertion of catheter" if they are listed. They will not force you to choose between the two. But they like these answers prior to giving medications. - Give **bethanechol** (muscarinic receptor agonist); this stimulates the detrusor muscle. - Making sure you don't confuse oxybutynin and bethanechol is pass-level for USMLE.

Cauda equina syndrome
- Will present as leg pain + urinary retention on USMLE. - Can be idiopathic, from disc herniation, or other trauma. - Metastases is HY cause (i.e., random guy over 50; or female with Hx of mastectomy 5 years ago). - "Saddle anesthesia" is an overrated detail and doesn't show up in the vast majority of NBME Qs, as per my observation. - Cauda equina is the collection of nerves at the end of the spinal cord. Impingement on these nerves is what causes the syndrome.

Miscellaneous	
Enuresis	- Nocturnal enuresis (i.e., wetting the bed) is normal on occasion up through age 5. - First step in ↓ recurrence is behavioral science-type answers (i.e., ↓ stressors + ↑ time spent with child). The question stem might mention a child wets the bed after a sibling is diagnosed with ALL (i.e., stressor). - After behavioral-type answers, the next step is star chart (positive reinforcement), where the child receives a star on a chart every time he/she doesn't wet the bed, where after, e.g., 10 stars, he/she gets an extra dessert. - If star chart is not listed, enuresis alarm is the next best step (i.e., literally an alarm that detects moisture in a child's underwear, typically using a sensor, where if moisture is detected, indicating urination, the alarm sounds or vibrates child awake, thereby encouraging him or her to stop urinating and go to instead use the toilet). - Answers such as imipramine, desmopressin, and water deprivation are wrong on USMLE.
Encopresis	- Shitting your pants. Normal on occasion up through age 5.

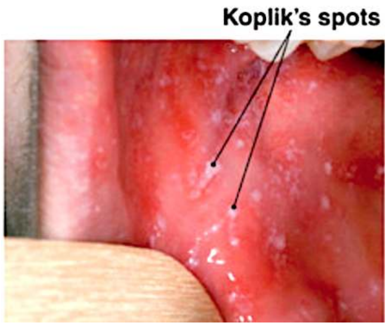
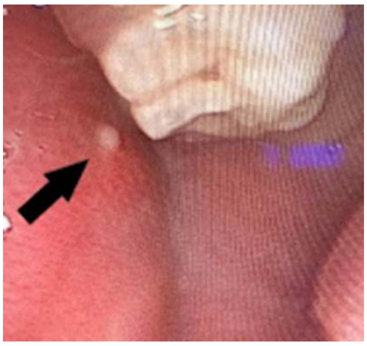
Urothelial malignancies for FM	
Renal cell carcinoma (RCC)	- Classic Q is a smoker over 50 with red urine and a painful flank mass. - Q need not mention smoking, but it is biggest risk factor. - Can cause **polycythemia** due to ↑ EPO and **hypercalcemia** due to PTHrp secretion (correct, same as squamous cell carcinoma of the lung). - For example, they say 55-year-old male + red urine + polycythemia + hypercalcemia + they ask for the source of the malignancy → answer "kidney," not lung. Squamous cell carcinoma of the lung won't cause hematuria, nor does it metastasize to the kidney.
Wilms tumor (Nephroblastoma)	- The answer for painless flank mass in a kid age 2-4 years. - If they give you a kid 2-4 years with a midline mass, that is neuroblastoma, not Wilms tumor.
Transitional cell carcinoma of the bladder	- Most common bladder cancer. - Classic vignette is hematuria in smoker *without* a painful flank mass or polycythemia/hypercalcemia (otherwise RCC). - USMLE wants you to know smoking is most common risk factor, but aniline dyes (industrial clothing dyes) and naphthylamine are important causes.

FM Gastro

Disorders affecting lips/oral cavity for USMLE	
Peutz-Jeghers syndrome	- Combination of perioral melanosis (fancy word for hyperpigmentation around the lips/mouth) + **hamartomatous** colonic polyps. 
Osler-Weber-Rendu	- Aka hereditary hemorrhagic telangiectasia; autosomal dominant. - NBME loves showing a mouth or fingernail picture of telangiectasias.  - Q will give nosebleeds + show you the above pic. There can be high-output cardiac failure due to pulmonary AV fistulae. - GI bleeding can occur leading to anemia.
Plummer-Vinson syndrome	- Triad of iron deficiency anemia + esophageal webs (dysphagia) + angular cheilitis (cracked corners of mouth).  - NBME Q can also mention pica (iron deficiency sign where patient eats clay, starch, or ice), or they can show spoon-shaped nails (koilonychia), which are a sign of severe iron deficiency.
Lip psoriasis	- Just know it's possible. USMLE can mention it on upper lip or forehead, and this somehow confuses students, where they think it has to be on extensors only.

	
Aphthous ulcer	- Aka canker sore. - Usually appears as very painful lesion, self-resolving lesion on labial mucosa. - Not infective; idiopathic; thought to have very mild autoimmune association; sometimes associated with certain triggers like spicy food. 
Kawasaki disease	- Vasculitis that causes **5+ days of fever** + injected (red) eyes and/or lips/tongue + cervical lymphadenopathy + edema of dorsa of the hands + **desquamation of palms/soles** (often mentioned as palms/soles "rash," but not true rash).  - Students obsess over coronary artery aneurysms as though they're so HY. They're not. USMLE basically never mentions them. Tx = aspirin + IVIG. Only thing we can give aspirin to kids for (Reye syndrome).
Perioral impetigo	- Often confused with herpes. 

	- If they show you image of young kid in particular with lip lesions, it's usually impetigo caused by *S. aureus* or Group A Strep (*S. pyogenes*). - Can lead to PSGN.
Herpes labialis	- Caused by HSV1 or 2. USMLE doesn't give a fuck about HSV1 being the lips and HSV2 being the genitals. Bunch of nonsense perpetuated by other resources.  - Primary infection presents with fever and regional lymphadenopathy. Recurrences aren't as severe. - HSV goes latent in sensory nerves. - Treat with acyclovir or valacyclovir.
Scarlet fever	- Caused by Group A Strep. - Causes strawberry tongue / red lips + salmon-pink maculopapular body rash.  - Treatment is penicillin to prevent rheumatic fever (type II HS); can also lead to PSGN (type III HS).
Hand-foot-mouth	- Caused by Coxsackie A. - Causes benign, but contagious, lesions on, you guessed it LOL! – the hands, feet, and mouth. - Usually pediatric, but can present in adults (i.e., daycare workers, parents).  - Don't confuse with coxsackie B, which can cause dilated cardiomyopathy, diabetes type I, and pleurodynia.
Herpangina	- Also caused by coxsackie A. - Presents as oropharyngeal vesicles or sores.

	 - Can occur with or without hand-foot-mouth.
Koplik spots	- Caused by measles. - White/blue-ish spots on buccal mucosa.  - One of the ways to distinguish measles (rubeola) from rubella (German measles). - The same way you can memorize Koplik spots as unique to measles, you should memorize sub-occipital and post-auricular lymphadenophaty as unique to rubella. - Both measles and rubella have a head-to-toe masculopapular body rash. - Both can present with fever, cough, coryza, conjunctivitis. These findings are non-specific for viral infection; it's been erroneously attributed to only measles over the years, but I've it show up in numerous NBME Qs for a variety of viruses).
Sialolithiaisis	- Stones within the ducts of the salivary glands. - Stensen duct is the opening of the parotid duct (from parotid gland) into the oral cavity, which is located near the 2^{nd} upper molar bilaterally.  - Sometimes a Q can say a patient has pain or inflammation on the buccal mucosa across from the second upper molar, and sialadenitis (inflammation) or sialolithiasis can be an answer.
Leukoplakia	- White-ish, painless, rough patch on lateral tongue.

	 - Precursor to squamous cell carcinoma of tongue. - Biggest risk factor is smoking / chewing tobacco.
Oral hairy leukoplakia	- Looks like leukoplakia, but caused by EBV. - Not considered premalignant (no dysplasia on biopsy).
Oropharyngeal candidiasis	- Painless white plaque on palate or tongue that bleeds when scraped.  - Seen in immunocompromised patients, such as AIDS or organ transplant recipients on immunosuppressant agents. - Can be seen in patients with asthma who use inhaled corticosteroids. Patients need to rinse their mouths with water after use to ↓ candida risk. - Tx with nystatin mouthwash. - Odynophagia (painful swallowing) in an immunocompromised patient is Candidal esophagitis until proven otherwise.

HY Esophageal conditions for USMLE	
Gastroesophageal reflux disease (GERD)	- Irritation of esophageal mucosa by gastric acid; usually caused by ↓ **lower esophageal sphincter (LES) tone.** - USMLE wants you to know **obesity** is a risk factor for GERD, likely due to ↑ stretching and pressure applied to the LES. - Hiatal hernia is also a risk factor. - Classic presentation is burning in the throat/chest after eating a meal. - Can present with **nocturnal cough** or **recurrent pneumonitis.** These are HY findings in patients who don't have classic esophageal irritation symptoms. - Can lead to Barrett esophagus, which leads to esophageal adenocarcinoma. - First step in diagnosis is 2-week trial of proton pump inhibitor (PPI), such as omeprazole, which will ↓ symptoms. - PPIs are more efficacious than H2-blockers (e.g., cimetidine). If you see both a PPI and H2-blocker as answers for a GERD Q, choose the PPI. - One 2CK form has "2-week trial of H2-blocker" as an answer, but PPI isn't listed, so just know the Q is out there. - If trial of PPI doesn't work, 24-hour esophageal pH monitoring is the answer. - Nissen fundoplication is used last resort, but is asked on NBMEs.

	- Pediatric GERD will present as coughing up milk 2-3x daily. It will not be described as high-energy or forceful (i.e., pyloric stenosis); it will not be bilious (i.e., duodenal atresia; annular pancreas); it will not occur as choking or spitting up during the first feed (i.e., tracheoesophageal fistula). - Answer to pediatric GERD Q is "immature lower esophageal sphincter". It's not complicated, but I've seen students say, "Wait, you can get GERD in kids?" - Tx for Peds GERD is thickened feeds (addition of dry rice to formula) and positional change prior to any use of PPIs.
Adenocarcinoma	- Affects lower 1/3 of esophagus. - HY pathogenesis is: GERD → Barrett esophagus → adenocarcinoma. - Will present in patient over 50 who has Hx of GERD with either 1) new-onset dysphagia to solids, or 2) dysphagia to solids *that progresses* to solids and liquids. - The "new-onset dysphagia" can refer to 3-6-month Hx in patient with, e.g., 10-20-year Hx of GERD. - 2CK wants immediate endoscopy in either of the above situations (i.e., don't choose barium first). - Sometimes rather than making you choose endoscopy straight up, they'll tell you in the last line an endoscopy was performed and shows a stricture, then they'll ask for next best step → answer = biopsy of the stricture.
Squamous cell carcinoma	- Affects upper 2/3 of esophagus. - Biggest risk factors for USMLE are **heavy smoking/alcohol use.** - Other risk factors like burns, chemicals, achalasia, etc., are nonsense. - Will present in patient over 50 who is heavy smoker/drinker who has 1) new-onset dysphagia to solids, or 2) dysphagia to solids that *progresses* to solids and liquids. - Same as with adenocarcinoma, USMLE wants immediate endoscopy as the answer, followed by biopsy of a stricture or lesion if present.
Zenker diverticulum	- Outpouching of esophagus above the cricopharyngeus muscle (just above the upper esophageal sphincter). - Often presents as overweight male over 40-50 with gurgling sounds when swallowing, or regurgitation of undigested food. - Halitosis (bad breath) is buzzy but only mentioned in maybe 1/3 of Qs. - Diagnosis is made with barium swallow. - Treatment on 2CK NBME is "diverticulectomy."
Achalasia	- Tightening (i.e., ↑ tone) of the LES due to loss of NO-secreting neurons in the Auerbach (myenteric) plexus. - Presents usually as dysphagia **to both solids and liquids from the start.** This is in contrast to cancer, which will be new-onset dysphagia to just solids, or *progression* from solids only to solids + liquids. The HY point is that neurogenic causes of dysphagia (i.e., achalasia) lead to solids + liquids dysphagia. - Almost always idiopathic; can rarely be caused by Chagas disease. - New 2CK NBME has "South American trypanosomiasis" as the answer for achalasia, which is aka Chagas disease. - First step in diagnosis is barium swallow, which shows a classic "bird's beak" appearance. - After the barium is performed, esophageal manometry (a pressure study) is confirmatory. Biopsy is not traditionally performed. - Treatment is pneumatic (balloon) dilation, followed by myotomy (cutting of muscle fibers of the LES).
Systemic Sclerosis (aka scleroderma)	- As discussed in detail in the Pulm section, this is an idiopathic autoimmune disease characterized by multi-organ system fibrosis and hardening of tissues (i.e., sclerosis). - Can lead to GERD. - Limited type = **CREST** = (**C**alcinosis, **R**aynaud's, **E**sophageal dysmotility, **S**clerodactyly, **T**elangiectasias).

	- Diffuse type = CREST + high BP (due to renal involvement). - Both limited and diffuse can cause pulmonary fibrosis leading to pulmonary hypertension. Some students erroneously think this is only seen in diffuse.
Mallory-Weiss tear	- Classically retching/vomiting in an alcoholic that causes stretching and tearing of the gastroesophageal junction leading to **small amounts** of blood in vomitus. - Can also occur in patients with eating disorders or hyperemesis gravidarum. - It's to my observation on NBME material that the highest yield point about MW tear is that it does **not** cause high-volume hematemesis. If the vignette mentions high-volume blood, that is ruptured varix instead (discussed below).
Esophageal varices	- Enlarged and tortuous superficial esophageal veins due to portal hypertension in patient with cirrhosis or splenic vein thrombosis. - Most Qs will give patient with **high-volume hematemesis.** In contrast, MW tear is 'just a little blood." This is not hard-rule, but it's most NBME Qs. - Ruptured varices are lethal ~50% of the time. The patient will vomit high volumes of blood. - Propranolol is prophylaxis, not for acute treatment. - Treatment is endoscopy + banding. - Octreotide can also be used for acute bleeding, but endoscopy + banding is best answer on USMLE if you are forced to choose.
Diffuse esophageal spasm	- Idiopathic spasm of the esophagus. - All you need to know is that this that causes pain that can mimic angina pectoris, but patient will be negative for cardiac findings/disease. - "Nutcracker esophagus" is a similar condition that is due to idiopathic hyperperistalsis and causes pain similar to DES.
Eosinophilic esophagitis	- Idiopathic allergic condition of the esophagus that presents with difficulty swallowing. - Patient might have history of asthma or atopy.

Peptic ulcers HY causes	
- "Peptic ulcer" is an umbrella term that refers to both duodenal and gastric ulcers. - Gastric ulcers cause pain immediately with meals (due to ↑ acid secretion). Patients can sometimes lose weight due to aversion to pain from eating. - Duodenal ulcers cause pain 1-2 hours after meals (due to ↑ acid entering duodenum). With meals, the pylorus tightens, thereby relieving any residual discomfort. Patients may gain weight since eating ↓ pain.	
Helicobacter pylori	- Responsible for almost all duodenal ulcers (>95%), whereas it causes a lower % (only >60%) of gastric ulcers. This is merely because the latter are caused by many other things as well, so we simply have ↓ fraction caused by *H. pylori.* In other words, there's no special tropism of *H. pylori* toward duodenal over gastric mucosa. - Mechanism for ulcers that shows up on USMLE is: "secretes proteinaceous substrates that damage mucosal lining." This is correct over "↑ gastric acid secretion" if both are listed side-by-side, even though *H. pylori* does ↑ gastrin levels, which ↑ acid secretion. - Produces urease, which causes ↑ ammonia production around the organism, allowing it to survive in the ↓ pH of the stomach. - Diagnose *H. pylori* with urease breath test or stool antigen. - Treat *H. pylori* with CAP → clarithromycin, amoxicillin, PPI (e.g., omeprazole). - USMLE really doesn't give a fuck about the treatment, but CAP is safe to know. Metronidazole, tetracycline, bismuth, and PPI tetrad is used if CAP fails (students ask about those other drugs).
Gastrinoma	- Zollinger-Ellison syndrome causes recurrent duodenal ulcers and sometimes jejunal or ileal ulcers. - Can be part of MEN1 or idiopathic.

	- *H. pylori* is more common than gastrinoma. As mentioned above, >95% of duodenal ulcers are due to *H. pylori*. There is an NBME Q where they give older male with a duodenal ulcer + no other information + they ask for most likely cause → answer = testing for *H. pylori*; gastrinoma is wrong. - Vignettes can be tricky with gastrinoma and tell you the patient has 8-10 watery stools daily, where you say, "That sounds like VIPoma." But they'll also tell you the patient has history of abdo pain after meals.
NSAIDs	- Cause gastric ulcers. I haven't seen these cause duodenal ulcers on USMLE. - Prostaglandins are necessary for stimulation of gastric alkaline mucous production and maintenance of the gastric lining. NSAIDs → ↓ prostaglandin production → disruption of gastric lining. This can lead to both ulcers and irritation (gastritis). - USMLE wants you to know that PPIs are first-line for ulcer treatment in general, but that **misoprostol** is a PGE1 analogue that is used in NSAID-induced ulcers (i.e., we're replenishing the ↓ prostaglandin from NSAIDs).
Smoking/Alcohol	- Not tested as overt causes of ulcers on USMLE. But you should know that smoking and alcohol are believed to *decrease healing* of pre-existing ulcers.

Gastritis causes + HY points	
- Acute gastritis will present as irritation leading to bleeding of gastric mucosa (e.g., from NSAIDs). - Chronic gastritis will be an atrophy or autoimmune destruction of the mucosa associated with ↓ mucosal thickness, ↓ HCl production (achlorhydria), ↑ gastrin production, and enterchromaffin-like cell hyperplasia (↑ histamine production to compensate for ↓ acid). NBME will give you this constellation of findings and then just simply have "chronic gastritis" as the answer, with acute gastritis not even listed. It's not hard, but I've seen students miss this a lot.	
H. pylori	- Causes what is referred to as "Type B gastritis," which is inflammation tending to affect the antrum of the stomach. - Can lead to pyloric channel ulcers + gastric outlet obstruction. This will present as a "succussion splash" on physical examination.
Pernicious anemia	- Causes what's referred to as type A gastritis," which affects mostly the fundus/body of the stomach. - Autoimmune antibody-mediated destruction of parietal cells. - Sometimes antibodies can be against intrinsic factor. - Ultra-HY cause of B12 deficiency (↑ MCV + hypersegmented neutrophils +/- neuropathy) on USMLE. - Associated with other autoimmune diseases, e.g., vitiligo. So if they give you ↑ MCV in patient with, e.g., type I diabetes, you should think, "Autoimmune diseases go together, so if the patient has one AA disease, he/she has ↑ propensity for others." Don't worry about strict HLA associations here.
NSAIDs	- Answer on USMLE for GI bleeding in someone taking, e.g., indomethacin or naproxen. - Causes type A gastritis, affecting the fundus/body. USMLE doesn't specifically give a fuck, but you should basically be like, "*H. pylori* causes antral gastritis, whereas other causes like NSAIDs and pernicious anemia are fundus/body of stomach." - Prostaglandins are necessary for stimulation of gastric alkaline mucous production and maintenance of the gastric lining. NSAIDs → ↓ prostaglandin production → disruption of gastric lining. This can lead to both gastritis (inflammation) and ulcers. - As mentioned above for ulcers, **misoprostol** can be used in these patients following PPI.

Gilbert syndrome
- Pronounced "Jeel-BEAR, not "GILL-burt." Gastroenterologist I met once tripped out over med students pronouncing this wrong. - Partial deficiency of bilirubin uptake enzyme at the liver (UDP glucuronosyltransferase). - Presents as isolated ↑ indirect bilirubin with yellow eyes in young adult with stress factor, such as studying for exams, or recent surgery/trauma. The patient will otherwise be completely healthy. - No treatment necessary.

Autoimmune liver conditions	
Primary biliary cirrhosis	- Inflammation of bile ducts within the liver, leading to their destruction. - Answer in a woman 20s-50s who has generalized pruritis, ↑ serum cholesterol, ↑ ALP, ↑ direct bilirubin. - USMLE loves to mention **Hx of autoimmune disease in the patient or family member** (because autoimmune diseases go together). So they'll say she has type I diabetes mellitus, SLE, or vitiligo; or her brother has RA, etc. - They can mention a stone is present in the gallbladder on ultrasound, which gets some students real emotional / confused, but it makes sense since patient has ↑ cholesterol. It's just a distractor point and doesn't relate to the actual diagnosis at hand. - Diagnose with anti-mitochondrial antibodies as first step, followed by liver biopsy to confirm. - Initial Tx = ursodeoxycholic acid (ursodiol).
Autoimmune hepatitis	- Young adult with ↑ LFTs who has (+) anti-smooth muscle antibodies.

HY Hepatobiliary conditions for FM	
Pancreatic cancer	- Head of pancreas cancer impinges on common bile duct, resulting in obstructive jaundice (↑ ALP + ↑ direct bilirubin) in smoker with weight loss, or in patient who had gallbladder taken out years ago (so it clearly can't be due to a stone in common bile duct). - Patient will not be febrile; can have dull abdominal pain. - Courvoisier sign is a painless, palpable gallbladder in an afebrile patient who's jaundiced. This is pancreatic cancer until proven otherwise and is pass-level.  - **Pancreatic enzymes are normal in pancreatic cancer.** - USMLE wants **CT of the abdomen** to diagnose.
Cholelithiasis	- Stones in the gallbladder. - Presents with biliary colic, which is acute-onset waxing/waning spasm-like pain in the epigastrium or RUQ.

	- Classic demographic is 4Fs = Fat, Forties, Female, Fertile for cholesterol stones. This is because estrogen upregulates HMG-CoA reductase, causing ↑ cholesterol synthesis and secretion into bile. NBME will give you standard vignette of 4Fs, and then the answer will just be “increased secretion of cholesterol into bile.” - Another NBME Q gives vignette of cholelithiasis, and then rather than asking for the diagnosis, they ask what the patient most likely has → answer = “lithogenic bile.” Slightly awkward, but means promoting the formation of stones. - ↑ cholesterol stones in pregnancy not just due to estrogen effect but also because progesterone slows biliary peristalsis (biliary sludging). Tangentially, progesterone also slows ureteral peristalsis, which is why there’s ↑ risk of pyelonephritis. - Cholesterol stones most common, but pigment (calcium bilirubinate) stones are exceedingly HY for hereditary spherocytosis and sickle cell, due to ↑ RBC turnover. - USMLE wants you to know that splenectomy is Tx for hereditary spherocytosis to ↓ incidence of cholelithiasis. Sometimes the vignette wants “cholecystectomy + splenectomy” as combined Tx. Or if they say this was in a parent’s Hx in patient with low Hb, you know the Dx is hereditary spherocytosis (autosomal dominant). - Diagnose with **abdominal ultrasound.** - Ursodeoxycholic acid (ursodiol) ↓ secretion of cholesterol into bile. USMLE just wants you to know this MOA + that it can be used in patients with asymptomatic cholelithiasis, those declining cholecystectomy, and in pregnancy.
Cholecystitis	- Cholelithiasis + fever. - Vignette will sound exactly like cholelithiasis, but if we add a fever on top of it, we now just call it cholecystitis. - Virtually always due to obstruction by a stone + infection as a result. - Diagnose with abdominal ultrasound, showing the stones. - If ultrasound is negative, do a HIDA scan. - Tx is cholecystectomy. - ALP and bilirubin will not be increased 19/20 questions. This is because inflammation of the gallbladder doesn’t mean we have any form of common bile duct obstruction. However there is one nonsense Q on a 2CK NBME where 4 of the answers are wildly wrong, with correct answer being cholecystitis in setting of high ALP + direct bilirubin. Since cholecystitis is caused by stones virtually always, the implication is patients can occasionally have concurrent choledocholithiasis.
Choledocholithiasis	- Stone anywhere within the biliary tree. Don’t confuse with cholelithiasis. - Obstructive jaundice (↑ ALP + ↑ direct bilirubin) in patient who has Hx of cholelithiasis or Hx of cholecystectomy performed within the past week. - Regarding the latter, the Q can say patient had cholecystectomy performed a week ago + “intra-operative cholangiography was not performed.” The implication is bile duct patency is supposed to be visualized during cholecystectomy to ensure there isn’t a stone retained within the biliary tree. However, the Q need not say “intra-operative cholangiography was not performed.” This makes the Q pass-level. - **USMLE wants abdominal ultrasound followed by ERCP** as the answer. ERCP tends to show up as what they want, but there is a Q floating around somewhere that asks for ultrasound first, so know that sequence. - Being able to discern choledocholithiasis from pancreatic cancer is vital for USMLE. If they give you obstructive jaundice but say the patient is a heavy smoker with weight loss, or hasn’t had a gallbladder for 25 years, you know it’s head of pancreas cancer, not choledocholithiasis. And then,

	once again, if it sounds like pancreatic cancer but the CT is negative, that's cholangiocarcinoma and we do ERCP.
Gallstone pancreatitis	- **Gallstone pancreatitis** is a type of choledocholithiasis in which the stone has descended distally enough in the common bile duct that it now obstructs the hepatopancreatic ampulla. This results in a backflow of pancreatic enzymes to the pancreas causing damage → acute pancreatitis. - Vignette will give obstructive jaundice *but also high pancreatic enzymes*. - In other words, the Q will give ↑ ALP, ↑ direct bilirubin, and ↑ amylase / lipase. - I repeat that pancreatic cancer will never have elevated enzymes on USMLE, so this is a pass-level means to distinguish. - New 2CK NBME has **CT of the abdomen** as done first, prior to ERCP. This is because even though we have a stone requiring removal from the biliary tree, CT of the abdomen first looks for pancreatic fluid collection.
Cholangitis	- Inflammation of the bile ducts. You must know **Charcot triad for cholangitis: 1) fever, 2) jaundice, 3) abdominal pain.** We classically learn this as "RUQ/epigastric pain," but I can tell you USMLE doesn't give a fuck and will just say "abdominal pain." I've had students get cholangitis Qs wrong because they're like, "I thought it had to be RUQ pain though." And I'm like, yeah, I agree, but USMLE doesn't give a fuck. - Usually occurs in setting of ascending bacterial infection (ascending cholangitis) or IBD (usually UC; primary sclerosing cholangitis). - USMLE wants antibiotics + **ERCP to diagnose and treat.** Biliary drainage by ERCP ↓ morbidity and mortality.
Acute pancreatitis	- ↑ amylase and/or lipase in patient who has abdominal pain. - Pain need not radiate to the back. - Alcoholism and stones are two biggest risk factors. - Can be caused by choledocholithiasis (gallstone pancreatitis; ↑ pancreatic enzymes in setting of ↑ ALP and ↑ direct bilirubin). - Low calcium + high glucose = HY poor prognostic indicators on USMLE. - Degree of lipase and/or amylase elevation doesn't relate to prognosis. - Family Med Qs don't focus on management; they just want you to be able to diagnose as a DDx of abdo pain.
Chronic pancreatitis	- **Steatorrhea in alcoholic** or patient with recurrent acute pancreatitis episodes → results in pancreatic burnout with ↓ production of pancreatic enzymes → ↓ proteases/lipases → malabsorption. - In other words, we have **normal** pancreatic enzymes – i.e., they are **not** elevated. - Question can say alcoholic has ↑ fat and protein seen in stool. - D-xylose test is normal. This is a monosaccharide that is readily absorbed by the small bowel insofar as the intestinal lining is intact. So if we get steatorrhea + normal D-xylose, this can be due to pancreatic insufficiency or lactose intolerance. If D-xylose test is abnormal, we know the malabsorption is due to an intestinal lining issue, such as Celiac or Crohn. - Tx = pancreatic enzyme supplementation (pancrelipase).
Primary biliary cirrhosis	- As mentioned earlier, this will be a woman 20s-50s who has generalized pruritis, ↑ cholesterol, ↑ ALP, ↑ direct bilirubin, and Hx of autoimmune disease in her or a relative. - Diagnose with anti-mitochondrial antibodies as first step, followed by liver biopsy to confirm. - Initial Tx = ursodeoxycholic acid (ursodiol).

Systemic inflammatory response syndrome (SIRS)	
- In the setting of stress (i.e., due to trauma, surgery, autoimmune flare, infection), catecholamines and ↑ sympathetic activity might shift the patient's vitals out of the normal range. - SIRS can also occur in patients **without** infection who have systemic stress (e.g., autoimmune flare).	
SIRS	- 2 or more of the following: - Temperature <36C or >38 (<96. 8F or >100.4). - HR >90. - RR >20. - WBCs <4,000 or >12,000.
Sepsis	- SIRS + source of infection.
Septic shock	- Sepsis + low BP.
- In other words, sometimes the stem might give you a patient's vitals slightly out of the normal range in the setting of trauma, surgery, or autoimmune flares, and you have to be able to say, "There's no infection. That's just SIRS from sympathetic activation." - Knowing if a patient is septic is important for management of patients on 2CK, where sometimes antibiotic regimens are stepped up. For example, when treating PID in Family Medicine, if the patient is septic, intravenous ceftriaxone and azithromycin is correct on one of the forms; IM ceftriaxone and oral azithromycin is wrong. This is because the latter is for most patients who have PID but aren't septic.	

HY Peds GI diagnoses for FM	
Pyloric stenosis	- Forceful/projectile **non**-bilious vomiting in neonate days to weeks old. Obstruction is above level of duodenum, so we won't see bile. - Students fixate on exact age of the kid for pyloric stenosis versus duodenal atresia. USMLE doesn't give a fuck and will give variable ages. - Hypertrophic pylorus; "olive-shape mass" in abdomen is buzzy but rarely seen in Qs. - **Ultrasound** done to diagnose; myotomy to treat. - USMLE wants you to know this is almost always a one-off / sporadic developmental defect, where the neonate will not have a broader syndrome. In contrast, duodenal atresia is usually Down syndrome.
Duodenal atresia	- Bilious vomiting in neonate. Bilious = obstruction at level of duodenum or lower. - Associated with **Down syndrome,** albeit not mandatory. - Will cause **double-bubble sign** on **abdominal x-ray.**
Intussusception	- Intussusception is telescoping of the bowel into itself. - Almost always under age 2. - Idiopathic, but can be triggered by viral infection or rotavirus vaccination, where mesenteric lymphadenopathy can act as a "lead point" for the intestinal telescoping. Underlying Meckel diverticulum is also a common lead point. - In ultra-rare scenarios, intussusception in elderly can occur due to colon cancer. - Classic presentation is intermittent/colicky abdominal pain, where the kid will have episodes of drawing the legs to the chest or squatting, with blood in the stool +/- bilious vomiting. - **An abdominal mass** (i.e., sausage-shaped mass) is a key feature, but not in all Qs. - Diagnosis is first done with ultrasound, which may show a sausage mass or target sign (I've never seen NBME show the image, so they don't care). After the ultrasound, **enema is both diagnostic and therapeutic.** USMLE doesn't care what kind of enema is used; air-contrast enema is classic, but you will see all different types of enemas as correct answers. - "Air enema with ultrasound" is the answer on one 2CK form.
Midgut volvulus	- Failure of rotation of proximal bowel. - Almost always under age 1. - Presents similarly to intussusception (often mistaken for it), but there is **no mass.** - Upper-GI series shows corkscrew appearance.

	 - If an abdominal x-ray is performed, it will show "dilated loops of small bowel with air fluid levels," since we have an obstruction. - Treatment is surgical.
Meckel diverticulum	- Outpouching near the terminal ileum. - Can contain heterotopic gastric or pancreatic mucosa, resulting in bleeding and **blood in the stool.** - Another important point is that Meckel **need not be pediatrics.** This notion of rule of 2s (i.e., 2 feet from terminal ileum, 2 types of heterotopic tissue, age 2) is garbage with respect to age for USMLE. There are Qs where it shows up in adults and students get them wrong saying, "Wait, I thought it was supposed to be kids?!" → No. USMLE will happily give you a 19-year-old with Meckel. - Diagnosed with Meckel (Tc99) scan, which localizes to terminal ileum. - Only treated surgically if symptomatic with bleeding. Otherwise can be left alone.
Cystic fibrosis	- Presents in FM as **exocrine pancreatic insufficiency.** - Secretions within the pancreatic ducts are inspissated (meaning desiccated / dried up within a lumen), making them sticky. This means the enzymes can't make their way to the duodenum → fat-soluble vitamin malabsorption → NBME exams love vitamin E deficiency in CF in particular (presents as neuropathy). - D-xylose test is normal, since the intestinal lining/architecture is intact. Any malabsorption that occurs is due to mere paucity of enzymes in the small bowel. Recall that abnormal D-xylose test would be Celiac and Crohn on USMLE.

Autoimmune intestinal absorptive disorders	
Celiac disease	- Intolerance to gluten (i.e., to gliadin proteins found in wheat, oats, rye, and barley; but not rice). - Causes type-IV hypersensitivity response where T cells attack the small intestinal villi, resulting in **flattening of the villi** and malabsorption. - Even though the disorder is T-cell-mediated destruction of the villi, patients still develop antibodies which are HY: anti-endomysial (aka anti-gliadin) and anti-tissue transglutaminase IgA. - Patients with concurrent IgA deficiency will have false (-) results for tissue transglutaminase IgA antibody. Since "autoimmune diseases go together," and "autoimmune and immunodeficiency syndromes go together" patients with IgA deficiency have 15x greater likelihood of developing Celiac. - Celiac presents as vague bloating/diarrhea in patients who might not be able to pinpoint what they're eating to cause the symptoms. - I'd say one of the highest yield points on USMLE regarding Celiac is that it can cause **iron deficiency anemia.** This is because iron absorption in the duodenum is impaired from the flattened villi. - For example, you might get a difficult/vague vignette where you're not sure if the diagnosis is Celiac or lactase deficiency, but you see that hemoglobin is low, and you're like "Boom. Celiac."

	- Associated with dermatitis herpetiformis, which presents as itchy, vesicular lesions on extensor areas. This causes "IgA deposition at dermal papillae." - D-xylose test is abnormal because the intestinal lining/architecture is abnormal (i.e., we have flattening of villi). - Diagnosis is made via antibody screening, followed by duodenal biopsy as confirmatory. - Treatment of Celiac is simply with gluten-free diet.
Lactose intolerance	- Deficiency of brush border disaccharidase (i.e., aka lactase deficiency). - Diarrhea/bloating in response to dairy. - In contrast to Celiac disease, there is **no iron deficiency anemia.** - ↑ incidence in Asians, but can also develop idiopathically starting in teenage years-onward. When a young adult has a new-onset diarrhea-related condition, it's usually lactose intolerance, as lactase production can ↓ with age. But once again, USMLE can trick you, so I'll be an asshole and reiterate that if you see ↓ Hb, choose Celiac instead. - Secondary lactose intolerance can occur following viral gastroenteritis (e.g., rotavirus or Norwalk virus) → sloughing of the tips of villi → vignette will say new-onset diarrhea/bloating following gastro illness 1-2 weeks ago. This is only transient and patient will recover. This is HY cause of lactose intolerance. - **Biopsy is normal.** This is in contrast to the flattened villi in Celiac. - Diagnosis is made using hydrogen breath test or detecting ↓ stool pH. - D-xylose test is normal, since the intestinal lining/architecture is intact. - Treatment is with avoidance of lactose-containing products, or with lactase pills.
Milk protein allergy	- Biggest risk factor is not being exclusively breastfed for the first 6 months of life. - Will present as **blood in the stool** in a child who they say is, e.g., 4 months old, who was started on formula 3 weeks ago. - Treatment is switching to **hydrolyzed casein formula.** Switching to soy-based formula is wrong fucking answer. There is up to 50% crossover of allergy cases with kids who have milk- and soy-protein allergy. - Vignette can say kid was started on either a cow-milk or soy formula when symptoms started. It doesn't matter. Just choose hydrolyzed casein as answer.

Inflammatory bowel disease (IBD)
- Refers to both ulcerative colitis (UC) and Crohn. - Family Med forms won't make you differentiate between Crohn and UC based on all of the nitpicky details. They just want you to diagnose IBD in general in a patient who has bloody, mucoid stools +/- other findings: - Both are associated with HLA-B27 → **PAIR** → **P**soriasis, **A**nkylosing spondylitis, **I**BD, **R**eactive arthritis. - For example, if a patient has psoriasis + bloody stools, you say, "The bloody stools are probably IBD." Or likewise, if patient with known IBD has lower back pain that's worse in the morning, you say, "that's probably ankylosing spondylitis." - Both UC and Crohn can be associated with other autoimmune diseases unrelated to HLA-B27, like vitiligo. - Both can cause anterior uveitis (non-specific finding seen in many autoimmune diseases). - Both have ↑ risk of colon cancer if the colon is involved. But if you're forced to choose, UC has > risk than Crohn because the colon is always involved in UC but not always in Crohn. - Both are treated with 5-ASA NSAID compounds **(mesalamine / sulfasalazine)** or steroids. If they ask first Tx, go with mesalamine or sulfasalazine, whichever they list (they won't list both) over steroids. Q on 2CK form has "prednisone therapy" as answer for Crohn, but a 5-ASA isn't listed. - USMLE wants you to know anti-TNF-α agents (i.e., infliximab, adalimumab, etanercept) are used in IBD in patients who fail initial Tx with 5-ASAs and steroids.

Irritable bowel syndrome (IBS)
- Don't confuse with IBD. IBS is psychosomatic (i.e., psych-related) condition. - Classic vignette will be a woman 20s-40s with stress factors who usually has alternating diarrhea / constipation; can also present as bloating or cramping. - Key detail is that symptoms are relieved with bowel motions. - NBME assesses "smooth muscle hypersensitivity" as the answer for the mechanism for IBS. - Treatment for diarrhea-predominant IBS is loperamide, which is an opioid that causes constipation. USMLE will give vignette of IBS with diarrhea, and then the answer is just "mu-opioid receptor agonist."

Other HY GI motility scenarios	
Diabetes	- Diabetic gastroparesis = ↓ peristalsis due to neuropathy of the GI tract. - Presents as GERD-like presentation in someone with Hx of advanced diabetes. - USMLE wants endoscopy first to rule-out physical obstruction. If endoscopy is negative, do gastric-emptying scintigraphy (aka gastric emptying scintigraphic assay) to confirm delayed gastric emptying. - Metoclopramide is used 1st for Tx. It is a prokinetic agent (i.e., ↑ peristalsis) and anti-emetic. It is a D2 antagonist but also an antagonist of serotonin 5HT3 and agonist of 5HT4 receptors. The effects on serotonin receptors ↑ gut peristalsis. - Erythromycin can also be used to agonize motilin receptors. - "2-week trial of PPI" is wrong answer. This is the answer for "regular GERD." If the patient has GERD-like presentation with advanced diabetes, however, the Dx is diabetic gastroparesis, not GERD.
Thyroid	- Hypothyroidism can cause constipation. - Hyperthyroidism can cause diarrhea. - USMLE will give vignette of a thyroid condition with GI disturbance, and then the answer will just be "motility disorder." This is same answer for diabetes.
Supplements	- Iron and Aluminum can cause constipation ("Aluminimum amount of feces.") - Magnesium can cause diarrhea.
Drugs	- Verapamil causes constipation. - Macrolides, orlistat, and α-glucosidase inhibitors (e.g., acarbose) cause diarrhea.

Viral hepatitis	
- Hep A and E cause acute hepatitis only. Hep B, C, and D can cause chronic hepatitis. - Hepatitis in general classically has ALT > AST, where #s can be in the hundreds to thousands, but I've seen plenty of variability on NBME Qs, which is why I don't consider this a foundational point.	
Hep A	**- The answer for acute hepatitis in the United States most of the time.** The Q might say the patient had recent travel to Mexico. - Fecal-oral; only causes acute hepatitis. - IgM against HepA means acute infection. - IgG against HepA means patient has cleared infection (because there is no chronic HepA). - For FM, you must know HepA vaccine is indicated for IV drug users and MSM. There's a 2CK NBME Q where they just mention otherwise healthy MSM, and answer is Hep A vaccination.

Hep B	- For Family Med, emigration status from China is HY.
	- Serology very HY:
	
	- (+) Surface antigen = patient currently has HepB.
	- (+) Surface antibody = patient is immune to HepB.
	- (+) Core antibody = patient has HepB now or did in the past.
	    - (+) Core antibody IgM = patient has acute infection.
	    - (+) Core antibody IgG = patient has chronic HepB, OR cleared HepB.
	- (+) Surface antibody / (-) Core antibody = Immune → Vaccinated against HepB.
	- (+) Surface antibody / (+) Core antibody = Immune → History of HepB / cleared it.

	- (-) Surface antibody / (-) Core antibody = Not immune / Susceptible → need to vaccinate. - (-) Surface antigen / (-) Surface antibody / (+) Core antibody IgM → window period (below). - Vaccination against HepB is at birth, 2 months, and 6 months (no longer at 4 months). - If patient has Hx of completed HepB vaccination but has titers that show susceptibility, the answer is just "give more vaccine." Sometimes people's immunity wanes. - Once a susceptible patient is exposed to HepB and the immune system attempts to clear it, sometimes Surface antigen will decline to the point that it is no longer detectible. But at the same time, the Surface antibody might not be high enough / at detectable levels yet. This is called the "window period," where both Surface antigen and antibody are negative, so it can appear as though the patient doesn't have an infection. However, Core antibody IgM will be (+). So the key point is that 1) you know the double-negative Surface antibody/antigen combo is seen in the window period, and 2) that Core antibody IgM is most reliable during the window period.
Hep C	- Transmitted almost exclusively from IV drugs/blood exposure. Not present in breastmilk and non-sanguineous body fluids (in contrast to HepB). - In contrast to HepB, HepC is not considered sexually transmitted. Large longitudinal study of couples with one HepC(+) partner showed sexual transmission almost nil (possibly due to menses exposure). If you're forced to choose for FM / behavioral science Qs, however, still inform that abstinence or barrier contraception minimizes risk. - No vaccine due to ↑↑ antigenic variation (i.e., >7 genotypes and 80 subtypes of HepC exist). - IgM against HepC means acute infection. - IgG against HepC means usually means chronic HepC.
Hep D	- Requires hepatitis B in order to infect, which can be due to co-infection (happening at the same time) or superinfection (occurs later in someone who already has HepB). - If USMLE asks how to prevent HepD infection, answer = vaccination against hepatitis **B.** There is no vaccine against HepD.
Hep E	- Causes fulminant hepatitis / ↑↑ risk of death in pregnant women. - Same as with HepA, only causes acute hepatitis.

Other HY hepatitis causes	
NASH	- Answer on FM shelf for liver condition in an overweight patient with normal labs. - Non-alcoholic steatohepatitis. - Patient will have metabolic syndrome (i.e., ↑ BMI + ↑ lipids) + either mild transaminitis or completely normal labs. I'd say ~50% of Qs will give ALT and AST a very tiny bit elevated; the other ~50% of Qs will give you completely normal labs where you say WTF. - For example, they'll give you 40-year-old patient with BMI of 35 who comes in for routine health maintenance exam who has no complaints + lab studies are all completely normal; then the answer is just "fatty liver" or "non-alcoholic steatohepatitis." Student says, "Wait, but the labs are completely normal though." → Exactly. There's nothing wrong. But it's assumed most overweight people have some degree of fatty liver. - The condition starts out with completely normal labs, then progresses to very mild transaminitis. Over time, LFTs can worsen. Rarely it can cause cirrhosis and carcinoma. - Theoretical Tx is just hit the treadmill more and stop eating your dumb fast food.
Alcoholic	- Classically AST > ALT. - The ratio need not be 2:1. I've seen variations on NBME forms where there can be very mild transaminitis (i.e., both AST and ALT are <100, with AST only slightly higher than ALT). If the vignette is hyper-easy, they'll say something like AST is 400 and ALT is 200. Both should normally be <50 U/L. - Causes steatosis (fatty liver).
Acetaminophen	- Acetaminophen overdose most important for USMLE. *N*-acetylcysteine is antidote. - Activated charcoal should apparently be administered to patients who present very acutely after massive ingestion, as this can ↓ acetaminophen absorption. But I've never seen this on NBME. It tends to be more Qbank that's gotchya-style this way. I've only ever seen *N*-acetylcysteine assessed across NBMEs.

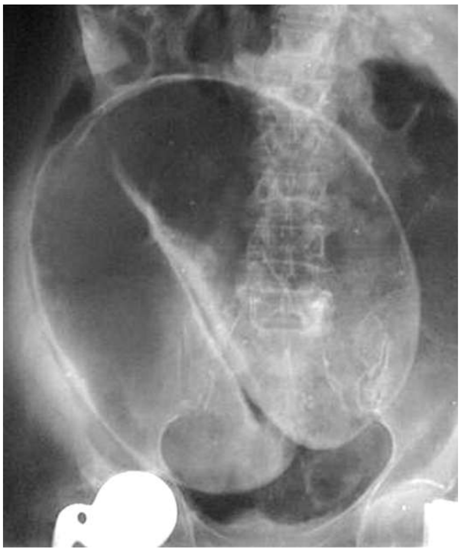
Other drugs	- I'd say some important drugs that can cause transaminitis are: - Statins and fibrates, methotrexate, lithium, valproic acid, propylthiouracil. - **Mild transaminitis** is normal and expected with these agents; the answer is you do not need to decrease dose. **- For statins, myopathy is more common than toxic hepatitis** (on NBME).

Heavy metal liver diseases	
Hereditary hemochromatosis	- Iron overload. - Mechanism for disease is "increased intestinal iron absorption." - Main iron regulation is via shutting off intestinal absorption; this is impaired in hemochromatosis. - Body has poor ability to excrete iron; otherwise there are minor losses via skin shedding. Since women excrete naturally via menses, presentation occurs earlier in men (who clearly don't have this mechanism of excretion). - Main iron regulation is via shutting off intestinal absorption; this is impaired in hemochromatosis. - Can present as "bronze diabetes" → hyperpigmentation due to hemosiderin deposition in skin + ↑ fasting sugars (iron deposition in tail of pancreas). - Miscellaneous other findings can be seen like infertility (iron deposition in hypothalamus, anterior pituitary, or gonads), cardiomyopathy, or arthritis (pseudogout). - Hereditary hemochromatosis, primary hyperparathyroidism, and hypothyroidism are 3 most important causes of pseudogout. I used to only discuss the former two with students over the years, but the latter shows up on a new 2CK NBME exam where they mention chondrocalcinosis (calcium deposition in cartilage). - USMLE wants you to know there is ↑ risk of hepatocellular carcinoma. There is easy NBME Q of hemochromatosis where they ask what patient is at increased risk for, and the answer is simply "hepatocellular carcinoma." - Diagnose with **ferritin >300 mg/dL.** Transferrin saturation will also clearly be ↑. - It is exceedingly rare that ferritin is >300 in other conditions, however this can occur in lymphoma and leukemia, where ↑ ferritin is a poor prognostic marker in non-Hodgkin lymphoma. There is one NBME Q on 2CK where ferritin is 300 where it's not hemochromatosis, but USMLE won't play gotchya. - Treat with **serial phlebotomy, not chelators.** - Chelators such as deferoxamine or deferasirox are for secondary hemochromatosis due to transfusional siderosis (i.e., repeated blood transfusions that contain iron, for e.g., β-thalassemia major).
Wilson disease	- Copper overload. - Inability to secrete copper into bile from the liver. Copper is normally excreted by the body via secretion into bile. - ↑ urinary copper + ↓ serum ceruloplasmin (copper-binding protein in the blood; in the case of copper overload, body tries to minimize amount carried in blood). - Buzzy / pass-level detail is Keiser-Fleischer rings, which is copper deposited in the cornea of the eye. Vignette can give you what sounds like Wilson disease, and then the answer is **"slit-lamp exam."** - Can cause ↑ LFTs with cirrhosis and Parkinsonism. - Parkinsonism in a young patient = Wilson until proven otherwise. - In old patient, Parkinsonism = Parkinson disease, normal pressure hydrocephalus, Lewy-body dementia, or progressive supranuclear palsy. - Treat with the copper chelator penicillamine.

Bowel ischemia	
Ischemic colitis	- Patient with cardiovascular disease (CVD) + blood in the stool. - Caused by bleeding at ischemic ulcers at watershed areas in the colon (i.e., splenic flexure and rectosigmoid junction). - Can occur randomly or due to inciting event like recent AAA surgery. - Diagnose with colonoscopy to visualize the ischemic ulcers.
Chronic mesenteric ischemia	- Patient with CVD + abdominal pain 1-2 hours after eating meals. - Caused by atherosclerosis of the SMA or IMA → consuming food ↑ oxygen demand of bowel → angina of bowel. - The timing of the pain in relation to meals sounds like duodenal ulcers, however instead of the vignette being a 29-year-old dude from Indonesia with *H. pylori*, it will be a 69-year-old dude with Hx of coronary artery bypass grafting, intermittent claudication, HTN, and diabetes. - Next best step is "mesenteric angiography" on USMLE.
Acute mesenteric ischemia	- Presents 3 different ways on USMLE: 1) Severe abdo pain in patient with AF → LA mural thrombus launches off to SMA or IMA. 2) Severe abdo pain in patient just cardioverted/defibrillated → can launch LA thrombus off to SMA or IMA. There's a Q on 2CK form where this is the case, where they don't mention AF in the stem. 3) Severe abdo pain in patient with Hx of chronic mesenteric ischemia (i.e., acute on chronic) → atheroma within SMA or IMA ruptures, effectively causing an "MI of the bowel." - USMLE wants "mesenteric angiography" as next best step. - Laparotomy is also answer on NBME form. - I've never seen medications or endarterectomy as answers.

Appendicitis
- Triad of RLQ pain, fever, and vomiting (Murphy's triad). - Pain starts at epigastrium (visceral pain) and migrates toward RLQ (parietal pain). USMLE will ask why there is movement of the pain → answer = "inflammation of the parietal peritoneum." - Pain at McBurney's point (1/3 of the way from the anterior superior iliac spine to the umbilicus). - (+) Rovsing sign (pain felt in RLQ upon palpation of LLQ). - USMLE wants you to know appendicitis pain can be RUQ in pregnancy due to displacement of bowel and a shift in appendiceal location. What they will do is tell you pregnant woman in 3rd trimester has RUQ pain, fever, and vomiting + negative abdo ultrasound (meaning not cholecystitis) → answer = "appendicitis." - "Inflammatory mass and fecalith in the cecum" is phrase used on 2CK NBME form to describe appendicitis. - Diagnosis is done via ultrasound first, and sometimes CT. However, I have not seen USMLE assess either of these, likely because their utility is debated. What I have seen is laparoscopic removal in the stable patient.

HY colorectal / proctocolonic diagnoses	
Angiodysplasia	- Tortuous, superficial vessels in the colonic wall that cause painless bleeding per rectum in elderly. - Classically associated with aortic stenosis (Heyde syndrome; possibly related to pressure backup to the colon). - NBME Q gives older guy arguing with his wife + gets chest pain and bleeding per rectum. Even though colonoscopy is used for diagnosis, they say in this Q that colonoscopy is negative, likely to tell you it's not cancer causing the bleeding.
Diverticulosis	- Mere presence of diverticula in the colon; >50% of people over age 60 in the US. - ↑ straining throughout life leads to herniation of mucosa + submucosa through the muscularis propria of the colonic wall.

	- Usually asymptomatic, but can bleed. Diverticular bleed is most common cause of painless bleeding per rectum in elderly, followed by colorectal cancer, followed by angiodysplasia. - Do not confuse with diverticulitis, which is when a diverticulum becomes inflamed.
Diverticulitis	- LLQ pain + fever in patient over 60. One of the highest yield diagnoses on USMLE. - Inflamed diverticulum, usually in the sigmoid colon. - CT of the abdomen with contrast is how to diagnose. - Do not scope acutely, as this can cause perforation. - After patient is treated with antibiotics, a colonoscopy should be scheduled weeks to months later to rule out malignancy. But once again, never scope acutely.
Sigmoid volvulus	- **Patient over 75** + 2-3 days of constipation + abdo pain. - Rotation around its mesentery causes "dilation of sigmoid colon" (answer on NBME as what is most likely to be seen in patient). - **Abdominal x-ray** is used to diagnose, which shows a **coffee bean sign,** which is one of the highest yield radiographic images on USMLE.  - Tx on NBME is "sigmoidoscopy-guided insertion of rectal tube."
Fecal impaction	- Patient over 70 + chronic constipation + **"hard stool palpated in the rectal vault."** - Can sometimes cause fecal incontinence and paradoxical overflow diarrhea leading to encopresis (i.e., word for shitting yo pants). - Idiopathic, but exacerbated by opioids. - Treatment is with enema and laxatives.
Hemorrhoids	- Bleeding per rectum that will be described as "blood on the toilet paper," or "blood that drips into the toilet bowl." - Hemorrhoids are technically normal vascular structures within the anal canal that facilitate/cushion the passage of stool. - Bleeding from internal hemorrhoids is painless. These are above the pectinate line in the anal canal. - Bleeding from external hemorrhoids is painful. These are below the pectinate line. - Pregnancy and cirrhosis are risk factors. - 2CK NBME wants rectal exam followed by anoscopy for diagnosis. - Hemorrhoids often self-resolve and can be managed conservatively with sitz bath, NSAIDs, and ↑ dietary fiber. However, if USMLE forces you to choose surgical management for more severe cases, the answer is rubber band ligation.
Anal fissure	- Young adult who has painful bowel movements +/- blood in the stool. - The key detail is they refuse the rectal exam because the pain is so bad.

	- For whatever reason, they can also say there's an associated skin tag. I've seen this more than once, where the student says, "What's with the skin tag?" No fucking idea. - Mechanism can be "increased anal sphincter tone." - Tx on 2CK Surg is sitz bath. - If the USMLE forces you to choose a med, topical nitrates or diltiazem is used.
Pilonidal cyst/abscess	- Answer on USMLE for cystic mass at the superior aspect of the gluteal cleft. - Contains hair; often caused by ingrown hairs. - Tx = incision and drainage.
Perianal abscess	- Answer on USMLE for painful, erythematous mass near anal verge. - Increased risk in Crohn and diabetes. - Tx = incision and drainage.

HY GI bacterial infections	
Gram-negative rods	
ETEC	- Enterotoxigenic *E. coli* causes traveler's diarrhea. - Will present as brown/green diarrhea in person who's gone to Mexico or Middle East classically.
EHEC	- Enterohemorrhagic *E. coli* causes bloody diarrhea 1-3 days after consumption of beef. - Produces shiga-like toxin, which can cause hemolytic uremic syndrome (HUS; triad of renal dysfunction, schistocytosis, and thrombocytopenia).
Shigella	- Bloody diarrhea 1-3 days after consumption of beef. - Also can cause HUS via shiga toxin. - Q on a 2CK form gives bloody stool in a kid + no mention of platelets, hemoglobin, or renal issues, and the answer is "shigellosis," where HUS is wrong.
Salmonella	- Food poisoning is caused by *Salmonella typhimurium* and *Salmonella enteritidis*. - Bloody diarrhea 1-3 days after infection from consuming poultry, or following exposure to eggs or reptiles (i.e., turtles, lizards, etc.). New NBME Q mentions bloody diarrhea in someone with a pet lizard. - *Salmonella typhi* causes typhoid fever, which will be rose spots on the abdomen in a patient who's "prostrated" (i.e., lying down in pain); can be either diarrhea or constipation. The reservoir for *S. typhi* is humans, not chickens/reptiles.
Vibrio	- *Vibrio cholerae* (cholera) presents as "liters and liters" of rice-water stool in someone who went traveling to, e.g., Mexico. The way you can differentiate this from ETEC traveler's diarrhea is that cholera is notably profusely high-volume. - Both ETEC and cholera vignettes can tell you the patient has 8-12 stools daily, so it's not the # of stools that matters; it's the emphasis on volume. Cholera causes death via severe dehydration and electrolyte disturbance. - Tx is oral rehydration on USMLE; if patient has low BP or altered mental status (i.e., confusion/coma), IV hydration is done. - *Vibrio parahemolyticus* doesn't cause profuse, watery diarrhea the same way cholera does. I've seen this organism asked once in an NBME vignette where the patient ate sushi (can be acquired from sushi and shellfish). - *Vibrio vulnificus* causes severe sepsis in half of patients. This is asked on offline NBME where a dude went running on a beach and got sepsis, with no mention of consumption of food. But it's apparently acquired from shellfish.
Yersinia enterocolitica	- Causes bloody diarrhea + either appendicitis-like (i.e., RLQ) pain or arthritis. - The RLQ pain is from mesenteric adenitis or terminal ileitis.
Campylobacter jejuni	- Bloody diarrhea 1-3 days after consumption of poultry. - Can cause Guillain-Barre syndrome (ascending paralysis + ↓ tendon reflexes + albuminocytologic dissociation in the CSF → Tx with IVIG + plasmapheresis).
Gram-positive rods	

Bacillus cereus	- Watery diarrhea and/or vomiting in patient who's consumed reheated or fried rice. The process of heating/re-heating causes germination of spores.
Clostridium perfringens	- Watery/secretory diarrhea following consumption of poultry. - Causes gas gangrene in necrotic tissues (e.g., necrotizing fasciitis).
Clostridium difficile	- Diarrhea (pseudomembranous colitis) ~7-10 days after commencing oral antibiotics. - Antibiotics kill off normal bowel flora, allowing *C. difficile* to overgrow. - *C. difficile* is not normal flora, however. It is acquired via consumption of spores. - Can be watery or bloody diarrhea on NBME. Can also cause LLQ cramping (not RLQ as with *Yersinia*). - There is NBME Q where they say 28-year-old with LLQ cramping and bloody diarrhea 7 days after starting oral antibiotics → answer = *C. diff*; wrong answer is *Yersinia.* - USMLE doesn't care which antibiotics cause it; can be any. - Diagnose with stool AB toxin test; stool culture is wrong answer. - Treat with oral vancomycin.
Gram-positive cocci	
Staph aureus	- *S. aureus* pre-formed heat-stable toxin is acquired from two main sources on NBME: 1) various meats sitting under a heat lamp / out for long periods of time, e.g., at a buffet; 2) dairy products like creams, custards, potato salad. - Notably causes vomiting 1-6 hours after consumption. This is notable, as the symptoms occur rather quickly. Diarrhea, both watery and bloody, can occur, but is not mandatory. The main crux is the vomiting.
Other GI bacteria	
Tropheryma whipplei	- Causes Whipple disease, a GI malabsorptive syndrome where the patient can also get arthritis and renal and cardiac disease. - 100% of Qs will say "PAS-positive macrophages in the lamina propria." You don't need to know the image for this. But for whatever magical reason, the USMLE gives a fuck about this detail. Old-school pathology docs who reminisce maybe. - New highly pedantic Q on 2CK NBME Q wants "ceftriaxone + daily TMP/SMX for one year" as the treatment. Organism isn't even HY. No idea why they care.
Tropical sprue	- Causes flattening of villi similar to Celiac, but considered infective/bacterial in origin. Literature says exact etiology not certain. - Answer on USMLE for patient living in tropical area with unknown malabsorptive disease, where you can easily eliminate the other answers. - Tx = tetracycline.

HY GI viral infections	
Rotavirus	- Most common cause of watery diarrhea in unvaccinated children < 5 years. - Vaccine normally given orally at 2, 4, and 6 months of age.
Norwalk virus	- Most common cause of watery diarrhea in adults and rotavirus-vaccinated children. - Cruise ships and business conferences are buzzy places to acquire (fecal-oral); basically any place with high density of people. - If the Q says a young child + family all have watery diarrhea, the answer is Norwalk, not Rota, since only the young child would get Rota, not the family also.
Herpes	- HSV1/2 causes herpes esophagitis → odynophagia + punched-out ulcers in esophagus.
CMV	- CMV esophagitis →odynophagia + linear (confluent) ulcers in esophagus. - CMV colitis → bleeding per rectum in AIDS patient with CD4 count <50. - CMV retinitis → blurry vision in HIV (or immunosuppressed) patient. - Most common organism transmitted via blood transfusions and organ transplants.

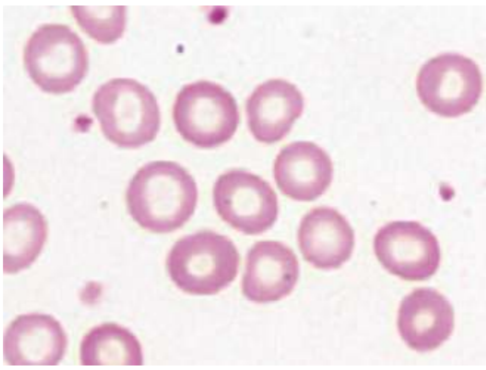
HY GI protozoal infections	
- A protozoan is a unicellular eukaryote. - **ECG** → ***E****ntamoeba histolytica,* ***C****ryptosporidium parvum,* and ***G****iardia lamblia* are all GI protozoa that are acquired via **cysts in water** (i.e., they are **water-borne**). If "water-borne" and "fecal-oral" are both listed as answers, USMLE wants "water-borne" as means of acquisition.	
Entamoeba histolytica	- Bloody diarrhea in person who went to Mexico. - Can cause "flask-shaped ulcers" in the small bowel and liver abscess. - Treat with metronidazole. - Iodoquinol kills intraluminal parasite.
Cryptosporidium parvum	- Watery diarrhea in person who went to Mexico. - Appears as acid-fast cysts (same stain as TB). - Self-limiting in immunocompetent persons → Tx = supportive care. - Chronic diarrhea in HIV → Tx = nitazoxanide.
Giardia lamblia	- Steatorrhea in person who went to Mexico. - Steatorrhea = bloating + extremely foul-smelling stool that floats. - The steatorrhea is due to *Giardia* causing malabsorptive diarrhea. - There is one NBME Q for *Giardia* where they say "foul-smelling *watery* diarrhea with bloating," which is audacious, since 9 times out of 10 it is not described as watery. But the "bloating" and "foul-smelling" are consistent with steatorrhea in this context. Just letting you know it exists. - Acquired via fresh water lakes / scuba diving. There is a *Giardia* NBME Q where they say a woman goes scuba diving in Mexico and then gets foul-smelling stools with bloating. Student says, "Wait but I thought *Giardia* was fresh water. If she went scuba diving in the ocean then how is that *Giardia*?" → People can go scuba diving in fresh water lakes too bro. Don't know what to tell you. USMLE wants you to know the images for both the cyst as well as the flagellated trophozoite. - Flagellated protozoan. - Tx = metronidazole.

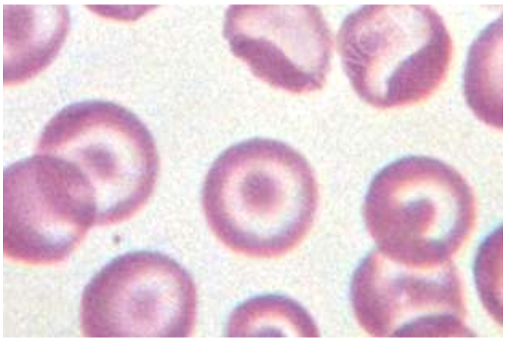
HY GI parasitic (helminth) infections	
- USMLE wants you to know routes of acquisition for the nematodes (i.e., ingested or through feet, etc.). - You do not need to obsess over exactly which anti-helminthic agent treats which organism. Waste of time. What USMLE wants you to know is the big-picture concept that -bendazoles (i.e., mebendazole) and pyrantel pamoate are the two agents that treat nematodes primarily. - Praziquantel is used for cestodes and trematodes. Albendazole is the odd one out that is used for neurocysticercosis (*Taenia solium*; cestode).	
Nematodes (roundworms)	
Ascaris lumbricoides	- Ingested. Giant roundworm; causes intestinal obstruction. (Ascariasis)
Enterobius vermicularis	- Ingested. Causes perianal itching / (+) tape test in children. (Enterobiasis)
Angiostrongylus	- Ingested by eating raw slugs/snails. Causes eosinophilic meningitis.
Trichinella spiralis	- Ingested by eating pork/**bear** meat. Causes triad of fever, periorbital edema, and myalgias. (Trichinosis)
Strongyloides stercoralis	- Acquired through soil/one's feet. Causes intestinal obstruction + pulmonary symptoms (apparently travels to lungs via blood, then travels up respiratory tree into esophagus, where it is swallowed).
Hookworms	- Acquired through soil/one's feet. "Hookworms" refers to *Ancylostoma duodenale* and *Necator americanus*. - Cause **microcytic anemia / ↓ hemoglobin** due to sucking blood in GI tract.
Cestodes (tapeworms)	
Diphyllobothrium latum	- Fish tapeworm. Nothing else you need to know.
Taenia solium	- Pork tapeworm. Causes cysticercosis (muscle pain/cysts) + neurocysticercosis ("Swiss cheese" appearance of brain or soap bubbles in ventricles).
Echinococcus granulosus	- Aka hydatid worm, or dog tapeworm; causes hydatid cyst disease.

	- Acquired from dogs (clearly). - Causes liver cysts. Only point you need to know is that you do not biopsy because this can cause anaphylaxis. Tx = surgically remove.
Trematodes (flukes)	
Schistosoma	- Penetrates skin when swimming; snail is host. - *Schistosoma mansoni/japonicum* can cause portal hypertension. Not HY but shows up once on NBME. Not to be confused with *Schistoma hematobium*, which causes squamous cell carcinoma of the bladder.
Paragonimus westermani	- Ingested from crab meat or crayfish. - Aka lung fluke. Most common cause of hemoptysis in the world.
Clonorchis sinensis	- Ingested from fish. Causes cholangiocarcinoma.

HY GI pharm for FM	
- You do not need to know MOAs for FM shelf. I just mention some of them here for context, which helps some of you guys remember the info more completely.	
PPIs	- Omeprazole. - Shut of proton pumps on parietal cells → ↓ acid secretion. - Irreversible and non-competitive; more efficacious than H2 blockers, which are reversible and competitive. - Choose PPIs over H2 blockers for Tx of most things, like GERD, ulcers, and *H. pylori.* - Diagnosis of GERD is 2-week trial of PPIs. - PPIs cannot be given with sucralfate or -azole antifungals (on NBME).
H2-blockers	- Cimetidine, ranitidine. - Not used often. PPIs more efficacious. - One 2CK form has "2-week trial of H2 blocker" as correct for diagnosis of GERD, but PPIs aren't listed. As I said earlier, if you're forced to choose between a PPI and H2 blocker, the PPI is correct basically always. - Cimetidine can cause gynecomastia (HY on USMLE) and inhibits P-450. - Ranitidine doesn't inhibit P-450. Sounds pedantic, but there's an NBME Q where they mention coma in someone taking diazepam + 2nd drug → answer = cimetidine as 2nd drug; ranitidine also listed but wrong (doesn't inhibit P-450).
Metoclopramide	- Anti-emetic + prokinetic agent (means ↑ peristalsis). - D2 antagonist but also an antagonist of serotonin 5HT3 and agonist of 5HT4 receptors. The effects on serotonin receptors ↑ gut peristalsis. - HY use on USMLE is **diabetic gastroparesis** (i.e., sounds like GERD but patient has severe diabetes, where answer is metoclopramide, not PPI). - Because it's a D2-antagonist, adverse effects are similar to the anti-psychotics – i.e., prolong QT interval, hyperprolactinemia, anti-pyramidal effects (acute dystonia, parkinsonism, akathisia, tardive dyskinesia). - There is 2CK Q where patient on metoclopramide has parkinsonism, where the answer is "discontinue metoclopramide" as next best step before simply giving the patient Parkinson disease meds.
Ondansetron	- Powerful anti-emetic classically used for nausea/vomiting from chemoradiotherapy. - 5HT3 antagonist.
Erythromycin	- Motilin receptor agonist that can be used for diabetic gastroparesis. - Also a 50S ribosomal subunit inhibitor (antibiotic).
Misoprostol	- PGE_1 analogue. Used for **NSAID-induced ulcers** after PPIs. - NSAIDs ↓ prostaglandins. Therefore misoprostol is the replenished prostaglandin. - Prostaglandins ↓ acid production, ↑ mucous and bicarb production, and ↑ gastric mucosal blood flow.
Magnesium	- Antacid. Causes diarrhea.
Aluminum.	- Antacid. Causes constipation ("Aluminimum amount of feces.")
Calcium	- Antacid. Causes milk-alkali syndrome (↑ Ca^{2+} + ↑ $HCO3^-$+ urolithiasis).

	- Bisphosphonates (e.g., alendronate) and tetracyclines (e.g., doxycycline) cannot be taken with calcium or iron supplements (i.e., divalent cations), since the latter chelate the drugs and ↓ absorption / oral bioavailability. USMLE loves this. - USMLE will give you, e.g., prostatitis treated with ciprofloxacin, + list all sorts of other meds the patient is on, as well as a calcium supplement, and then tell you the patient's condition isn't improving + ask why → answer = calcium carbonate.
Bismuth	- Can be used as part of *H. pylori* regimens (but not first-line).
Sucralfate	- Can be used to coat ulcers / form a barrier to protect against acid and ↓ pain. - Cannot take with PPIs. The latter ↓ ability of sucralfate to crosslink/work correctly.
5-ASA compounds	- Mesalamine, sulfasalazine; NSAIDs used for IBD.
TNF-α blockers	- Infliximab, adalimumab, and etanercept. - USMLE likes these for IBD *after* 5-ASA compounds and steroids are attempted. - Infliximab and adalimumab are monoclonal antibodies against TNF-α. - Etanercept is a recombinant receptor that mops up soluble TNF-α.
Octreotide	- Somatostatin analogue used for esophageal varices Tx after banding. - Acts by ↓ portal blood flow.
Propranolol	- Beta-blocker that can be used for esophageal varices prophylaxis (not Tx).
Lactulose	- Used to Tx hyperammonemia in cirrhosis. - Carbohydrate that is metabolized by gut bacteria into acidic substrates that trap ammonia as ammonium. The latter is then excreted (ions aren't absorbed as easily).
Neomycin	- Used to Tx hyperammonemia in cirrhosis. - Antibiotic that kills ammonia-producing bacteria.
Loperamide	- Mu-opioid receptor agonist used to treat diarrhea-predominant IBS. - Opioids cause constipation, so "this is a good thing" in this case.
Ezetimibe	- Blocks cholesterol absorption in the small bowel.
Cholestyramine	- Bile acid sequestrant. Causes reduced enterohepatic circulation of bile acids at terminal ileum → liver must now convert more cholesterol into bile acids in order to replenish them → liver pulls cholesterol out of the blood to accomplish this.
Orlistat	- Pancreatic lipase inhibitor used for obesity; can cause steatorrhea.
Nystatin	- Mouth wash used for oropharyngeal candidiasis. - Forms pores in fungal ergosterol membrane.
Fluconazole	- Used for candidal esophagitis. - Inhibits ergosterol synthesis (i.e., lanosterol → ergosterol).
Acyclovir	- HSV esophagitis; HSV1/2 causes "punched out" ulcers. - DNA polymerase inhibitor.
Ganciclovir	- CMV esophagitis and colitis; causes linear/confluent ulcers. Same MOA as acyclovir.
CAP	- Clarithromycin, Amoxicillin, PPI. - First-line Tx for *H. pylori*. - If patient still has (+) urease breath test after 4 weeks, assume antibiotics resistance and switch out the CA and add tetracycline, metronidazole, and bismuth (keep PPI).
Vancomycin	- Given orally to treat *C. difficile.*
Metronidazole	- "Anaerobes below the diaphragm." - Used for diverticulitis in combo with a fluoroquinolone. - Tx for *Giardia* + *Entamoeba.*
Imipenem	- Carbapenem antibiotic that has fantastic penetration for pancreatitis.
Mebendazole	- Used for most nematodes (i.e., *Ascariasis, Enterobiasis,* hookworms, etc.).
Albendazole	- Same as mebendazole, but for whatever reason it's known as a preferred agent for neurocysticercosis, which is caused by a cestode (*Taenia solium*).
Pyrantel pamoate	- Agent used against nematodes (all of the roundworms + hookworms).
Praziquantel	- Used for cestodes (tapeworms) and trematodes (flukes).

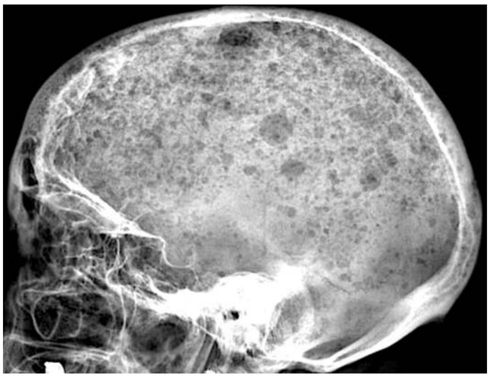
Iron deficiency anemia vs thalassemia			
Variable	**IDA**	**Thalassemia**	**HY points**
Hb	↓	↓	- Thalassemia vs IDA = exceedingly HY for Family Med. In real life, adults with thalassemia are often misdiagnosed as having IDA, so USMLE wants you to distinguish. - Thalassemia means abnormal synthesis of either the α or β globin chain. It presents two ways on USMLE: 1) As a microcytic anemia ***despite*** normal iron + ferritin; or 2) As a patient who has a microcytic anemia ***despite*** iron supplementation. - Most IDA is seen in menstruating women, even if their menses aren't heavy. FM forms will actually explicitly say menses are normal/not heavy, but Dx is still IDA. - FM form will give you microcytic anemia in a woman who's 32, and then the answer is just "check serum iron and ferritin." This is because thalassemia is often misdiagnosed as IDA, where if we get the iron studies back and see iron + ferritin are normal, we say, "Ok that's thalassemia, not IDA." → next best step = hemoglobin electrophoresis. - Obgyn forms like to give pregnant women at first antenatal screening with microcytic anemia who are given iron for 3 weeks but show no improvement. Q won't mention anything about patient's iron or ferritin levels. So you say, "Ok, this is a microcytic anemia *despite* iron supplementation, meaning the iron + ferritin are actually normal, so this is thalassemia." → next best step = hemoglobin electrophoresis.
MCV	↓	↓	
Serum iron	↓	Normal	
Ferritin	↓	Normal	
Hb electrophoresis	Normal	**α: normal;** **β: ↑ HbA2 + HbF** (See discussion at bottom of table)	
What happens if we give iron?	Improvement	**No improvement (HY)**	
RDW	↑ (HY)	↓/Normal (HY)	- Red cell distribution width. - All you need to know is, basically always, ↑ RDW means IDA straight-up 9/10 Qs. - The only Q where ↑ RDW is seen in thalassemia is on CMS FM form 4, where IDA is wrong, but they give ferritin within the normal range, so IDA isn't possible. But the ↑ RDW is essentially an erratum, as this is rare as fuck. - Therefore we can essentially say that ↑ RDW is sensitive but not specific for IDA, where if RDW isn't ↑, we can rule-out IDA, but be careful about automatically saying "must be IDA" if RDW is ↑.
Blood smear	Pale RBCs	Target cells	- IDA causes pale RBCs (i.e., ↑ central pallor). Not dramatic. 

<table>
<tr><td></td><td></td><td></td><td>- Target cells are highly buzzy and pass-level for thalassemia. Probably 14/15 Qs on USMLE mentioning target cells are thalassemia (α or β).

- There is one NBME Q with target cells in a splenectomy patient, but they're a background finding and don't facilitate answering the Q in any way.</td></tr>
<tr><td>Tx</td><td>Iron</td><td>Transfusions if severe</td><td>- Oral iron is ideal over parenteral (IV) or intramuscular iron always. I mention this because answers such as "parenteral iron" are wrong/distractors basically always. I'll say to a student, "You want to give this patient IV iron?"
- Oral iron can cause constipation and black stools. USMLE wants you to know iron, aluminum (antacid), and verapamil all cause constipation.
- Transfusional siderosis (secondary hemochromatosis) is sophisticated way of saying iron overload due to repeated blood transfusions, e.g., in β-thalassemia major. Each RBC transfusion contains iron. Use chelators (e.g., deferoxamine) to treat.</td></tr>
<tr><td colspan="4">- Thalassemia is impaired synthesis of either the α- or β-globin chain in hemoglobin.
- α-thalassemia can have 1-4 alleles mutated; β has 2. Don't worry about obscure intermediate types, etc.
- USMLE overwhelmingly focuses on adults for both α- and β-thalassemia, since these patients are usually misdiagnosed as having iron deficiency anemia. That's why you need to know about thalassemia, so you don't misdiagnose patients with IDA.
- Adults with α-thalassemia will be asymptomatic (1 mutation), where they have an incidental microcytic anemia picked up on antenatal screening (hence Obgyn forms love thalassemia). Adults with 2 mutations have symptoms akin to IDA, but they'll fail to improve with iron supplementation. 3 α mutations (HbH disease; β4 tetramer) will present as a sick kid; 4 α mutations (Hb Barts; γ4 tetramer) will be fatal in utero.
- β-thalassemia minor will be 1 mutation in an adult who presents similar to 2 α-thalassemia mutations (i.e., similar to IDA).
- β-thalassemia major is 2 mutations and will present as a sick kid similar to 3 α-thalassemia mutations.
- It is exceedingly HY you know that β-thalassemia has $\uparrow$ HbA2 (α2/δ2) and HbF (α2/γ2); α-thalassemia (1 or 2 mutations) has a normal hemoglobin electrophoresis.
- An NBME Q gives two asymptomatic Vietnamese parents and then asks what their child is at increased risk for $\rightarrow$ answer = α thalassemia trait (1 mutation); wrong answer is β-thalassemia minor. Students think this Q is a guessing-game about whether α or β is more likely in Vietnamese; it's not. The reason it's α thalassemia trait is because both parents are asymptomatic, which is possible with 1 α mutation. If either of the parents had a β mutation, he or she would be symptomatic.
- Sick children with thalassemia are known to get "chipmunk facies/skull" and hepatosplenomegaly occur due to $\uparrow\uparrow$ extramedullary hematopoiesis.</td></tr>
</table>

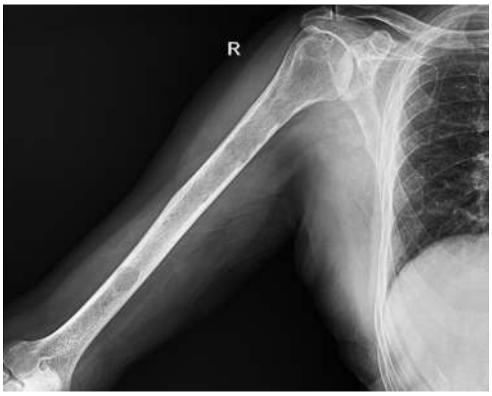
Anemia of chronic disease (AoCD)
- The answer on USMLE for anemia (↓ Hb) in patient who has **renal failure,** an autoimmune disease (i.e., RA, JRA, IBD, SLE, etc.), hepatitis B/C, HIV, or malignancy. - Patients will have low iron but **normal ferritin** (or sometimes slightly high). Ferritin is the most sensitive marker of iron stores, which means these patients are not iron deficient. - There's two main reasons iron is low in AoCD: 1) RBC production is decreased because cytokines (namely IL-6 and TNF-α) block iron release from storage sites, and 2) hepcidin, a molecule produced by the liver, decreases gut absorption of iron and promotes its storage in cells instead. - It's to my observation that **renal failure** is the highest yield cause of AoCD on USMLE. This is due to "cytokine-mediated EPO deficiency." This is an answer on an NBME form. EPO is the treatment for AoCD only if renal failure is the etiology. Otherwise the Tx is merely addressing the underlying condition (e.g., the RA), and EPO is a wrong answer. - There is an AoCD NBME Q for RA where they ask how to treat and answer is "no specific measures indicated." Student says, "Wait, I thought we were supposed to address the underlying condition, so I was confused by this answer choice." I agree. So don't take it up with me. Take it up with NBME. - One of the highest yield points I can communicate is that even though classic MCV is normal in AoCD, it will be ↓ in ~50% of NBME Qs. I've seen students say, "Oh it can't be AoCD cuz MCV is low." No. USMLE loves giving low MCV in AoCD, especially on 2CK forms. I've seen MCVs of 72 and 75 in JRA Qs.

Multiple myeloma

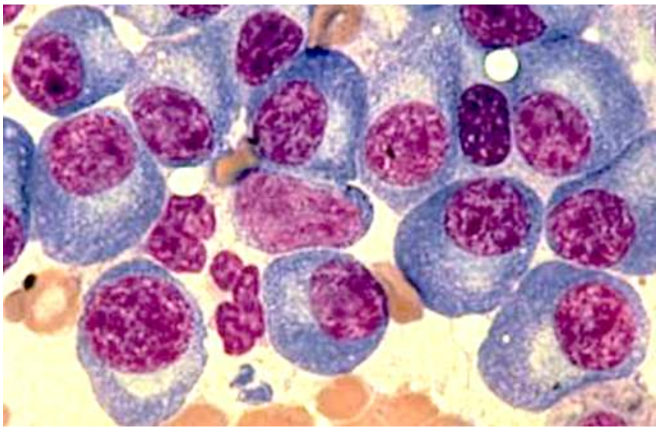
- Cancer of plasma cells.
- Plasma cells normally secrete immunoglobulins, so we have ↑↑ serum immunoglobulins.
- Multiple myeloma buzzy presentation is patient over 50 with mid-back pain + hypercalcemia.
- ↑ serum Ca^{2+} occurs due to lytic lesions of bone caused by the proliferating plasma cells. This can present as pathologic rib fractures or "pepper pot skull."



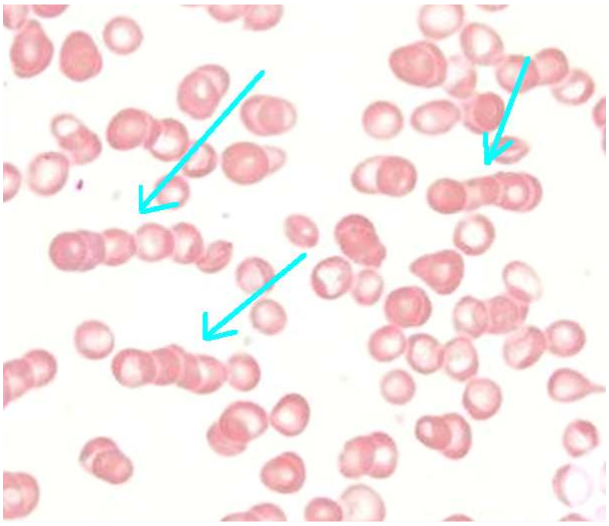
- There is NBME Q floating around where they give lytic lesions of the humerus as well; not buzzy/textbook in comparison to rib/skull lesions, but they show image similar to following:



- First step in diagnosis is **serum protein electrophoresis (SPEP),** which shows ↑↑ serum IgG kappa or lambda light chains. This is known as an M-spike (monoclonal spike; this does not mean IgM). Then **bone marrow biopsy** confirms the diagnosis, showing >10% plasma cells.
- Smear in multiple myeloma will show plasma cells (blue cells below) with “clockface chromatin,” which is the appearance ascribed to the nuclei (purple below).



- Urinalysis will show Bence Jones proteinuria, which is simply the IgG light chains in the urine.
- Amyloidosis is proteins depositing where they shouldn’t be depositing. Immunoglobulins are proteins. Since these are flying around in the blood, they deposit in the heart (**cardiac amyloidosis;** S4 heart sound with diastolic dysfunction) and kidney (**renal amyloidosis;** nephrotic syndrome).
- The IgG light chains stick to RBCs, causing the RBCs to stick to each other as stacks (i.e., Rouleaux). These are heavy and sediment quickly on centrifugation, hence ↑ ESR. USMLE can show you a picture of the Rouleaux rather than the clockface chromatin, where you need to know instantly the diagnosis is MM.



- MCV can sometimes be ↑ in MM (e.g., 108; NR 80-100). Don’t be confused by this. It is common. This is caused by 1) ↑ immunoglobulins being transported by RBCs (thereby ↑ RBC size); 2) ↓ RBC production within bone marrow due to infiltrating plasma cells, causing ↑ RBC size to compensate; and 3) B9 and B12 deficiencies can be caused by MM.
- Tx is initially IV bisphosphonate to ↓ risk of bone lysis.
- Should be noted that patients with bone marrow biopsy showing <10% plasma cells have a condition known as MGUS (monoclonal gammopathy of undetermined significance). There is small annual risk of progression to multiple myeloma. These patients might have fatigue and/or ↑ MCV but no hypercalcemia or renal dysfunction.

Waldenstrom macroglobulinemia
- Cancer of plasmacytoid cells. - "-Oid" means looks like but ain't. So these are cells that look like plasma cells but they ain't plasma cells. - Causes IgM M-protein spike on SPEP (MM is IgG M-protein spike). - In contrast to MM, does not cause hypercalcemia, lytic lesions, Bence-Jones proteinuria, or amyloidosis. - Waldenstrom causes **hyperviscocity syndrome,** which will present as headache, blurry vision, tinnitus, or Raynaud phenomenon.

Polycythemia vera vs secondary polycythemia	
Polycythemia vera	- JAK2 mutation causing "proliferation of hematopoietic stem cells" (answer on NBME). - Qs will give ↑ Hb at 18+ (NR 12-17.5 g/dL) *and* ↑ WBCs and/or platelets. - In other words, at least 2/3 cell lines will be ↑, with RBCs always ↑. - ↑ Hb can cause hyperviscosity syndrome. - Basophilia can cause pruritis after a shower. - EPO is suppressed due to ↑ Hb. - USMLE wants **phlebotomy** as the answer to ↓ acute symptoms associated with hyperviscosity syndrome (i.e., "What is next best step to ↓ patient's headache / blurry vision?"). - Serial phlebotomy and hydroxyurea can be answers to ↓ recurrent symptoms; the USMLE will not give both at the same time.
Secondary polycythemia	- Isolated ↑ in RBCs due to ↑ EPO – i.e., WBCs and platelets are normal. - Seen in those with lung disease where arterial O2 tension is low → kidneys sense ↓ O2 and secrete EPO → acts on bone marrow to stimulate erythroid precursors. - Also seen in **renal cell carcinoma,** which can secrete EPO (and PTHrp causing hypercalcemia). - Exogenous administration (e.g., cyclists).

Brief clotting factor pathway overview
- Prothrombin time (PT) reflects the functioning of the extrinsic pathway; activated partial thromboplastin time (PTT; aPTT) reflects intrinsic pathway. - In other words, if PT alone is high, then the extrinsic pathway is fucked up. - If PTT alone is high, then the intrinsic pathway has a problem. - If both PT and PTT are elevated, then the common pathway has an issue. **- Normal PT is 10-15 seconds.** **- Normal PTT is 25-40 seconds.** **- Normal bleeding time is 2-7 *minutes*.** This relates to **platelets** and has nothing to do with clotting factors. - You must know the green numbers for FM shelf. Time to take your training wheels off. Know the values. - If you're confused by bleeding time, PT, and PTT, you can go to my HY Heme/Onc PDF where I have a diagram of the clotting cascade, etc. But I want to stay short here because this is a 2CK PDF.

HY Core DDx	
ITP	- Idiopathic/immune thrombocytopenic purpura. - Antibodies against glycoproteins IIb/IIIa on platelets (asked on a 2CK form somewhere even though this sounds Step 1-esque). **- Causes ↓ low platelets; ↑ bleeding time; no change PT and PTT.** - Bleeding time is high because platelets are fucked up. PT and PTT and normal because these refer to clotting factors, which have nothing to do with ITP. - Normal platelet count is 150-450,000/µL. Platelets in ITP Qs are usually <100,000.

	- Precipitated by viral infection, where antibodies against viral proteins cross-react with platelets (disrupts platelet aggregation); type II hypersensitivity. - Viral infection **will be asymptomatic in ~50% of vignettes.** In other words, the Q need not mention the viral infection, so don't get confused if you don't see it. - Will present two ways on USMLE: 1) School-age kid (i.e., 12) who has coryza for a few days followed by petechiae and/or nosebleeds. 2) Woman 30s-40s + no mention of viral infection + random bruising + bleeding time 9 minutes. - For the latter vignette, if they say exact same Q but bleeding time 6 minutes (normal), then answer is domestic abuse, not ITP. This is important for FM. - There is an unusual 2CK Q on new NBME where they say girl only has heavy menses (classically a clotting factor presentation for vWD) and the answer is ITP, but vWD isn't listed and the other answer choices are clearly wrong. Just letting you know this exists. - Tx = steroids, then IVIG, then splenectomy, in that order. - If splenectomy is performed and platelets return to normal range, but then 6 months later they fall again, the answer is "accessory spleen." Sounds weird, but a small % of people have a second spleen the size of a pea that can grow in size if the main spleen is removed.
vWD	- von Willebrand disease; **autosomal dominant.** - von Willebrand factor (vWF) normally bridges glycoprotein Ib on platelets to vascular endothelium / underlying collagen (platelet adhesion). **- ↑ bleeding time; ↑ PTT (only half the time); no change PT; normal platelets.** - ↓ vWF means ↓ platelet function (and hence ↑ BT), but platelet count is unchanged. - vWF also has secondary role of stabilizing factor VIII in plasma; since we have ↓ vWF, the intrinsic pathway gets fucked up, so we have ↑ PTT. But bear in mind, this effect on factor VIII *is only a mere secondary* role, so PTT isn't always ↑. It's to my observation that only ~50% of vWD vignettes give ↑ PTT. PT is normal because the extrinsic pathway is fine. - 8/10 Qs presents as a teenager or young adult with a mix of one platelet problem and one clotting factor problem. - Platelet problem = mild, cutaneous findings (i.e., petechiae, bruising) + epistaxis. - Clotting factor problem = excessive bleeding with tooth extraction + heavy menses. (Hemarthroses are also clotting factor but are seen in hemophilia, not vWD, as I explain below). - For example, the Q can give 17-year-old girl who has nosebleeds + heavy periods; or 17-year-old girl who has bruising + Hx of excessive bleeding with tooth extraction (i.e., we have one platelet problem + one clotting factor problem). This is 8/10 Qs I'd say. - 1/10 Qs might give only isolated platelet or clotting factor problem (i.e., only say the patient has Hx of excessive bleeding with tooth extraction), but they'll also say *one of the parents* has Hx of nosebleeds → so you say, "Oh ok. vWD is AD, so that's your combo of platelet problem + clotting factor problem linked via autosomal dominant inheritance." - 1/10 Qs will say the patient has a cut on the finger that takes longer to heal than normal + has heavy menses + normal platelets + has normal PT and PTT + they don't mention BT. The cut on the finger is their way of saying BT is elevated. Despite only an isolated ↑ in BT here, since platelets are normal, we know it's not ITP and is instead vWD. - Ristocetin cofactor assay is abnormal. All you need to know is that this is some test that is abnormal if platelet adhesion is fucked up. Tx = desmopressin (DDAVP), which can in theory help boost functional vWD levels.
Hemophilia A+B	- Deficiency of factors VIII and IX, respectively. - **Both X-linked recessive.** **- Isolated ↑ PTT; BT and PT are normal.** - Factors VIII and IX are in the intrinsic pathway, so PTT is ↑. Extrinsic pathway is unaffected so PT is normal. Platelets have nothing to do with this condition, which is why bleeding time is normal.

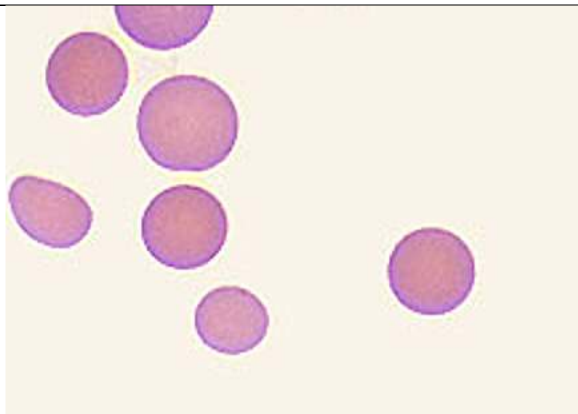
	- Will present two ways on USMLE: 1) School-age boy with hemarthrosis who has isolated ↑ PTT. They can also say he has a maternal uncle who died years ago from minor head trauma. 2) Neonate who has excessive bleeding with circumcision. - Hemophilia A is treated with DDAVP and factor VIII replacement. - Hemophilia B is treated with factor IX replacement. DDAVP doesn't work apparently. - If the Q tells you a patient with hemophilia A has persistently elevated PTT despite factor VIII supplementation, the answer is "check for antibodies against factor VIII." Sometimes patients can develop resistance against exogenous factor replacements.
Vitamin K deficiency	**- Causes ↑ PT, ↑ PTT, no change bleeding time.** - Vitamin K is cofactor for gamma-glutamyl carboxylase, which is an enzyme that activates clotting factors II, VII, IX, and X, as well as anti-clotting proteins C and S. - Bleeding time is normal because platelets aren't affected. - Will present two ways on USMLE: 1) Neonate who has bleeding from umbilical stump +/- retinal hemorrhages. The latter can sound like shaken-baby syndrome, but the bleeding from the umbilical stump is highly buzzy for vitamin K deficiency. 2) Adult who's been on broad-spectrum antibiotics for many weeks who now requires a lower dose of warfarin → answer = "depletion of bowel normal flora" (Vitamin K is normally synthesized by colonic normal flora). - Warfarin inhibits vitamin K function, so if Vit K is ↓, we don't need as much warfarin.
HUS	- Hemolytic uremic syndrome. - Presents as triad of: 1) thrombocytopenia; 2) hemolytic anemia with **schistocytes;** and 3) renal insufficiency with or without hematuria. - The combination of thrombocytopenia + schistocytosis = microangiopathic hemolytic anemia (MAHA). - Mechanism is: E. coli (EHEC O157:H7) and Shigella both secrete toxins (Shiga-like toxin and Shiga toxin, respectively) that cause inflammation and damage of renal microvasculature → platelets adhere to plug damaged endothelium → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells).
TTP	- Thrombotic thrombocytopenic purpura. - Presents as HUS triad + fever and neurologic signs. In other words: - Presents as pentad of 1) thrombocytopenia; 2) **schistocytes;** 3) renal insufficiency; 4) fever; 5) neurologic signs. - TTP is caused by antibodies against, or a mutation in, a protein called ADAMTS13, which is a metalloproteinase that breaks down vWF multimers → unable to break down platelet aggregates that plug renal microvasculature → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells). - Can present as a stroke-like presentation in young woman, where they just want "schistocytes" as the answer (Step 1 NBME). - 2CK NBME has "plasmapheresis" as Tx.
Sickle cell	- **Autosomal recessive.** - Carrier status (one mutation) is referred to as sickle cell trait, which is less severe than sickle cell anemia (two mutations). - Sickle cell crises can present as **abdominal or chest pain,** as well as dactylitis (inflammation of the fingers). - The abdominal pain, in particular, as the initial presentation for sickle crisis has become HY on NBME exams. A new 2CK form gives RLQ pain + low Hb in child from Libya, where "blood smear" is the answer for what will confirm the diagnosis. The Q is tricky because it almost sounds like appendicitis. - Diagnosed with blood smear, followed by hemoglobin electrophoresis to detect HbS. - **Hydroxyurea** ↑ **HbF** and can ↓ recurrence of sickle crises. Sounds weird, but the mere presence of more HbF simply means there's fractionally less HbS floating around.
Myelofibrosis	- Usually JAK2 mutation (same as polycythemia vera).

	- The answer on USMLE if they say "tear-drop-shaped RBCs" or "dry tap on bone marrow aspiration." - Massive splenomegaly seen in 100% of questions. - I've seen NBME also write "tear-drop shaped poikilocytes" (fancy word that means abnormal RBCs seen as >10% on a smear). - In myelofibrosis, the proliferation of abnormal cells in the bone marrow, notably megakaryocytes and granulocytes, leads to the release of growth factors and cytokines that stimulate deposition of collagen, resulting in bone marrow fibrosis.
Essential thrombocytosis (ET)	- Aka essential thrombocythemia. - Usually JAK2 mutation (same as myelofibrosis and polycythemia vera) causing over-proliferation of platelets. - USMLE Q will tell you patient has pain in fingertips, Raynaud phenomenon, or headache (hyperviscosity symptoms) + platelet count is, e.g., 1.4 million/µL (NR 150,000-450,000/µL). - ET is a myeloproliferative neoplasm where platelets **can be >1,000,000/µL.** **-** The patient will present with no specific triggering event. Compare this to reactive thrombocytosis below. - Tx = aspirin + plateletpheresis (removing platelets from blood).
Reactive thrombocytosis (RT)	- Aka reactive thrombocythemia. - Triggering event (i.e., inflammation, infection, surgery) can stimulate thrombopoietin production and increased synthesis and release of platelets. - It is to my observation the USMLE loves to give reactive thrombocytosis in the setting of **infective endocarditis,** where they'll tell you platelet count is 900,000, and the student is like, "why the fuck are the platelets 900,000." It's just RT. - ET is usually women >50 and is symptomatic; RT is anyone and is usually asymptomatic. When RT occurs, it will be a super-high platelet count you incidentally notice as part of the stem, but the focus of the Q isn't the patient's symptoms related to the ↑ platelets. - For example, there is a difficult offline NBME Q where they give neutrophilic shift with platelets of 1.4 million, where it looks like it could be RT due to infection (i.e., neutrophilic shift usually indicates bacterial infection), but the answer is ET on this form, not RT, presumably because the platelet count is >1 million (rare in RT to be this high) and the patient has symptoms (pain in fingertips). - Tx = address underlying infection + give aspirin. - I haven't seen RT as an actual correct answer on an NBME form; I've only seen it baked into questions as an incidental finding where the labs show high platelets in the setting of infection, and you, as the student, need to not be thrown off by that or think it's weird. You just say, "Oh that's probably just RT because we have infection/sepsis here."
SLE	- Systemic lupus erythematosus; classically women 20-40s. - Most common presenting feature is arthritis (>90%). - Malar rash too easy for most Qs, although you should know this is a type III HS. - **Thrombocytopenia** is mega-HY for Step – i.e., 32-year-old woman with arthritis + low platelets = SLE. - RBCs and WBCs can also be ↓, but it's the ↓ platelets that's highest yield. The cell lines are down due to antibodies. In other words, if you get a patient with SLE where all cell lines are down, this is due to "increased peripheral destruction," not "deficient bone marrow production." - Discoid lupus = skin lesions. Mucositis = mouth ulcers. - ↑ risk for malignancy (i.e., non-Hodgkin lymphoma) → i.e., 44-year-old woman with SLE has seizure + ring-enhancing lesion seen on CT of head → answer = primary CNS lymphoma; Toxo is wrong answer. - ↑ BUN + Cr = lupus nephritis. Send patient for renal biopsy as next best step. - Treat flares with steroids (i.e., prednisone).

HY Neutropenia conditions	
- A HY general point for USMLE is that neutropenia can present as **mouth ulcers, fever, and sore throat.** - Questions will not usually come out and say, "Yes, the kid has neutropenia." The situation might be, e.g., a kid who's undergoing chemotherapy for AML who now has mouth ulcers and/or sore throat.	
Neutropenic fever (Febrile neutropenia)	- Neutropenia + fever, LOL! - Can occur in the setting of patient who has chemo- or radiotherapy-induced pancytopenia (i.e., RBCs, WBCs, and platelets are all ↓), or viral-induced neutropenia (i.e., Parvovirus B19), where you see that the patient has fever and neutrophils are low (+/- RBCs and platelets). - Medical emergency (i.e., the patient has infection but no ability to fight it). USMLE wants "IV broad-spectrum antibiotics" as the next best step. - After Abx, the USMLE wants granulocyte colony-stimulating factor (i.e., G-CSF, or filgrastim) as the answer. - I've never seen granulocyte transfusion as a correct answer on USMLE.
Viral-induced neutropenia	- Parvovirus B19 can infect myeloid precursors and cause neutropenia – i.e., viral-induced neutropenia. - Other viruses can cause formation of antibodies against neutrophils, similar to how they can cause antibodies against platelets in ITP.
Cyclic neutropenia	- Gene mutation causing episodes of fever + mouth ulcers in a kid age 1-3. These episodes typically occur every 21 days, but the Q might not specify this number and will just say, "There have been several episodes like this before."

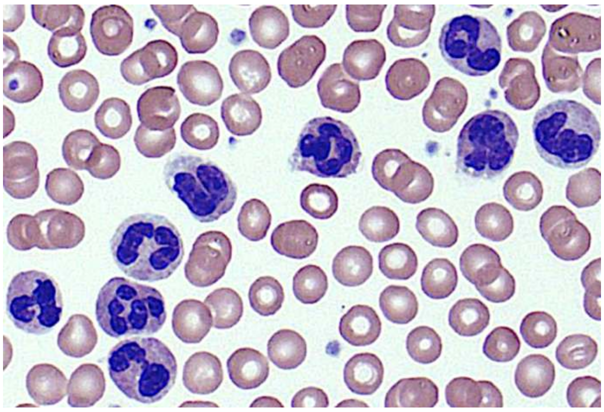
Other anemias	
Aplastic anemia	- Means all cell lines (i.e., RBCs, WBCs, platelets) are ↓. - Can be due to Parvo B19, where it infects not just myeloid progenitors, causing neutropenia, but also erythroid progenitors and megakaryocytes, causing ↓ RBCs and platelets, respectively. - Can also be due to other viruses, such as HepA CMS Peds 5) or HIV (Free 120). - Can be chemo- and/or radiotherapy-induced. - Fanconi anemia is rare, AR aplastic anemia that, for whatever reason, also presents with hypo-/aplastic thumbs/radii.
Pure-RBC aplasia	- Means only RBCs are ↓. - Can be caused by Parvo B19. - Thymoma can cause pure-RBC aplasia (as well as myasthenia gravis). - Diamond-Blackfan anemia is rare pure-RBC aplasia that presents with triphalangeal thumbs.

Spherocytosis	
Hereditary	- **Autosomal dominant** (one of the highest yield inheritance patterns on USMLE). - Heterozygous mutation in ankyrin, spectrin, or band proteins, causing an RBC cell membrane/cytoskeletal defect. This causes the RBC to deform away from the normal biconcave disc and into a spherical shape. - The USMLE might give you hereditary spherocytosis and then the answer is just "cell membrane," or "cytoskeleton," or "cytoskeletal defect." - Can be described as RBCs lacking central pallor. If they show you a smear, they look full and red.

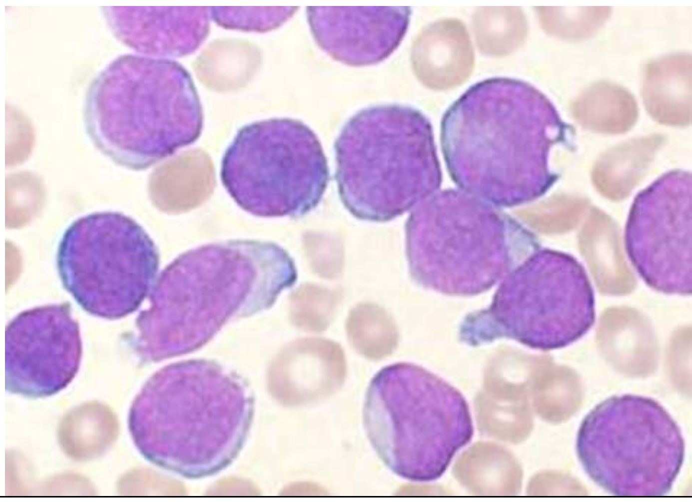
	 - Causes a normochromic, normocytic anemia (asked on an NBME directly). - Can cause **pigment stone cholelithiasis** due to the spleen identifying the spherocytes as abnormal and increasing RBC turnover. - RBC mean corpuscular hemoglobin concentration (MCHC) can be ↑. This is not specific to spherocytosis but is nevertheless a rare lab parameter to see in Qs. It's to my observation that if they mention it, 4/5 times the Dx is spherocytosis. - Diagnose via osmotic fragility test and eosin-5-maleimide. - Coombs test is negative (meaning we don't have antibodies against RBCs). This is important in terms of contrasting with spherocytosis due to hemolysis (as I discuss below). - Treatment is splenectomy (to ↓ RBC turnover). The answer can appear on USMLE as "splenectomy and cholecystectomy" together. - USMLE can mention a child who has a low Hb + one of the parents had splenectomy +/- cholecystectomy performed in the past for "an anemia" → answer = hereditary spherocytosis. I also point out to the student the AD inheritance pattern based on one of the parents having the condition as well. - Q can also just ask straight-up which heme condition is often treated with splenectomy, and the answer is hereditary spherocytosis; students sometimes erroneously choose sickle cell. Remember, sickle cells causes autosplenectomy due to repeated microinfarcts within splenic microvasculature; it is not treated with splenectomy.
Hemolytic anemia	- Spherocytosis is not always due to hereditary spherocytosis and can sometimes be seen with drug- or infection-induced hemolysis. - In this case, the Coombs test is (+), meaning we have antibodies against RBCs. In contrast, with hereditary spherocytosis, the mechanism has nothing to do with antibodies, so Coombs test is (-). - Antibodies binding to RBCs can trigger complement-mediated damage, which can lead to deformation of the RBC membrane and cytoskeleton. - USMLE will give kid with recent viral infection who has low Hb and spherocytes on a smear + positive Coombs; answer = hemolytic anemia, not hereditary spherocytosis (i.e., answer = "no specific mutation").

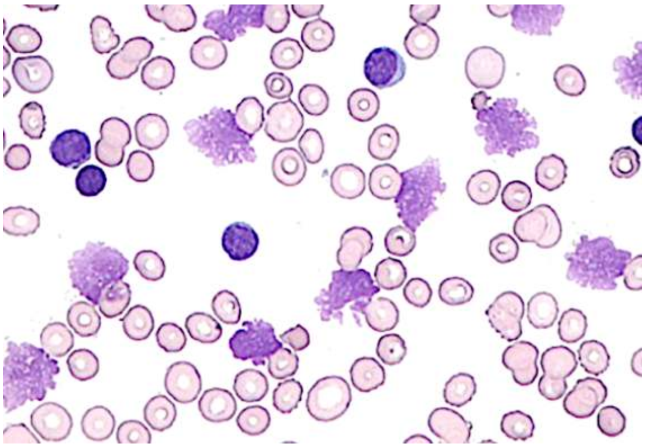
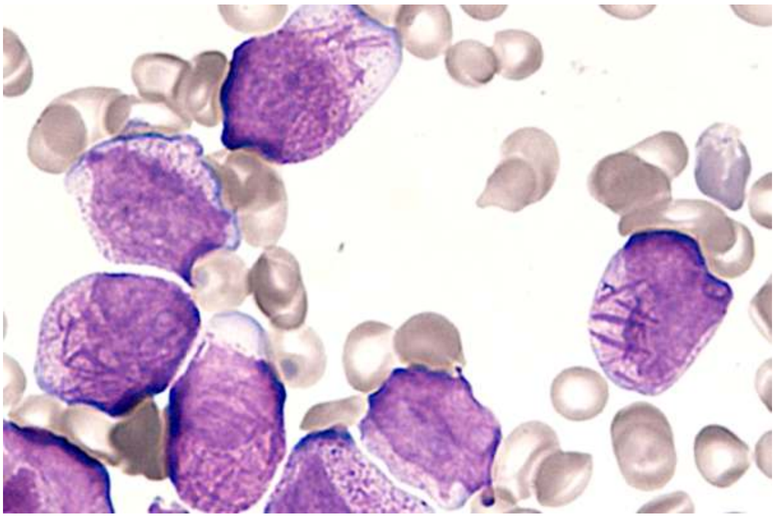
Leukemoid reaction
- Exaggerated immune response to infection in which WBCs go ↑↑ and can hence resemble leukemia. - As I mentioned before, "-oid" means "looks like but ain't," so the patient's lab can look like leukemia (because the WBCs are so high), but it's not leukemia. - Normal leukocyte count is 4-11,000/µL. Typical infections might cause WBCs to go ↑ to the teens or 20,000s/µL. But when WBC levels creep into the 30s, this becomes increasingly rare, even for patients who are septic, so we want to start thinking about leukemia. This is why if the patient doesn't turn out to have leukemia we call it leukemoid reaction.

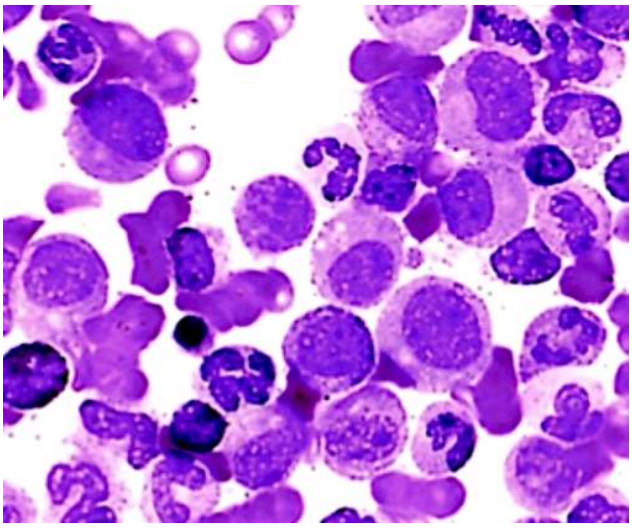
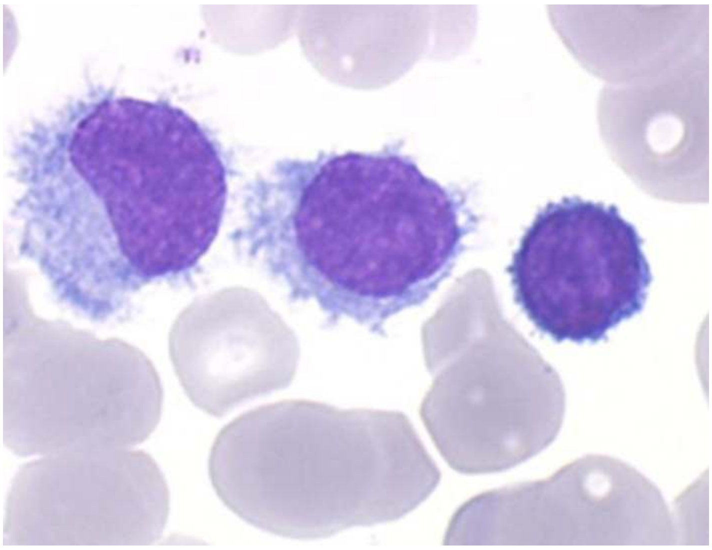
- This will be the answer on USMLE for patient who has **infection + WBCs >30,000/µL, PLUS** they give you a blood smear that shows neutrophils or they say leukocyte ALP is elevated (in CML it's low).



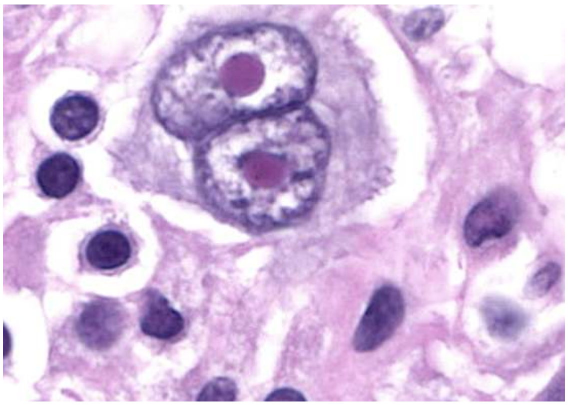
- This is the smear the USMLE will show you if they want leukemoid reaction. This is showing you **neutrophilia** (i.e., just lots of neutrophils). They can also write the answer as **reactive granulocytosis,** or as "increased release of leukocytes from bone marrow post-mitotic reserve pool."
- A typical leukemoid reaction Q might give you a UTI + WBCs 32,000/µL + the above smear. Not hard.

HY Leukemias	
- A leukemia is a cancer of white blood cells in the blood. - They are almost always B-cell in origin. - Apart from Sezary syndrome and mycosis fungoides (discussed below), if one of the HY lymphocytic leukemias (i.e., ALL, CLL) is T-cell in origin, USMLE will give an SVC-like syndrome (i.e., positive Pemberton sign with congested neck veins and/or facial erythema/swelling) due to a thymic lesion.	
ALL	- Acute lymphoblastic leukemia. - The answer on USMLE for a kid with leukemia (i.e., pre-adolescent). - Q will give you kid who's, e.g., 6-years-old, and has leukocyte count of 60,000/µL, where it's 90% lymphocytes. - **An important DDx of ALL is pertussis,** that for whatever reason, can cause an ALL-like picture where WBCs can be 30-40,000+, where it's 90% lymphocytes. This is called reactive lymphocytosis. The distinction with ALL is that, in pertussis, they'll say there's a cough and/or post-tussive emesis (vomiting after cough) and/or hypoglycemia (HY finding in pertussis). - ↑ Risk with Down syndrome. The Q can give you standard vignette of Downs and then ask for what child is at greatest risk of developing → answer = "excess lymphoblasts." 

	ALL; lymphoblasts appear smooth and relatively uniform.
CLL	- Chronic lymphocytic leukemia. - Vignette will sound just like ALL but not in a kid (usually older adults). - The Q might say 70-year-old + WBCs 87,000/µL (90% lymphocytes), where they ask for the next best step → answer = "quantitative immunoglobulin assay," which can show ↑ monoclonal immunoglobulin sometimes in CLL. Sounds nitpicky, but it's an answer on an NBME. - CLL smear can show smudge cells, which are fragile cells on LM.  - Leukemic cells are CD5 and CD23 positive (asked on an NBME). - Can be associated with warm autoimmune hemolytic anemia (IgG against RBCs).
AML	- Acute myeloblastic leukemia. - Will show Auer rods on a smear.  - If an AML Q doesn't give you the buzzy image of Auer rods, they'll say there's "30% blasts" or "50% blasts." - Tx is all-*trans* retinoic acid (vitamin A). This causes the maturation of leukemic cells into mature granulocytes. - Tx of AML also leads to leukemic cell lysis, where release of Auer rods into the circulation can precipitate DIC.
CML	- Chronic myelogenous leukemia. - Caused by t(9;22) translocation (Philadelphia chromosome). - Results in formation of bcr/abl tyrosine oncogenic tyrosine kinase, which is a "fusion protein."

	- The stem will say there's all sorts of myelo-sounding cells – i.e., myelocytes, promyelocytes, metamyelocytes. 9/10 Qs that mention these cells are CML. - Leukocyte ALP is low. - Smear shows what I refer to as a "motley mix" or "soup," which shows all different types of cells.  - This is HY smear that differentiates it from leukemoid reaction (which instead shows neutrophilia). - Tx is with imatinib, which targets the bcr/abl tyrosine kinase.
Hairy cell	- Q will merely say patient has leukemia characterized by "cytoplasmic projections" that stains positive for tartrate-resistant acid phosphatase (TRAP).  - The stem can give low WBCs, which is unusual for leukemia. 9 out of 10 questions for leukemia will give ultra-HY WBCs. The NBME Q I've seen on hairy cell gives image similar to above + gives WBCs 3,500 + they say WBCs with cytoplasmic projections" + stains positive for TRAP.
Sezary Syndrome	- Cutaneous T-cell lymphoma that extends to the blood as a T-cell leukemia. - Caused by human T-cell lymphotropic virus (HTLV). - Has cerebriform-shaped T cells in blood stream.

	 - Presents classically as erythroderma (red skin).

Lymphomas	
- A lymphoma is a cancer of the lymphatic system. - Usually starts in lymph nodes, but can also be present in other areas of the lymphatic system, such as the spleen and bone marrow. - Almost always B-cell in origin. - Lymphomas are categorized as Hodgkin or non-Hodgkin (NHL). - Epstein-Barr virus, HIV, general immunosuppression (i.e., chronic corticosteroids or immunosuppressant drugs), and autoimmune disease (e.g., SLE, Hashimoto) increase the risk for lymphomas.	
Hodgkin	- Characterized by Reed-Sternberg cells.  - 100% of questions give a painless lateral neck mass or facial swelling (lymphadenopathy). - The USMLE Q will then give at least one additional finding of: 1) hepatomegaly; 2) mediastinal mass (not thymoma; this is mediastinal lymphadenopathy); 3) Virchow node (Troisier sign of malignancy; palpable left supraclavicular lymph node); this is not limited to gastric cancers; I've seen an NBME Q where they give this latter finding for Hodgkin. - B symptoms (i.e., fever, night sweats, weight loss) are common but not mandatory in vignettes. - Question can give normal leukocyte count and differential. An NBME Q gives WBCs of 10,000/μL (NR 4-11,000) and lymphocytes of 33% (NR 25-33). This is because lymphomas are cancers of lymphatic tissue. They're not leukemias, which are cancerous WBCs floating around the blood.
Burkitt	- t(8;14) translocation of c-myc gene. - Presents as jaw or abdominal lesion.
Follicular	- t(14;18) translocation of *Bcl-2* gene. - Most common indolent NHL (which means less aggressive).

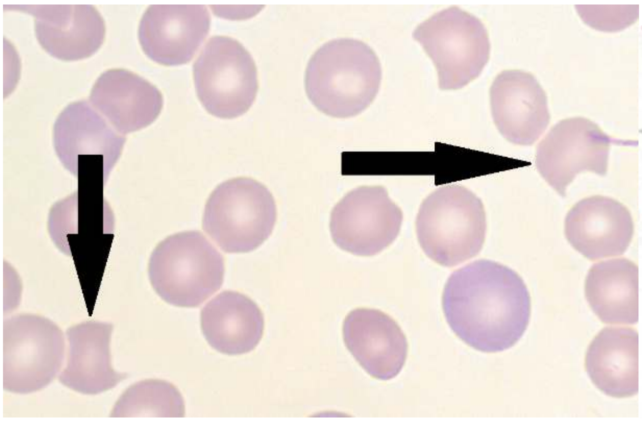
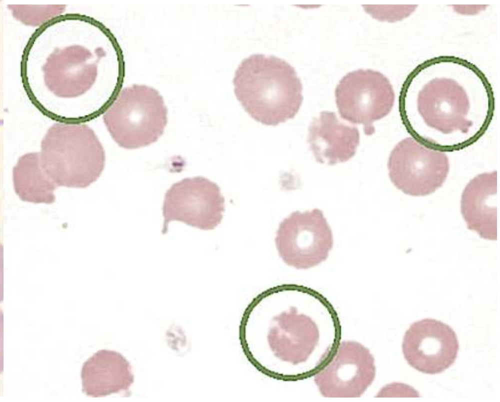
	- Presents as waxing/waning painless neck mass over 1-2 years.
DLBCL	- Diffuse large B cell lymphoma. - Most common aggressive NHL.
Mycosis fungoides	- Cutaneous T-cell lymphoma characterized by cerebriform-shaped cells. - Less aggressive than the T-cell lymphoma of Sezary syndrome and does not extend to the blood as a T-cell leukemia.  - An important differential for “skin rash.”

Warm vs cold autoimmune hemolytic anemia (AIHA)	
- Both present with (+) Coombs test, which just means antibodies against RBCs. - In other words, if a Q says Coombs is (+), we have Abs against RBCs; if it’s (-), we don’t.	
Warm	- IgG antibodies against RBCs. - “Warm” because agglutination of RBCs occurs at warmer temperatures. The stem can say agglutination occurs when the nurse holds the tube of blood by hand. - Seen on USMLE for autoimmune diseases like SLE or RA; CLL; or can be drug- or infection-induced.
Cold	- IgM antibodies against RBCs. - “Cold” because agglutination of RBCs occurs at cooler temperatures. The stem can say agglutination occurs while the tubes are transported en route to the laboratory. - Seen on USMLE for *Mycoplasma* pneumonia and CMV mononucleosis.

Infectious mononucleosis
- Usually caused by EBV, but can also be CMV. - Primary infection presents usually in teenager or young adult with fever >38 C, lymphadenopathy, tonsillar exudates, and lack of cough, making the presentation appear bacterial (i.e., presents with all four CENTOR criteria, as discussed earlier). As a result, it is often misdiagnosed as *Strep* pharyngitis. - If amoxicillin or penicillin is given to treat EBV, this can cause a rash. This is not to be confused with a rash caused by allergy to beta-lactams. If pre-adolescent receives beta-lactam and gets a rash, that is likely beta-lactam allergy. If a patient adolescent or older gets a rash, we do a heterophile antibody (Monospot) test as next best step in diagnosis, as EBV mono is more likely. - Following the primary infection, mono can present as recurrent episodes of extreme fatigue that arise at interval-separations of months to years.

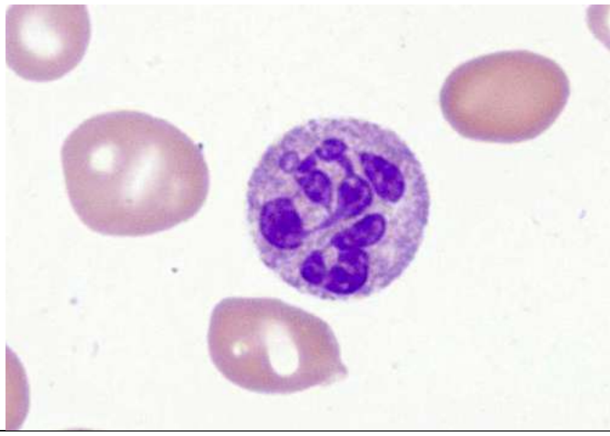
Glucose-6-phosphate dehydrogenase (G6PD) deficiency

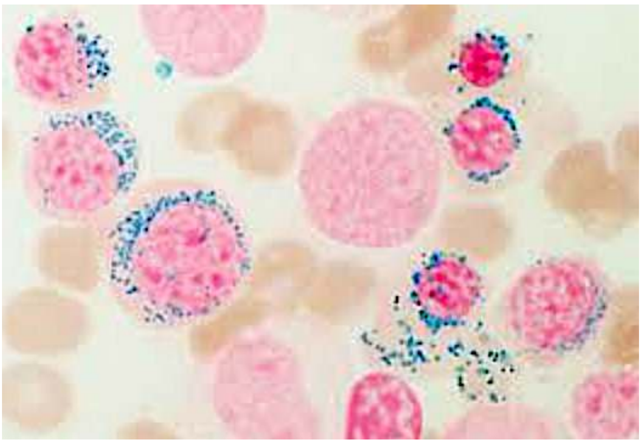
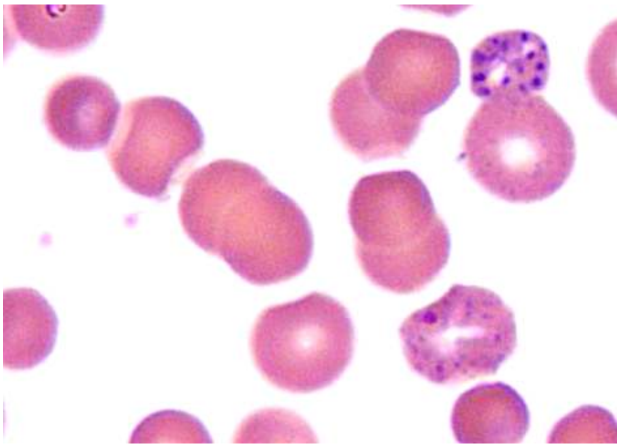
- Glucose-6-phosphate dehydrogenase deficiency; **X-linked recessive.**
- Most common cause of hemolysis due to an enzyme deficiency.
- Presents as hemolysis and jaundice in boy who recently received a drug or had a viral infection.
- HY agents that can precipitate oxidative damage of RBCs in G6PD are primaquine, dapsone, and sulfa drugs, and fava beans.
- Vignette might be 12-year-old boy with scleral icterus following a viral infection. Labs might show increased unconjugated hyperbilirubinemia and LDH (due to hemolysis). RBCs are packed with LDH. High LDH can be the USMLE's way of saying there is hemolysis.
- G6PD is an enzyme necessary to produce NADPH, which is a reducing agent that protects RBC membranes from oxidative damage.
- Bite cells (left image) and Heinz bodies (right image) are buzzy and HY. Bite cells are RBCs with bite-like notches caused by oxidative damage. Heinz bodies are clumps of oxidized/denatured hemoglobin.

Macrocytosis HY causes

- Normal RBC mean corpuscular volume (MCV) is 80-100 fL (femtoliters).
- Macrocytosis refers to large RBCs (>100 fL).

B9 (folate) / B12 deficiencies	- Deficiency of either vitamin leads to impaired DNA synthesis, where RBCs remain in the bone marrow for a longer time period, leading to RBCs that are larger and less mature. - High MCV as a result of impaired DNA synthesis is called megaloblastic anemia. - Impaired DNA synthesis also causes hypersegmented neutrophils, where longer time spent in the bone marrow results in an increased number of nuclear lobes and segments. 
Alcohol	- An important cause of high MCV on USMLE in the setting of normal B9/B12 levels. - Alcohol can also cause other types of anemia (i.e., sideroblastic).

Heme synthesis disorders	
- Family Med won't ask about enzyme deficiencies or substrates that buildup the way Step 1 will, but they expect you to be able to diagnose these conditions based on patient presentation.	
Sideroblastic anemia	- Can be X-linked recessive, but also can be caused by alcohol. - Failure to incorporate iron into heme. This leads to "ringed sideroblasts," which are erythroblasts (RBC precursors) where iron accumulates inside the mitochondria, where the latter form a ring around the nucleus. - The iron-laden macrophages stain blue with Prussian blue stain, making this a very buzzy and pass-level image for USMLE. 
Lead poisoning	- The stereotypical vignette of a 2-year-old eating paint chips in an old house leading to mental deterioration is too easy and maybe only ~1/3 of lead questions. - The other 2/3 will be an adult who has **microcytic anemia** and mental decline. Buzzy hobbies can be hunting (bullets containing lead) or home-brewing of alcohol (casks made of lead).  - Can cause basophilic stippling of RBCs (rRNA precipitates). - Tx is calcium EDTA, dimercaprol, or succimer.
AIP	- Acute intermittent porphyria. - Answer on USMLE for the heme synthesis disorder where patient has red/pink/port wine colored urine and abdominal pain. - Patients can sometimes have miscellaneous neurologic findings, but not mandatory. - Treatment is hematin and glucose.
PCT	- Porphyria cutanea tarda. - Answer on USMLE for the heme synthesis disorder where patient has photosensitivity / blistering. - Urine can appear "tea-colored" / dark. Don't confuse this with the red/pink/port wine colored urine in AIP.

HY Heme/onc pharm points	
Aspirin	- Irreversible cyclooxygenase (COX) 1&2 inhibitor. - Non-steroidal anti-inflammatory drug (NSAID). - Blocks synthesis of prostaglandins and platelet thromboxane A2. - ↓ prostaglandin means ↓ inflammation, pain, and body temperature. - ↓ thromboxane A2 means ↓ platelet clumping. - Many HY use cases for Family Med: - Atrial fibrillation (with low CHADS2 score – i.e., Congestive heart failure, Hypertension, Age >75, Diabetes, Stroke/TIA/Hx of emboli, with the S being worth 2 points). If 0 or 1 points, give aspirin to AF patient; if 2+ points, give warfarin. USMLE doesn't obsess over valvular vs non-valvular AF in terms of giving warfarin vs NOACs (e.g., dabigatran, apixaban); they will either give aspirin or warfarin as the answer choices. - Peripheral vascular disease and carotid stenosis → USMLE wants triad of anti-platelet therapy (aspirin alone sufficient on USMLE), statin, and ACEi or ARB. - Ischemic heart disease (i.e., patients who have angina pectoris) → nitrates are used for acute angina, but daily aspirin is also given to these patients. - Aspirin can cause GI bleeds (↓ prostaglandin → ↓ gastric mucosal barrier mucous production and protection), nephropathy (pre-renal, interstitial nephropathy, nephrogenic diabetes insipidus), asthma (inhibition of COX causes shunting of arachidonic acid down the lipoxygenase pathway → leukotriene synthesis → bronchoconstriction). - Never give aspirin to children due to Reye syndrome risk (defect in beta-oxidation, with cerebral edema and hepatitis). - Misoprostol is given to patients following PPIs if they have NSAID-induced gastric ulcers (misoprostol is PGE1 analogue). - Aka salicylic acid (salicylate); can cause high-anion gap mixed metabolic acidosis-respiratory alkalosis (S in MUDPILES).
Other NSAIDs	- "Other NSAIDs" = reversible COX inhibitors, in contrast to aspirin (irreversible). - Indomethacin → used for acute gout and pericarditis. - Ibuprofen → general NSAID used broadly for pain, fever, and autoimmune diseases (i.e., RA, IBD, SLE, etc.). - Naproxen → USMLE-favorite drug for causing GI bleeds and nephropathy; shows up on 2CK forms a lot. - Diclofenac, ketorolac (just know these names = NSAIDs). - 5-ASA compounds (mesalamine, sulfasalazine) used for IBD.
Acetaminophen	- Central-acting COX inhibitor. - Not considered an NSAID because it doesn't act peripherally. - Decreases pain and fever, but is not anti-inflammatory in the periphery. - First-line med for osteoarthritis (weight loss first Tx). NSAIDs not ideal first-line because they kill the kidneys + OA is non-inflammatory, so NSAIDs don't provide added benefit over acetaminophen. Patients with chronic pain who take NSAIDs frequently are at risk of nephropathy and GI bleeds. - Metabolite NAPQI causes massive hepatotoxicity. Activated charcoal can be given immediately after consumption, otherwise *N*-acetylcysteine HY as Tx.
Celecoxib	- Selective COX-2 inhibitor that does not cause GI bleeds. - Still can cause nephropathy. - NBME asks why celecoxib can increase the risk of MI → answer = decreases prostaglandin production without inhibiting platelets. - Can in theory be used for pain and inflammation.
Clopidogrel	- Anti-platelet agent; inhibits ADP P2Y12 receptor on platelets. MOA is HY. - Can be used to fulfill the anti-platelet component of the peripheral vascular disease and carotid stenosis triad (as mentioned before: 1) anti-platelet therapy; 2) statin; 3) ACEi or ARB). - In theory can be used for stenting procedures, but I haven't seen USMLE give a fuck. It's more that this use case is perpetuated in resources.

	- Can also be used in patients with ischemic heart disease who can't tolerate aspirin.
Cilostazol	- Phosphodiesterase-3 inhibitor; prevents breakdown of cAMP. - Used for intermittent claudication / peripheral vascular disease **after a walking/exercise program is already prescribed.** Do not choose cilostazol as a first treatment for peripheral vascular disease. Choose walking/exercise program first. Then, if they force you to choose a drug for additional management, choose cilostazol.
Heparin	- Upregulates antithrombin III. - Not too relevant for Family Med because it's subcutaneous injection + usually used in hospitals for DVT, PE, and superficial thrombophlebitis. - Used as bridging agent during the commencement of warfarin.
Warfarin, dicoumarol	- Vitamin K epoxide reductase inhibitors leading to impaired clotting factor activation. - Highest yield use cases on USMLE are for maintenance anticoagulation therapy in patients who have DVT, prosthetic valves, or atrial fibrillation with elevated CHADS2 score. - Warfarin levels are sensitive to drugs that inhibit or stimulate P-450. For example, P-450 inhibitors can ↑ INR and cause bleeding diathesis; P-450 inducers can ↓ INR and cause thromboses. - Teratogenic; must be avoided in pregnancy.
Hydroxyurea	- Increases HbF in sickle cell to ↓ recurrence of painful crises. - Can be used to reduce bone marrow cell production in polycythemia vera to reduce hyperviscosity episodes. As discussed earlier, if USMLE gives you acute hyperviscosity syndrome, choose phlebotomy as best answer. Hydroxyurea is merely to decrease recurrence.
Methotrexate	- Dihydrofolate reductase inhibitor (enzyme required in folate pathway). - First-line disease-modifying anti-rheumatic drug (DMARD) in rheumatoid arthritis. - First-line oral agent in plaque psoriasis that is non-responsive to topical agents, or for systemic psoriasis (arthritis). - Causes pulmonary fibrosis, mucositis (mouth ulcers due to neutropenia), and hepatotoxicity. - Toxicity mitigated with folinic acid (not folic acid), which is aka "leucovorin rescue."

FM Repro/Obgyn

Penile/testicular disorders	
Hypospadias	- Urethral meatus opens on underside of penis (ventral aspect).
Epispadias	- Urethral meatus opens on top of penis (dorsal aspect).
Phimosis	- Cannot retract the prepuce (foreskin). - In other words, cannot pull it back to expose the glans.
Paraphimosis	- Cannot reduce the prepuce. - In other words, once it is retracted, cannot replace it back over the glans. - Just remember that paraphimosis is the bad one / emergency.
Cryptorchidism	- Undescended testis. - Increased risk of infertility (↑ temperature within abdomen → ↓ spermatogenesis). - Increased risk of testicular carcinoma. - USMLE wants no treatment prior to age 6 months; after 6 months, perform orchidopexy.
Testicular torsion	- Acutely painful testis; medical emergency; can lead to ischemia and loss of testis. Can be caused by abnormality of the gubernaculum, which is a ligamentous cord that anchors the testis within the scrotum. This leads to "bell-clapper deformity," where improper anchoring = ↑ risk of rotation. - Negative Prehn sign, which is failure of relief of pain upon lifting of the scrotum. Apparently this is an unreliable test in real life, but NBME vignettes can still mention it. In contrast, a (+) Prehn (relief of pain upon lifting the scrotum) is more indicative of epididymitis. - Negative cremasteric reflex, which is failure of the scrotal skin to retract upon palpation of the medial thigh. In contrast, a (+) cremasteric reflex (scrotal skin retracts) is more indicative of epididymitis. Normally, when the inner thigh is stroked, the cremaster muscle contracts and pulls up on the ipsilateral testis. In torsion, the twisted spermatic cord can impair the nerve fibers responsible for triggering the reflex. - USMLE wants Doppler ultrasonography as next best step, followed by surgery.
Epididymitis	- Inflammation/infection of epididymis. - (+) Prehn sign and (+) cremasteric reflex. - Chlamydia and gonorrhea are most common causes in males <35. - Over 35, *E. coli* is most likely. - The USMLE will not play games where they give you a borderline case and force you to guess. They will give you either a 25- or 45-year-old male. This also goes for prostatitis (same organisms). There is an NBME Q of a 45-year-old where the answer is *E. coli*, and chlamydia is wrong.
Torsion of appendix testis	- Not the same as torsion of testis. - Asked on a 2CK form. - Q will tell you there's a kid with a painful testis + the superior pole is blue + there's an intact cremasteric reflex ("regular torsion" it's not intact). - "Blue dot" sign = blue superior pole of testis. - Not my opinion if you think it's nitpicky or weird. Take it up with the NBME exam.
Peyronie disease	- Curvature deformity of penis caused by fibrous plaque within tunica albuginea. - Doesn't have to be treated unless causing functional impairment.
Orchitis	- Inflammation/infection of the testis. - Mumps is HY cause in unvaccinated / immigrants (sometimes implies unvaccinated). - If not caused by mumps virus, then the bacterial causes are same as epididymitis and prostatitis for young vs old (i.e., STIs vs *E. coli*, respectively).
Priapism	- Erection lasting longer than 4 hours. - Can be caused by sickle cell or by PDE-5 inhibitors (i.e., tadalafil/sildenafil). - Injection of alpha-1 agonist (i.e., phenylephrine) into the erectile tissue is treatment → decreases blood flow due to blood vessel constriction.
Hydrocele	- Serous fluid collection within the scrotum between the layers of the tunica vaginalis surrounding the testis. - Mechanism is patent processes vaginalis.

	- The processus vaginalis is an embryonic outpouching of peritoneum that descends into the scrotum. It normally obliterates shortly after birth, but if it remains open, can cause hydrocele or indirect inguinal hernia. - Transilluminates when a light is shone to it. - USMLE wants no treatment before the age of 1; most spontaneously close.
Spermatocele	- Obscure serous fluid collection within the epididymis containing sperm. - Occurs usually in adult men (rather than neonates as with hydrocele). - Transilluminates similar to hydrocele. But once again, it's an older adult, not neonate.
Varicocele	- Dilation/congestion of pampiniform venous plexus draining the scrotum. - Does not transilluminate (in contrast to hydro- and spermatoceles). - Can be described as buzzy "bag of worms." Sounds too easy, but an NBME Q literally uses this colloquialism. - Can also be described as a heavy, dragging scrotum lower on the left. - Occurs almost always on the left side due to the anatomy of venous drainage of the testis. - On the left, we have left testicular vein → left renal vein → IVC. - On the right, we have right testicular vein → IVC ("right to IVC"). - The left testicular vein drains at a 90-degree angle into the left renal vein, creating a proclivity for hydrostatic pressure backup ipsilaterally. - There is a 2CK NBME Q where bilateral varicocele is the answer. Highly obscure, but just be aware it's somehow possible/exists. - Can occur due to left renal vein thrombosis in setting of nephrotic syndrome (loss of antithrombin III in the urine → hypercoagulable state).
Testicular cancer	- Seminoma most likely in teens and older. - Yolk sac tumor most likely in school-age kids and younger. - Hard nodule on testis is buzzy, but be aware of it as DDx for non-transilluminating testicular mass that does not present as classic hard nodule.

Endometrial cancer
- Highest yield point on USMLE is that it is caused by "unopposed estrogen," usually due to anovulation/PCOS. - Estrogen normally stimulates growth of endometrium; progesterone prevents overgrowth. - In anovulation/PCOS, no corpus luteum is formed, since the latter is the follicular remnant. - The corpus luteum normally secretes progesterone to maintain the endometrial lining if fertilization occurs. If no corpus luteum is formed, there's no progesterone produced to balance estrogen. This leads to endometrial hyperplasia, which increases the risk of endometrial adenocarcinoma. - USMLE can give an overweight female who's perimenopausal who has breakthrough bleeding (i.e., mid-cycle bleeding), or a post-menopausal woman with any vaginal bleeding, and they want endometrial biopsy as the answer. The fact that the patient has high BMI insinuates that anovulation/PCOS may have been a part of her past, where unopposed estrogen is a part of her history. - Can occur secondarily as a result of granulosa cell tumor of the ovary (as discussed above). This is a rare cause of unopposed estrogen. - Can occur in women who take hormone-replacement therapy (HRT) who stop taking the progesterone component (mentioned in one NBME Q). - Can be associated with hereditary non-polyposis colorectal cancer (HNPCC; Lynch syndrome).

Vulvovaginal lesions for FM	
Vulvovaginal carcinoma	- Cancer of the vagina and/or vulva. - Almost always squamous cell carcinoma due to HPV 16 or 18. - Can occur rarely due to lichen sclerosus.
Condylomata acuminata	- Aka warts; caused by HPV 6/11.

	- Present as painless, skin-colored or slightly hyperpigmented cauliflower-like popular lesions. - I discuss all of the STDs in more detail later.
Lichen sclerosus	- Presents as whitish-grey, rough, irritated or scratchy patch on the vulva or perineum. - Thought to be caused by a mix of chronic irritation and autoimmunity; not HPV-related. - Characterized by atrophy, hyperkeratosis, and a band of lymphocytes in the dermis. It is not neoplasia, dysplasia, or metaplasia. - USMLE wants biopsy as next best step to rule out SCC (on an old Step 1 Free 120 even though NBS Qs are classically 2CK). If SCC is negative and the diagnosis is LS, USMLE wants topical steroid as treatment.
Bartholin gland cyst	- Shows up as a tender/painful bump at a 4- or 8-o'clock position on the vulva. - If the cyst becomes infected (i.e., warm and red), we call it Bartholin gland abscess. - USMLE wants "polymicrobial" as the most likely organism. - For cysts, sitz bath + warm compresses are first Tx. - For overt abscesses, drainage is the answer.
Vaginal foreign body	- Vignette will be a child with foul-smelling discharge from the vagina. - There will be no signs of physical trauma or lacerations (means not abuse).
Sexual abuse	- Can occur in children or elderly (I've seen both on NBME). - Vignette likes to mention lacerations. - There may or may not be discharge.

Cervical cancer + Pap smear screening
- Squamous cell carcinoma; occurs at the transitional zone between the stratified squamous ectocervix and columnar endocervix. - Caused by HPV 16/18. There are other strains students get pedantic about, but USMLE doesn't give a fuck. - Can cause vaginal bleeding + uni- or bilateral hydroureter / -nephrosis. - Pap smears should be started at age 21. If the vignette gives you a teenager who's sexually active, the answer is you test for STDs but do not do a Pap smear. - Perform Pap smear age 21-65. - From age 21-30, they are done every 3 years without HPV co-testing. - From age 30-65, the female has the option of continuing Paps every 3 years, or doing Pap-HPV co-testing every 5 years. - Pap smears are stopped at age 65. - If the female has a history of a high-grade squamous intra-epithelial lesion (HSIL) or cervical intra-epithelial neoplasia (CIN), Pap smears are continued after age 65. - Patients who have had hysterectomy due to cervical cancer should have continued Pap smears of the vaginal cuff. Patients with hysterectomy due to endometrial cancer do not. - Patients with HIV require two Pap smears in the first year following diagnosis (6 months apart), followed by annual Pap smears thereafter. The vignette will tell you HIV patient had normal Pap smear last year, and student is like, "Cool, she doesn't need one for another 2 years." No. This is important. You need to know HIV patients need once/year Paps. - Low-grade squamous intra-epithelial lesions (LSIL) and atypical squamous cells of undetermined significance (ASC-US) are managed with either repeat Pap smear or HPV co-testing, where (+) vs (-) HPV test determines whether colposcopy + biopsy is performed. **USMLE will not** assess the algorithmic specifics. If you memorize the algorithm based on the female's age, you're wasting your time for USMLE purposes. - HSIL on Pap smear is managed with immediate colposcopy + biopsy. - CIN I is what we call LSIL that is confirmed on colposcopy + biopsy. - CIN II/III is what we call HSIL that is confirmed on colposcopy + biopsy. - CIN I has high rate of spontaneous regression. Repeat Paps are done in a year.

- CIN II/III requires excisional or ablational treatment due to higher risk of progression to invasive cervical cancer. Such treatments include LEEP, coning, cryotherapy, or laser ablation. - Cone biopsy is also an answer on NBME for next best step in patient who has colposcopy performed for HSIL on Pap + “the entire squamocolumnar junction cannot be visualized.” In other words, do a cone biopsy if colposcopy is insufficient. - CIN III is **not** excised/ablated during pregnancy. HPV lesions (including warts) are known to get temporarily worse during pregnancy due to relative immunosuppression (biologic attempt to minimize attack against fetal antigens). In the post-partum period, CIN III lesions are evaluated for signs of regression over the course of weeks; if they do not regress, they are excised same as non-pregnant women with CIN II or III.

Breast cancer screening
- For USMLE, mammography is done every two years between ages 50-74. - Since committees / cancer task forces have varying recommendations, the USMLE doesn’t play trivia as far as “oh em gee do we start at 45 vs 50, etc.” They won’t be borderline about it. They will just give you a patient who’s, e.g., 56 who needs a mammogram since she hasn’t had one for two years. - If first degree family member (i.e., parent or sibling) has Hx of breast cancer, then start mammography at age 40, not 50. - Any breast screening done for a women under 30, do not do mammography on USMLE. Ultrasound is typically done. This is because younger women have denser breast tissue, making mammography less specific (i.e., more false-positives) and hence less reliable of a diagnostic modality. - Estrogen and progesterone receptor (ER/PR) positivity is correlated with better prognosis for breast cancers. This is partially because SERMs and aromatase inhibitors can be used as Tx. - HER2/neu positivity is correlated with worse prognosis. Trastuzumab (Herceptin) can be used to target HER2/neu. - Triple-negative ER/PR/HER2/neu is associated with bad prognosis due to ↑ aggressiveness, ↑ rates of recurrence, and ↑ risk of metastasis to lungs/brain. - 2CK NBME wants “bilateral mastectomy + oophorectomy” in patients who have confirmed *BRCA* mutation. - Breast cancer risk is ↑ as a result of HRT, even when progesterone is given as part of it. Breast cancer risk is ↑ as a result of ↑ absolute estrogen exposure the female has in her life. This has nothing to do with unopposed estrogen, which is the major risk factor for endometrial cancer. - Combined OCPs have been suggested in some literature as ↑ the risk of breast cancer, but there is no significance. They won’t assess this on USMLE either way. But they do assess ↑ breast cancer risk as a result of HRT. - The reason the only approved indication of HRT is severe vasomotor Sx (i.e., and not to maintain bone density, etc.) is because HRT ↑ the risk of breast cancer, thromboembolic events, MI, and stroke. This is because estrogen upregulates fibrinogen and factors V and VIII.

Breast conditions for FM	
Fibrocystic change	- Benign cystic alteration in breast tissue in women 20s-50s. - Shows up three ways on NBMEs: 1) Highly buzzy/classic presentation is a woman 20s-30s who has bilateral, waxing/waning breast tenderness over the course of her menstrual cycle; 2) Unilateral tenderness or sharp pain in one breast (students are often surprised that it’s not always bilateral). 3) Painless, unilateral breast cyst in woman in her 40s that drains brown/greenish fluid → answer = fibrocystic change straight-up on the NBME. Call it weird all you want. Take it up with NBME, not me. - You do not need to biopsy or do ultrasound. It is a clinical diagnosis and Tx is symptomatic only (i.e., warm showers, evening primrose oil).
Simple breast cyst	- “Simple” means hypoechoic, or black on ultrasound. It reflects a fluid-filled cyst. - “Complex” means at least in some part hyperechoic, or white on ultrasound. It reflects a solid component to the cyst.

	- Simple cysts are observed and do not require FNA drainage unless they cause discomfort to the patient (e.g., large cysts). There is no specific size that requires drainage on USMLE. - There is an NBME Q where they say a woman was started on HRT three months ago and now has a simple cyst → answer = "biopsy of the lesion." So you should be aware that sudden growth in size can be indication for biopsy. - All complex cysts require FNA biopsy, similar to fibroadenoma.
Fibroadenoma	- Most common breast tumor overall; benign. - Presents as rubbery, mobile, non-tender breast lump in woman 20s-30s. - Will appear as solid breast mass without calcification on ultrasound. - FNA is next best step to confirm Dx. If confirmed fibroadenoma, they do not need to be excised.
Mastitis	- Classically presents as red, cracked, fissured nipple in breastfeeding woman. - Usually caused by *S. aureus*, but Group A Strep also possible. - USMLE wants oral dicloxacillin (not doxycycline) + continue breastfeeding through the affected breast as treatment. There is no risk of harm to the neonate. Methicillin-class beta-lactams (e.g., dicloxacillin) and first-generation cephalosporins (e.g., cephalexin) are equivalent on USMLE for all intents and purposes. They both treat *S. aureus* skin infections. - High-yield point is that mastitis need not affect the nipple and can present as a red, warm, tender, ***non***-fluctuant mass of the breast. When we talk about infection of the breast, non-fluctuant = mastitis; fluctuant = abscess. This is asked insidiously on a 2CK form and students get it wrong all the time thinking it's abscess – i.e., "They say it's a warm, red, tender mass of the breast though!" Yeah, but they say non-fluctuant, not fluctuant. Not my issue if you get emotional over dumb shit.
Abscess	- Warm, red, tender *fluctuant* mass of the breast. - Drain as next best step.
Galactocele	- Aka milk retention cyst; usually seen in recently breastfeeding women. - Often confused with breast abscess. - Can present as painless or tender fluctuant mass of the breast that is peri-areolar. - USMLE will say the patient is afebrile and that the lesion is not warm or red. - Tx is warm compresses followed by drainage.
Virginal breast hypertrophy	- Virginal breast hypertrophy is benign asymmetric growth of the breasts that can occur in young women. USMLE will try to trick you into thinking it's sinister by saying a first-degree relative had breast cancer. - They will say 16-year-old girl has large breast lump under the nipple on one side. Answer is just follow-up / observe.
Gynecomastia	- Physiologic gynecomastia can occur in teenage boys on USMLE, where there can be ↑ breast tissue, either tender or not, occurring uni- or bilaterally. Just observe. - Can occur as a result of ↑ estrogen in liver failure, or due to certain drugs like cimetidine, spironolactone, ketoconazole, or even marijuana.

Ovarian conditions for FM	
Ruptured ovarian cyst	- DDx for sudden, sharp pain in an adnexa. - The Q will say there is no mass / will not mention mass. - Treated symptomatically (i.e., NSAID). Do not do laparoscopy.
Adnexal/ovarian torsion	- 3/4 Qs will give intermittent adnexal pain that becomes constant over hours to weeks (I've seen NBME give both time frames). - 1/4 Qs will just say few-hour Hx of constant pain without the intermittent detail first. - Presents as "large" adnexal mass (i.e., 8 x 12-cm). - Treatment is laparoscopic detorsion.

	- Tricky detail is that the biggest risk factor is a pre-existing anatomic abnormality (e.g., cyst). So if they say Hx of cyst + now has large adnexal mass and pain, that is torsion, not ruptured cyst.
Theca-lutein cyst	- Arises from theca cells of the ovary; usually bilateral. - Frequently associated with conditions where hCG is very high, such as moles, multiple gestation pregnancies, or fertility drug use. - Can occasionally bleed, where the answer is "hemorrhagic theca-lutein cyst" as a diagnosis of exclusion (i.e., we eliminate to get there), where endometrioid cyst can be eliminated because the Q won't mention anything about dysmenorrhea or signs of endometriosis.
Endometroid cyst	- Aka endometrioma, or "chocolate cyst." - Cyst filled with endometrial tissue that bleeds + causes pain. - Occurs in endometriosis (endometrial tissue growing outside the myometrium, usually on the ovary, as with this case; discussed more later).

Prostate cancer vs BPH points	
Prostate cancer	- Adenocarcinoma (i.e., glandular); most common cancer in men. - Stimulated by DHT. - Causes osteo**blastic** metastases (highest yield point on USMLE). - Can present as random guy over 50 with no PMHx who just suddenly has neurologic symptoms in the legs and two blastic lesions in his spine picked up on MRI. - Prostate-specific antigen (PSA) will often be elevated. Free-PSA (i.e., not bound to protein) is low compared to bound-PSA in prostate cancer. - Gleason scoring is used to grade prostate cancer.
BPH	- Benign prostatic hyperplasia. - All old dudes have enlarged prostates due to lifetime exposure to DHT, which causes prostatic enlargement. - Presents as dribbling, hesitancy commencing urination, or interrupted stream in a male over 60 (but usually 70s+). - "Old dude + high creatinine = BPH till proven otherwise." - Discontinue anti-cholinergic meds because they cause urinary retention. This concept shows up a lot on NBMEs. Three classes of drugs in particular: 1) First-gen H1 blockers (i.e., diphenhydramine, chlorpheniramine); 2) TCAs (i.e., amitriptyline). 3) Anti-psychotics. - You'll get a Q where an older male has high Cr and then the answer is just "discontinue anticholinergic meds," or "discontinue the diphenhydramine." - Can be described as "nodular" on palpation. - Prostate-specific antigen (PSA) will often be elevated. Free-PSA is high compared to bound-PSA in BPH. - Tx is alpha-1 blocker, such as tamsulosin/terazosin; or the 5-alpha-reductase inhibitor, finasteride. - Transurethral resection of prostate (TURP) for refractory cases.

HY STDs	
- STDs are exceedingly HY for Family Med.	
Human papilloma virus	- HPV 6/11 cause condylomata acuminata (warts). This is not limited to the genitalia and can cause laryngeal papillomatosis in neonates (warts of the vocal cords), which is asked on NBME. - HPV 16/18 cause squamous cell carcinoma of genitalia/anus; risk of overt SCC is ↑ in immunocompromised (i.e., HIV in MSM) and heavy smoking.

Chlamydia trachomatis	- The "regular" STD are *Chlamydia* D-K strains. - Causes mucopurulent discharge. - Can advance to pelvic inflammatory disease (PID) in females, which is when infection causes inflammation and scarring of Fallopian tubes, leading to ↑ risk of ectopic pregnancy. - Obligate intracellular, so cannot be grown. Discharge will show WBCs under light microscopy with **no organisms.** If the vignette tells you no organisms grow, this is pass-level for *Chlamydia*. - Treat with stat/one-off oral dose of **azithromycin;** can cause GI disturbance. - Can also be treated with one-week of BID (i.e., twice/day) **doxycycline;** cannot be taken with dairy or divalent cations (impaired absorption); can also cause photosensitivity; considered to be slightly more efficacious than azithromycin but much more annoying and arduous to take. - USMLE won't play trivia as to which drug is first-line; there will only be one correct answer. For example, doxy is not given in pregnancy, so if you're forced to choose between the two, in this case you know it's azithromycin (or even sometimes erythromycin in pregnancy). - Doxy isn't given in pregnancy because it can cause teeth discoloration in the eventual neonate. - *Chlamydia* can cause reactive arthritis (triad of urethritis, arthritis, and eye-itis – i.e., any inflammation of the eye, e.g., conjunctivitis, anterior uveitis, etc.). - To prevent ophthalmia neonatorum (neonatal conjunctivitis), treat the female while she is pregnant; to Tx the actual condition in neonates, give oral erythromycin. Conjunctivitis in a neonate can lead to *Chlamydia* pneumonia. - *Chlamydia* A-C are not STDs and cause trachoma (cause of blindness in Africa). - *Chlamydia* L1-3 cause lymphogranuloma venereum (anal strictures).
Neisseria gonorrhea	- Causes mucopurulent discharge, same as *Chlamydia.* - Gram-negative diplococci on light microscopy. - Same as with *Chlamydia*, can advance to PID, with ↑ risk of ectopic pregnancy. - Doesn't cause reactive arthritis; it causes gonococcal arthritis, which will present one of two ways on NBME: 1) monoarthritis of the knee; or 2) triad of mono- or polyarthritis + **cutaneous papules/pustules** + tenosynovitis (inflammation of tendon sheaths; stems like to give deQuervain tenosynovitis). - Treatment is IM ceftriaxone. - Always cotreat for *Chlamydia.* In other words, if the gram-(-) diplococci are seen under LM, there's no way to know if *Chlamydia* is also there or not since the latter shows no organisms, so if a patient has *Gonorrhea*, the proper Tx is IM ceftriaxone, PLUS either oral azithromycin or doxycycline. - If patient develops PID despite having been treated early with ceftriaxone for *Gonorrhea*, the answer for why this happened can be "Hx of improper antibiotic treatment," where the patient was supposed to be cotreated for *Chlamydia* with azithromycin but was only given the ceftriaxone for *Gonorrhea.* - If patient presents with PID who's septic (i.e., high fever, tachy, high WBCs), USMLE wants "admit to hospital + IV antibiotics," not the outpatient combo of IM + oral. - 2CK Obgyn form assesses that if an asymptomatic patient comes in after a partner tested positive for *Gonorrhea* or *Chlamydia*, the answer is give treatment without waiting for test results. - Erythromycin ointment is used on neonates as prophylaxis for conjunctivitis; if neonate already has it, give IM cefotaxime (preferred 3rd-gen cephalosporin in peds if listed).
Trichomonas vaginalis	- Causes trichomoniasis. - Flagellated protozoan seen on wet mount. - Presents as yellow-green discharge.

	- Can cause "strawberry cervix," or punctate hemorrhages on the cervix. If they don't say this, they can sometimes say yellow-green discharge + a vaginal canal that is erythematous. - Treat with metronidazole for patient **and partner** (high rate of reinfection).
Gardnerella vaginalis	- Causes bacterial vaginosis. - Gram-negative rod that causes a thin grey/watery discharge. - Positive whiff test (KOH prep causes fish-like odor). - Clue cells exceedingly HY (squamous epithelial cells studded with bacteria). They want you to know this image for USMLE: 
Candida spp.	- Causes candidiasis. - Buzzy thick, white, cottage cheese-like discharge in ~2/3 of questions. - The other ~1/3 of Qs will mention an itchy/erythematous vaginal canal without any overt discharge (in contrast to trichomoniasis which can present with erythema of the vagina but has characteristic yellow-green discharge). - Treat with topical nystatin or oral fluconazole.
Treponema pallidum	- Causes Syphilis. - Spirochete (spiral-shaped bacterium) visible under dark-field microscopy. - Primary syphilis = painless chancre (painless ulcer) on genitalia. - Secondary syphilis = 6 weeks to 6 months after appearance and disappearance of the initial chancre, patient can get body rash that **includes palms + soles,** and condylomata lata (painless genital plaques). - Tertiary syphilis = years later, patient can get gummas (appear as painless chancres but are on other areas of the body such as the face/nose), arthritis, and ascending aortitis (tree-barking of vasa vasorum). - Neurosyphilis **can occur at any stage;** it is not sequential where we have 1 → 2 → 3 → neurosyphilis. There is a 2CK Q that gives neurosyphilis in an 18-year-old. - Neurosyphilis presents as tabes dorsalis (obliteration of dorsal columns, with loss of vibration/proprioception + a positive Romberg sign, where patient falls over when standing with eyes closed), Argyll-Robertson pupil (i.e., "prostitute pupil"; accommodates but doesn't react), and "stroke without hypertension" (i.e., sometimes findings akin to stroke but in a younger patient). - Diagnosis of primary syphilis is made via visualizing the spirochetes from a chancre scraping under dark-field microscopy. - Diagnosis of secondary, tertiary, and neurosyphilis can be done with serology, where a VDRL and/or RPR is done first (sensitive but not specific); an FTA is done as confirmatory (specific but not sensitive). - VDRL test mixes the patient's serum with cardiolipin antigen. If antibodies against *T. pallidum* are present, the test demonstrates clumping/flocculation. The

	RPR enhances this reaction by using charcoal particles. There is an NBME Q floating around where they say something about a patient whose test results demonstrate clumping with charcoal particles, and the answer is SLE. - Patients with SLE who have anti-phospholipid syndrome can get false-positive VDRL tests because this syndrome is often caused by antibodies against cardiolipin (in SLE, we simply call these antibodies "lupus anticoagulant"). - FTA mixes a patient's serum with fluorescent *Treponema* antibodies. If binding occurs, this confirms the diagnosis of syphilis. - USMLE will show you 24-year-old male with rash on his back + KOH prep is negative + ask what's most likely to diagnose → answer = FTA. - Treatment for all syphilis types is penicillin. - If patient has Hx of anaphylaxis to beta-lactams but is pregnant or has tertiary or neurosyphilis, the answer is desensitize + give penicillin. This is because penicillin is the most efficacious and needs to be given in severe cases. - If patient has Hx of mere rash to beta-lactams, but not anaphylaxis, then the beta-lactam can be given anyway.
HSV 1/2	- Herpes simplex virus 1/2. - Causes painful vesicular lesions that recur at varying intervals (usually months). - Primary infection is most severe, often with fever, regional lymphadenopathy, burning/stinging/itching pain (herpetic neuralgia), and many vesicles. - Recurrences are often less severe and preceded by herpetic neuralgia. - HSV1/2 can also cause encephalitis (confusion + blood in CSF due to temporal lobe hemorrhage) and herpetic whitlow (vesicle[s] on the finger). - Viral culture can be negative in stem (not 100% sensitive). - Treat with acyclovir (or valacyclovir).
Haemophilus ducreyi	- Causes chancroid, which is a painful ulcer. - Gram-negative rod. - "-Oid" means looks like but ain't. So it looks like a syphilitic chancre, but it's not. - The syphilitic chancre is painless; the *H. ducreyi* lesion is painful. - Often a wrong/distractor answer for HSV Qs, where students get trigger-happy and erroneously choose the weird answer (*H. ducreyi*). - Chancroid will be the answer if they tell you there's a single painful genital lesion in someone who went abroad, classically backpacking in Africa or South America. - If they tell you there's a single, small painful lesion, *but that it's a recurrence*, this is HSV, not *H. ducreyi*. The latter is bacterial and doesn't cause recurrences the way HSV does; HSV can rarely appear as a single vesicle. - There is a 2CK NBME Q where answer is actually *H. ducreyi*, but I once again caution that this is usually a wrong answer, so be careful. But I have seen it correct as a one-off. - USMLE won't assess treatment, but either azithromycin or ceftriaxone is considered first-line.

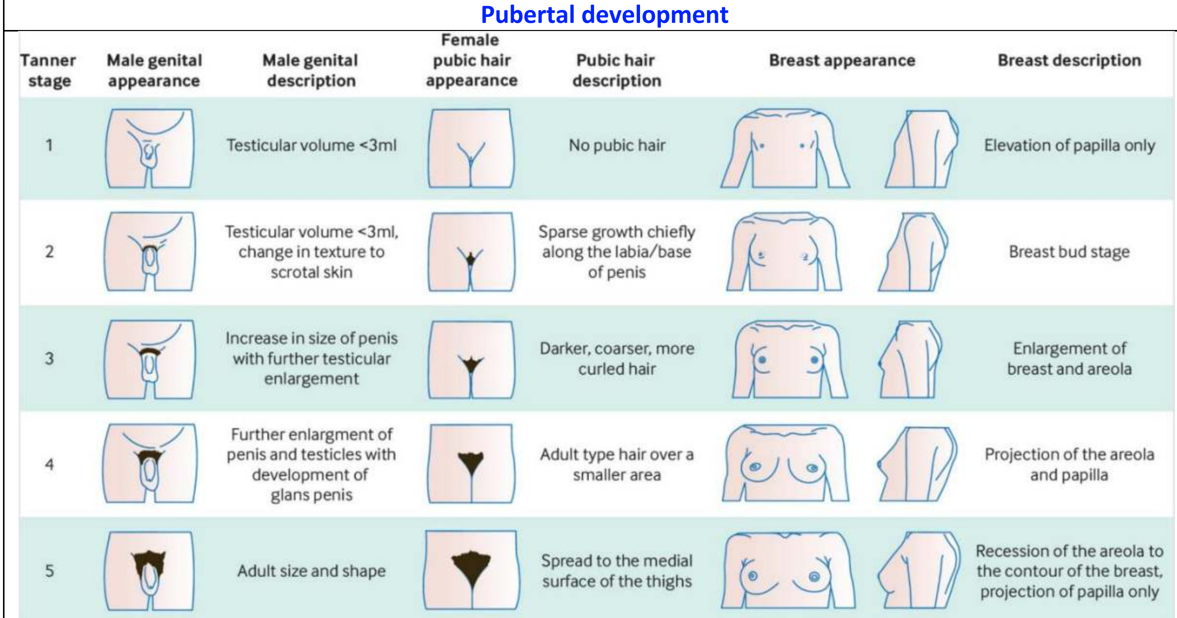
Pelvic inflammatory disease (PID)
- As discussed earlier, is caused by either *Chlamydia* or *Gonorrhea*. - When the infection ascends the uterus and Fallopian tubes, it can lead to inflammation and scarring, thereby increasing the risk for ectopic pregnancy (i.e., loss of Fallopian tube cilia leads to premature implantation usually in the ampulla of the Fallopian tube). - Adnexal and **cervical motion tenderness** are buzzy. But particularly the latter. If they say cervical motion tenderness, you know right away they're talking about PID. - If the Q tells you a girl has PID and is treated with antibiotics but has persistent fever and adnexal pain, next best step = ultrasound to look for **tubo-ovarian abscess,** which is a potential sequela of PID.

Ectopic pregnancy points
- FM shelf merely wants you to be able to diagnose ectopic pregnancy. Shows up on the forms as adnexal pain in the setting of a (+) β-hCG. If the woman's vitals are unstable, this is ruptured ectopic. - FM shelf will give you this scenario + list other DDx alongside (e.g., threatened abortion, etc.), and you just need to know the answer is ectopic pregnancy. It's not hard, but it's asked. - Ectopic is implantation outside the uterus, including the parametrium of the uterus and cervix. - β-hCG levels will be much lower than expected for the gestational age. You don't have to be an obstetrician and know exact values. - β-hCG should double approximately every 2-3 days in early pregnancy, so the Q can also mention something about poor rate of change of increase. - Highest yield point for USMLE is that ectopic risk is ↑↑ in women who have Hx of PID. As discussed earlier, this is due to scarring of the Fallopian tubes and disruption of the cilia. Ectopic implantation most frequently occurs in the ampulla (70-80%). - Methotrexate is given for Tx (asked once on 2CK form) for "small, stable ectopics" – i.e., mother is hemodynamically stable, there is no evidence of tubal rupture / fluid in the peritoneal cavity, the ectopic is <3.5 cm, and β-hCG is <5000 mIU/mL. For your Obgyn rotation, I would know those criteria. - Laparoscopic salpingostomy/salpingectomy is done if methotrexate cannot be given. - If the female is hemodynamically unstable (i.e., low BP), the answer is laparotomy, not laparoscopy.

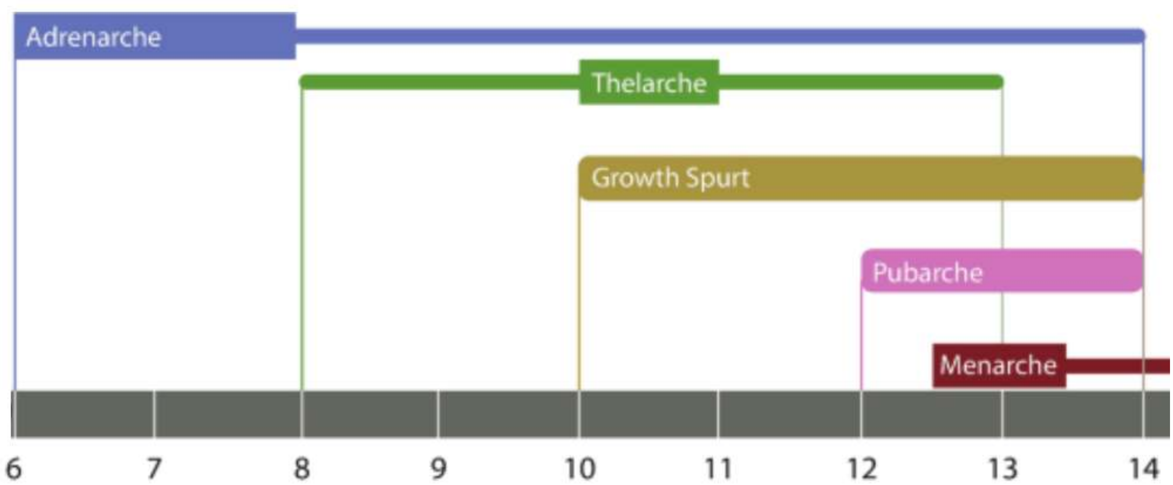
Miscarriage types	
- The new FM shelves and CMS forms are giving miscarriage types as answer choices.	
Threatened	- Means miscarriage may occur, but not inevitable. - Vaginal bleeding from closed cervix, with or without cramping, usually during first trimester. - Bleeding can be heavy. "Heavy" vaginal bleeding on USMLE can be described as passage of "clots" per vaginum. This is the same for menses. - Different from implantation bleeding, which is light vaginal bleeding that sometimes occurs 1-2 weeks after ovulation. - Classic Tx is bed rest, but new guidelines suggests evidence for benefit is lacking.
Inevitable	- As the name implies, means miscarriage that is ensuing / unavoidable. - Vaginal bleeding from open cervix, usually during first trimester. - Labor during first trimester usually presents as severe cramping. - Treatment is expectant management – i.e., allow for spontaneous first-trimester labor to occur. Misoprostol (PGE1 analogue) can be given to ripen the cervix. Surgical interventions like dilation & curettage (D&C) or suction curettage are also sometimes indicated.
Complete	- All products of conception have been expelled; no gestational sac seen. - As the name implies, the abortion is completed.
Missed	- Miscarriage that the body has not detected or attempted to expel yet. - Picked up on ultrasound as absent fetal heartbeat. - Cervix is closed. - Treatment similar to inevitable abortion (i.e., expectant, prostaglandin, surgical).
Incomplete	- Only some of the uterine contents have been expulsed. - Most difficult abortion type on USMLE because it will sound like missed, where they say a fetus with no heartbeat is visualized on ultrasound, but the difference is that they will say there's been passage of clots through an open cervix. - As mentioned before, passage of clots can indicate "heavy" passage of material, indicating components of the fetal material has already passed. - Even though heavy bleeding / passage of clots can occur with threatened abortion as well, in the latter, there is a closed cervix and the fetus is viable. - Treatment is expectant, prostaglandin, surgical.
Septic	- Miscarriage as a result of chorioamnionitis (infection of the gestational sac during pregnancy).

	- USMLE Q will give fever and discharge in a pregnant woman and then say buzzy details such as that it was an unwanted pregnancy + there is a laceration visualized on the cervix, indicating an attempt to self-abort with a hanger, which inoculated / seeded the infection. - Treatment is usually IV antibiotics and surgical evacuation.
Blighted ovum	- Aka anembryonic pregnancy. - Obscure diagnosis, but is asked on 2CK NBME and confuses people. - Fertilized egg that implants onto uterine wall, a gestational sac starts to develop, but then the embryo fails to develop + resorbs, resulting in an empty gestational sac. - β-hCG starts to rise normally because the placenta is present but the body doesn't yet recognize that the embryo is unviable. - The answer on USMLE for early pregnancy where they say ultrasound shows a gestational sac but no yolk sac or embryo.
Recurrent/habitual	- Aka repeated pregnancy loss. - Defined as 2 or more consecutive spontaneous abortions. - Distinct from isolated miscarriages, which occur in 15-20% of pregnancies. - Most important cause on USMLE is anti-phospholipid syndrome in SLE.

Pubertal development



Tanner stage	Male genital appearance	Male genital description	Female pubic hair appearance	Pubic hair description	Breast appearance	Breast description
1		Testicular volume <3ml		No pubic hair		Elevation of papilla only
2		Testicular volume <3ml, change in texture to scrotal skin		Sparse growth chiefly along the labia/base of penis		Breast bud stage
3		Increase in size of penis with further testicular enlargement		Darker, coarser, more curled hair		Enlargement of breast and areola
4		Further enlargment of penis and testicles with development of glans penis		Adult type hair over a smaller area		Projection of the areola and papilla
5		Adult size and shape		Spread to the medial surface of the thighs		Recession of the areola to the contour of the breast, projection of papilla only



- A mnemonic to remember the above sequence is "Tap em" (i.e., TAPM).
- Thelarche: Onset of breast development.

- Adrenarche: ↑ production of adrenal androgens, leading to the development of pubic and axillary hair. - Pubarche: Pubic hair growth - Menarche: Onset of menstruation. - USMLE asks straight-up what is most likely to occur first in a pre-pubescent female → answer = breast bud development. - USMLE wants you to know that at Tanner stage 3, "menarche is imminent." That is an answer straight-up on one of the 2CK forms, but there are also various Repro Qs where mom will bring in her daughter who's 13/14, who's never had a menstrual period, and they'll say she's either Tanner stage 2 or 3, and the answer is just "schedule follow-up in 6 months." - Turner syndrome can present with Tanner stage 1-2 breasts and 4-5 pubic/axillary hair. The latter are due to the effects of adrenal, rather than ovarian, androgens. In other words, don't be confused if you get a Turner syndrome Q and they say Tanner stage 4 pubic hair. - In androgen insensitivity syndrome (46XY karyotype, but female phenotype), breast development is normal (i.e., Tanner 4-5), but pubic/axillary hair are Tanner stage 1 (i.e., scant/absent). - Bone age refers to degree of maturation of a child's bones via radiographic examination. - USMLE wants you to know that bone age < chronologic age means the short child will catch up (i.e., the growth chart is merely right-shfited). This is called constitutional short stature. The vignette will usually say both parents are average height. For example, a 14-yr-old boy is shortest in his class but has a bone age of 12, meaning is skeleton is aged 12, so he'll catch up later. - For constitutional short stature, sometimes instead of mentioning bone age, they'll mention Tanner stage. For example, a 14-yr-old boy is shortest in his class + pubic hair is Tanner stage 2. The implication is he's still very early in development and will probably catch up. - Bone age = chronologic age means genuine short stature. I've seen this in Turner syndrome Qs, where the female is usually < 5 feet.

Sex disorders for FM	
Turner syndrome	- 45XO karyotype; female phenotype. - Infertile due to streak ovaries. In Turner, this is colloquially referred to as "menopause before menarche." - Can still have children with IVF using donor egg + exogenous hormones (asked sometimes on behavioral/psych Qs). - Short stature (usually < 5 feet), Tanner stage 1 breast development ("shield chest"), cystic hygroma (webbed neck due to lymphatic insufficiency; asked on NBME); scattered nevi (confuses students for things like NF1, but I don't know what to say; you need to know scattered nevi are seen in Turner); normal pubic and axillary hair (Tanner 4-5). - Coarctation of the aorta + bicuspid aortic valve (aortic stenosis) HY.
Klinefelter syndrome	- 47XXY karyotype; male phenotype. - Tall, eunuchoid body shape; small testes; gynecomastia; reduced IQ.
Androgen insensitivity syndrome (AIS)	- 46XY karyotype; female phenotype. - Defect in androgen receptor. Since DHT is needed for external male development, the patient appears female. - Testosterone and DHT are normal or elevated due to ↓ negative-feedback at the hypothalamus and anterior pituitary → ↑ LH production → ↑ testicular androgen production. - Breast development is Tanner 4-5 but pubic/axillary hair is Tanner 1. - USMLE vignette will say: 15-year-old girl is brought to the physician by her mother for not yet having a menstrual period + vagina ends in blind pouch + pubic/axillary hair is scant/absent → answer = AIS. - Next best step = ultrasound first if listed, then karyotype.
Müllerian agenesis	- 46XX karyotype. - Presents identical to AIS on USMLE except they'll say pubic/axillary hair is normal/coarse.

	- In other words, 15-year-old girl + never had a menstrual period + vagina ends in blind pouch + pubic/axillary hair is coarse → answer = Müllerian agenesis. - Next best step = ultrasound first if listed, then karyotype.
5α-reductase deficiency	- 46XY karyotype; phenotypically female, or neonate with ambiguous genitalia. - 5α-reductase is enzyme that converts testosterone → DHT. If there is a deficiency and DHT is ↓, male external structures don't develop, so phenotype is female. - Presents as "phallus (penis) at age 12." This is buzzy. They will say a 12-year-old girl has grown 4 inches in the past 4 months + has acne + hair on upper lip + clitoral hood is 3-4 cm → answer = 5α-reductase deficiency. - At puberty, the ↑↑ in testosterone can partially override the enzyme deficiency, leading to threshold production of DHT where male secondary sex characteristics can develop.

Imperforate hymen
- Blood accumulation within the vaginal canal due to imperforate hymen is called hematocolpos. - USMLE Q will give a teenage girl who's never had a menstrual period. Physical exam shows a "blueish bulge" behind the hymen. The Q might say the girl has had monthly pain (i.e., she has in fact had menses, but blood is blocked from leaving the vagina due to the imperforate hymen). - Answer is either hematocolpos or "cruciate incision of the hymen."

Polycystic ovarian syndrome (PCOS)
- High BMI female → insulin resistance. - Insulin resistance causes abnormal GnRH pulsation → leads to ↑ LH and ↓ FSH. This is often truncated as just saying there's an ↑ LH/FSH ratio. Some students think FSH is also ↑, but it's just the ratio that's ↑. That's wrong. FSH is low. - LH normally acts on the theca lutein cells of the ovaries to make androgens. Since LH is ↑, we get hirsutism. - FSH normally stimulates follicular development. Since FSH is ↓, we have poor follicular development, leading to failure of a Graafian follicle to rupture during ovulation. The unruptured follicle is retained as a follicular cyst. - Failure of ovulation is called anovulation. This term is exceedingly HY on USMLE. It presents as a female with irregular periods. - Anovulation is a broader term than PCOS, as it can be due to other conditions as well, such as hypothyroidism and Cushing syndrome. But when you hear the word "anovulation" alone, it is usually used as synonymous for the same mechanism as PCOS – i.e., high-BMI female who has irregular periods due to abnormal GnRH pulsation causing an ↑ LH/FSH ratio, with the ↓ FSH causing failure of follicular rupture. - So high-BMI female + irregular periods = anovulation till proven otherwise. - Anovulation + hirsutism = PCOS. - In clinical practice, part of the PCOS diagnosis requires ultrasounds showing 11+ cysts bilaterally (Amsterdam criteria). But USMLE doesn't assess this. You just need to know anovulation + hirsutism = PCOS. - Normally when a follicle ruptures, the remnant is called the corpus luteum, which secretes progesterone. - Progesterone inhibits growth of endometrium; estrogen stimulates growth of endometrium. - Women who have anovulation have ↓ progesterone production because they don't form a corpus luteum. - This means they have ↑ estrogen in comparison to progesterone. We call this **unopposed estrogen.** This is one of the highest yield phrases for USMLE. - Unopposed estrogen means ↑ risk of endometrial hyperplasia and, in turn, ↑ risk of endometrial carcinoma. - The Q can give you a high-BMI female who's post-menopausal + has vaginal bleeding. Answer is just straight-up endometrial biopsy. Student asks how we know it's endometrial cancer. My response is, if she's overweight, this implies she was probably overweight in the past, which implies she's had history of anovulatory cycles and endometrial hyperplasia, leading to ↑ endometrial cancer risk.

- Because insulin resistance is the basis for PCOS, patients are at ↑ risk of developing type II diabetes.
- As mentioned above, hypothyroidism and Cushing syndrome are also HY causes of anovulation on USMLE. - ↑ Glucocorticoids in Cushing syndrome can cause insulin resistance and anovulation, where the diagnosis can appear like PCOS. The difference is PCOS is idiopathic in response to high BMI – i.e., it is not caused by a known secondary etiology like Cushing syndrome, even though the presentations can be similar.

Primary dysmenorrhea vs endometriosis vs adenomyosis	
Primary dysmenorrhea	- Dysmenorrhea = period pain. - Primary dysmenorrhea = "normal period pain"; benign. - Mechanism is prostaglandin PGF2α hypersecretion. - During menstruation, the endometrial cells release high levels of prostaglandins, which cause uterine contractions and pain. The prostaglandins can also make their way into the systemic circulation, leading to headache and nausea. - This is the answer on USMLE for period pain under the age of 20. The Q can say, girl who is 17 has menstrual pain so bad she has to miss class + physical exam shows no abnormalities → answer = primary dysmenorrhea. - The answer can also be written simply as "prostaglandin secretion." - The key detail regarding primary dysmenorrhea is that **the physical exam is normal.** - In endometriosis, the physical exam is abnormal + is always over age 20. - Treatment is NSAID. OCP is wrong answer if listed alongside NSAID (on NBME).

Endometriosis	- Growth of endometrial tissue outside the uterus. Most common location is the ovary, but can also grow in other locations like the pouch of Douglas.
	- Mechanism is thought to be retrograde menstrual flow through the Fallopian tubes + seeding onto the ovaries or peritoneum.
	- Causes severe menstrual pain over age 20 + **an abnormal physical exam.**
	- The abnormal physical exam can be any miscellaneous finding – i.e., there is no specific finding you have to memorize. They can say findings such as nodularity of the uterosacral ligaments, a fixed retroverted uterus (due to adhesions from lesions), or pelvic tenderness.
	- In contrast, severe period pain + normal physical exam = primary dysmenorrhea.
	- In other words:
	    - 23F + period pain so bad she has to miss work + normal physical exam → answer = prostaglandin secretion / primary dysmenorrhea.
	    - 23F + period pain so bad she has to miss work + physical exam shows nodularity of the uterosacral ligaments → answer = endometriosis.
	- Descriptors such as pain with defecation (due to pouch of Douglas lesions) or dyspareunia (pain during sex) are highly buzzy and pass-level but too easy, so are often omitted from NBME questions, as per my observation.
	- USMLE wants diagnostic laparoscopy as next best step.
	- NSAIDs and OCPs are short-term measures, but definitive Tx is laparoscopic removal of lesions.

Adenomyosis	- Growth of endometrial tissue within the myometrium. - Presents as a diffusely enlarged uterus + vaginal bleeding in woman 30s-50s. - May or may not be painful. - Q will say woman had tubule ligation two years ago + now has uterus that is 8 weeks' gestation in size + vaginal bleeding → answer = adenomyosis. Student is confused and says "how can she be pregnant if she had tubule ligation?" She's not. Use your head. It's just how the Q can describe the ↑ size. - Tx = NSAIDs + OCPs. - Leuprolide can be used in theory, but I haven't seen it assessed.

Contraception for FM	
Barrier (condoms)	- Self-explanatory, but apart from abstinence, male condoms are #1 way to ↓ pregnancy and STD risk. - Recommended for use in women on isotretinoin, even if they are already on other contraceptive method. - OCPs ↑ risk of cervical cancer slightly, not as a direct effect, but because of ↓ barrier contraception use, where HPV exposure is ↑.
Combined oral contraceptive pills	- Contain both estrogen and progesterone. - Estrogen helps to ↓ breakthrough bleeding risk (i.e., metrorrhagia). - Estrogen ↓ probability of ovulation; progesterone ↓ penetration of sperm by ↑ thickening of cervical mucous. In some women, and depending on dose, progesterone can also ↓ ovulation. - Contraindicated in smokers >35 (HY on USMLE), migraine with aura, Hx of thrombotic disorders, or current breast/gynecologic cancer due to the estrogen-containing component. - ↓↓ Risk of ovarian cancer the most due to synchronization of cycles (asked on NBME). Also ↓ risk of endometrial (probably also due to synchronization). - ↑ risk of cervical cancer due to ↓ barrier contraception use (as mentioned above); some studies have suggested slight ↑ in breast cancer risk, albeit without significance.
Progestin-only pills	- Aka "mini-pill"; only contain progesterone analogue. - Thicken cervical mucous. - Must be taken precisely at same time every day, so require female is well-adherent / compliant.
Copper IUD	- Copper intrauterine device; releases (you wouldn't have guessed it) copper ions into the uterine cavity, creating an environment that is toxic to sperm + inhibits their motility, thereby preventing fertilization. - Known for being one of the most effective forms of emergency contraception when inserted within five days post-coitally and can provide up to 10 years of contraceptive protection. - IUDs have risk of migration through uterine wall.
Levonorgestrel IUD	- Release progesterone analogue; usually last 3-7 years and are ideal in women who desire longer-term contraception or have poor medication compliance. - Thicken cervical mucous. - IUDs have risk of migration through uterine wall.
"Depo shot"	- Depot medroxyprogesterone acetate injection form of contraception. - Administered every 3 months. - Thicken cervical mucous. - Known to ↓ bone mineral density; rebounds with discontinuation. - Known to cause erratic bleeding in 1/3, no changes in cycle bleeding in 1/3, and complete amenorrhea is 1/3. - Usually a distractor/wrong answer on USMLE. They'll often give a woman who wants a more reliable form on contraception because she forgets to take pills, where the Depo shot seems like reasonable answer in comparison, but IUD will be the answer (because it's even better).
Implantable rod	- Aka Implanon/Nexplanon. - Releases progesterone analogue; thickens cervical mucous; lasts 3 years.

Hormone replacement therapy (HRT)
- Estrogen combined with progesterone; only approved for severe perimenopausal vasomotor symptoms (i.e., hot flashes, urge incontinence, atrophic vaginitis). - Not given to help preserve bone density or for positive role on mood and neurocognition, since ↑ absolute estrogen exposure in women ↑ risk of breast cancer, MI, and thromboembolic events (i.e., DVT, PE, stroke). The latter is because estrogen upregulates fibrinogen and factors V and VIII. - Can cause unopposed estrogen and ↑ endometrial cancer risk only if the woman inadvertently stops taking the progesterone component (i.e., endometrial hyperplasia → ↑ endometrial adenocarcinoma risk). - Estrogen alone can considered only if the woman has Hx of hysterectomy (i.e., no ↑ endometrial adenocarcinoma risk). - **Can cause ↓ libido due to ↓ endogenous androgen production** (i.e., ↑ negative-feedback at hypothalamus / anterior pituitary → ↓ GnRH → ↓ LH and FSH).

Repro/obgyn drugs for FM	
- Not meant to be exhaustive list; just HY ones relevant to Family Med.	
Aromatase inhibitors	- ↓ Conversion of androgens to estrogens. - Anastrozole and exemestane are used for breast cancer in post-menopausal women.
BPH meds	- Finasteride is 5α reductase inhibitor used for BPH. ↓ conversion of testosterone to DHT. Since DHT causes prostatic growth, ↓ DHT means ↓ prostatic growth. - Tamsulosin/terazosin are α1 blockers that ↓ constriction of internal urethral sphincter of the bladder and help promote urinary outflow.
Clomiphene	- Selective-estrogen receptor modulator (SERM) used to stimulate ovulation in those with irregular cycles (in particular those PCOS). - Has partial-agonist effects at the hypothalamus that are weaker than endogenous estrogen, so the hypothalamus interprets this as ↓ estrogen is present, so negative-feedback also ↓ → GnRH ↑.
Minoxidil	- Arteriolar dilator that promotes hair growth in androgenetic alopecia. - USMLE wants you to know the latter is polygenic and risk is ↑ with anabolic steroids.
SERMs	- SERMs are agents that have different effects depending on the tissue. - Tamoxifen and raloxifene are SERMs used for estrogen-receptor (+) breast cancer, where they both are antagonistic at breast and agonistic at bone. - Tamoxifen causes ↑ risk of endometrial cancer due to agonistic effects at endometrium.

FM Derm

HY Derm conditions for Family Med	
Tinea capitis	 - Cause is dermatophytes (i.e., *Microsporum*; *Trichophyton*). - Classically circular, scaly area of scaly alopecia. - Tx = oral griseofulvin for patient only; "patient + classmates" is wrong answer on NBME. - Another NBME Q asks how to prevent; answer = "avoidance of sharing of hats"; "use of medicated shampoo" is wrong answer.
Seborrheic dermatitis	 - Inflammatory response to over-colonization with *Malassezia* yeast. - Presents as itchy scalp with flakes. If severe, presents as "weeping papules" / scaling of the hairline. - More common in adults. - Don't confuse with tinea capitis above, which is a circular area of scaling alopecia more common in children. - Tx = Topical selenium or ketoconazole shampoo;); cause. - Can sometimes occur in HIV patients; sudden onset in high-risk group → answer = do HIV test.
Tinea corporis	- Aka ringworm. - Q will often mention a dog at home. - Tx = topical miconazole or clotrimazole. - Tx for tinea pedis (athlete's foot) is topical terbinafine or topical -azole.

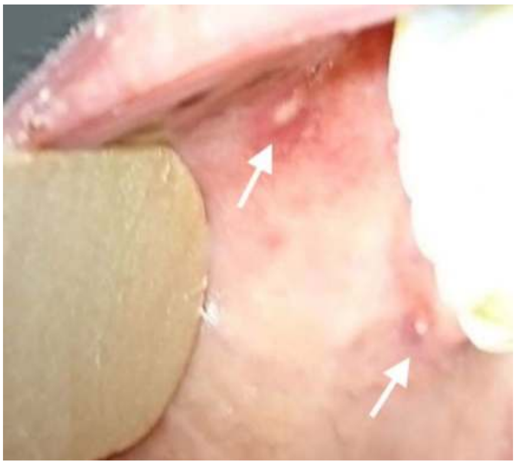
	
Lymphangitis	 - Above image shows sporotrichosis (*Sporotrix schenckii*), but similar pattern can also be seen for miscellaneous burns or puncture wounds of the hand that cause ascending infection toward the axilla. - Students early in their prep know that *Sporothrix* is classically a papule on the finger caused by rose thorns; this presentation is usually too easy for real USMLE though. - Treatment for sporotrichosis is oral itraconazole. - USMLE can show image similar to the above, where they list answers like phlebitis, arteritis, lymphangitis, etc.; answer is just "lymphangitis."
Cellulitis	 - *Staph aureus* exceeds *Strep pyogenes* (Group A Strep) as causal organism; must give beta-lactamase-resistant beta-lactam in the methicillin class (i.e., dicloxacillin, flucloxacillin) or first-generation cephalosporin (i.e., cephalexin, cephazolin), or Augmentin (amoxicillin-**clavulanate**). - Amoxicillin and penicillin alone are wrong answers.

<table>
<tr><td></td><td>- 90% of community Staph (i.e., MSSA) produces beta-lactamase, so amoxicillin and penicillin alone will not work if Staph is the cause.</td></tr>
<tr><td>Erysipelas</td><td>
- Infection of upper dermis and superficial lymphatics.
- Caused by Strep pyogenes (Group A Strep), which eclipses Staph aureus for erysipelas.
- Looks worse than cellulitis but is more superficial / “not as bad”; has characteristic “fiery red” appearance and may appear well-demarcated with raised edges.
- Although Group A Strep > Staph for erysipelas, Tx is same as cellulitis (oral dicloxacillin, cephalexin, or Augmentin) because Staph can still cause it. Penicillin alone can be used for Strep pharyngitis.</td></tr>
<tr><td>Acne</td><td>
- Caused by Propionibacterium acnes; not difficult, but I’ve seen enough students select tinea faciei (fungal infection of face) for simple acne.
- First-line Tx for acne on USMLE is topical retinoids (i.e., topical tretinoin; not oral isotretinoin; latter is only for severe acne). Topical retinoids (vitamin A) inhibit sebum production; they cause photosensitivity and desquamation (peeling).
- Topical benzoyl peroxide is second-line for acne (although often co-administered with topical tretinoin). It clears pores and kills bacteria.
- Topical clindamycin can be used if topical retinoids and benzoyl peroxide are insufficient; if topical antibiotic is insufficient, oral tetracycline is used; the latter causes blistering photosensitivity.
- Last resort is oral isotretinoin; must do beta-hCG (pregnancy test) before commencement due to teratogenicity; oral isotretinoin does not cause problems with sperm in men; topical retinoids in both men and women do not cause teratogenicity.</td></tr>
</table>

	 - Above image shows the HY photosensitivity due to topical retinoids; answer = "avoidance of sun exposure"; tetracycline photosensitivity, in contrast, tends to be blistering; do not choose answers such as "avoidance of spicy/sweet foods" for acne questions.
Tinea faciei	 - Fungal infection of face. - Tx = topical -azoles (clotrimazole, miconazole).
Meningococcus	 - *Neisseria meningitidis;* gram-negative diplococcus. - Causes meningitis + characteristic non-blanching rash. - Low BP can be endotoxic shock, but student should bear in mind Waterhouse-Friderichsen syndrome is often asked; give hydrocortisone to increase BP after normal saline is administered.

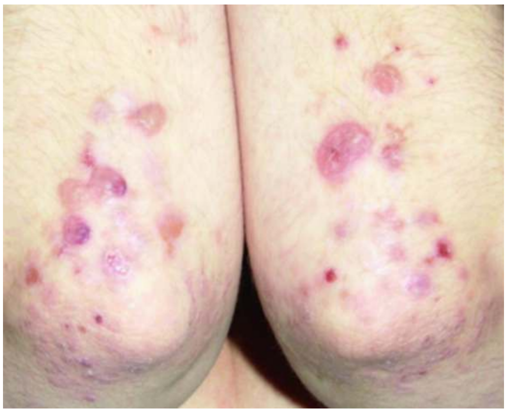
Herpetic whitlow	 - One of the highest-yield spot-diagnoses on USMLE. - HSV1/2 infection of the finger. - In children, can be acquired via touching mother's cold sore (e.g., during breastfeeding). - In adults, classically dental workers. - Tx = topical and oral acyclovir.
Lyme disease	 - *Borellia burgdorferi;* a spirochete. - Spirochete; spiral/corkscrew-shaped. - Causes lyme disease; spread by *Ixodes* tick (same as *Ehrlichia, Babesia*, and *Anaplasma*). - Primary Lyme causes a classic target rash known as erythema chronicum migrans, but a HY point is that the rash need not be a target on USMLE. It can merely be circular with no clearing, but the target is classic. - Bells palsy can also be seen in primary Lyme. What the USMLE will do is give two side by side images: 1) circular rash on limb that is not a target; 2) Bells palsy, where the student needs to infer this is Lyme disease even though rash isn't a target, since Bells palsy is HY for Lyme. - Secondary Lyme tends to cause arthritis. Some sources say Bell's palsy is part of secondary Lyme (occurs at least one month after initial infection), but I've seen USMLE give it as part of initial/primary infection. - Tertiary Lyme can cause CNS and/or heart problems. - Treatment is doxycycline for most cases of Lyme. - Ceftriaxone is given for advanced Lyme involving the CNS or heart. - For children <8 and pregnant women, give amoxicillin in place of doxycyline. - There is an NBME Q of a pregnant woman with non-disseminated Lyme, where ceftriaxone is correct over doxycycline, and amoxicillin isn't listed. In other words, if

	USMLE doesn't want doxy as the answer, they will not play trivia as to whether it's ceftriaxone or amoxicillin to be used as the alternative. But you could be aware that, in theory, ceftriaxone is harder-hitting and preferred if cases are more severe.
Scabies	 - *Sarcoptes scabiei.* - Lesions classically described as "linear burrows". - Can become superinfected with *Staph aureus.* - Tx = topical permethrin. - Disseminated scabies can occur in HIV patients; Tx is oral ivermectin. - Topical permethrin is also treatment for pediculosis (lice).
Bed bugs	 - Caused by an insect called *Cimex*. - Grows as clusters of itchy lesions on the limbs or torso.
Cutaneous anthrax	 - Caused by *Bacillus anthracis.* - Classically causes a black "eschar." - Seen in postal workers + farmers. - Pulmonary anthrax leads to hemorrhagic mediastinitis. - Tx = ciprofloxacin.

Tularemia	 - Caused by *Francisella tularensis.* - cutaneous tularemia can present as ulcerative lesions; can also cause bilateral atypical pneumonia; rabbits are classic source. - Tx = doxycycline.
Hot tub folliculitis	 - Rash caused by *Psuedomonas* from water sources. - Increased risk in diabetics.
Measles	 - Aka rubeola; causes a head-to-toe macular popular rash. - Image shows Koplik spots (pathognomonic whiteish lesions on buccal mucosa). - Immigrant Hx on USMLE sometimes implies unvaccinated status. - Can rarely cause subacute sclerosing panencephalitis (reactivation of latent infection in the CNS in teenagers).

<table>
<tr><td></td><td>- Rubella also causes a head-to-toe macular popular rash, but rather than Koplik spots, suboccipital and post-auricular lymphadenopathy (tenderness on back of head and behind ears) is characteristic.</td></tr>
<tr><td>Mumps</td><td>
- Causes POM → Parotitis, Orchitis, Meningitis.
- Doesn’t typically cause rash.</td></tr>
<tr><td>Parvovirus B19</td><td>
- Fifth disease = “slapped cheek” facial erythema caused by Parvo B19 in Peds.
- Next best step = check serum IgM titers; if IgM titers are not listed, choose bone marrow biopsy.
- Can cause pure-RBC aplasia (i.e., only RBCs are low) or full-blown aplastic anemia (where all cell lines - RBCs, WBCs, and platelets - are down). There is increased risk of pure-RBC aplasia and aplastic anemia in sickle cell.
- Once child has developed the red cheeks, he/she has immunologically cleared the illness (i.e., if they turn it into a behavioral science Q, tell parents to chill the fuck out / relax because the child has cleared the virus).

- Can cause body rash +/- arthritis in adults; same as with Fifth disease, next best step is check serum IgM titers.</td></tr>
</table>

Roseola	 - Caused by HHV6; described as "spiking fever followed by a rash" – i.e., child will have high fever for 2-3 days, followed by a rash.
Eczema herpeticum	 - HSV1/2 infection superimposed on eczema. - Often self-inoculated from touching cold sore then cracked skin of eczema. - Herpes infection can present with fever, lymphadenopathy, and tingling, burning, and painful rash (herpetic neuralgia). - Tx = acyclovir.
Dermatitis herpetiformis	 - Extra-intestinal cutaneous eruption that can occur in patients with Celiac disease. - Despite the name, this is not actual herpes. It is IgA deposition at dermal papillae. - Don't confuse with eczema herpeticum.

Hidradenitis suppurativa	- Skin condition characterized by painful abscesses and sinus tracts, most commonly in the groin, axilla, and under the breasts; frequently affects apocrine glands. - Cause is a combination of genetic and environmental factors (idiopathic). - NBME answer is "surgical excision of lesions." 
Pediatric shingles	 - You need to know that pediatric shingles "is a thing." - Classically seen in immunocompromised children (e.g., due to chemotherapy). - Shingles is aka "herpes zoster" and is caused by varicella zoster virus (VZV); herpes zoster is **not** a virus name; herpes zoster literally is just another name for shingles, which is caused by VZV (human herpes virus 3; HHV3), not HSV1/2 (HHV1/2).
Herpes zoster oticus	 - VZV (shingles) causing periaural lesions. - Can concurrently affect CN VII leading to Bells palsy (Ramsay-Hunt syndrome II).
Kaposi sarcoma	

	 - Caused by HHV 8 (Kaposi sarcoma-associated herpes virus) in immunocompromised patients (i.e., AIDS, chemotherapy). - Kaposi sarcoma are violaceous, tumorous lesions of vascular-lymphatic origin. - Q can show you image such as above and then the answer is just "anti-neoplastic" for the Tx (whereas answers like anti-fungal, anti-bacterial, etc., are wrong).
Pityriasis rosea	 - Caused by HHV6 or 7; self-limiting. - Starts as Herald patch (larger pink ellipse above), usually on the back or trunk, then spreads upward onto the shoulder blades ("Christmas tree distribution"); USMLE will show you image and expect you can make spot-diagnosis. - Can occur in teenagers, although more common in the 20s.
Tinea versicolor	 - Caused by *Malassezia furfur*; fungus causes spotty hypopigmentation; patient usually lives in sub-tropical areas like Florida. - Can occur in teenagers who surf / go to the beach. - Tx = topical selenium.

Molloscum contagiosum	 - Very HY spot-diagnosis for Peds; caused by Poxvirus. - Classically skin-colored or reddish papules with central umbilication. - USMLE likes giving vignette where kid went to a recent pool party.
Mycobacterium marinum	 - Unusual diagnosis but asked. - Vignette will say kid recently went to aquarium or water park + now has red lesions on the hand/arm. - Can occur in adults as well who work at aquariums or water parks.
Leishmaniasis	 - *Leishmania donovani*; protozoan; unicellular eukaryote; spread by sandfly bite.

	- Can cause skin ulcer; USMLE likes Middle East as endemic (I've seen Iraq on NBME). - Visceral disease known as kala azar, which causes ↑ LFTs and pancytopenia.
MAI	 - *Mycobacterium avium intracellulare* (MAI); can cause lymphadenitis with classic reddish/violaceous lesion on the neck or pre-auricularly.
Secondary syphilis	 - *Treponema pallidum* (spirochete); image shows maculopapular/nodular body rash; palms and soles are classically affected in 2ndary syphilis but Q need not mention it. - Diagnosed with serology; VDRL/RPR ordered before FTA, but latter more specific. - VDRL can be falsely positive in patients with antiphospholipid syndrome (usually SLE patients with lupus anticoagulant). - USMLE Q can also show you picture of condylomata lata (wart-like lesions on the genitals) + tell you there's palms/soles rash, then answer = *Treponema pallidum*. Tx = penicillin.
Arsenic poisoning	

	- Mees lines are white lines seen on fingernails in arsenic toxicity. - Arsenic is present in small amounts in fertilizers, which causes plants to flourish; gardeners at increased risk. - Arsenic can also cause palms/soles rash (arsenical keratosis).
Eczema	 - Type I (immediate); can occur as part of atopy (i.e., asthma in winter, hay fever / rhinoconjunctivitis in spring; eczema in summer). - Remember that 1/3 of asthma patients have cough-variant asthma (i.e., only have dry cough, usually worse in the winter or with exercise). - Treat with oil-based emollient and topical corticosteroids; if steroids used >5-7 days continuously, thinning of the dermis may occur.
Urticaria	 - Aka hives; type I hypersensitivity; can be precipitated by various allergens, including pollen, pet dander, and drugs.
Erythema multiforme	

	- Immune complex-mediated rash (type III hypersensitivity) with many etiologies; HSV1/2 infection is classic cause.
Erythema nodosum	 - Panniculitis (inflammation of subcutaneous fat) that resembles a rash. - Seen with autoimmune diseases like sarcoidosis and Crohn, as well as part of serum sickness due to medications (e.g., sulfa).
Dermatomyositis	 - Image shows heliotrope rash (left) and shawl rash (right) seen in dermatomyositis. - Anti-Jo1 antibodies and electromyography + nerve conduction studies = diagnostic. - Do not confuse heliotrope rash of dermatomyositis with malar rash of SLE. - Give steroids for flares.  - Patients can also get Gottron papules (above image) and “mechanics’ hands” (i.e., rough-surfaced / scaly hands).

Vitiligo	 - T cell-mediated destruction of melanocytes. - Relevance for Family Med is that it can be associated with other autoimmune polyglandular syndromes, as well as IgA deficiency. - For example, B12 deficiency + vitiligo = possible pernicious anemia; or weight gain and fatigue + vitiligo = possible Hashimoto. Etc.
Lentigo	- "Age spot." - ↑ melanocyte #; no change melanin production. - Common on zygoma due to sunlight; not malignant. 
Latex allergy	 - Q will give you a nurse or laboratory technician. The implication is that he/she has been using gloves resulting in contact dermatitis due to latex. - Can also occur due to adhesives (i.e., from bandages applied).

Rosacea	 - Idiopathic patchy facial erythema that worsens with spicy foods and alcohol; slight pain of rash may occur in cold weather. - Occurs usually in middle-age and older.
Chilblains	 - Aka perniosis; painful inflammation of distal capillaries due to repeated exposure to cold air, followed by immersion in hot water (i.e., from bath/shower). - This is different from frostnip and frostbite. Frostnip is cold-exposed skin (effects quickly reversible; no skin damage); frostbite is more severe and can result in damage such as blistering and necrosis. - Chilblains can be confused with tinea pedis, since they can be extremely itchy. The patient might self-attempt topical antifungal sprays/foams to no avail. - Tx = keep feet warm in winter; avoid abrupt temperature changes.
Aphthous ulcer	 - Aka aphthous stomatitis; no Tx necessary; **not** related to HSV1/2; etiology is multifactorial; can be precipitated by allergens such as spice; T cell-mediated.

Lichen planus	 - Classically described as "The Ps" → purple, pruritic, polygonal papules; however Qs need not mention they're pruritic; you just need to know hepatitis C + red/purple skin lesions = lichen planus.
Aplasia cutis congenita	 - Absence of skin on an area of scalp; can be caused by teratogens such as methimazole; dumb/seemingly pedantic detail, I know, but it shows up on real deal.
Hereditary angioedema	 - Deficiency of C1 esterase *inhibitor* (not C1 esterase alone). - Characterized by recurrent swelling of various bodily regions. - Tx = Danazol (androgen receptor partial agonist causes liver to produce more C1 esterase inhibitor).

NF1	 - Neurofibromatosis type I; autosomal dominant; one of the phakomatoses (neurocutaneous disorders → NF1/2, VHL, TSC, Sturge-Weber). - Image shows café au lait spot. - NF1 → café au lait spots, neurofibromas, axillary/groin freckling, optic glioma, pheochromocytoma.
Tuberous sclerosis	- Autosomal dominant; one of the phakomatoses. - Image shows adenoma sebaceum (angiofibromas). - Intracranial/periventricular nodules ("tubers" – i.e,. hamartomas), adenoma sebaceum (angiofibromas; pictured below), cardiac rhabdomyoma, lymphangeoleiomyomatosis, subungual fibromas, renal angiomyolipoma. 
Sturge-Weber	 - Classically Port Wine-stain birthmark (nevus flammeus), but can be described as "violaceous papules in a temporal distribution." - Associated with leptomeningeal angioma (causing seizure) and glaucoma. - Condition is not inherited and is due to somatic mosaicism.

McCune-Albright syndrome	 - Triad of “coast of Maine” café au lait spots (above image), polyostotic fibrous dysplasia (bone is replaced by fibrous tissue), and endocrine hypersecretion (classically precocious puberty).
Acanthosis nigricans	 - Brown/black velvety skin, usually along nape of neck. - Associated with hyperinsulinemia and type II diabetes almost always. - Can also very rarely be seen in patients with underlying, visceral malignancy.
Basal cell carcinoma (BCC)	 - Classically pearlescent / slightly translucent; talengiectasias common; borders can sometimes be described as “heaped up” or “rolled”; may or may not be ulcerated. - Treatment for skin cancers on cosmetically sensitive areas such as the eyelid or nose can be managed with Mohs micrographic surgery (on NBME).

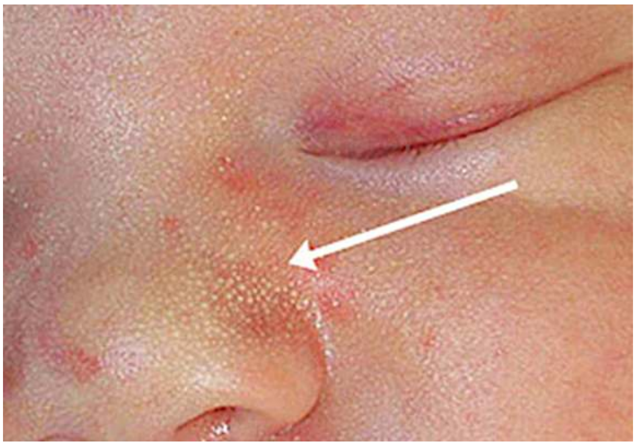
Actinic keratoses	  - Aka solar keratoses; precursor to squamous cell carcinoma (SCC); classically described as red/scaly lesions on forehead, ear, or arms of fisherman, farmers, or construction workers. Cryotherapy is usual treatment.
Squamous cell carcinoma (SCC)	- Ulcerated lesion in the setting of a patient who's had actinic keratoses – i.e., Q says patient has had scaly red areas on forehead for many years + now has recent ulceration → answer = SCC. - SCC will not have telangiectasias or pearlescence/translucency (as with BCC). 
Keratoacanthoma	 - Can be confused with SCC; described as dome-shaped with a hyperkeratotic core, surrounded by a wall of inflamed skin. - Tx is surgical excision.

Melanoma	 - Depth is most important for prognosis. - Answer on NBME = excisional biopsy; suspected malignant lesions in non-cosmetically sensitive areas (i.e., not on the head/neck) can simply be excised with narrow margins; excision of additional tissue is important if margins are positive (i.e., entire lesion wasn't excised). - "Full-thickness biopsy" answer on USMLE for suspected melanoma on back of neck (NBME Q gives lesion similar to above on neck, with full-thickness biopsy as answer). Excision of lesion is management for neck lesions if full-thickness biopsy confirms melanoma; do Mohs micrographic surgery for facial lesions. - Punch biopsy is a type of full-thickness biopsy; if Q asks, choose side of lesion for biopsy, rather than middle of lesion. - Students will sometimes ask about shave biopsy. I have never seen this assessed on NBME, but literature says it can sometimes be used to remove superficial non-pigmented lesions where the clinician does not suspect melanoma (performing shave biopsy on melanoma can create problems for assessing depth, prognosis, and therapy).  - Acral lentiginous melanoma occurs on the palms/soles (areas not usually exposed to sun); more common in persons of African and Asian descent.  - Can also occur under the finger nail.

Seborrheic keratoses	 - Sun exposure and old age are biggest risk factors; also seen in smokers. - Described as “greasy,” or waxy, skin growths that appear like they can be “peeled off”; not malignant. - Sign of Leser-Trélat = sudden, eruptive seborrheic keratoses secondary to underlying, visceral malignancy (i.e., unrelated to age or sun).
Blue nevus	- Aka Mongolian spot. - Type of benign birthmark where dermal melanocytes fail to migrate superficially to stratum basale. - More common in Asians and blacks. - Often mistaken for child abuse. - Schedule routine follow-up; wrong answer is contacting child protective services. 
Strawberry hemangioma	 - Benign capillary tumor that will grow slightly then regress spontaneously within a few years. - No Tx necessary unless causing functional impairment. - Don’t confuse with cherry hemangioma, which is due to sun exposure in elderly and presents as a tiny cherry-colored papule.

Cherry hemangioma	 - Benign capillary tumors caused by sunlight; benign; greater prevalence as age ↑. Students confuse with strawberry hemangioma in peds.
Preauricular pit	 - Preauricular pits are benign finding, usually not associated with any congenital disorder. - No Tx necessary. Can become infected; do surgical closure if frequent infections.
Kasabach-Merritt syndrome	 - Aka infantile hemangioma with thrombocytopenia; asked several times on 2CK Peds assessment. - Child will have thrombocytopenia due to platelet sequestration within the lesion. - Large and deforming; don't confuse with strawberry hemangioma. - Tx is surgical.

Cushing syndrome	 - USMLE wants you to know purple striae are HY physical exam finding. - Weakening of dermal collagen + capillary walls → micro bleeding into skin.
Cigarette burn	 - Q will show you image similar to above and then the answer will be "contact child protect services"; can also be on face / resemble impetigo if at different stages of healing.
Keloids	- Disorganized growth of dermal collagen. - Tx is surgical excision, although recurrence is common. Benign. 

Pseudofolliculitis barbae	 - Aka razor bumps; increased prevalence in African descent; curly beard hair grows back into the skin; Tx is to allow the beard to grow; USMLE Q will show you image and just ask the spot-Dx.
Milia	 - Aka “milk spots”; clogged eccrine ducts; common, benign finding in babies. - No Tx necessary; - “Exfoliative cleanser” and “topical low-dose corticosteroid” = wrong answers.
Varicose veins	- Q will ask next best step in diagnosis → answer = duplex venous ultrasonography for venous disease / valvular insufficiency. - Tx is compression stockings. - Do not choose surgical interventions for varicose veins on USMLE. 

Superficial thrombophlebitis	 - Similar to DVT, but in more superficial vein; presents as painful, warm, "palpable cord" at the ankle that may or may not track up to the knee. - Treated with subcutaneous enoxaparin (heparin); wrong answer is compression stockings; very difficult Q since compression stockings are common answer for venous disease, but if patient has active venous occlusion (i.e., DVT or superficial thrombophlebitis), give heparin. - In other words, for simple venous disease / varicose veins, choose compression stockings, but if the patient has an active clot presenting as a palpable vein with pain, give heparin instead.
Buerger disease	 - Aka thromboangiitis obliterans; condition is digital gangrene in males who are heavy smokers; contrasts with gangrene due to diabetes, which is pedal. - Tx = smoking cessation.
Charcot joint	- Aka neurogenic joint; mechanism is "lack of appropriate joint sensation" – i.e., patient can feel his or her feet due to neuropathy → damage → can't heal the area due to vasculopathy. - Q will be diabetic with poor HbA1c + foot ulcer + x-ray of the shows disorganization of the tarsals/metatarsals.

	
Cryoglobulinemia	- Cryoglobulins are immune complexes that precipitate at cold temperatures.  - Often presents as livedo reticularis, which is a mottled, reticulated vascular pattern that has many etiologies; associated with hepatitis C. - Cryoglobulinemia is associated with decreased serum complement protein C4 (in contrast, C3 is sometimes down in SLE flares and Group A Strep infections).
SVC syndrome	 - Image shows Pemberton sign (facial erythema due to diminished venous return when arms are raised above head). - Seen classically in Pancoast (i.e., apical) tumors of the lung.

<table>
<tr><td>Actinic purpura</td><td>
- Aka senile or solar purpura.
- Flat, purple discolorations caused by damage to dermal collagen and blood vessels due to chronic sun exposure.
- Commonly appear on the back of the hands and forearms in elderly.
- Benign; may be mistaken for bruises.
- NBME Q describes actinic purpura and then asks for mechanism. The answer is "solar elastosis," which is a term that refers to degenerative changes to the dermis due to chronic sun exposure.</td></tr>
<tr><td>Keratosis pilaris</td><td>- Benign bumps + rough patches on the skin due to build-up of excess keratin.
- The excess Keratin can block hair follicles and pores, causing small, hard bumps.
- No treatment necessary. Emollients may be used.
- Not obscure condition I'm mentioning for fun. Shows up on NBME.
</td></tr>
</table>

FM Neuro

Stroke risk factors
- The two most important risk factors for stroke (cerebral infarction) on USMLE are hypertension and atrial fibrillation. - Regarding HTN, a strong systolic impulse pounds the carotid arteries, leading to endothelial damage and increased development of atherosclerotic plaques (carotid stenosis), which then launch off to the brain/eye. - Atrial fibrillation results in turbulence and stasis, leading to left atrial mural thrombus, which launches off. - Hypertension is most common in the population for causing stroke. Blood pressure control is more important than smoking cessation for decreasing stroke risk. - If the USMLE Q gives you a patient who has both AF and HTN, they want AF as the most important risk factor. In other words, even though HTN is the most common risk factor in the population for stroke, AF is still more likely to cause stroke if the patient has both. - There are other details regarding stroke I will discuss in this PDF, but for starters, these risk factors are pass-level points for understanding how/why most strokes occur – i.e., "How do most strokes occur?" → "Oh, well the patient will usually have carotid stenosis from HTN, where an atheromatous plaque has launched off, or will have AF where an LA mural thrombus has launched off." Simple. - I discuss carotid stenosis and AF management for stroke prevention in detail in the HY Cardio PDF.

Circle of Willis HY strokes	
Affected vessel	**HY Points**
ACA	- Anterior cerebral artery. - Motor/sensory abnormalities of contralateral leg.
MCA	- Middle cerebral artery. - Motor/sensory abnormalities of contralateral arm + face. - Dominant MCA stroke (usually left) can lead to Wernicke or Broca aphasia (discussed below). - Non-dominant MCA stroke (usually right) can cause hemispatial neglect (inability to draw clockface).
PCA	- Posterior cerebral artery. - Contralateral homonymous hemianopsia with macular sparing. - Prosopagnosia (inability to recognize faces).

Other stroke syndromes	
Lateral medullary syndrome	- Aka Wallenberg syndrome. - The answer for **dysphagia + ipsilateral Horner syndrome** after a stroke. - Due to stroke of posterior inferior cerebellar artery (PICA) or vertebral artery. - PICAchew (Pikachu) → PICA stroke causes dysphagia. - Students might know Horner syndrome (ipsilateral miosis, partial ptosis, anhidrosis) can be caused by Pancoast tumor, but another HY cause is lateral medullary syndrome.
Medial medullary syndrome	- The answer for **ipsilateral tongue deviation** after a stroke. - Due to stroke of anterior spinal artery.
Lateral pontine syndrome	- The answer for **ipsilateral Bells palsy** after a stroke. - Due to stroke of anterior inferior cerebellar artery (AICA). - FACIAL (contains AICA backwards) → "That's the one that's Bells palsy."
Weber syndrome	- Midbrain stroke. - The answer for **ipsilateral CN III palsy** (i.e., down and out eye) + contralateral spastic paraparesis (weakness).
Locked-in syndrome	- Basilar artery stroke. - The answer for **inability to move entire body** except for eyes.

Gerstmann syndrome	- Stroke of angular gyrus of parietal lobe. - Tetrad of 1) agraphia (inability to write); 2) acalculia (cannot do math); 3) finger agnosia (can't identify fingers); 4) left-right disassociation (cannot differentiate between left and right sides of body).
Hemiballismus	- Stroke of subthalamic nucleus. - Causes "ballistic" flailing of contralateral arm and/or leg. - Weird condition, but for some reason asked. For example, if they say patient has random flailing of left arm following a stroke, answer = right subthalamic nucleus.

HY Aphasia types	
Wernicke	- Fluent aphasia (i.e., patient can speak with normal pace + use lots of words), but none of it makes sense ("word salad"). - Ability to comprehend is similarly impaired (i.e., patient makes no sense in terms of what comes out + inability to make any sense of what comes in). - Repetition is impaired (i.e., patient cannot repeat back a sentence spoken to him/her). - Wernicke area is located in the temporal lobe; caused by L-sided MCA infarct.
Broca	- Non-fluent aphasia (i.e., patient has "telegraphic speech"), where there is frustration in not being able to communicate despite comprehending normally, akin to trying to communicate in a second language. - Repetition impaired. - Broca area is located in frontal lobe; caused by L-sided MCA infarct.
Conductive	- Patient only has repetition impaired. - Stroke of arcuate fasciculus, which is bundle of nerve fibers that connects Wernicke and Broca areas.
Global	- Presentation akin to having Broca and Wernicke aphasias at the same time, in addition to repetition being impaired. - Stroke of Broca + Wernicke areas + arcuate fasciculus.
Transcortical sensory	- Presentation same as Wernicke but repetition intact.
Transcortical motor	- Presentation same as Broca, but repetition intact.

Pseudotumor cerebri
- Term that refers to increased intracranial pressure in the setting of no clear structural cause. - Some notable causes are obesity (idiopathic intracranial hypertension), OCPs, isotretinoin, and danazol.

Reye syndrome
- Cerebral edema and hepatotoxicity seen with administration of aspirin to children (and sometimes adolescents) during a viral infection. - Thought to be due to impairment of beta-oxidation, but USMLE doesn't care.

Pass-level spinal tract functions	
Spinothalamic tract	- **Pain and temperature sensation from contralateral body.** - Cross-over (decussation) point is in spinal cord, meaning if, e.g., the left spinothalamic tract in the spinal cord is fucked up, we lose pain and temperature on the right side of the body below the lesion.

	- Decussation point is called anterior white commissure. A lesion at this point is called syringomyelia, where there is bilateral loss of pain and temperature below the lesion.
Corticospinal tract	- **Motor function from ipsilateral body.** - In other words, if left-sided corticospinal tract is fucked up, we lose motor function on the left side below the lesion.
Dorsal columns	- **Carry vibration + proprioception from ipsilateral body.** - Tabes dorsalis is dorsal column lesion caused by neurosyphilis. - If the dorsal columns are disrupted, patient will have (+) Romberg test, which is where he/she falls over when standing with eyes closed. I discuss this more below.

Upper- vs lower motor neuron findings	
UMN	- Lesion of the corticospinal tract (i.e., lesion from the cerebral cortex down until the anterior horn in the spinal cord). - Present as hyperreflexia, hypertonia, clonus, Babinski sign.
LMN	- Lesion involving the motor neuron from the anterior horn until the muscle itself. If the USMLE gives you an anterior horn lesion, this will be LMN, not UMN. - Present as hyporeflexia, hypotonia/flaccidity, fasciculations.

HY Spinal conditions	
Tabes dorsalis	- Loss of dorsal columns due to neurosyphilis. - Bilateral loss of vibration + proprioception. - Ataxia with (+) Romberg test → indicates dorsal columns are fucked up as the cause of the ataxia, since proprioception is lost. In contrast, if the cause of ataxia is cerebellar, for instance, Romberg test is (-).
Syringomyelia	- 9/10 Qs will give bilateral loss of pain and temperature due to lesion of anterior white commissure (decussation point for spinothalamic tract). - Maybe 1/3 of Qs will also involve corticospinal tract, where you'll see some impaired motor function in addition to the bilateral loss of pain/temperature. - 1/10 Qs will not mention pain/temperature loss but instead just give impaired motor function, where it sounds nothing like syringomyelia, but you're forced to eliminate to get there (on a 2CK Neuro CMS form).
Brown-Sequard syndrome	- Hemisection of spinal cord. - Ipsilateral loss of vibration + proprioception (dorsal columns). - Ipsilateral loss of motor function (corticospinal tract). - Contralateral loss of pain + temperature (spinothalamic). - Not typically caused by stab wound where half the spinal cord is perfectly transected. It's to my observation on NBME forms (especially 2CK Neuro CMS) that they like viral infection or SLE causing transverse myelitis.
Transverse myelitis	- Inflammation of spinal cord resulting in variable deficits. - As mentioned above, USMLE likes viral infection and SLE (autoimmune). - Brown-Sequard syndrome is one way transverse myelitis presents.
Central cord syndrome	- The answer when weakness in the upper limbs is greater than weakness in the lower limbs. - Typically follows hyperextension injury of the neck during whiplash in a rear-end MVA.
Anterior cord syndrome	- The answer when there's preservation of vibration/proprioception but loss of motor + pain/temperature, classically in patient with arterial disease. - In other words, everything but the dorsal columns is fucked up. Essentially the opposite of tabes dorsalis.

Mononeuritis multiplex	
- Weird neuropathy that means neuropathy of "one large nerve in many locations" – e.g., wrist drop + foot drop in same patient. - Presents in patients with one of the vasculitides (i.e., Wegener, Churg-Strauss, microscopic polyangiitis). - In other words, you'll get a vignette of Wegener, and then you're like, "Wait, why does he/she have wrist drop though?" That's just mononeuritis multiplex.	
Granulomatosis with polyangiitis	- Formerly known as Wegener granulomatosis. - Answer on USMLE for adult with triad of 1) hematuria, 2) hemoptysis, and 3) "head-itis" – i.e., any problem with the head, such as nasal septal perforation, mastoiditis, sinusitis, otitis. - Associated with cANCA and anti-proteinase 3 (anti-PR3) antibodies. - Causes "necrotizing glomerulonephritis" that can lead to rapidly progressive glomerulonephritis (RPGN).
Eosinophilic granulomatosis with polyangiitis	- Formerly known as Churg-Strauss. - Presents as combo of asthma + eosinophilia +/- head-itis. - Head-itis always seen in Wegener vignettes, but maybe only ~50% of CS Qs. - Renal involvement rare for CS. - Associated with pANCA and anti-myeloperoxidase (anti-MPO) antibodies.
Microscopic polyangiitis	- Will just present as hematuria in a patient who is pANCA / anti-MPO (+). - Similar to Wegener, can cause RPGN.

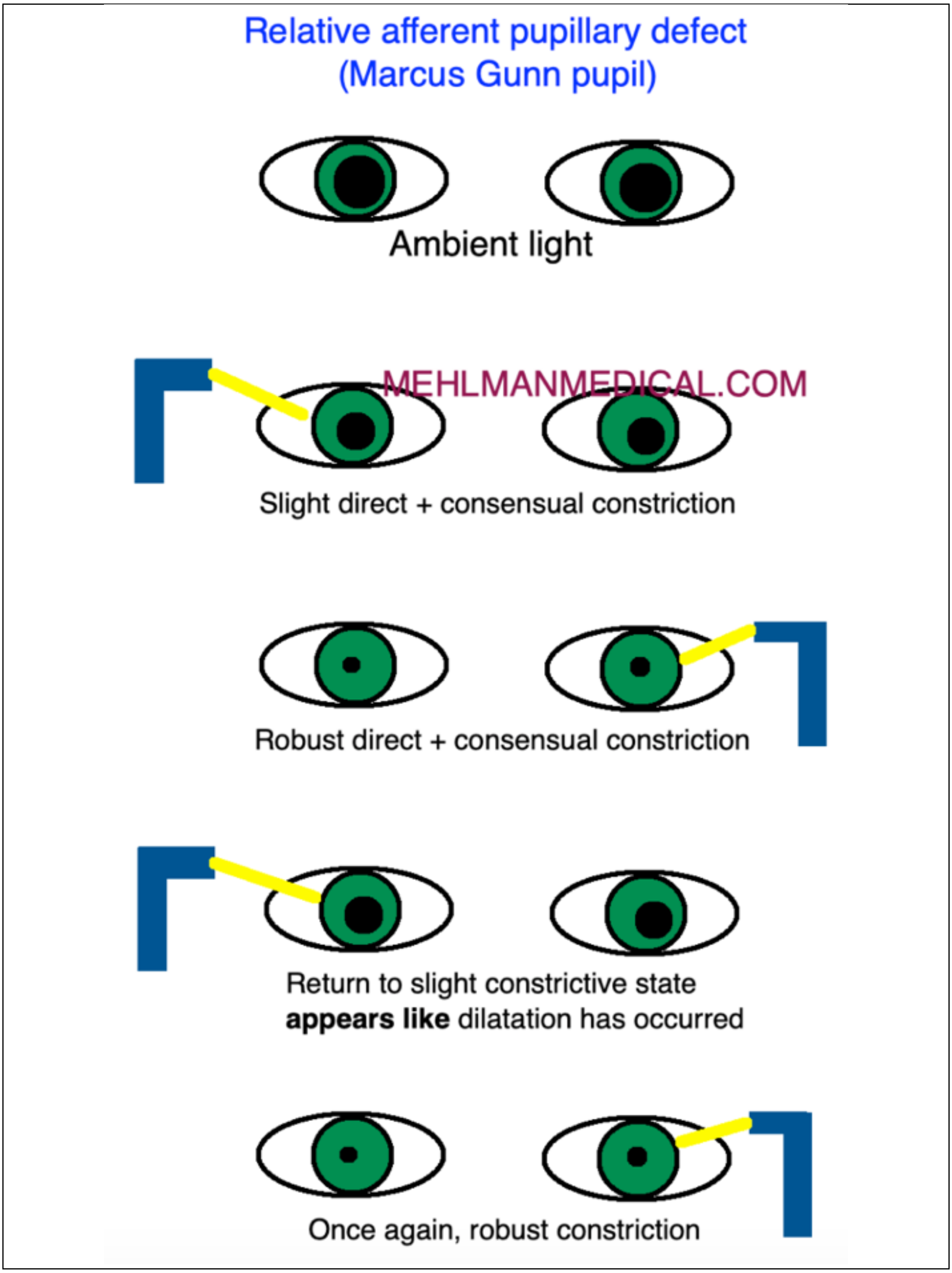
HY dementia types for FM	
Alzheimer	- Gradual-onset idiopathic cognitive decline. - Patient must have normal neurologic exam (i.e., no motor or sensory abnormalities). - MMSE score will be low (i.e., low-20s out of 30) on USMLE. If the diagnosis is instead benign senility, USMLE will give MMSE usually 28+. - Beta-amyloid plaques and neurofibrillary tangles (hyperphosphorylated tau protein) seen on brain biopsy. - Early-onset Alzheimer in Down syndrome (amyloid precursor protein gene is located on chromosome 21). - Presenilin gene mutations can cause Alzheimer (on NBME exam). Presenilin is a protein involved in the cleavage of amyloid. - Tx = cholinesterase inhibitors (donepezil, galantamine, rivastigmine); memantine (NMDA glutamate receptor antagonist) can also be used. - Sundowning is worsening of dementia at night that can resemble delirium. - 1^{st}-line Tx for sundowning on NBME is "decrease ambient noise and distractions." "Bright illumination of the room at all times" is wrong answer.
Frontotemporal dementia	- Aka Pick disease. - Presents usually as triad of 1) personality change, 2) apathy, and 3) disinhibition. - Accumulation of hyperphosphorylated tau protein (similar to Alzheimer), except rather than accumulating as neurofibrillary tangles, it accumulates as round, silver-staining inclusions knowns as Pick bodies.
Lewy body dementia	- Dementia + visual hallucinations + Parkinsonism. - Lewy bodies are collections of alpha-synuclein. This protein is deposited throughout the brain in Lewy-body dementia. In Parkinson disease, in contrast, it is deposited primarily in the substantia nigra pars compacta of the midbrain.
Vascular dementia	- Aka multi-infarct dementia. - The answer for dementia + motor/sensory abnormalities. - Seen in patients who have repeated mini-strokes (cerebral infracts) due to hypertension. - Resources tend to focus on this notion of "step-wise decline" (i.e., concrete timepoints at which deficits started), but it's to my observation on NBME exams that this is rarely a salient aspect of Qs. What USMLE likes is giving motor and/or

	sensory deficits – i.e., you'll get a big paragraph with dementia, and you'll notice somewhere in the stem that the patient has, e.g., 3/5 strength in the right upper extremity. This indicates Hx of stroke.
AIDS	- Just be aware AIDS can cause dementia, known as AIDS complex dementia. - Can present as "wet, wobbly, wacky," similar to normal pressure hydrocephalus.
Pseudodementia	- Not actual dementia. This is depression that presents as cognitive decline. - Patients with depression who have apathy will perform poorly on the MMSE. - The Q might say the patient is unable to draw a clockface, but when prompted, is able to finish it quickly. They might also say patient remembers 0 out of 3 objects after 5 minutes. - Look for obvious signs of depression, such as short, quiet answers, and low mood.
Subacute combined degeneration	- The fancy name for neurologic degeneration seen in B12 deficiency. - Can present sometimes as a reversible cause of dementia. In elderly patients on tea and toast diets, or those in high-risk groups (i.e., vegans, pernicious anemia), B12 must be considered as cause of cognitive decline. - The patient can have peripheral neuropathy as a result of deficits to the 1) corticospinal tracts, 2) dorsal columns, and 3) spinocerebellar tracts. - The easy way to remember those three is to start by saying, "The spinothalamic tract is not involved." Then you say, "Well what are other ones I can think of?"
Neurosyphilis	- Just be aware that neurosyphilis is a reversible cause of dementia and should be considered. There's an NBME Q floating around for 2CK where they give (+) VDRL in 82-year-old woman with cognitive decline, and the treatment is penicillin.

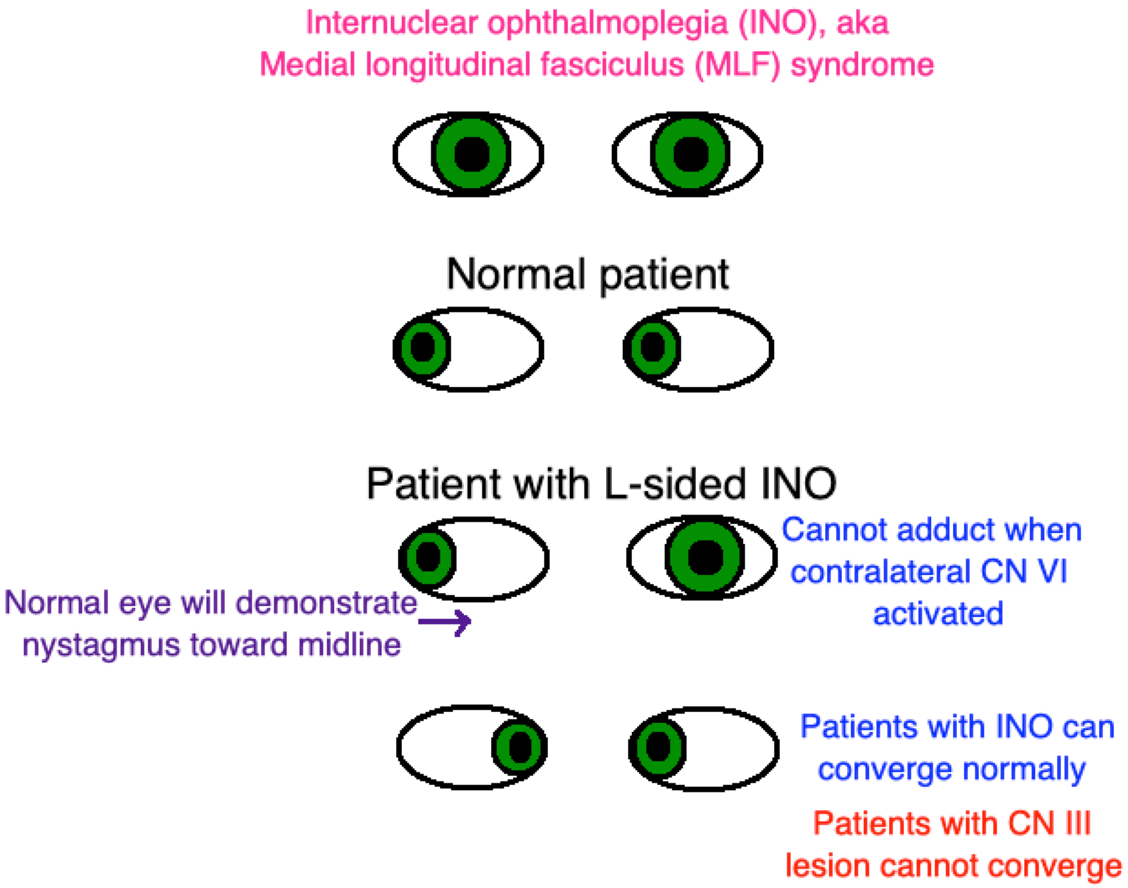
Parkinson disease / Parkinson-plus disorders	
- A Parkinson-plus disorder is a disease that presents similarly to Parkinson disease, but it's not.	
Parkinson disease	- Loss of dopamine-secreting neurons in the pars compacta of the midbrain, with deposition of alpha-synuclein on biopsy. - Presents with classic features of bradykinesia/akinesia, resting tremor, shuffling, short-steppage gait, micrographia, and cogwheel rigidity. - Alpha-synuclein gene mutation most common; many genes implicated. - Carbidopa-levodopa is combo frequently used for Tx. Levodopa crosses the BBB to be converted to dopamine centrally. However, levodopa is subject to fast metabolism when administered alone. The addition of carbidopa functions as a competitive inhibitor of breakdown enzymes, resulting in increased levodopa availability for passage across the BBB. Do not confuse this mechanism with direct COMT inhibitors (tolcapone, entacapone), which prevent breakdown of L-dopa. - Carbidopa-levodopa can cause psychosis if administered in too-high a dose. This is assessed on 2CK Psych forms, where if patient gets psychotic episodes following recent addition of C-L to regimen, or following an increase in dose, the answer is simply "decrease dose of carbidopa-levodopa." "Discontinue carbidopa-levodopa is the wrong answer." - Be aware of D2 agonist names – i.e., ropinirole, pramipexole, cabergoline, pergolide, and bromocriptine, which can all in theory be used as treatments. - Amantadine increases presynaptic release of dopamine. - Selegiline inhibits monoamine oxidase B, which is an enzyme that preferentially breaks down dopamine. USMLE wants you to know this can cause serotonin syndrome, either alone, or in combo with drugs like St John Wort or SSRIs.
RLS	- Restless leg syndrome. - Idiopathic, irresistible urge to move legs while in bed/sleeping. - Most common cause is iron deficiency anemia. First step is checking the patient's serum iron and ferritin. - If iron studies are normal, gabapentin and D2 agonists (ropinirole, pramipexole) can be used.

	- USMLE wants you to know that patients with RLS have increased risk of developing Parkinson disease, which makes sense since D2 agonists help, indicating a potential problem with dopamine signaling or production in some patients.
NPH	- Normal pressure hydrocephalus presents as “wet, wobbly, wacky” +/- Parkinsonism. - The parkinsonism is the most overlooked detail by students, who will usually only know the wet, wobbly, wacky part. - What the USMLE will do is give you an older male who has WWW triad + they give you 2-3 more sentences describing what sounds like Parkinson disease, where you’re like “What the hell? Is this Parkinson disease?” No. It’s just NPH, which is a Parkinson-plus disorder, where it can look like Parkinson disease but it ain’t. - Once again, due to impingement on the corona radiata, and the mechanism they want for urinary incontinence = “failure to inhibit the voiding reflex.”
Wilson disease	- Parkinsonism in a young patient is Wilson disease till proven otherwise. - Excessive copper accumulation in tissues, including the basal ganglia and liver, due to inability to excrete it into bile. - I discuss this stuff in more detail in the HY Gastro PDF.
Lewy body dementia	- As discussed before, this is a Parkinson-plus disorder. - The patient will have dementia + visual hallucinations + Parkinsonism.
Progressive supranuclear palsy	- Obscure condition that gets asked on 2CK. - You just need to know that 100% of questions will say “axial dystonia + Parkinsonism,” where they’ll just ask for diagnosis straight-up. - Axial dystonia is a type of muscle condition resulting in abnormal posture and movement of the spine and torso.
MPTP	- Aka synthetic heroin. - Just know it is a cause of Parkinsonism. - Shows up on a 2CK Psych form. Students are like wtf?

Optic neuritis
- Inflammation of CN II (optic nerve) that presents variably as change in visual acuity, color vision, central scotoma, etc., and **is basically always seen in multiple sclerosis on USMLE.** I discuss MS more later. - Causes Marcus Gunn pupil (aka relative afferent pupillary defect). In this case, the intensity of light is under-sensed by the affected eye (afferent signal), so the resultant efferent CN III (oculomotor) parasympathetic output (back to the eyes for pupillary constriction) is attenuated; this means when you shine a light in the affected eye, both pupils constrict less in comparison to when the light is shone in the unaffected eye; this **gives the mere impression** that the pupils dilate when light is shone in the affected eye. Still confusing? Take a look at the following illustration (assuming a lesion of the patient’s right eye [left side of illustration]):


Relative afferent pupillary defect
(Marcus Gunn pupil)
Ambient light
MEHLMANMEDICAL.COM
Slight direct + consensual constriction
Robust direct + consensual constriction
Return to slight constrictive state
appears like dilatation has occurred
Once again, robust constriction

MLF syndrome
- Medial longitudinal fasciculus syndrome (MLF syndrome; aka internuclear ophthalmoplegia; INO). - Pathognomonic for multiple sclerosis. MS patients can also get optic neuritis (as mentioned above), but this is not pathognomonic for MS (i.e., it can be seen in other circumstances, such as with ethambutol or sildenafil use). - Mechanism for pathology: when you look to one side (let's say the right), CN VI on the right and CN III on the left are activated. This unison of activation is accomplished when the left MLF in the midbrain is activated. In the case of MLF syndrome, the right eye can abduct without a problem, but the left eye will fail to adduct, and the right eye will exhibit nystagmus back toward the midline. - What you need to remember is: in INO, the side that cannot adduct is the side that's fucked up. So if you look to the right and the left eye doesn't adduct, then the left MLF is where the lesion is.



Autoimmune neuropathies	
Multiple sclerosis	- T-cell-mediated attack against oligodendrocytes and myelin-basic protein. - Idiopathic autoimmune disorder classically in white women 20s-30s who live far from the equator. - Relapsing-remitting type is most common, where the patient will have repeated episodes of neurologic disturbances over many years at varying intervals. - Highest yield findings are optic neuritis and INO, as discussed above. - Urge incontinence is exceedingly HY. Some students ask about overflow incontinence due to MS; I've never seen this on NBME. Urge incontinence is what shows up all over the NBME forms for MS. I discuss the incontinences in extensive detail in my HY Renal and Repro/Obgyn PDFs. - MRI is gold standard for diagnosis, showing scattered white matter lesions within the CNS (i.e., both brain and spinal cord). - CSF analysis shows IgG oligoclonal bands.

	- Clonus and (+) Babinski sign (UMN findings) also HY. - Tx is steroids for acute flares (IV methylprednisolone). - β-interferon is what USMLE wants between flares to decrease recurrences. - Baclofen (GABA-B receptor agonist) is used for spasticity.
Neuromyelitis optica	- Aka Devic disease. - Sounds exactly like MS but they will say CSF oligoclonal bands are negative, and then the answer is just "antibodies against aquaporin-4." - You say Wtf? Not my opinion. It's asked on a 2CK Neuro form.
ALS	- Amyotrophic lateral sclerosis. - In ALS, the lateral corticospinal tracts (UMN) and anterior horns (LMN) degenerate. - There are two things you will see in all ALS Qs, making the Dx very easy: 1) The vignette will always have combination of UMN and LMN findings. Fasciculations, decreased reflexes, decreased tone, and muscle atrophy are classic for LMN. Babinski reflex, clonus, increased (brisk) reflexes, increased tone are classic for UMN. Serum CK can be elevated due to increased tone. 2) Must have **no** sensory findings. If any sensory abnormalities (i.e., paresthesias, numbness, etc.) are present in the vignette, ALS is the wrong answer. USMLE will often give vignette of ALS and then the answer is simply "motor neurons." - Knowing this second point in particular will help on difficult 2CK Qs, where you'll get a big paragraph + they say somewhere in the stem something about a sensory abnormality, and you can say, "Cool, not ALS." For 2CK, next best step in Dx = "electromyography and nerve conduction studies."
GBS	- Guillain-Barre syndrome (aka acute inflammatory demyelinating polyneuropathy). - T-cell- and antibody-mediated destruction of Schwann cells and myelin of peripheral nerves. - In other words, MS = CNS/oligodendrocytes; GBS = PNS/Schwann cells. - Can be described as "segmental and inflammatory demyelination." - Presents as ascending paralysis and loss of deep tendon reflexes. - Can affect both upper and lower limbs. - Symptoms can start as tingling in the hands and/or feet. - USMLE can say "weakness of proximal and distal muscles in lower limbs and weakness of distal muscles in upper limbs" → the implication is that the weakness has already ascended in the legs but not yet in the arms. - Textbook cause is *Campylobacter jejuni*, but in almost all Qs they will not give a patient who has bloody diarrhea after a BBQ. In other words, you can't use this as a crutch. They will just give you the symptoms alone and you have to know it's GBS.
CIDP	- Chronic inflammatory demyelinating polyneuropathy. - Vignette will sound exactly like GBS, but symptoms will be present for months and progress slowly. - GBS, in contrast, develops quickly over days and peaks within weeks. - In other words, if you get a vignette where it sounds like GBS, but they say the symptoms have been present for 4 months, that's CIDP, not GBS. There's one Q on a 2CK form on it.
CMTD	- Charcot-Marie-Tooth disease. - Obscure peripheral nerve autoimmune disease that is the answer on USMLE if they give you high-arched feet (pes cavus), hammer toes, and foot drop.
Bell's palsy	- LMN lesion of CN VII (facial nerve). - Paralysis of ipsilateral facial muscles. - Usually idiopathic/autoimmune following viral infection, shingles (i.e., Ramsay-Hunt syndrome II), or Lyme disease. - If the Q gives you Bell's palsy and asks for next best step, the answer is do serology for Lyme disease. If this isn't listed (or the patient doesn't live in endemic area, such as Africa), choose "no further diagnostic studies indicated." Nerve conduction studies are wrong (this is on NBME). Bell's palsy is a clinical diagnosis.

	- IV steroids, followed by a 10-21-day taper of oral steroids is indicated to minimize the immune response. - Ipsilateral hyperacusis (due to paralysis of stapedius muscle) and/or loss of taste to the anterior 2/3 of the tongue are basically nonexistent in questions. - There is a Q on NBME where they say patient has loss of taste following trauma + neurologic exam shows no abnormalities → answer = olfactory nerve (CN I) palsy, not CN VII, due to cribriform plate fracture (loss of smell means loss of most taste). If the Q wanted CN VII lesion, they would give a concurrent Bell's palsy presumably, since loss of taste wouldn't happen in isolation with CN VII palsy. - In contrast to CN VII lesion that is 19 times out of 20 LMN causing Bell's palsy, you should be aware that 1/20 are UMN, which results in loss of motor function of the contralateral middle and lower face, with the forehead spared.
PANDAS	- Pediatric Autoimmune Neuropsychiatric disorder Associated with Streptococci. - In rare cases, Group A Strep pharyngitis can cause a tic disorder, OCD, or ADHD in the weeks following infection. - USMLE will give you kid who had sore throat two weeks ago + now has a new-onset tic, ADHD, or OCD, and they will ask will most likely diagnose etiology for the disorder → answer = "anti-streptolysin O titers."
Sydenham chorea	- Autoimmune movement disorder seen as part of rheumatic fever. - Chorea = fast, purposeless, jerky movements. - Vignette will give child who has rheumatic fever and then ask about the cause of the movement disorder, and the answer is just "autoimmune." - I discuss rheumatic heart disease in detail in the HY Cardio PDF.
Myoneural junction disorders	
Myasthenia gravis	- Autoantibodies against post-synaptic nicotinic acetylcholine receptors. - Classic vignette is female office worker in her 40s who has triad of 1) diplopia, 2) dysphagia, and 3) ptosis that worsens throughout the day. - Gets worse with recurrent stimulation of muscle (vignette might say the patient cannot perform upward gaze for 60 seconds). - Tensilon test is administration of edrophonium (short-acting acetylcholinesterase inhibitor) → patient experiences significant improvement in symptoms. - Tx = pyridostigmine (longer-acting acetylcholinesterase inhibitor). - MG can be a paraneoplastic syndrome of thymoma. - 10-15% of patients with MG have thymoma. - If MG is diagnosed, a chest x-ray should be performed to look for thymoma. If the x-ray is abnormal, a CT scan is ordered. - Removal of the thymoma in these patients can significantly improve/cure the MG.
Lambert-Eaton	- Autoantibodies against pre-synaptic voltage-gated calcium channels. - Presents as proximal muscle weakness that improves with activity (vignette might say the patient tries to get up from a chair a few times before finally being able to). - Tensilon test does not significantly improve Sx in comparison to MG. - Highest-yield point is that it is a paraneoplastic syndrome of small cell lung cancer.

Vertigo	
BPPV	- Benign paroxysmal positional vertigo. - Vertigo is the feeling of the room spinning. - Brief episodes of dizziness, usually 30-60 seconds, sometimes with vomiting. - Caused by a semicircular canal otolith (ear stone; aka otoconia). - Otoconia made of calcium carbonate are normally found within the utricle and saccule of the inner ear and play a role in detecting acceleration and motion. If one becomes dislodged and enters the semicircular canals, BPPV results. - Diagnosis is made via Dix-Hallpike maneuver. In this test, nystagmus is induced when the patient is rapidly moved from a seated position to lying down with the head tilted backward and turned to one side. If nystagmus occurs, Dx = BPPV.

	- Tx is Epley maneuver, which is a series of head movements aimed at moving the semicircular canal otolith back to the utricle, thereby curing the BPPV.
Vestibular neuritis	- Viral infection + vertigo. - Inflammation of the vestibular nerve (branch of CN VIII, vestibulocochlear nerve).
Labyrinthitis	- Viral infection + vertigo +/- tinnitus. - Inflammation of both the vestibular nerve and cochlear nerve. - This distinction of "VN is just vertigo whereas labyrinthitis is also tinnitus" is perpetuated on the internet, but I can tell you there is an NBME Q for Step 1 (that is repeated across forms) where they don't mention anything about tinnitus and the answer is labyrinthitis (where vestibular neuritis isn't listed). This is why I write "+/-" for tinnitus above. - The tympanic light reflex can be abnormal, which can reflect increased middle ear pressure (sometimes seen with viral infection). I've seen the NBME write in the stem "abnormal light reflex," which is a confusing point, since this could be misconstrued as referring to the eyes, not the ear. The vignette in which they mention this they also have multiple sclerosis as a wrong answer + mention the patient has a sibling with MS. So the Q is dumb/gotchya-style overall. The key sentence in this NBME Q is that the patient had nausea and reduced ability to eat foods past few days, which implies recent viral infection.
Meniere	- Waxing and waning, asymmetric tinnitus and vertigo that has a slowly progressive course over many years. - Has a familial component but no strict inheritance (i.e., the vignette can sometimes mention a relative with similar findings). - Due to defective endolymphatic drainage. - Can cause low-frequency hearing loss (i.e., patient says it is difficult to hear conversations at the dinner table with many people). This is in contrast to presbycusis, which is high-frequency hearing loss in elderly.
Drug-induced	- Aminoglycosides (i.e., gentamicin, amikacin), chemo agents like cisplatin, and loop diuretics (furosemide, ethacrynic acid) are known for ototoxicity. - Keep point is that this ototoxicity need not be hearing loss and can be vertigo – i.e., the patient is receiving IV Abx for endocarditis treatment and feels like the room is spinning (ototoxicity due to gentamicin).

HY headache types for FM	
Trigeminal neuralgia	- 11/10 lancinating, knife-like pain that occurs as episodes lasting usually <1 minute. - Classically brought on by minor stimuli, such as brushing one's hair/teeth, or a gust of wind. - Thought to be caused by trigeminal nerve irritation or impingement (i.e., at the exit points from the skull). - There is an NBME Q where they ask what part of the brain is fucked up (i.e., they don't list any nerves), and the answer is pons, since CN V originates from the pons. - Prophylaxis is carbamazepine. - There is no Tx for acute episodes since pain lasts such short duration. - Often confused with cluster headache if occurring in the V1 (ophthalmic branch of trigeminal nerve) distribution.
Cluster headache	- Male 20s-40s who wakes up from sleep with 11/10 lancinating, knife-like headache; often associated with ipsilateral lacrimation, rhinorrhea, and sometimes pupillary changes (I've seen NBME say ipsilateral miotic pupil). - Episodes last minutes to half hour (longer than trigeminal neuralgia), where the Q will say the guy wakes up from sleep + paces around his room until pain eventually goes away. - Prophylaxis is verapamil. - Tx for acute episodes is oxygen (bizarre, unless patient has O2 tank lying around).
Migraine	- Unilateral throbbing/pounding headache that lasts hours.

	- Can be associated with prodromal visual aura or sounds. - Prophylaxis is propranolol. - Tx is NSAIDs first, followed by triptans (e.g., sumatriptan). - Triptans are serotonin (5HT) receptor agonists. - Triptans are not prophylactic meds; they are only used as abortive therapy. - USMLE will give 30s female with hypertension who has migraine Hx, and the treatment is "beta blockade" for her HTN management. - Estrogen-containing contraceptives are contraindicated in patients who have migraine with aura.
Tension headache	- Bilateral, dull, band-like headache. - Treated with sleep and acetaminophen.
Temporal arteritis	- Aka giant cell arteritis. - 9/10 Qs will be painful unilateral headache in patient over 50. I've seen one Q on NBME where it's bilateral. - Flares can be associated with low-grade fever and high ESR. - Patients can get proximal muscle pain and stiffness. This is polymyalgia rheumatica (PMR). The two do not always go together, but the association is HY. (Do not confuse PMR with polymyositis. The latter will present with ↑ CK and/or proximal muscle weakness *on physical exam*. PMR won't have either of these findings. I talk about this stuff in detail my MSK notes.) - Patients can get pain with chewing. This is jaw claudication (pain with chewing). - Highest yield point is we give **steroids before biopsy** in order to prevent blindness. - An NBME has "ischemic optic neuropathy" as the answer for what complication we're trying to prevent by giving steroids in temporal arteritis. - IV methylprednisolone is typically the steroid given, since it's faster than oral prednisone. - It's to my observation many 2CK NBME Qs will give the answer as something like, "Steroids now and then biopsy within 3 days," or "IV methylprednisolone and biopsy within a week." Students ask about the time frames, but for whatever reason USMLE will give scattered/varied answers like that. - Another 2CK Neuro CMS Q gives easy vignette of temporal arteritis and then asks next best step in diagnosis → answer = biopsy. Steroids aren't part of the answer. Makes sense, since they're asking for a diagnostic step.

WKS and ataxia	
- Ataxia is a neurological disorder characterized by a lack of coordination of voluntary muscle movements, leading to unsteady gait and difficulty with balance.	
WKS	- Wernicke-Korsakoff syndrome. - Damage to primarily the mamillary bodies due to thiamine (B1) deficiency. - Wernicke encephalopathy = **A COW** = **A**taxia, **C**onfusion, **O**phthalmoplegia, **W**ernicke. - Korsakoff psychosis = retrograde amnesia; causes confabulations, which means making up stories about the past because of loss of memory regarding prior events. - Therefore, WKS = A COW + confabulations. - Application to USMLE is that alcoholics presenting to hospital with confusion, ataxia, or eye findings require thiamine. - One NBME Q asks what giving thiamine will help reduce risk of in alcoholic → answer = anterograde amnesia (apparently by reducing risk of confusion due to Wernicke), where retrograde amnesia isn't listed as answer.
General Ataxia	- USMLE wants you to be able to differentiate ataxia caused by cerebellar vs dorsal column lesions. - Dorsal column lesions cause a (+) Romberg test, meaning the patient falls over when standing with the eyes closed due to loss of proprioception. - Cerebellar lesions won't cause a (-) Romberg test almost always due to maintenance of proprioceptive capacity.

Cerebellar lesion signs	
Asterixis	- Aka "hepatic flap." - Motor disturbance characterized by brief, sudden lapses of muscle tone, presenting as the patient demonstrating a "flapping" motion of the hands when he or she extends the arms and hands anteriorly.
Dysdiadokokinesia	- Impaired ability to perform rapid alternating movements; presents as difficulty rapidly tapping a finger or quickly pronating and supinating the hand.
Intention tremor	- Kinetic tremor that worsens as the patient's hand approaches a target.
Truncal ataxia	- Tends to occur with more midline cerebellar lesions (i.e., of the vermis). - Can be seen, e.g., in pediatrics with medulloblastoma, where a kid has morning vomiting (often indicates brain tumor) and truncal ataxia.
Limb ataxia	- Tends to occur with more lateral cerebellar lesions. - An important rule is that cerebellar lesions cause ipsilateral deficits, unlike many brain lesions that cause contralateral limb findings.

Neuro conditions caused by vitamin deficiencies	
Vit B12	- As discussed earlier, causes subacute combined degeneration, with reversible dementia and/or systemic neurologic findings. - Hx of gastrectomy, pernicious anemia, veganism, or strict vegetarianism is buzzy. - If patient has B12 deficiency + an unrelated autoimmune disorder, e.g., vitiligo, Hashimoto, etc., the cause is probably pernicious anemia, since "autoimmune diseases go together," where having one autoimmune disease ↑ risk of others. - Patients need not have neurologic dysfunction in B12 deficiency. This is a common misconception. - Causes ↑ MCV (megaloblastic anemia) and hypersegmented neutrophils. - If Q gives you older patient on tea and toast diet + high MCV + no other information, folate (B9) deficiency is more common than B12 deficiency (on FM).
Vit B1	- Can cause WKS, dry beri beri (neuropathy), and wet beri beri (dilated cardiomyopathy). - Two new 2CK NBME Qs assess thiamine deficiency causing neuropathy (dry beri beri) *post-gastrectomy*, which is bizarre, since we classically think of this as causing B12 deficiency. - A helpful detail that B1 deficiency is the answer post-gastrectomy instead of B12 is if they give *confusion*, which is characteristic of Wernicke encephalopathy, not subacute combined degeneration. - One of the Qs also mentions a (+) Romberg test for B1 deficiency as well. Everyone gets these Qs wrong because the notion of B1 deficiency as correct post-gastrectomy, as opposed to B12, is recondite and obscure.
Vit E	- Can cause peripheral neuropathy. - Weirdly an extremely HY vitamin deficiency on USMLE, despite it not seeming that common. Shows up in **cystic fibrosis questions.** - Patients with CF have exocrine pancreatic insufficiency leading to fat-soluble vitamin malabsorption. So CF patient + neuropathy = vitamin E deficiency. - Can in theory be seen with any cause of exocrine pancreatic insufficiency (i.e., chronic pancreatitis due to alcoholism, pancreatectomy, severe diabetes), but it's HY for CF Qs as I said. - Likewise, bowing of tibias / low Ca^{2+} in CF patients would be vitamin D deficiency.
Vit B6	- Neuropathy and/or seizures in patients taking isoniazid for tuberculosis.

Tremor types for USMLE	
Essential	- Autosomal dominant intention tremor. - Can occur at any age, but usually middle age and older. - Patients self-medicate with alcohol, which ↓ tremor. - Propranolol (beta-blockade) is treatment.

Resting	- Parkinson disease (or Parkinson-plus disorder) in older patients. - Wilson disease till proven otherwise in younger patients.
Intention	- Cerebellar lesions or essential tremor.
Drug-induced	- Lithium and valproic acid can both cause tremor. - Lithium is classic for tremor, but an NBME Q gives valproic acid as correct answer (and lithium wrong answer) for patient who has tremor + abnormal liver function tests, since valproic acid is known for hepatotoxicity but lithium is not.

Meningitis	
- Presents as high fever + **stiff neck** and photophobia.	
Bacterial	- Usually Group B Strep in neonates (gram-positive cocci). - *S. pneumo* and *N. meningitides* most common otherwise. - CSF shows ↑ Protein, ↓ glucose, ↑ neutrophils (polymorphonuclear cells).
Fungal	- *Cryptococcus neoformans* in immunocompromised patients (i.e., HIV). - CSF shows ↑ Protein, ↓ glucose, ↑ lymphocytes.
Viral	- Aka aseptic meningitis. - Commonly echovirus (if vaccinated); can also be mumps (if unvaccinated). - CSF shows normal protein (or a tad elevated), normal glucose, ↑ lymphocytes.
Encephalitis	
- Presents as high fever + **confusion.**	
HSV ½	- Herpes encephalitis can occur following vertical transmission. - **RBCs in the CSF** due to temporal lobe hemorrhage. - Non-contrast CT of the head is often negative, so don't be confused by this. However they can say EEG shows abnormal spikes over the temporal region. - Other CSF findings same as viral meningitis. - I've only ever seen one NBME Q on 2CK Neuro CMS form where RBCs were absent in CSF in herpes encephalitis, but you could still get the answer because other CSF findings were viral + no other virus listed.
HY points	
- Do lumbar puncture before antibiotics, as the latter first can interfere with culture results. - Four indications for doing CT before LP in suspected meningitis / encephalitis: 1) Focal neurologic signs. 2) Seizure. 3) Papilledema, or the optic fundi cannot be visualized. 4) Confusion that interferes with neurologic exam / reduced Glasgow score. - The reason for CT before LP in the above scenarios is because of the risk of intracranial mass lesion, where if LP is performed, which acutely ↓ intracranial pressure, herniation and death might occur. - 2CK NBME Q gives point #3 above, where answer is CT, not LP, and many students get it wrong. - Empiric Tx for meningitis is ceftriaxone + vancomycin. Steroids can be added if confirmed *S. pneumo*. - For Group B Strep only, choose ampicillin + gentamicin, or ampicillin + cefotaxime; the combo of vancomycin + ceftriaxone is wrong on the NBME for GBS. - 2CK NBME Q has amp + gent as correct over vanc + ceftriaxone for GBS. - We normally avoid ceftriaxone in the first month of life and prefer cefotaxime as the 3rd-gen cephalosporin. The exact age of the transition is not so important. What you need to know is, if you get a Peds Q and both cefotaxime and ceftriaxone are listed, cefotaxime is what we generally prefer. - Ceftriaxone, compared to cefotaxime, displaces bilirubin from albumin, thereby ↑ risk of neonatal hyperbilirubinemia. It can also cause calcium deposits in the lungs and kidney. - HY that you know meningitis can cause sensorineural hearing loss. If they tell you, e.g., 6-year-old boy isn't paying attention in class + isn't doing well in school + has Hx of meningitis; next best step = air-conduction audiometry. Do not confuse with tympanometry, which evaluates mobility of the tympanic membrane.	

Other CNS infections	
Toxoplasmosis	- *Toxoplasma gondii*; protozoan (unicellular eukaryote). - Causes ring-enhancing lesion(s) on CT head. - Doesn't need to be in patient who has exposure to cats. - Can be acquired from pork. - Prophylaxis is trimethoprim-sulfamethoxazole (TMP-SMX) in HIV patients with CD4 count <100/μL. - Since patients with HIV are already commenced on TMP-SMX for *Pneumocystis jirovecii* pneumonia (PJP) prophylaxis once CD4 count is <200, the latter functions as two bird with one stone for eventual Toxo prophylaxis. - It is important for USMLE that you know the prophylaxis for PJP and toxo is the same, since if they give you ring-enhancing lesion on CT in HIV patient with CD4 count <100 who's on TMP-SMX, you know it's not Toxo. In this case, the diagnosis is primary CNS lymphoma. - The treatment for toxo is sulfadiazine + pyrimethamine. This distinction is asked on a 2CK NBME, where TMP-SMX is also listed and wrong.
AIDS	- Can cause AIDS complex dementia, as discussed earlier. This presents as "wet, wobbly, wacky" (similar to NPH) in an AIDS patient. - Can cause primary CNS lymphoma, especially when CD4 count is <100/μL.
Neurocysticercosis	- *Taenia solium*; cestode; pork tapeworm. - Causes lesions within the brain in patient who went abroad, e.g., to Mexico. - Treated with albendazole or praziquantel.
Mastoiditis	- Complication of untreated or severe otitis media. - Q will tell you a child has tenderness behind the ear + the pinna is displaced upward and outward. - Next best step is CT or MRI of the temporal bone to visualize a potential fluid collection, since if this is not drained it can lead to brain abscess. - Peds CMS form has CT of temporal bone as answer for mastoiditis; sounds wrong to do head CT in a kid, but it's what they want on the form. - X-ray is wrong answer, as it doesn't adequately visualize a potential fluid collection.
Brain abscess	- Abscess in the brain. Not rocket science. - Can be caused by septic embolus due to endocarditis. - Q will give you murmur + fever + neurologic findings → vegetation from heart valve has launched off to the brain, causing stroke-like presentation. - Tx = drainage of abscess + IV antibiotics.
Epidural abscess	- Usually presents as "point tenderness over a vertebra," which on USMLE means epidural abscess, compression fracture (osteoporosis), or metastases. - Patient may or may not have neurologic findings. Fever is not mandatory. - Three demographics I've seen on NBME exams for epidural abscess: 1) IV drug user who now has point tenderness over vertebra; 2) Patient who's had some sort of cellulitis of the legs / lower body that has spread to the spine, where you've eliminated the other answers to get there; 3) Patient with active endocarditis (i.e., murmur + fever) who has neurologic findings in the legs (septic embolus also possible, but this tends to produce stroke-like findings rather than spinal pain + lower leg findings). - Tx = drainage of abscess + IV antibiotics.
Pott disease	- Infection / granulomatous destruction of the vertebral column by tuberculosis. - Just know that this is possible, the same was psoas abscess is also possible.
Rabies	- RNA virus that ascends peripheral nerves to CNS; near-100% fatality rate. - If patient comes into contact with rabid animal, give rabies IVIG + multiple courses of vaccine. - Qbank will focus on exact days/spacing to give vaccine. Waste of time/garbage.
Amoebic meningoencephalitis	- *Naegleria fowleri*; amoeba that lives in lakes, streams, hot springs that enters through the cribriform plate. - Causes rapid deterioration and death within days.

	- You'll see a news article on this every 6 months or so about some "brain-eating amoeba" that killed person who swam in lake.
Eosinophilic meningoencephalitis	- *Angiostrongylus* (rat lung worm). - Consumption of raw slugs/snails. - Caused paralysis and death in an Australian teenager who ate raw garden slug after being dared by friends.

Cavernous sinus thrombosis
- Severe headache, diplopia, proptosis, and/or ophthalmoplegia in patient with recent sinus infection. - Bacterial infection can ascend the valveless facial veins to the cavernous sinus, leading to clotting of blood. - Usually *S. aureus*. - Tx = IV antibiotics + heparin. - CN VI palsy can occur with pathologies of the cavernous sinus.

Primary CNS lymphoma
- HY and underrated diagnosis for USMLE. - Causes ring-enhancing lesion on head CT that can be mistaken for Toxo. As discussed above, if AIDS patient has CD4 count <100 and is already on TMP-SMX, the Dx is primary CNS lymphoma, not Toxo. - Can be seen in non-AIDS patients who have immunosuppression due to other causes (i.e., organ transplant recipients on immunosuppressant meds; patients undergoing radio- and/or chemotherapy). - Can be caused by autoimmune diseases. For example, if they give you a 1-liner where they say, "44-year-old woman with SLE has ring-enhancing lesion on CT" → answer = CNS lymphoma, not Toxo. - Many autoimmune diseases (e.g., SLE, dermatomyositis, Hashimoto) ↑ the risk of non-Hodgkin lymphoma.

Encephalopathy	
- General term that refers to altered brain function (usually presenting as confusion) due to various causes.	
Hepatic	- Seen in alcoholism due to hyperammonemia. - Classically occurs on USMLE when there is an ↑ in blood in the GI tract, such as following acute variceal bleed. Blood is broken down in the GIT into proteins that liberate amino acids, which are converted by gut bacteria into ammonia that is absorbed. - Lactulose is important drug for USMLE for treatment of hepatic encephalopathy. It is a sugar that is broken down by gut bacteria into acidic products. This leads to the conversion of ammonia into ammonium. The latter is ionic and not readily absorbed. The USMLE wants "acidification; ammonium" as the answer for what happens when we give lactulose. - Neomycin is another drug for hepatic encephalopathy. It is an aminoglycoside antibiotic that kills ammonia-producing gut bacteria.
Uremic	- Confusion in the setting of renal failure due to elevated blood urea nitrogen.
Metabolic	- Confusion due to electrolyte imbalances, hypoglycemia, and thyroid storm.
Toxic	- Mental status changes due to toxins. Reye syndrome and lead poisoning are HY.

Inflammatory	- Herpes encephalitis (infectious).
	    - Encephalitis is infection of the substance of the brain, causing confusion (encephalopathy).
	    - As discussed earlier, infection + confusion = encephalitis; infection + stiff neck and/or photophobia = meningitis.
	- Lupus cerebritis (autoimmune).

Anoxic	- Confusion caused by diminished oxygen delivery to the brain.

Neuropathic pain disorders	
Diabetic neuropathy	- Presents as pain, paresthesia, and/or numbness in diabetic. - USMLE likes TCAs (i.e., amitriptyline) or gabapentin as first-line for Tx. - Use nortriptyline in elderly if a TCA is given (fewer side-effects).
Herpetic neuralgia	- Pain, burning, tingling experienced prior to, during, or after herpes or varicella (shingles) episode. - Acyclovir is used to treat the virus; gabapentin used classically for the pain.
Post-trauma	- Patients who have injuries resulting in nerve trauma can sustain chronic neuropathic pain. Treat with gabapentin or TCAs. - Trauma can result in complex regional pain syndrome, which is neuropathic pain that occurs as sequela of injury that is disproportionate to the injury itself.
Thalamic pain syndrome	- Damage to the thalamus due to a stroke that results in body pain, usually contralateral to the side of the infarction, occurring many months post-stroke. - The answer on USMLE for severe limb pain many months following a stroke.

TNR disorders	
Huntington	- Autosomal dominant CAG trinucleotide repeat (TNR) expansion on chromosome 4 resulting in neurodegeneration and choreoathetosis. - Chorea = fast, purposeless, jerky movements; athetosis = slow, writhing movements. - Anticipation = disorder occurs earlier and more severe with each successive generation; caused by expansion of the TNR repeat. - USMLE likes caudate nuclei as part of the brain that's fucked up.
Friedreich ataxia	- Autosomal recessive GAA TNR disorder. - Causes ataxia, kyphoscoliosis, cardiomyopathy, pes cavus (high-arched feet), hammer toes, and early-onset type II diabetes.
CJD	- Creutzfeldt-Jacob disease. - Prion disease; a prion is a misfolded protein that causes the misfolding of other proteins. - CJD is abnormally folded protein within the CNS ($PrP^{C} \rightarrow PrP^{SC}$), where the former is an α-helix that becomes a misfolded β-pleated sheet. Sounds absurdly pedantic, but it's known to be asked for whatever reason. - If the misfolded PrP^{SC} variant of the protein comes into contact with normal PrP^{C} variants, it causes them to misfold into additional PrP^{SC}, with the process functioning exponentially. - Presents as neurodegeneration over weeks to months + **myoclonus.** - The myoclonus is the highest yield detail, since you will sometimes see CJD as a distractor answer, where I'll say to the student, "The patient doesn't have myoclonus though."

Infective neuropathies	
Polio	- RNA virus that infects the anterior horns of the spinal cord and causes polio myelitis (inflammation of spinal cord). - Classically causes degeneration of motor neurons in one leg, leading to one leg much smaller than the other leg. - Highest yield points about Polio actually have zero to do with the infection and almost all to do with the vaccines. - Sal**k** vaccine = **K**illed intramuscular. - Sabin vaccine = live-attenuated oral. - USMLE will ask what is a common feature of these vaccines that accounts for their efficacy? → answer = "neutralizing antibodies in the circulation." - Only the live-attenuated vaccine is capable of generating a CD8+ T-cell response and IgA secretion in the gut. - However, both the killed and oral are capable of generating IgG antibodies in the circulation. - If you're confused about the immuno, I talk about this stuff in my HY Immuno PDF.

PML	- Progressive multifocal leukoencephalopathy. - Caused by JC polyoma virus. - Presents as neurodegeneration over weeks to months in immunocompromised patient, i.e., AIDS patient with CD4 count <100, patients undergoing chemoradiotherapy, or those on immunosuppressant drugs. - USMLE wants you to know this condition is due to "reactivation of latent infection," which means the patient is infected at some point during life years ago, but the infection now manifests due to immunosuppression. "Acute infection in immunocompromised patient" is the wrong answer.
TSP	- Tropical spastic paraparesis. - Obscure neurologic condition caused by HTLV-1/2 (the same virus that causes mycosis fungoides and Sezary syndrome; if you are like wtf?, I discuss these in my HY Derm PDF). - Development of anti-neuronal antibodies that presents as neurodegeneration.
SSPE	- Subacute sclerosing panencephalitis. - Obscure condition characterized by cognitive decline and neuropathy due to reactivation of certain strains of measles in the CNS. - Will present as teenager experiencing cognitive decline + CSF analysis shows measles.

Toxin-induced neuropathies	
Botulism	- Causes flaccid paralysis. - Botulin toxin of *Clostridium botulinum* inhibits presynaptic acetylcholine, which is normally stimulatory at muscles. - Can present as floppy baby syndrome; can also cause cranial nerve palsies. - USMLE can be weird about the answer, where I've seen them write on an NBME exam something along the lines of "prevents acetylcholine from binding to its receptor" as the MOA of the toxin, even though this isn't technically the direct effect. - Acquired as spores in honey in infants under 1; acquired as pre-formed toxin from canned goods in anyone older.
Tetanus	- Causes spastic paralysis. - Tetanus toxin of *Clostridium tetani* inhibits presynaptic neurotransmitters GABA and glycine, which are normally inhibitory. - Can present as opisthotonos (arched back) and trismus (lock-jaw). - Presents in two patients on USMLE: 1) neonate born at home whose umbilical cord was cut with a kitchen knife + tied with twine; 2) random dude who cut himself in back yard. **Steps 2CK/3:** - DtaP given at 2, 4, 6 months, then again at 15-18 months, then again at 4-6 years. - School-age kids require Tdap booster at 11-12 years, followed by Td booster every 10 years thereafter. - Pregnant women should get Tdap at 27-36 weeks to protect neonate from pertussis. For cuts/wounds post-vaccine: - 0-5 years post-vaccine: no Tx is necessary. - 6-9 years post-vaccine: if clean wound: no Tx; if dirty wound: give Td booster. - 10+ years post-vaccine: if clean wound: give Td booster; if dirty wound: IVIG + vaccine. - In other words, only ever give IVIG if it's a dirty wound + has been 10+ years.
Tetrodotoxin	- Toxin acquired by pufferfish. - Causes paralysis and death.
Ciguatera	- Toxin acquired by various meaty fish. - The answer on USMLE for temperature dysesthesia ("hot feels cold; cold feels hot").

<table>
<tr><th colspan="2">Brain bleeds</th></tr>
<tr><td>Epidural hematoma</td><td>- Rupture of middle meningeal artery (branch of external carotid); classically occurs following head trauma to the temple.
- Lucid interval – i.e., patient loses consciousness briefly, followed by arousing and returning to normal (i.e., has a period of lucidity), then goes home thinking he/she is okay, then goes to sleep, then dies.
- Bleed appears as lens (biconvex) shape; fast accumulating.
- If low GCS, do “intubation + hyperventilation,” then do craniotomy. If normal GCS, just do craniotomy.
- Hyperventilation mechanism: decreased CO_2 → decreased cerebral perfusion → reduces intracranial pressure.
</td></tr>
<tr><td>Subdural hematoma</td><td>- Rupture of bridging (superior cerebral) veins.
- No lucid interval (i.e., patient did not lose consciousness).
- Crescent-shaped; can be slowly accumulating and heterogenous in appearance due to combination of dried + fresh blood.
- Increased risk in elderly/dementia and alcoholics; answer in acceleration-deceleration injuries (shaken baby syndrome; motor vehicle accidents).
- NBME wants craniotomy as the answer for Tx (no evidence for mere observation as the answer on any of the USMLE material).
- If the patient has decreased Glasgow score, do “intubation + hyperventilation” as answer before craniotomy.
</td></tr>
</table>

<table>
<tr>
<td>Subarachnoid hemorrhage</td>
<td>- Rupture of anterior communicating artery (AcoM) or posterior communicating artery (PcoM) saccular/berry aneurysm.
- AcoM > PcoM in terms of location.
- Patient often reports “worst headache of his/her life.”
- Can present with stiff neck (meningism) the same way meningitis does, due to irritation of the meninges by the hemorrhage. This often confuses students and is a very HY detail.
- PcoM aneurysm classically associated with ipsilateral blown pupil (mydriasis) due to CN III impingement.
- HTN is most common risk factor in population and accounts for most cases.
- Ehlers-Danlos and autosomal dominant polycystic kidney disease (ADPKD) are HY non-HTN-associated specific causes of saccular aneurysms.
- A patient with HTN is still more likely to die from myocardial infarction than berry aneurysm (asked on NBME). But if patient has berry aneurysm, it is most likely from HTN.
- Bleed appears like a “star fish” or “sand dollar.” Q can say there is blood visualized in the basal cisterns.

- Treatment is usually supportive. HTN must be controlled. Anticoagulants must be reversed (i.e., FFP for patients on warfarin). Nimodipine can prevent vasospasm and neurologic sequelae during convalescence.</td>
</tr>
<tr>
<td>Intracerebral hemorrhage</td>
<td>
- Rupture of Charcot-Bouchard microaneurysms within the lenticulostriate arteries in patients who have chronic HTN.</td>
</tr>
</table>

	- Can present on USMLE as a 1-2-line stem, without any image or patient info, where they just say, "X person has bleed + decorticate posturing; what's the diagnosis?" → answer = intracerebral hemorrhage. - In other words, USMLE wants you to know intracerebral hemorrhage has high likelihood of causing brainstem compression. - Associated with Alzheimer (i.e., amyloid angiopathy, which presents in Alzheimer as recurrent hemorrhagic stroke). - Can be associated with brain cancer as well (e.g., glioblastoma multiforme).

Post-concussion syndrome
- A concussion is a type of traumatic brain injury caused by a blow to the head or by a sudden movement that results in the brain moving rapidly inside the skull. The rapid movement can cause brain cells to stretch and become damaged. - Post-concussion syndrome (aka post-concussive disorder) comprises a range of symptoms that can persist for weeks, months, or even years after a concussion. - Such symptoms include cognitive disturbance, memory loss, headache, and sensitivity to light or sound. - USMLE will ask for the diagnosis straight-up on Neuro forms.

Seizure terminology	
FOAS	- Focal-onset aware seizure (FOAS). - Formerly known as simple seizures. - No loss of consciousness (LoC).
FOIAS	- Focal-onset impaired awareness seizure (FOIAS). - Formerly known as complex seizures. - LoC. - This includes staring into space blankly, as with absence seizures (discussed below).
Partial	- One part of the brain is affected. - Simple partial = no LoC + only affects one part of brain. - Complex partial = LoC + only affects one part of brain.
Generalized-onset	- Formerly known as generalized. - Dumb changes in nomenclature. But I don't know what to tell you. - Involves both cerebral hemispheres.
Tonic-clonic	- Aka "grand mal" seizure; type of generalized seizure. It is characterized by two phases: - Tonic phase: lasts a few seconds; muscles stiffen and the patient falls to the ground and loses consciousness. - Clonic phase: rhythmic jerking movements of the limbs; usually lasts several minutes. - Following the seizure (i.e., during the postictal phase), the patient may be confused, drowsy, and have no recollection of the seizure. - Tongue biting/lacerations are a HY post-seizure finding. If the Q tells you explicitly that the tongue is normal in this setting, they are telling you it is not a seizure. - Tx for recurrent seizures are agents such as valproic acid, carbamazepine, and phenytoin. I discuss these later in this PDF in the pharm section. - There is a difficult NBME Q where they tell you a patient has twitching of one arm prior to falling to the ground and having a tonic-clonic seizure. The answer on the NBME form is "complex partial," not generalized tonic-clonic. This type of seizure is called FOIAS with secondary generalization, or complex partial with secondary generalization. - In other words, if a patient has focal neurologic signs preceding a tonic-clonic seizure, this indicates an origin in one location prior to spreading to other areas.
Status epilepticus	- The updated definition is a seizure lasting >5 minutes, or 2 seizures within 5 minutes. - Definition used to be a seizure lasting >30 minutes, or 2 seizures within 30 minutes.

	- First-line Tx is a benzo (IV lorazepam is usually 1st-line, but USMLE doesn't care). - If benzo doesn't work, phenytoin (or fosphenytoin) is next, followed by barbiturates. - In other words, for USMLE: benzo → phenytoin → barbiturate (e.g., phenobarbital).
Myoclonic	- A type of generalized seizure that causes muscle jerks lasting less than a second. - A succession of jerks can be seen over a short time period. - The patient will not have loss of consciousness. - Can present similarly to simple partial, despite the EEG showing generalized activity.
Febrile seizure	- Fever can precipitate idiopathic seizure in 2-4% of children ages 6 months - 5 years. - About a two-fold risk progression to epilepsy compared to general population. - Tx is with benzodiazepine. - Febrile seizures lasting longer than 10 minutes, seizures that are recurrent within 24 hours, or focal neurologic signs more significantly ↑ risk of progression to epilepsy.
Absence seizure	- Vignette will be a kid staring off into space in class for 30 seconds spacing out, sometimes with rapid blinking. - This is considered loss of consciousness. - EEG shows symmetric 3-Hz spike-and-wave discharges. - Tx is ethosuximide (thalamic calcium channel blocker).

Epilepsy disorders	
Temporal lobe epilepsy	- Most common epilepsy disorder. Originates at the medial temporal lobe. - Seizures are often precedes by visual auras, or warning signs. These can manifest as gustatory/olfactory sensations or déjà vu. - FOIAS are most common type, with or without secondary generalization. - For USMLE, pick "medial temporal lobe," or just "temporal lobe," if they ask for the origin of a seizure in the absence of any preceding neurologic findings.
West syndrome	- Aka infantile spasms; X-linked recessive. - Epilepsy syndrome in infants characterized by – you'd never guess it – spasms. - Causes an abnormal EEG pattern called hypsarrhythmia, which is chaotic pattern. - Leads to intellectual disability. - Treatment is with ACTH (obscure, but apparently ↑ endogenous cortisol, which can mitigate the progression).
Lennox-Gastaut	- Severe childhood-onset epilepsy characterized by near-daily seizures and cognitive decline (hyperoralism is a sign of cognitive regression [babies put things in their mouths]). - Poor prognosis, with 5% mortality rate in childhood; 80-90% persistence of seizures into adulthood.
JME	- Juvenile myoclonic epilepsy. - Characterized by myoclonic jerks (usually hypnagogic and/or hypnopompic) that progress to tonic-clonic seizures after several months. - Age of onset is usually 10-16, but can also start in adulthood. - Tx is valproic acid.

Conditions misconstrued as seizures	
Adams-Stokes attack	- Idiopathic arrhythmia disorder in Peds that causes transient hypoxia to the brainstem, resulting in seizure-like episodes. - EEG does not show any seizure activity. - Q will give you kid with miscellaneous arrhythmia description + describe twitching that sounds like seizure → answer = Adams-Stokes attack.
Breath-holding spell	- They'll say 3-year-old was having a tantrum followed by falling on the floor + appearing blue. - Child will involuntarily stop breathing following a trigger – e.g., being upset, frightened, or experiencing pain.

	- The child often cries or becomes upset, exhales forcefully, stops breathing, then develops cyanosis, which can sometimes be followed by brief loss of consciousness and jerking (can be mistaken for seizure). - Shows up on Peds forms so you need to know it exists.
Syncope	- Just be aware that syncope in elderly (e.g., from aortic stenosis or atrial fibrillation) can sometimes cause twitching of muscles that accompanies fainting, where it resembles a seizure. - The muscular activity is due to hypoxia to the brain, similar to Adams-Stokes attack. - Shows up on an NBME form as elderly patient with cardiovascular Hx who has seizure-like episode. - Vignette might say there is no evidence of tongue-biting (meaning not a seizure).

Peds neuro conditions for FM	
Spinal muscular atrophy	- Aka Werdnig-Hoffman syndrome. - Genetic disorder affecting anterior horns. - Presents with profound hypotonia, absent reflexes, and **tongue fasciculations.**
Cerebral palsy	- Motor dysfunction and intellectual disability of varying severity, posited to be due to intrapartum conditions such as hypoxia. - Will show up on USMLE as spastic paraparesis in a child who has “scissoring” of the legs.
Fetal alcohol syndrome	- Most common cause of mental retardation. - Heart/lung defects (defective neural crest migration). - **Smooth, flat, and/or elongated philtrum.** - Thin vermillion border (thin upper lip). - Widely spaced eyes (hypertelorism), midface hypoplasia, short nose.
Phenylketonuria	- Autosomal recessive; deficiency of phenylalanine hydroxylase (or rarely THB, which is cofactor for the enzyme), causing inability to convert phenylalanine into tyrosine. - Screened for on heel-prick test at birth, since failure to detect PKU will result in mental retardation. - Phenylalanine-deficient diet must be given. - Presents as child with “mousy” or “musty” body odor + appears lighter skin than siblings (can cause partial albinism, since tyrosine is normally converted into melanin, and tyrosine can’t be synthesized).
Cretinism	- Aka congenital hypothyroidism. - Most common causes are iodine deficiency in the mother during pregnancy (worldwide) and fetal thyroid dysgenesis (western countries). - Leads to mental retardation (poor myelin sheath development), impaired bone growth, hypotonia, macroglossia, and protuberant abdomen (due to umbilical hernia; can be confused with kwashiorkor, which is protein-calorie malnutrition causing ascites). - Screened for at birth using the heel-prick test to prevent exacerbation.
Rett syndrome	- Neurologic degeneration that occurs exclusively in female infants. - “Hand-wringing” or “hand-flapping” is buzzy finding. - Mental regression can present as putting objects in one’s mouth (something babies normally do).
Friedreich ataxia	- Autosomal recessive GAA TNR disorder. - Causes ataxia, kyphoscoliosis, cardiomyopathy, pes cavus (high-arched feet), hammer toes, and early-onset type II diabetes.
Angelman	- “Happy puppet” – i.e., child presents with laughter.
Prader-Willi	- Mental retardation + hyperphagia.

Phakomatoses (neurocutaneous disorders)		
- I discussed NF1 and TSC a briefly earlier, but these conditions are exceedingly HY and need to be reinforced.		
Condition	**Genetics**	**HY points**
Neurofibromatosis type I (NF1)	*NF1*; AD; chromosome 17	- Neurofibromas, café au lait spots (hyperpigmented macules), axillary/groin freckling, Lisch nodules (iris hamartomas), pheochromocytoma, **optic nerve glioma,** oligodendroglioma, ependymoma.
Neurofibromatosis type II (NF2)	*NF2*; AD; chromosome 22	- In kids, sometimes causes bilateral cataracts. - Bilateral acoustic schwannomas + meningioma in adults.
Tuberous sclerosis (TSC)	*TSC1*; AD; chromosome 9; hamartin *TSC2* ; AD; chromosome 16; tuberin	- Periventricular nodules (tubers) on MRI of head, seizures, adenoma sebaceum (angiofibromas), subungual fibromas, renal angiomyolipoma, cardiac rhabdomyoma, hypopigmented macules ("ashleaf spots"), hyperpigmented velvety lesions ("shagreen patches").
Von Hippel-Lindau	*VHL*; AD; chromosome 3	- Cerebellar/retinal hemangioblastomas, bilateral renal cell carcinoma, pancreatic cysts. - Mutation causes constitutive activation of hypoxia-inducible factor.
Sturge-Weber	Not inherited; somatic mosaicism of *GNAQ* gene	- Port wine stain birth mark (nevus flammeus; may also present as violaceous papules in temporal distribution), leptomeningeal angioma (presenting as seizure); glaucoma.

Rinne + Weber tests	
Rinne test	- A vibrating tuning fork is placed against the mastoid bone (bone conduction; BC). Once the patient can no longer hear it, it is removed from the bone and placed in front of the ear canal (air conduction; AC). - For both normal hearing and sensorineural hearing loss, AC > BC. - For conductive hearing loss, BC > AC.
Weber test	- A vibrating tuning fork is placed on the center of the forehead or top of the head, and the patient is asked if they hear the sound equally in both ears or if it's louder in one ear. - If it is louder in one ear, this is called "lateralization" to that ear.
Normal hearing	- Rinne shows air > bone conduction; Weber does not lateralize.
Conductive on left	- Rinne shows bone > air conduction; Weber lateralizes (is louder) to left.
Conductive on right	- Rinne shows bone > air conduction; Weber lateralizes to right.
Sensorineural on left	- Rinne shows air > bone conduction; Weber lateralizes to right.
Sensorineural on right	- Rinne shows air > bone conduction; Weber lateralizes to left.

Cholinergic pharm points for FM
- If an agent is cholinergic, it will cause **DUMBBELSS** signs/symptoms –> **D**iarrhea, **U**rination, **M**iosis, **B**radycardia, **B**ronchoconstriction, neuromuscular **E**xcitation, **L**acrimation, **S**alivation, **S**weating. - If an agent is anti-cholinergic, it will cause the opposite of DUMBBELSS, or "anti-DUMBBELSS" –> Constipation, Urinary retention, Mydriasis, Tachycardia, Bronchodilation (in reality, β2 agonism, but not M antagonism, will bronchodilate), Skeletal muscle relaxation, Dry eye, Dry mouth, Anhydrosis. - The reason knowing this stuff is important is because TCAs (e.g., amitriptyline), antipsychotics, and 1^{st}-generation H1 blockers (i.e., diphenhydramine, chlorpheniramine) have strong anti-cholinergic side-effects,

which means they can cause "anti-DUMBBELSS" in any patient. In addition, anti-cholinergic effects promote delirium and confusion, especially in the elderly. - For example, old dude + high creatinine + on amitriptyline = post-renal azotemia due to a combo of BPH (which all old dudes have) and the TCA, which is anti-cholinergic. You want to take him off any anti-cholinergic meds he's on. Likewise, if you get a psych Q where a patient has confusion/delirium and one of his or her drugs is diphenhydramine, the answer might be, "discontinue anti-cholinergic medications."

Pro-cholinergic agents for FM	
Muscarinic receptor agonists	
Bethanechol	- Stimulates detrusor muscle. - Used for overflow incontinence in diabetic patients with neurogenic (i.e., hypo-contractile) bladder.
Carbachol	- Constricts pupil and lowers intraocular pressure. - Used for glaucoma. - See my HY Ophthalmology PDF for more detail on that stuff.
Pilocarpine	- Constricts pupil and lowers intraocular pressure. - Powerful agent used for closed-angle glaucoma.
Methacholine	- Causes bronchoconstriction. - Used to help diagnose asthma. If administered to patient (not during acute attacks), spirometry will worsen and asthma symptoms may be replicated.
Acetylcholinesterase inhibitors	
Donepezil	- Increases acetylcholine available in synaptic cleft (i.e., pro-cholinergic). - Used 1st-line for Alzheimer disease. - Galantamine and rivastigmine have same MOA and use-case, but LY.
Edrophonium	- Aka Tensilon (the brand name). - Short-acting acetylcholinesterase inhibitor. - Administration to patients with myasthenia gravis causes marked improvement in symptoms and muscle function. - Administration to patients with Lambert-Eaton syndrome does not result in as marked improvement of symptoms.
Pyridostigmine	- Treatment for myasthenia gravis.
Physostigmine	- Antidote for Jimson weed poisoning (asked on NBME). - Jimson weed contains atropine-like compounds that are anti-cholinergic.
Organophosphates	- Vignette will either be backpacker in his/her 20s picking fruit on a farm, or someone who drank fluid in a suicide attempt. - The phosphate group binds to acetylcholinesterase, inhibiting it. - NBME Q asks how toxicity could best be prevented → answer = "wearing gloves," where "wearing mask" is wrong answer. So acquisition occurs mostly through skin. - USMLE wants atropine first, followed by pralidoxime as Tx (discussed below).

Anti-cholinergic agents for FM	
Muscarinic receptor antagonists	
Benztropine	- Used for acute dystonia due to anti-psychotics. - In other words, if patient starts anti-psychotic and then within hours to days gets stiff neck (torticollis), muscle rigidity (without fever), and/or abnormal eye movements (oculogyric crisis), this is 1st-line Tx. - Muscle rigidity with fever, in contrast, suggest neuroleptic malignant syndrome.
Diphenhydramine	- 1st generation H1-blocker that has strong anti-cholinergic side-effects to the point that it is often just characterized straight-up as a primary anti-cholinergic med. - Can be used same as benztropine for acute dystonia (USMLE won't list both).

	- Can be used for motion sickness (anti-cholinergic effects treat motion sickness). If USMLE tells you it is used in this setting + they ask you antagonism of which receptor enables its anti-motion sickness effect, the answer is M3. The wrong answer is H2, since this refers to stomach acid secretion. Diphenhydramine is an H1 blocker, not H2. - Most Qs on USMLE regarding diphenhydramine revolve around you knowing it should be discontinued from a patient's regimen, rather than added.
Prochlorperazine	- Anti-psychotic used as an anti-nausea med. - Even though it is an D2-receptor antagonist, as I mentioned before, anti-psychotics have anti-cholinergic side-effects. In this case, the anti-muscarinic side-effects are actually a "good" thing since these treat nausea and motion sickness.
Ipratropium	- One of the 1st-line agents used for COPD (asked on 2CK FM forms). - COPD patients can be started on either a muscarinic receptor antagonist, such as ipratropium, or a beta-2 agonist, such as olodaterol. - New literature gets pedantic about the duration (i.e., long- vs short-acting) anti-muscarinics and beta-2 agonists, but USMLE doesn't (i.e., some sources say tiotropium is recommended over ipratropium 1st-line now, but I've had students come out of the exam saying ipratropium showed up, but that it didn't matter since it was the only drug listed that worked anyway).
Oxybutynin	- Used for urge incontinence.
Scopolamine	- Scopolamine patches are used for motion sickness (i.e., put it on the arm before getting on a boat or plane). This drug is an anti-cholinergic straight-up. - As mentioned above, oral diphenhydramine can also be used for motion sickness, but scopolamine is the classic treatment.
Hyoscyamine	- Used sometimes for irritable bowel syndrome.
Smoking cessation agents	
Bupropion	- Dopamine + norepinephrine reuptake inhibitor; has anti-depressant effects. - Also antagonizes nicotinic receptors, which disincentivizes smoking, since doing so won't produce rewarding effects. - Lowers seizure threshold. Don't give to patients with eating disorders (electrolyte abnormalities increase seizure risk). - Doesn't cause sexual side-effects, unlike SSRIs.
Varenicline	- Used for smoking cessation. - Partial agonist at nicotinic receptors, but since this effect is less than endogenous ACh, it functions like an antagonist.

Epilepsy/hypnotic agents	
Valproic acid	- Used as an alternative to lithium in Tx of bipolar disorder. - Causes neural tube defects in pregnancy due to interference with folate metabolism.
Carbamazepine	- Generic anti-epileptic. - Causes neural tube defects in pregnancy due to interference with folate metabolism. - Can cause aplastic anemia and SIADH.
Phenytoin	- Generic anti-epileptic. - Causes neural tube defects in pregnancy due to interference with folate metabolism. - Can cause fetal hydrantoin syndrome (finger nail hypoplasia + abnormal facies). - Used for status epilepticus if benzo fails.
Ethosuximide	- Used for absence seizures.
Benzodiazepines	- Diazepam, lorazepam, midazolam, chlordiazepoxide, etc. - Hypnotic (induces sleep); anxiolytic (reduces anxiety); anti-convulsant. - Given 1st-line for status epilepticus. - Used for alcohol withdrawal (delirium tremens / alcoholic hallucinosis). - Clonazepam used for insomnia/anxiety (shows up on NBME). - Used for stimulant intoxication (i.e., cocaine, amphetamine, PCP). - Used for specific phobia (e.g., fear of flying).

	- Used for social phobia 2nd-line if patient has asthma, since we don't give propranolol in this setting (asked on NBME). - Can cause respiratory depression. - Treat overdose with flumazenil (antagonist of benzo receptor).
Non-benzo hypnotics	- Zolpidem, zaleplon. - Used occasionally for anxiety or insomnia. - Can cause addiction similar to benzos.

Drugs for neuropathic pain	
TCAs	- Amitriptyline, nortriptyline, imipramine. - Used for neuropathic pain, i.e., diabetic peripheral neuropathy. - Causes CCC → Coma, Convulsions, Cardiotoxicity (QT prolongation on ECG). - Have nasty anti-cholinergic side-effects (cognitive dysfunction and urinary retention, especially in elderly).
Gabapentin	- Used for neuropathic pain, e.g. , diabetic peripheral neuropathy.

Headache drugs	
Migraine	- Abortive therapy: NSAIDs first, then sumatriptan (serotonin receptor agonist). - Prevention: propranolol (beta-blockade).
Cluster	- Abortive therapy: oxygen. - Prevention: verapamil (non-dihydropyridine calcium channel blocker).
Tension	- Rest + acetaminophen.
Trigeminal neuralgia	- Abortive: goes away on its own because episodes last maximum 30-60 seconds. - Prevention: carbamazepine (sodium channel blocker).

Parkinson drugs	
Carbidopa-levodopa	- Combo frequently used for Tx of Parkinson disease. - Levodopa crosses the BBB to be converted to dopamine centrally. However, levodopa is subject to fast metabolism when administered alone. The addition of carbidopa functions as a competitive inhibitor of breakdown enzymes, resulting in increased levodopa availability for passage across the BBB. Do not confuse this mechanism with direct COMT inhibitors (tolcapone, entacapone), which prevent breakdown of L-dopa. - Carbidopa-levodopa can cause psychosis if administered in too-high a dose. This is assessed on 2CK forms, where if patient gets psychotic episodes following recent addition of C-L to regimen, or following an increase in dose, the answer is simply "decrease dose of carbidopa-levodopa." "Discontinue carbidopa-levodopa is the wrong answer."
D2 receptor agonists	- Ropinirole, pramipexole, cabergoline, pergolide, and bromocriptine. - Used for Parkinson disease management. - Bromocriptine classic agent used for prolactinoma (dopamine agonism ↓ prolactin). - Ropinirole + pramipexole classically show up as restless leg syndrome treatment following making sure the patient is iron replete. New literature suggests gabapentin is first-line over D2 agonists now for RLS, but USMLE won't play trivia. They will give patient with RLS + say, "In addition to ropinirole, what else could be added?" And gabapentin will be the only answer listed that works.
Amantadine	- Used for Parkinson disease. - Increases presynaptic release of dopamine.
Selegiline	- Used for Parkinson disease. - Inhibits monoamine oxidase B, which is an enzyme that preferentially breaks down dopamine.

	- Can cause serotonin syndrome, either alone, or in combination with drugs like St John Wort or SSRIs.

Alzheimer drugs	
Donepezil	- As mentioned earlier, this is a cholinesterase inhibitor used in Alzheimer. - Rivastigmine and galantamine you can be aware of and have same MOA, but are LY.
Memantine	- NMDA glutamate receptor antagonist. - Glutamate receptor activation normally causes Ca^{2+} to flow into the neuron, leading to depolarization and neuroexcitation. In other words, USMLE wants you to know glutamate receptor is a ligand-gated calcium channel (where glutamate is the ligand that activates the channel). Antagonism causes neuroinhibition.

Neurotoxic drugs relevant to FM	
Aminoglycosides	- 30S ribosomal subunit inhibitors used for gram-negative rods. Highest yield use-case is in combination with vancomycin as empiric Tx for endocarditis. - Gentamicin, tobramycin, amikacin (latter sounds unusual but shows up on an NBME). - Cause ototoxicity (sensorineural hearing loss and/or vertigo). - Cause nephrotoxicity.
Loop diuretics	- Block the $1Na^+/1K^+/2Cl^-$ symporter on the apical membrane of the thick ascending limb. - Furosemide, ethacrynic acid, etc., can cause sensorineural hearing loss and/or vertigo.
Aspirin	- Irreversible cyclooxygenase 1 and 2 inhibitor. - Most common symptom is tinnitus (ear-ringing).
Quinidine	- 1a sodium channel blocker used occasionally as anti-arrhythmic. - Causes cinchonism, which means headache + tinnitus.
Digoxin	- Na^+/K^+-ATPase inhibitor used for advanced heart failure. - Optic neuritis presenting as yellow wavy, “Vincent van Gogh vision.”
Sildenafil	- Phosphodiesterase-5 inhibitor used for erectile enhancement (Viagra). - Can cause blue haze. Apparently this is due to its ability to also inhibit PDE-6 found in retinal receptors (holy shit). Tadalafil (Cialis) doesn’t inhibit PDE-6 the same way, so this effect is more just with Viagra. - USMLE wants you to know that pilots cannot take this within 3 hours of getting on an airplane due to possible blue haze. Unusual/obscure factoid, as if one would require a boner before getting in the cockpit, no pun intended.

FM Psych

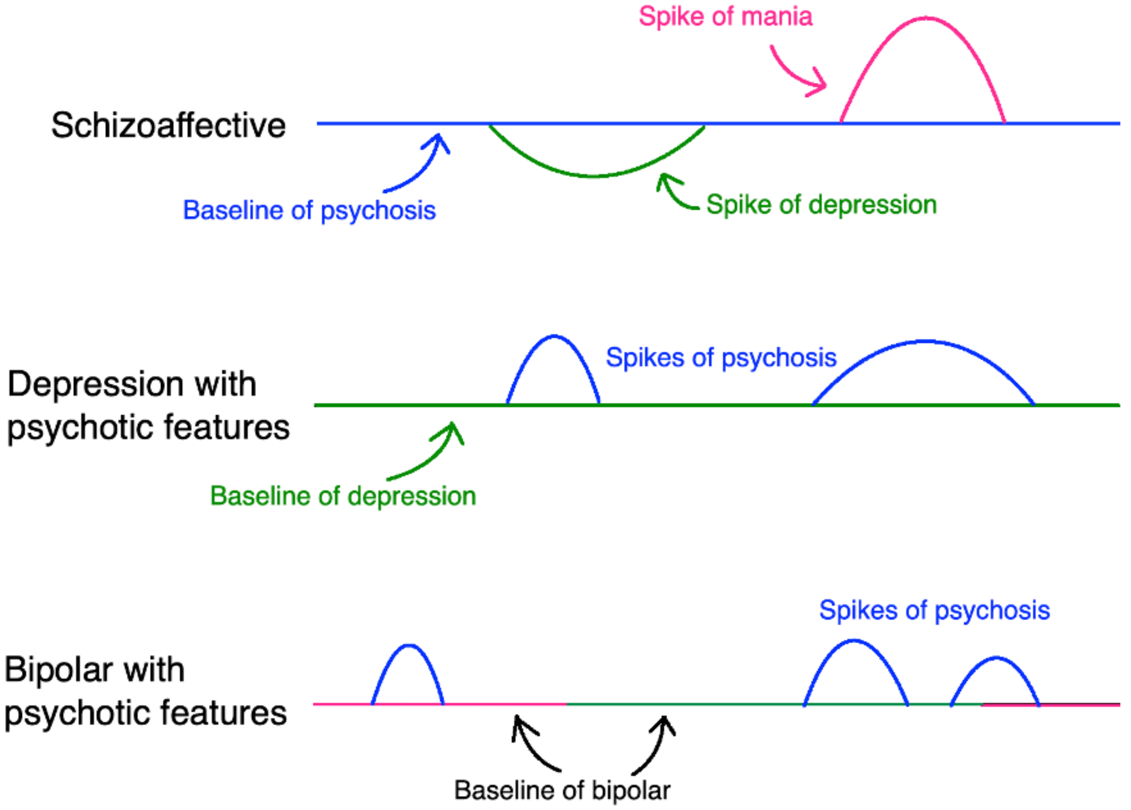
Depressive disorders	
Major depressive disorder (MDD)	- At least 5 out of 9 SIGECAPS must be present for at least a 2-week period. S – Sleep disturbance (insomnia or hypersomnia). I – Interest loss. G – Guilt or feelings of worthlessness. E – Energy loss. C – Concentration problems. A – Appetite changes (usually causes weight loss, but sometimes weight gain). P – Psychomotor agitation (restlessness; slowed speech or movements). S – Suicidal ideation. - The symptoms must cause socio-occupational impairment and must not be attributed to a substance or another medication condition. - As far as NBME Qs go, I can say that USMLE doesn't actually give a fuck about fulfilling the 5/9 criteria. They'll give you an elderly male who's a bit quiet + teary-eyed + who's had weight loss, and the answer is just depression. Students get confused because they don't count 5/9, but as I said, USMLE doesn't care. - It's to my observation that *weight loss* is one of the most buzzy indicators in a psych vignette that MDD is likely the answer. - Terminal insomnia (patient wakes up too early in the morning) can be seen. - Can cause ↓ libido, but patient still has nocturnal erections (asked on NBME). - Can be caused by beta-blockers (propranolol); asked on NBME. - Tx is with drugs such as SSRIs, CBT, and occasionally ECT.
Depression due to a medical condition	- Hypothyroidism is important cause of low mood and apathy. - USMLE likes post-MI and cancer diagnoses as causes of depression. - Q might give guy who's had weight loss + teary-eyed post-MI; answer on an NBME form is just sertraline (an SSRI).
Atypical depression	- Hypersomnolence + hyperphagia in the setting of depression that is reactive / ameliorates when positive circumstances present themselves. - This latter aspect is called "mood reactivity," where positive events of experiences can significantly improve mood; this distinguishes atypical depression from other forms of depression, where mood usually remains persistently low regardless of external circumstances. - Tx = CBT, SSRIs, or monoamine oxidase inhibitors (MAOIs). - MAOIs are considered to be highly efficacious but are not usually used first-line because of serotonin syndrome risk (discussed later).
Pseudodementia	- Not actual dementia. This is depression that presents as cognitive decline, usually in elderly. - Patients with depression who have apathy will perform poorly on the MMSE. - The Q might say the patient is unable to draw a clockface, but when prompted, is able to finish it quickly. They might also say patient remembers 0 out of 3 objects after 5 minutes. - Look for obvious signs of depression, such as short, quiet answers, and low mood.

Dysthymia and cyclothymia	
Dysthymia	- At least 2 years of depressed mood without fulfilling MDD criteria.
Cyclothymia	- At least 2 years of oscillation between depressed mood and hypomania (discussed below).

<table>
<tr><th colspan="2">Bipolar disorder I and II</th></tr>
<tr><td colspan="2">- DIGFAST is the mnemonic for remembering how mania and hypomania present:
- D – Distractibility (easily distracted / not able to focus).
- I – Impulsivity (e.g., shopping sprees, reckless driving).
- G – Grandiosity (inflated self-esteem / having sense of special powers or gifts).
- F – Flight of ideas (rapidly shifting thoughts/ideas, sometimes with lacking coherence).
- A – Activity increase (goal-directed activity; sexual activity; psychomotor agitation).
- S – Sleep deficit (decreased need for sleep).
- T – Talkativeness (pressured speech; term means haphazard loquaciousness, sometimes without pausing).</td></tr>
<tr><td>Mania</td><td>- State of intensely elevated, expansive, or irritable mood, often with marked impairment in judgment.
- Usually lasts longer than a week and causes socio-occupational dysfunction (i.e., the patient works as dentist and can’t go to work / has problems at work).
- Can sometimes cause psychotic features and require hospitalization.
- ↑↑ risk of suicide.</td></tr>
<tr><td>Hypomania</td><td>- Usually lasts longer than 4 days and does not cause socio-occupational dysfunction (i.e., the Q says patient works as dentist + maintains the job no issues).
- Does not cause psychotic features and does not require hospitalization.</td></tr>
<tr><td>Bipolar I</td><td>- Diagnosed if patient has at least one manic episode, with or without a depressive episode.</td></tr>
<tr><td>Bipolar II</td><td>- Diagnosed if patient has at least one hypomanic episode + at least one major depressive episode. The patient must have never had a full-blown manic episode.
- The reason at least one major depressive episode is required for diagnosis is because a hypomanic episode might be part of cyclothymia, rather than bipolar II, if the patient has merely experienced depressed mood in the past without a full-blown depressive episode.</td></tr>
<tr><td colspan="2">- Treatment on USMLE is lithium or valproic acid.
- Lithium can cause nephrogenic DI, hypothyroidism, and tremor. It is also a teratogen (causes Ebstein anomaly). New 2CK NBME has lithium causing serotonin syndrome. Very unusual/odd, but asked.
- Valproic acid causes neural tube defects (interference with folate metabolism), tremor, and hepatoxicity. If the Q gives you increased LFTs + tremor, choose valproic acid over lithium.</td></tr>
</table>

<table>
<tr><th colspan="2">Psychotic disorders</th></tr>
<tr><td>Schizophrenia</td><td>- Four points for diagnosis.
1) At least 6 months in duration.
2) Two or more of the following:
- Hallucinations (always auditory on USMLE; hearing voices that aren’t there).
- Delusions (false beliefs; can be “bizarre,” which refers to stuff like demons, aliens, and “the lord”).
- Disorganized speech (often nonsensical or muffled speech).
- Disorganized or catatonic behavior.
- Negative symptoms (flattened/blunted affect, anhedonia [inability to feel pleasure], diminished speech).
3) Must have socio-occupational dysfunction (i.e., significant impairment in work, interpersonal relationships, or self-care).
4) Not attributed to a mood disorder (i.e., depression or bipolar), substance use, or another medical condition.</td></tr>
<tr><td>Schizophreniform</td><td>- Same as schizophrenia but 1-6 months in duration.</td></tr>
<tr><td>Brief psychotic disorder</td><td>- Same as schizophrenia but <1 month in duration.</td></tr>
</table>

Commonly confused conditions



Schizoaffective	- At least two weeks of psychosis without a mood disorder (i.e., neither depression nor mania/hypomania), followed by periods of time with simultaneous psychotic and mood disorder. - For example, the Q will say a college student has been hearing voices for the past 6 weeks + has been staying up all night for the past month (i.e., 4 weeks of mania or hypomania + at least 2 weeks of psychosis alone). - The name of the condition starts with "schizo," so you can remember that psychosis is the baseline of the disorder (i.e., what the patient will have as his/her underlying baseline). Then the mood disorder spikes on top of it.
Depression with psychotic features	- Depression, rather than psychosis, is the baseline. The patient will have periods of time in which he/she only experiences major depression without any type of psychosis. Then the psychosis periodically spikes on top of the depression.
Bipolar with psychotic features	- Bipolar is the baseline, where the patient will have periods of either depression *or* mania alone, followed by psychosis spiking on top of it. - If the patient has periods of mania alone followed by a simultaneous spike of psychosis, this makes the diagnosis easier. - If the patient has a period of depression alone followed by spiking psychosis, it can appear like depression with psychotic features, except the patient will *also* have a history of at least one manic episode in the past.

Delusion disorder

- One or more non-bizarre delusion, without other signs of psychosis.
- By non-bizarre, as discussed earlier, this refers to potentially plausible scenarios (e.g., people at work are stealing from or trying to undermine you) that are not true, whereas bizarre on USMLE is anything that involves aliens, demons, the heavens, the lord, etc.
- The person's behavior and thinking, apart from the delusion itself, tend to be normal.
- Tx = CBT, anti-psychotics (e.g., risperidone), mood-stabilizers (lithium), and/or anti-depressants (SSRIs).

Grief (Bereavement)	
Normal	- No specific time frame, but usually < 6 months. - Normal progression is: Denial, Anger, Bargaining, Lamentation, Acceptance. - Gradual improvement (decreases over months). - Functional coping (maintains basic self-care, responsibilities, and socio-occupational function). - USMLE wants you to know that hearing voices of loved ones who've past away recently is normal (i.e., a 70-yr-old man sometimes wakes up in the middle of the night hearing his wife talking to him). But the voices will never be of the nature in which the patient is told to feel guilty or worthless.
Pathologic	- Can be >6-12 months. - Persistence / doesn't abate with time. - Functional impairment of daily tasks, neglect of self-care, or thoughts of suicide. - Hearing voices telling the patient he/she should feel guilty/worthless (i.e., the patient hears his uncle, who lives in another state, tell him he's worthless).

Anxiety disorders	
Generalized anxiety disorder (GAD)	- 6+ months of excessive and persistent worry about various aspects of life, often accompanied by physical symptoms like restlessness, fatigue, and muscle tension. - USMLE might give 40-year-old woman who is worried about many things (e.g., her son going to college, her work, her marriage, etc.) for 6+ months. - Patient can also have concurrent mood or psychotic disorder. In this case, the diagnosis becomes, e.g., "GAD with comorbid MDD." - Tx for GAD is CBT and SSRIs. - Buspirone (serotonin receptor agonist) can also be used.
Panic disorder	- Panic disorder is diagnosed if patient has 2 or more panic attacks + at least one month of worry about having more attacks. - A panic attack is an episode of intense fear that triggers severe physical reactions, even though there is no real danger or apparent cause. - USMLE will usually give patient with hyperventilation, where he/she feels like he/she is going to die or is having an MI. The USMLE is obsessed with making you think panic attack is cardiac. Don't get fooled. - Vignette can mention a mid-systolic click (mitral valve prolapse) + ask you the cause of the patient's symptoms → answer = panic disorder, not mitral valve prolapse. Student is confused because they say, "But wait, the patient has a mid-systolic click though." You're right. But the MVP itself isn't the cause of the patient's presentation. MVPs are common in the population and almost always asymptomatic. - Tx = breathing exercises to encourage the patient to stop hyperventilating. Breathing into a paper bag is wrong answer. Often times they won't have breathing exercises as an answer, where benzo is correct.
Social phobia	- Fear of being judged, negatively evaluated, or rejected in a social or performance scenario. Presents on USMLE as fear of public speaking. - USMLE wants beta-blocker (propranolol or atenolol) as 1st-line Tx. - If Q gives you an asthma patient, choose benzo. They make this distinction on NBME. - CBT for longer-term management.
Specific phobia	- Fear of a specific object or situation leading to significant distress or functional impairment. High-yield example on USMLE is fear of flying. - USMLE wants benzo for acute relief; CBT for longer-term.
Agoraphobia	- Fear of being in places or situations from which escape might be difficult or embarrassing, or where help might not be available if one were to have a panic attack. - This usually refers to crowded, open spaces.
Separation anxiety disorder	- Excessive distress with separation from home or major attachment figure. - Peaks at 12-18 months of age and usually subsides by 2-3 years, but children can still get this up into their teenage years.

	- USMLE can give a vignette of a child going to summer camp or school who gets stomach aches on arrival.
Selective mutism	- Disorder where patient fails to speak in certain situations, such as when confronted with new people, despite being able to speak in other situations.

PTSD vs Acute stress disorder	
PTSD	- Post-traumatic stress disorder; presents as at least 1 month of reliving a traumatic event, usually with flashbacks or nightmares about the event. - Treatment includes CBT and group therapy. - SSRIs if medication used.
Acute stress disorder	- Same as PTSD, but presenting for 3 days to <1 month since the traumatic event.

Adjustment disorder
- Socio-occupational dysfunction starting within 3 months of one specific stressor (i.e., a breakup, loss of job, death, illness). - Usually lasts <6 months. If longer than 6 months, it is called chronic adjustment disorder. - The key is the single stressor, whereas GAD is many, or no one specific, stressor. - The patient cannot have any psychotic symptoms but can have a mood disorder, where we call it "adjustment disorder with depressed mood," or "adjustment disorder with anxious mood," etc. I've seen NBMEs, particularly on 2CK psych forms, write answers like this. - First-line Tx is CBT and SSRIs.

Obsessive-compulsive disorder (OCD)
- Obsessions = Unwanted and intrusive thoughts, images, or urges that cause significant anxiety or distress. - Example is constantly feeling the need to touch a doorknob correctly before opening the door. - Compulsions = Repetitive behaviors a person performs in response to an obsession. - Example is touching the doorknob repeatedly in order to make sure it is touched "correctly." - OCD is persistent obsessions and/or compulsions causing the patient significant distress or interfering with the patient's socio-occupational functioning. The patient need not have both in order to be diagnosed – i.e., a patient can just have either obsessions or compulsions alone. - A key point that distinguishes OCD from OCDP (obsessive-compulsive personality disorder, which I discuss below), is that OCD is ego-dystonic, whereas OCDP is ego-syntonic – meaning, in OCD, the patient doesn't like it / feels distressed; in OCDP, the patient is content / doesn't view anything as wrong. - Tx = CBT. If drugs, use SSRI or clomipramine (a TCA). The latter shows up on 2CK Psych form and is a mini-factoid about OCD (i.e., you can use the TCA clomipramine to treat it).

Conduct vs oppositional defiant disorder	
Oppositional defiant	- Argumentative and defiant behavior, often leading to problems at school with teachers and/or problems with grades. - The vignette can sound like normal teenage behavior, but the emphasis will be that there is impediment to social and/or scholastic progression. - Another key point is that there is no law-breaking. If the vignette mentions anything about crimes, then the USMLE wants conduct disorder instead.
Conduct disorder	- >6-month pattern of law-breaking starting age 17 or younger. - Vignette might give a teenager who engages in criminal behavior, such as killing a cat, destroying property, or engaging in theft. - USMLE won't necessarily present to you a pattern of ongoing behavior, but rather just a snapshot of a child + ask for the diagnosis – i.e., 14-year-old killed a cat;

	what's the most likely diagnosis → answer = conduct disorder (because what he did is a crime); oppositional defiant disorder is wrong answer.

Personality disorders	
Cluster A (odd, eccentric)	
Paranoid	- Suspicion of others intending to cause harm or deceive. - Reluctance to confide in others due to unwarranted fears that the information will be used against him or her. May bear grudges persistently. - Projection is a common defense mechanism.
Schizoid	- Pervasive pattern of detachment from social relationships and restricted range of emotional expression. The patient prefers being solitary, showing little interest in close relationships, and usually not wanting children. - The patient is ego-syntonic about not having social interaction, in contrast to avoidant personality disorder (discussed below), where the patient is ego-dystonic about it.
Schizotypal	- Cognitive and perceptual distortions accompanied by unusual beliefs. - Patient may have superstitious thinking or fixation on astrology or Buddhist temples, but he/she doesn't qualify for any psychotic disorder. - Often experience socio-occupational dysfunction, where they might appear odd or peculiar to coworkers, clients, or others.
Cluster B (dramatic, emotional, erratic)	
Borderline	- Unstable personal relationships, self-image, emotions, and impulsivity. - Patient can demonstrate parasuicidal behavior, which are suicidal gestures without true intent to commit suicide, such as slicing/cutting of one's wrists and legs. - Patient may have history of being engaged to multiple prior partners. - Tx = dialectal behavioral therapy (DBT).
Histrionic	- Attention-seeking, eccentricity, sexual suggestibility. - Patient is uncomfortable when not center of attention, often using their appearance to draw attention. - USMLE might give a female who is highly flirtatious and sexually provocative with her physician. They can also give a male who wears bright colors (i.e., a 44-year-old man presents to emergency high-energy and dressed all in yellow).
Narcissistic	- Inflated sense of self-importance; deep need for excessive attention and admiration; lack of empathy for others. - USMLE can give vignette of a patient who castigates medical staff for a doctor not starting an appointment at the scheduled time.
Antisocial	- Persistent law-breaking and violation of the rights of others. - Antisocial means criminal, not aversion to socializing, and is frequently misused. - Diagnosis of antisocial personality disorder is the same as conduct disorder but age 18+, where the patient breaks the law / commits crime. Diagnosis also requires evidence of conduct disorder age 15 or younger. - Person who commits singular crime (e.g., murder, theft) demonstrates antisocial behavior but doesn't have the personality disorder, unless there is evidence of persistence since age 15 or younger. - Cases that are borderline (i.e., CD starting at age 16 or 17) won't get asked on USMLE, but in real life, either further history is obtained to determine any law-breaking activity age 15 or younger, or alternative diagnoses are considered. - USMLE can make the vignette a little more tricky, e.g., by saying a guy cheated on his bar exam to become a lawyer (which is a crime). - Same as with conduct disorder, USMLE might give you a mere snapshot of a person at one point in time, without giving you a longer-term pattern of behavior, and the answer is still ASPD. But for real-life diagnosis, the patient must have longer-term pattern of behavior.
Cluster C (anxious, fearful)	
Avoidant	- Extreme social inhibition, inadequacy, and sensitivity to rejection or negative criticism.

	- Patient might be reluctant to engage in new activities or meet strangers due to fears of inadequacy; patient might view him or herself as socially inept or unappealing. - Ego-dystonic, as mentioned before, where the patient desires social interaction but is unable to engage in it. This contrasts with schizoid, which is ego-syntonic, where the patient prefers not to have social interaction.
Dependent	- Pervasive/excessive need to be taken care of; clinging behavior and fear of separation. - Patient will often go to great lengths to avoid being alone, constantly seeking to be physically close to intimate partner.
Obsessive-compulsive (OCPD)	- Preoccupation with orderliness, perfectionism, and control; patient is overly focused on productivity to the detriment of personal relationships, often with insistence that his/her way of doing things is the only right way. - The key here is that the pattern of behavior causes detriment to socio-occupational functioning – e.g., a secretary insists on reading every letter she prints a minimum of 9 times before sending it off to check for spelling errors, but this creates inefficiency at the office. - Ego-syntonic, whereas OCD is ego-dystonic.

Autism spectrum disorder (ASD)
- Neurodevelopmental condition characterized by repetitive patterns of behavior, interests, or activities, and persistent challenges in social communication and interaction. - Usually starts before age 3; more common in boys. - Child demonstrates difficulty understanding social cues, displays atypical speech patterns or body language, and/or intensely focus on specific activities. - Interest in linear patterns – i.e., trains, train tracks, and organizing toys in straight lines, can be seen. - Repetitive motions (stimming), such as rocking back and forth, are characteristic. - Patients with ASD might have reduced IQ at the expense of enhanced abilities elsewhere (e.g., painting). - Asperger disorder is a variant of ASD where IQ is normal or increased.

Eating disorders	
Anorexia	- Intense preoccupation with maintaining low body weight and an intense fear of gaining weight. BMI must be low (i.e., <18.5) for diagnosis. - Purging can be seen (not limited to bulimia). - Vignette can give girl who runs 12 miles per day and barely eats. - USMLE likes metatarsal stress fractures due to decrease bone density. - The Q can ask what the patient is at risk for later in life → answer = osteoporosis (reduced adipose → reduced estrogen). - Amenorrhea in females can be seen but is no longer mandatory for diagnosis. USMLE wants “abnormal GnRH pulsation” as the mechanism for amenorrhea in anorexia, where LH and FSH are both decreased. This is called hypogonadotropic, or central, amenorrhea. - BMI <15 or overt malnutrition is indication for hospital admission. - Tx is CBT and/or SSRI. Mirtazapine is an α2-antagonist that stimulates appetite and is used for patients with depression and anorexia. - Avoid bupropion in patients with eating disorders (lowers seizure threshold). - Most common cause of death is ventricular fibrillation due to hypokalemia. - Olanzapine can be used in theory (anti-psychotic that ↑ appetite).
Bulimia	- Binge eating followed by compensatory behaviors such as vomiting, excessive exercise, or the use of laxatives. - BMI is normal or increased (unlike anorexia, where it must be decreased). - Tx is CBT and/or SSRI.
Binge-eating disorder	- Frequent episodes of consuming large amounts of food in a short time period, often in secret; usually accompanied by feelings of guilt or shame.

	- Unlike bulimia, these episodes are not regularly followed by purging, excessive exercise, or other compensatory behaviors. - Tx is CBT and/or SSRI.

Dissociative disorders	
Dissociative identity disorder (DID)	- Presence of two or more distinct identities that take control of the patient's behavior, with memory gaps occurring during these transitions. - Individuals with DID often report experiences of "time loss" or being told of actions they don't recall performing.
Dissociative amnesia	- Inability to recall important personal information, typically related to a traumatic or stressful event, that goes beyond repression. The amnesia can result in loss of identity and/or life history. - Dissociative fugue is a subtype of dissociative amnesia. - Fugue = memory loss + travel, where the patient will have traveled to a destination (e.g., parking lot 50 miles from his/her home + doesn't know who he/she is when the police arrive).
Depersonalization / Derealization Disorder	- Patients feel that they are detached from themselves (depersonalization) or that the world around them is unreal or dreamlike (derealization). - Experiences can be distressing, but the patient is aware that these perceptions are not reflective of reality.

Somatic symptom and related disorders	
- Symptom generation and motive are both unconscious.	
Somatic symptom disorder	- One or more chronic somatic symptoms (e.g., GI disturbance/pain, adnexal pain, shortness of breath, neuropathic pain) that are distressing or result in socio-occupational dysfunction, but in which medical examination and diagnostic interventions cannot find any abnormalities.
Conversion disorder	- Neurologic deficits (i.e., paralysis, loss of function, numbness) usually following a stressor, e.g., loss of job, argument with spouse, etc. - Patient has La belle indifference, which refers to apparent lack of care or concern about the deficit (i.e., the patient supposedly can't move his arm but doesn't seem fazed by it). - Example is a child being dropped off at summer camp + says he can't move his finger → child is likely stressed by being separated from parents. - Hoarseness of voice shows up on NBME form where answer is conversion disorder. - NBME also gives guy with blindness in one eye, but there's no relative afferent pupillary defect (indicating CN II isn't fucked up), so Dx = conversion disorder. If you're confused about RAPD, go to HY Neuro PDF.
Illness anxiety disorder	- Formerly known as hypochondriasis. - Preoccupation with fear of having or acquiring a serious illness; worries persist even after medical reassurances of no significant health concern/risk.

Factitious disorders	
- Symptom generation is conscious/intentional; motive is unconscious. - Production of symptoms in factitious disorders is driven by a need to be seen as ill or injured, rather than for a tangible benefit such as drugs, money, or time out of prison. - Seeking medical attention/care is known as attempting to seek "primary gain," whereas something concrete like cash or time out of prison is "secondary gain."	
Factitious disorder imposed on self	- Formerly known as Munchausen syndrome. - Intentional production / feigning of physical symptoms (e.g., GI pain).

	- The primary motivation is to assume the sick role and receive medical attention, care, or sympathy. - Willing to undergo invasive procedures. The Q might say patient has history of cholecystectomy or appendectomy.
Factitious disorder imposed on another	- Formerly known as Munchausen syndrome by proxy. - Feigning symptoms in another person in order to achieve medical attention. - Considered a form of abuse. Can occur with parents or caretakers toward their children, elderly, or those in their care. - Needs to be reported.
Factitious thyrotoxicosis	- Injection of thyroid hormone. Aka surreptitious thyrotoxicosis. - USMLE vignette will usually make this an endocrine-type Q (see HY Endocrine PDF), where it need not be attached to the patient's desire to seek medical care (i.e., the patient could just be attempting to lose weight), however some patients will inject as a means to acquire medical attention. - Q need not say patient is a pharmacist (students use that as crutch).
Factitious use of glycemic meds	- Use of insulin or hypoglycemic meds (e.g., sulfonylureas) to achieve hypoglycemia or weight loss. - USMLE will usually ask this in the setting of C-peptide levels (i.e., low C-peptide indicates exogenous insulin; high C-peptide means either secretagogue abuse like sulfonylureas, or insulinoma). - Some patients use these meds to achieve medical attention.

Malingering
- Symptom generation and motive both intentional. - Aim is secondary gain – i.e., money, drugs, time out of prison/school. - Patient is not willing to undergo invasive procedures. For example, a prisoner complains of severe RLQ pain and feigns a rigid abdomen, but once at hospital, refuses to undergo laparoscopic appendectomy.

Sleep disorders	
Narcolepsy	- Chronic sleep disorder affecting the brain's ability to regulate sleep-wake cycles normally. - Characterized by excessive daytime sleepiness, sudden episodes of muscle weakness (cataplexy), sleep paralysis, and hallucinations. The latter tend to occur prior to sleep (hypnagogic) or upon waking up (hypnopompic). - Thought to be due to deficiency of a neurotransmitter called orexin (aka hypocretin), which normally promotes wakefulness. - Treatment is modafinil (dopamine reuptake inhibitor + promotes release of orexin from hypothalamus).
Sleep apnea	- Can be obstructive (i.e., usually from obesity) or central (i.e., brain-related). - Chronic fatigue and poor oxygenation can lead to dysthymia / depression. The answer on NBME is "mood disorder due to a medical condition." - Polysomnography (sleep study) is what USMLE wants to diagnose. - When obstructive sleep apnea progresses to the point that the patient is a chronic CO_2 retainer with pulmonary hypertension and/or cor pulmonale, we call it obesity hypoventilation syndrome (Pickwickian syndrome). If you're confused about the cardio, go to the HY Cardio PDF.
Circadian rhythm sleep-wake disorder	- Misalignment between the individual's sleep pattern and the societal norm or natural environment. - Common types include delayed sleep phase disorder (going to bed / waking up later than desired), advanced sleep phase disorder (going to

	bed / waking up earlier than desired), and shift work disorder (struggle with sleep due to abnormal work hours).
Somnambulism	- Sleep walking. - Occurs during Stage 3 (deep) sleep.
Nightmare disorder	- Occurs during REM sleep. - Dreams often involve threats to survival or self-esteem, leading to awakenings and distress. - Episodes can be remembered by the patient after awakening since they occur during REM sleep.
Sleep terror disorder	- Often mistaken for nightmares; more common in male children and abates by pre-adolescence. - Occur in Stage 3 (delta wave) sleep. - Episodes where the patient somnambulates and can wake up at different location, often in a panic; can demonstrate violent behavior, such as pulling on the mother's arm to go outside, followed by awakening from the episode with no memory of what happened. - This is in contrast to nightmare disorder, where the patient remembers a dream + cannot somnambulate due to muscle paralysis during REM.
Restless leg syndrome	- Idiopathic, irresistible urge to move legs while in bed/sleeping. - Most common cause is iron deficiency anemia. First step is checking the patient's serum iron and ferritin. - If iron studies are normal, gabapentin and D2 agonists (ropinirole, pramipexole) can be used. - USMLE wants you to know that patients with RLS have increased risk of developing Parkinson disease, which makes sense since D2 agonists help, indicating a potential problem with dopamine signaling or production in some patients.
Insomnia	- Inability to obtain adequate sleep, either due to difficulty falling asleep, staying asleep, or waking up too early, leading to impaired daytime functioning. - USMLE wants benzo (usually clonazepam) for acute insomnia. - Terminal insomnia can occur in depression (waking up too early).

Miscellaneous body disorders	
Trichotillomania	- Pulling out one's hair; can be anywhere (i.e., eyebrows, scalp, etc.). - Patient may sometimes also eat his/her hair, leading to gastric bezoar (asked on NBME exam). This is an undigestible mass that can lead to bowel obstruction.
Skin-picking disorder	- Aka excoriation disorder. - Patient picks at his/her skin, leading to skin lesions.
Body dysmorphic disorder	- Excessive preoccupation with one or more aspects of one's appearance, leading to socio-occupational dysfunction.
- CBT can be attempted for all of these conditions, otherwise anti-depressants like SSRIs can be tried.	

Tic disorders	
Tourette	- Neurologic disorder characterized by repetitive, involuntary movements and vocalizations (i.e., tics). - Motor tics range from mere eye blinking, head jerking, or shoulder shrugging to more complex movements such as hopping, twirling, or even mimicking others' actions. - Vocal (phonic) tics include throat clearing, sniffing, or humming. More complex vocal tics can involve repeating words or phrases, or shouting inappropriate words (coprolalia).

	- Haloperidol (typical anti-psychotic; D2 antagonist) and clonidine (α2 agonist) are two agents used to Tx that show up on NBME.
Provisional tic disorder	- In pediatrics it is normal for the kid to sometimes have a singular tic that lasts up to a year (e.g., unusual tongue or lip gesture). - There is very low chance of it progressing to chronic tic disorder / Tourette. - USMLE wants follow-up in 3-6 months as the next best step (i.e., observe).
PANDAS	- Pediatric Autoimmune Neuropsychiatric Disorder Associated with Streptococci, where Group A Strep pharyngitis can cause a tic disorder, OCD, or ADHD in the weeks following infection. - USMLE will give you kid who had sore throat two weeks ago + now has a new-onset tic, ADHD, or OCD, and they will ask what will most likely diagnose etiology for the disorder → answer = "anti-streptolysin O titers."
Drug-induced	- Stimulants such as methylphenidate, amphetamines, and caffeine can cause tics. - Tardive dyskinesia due to anti-psychotics is considered a type of tic disorder.

Important developmental milestones

- What USMLE will do is ask "normal" or "delayed" for Gross motor, Fine motor, Verbal, and Social.
- Highest yield points regarding milestones:
  - Gross motor:
    - Head up at 3 months (can support his/her own head unassisted).
    - Turn over front to back at 4-5 months.
    - Sits up at 6 months.
    - Crawls by 9 months.
    - Stands/walks alongside sofa at 10-11 months.
    - Walks at 12-15 months.
  - Fine Motor:
    - Pincer grasp at 9 months.
    - Stacks 3, 6, 9 blocks at age 1, 2, 3, respectively.
    - Copies circle at 3-4 years.
    - Copies square at 4-5 years.
    - Copies triangle at 5-6 years.
  - Language development:
    - "Mama" and "dada" by 1 year.
    - Recognizes own name by 1 year.
    - 2-word sentences at age 2.
    - 900 words at age 3.
  - Social:
    - Social smile at 2-3 months.
    - Stranger anxiety at 6-9 months.
    - Separation anxiety at 9-14 months.
    - Imaginary friends are normal at age 4-6 (Peds shelf likes this; no this is not psychosis).
    - Repeated questioning about death is normal age 8-10.
    - Parallel play age 2-3 (kids play side by side); cooperative play age 3-4.

Important child abuse findings

- Spiral fractures (rotational/twisting force applied to bone).
- Posterior rib fractures (squeezing).
- Circular burns (cigarettes).
- Burns sparing flexor regions (from being dipped in hot water [child flexes limbs to ↓ exposed surface area).
- Retinal detachment / hemorrhages.
- Subdural hematoma (shaken baby syndrome).
- **Avoidance of eye contact** (if they say Hx of meningitis, think instead neurosensory hearing loss).

Peds bladder/bowel control	
Enuresis	- Nocturnal enuresis (i.e., wetting the bed) is normal on occasion up through age 5. - First step in ↓ recurrence is behavioral science-type answers (i.e., ↓ stressors + ↑ time spent with child). Q might say child wets the bed after a sibling is diagnosed with ALL (i.e., stressor). - After behavioral-type answers, the next step is star chart (positive reinforcement), where the child receives a star on a chart every time he/she doesn't wet the bed, where after, e.g., 10 stars, he/she gets an extra dessert. - If star chart is not listed, enuresis alarm is the next best step (i.e., literally an alarm that detects moisture in a child's underwear, typically using a sensor, where if moisture is detected, indicating urination, the alarm sounds or vibrates child awake, thereby encouraging him or her to stop urinating and go to instead use the toilet). - Answers such as imipramine, desmopressin, and water deprivation are wrong on USMLE.
Encopresis	- Shitting your pants. Normal on occasion up through age 5.

Repro/Obgyn Psych conditions	
Depression during pregnancy	- Sertraline is first SSRI used due to lower risk profile. - Electroconvulsive therapy (ECT) is used for women who have suicidal/homicidal ideation, or who are refusing to eat or drink. - ECT is generally considered to be safe for the fetus when performed appropriately and with careful precautions.
Bipolar during pregnancy	- Avoid lithium because of its teratogenic effects (Ebstein anomaly). - Avoid valproic acid because it causes neural tube defects and cognitive impairment. - Bipolar is Tx in women planning to get pregnant, or during pregnancy, with agents such as lamotrigine, olanzapine, or quetiapine.
Postpartum blues	- Aka "baby blues"; normal, transient postpartum psychologic state characterized by sadness, anxiety, mood swings, and insomnia. - Peaks in first week. If >10 days on USMLE, think postpartum depression instead. - Resolves without intervention. - Thought to be related to the sudden hormonal shifts and physical and emotional adjustments postpartum.
Postpartum depression	- Characterized by persistent sadness, anxiety, and exhaustion that can hinder daily care activities and bonding with the baby. - The answer on USMLE if the vignette mentions thoughts of worthlessness or guilt, especially after 10 days postpartum. - Vignette might say the woman leaves her baby in crib alone crying for long periods of time, or the child has soiled diapers (i.e., not catered to) → answer = "immediate psychiatric referral." - If SSRI is used, sertraline or paroxetine are often used because of ↓ concentration in breast milk.
Postpartum psychosis	- Extreme mood swings, hallucinations, paranoia, and attempts to harm oneself or the baby; occurs in first few weeks postpartum. - Answer = immediate psychiatric referral. - Medications used are anti-psychotics, usually olanzapine or risperidone.
Premenstrual syndrome	- Mood swings, breast tenderness, and irritability occurring in the luteal phase (i.e., the two weeks prior to menses). - Common; benign; thought to be due to hormonal fluctuations.
Premenstrual dysphoric disorder	- Severe, sometimes incapacitating, form of premenstrual syndrome. - Characterized by significant mood disturbances and physical symptoms that dramatically interfere with socio-occupational functioning. - Also thought to be due to hormonal fluctuations. - SSRIs are often first-line.

HY dementia types	
- General term for loss of memory, language, and problem-solving abilities that interfere with daily life.	
Alzheimer	- Gradual-onset idiopathic cognitive decline. - Patient must have normal neurologic exam (i.e., no motor or sensory abnormalities). - MMSE score will be low (i.e., low-20s out of 30), and the patient will put in effort during the test. - Beta-amyloid plaques and neurofibrillary tangles (hyperphosphorylated tau protein) seen on brain biopsy. - Early-onset Alzheimer in Down syndrome (amyloid precursor protein gene is located on chromosome 21). - Presenilin gene mutations can cause Alzheimer (on NBME exam). Presenilin is a protein involved in the cleavage of amyloid. - Tx = cholinesterase inhibitors (donepezil, galantamine, rivastigmine); memantine (NMDA glutamate receptor antagonist) can also be used. - Sundowning is worsening of dementia at night that can resemble delirium. - 1st-line Tx for sundowning on NBME is “decrease ambient noise and distractions.” “Bright illumination of the room at all times” is wrong answer.
Frontotemporal dementia	- Aka Pick disease. - Presents usually as triad of 1) personality change, 2) apathy, and 3) disinhibition. - Accumulation of hyperphosphorylated tau protein (similar to Alzheimer), except rather than accumulating as neurofibrillary tangles, it accumulates as round, silver-staining inclusions knowns as Pick bodies.
Lewy body dementia	- Dementia + visual hallucinations + Parkinsonism. - Lewy bodies are collections of alpha-synuclein. This protein is deposited throughout the brain in Lewy-body dementia. In Parkinson disease, in contrast, it is deposited primarily in the substantia nigra pars compacta of the midbrain.
Vascular dementia	- Aka multi-infarct dementia. - The answer for dementia + motor/sensory abnormalities. - Seen in patients who have repeated mini-strokes (cerebral infracts) due to hypertension. - Resources tend to focus on this notion of “step-wise decline” (i.e., concrete timepoints at which deficits started), but it’s to my observation on NBME exams that this is rarely a salient aspect of Qs. What USMLE likes is giving motor and/or sensory deficits – i.e., you’ll get a big paragraph with dementia, and you’ll notice somewhere in the stem that the patient has, e.g., 3/5 strength in the right upper extremity. This indicates Hx of stroke.
AIDS	- Just be aware AIDS can cause dementia, known as AIDS complex dementia. - Can present as “wet, wobbly, wacky,” similar to normal pressure hydrocephalus.
Subacute combined degeneration	- The fancy name for neurologic degeneration seen in B12 deficiency. - Can present sometimes as a reversible cause of dementia. In elderly patients on tea and toast diets, or those in high-risk groups (i.e., vegans, pernicious anemia), B12 must be considered as cause of cognitive decline. - The patient can have peripheral neuropathy as a result of deficits to the 1) corticospinal tracts, 2) dorsal columns, and 3) spinocerebellar tracts. - The easy way to remember those three is to start by saying, “The spinothalamic tract is not involved.” Then you say, “Well what are other ones I can think of?”
Neurosyphilis	- Just be aware that neurosyphilis is a reversible cause of dementia and should be considered. There’s an NBME Q floating around for 2CK where they give (+) VDRL in 82-year-old woman with cognitive decline, and the treatment is penicillin.
Pseudodementia	- Not actual dementia. This is depression that presents as cognitive decline. - Patients with depression who have apathy will perform poorly on the MMSE. - The Q might say the patient is unable to draw a clockface, but when prompted, is able to finish it quickly. They might also say patient remembers 0 out of 3 objects after 5 minutes. - Look for obvious signs of depression, such as short, quiet answers and low mood.

Benign senility	- If the diagnosis is benign senility, MMSE will usually be 28+ on USMLE. - This contrasts with true dementia, where USMLE will give MMSE low-20s. - One of the highest yield points is you knowing that *if the patient complains, it's not dementia.* In real life, this is probably not a 100% rule, but on USMLE, it's HY way to distinguish. - Patients with true dementia either won't complain about it or just simply won't be aware of their cognitive decline. If the Q tells you the patient is concerned because she walked into a room and doesn't know why she went in there, it's benign senility, not dementia. - For true dementia, the vignette might say the adult daughter reports her mother left the stove burner on the other day and is unconcerned about it, or that she went for a walk the other day and she got lost and it took her hours to come home, or that the police brought her home.

Parkinson disease / Parkinson-plus disorders	
- A Parkinson-plus disorder is a disease that presents similarly to Parkinson disease, but it's not.	
Parkinson disease	- Loss of dopamine-secreting neurons in the pars compacta of the midbrain, with deposition of alpha-synuclein on biopsy. - Presents with classic features of bradykinesia/akinesia, resting tremor, shuffling, short-steppage gait, micrographia, and cogwheel rigidity. - Alpha-synuclein gene mutation most common; many genes implicated. - Carbidopa-levodopa is combo frequently used for Tx. Levodopa crosses the BBB to be converted to dopamine centrally. However, levodopa is subject to fast metabolism when administered alone. The addition of carbidopa functions as a competitive inhibitor of breakdown enzymes, resulting in increased levodopa availability for passage across the BBB. Do not confuse this mechanism with direct COMT inhibitors (tolcapone, entacapone), which prevent breakdown of L-dopa. - Carbidopa-levodopa can cause psychosis if administered in too-high a dose. This is assessed on 2CK Psych forms, where if patient gets psychotic episodes following recent addition of C-L to regimen, or following an increase in dose, the answer is simply "decrease dose of carbidopa-levodopa." "Discontinue carbidopa-levodopa is the wrong answer." - Be aware of D2 agonist names – i.e., ropinirole, pramipexole, cabergoline, pergolide, and bromocriptine, which can all in theory be used as treatments. - Amantadine increases presynaptic release of dopamine. - Selegiline inhibits monoamine oxidase B, which is an enzyme that preferentially breaks down dopamine. USMLE wants you to know this can cause serotonin syndrome, either alone, or in combo with drugs like St John Wort or SSRIs.
RLS	- Restless leg syndrome. - Idiopathic, irresistible urge to move legs while in bed/sleeping. - Most common cause is iron deficiency anemia. First step is checking the patient's serum iron and ferritin. - If iron studies are normal, gabapentin and D2 agonists (ropinirole, pramipexole) can be used. - USMLE wants you to know that patients with RLS have increased risk of developing Parkinson disease, which makes sense since D2 agonists help, indicating a potential problem with dopamine signaling or production in some patients.
NPH	- As discussed earlier, normal pressure hydrocephalus presents as "wet, wobbly, wacky" +/- Parkinsonism. - The parkinsonism is the most overlooked detail by students, who will usually only know the wet, wobbly, wacky part. - What the USMLE will do is give you an older male who has WWW triad + they give you 2-3 more sentences describing what sounds like Parkinson disease, where

	you're like "What the hell? Is this Parkinson disease?" No. It's just NPH, which is a Parkinson-plus disorder, where it can look like Parkinson disease but it ain't. - Once again, due to impingement on the corona radiata, and the mechanism they want for urinary incontinence = "failure to inhibit the voiding reflex."
Wilson disease	- Parkinsonism in a young patient is Wilson disease till proven otherwise. - Excessive copper accumulation in tissues, including the basal ganglia and liver, due to inability to excrete it into bile. - I discuss this stuff in more detail in the HY Gastro PDF.
Lewy body dementia	- As discussed before, this is a Parkinson-plus disorder. - The patient will have dementia + visual hallucinations + Parkinsonism.
Progressive supranuclear palsy	- Obscure condition that gets asked on 2CK. - You just need to know that 100% of questions will say "axial dystonia + Parkinsonism," where they'll just ask for diagnosis straight-up. - Axial dystonia is a type of muscle condition resulting in abnormal posture and movement of the spine and torso.
MPTP	- Aka synthetic heroin. - Just know it is a cause of Parkinsonism. - Shows up on a 2CK Psych form. Students are like wtf?

Psychosomatic disorders	
Irritable bowel syndrome (IBS)	- Don't confuse with inflammatory bowel disease (IBD). - IBS is psychosomatic (i.e., psych-related) condition. - Classic vignette will be a woman 20s-40s with stress factors who usually has alternating diarrhea / constipation; can also present as bloating or cramping. - Key detail is that symptoms are relieved with bowel motions. - NBME assesses "smooth muscle hypersensitivity" as the answer for the mechanism for IBS. - Treatment for diarrhea-predominant IBS is loperamide, which is an opioid that causes constipation. USMLE will give vignette of IBS with diarrhea, and then the answer is just "mu-opioid receptor agonist."
Chronic interstitial cystitis	- Not an infection. - This is >6 weeks of suprapubic tenderness + dysuria (pain with urination) that is unexplained, where laboratory and urinary findings are negative. - They can mention **anterior vaginal wall pain** (bladder is anterior to vagina). - USMLE wants you to know you **don't treat.** Steroids are wrong answer. "Treatment" is standard placating placebo nonsense such as "education," "self-care" and "physiotherapy."
Fibromyalgia	- This is a psych condition, not an actual muscle disorder, but is often confused with polymyositis and polymyalgia rheumatica (see HY Anatomy/MSK PDF). - Labs will be normal. ESR will not be elevated. Patient will not have fever. - Will be described as woman 20s-50s with multiple (and often symmetric) muscle tenderness points. - Treatment is usually SNRIs or TCAs. USMLE can write this as "anti-depressant therapy." This confuses students ("But she doesn't have depression though.") → Right. But SNRIs and TCAs are still anti-depressant medication.
Pseudocyesis	- Aka false or phantom pregnancy. - Woman believes she is pregnant when she's not, where she will exhibit many of the signs and symptoms of pregnancy, such as amenorrhea, abdominal bloating, and breast enlargement/tenderness. - Can be idiopathic or induced by stressor (e.g., trauma of prior miscarriage).

Delirium	
- Acute disturbance in attention and cognition, usually over hours to days. - Most commonly presents as hyperactive delirium, which is where the patient appears confused, restless, agitated, and combative. - Can also present as hypoactive delirium, where the patient mute, lethargic, or slow to respond to stimuli. - Important causes are infection, electrolyte disturbance (e.g., hypercalcemic crisis), medications (especially first-gen H1 blockers like diphenhydramine), and metabolic causes (e.g., hypercarbia in COPD exacerbation).	
Hypercalcemic crisis	- Refers to cognitive dysfunction / a delirium-like state in the setting of severe hypercalcemia, often due to malignancy or primary hyperparathyroidism. - I've also seen this once in a patient on a thiazide (can cause hypercalcemia). - USMLE wants you to know that high calcium, as well as any sodium disturbance, can cause delirium. - First step on USMLE for Tx of hypercalcemia is normal saline. - After normal saline, USMLE wants bisphosphonate therapy (I've seen pamidronate listed on NBME). - I've never seen calcitonin or loop diuretics as correct answers for hypercalcemia Tx. They're always wrong.
Medication-induced delirium	
Reye syndrome	- Acute encephalopathy and hepatic dysfunction caused by giving aspirin in the setting of a viral illness and/or fever. - Aspirin should be avoided under age 12, and some sources say avoid in teens for that matter. - Mechanism is obscure and thought to be related to impairment of β-oxidation.
Anti-cholinergic delirium	- Anti-cholinergic medications can cause confusion and a delirium-like presentation. Children have ↑ susceptibility. - Q can mention a kid was given an over-the-counter medication and now has low-grade fever and confusion. - Details such as flushed, dry, warm skin, or enlarged pupils, suggest anti-cholinergic delirium over Reye. - Can be confused with Reye syndrome if they just say "over-the-counter" med. - 1st-gen H1-blockers (i.e., diphenhydramine) are notoriously anti-cholinergic.
Dextromethorphan	- Can cause delirium and psychosis in children when taken in excessive amounts. - An opioid that is an anti-tussive (cough suppressant).
Alcohol withdrawal	
Alcoholic hallucinosis	- Presents 12-48 hours after the last drink. - Patient can have hallucinations, usually tactile (i.e., bugs crawling on skin) or visual (bugs crawling up wall). - Treat with benzo + thiamine.
Delirium tremens	- Presents 2-4 days following last drink. - Patient will have tremulousness, tachycardia, diaphoresis, and restlessness. - Seizures can also sometimes occur. - USMLE-favorite vignette is 40s male who gets tremulousness and tachycardia while in hospital 2ish days after surgery. Answer is just benzo. - Can also show up on NBME as a guy who goes from drinking 12 beers a day to suddenly only 2 beers a day. - Treat with benzo + thiamine.

Indications for treating minors without parental consent	
- It should be noted that in the USA, policies differ by state. The following are generic rules for Family Med.	
Sexual health	- STI treatment; birth control; pregnancy-related care.
Substance use	- Substance abuse Tx (i.e., drugs and alcohol).
Mental health	- Access to counseling. - Only some US states allow prescription psych meds without parental consent.
Emergency care	- Parental consent not required for exigent care. - Blood transfusions are given to Jehovah witness minors in life-saving situations.
Emancipated, Mature minors	- Refers to minors who can consent to medical Tx in the same manner as adults. - Emancipation can occur through the legal process, marriage, military service, or other specific state-defined circumstances. - Some states allow practitioners to determine if a minor has the maturity to consent to certain medical Tx; requires assessment of the minor's age, ability to understand the Tx, and the nature and risks of the proposed Tx.

Drug abuse	
Glue	- Ataxia + cognitive decline in teenager. - Will sound a bit like alcohol abuse but the effects of alcohol don't occur so young.
Paint	- Cognitive decline. - Q can say teenager is seen with gold or silver coloration around the nose/mouth.
Butane (inhalant)	- Cognitive decline. - Classically inhaling computer cleaner (dusters) or whipped cream bottles. - Q will say high schooler found on floor in school bathroom + is brought into ED sluggish + pupils and vitals all normal → answer = butane.
Caffeine	- Most common drug addiction in the world. - Adenosine accumulates in the brain throughout the day and causes sense of fatigue. - Caffeine blocks adenosine receptors, promoting sense of wakefulness. - Intoxication can cause sense of over-stimulation, panic, and palpitations. - Withdrawal can cause headache, sense of depression, fatigue, anxiety (i.e., sense of worry/doom), and inability to concentrate.
Smoking/vaping	- Nicotine can promote sense of euphoria. - Withdrawal can cause many symptoms, including anxiety, depression, difficulty concentrating, and weight gain due to increased appetite.
Marijuana	- Injection (redness) of conjunctivae + dry mouth. - ↑ risk of developing psychosis and schizophrenia.
Cocaine	- Mydriasis, tachycardia. - High BP causing aortic dissection; can cause chest pain (coronary vasospasm). - Abruptio placentae if pregnant teens. - Give benzo if acutely intoxicated + observe in emergency.
Amphetamine	- Mydriasis, agitation, **insomnia** / staying up all night. - Can cause tactile hallucinations. - Give benzo if acutely intoxicated + observe in emergency.
PCP	- Bellicosity / pugnacity (if you're ESL, those mean wanting to fight + aggressive). - **Nystagmus** +/- mydriasis. - One 2CK NBME Q gives mutism + constricted pupils for PCP + nothing about pugnacity, so just be aware this presentation is rare but possible. - Give benzo if acutely intoxicated + observe in emergency.
MDMA (ecstasy)	- Euphoria, heightened sensory perception, low-grade fever, bruxism (teeth grinding). - An NBME Q floating around gives increased creatine kinase, so this is also possible. - Give benzo if acutely intoxicated + observe in emergency.
LSD (acid)	- Visual hallucinations. - Give benzo if acutely intoxicated + observe in emergency.
MPTP	- Synthetic heroin. - Causes Parkinsonism.

	- Shows up on a 2CK form, so if you think it's weird, take it up with NBME, not me.
Heroin/opioids	- E.g., oxycodone or dextromethorphan. - Respiratory depression + constricted pupils + constipation. - Naloxone (opioid receptor antagonist) for acute toxicity. - Methadone (opioid receptor agonist) to ↓ relapses.
Benzos	- E.g., diazepam. - Respiratory depression. - Flumazenil to treat acute toxicity (benzodiazepine receptor antagonist).
Barbiturates	- Respiratory depression. - Q will say naloxone and flumazenil had no effect, so you eliminate to get to barbiturates.
Acetaminophen	- Fulminant liver failure. - Give activated charcoal if ingested within 1-2 hours. - *N*-acetylcysteine must be given after to regenerate reduced glutathione to prevent liver damage from NAPQI (acetaminophen metabolite).
Aspirin	- Tinnitus + mixed metabolic acidosis-respiratory alkalosis. - Give sodium bicarb to treat (↑ excretion through urinary alkalinization).
TCAs	- E.g., amitriptyline. - CCCs (coma, convulsions, cardiotoxicity). - ECG changes seen frequently in vignettes. For example, you'll get a big vague paragraph about some drug overdose + they say in last line QT is prolonged → answer = the TCA. Then you remind student about cardiotoxicity and they're like Oh yeah. - Anti-cholinergic effects (i.e., delirium + hot, red, dry patient).

Depression meds	
SSRIs	- Selective-serotonin reuptake inhibitors. - Fluoxetine, escitalopram, sertraline, etc. - Cause sexual dysfunction (anorgasmia) and sleep disturbance. - The fact that they cause anorgasmia actually makes them the Tx for premature ejaculation. - Do not combine with drugs such as monoamine oxidase inhibitors or St John wort, as this can cause serotonin syndrome (discussed below). - There are unique side-effects of various SSRIs – e.g., sertraline is more likely to cause diarrhea; fluoxetine has a stimulating effect and is more likely to cause insomnia; citalopram can prolong QT interval at higher doses. But USMLE doesn't give a fuck. - Used for a variety of psych conditions external to depression, e.g., fibromyalgia, OCD, etc. - It can take 4-6 weeks for an SSRI to achieve desired effect. If after this time point the drug isn't working, the first step is increasing the dose. If this doesn't work, the next step is switching to a different SSRI, followed by switching to a different class agent.
SNRIs	- Serotonin and norepinephrine reuptake inhibitors. - Desvenlafaxine, duloxetine, etc. - Can increase blood pressure at higher doses.
TCAs	- Block reuptake of both serotonin and norepinephrine. - Amitriptyline, nortriptyline, clomipramine, doxepin, etc. - High-yield on USMLE as 1st-line for diabetic neuropathic pain; can also be used for neuropathic pain in general (e.g., from trauma). Gabapentin is otherwise frequently used. - Have nasty anti-cholinergic side-effects (anti-DUMBBELSS; see HY Neuro PDF for discussion of anti-cholinergic vs pro-cholinergic effects if you're confused). - Three HY anti-cholinergic side-effect vignettes are 1) palpable suprapubic mass in an older male → full bladder as a result of anti-cholinergic med + BPH. 2) Hot, red, dry patient (as a result of anhidrosis). 3) Confusion + dilated pupils (anti-cholinergic delirium + mydriasis). - Coma, convulsions, cardiotoxicity.

	- Sometimes the Q can just mention prolonged QT interval in patient on an anti-depressant, and the answer is the TCA, since they're cardiotoxic. - If elderly, use nortriptyline, since ↓ BBB penetration and anti-cholinergic side-effects. - TCAs, such as imipramine, for nocturnal enuresis are wrong on USMLE. - Doxepin is asked on 2CK Psych form, where the Q rides on you knowing it's a TCA to get it right (i.e., the only drug listed that causes anti-cholinergic effects). I mention this because students get the Q wrong and then are like wtf is doxepin.
MAOIs	- Monoamine oxidase inhibitors. - Phenelzine, tranylcypromine. - Avoided 1st-line overwhelming majority of the time because of ↑ risk of serotonin syndrome. This can be in isolation, but also when commenced too soon following discontinuation of SSRI. - Selegiline is MAO-B inhibitor used for Parkinson disease. Answer on NBME where they ask which Parkinson med caused serotonin syndrome in a patient.
Bupropion	- Dopamine + norepinephrine reuptake inhibitor; has anti-depressant effects. - Also antagonizes nicotinic receptors, which disincentivizes smoking, since doing so won't produce rewarding effects. - Lowers seizure threshold. Don't give to patients with eating disorders (electrolyte abnormalities increase seizure risk). - Doesn't cause sexual side-effects, unlike SSRIs.
Mirtazapine	- α2-antagonist; stimulates appetite; used for patients with depression and anorexia.
Trazodone	- Serotonin antagonist and reuptake inhibitor (SARI). - Used for depression, but is frequently used for insomnia due to strong sedative effects. - Can cause serotonin syndrome.

Serotonin syndrome
- Flushing, tachycardia, diarrhea following drug-drug interactions (i.e., MAOI + SSRI; SSRI + St John wort) or high-risk drugs in isolation (e.g., MAOI, trazodone, lithium). Lithium sounds unusual, but on new 2CK NBME. - Can cause high fever, i.e., 105F+. - Diagnose with urinary 5-hydroxyindole acetic acid (5-HIAA). - Can be treated with cyproheptadine (blocks serotonin receptors). - Often confused with carcinoid syndrome, which is due to carcinoid tumors (serotonin-secreting tumors) of the lung, small bowel, or appendix. Carcinoid syndrome isn't due to drugs. Additionally, it can cause tricuspid vegetations, whereas serotonin syndrome does not. It is still diagnosed with urinary 5-HIAA.

Anti-psychotic meds
- Categorized as typical (antagonize D2 receptors) or atypical (antagonize mostly D2 receptors, but also have additional binding effects at other receptors such as D4 or serotonin). - HY Typicals: Haloperidol, Chlorpromazine, Prochlorperazine, Thioridazine. - HY Atypicals: **O**lanzapine, **C**lozapine, **Q**uetiapine, **R**isperidone, **A**ripiprazole, **Z**iprasidone – i.e., **O**ld **C**losets **Q**uietly Whisper (**R**isper) from **A** to **Z.** - All anti-psychotics can cause extra-pyramidal side-effects (EPS), anti-cholinergic side-effects, neuroleptic malignant syndrome (NMS), QT prolongation, and hyperprolactinemia (D2 agonism normally inhibits prolactin). - Typicals are older and more known for causing these effects. As a result, atypicals are usually the preferred first-line agents for schizophrenia and psychotic disorders. - Typicals like haloperidol are still occasionally used acutely for violent delirium, Tourette, or psychosis patients who have poor medication compliance. A 2CK form wants haloperidol decanoate as the answer for what's good to use for schizophrenia patient with poor compliance. - Prochlorperazine can be used as an anti-nausea medication despite being a typical anti-psychotic. - **Clozapine can cause neutropenia (agranulocytosis).** This is exceedingly HY. - Olanzapine and clozapine can worsen obesity. Aripiprazole or ziprasidone are better for high-BMI patients.

- Chlorpromazine can cause Corneal deposits. Thioridazine can cause reTinal deposits.

Extrapyramidal side-effects
- The name for the movement disorders associated with anti-psychotic use.
- "Rule of 4s" → After a patient is started on an antipsychotic, a general trend is seen in terms of the onset of particular symptoms. The time frame is not strict/rigid; use it as a general trend – i.e., acute dystonia wouldn't just start at 4 months; tardive dyskinesia wouldn't occur as early as 2 weeks. **- Acute dystonia at 4 hours** → torticollis, oculogyric crisis, muscle rigidity **without** fever. - Torticollis = stiff / crooked neck. - Oculogyric crisis = weird eye movements (don't confuse with tongue movements of tardive dyskinesia). - Muscle rigidity **without** fever = acute dystonia. Muscle rigidity **with** fever = neuroleptic malignant syndrome. - Treat acute dystonia with **benztropine or trihexyphenidyl** (muscarinic receptor antagonists, which decrease muscle tone) or a 1st generation H1 blocker (**diphenhydramine** or **chlorpheniramine**). The latter have nasty anti-cholinergic (anti-muscarinic) side-effects that are *actually what we want* when we're treating acute dystonia. Maybe 2/3 of acute dystonia Tx Qs will have benztropine as the answer; ~1/3 will have one of the 1st gen H1 blockers as correct. **- Akathisia at 4 days** → restlessness. - Treat with propranolol (beta-blockade). **- Parkinsonism at 4 weeks** → akinesia / bradykinesia. - Treat with amantadine. **- Tardive dyskinesia at 4 months** → abnormal facial movements (notably tongue). - Risk is greater with typicals compared to atypicals, but TD can be seen in the latter on NBME. - Treatment is stop the typical and give an atypical. - Couple of weird Psych NBME Qs on this: - Patient on *atypical* + gets TD; Tx? → stop the atypical and give another atypical. (Just stop the drug and switch to yet another atypical.) - Patient on typical for 15 years and has no problems whatsoever (i.e., does not have TD); patient asks the psychiatrist what can be done to decrease his risk of developing TD; the correct answer is "stop the typical and give an atypical"; wrong answer is "maintain current drug regimen." Apparently even if the patient has been on a typical long term without an issue, switching to an atypical *still* confers a reduction of risk of TD. - Metoclopramide (D2 antagonist used as anti-emetic / pro-kinetic) can also cause EPS side-effects and prolong QT interval, same as the anti-psychotics. There is 2CK Psych Q where they give Parkinsonism in patient on metoclopramide, and the next best step is "discontinue metoclopramide."

Anti-cholinergic meds for FM	
Muscarinic receptor antagonists	
Benztropine, trihexyphenidyl	- Used for acute dystonia due to anti-psychotics. - In other words, if patient starts anti-psychotic and then within hours to days gets stiff neck (torticollis), muscle rigidity (without fever), and/or abnormal eye movements (oculogyric crisis), this is 1st-line Tx. - Muscle rigidity with fever, in contrast, suggest neuroleptic malignant syndrome.
Diphenhydramine	- 1st generation H1-blocker that has strong anti-cholinergic side-effects to the point that it is often just characterized straight-up as a primary anti-cholinergic med. - Can be used same as benztropine for acute dystonia (USMLE won't list both). - Can be used for motion sickness (anti-cholinergic effects treat motion sickness). If USMLE tells you it is used in this setting + they ask you antagonism of which receptor enables its anti-motion sickness effect, the answer is M3. The wrong answer is H2, since this refers to stomach acid secretion. Diphenhydramine is an H1 blocker, not H2.

	- Most Qs on USMLE regarding diphenhydramine revolve around you knowing it should be discontinued from a patient's regimen, rather than added.
Prochlorperazine	- Anti-psychotic used as an anti-nausea med. - Even though it is an D2-receptor antagonist, as I mentioned before, anti-psychotics have anti-cholinergic side-effects. In this case, the anti-muscarinic side-effects are actually a "good" thing since these treat nausea and motion sickness.
Scopolamine	- Scopolamine patches are used for motion sickness (i.e., put it on the arm before getting on a boat or plane). This drug is an anti-cholinergic straight-up. - As mentioned above, oral diphenhydramine can also be used for motion sickness, but scopolamine is the classic treatment.
Hyoscyamine	- Used sometimes for irritable bowel syndrome.
Smoking cessation agents	
Bupropion	- As mentioned earlier, D + NE reuptake inhibitor; has anti-depressant effects. - Also antagonizes nicotinic receptors, which disincentivizes smoking, since doing so won't produce rewarding effects. - Lowers seizure threshold. Don't give to patients with eating disorders (electrolyte abnormalities increase seizure risk). - Doesn't cause sexual side-effects, unlike SSRIs.
Varenicline	- Used for smoking cessation. - Partial agonist at nicotinic receptors, but since this effect is less than endogenous ACh, it functions like an antagonist.

Epilepsy/hypnotic agents for FM	
- USMLE doesn't care about MOAs for these. They care about side-effects mostly, as per my observation.	
Valproic acid	- Blocks sodium channels and ↑ GABA. - Used as an alternative to lithium in Tx of bipolar disorder. - Causes neural tube defects in pregnancy due to interference with folate metabolism.
Carbamazepine	- Blocks sodium channels. - Generic anti-epileptic. - Causes neural tube defects in pregnancy due to interference with folate metabolism. - Can cause aplastic anemia and SIADH.
Phenytoin	- Blocks sodium channels. - Generic anti-epileptic. - Causes neural tube defects in pregnancy due to interference with folate metabolism. - Can cause fetal hydrantoin syndrome (finger nail hypoplasia + abnormal facies). - Used for status epilepticus if benzo fails.
Ethosuximide	- Blocks thalamic calcium channels. Only anti-epileptic USMLE might ask MOA. - Used for absence seizures.
Benzodiazepines	- Diazepam, lorazepam, midazolam, chlordiazepoxide, etc. - Bind to GABA-A receptor, which ↑ frequency of Cl^- channel opening, thereby allowing Cl^- to flow into the neuron, hyperpolarizing it and inhibiting it (i.e., ↓ firing). - Hypnotic (induces sleep); anxiolytic (reduces anxiety); anti-convulsant. - Given 1st-line for status epilepticus. - Clonazepam used for insomnia/anxiety (shows up on NBME). - Used for stimulant intoxication (i.e., cocaine, amphetamine, PCP). - Used for specific phobia (e.g., fear of flying). - Used for social phobia 2nd-line if patient has asthma, since we don't give propranolol in this setting (asked on NBME). - Used for alcohol withdrawal (delirium tremens / alcoholic hallucinosis). - USMLE wants you to know that alcohol also binds to and activates GABA-A receptor, but at a different location from benzos. There's an NBME Q where you have to select the illustration of an alcohol molecule and benzo binding to the same receptor, but at different sites. - Can cause respiratory depression.

	- Treat overdose with flumazenil (antagonist of benzo receptor).
Barbiturates	- Phenobarbital, thiopental, etc. - Bind to GABA-A receptor; ↑ duration of chloride channel opening, making them more efficacious (and dangerous) than benzos. - Used for general anesthesia. - Used for unremitting status epilepticus if benzos and phenytoin fail.
Non-benzo hypnotics	- Zolpidem, zaleplon. - Activate GABA-A receptor, but bind to a different subunit from benzos. - Can cause addiction similar to benzos.
Magnesium	- Used for Tx and prevention of eclamptic seizures in pregnancy. - Given to pregnant women giving birth <32 weeks' gestation for neuroprotection for the fetus. - Used for torsades arrhythmia and digoxin toxicity.
Lamotrigine	- Blocks sodium channels. - General anti-epileptic. - Can cause Stevens-Johnson syndrome.

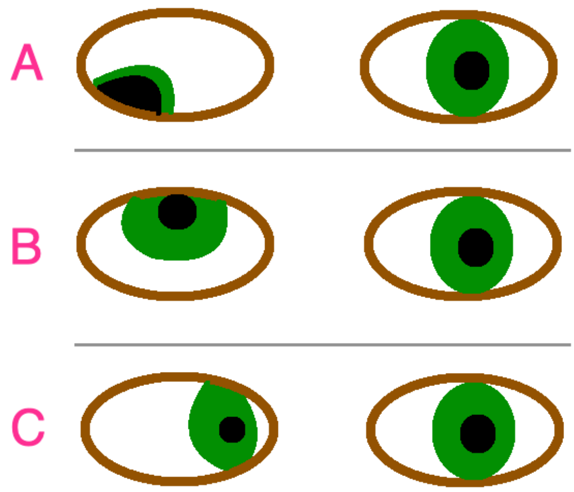
Parkinson drugs for FM	
Carbidopa-levodopa	- Combo frequently used for Tx of Parkinson disease. - Levodopa crosses the BBB to be converted to dopamine centrally. However, levodopa is subject to fast metabolism when administered alone. The addition of carbidopa functions as a competitive inhibitor of breakdown enzymes, resulting in increased levodopa availability for passage across the BBB. Do not confuse this mechanism with direct COMT inhibitors (tolcapone, entacapone), which prevent breakdown of L-dopa. - Carbidopa-levodopa can cause psychosis if administered in too-high a dose. This is assessed on 2CK forms, where if patient gets psychotic episodes following recent addition of C-L to regimen, or following an increase in dose, the answer is simply "decrease dose of carbidopa-levodopa." "Discontinue carbidopa-levodopa is the wrong answer."
D2 receptor agonists	- Ropinirole, pramipexole, cabergoline, pergolide, and bromocriptine. - Used for Parkinson disease management. - Bromocriptine classic agent used for prolactinoma (dopamine agonism ↓ prolactin). - Ropinirole + pramipexole classically show up as restless leg syndrome treatment following making sure the patient is iron replete. New literature suggests gabapentin is first-line over D2 agonists now for RLS, but USMLE won't play trivia. They will give patient with RLS + say, "In addition to ropinirole, what else could be added?" And gabapentin will be the only answer listed that works.
Amantadine	- Used for Parkinson disease; increases presynaptic release of dopamine.
Selegiline	- Used for Parkinson disease. - Inhibits monoamine oxidase B, an enzyme that preferentially breaks down dopamine. - Can cause serotonin syndrome, either alone, or in combo with drugs like St John Wort or SSRIs.

Alzheimer drugs	
Donepezil	- As mentioned earlier, this is a cholinesterase inhibitor used in Alzheimer. - Rivastigmine and galantamine you can be aware of and have same MOA, but are LY.
Memantine	- NMDA glutamate receptor antagonist. - Glutamate receptor activation normally causes Ca^{2+} to flow into the neuron, leading to depolarization and neuroexcitation. In other words, USMLE wants you to know glutamate receptor is a ligand-gated calcium channel (where glutamate is the ligand that activates the channel). Antagonism causes neuroinhibition.

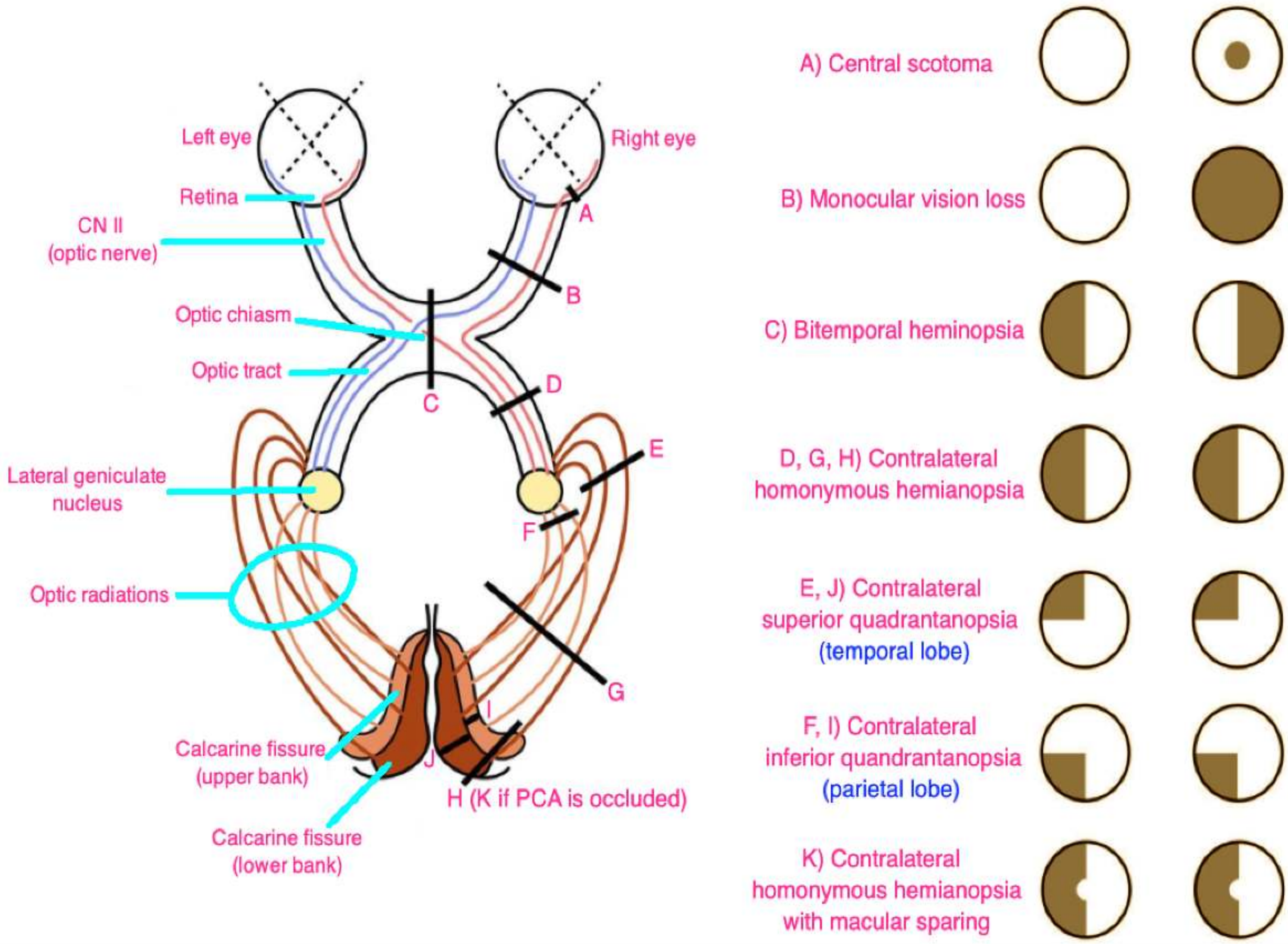
Opioids for USMLE	
- Mu (μ) and kappa (κ) are most well-studied, although delta (δ) and NOP forms exist. - Mu agonism is responsible for most of the classic effects of opioids – i.e., analgesia, euphoria, respiratory depression, and dependence. Agonism results in inhibition of neuronal firing and neurotransmitter release via opening of K^+ channels and closing of Ca^{2+} channels. - Kappa has analgesic effects but can also produce dysphoria. - Opioids can cause respiratory depression, miosis, and constipation. - Opioid withdrawal can cause flu-like symptoms, irritability, piloerection, and yawning.	
Opioid receptor agonists	
Morphine	- Gold standard opioid; used for moderate to severe pain. - USMLE wants you to know that morphine is "metabolized into active metabolites that accumulate." This is NBME answer for how overdose can occur in patients who use self-controlled pumps, since it can take time to work, so patient uses too much. - Patients who are in severe pain can sometimes have hypertension as a result. Surgery Q has morphine as answer for how to control BP post-op. - If patient has history of opioid abuse but has severe pain, e.g., due to trauma, do not withhold using opioids. Always treat pain fully.
Codeine	- Less potent than morphine; used for mild to moderate pain; in many antitussives.
Oxycodone	- Potent opioid often combined with acetaminophen. - USMLE wants you to know that oxycodone/acetaminophen combo is preferred initially in patients who are post-op. Do not just jump to morphine.
Fentanyl	- Potent synthetic opioid with high abuse potential.
Meperidine	- Potent opioid with high abuse potential. - I've noticed NBME likes this opioid for malingering Qs, where they'll say, e.g., a 47-year-old man comes into the ER in severe pain requesting meperidine.
Dextromethorphan	- Opioid used in antitussives. - Common cause of delirium from over-the-counter cold meds.
Loperamide, eluxadoline	- Opioids used for diarrhea-predominant irritable bowel syndrome. - Low addictive potential. - The fact that opioids cause constipation as an adverse effect "is a good thing" in the setting of wanting to help limit diarrhea in IBS.
Buprenorphine	- Mu partial agonist and kappa antagonist. - The kappa antagonism can help cause less dysphoria. - NBME Q floating around somewhere wants you to know it has the kappa effects.
Tramadol	- Unique MOA where it is weak mu opioid agonist + also an SNRI.
Methadone	- Long-acting full mu opioid agonist that is used for opioid/heroin withdrawal. - Reduces opioid cravings.
Opioid receptor antagonists	
Naloxone	- Opioid receptor antagonist used for acute overdoses.
Naltrexone	- Opioid receptor antagonist used for alcohol dependence (i.e., ↓ alcohol cravings). - Can also be used for opioid dependence in patients who are already detoxified in order to prevent relapse (i.e., by blocking opioid receptors, if the patient abuses any opioids, he/she won't feel the euphoric effects).
Suboxone	- Buprenorphine + naloxone combo given orally for opioid addiction in those who have high risk of abuse, or those who have already abused methadone. - Buprenorphine's partial agonist effects at mu receptors help with cravings, but the naloxone is added so that if the patient crushes the suboxone and injects it, the naloxone will antagonize the opioid receptors and cause withdrawal symptoms. However, when taken orally, the naloxone has negligible effect due to poor oral bioavailability.

FM Ophthal

Extraocular nerve palsies



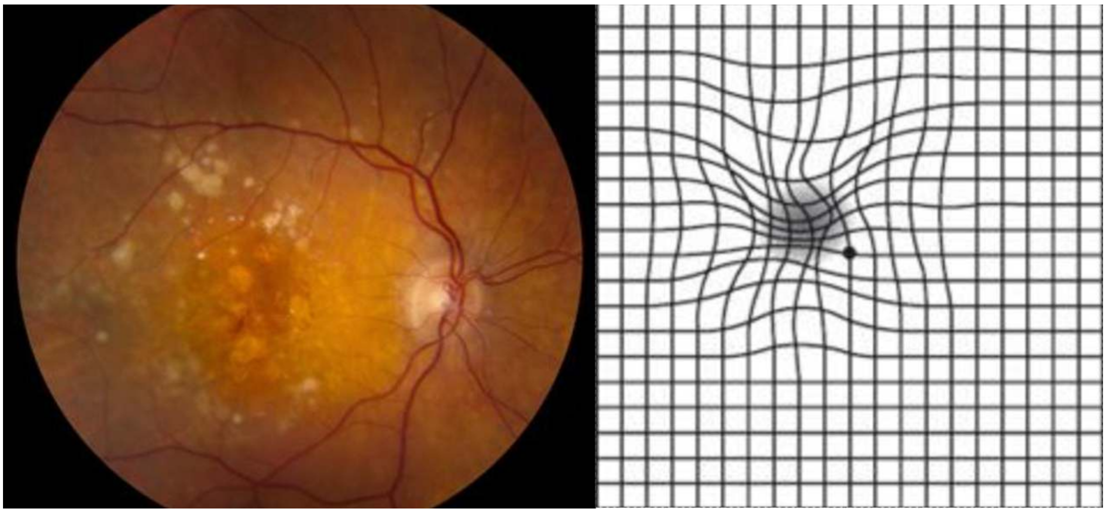
- A = CN III (oculomotor) lesion → presents as down and out.
- B = CN IV (trochlear) lesion → presents as upward gaze + patient will tilt head toward unaffected side.
- C = CN VI (abducens) lesion → medially directed eye.



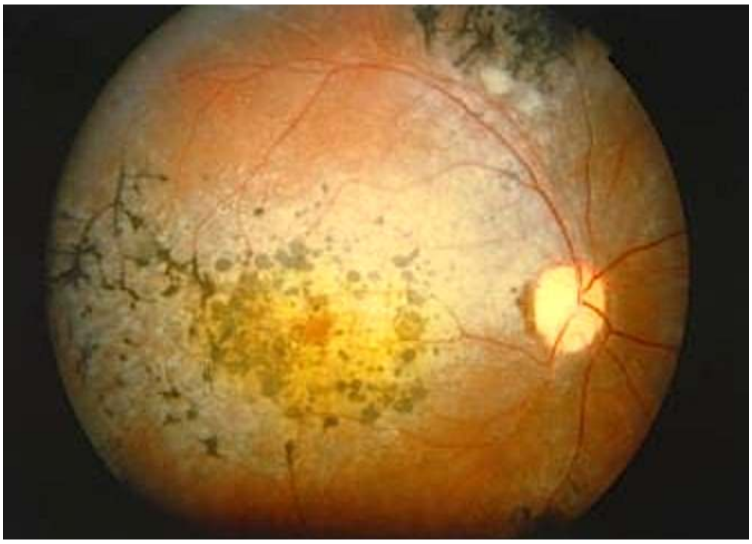
Glaucoma	
Open-angle	- Chronic eye condition characterized by increased intraocular pressure due to impaired drainage of aqueous humor. - Unlike acute angle-closure (closed-angle) glaucoma, the drainage angle (canal of Schlemm) between the cornea and iris remains open. - Progresses slowly and is usually painless. - As it advances, causes peripheral vision loss (tunnel vision) and results in optic nerve damage characterized by "cupping of the optic disc" (i.e., an increased cup-to-disc ratio). - The USMLE wants **tonometry** as the first step in management (measures intraocular pressure). - Treatment for open-angle glaucoma for USMLE is not going to be a trivia game of "which drug first." They just want you to know that topical prostaglandins, such as **latanoprost,** can increase outflow of aqueous humor, thereby reducing IOP. - Topical beta-blockers, such as **timolol,** decrease production of aqueous humor. **- Carbonic anhydrase inhibitors,** such as dorzolamide, decrease production of aqueous humor. - **Pilocarpine** is a muscarinic receptor agonist that constricts the pupil and increases outflow of aqueous humor.
Closed angle	- Presents on USMLE as a very buzzy red, teary, fixed mid-dilated pupil. - Due to closure of the drainage angle (canal of Schlemm) between the cornea and the iris, leading to a rapid increase in IOP. - Can lead to compression of the optic nerve and blood vessels, causing severe symptoms such as intense eye pain, headache, nausea, vomiting, blurred vision, and seeing colored rings or halos. - Similar to open-angle glaucoma, the USMLE wants **tonometry** as the first step to confirm increased IOP. - Treatment is IV mannitol (helps draw excess aqueous humor out of the eye), timolol, pilocarpine (muscarinic receptor agonist), or alpha-2 agonists such as brimonidine.

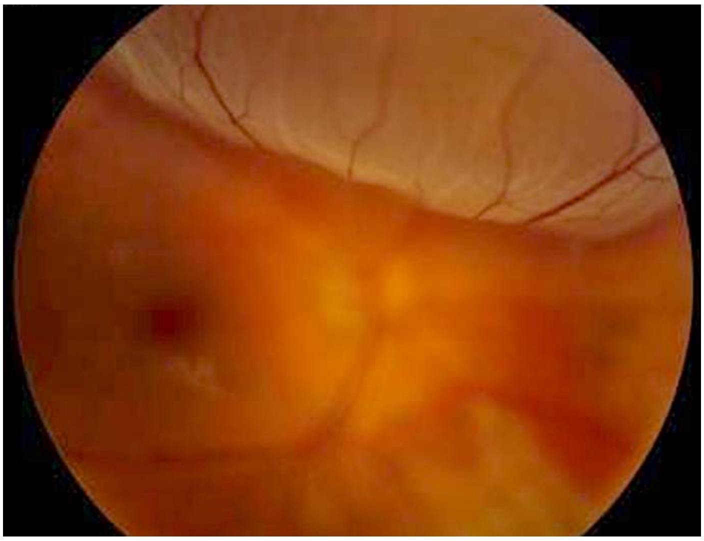
Macular degeneration

- USMLE wants you to know that "wavy lines" on a straight-line Amsler test = macular degeneration.



- It is characterized by loss of central vision.
- Fundoscopy shows drusen, which are yellow lipoproteinaceous deposits.
- There are dry (atrophic) and wet (neovascular) types of macular degeneration.
- Most commonly occurs idiopathically due to age (i.e., age-related macular degeneration).
- Dry is treated supportively; wet is treated with anti-VEGF agents and sometimes laser therapy.

Retinitis pigmentosa
- Presents on USMLE as a combo of nyctalopia (loss of night vision) and loss of peripheral vision. - There will often be a family history. Fundoscopy shows characteristic pigmentation. 

Retinal detachment
- Occurs classically in those with sudden trauma to the eye or severe myopia. - The fundoscopy is HY and important for USMLE, especially 2CK.  - Treatment is surgical.

Vitreous hemorrhage

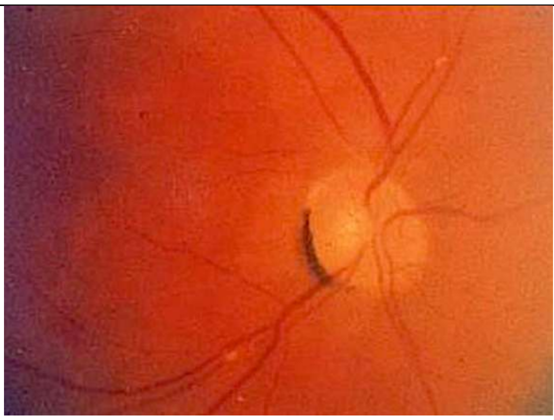
- USMLE wants you to know that the biggest risk factors for bleeding into the vitreous are diabetes and hypertension.

- Vitreous hemorrhage will usually be a distractor, where they can tell you a young boxer was hit in the eye, but the answer is retinal detachment, not vitreous hemorrhage.

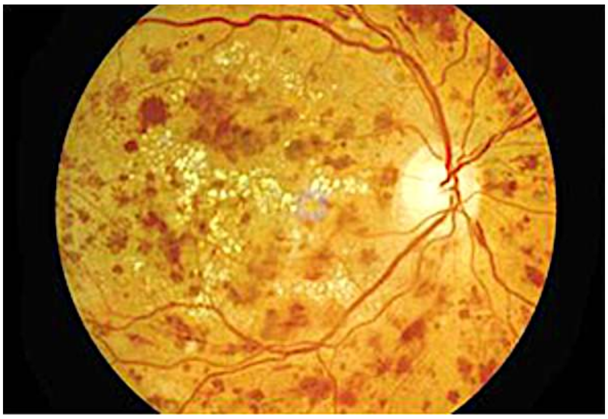
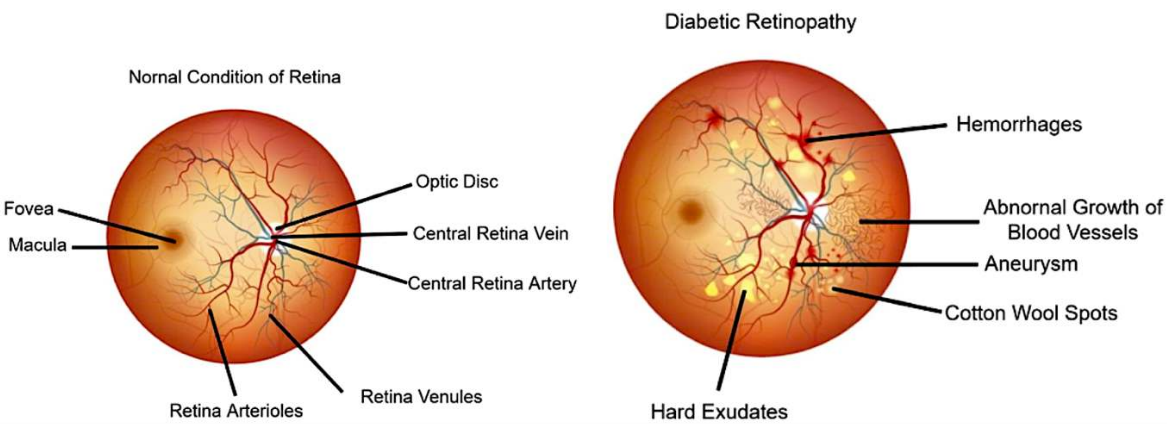
Central retinal artery occlusion

- Presents as sudden, painless vision loss in a patient where a "curtain comes down."
- Due to a carotid atheroma (in setting of HTN) or left atrial mural thrombus (in setting of atrial fibrillation) that's launched off to the retinal vessels.
- A pale retina will be seen on fundoscopy with a cherry red spot due to maintenance of perfusion from the choroidal artery.

- Ocular massage can be attempted as Tx, where the embolus may be dislodged, restoring perfusion.

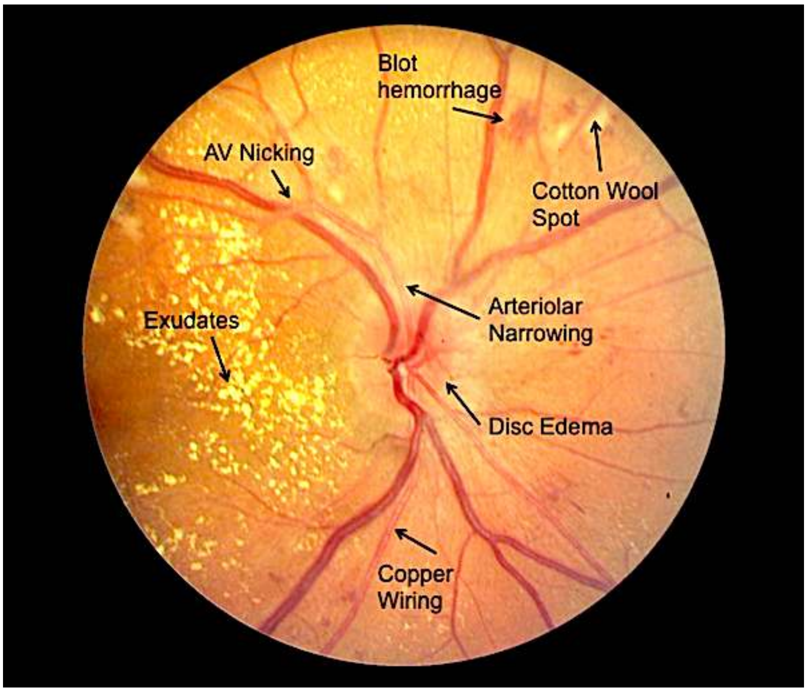


- USMLE can show image that looks similar to this one. They want you to know that weird crescent-shaped finding is called a "refractile body," and refers to an embolus.

Diabetic retinopathy
- Can present in numerous ways, such as with blurred or fluctuating vision, and difficulty recognizing colors. - The USMLE vignette will usually say things like cotton wool spots (soft exudates), hard exudates, and retinal hemorrhages. - Soft exudates are small, whitish lesions on the retina caused by microinfarctions and are axoplasmic material; hard exudates are lipoproteinaceous deposits due to leaky retinal vessels. - Soft/hard exudates and retinal hemorrhages are seen classically in both diabetic and hypertensive retinopathies.   - Can be dry or wet (neovascular). Wet type can be treated with anti-VEGF agents.

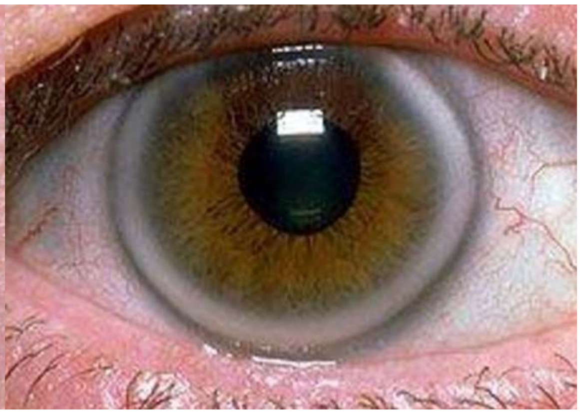
Hypertensive retinopathy

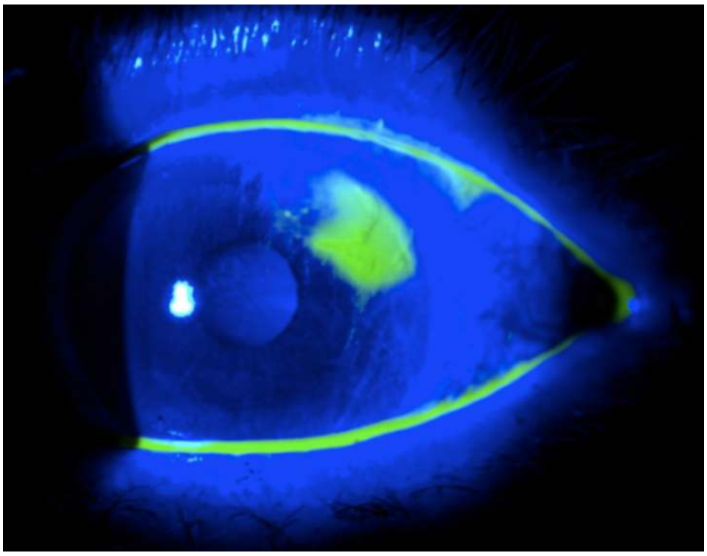
- Buzzy findings like AV nicking and arteriolar narrowing.
- Q can mention "sausage-like" appearance of retinal arterioles.



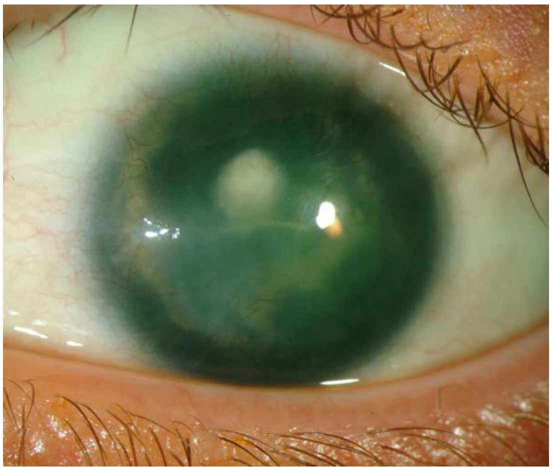
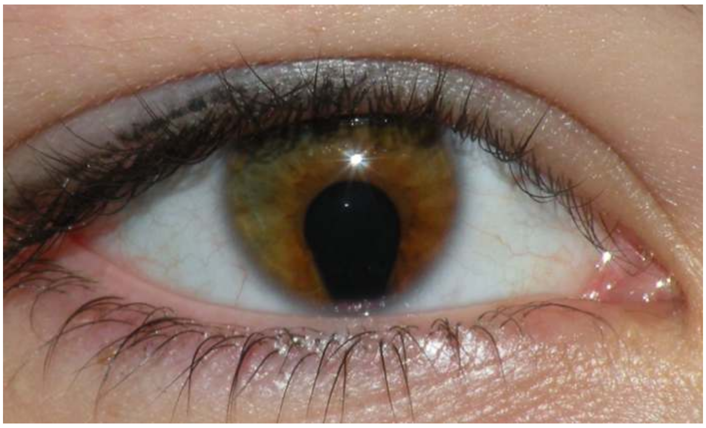
Dyslipidemia

- Orbital xanthoma (cholesterol deposits; left image) and corneal arcus (lipid deposits; right image).
- Both can occasionally be seen idiopathically in patients without lipid disorders (the latter in elderly).

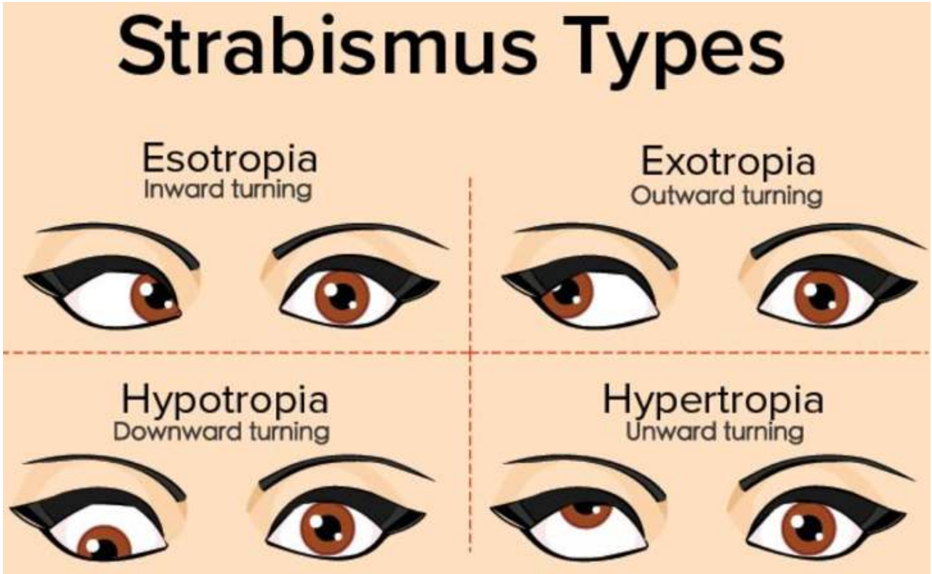



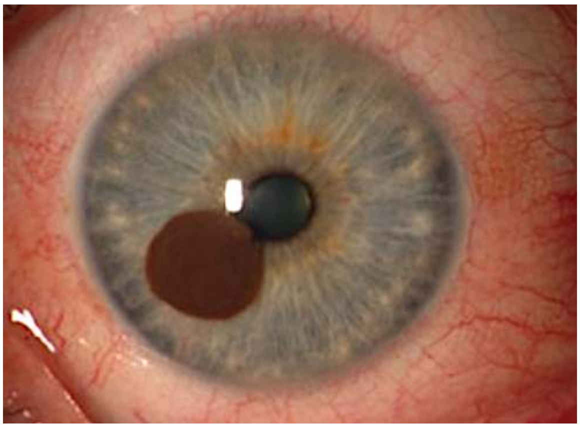
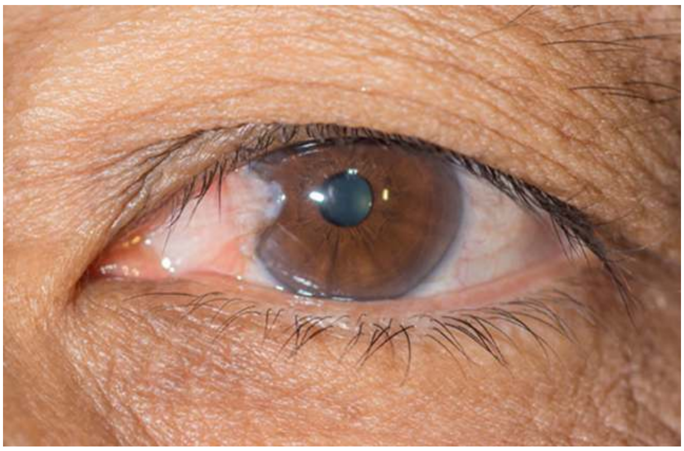
Corneal abrasion
- Scratchy or tearing sensation of the eye. Can be caused by contact lens-use or coming into contact with irritants such as sand (i.e., playing in sandbox with son/daughter). - USMLE will either ask for the next best step, where the answer is "instillation of fluorescein into the eye," or they will show you the following image (which is a fluorescein instillation) and then just ask for the diagnosis (i.e., corneal abrasion).  - Fluorescein is an orange dye that when dripped into the eye appears blue-green under violet light. - Anything blue is normal. Anything green is abnormal.

Subconjunctival hemorrhage
- Buzzy image / spot-diagnosis for USMLE. - Caused by bursting of superficial conjunctival vessels. Looks sinister but self-resolves uneventfully. - Can be caused by trauma or brief periods of increased pressure (i.e., severe vomiting or coughing). 

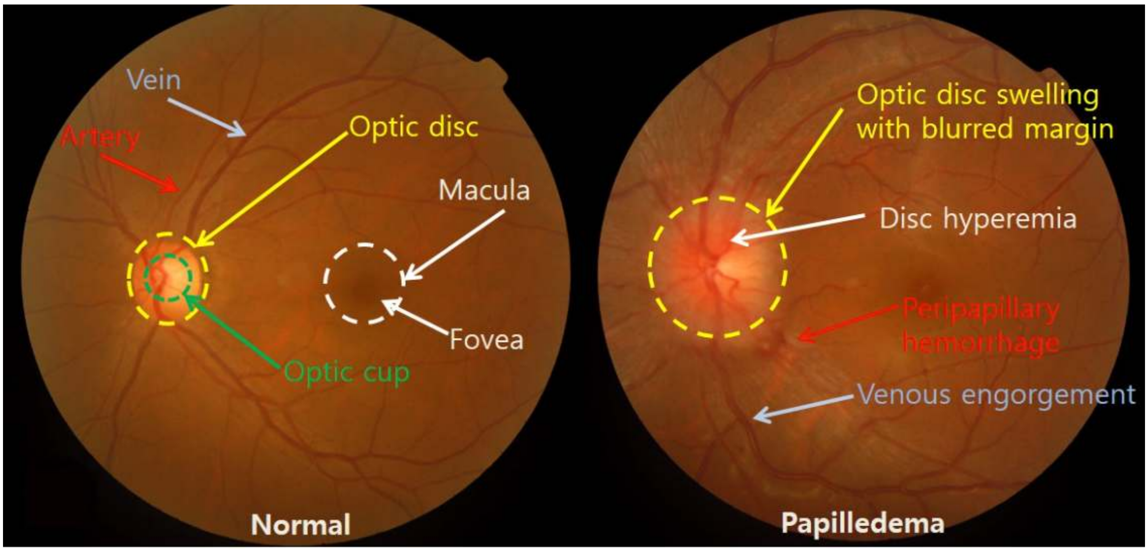
Iris abnormalities	
Aniridia	- Seen as part of **WAGR** complex with *WT1* mutations – i.e., **W**ilms tumor, **A**niridia, **G**enitourinary anomalies, **R**etardation. 
Coloboma	- Means hole in the eye. Seen in **CHARGE** syndrome, → **C**oloboma of the eye, **H**eart defects, **A**tresia of the choanae, **R**etardation of growth and development, **G**enitourinary anomalies, **E**ar defects. 
Albinism	- Can cause pale blue or pink irides (pleural for iris). 

<table>
<tr><th colspan="2">HY Peds Ophthal DDx for FM</th></tr>
<tr><td>Orbital cellulitis</td><td>
- Infection involving tissues posterior to orbital septum; Staph aureus most common cause; rare sequela of adjacent infection, e.g., from the paranasal sinus; medical emergency; requires IV antibiotics; preseptal cellulitis is a less severe version of orbital cellulitis.</td></tr>
<tr><td>Hordeolum (stye)</td><td>- Painful bump on the eyelid.
- Staph aureus infection of sweat gland or oil duct; treat with warm compresses.
- Can occur any age.
</td></tr>
<tr><td>Chalazion</td><td>
- Painless bump on the eyelid.
- Blocked oil duct; not an infection; treat with warm compresses.</td></tr>
</table>

Retinoblastoma	 - The image shows leukocoria, which is an abnormal white appearance of the eye on light reflex (should normally appear red). - USMLE wants you to know that *RB* gene mutations not only cause congenital retinoblastoma, but also ↑ risk of osteosarcoma. - In other words, classic Q is they show you image above + ask what child is at risk for later → answer = osteosarcoma.
Strabismus	- Means "lazy eye," or misalignment of the eyes, due to weakened muscles in one eye.  - Can lead to amblyopia, which is reduced visual acuity in the weak eye due to the brain favoring the stronger eye during development. - Strabismus, however, is not reduced vision in the weak eye. It precedes amblyopia if not treated with eye exercises and/or penalization of (patching) the strong eye.

UV light-induced Ophthal pathologies	
Cataracts	- Progressive clouding or opacification of the lens, leading to blurred or hazy vision, particularly in bright light, with increased glare sensitivity.  - Can be caused by sunlight, but the most important cause on USMLE is diabetes mellitus, since high glucose entering the lens → converted to sorbitol via aldose reductase → sorbitol has high osmotic pull, causing osmotic damage to the lens and cataracts. - Recent changes in visual acuity can suggest new-onset or worsening diabetes. - Can be seen in Peds in congenital rubella, congenital syphilis, and NF2.
Melanoma	- Melanomas can grow from the conjunctiva and cornea. Just be aware they exist. 
Pterygium	- Non-cancerous overgrowth of non-keratinized stratified squamous epithelium growing from the conjunctiva. - Crosses over the cornea (clear part of the eye covering the iris and pupil). - History of sun exposure is the major risk factor. 

Pinguecula	- Yellow-white benign lesion on the conjunctiva composed of protein and fat. - Does not cross over the cornea. - Sunlight is the major risk factor. 

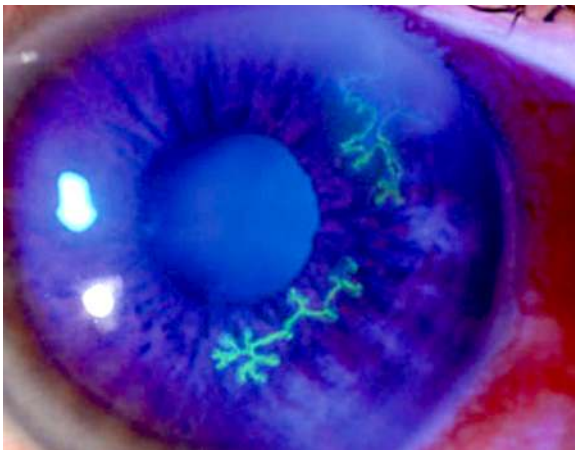
Papilledema
- Means increased intracranial pressure on USMLE. - Papilledema refers to blurring of the optic disc margins due inflammation of the optic nerve head in the setting of increased intracranial pressure. 

HY Autoimmune eye pathologies	
Anterior uveitis	- WBCs in the anterior chamber sometimes seen in flares of autoimmune diseases, especially sarcoidosis and ankylosing spondylitis. - USMLE might sometimes just give you patient with sarcoidosis and then ask what needs to be done in terms of follow-up for the patient → answer = "slit-lamp examination" to check for anterior uveitis.
Keratoconjunctivitis sicca	- Dry eye (xerophthalmia) caused by decreased tear production.

	- Classically autoimmune, either a stand-alone condition or as part of Sjögren syndrome. - Latter presents additionally with dry mouth (xerostomia) and systemic findings such as arthritis.
Dermatomyositis	- Causes heliotrope rash (violaceous eyelids). - Don't confuse with the malar rash of lupus.  - Violaceous eyelids can also be seen in neuroblastoma in Peds, albeit completely unrelated. I just mention it cuz it shows up on an NBME form.

Miscellaneous hereditary conditions	
Leber hereditary optic neuropathy	- USMLE wants you to know that mitochondrial disorders, e.g., LHON, present as a triad of 1) eye/ear problems, 2) hypotonia, and 3) lactic acidosis (low bicarb; metabolic acidosis). - Other mitochondrial disorder names you could be aware of are MELAS (Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke-like episodes) and MERRF (Myoclonic epilepsy with ragged red fibers). - Heteroplasmy is variability of phenotype severity across offspring (Step 1 term).
Osteogenesis imperfecta	- Collagen I mutation. - Can cause the very buzzy blue sclerae. 
Alport syndrome	- Mutation in type IV collagen; X-linked recessive. - The answer on USMLE for eye and/or ear problem + red urine.
Marfan syndrome	- Can cause lens dislocations.
Homocystinuria	- Can cause lens dislocations.

Ophthalmia neonatorum	
- Fancy way of saying neonatal conjunctivitis.	
Chlamydia	- Caused by *C. trachomatis* D-K (the STD). - Presents usually >1 week post-birth and can lead to **pneumonia** with a lymphocytic shift (since chlamydia is obligate intracellular, so requires lymphocytes for cell-mediated immunity to clear them, unlike extracellular bacteria which can be cleared humorally with neutrophils). - The chlamydia can drain through the nasolacrimal duct down to the lungs, causing the pneumonia. - Prophylaxis is treating the mom while pregnant if she has chlamydia; treatment in the neonate who already has the conjunctivitis is **oral** erythromycin. - NBME can give you vignette of wheezes, crackles, and fever in a 3-week-old who they say was treated for chlamydia conjunctivitis 2 weeks ago with topical erythromycin at birth. But this is not the correct treatment since topical will not cover any organism that has already entered the nasolacrimal duct.
Gonorrhea	- Gonococcal conjunctivitis will present in the first week of life with yellow discharge from the eye(s). - The USMLE vignette will be vague, where the student says, "But how are we supposed to know it's not chlamydia, just because it's first week of life, that's it?" And my response is: if they give you chlamydia ophthalmia neonatorum, they'll **always** tie it to subsequent pneumonia somehow, since that is such a HY point. - Prophylaxis for gonococcal is erythromycin ointment; treatment is IM 3^{rd}-gen cephalosporin (usually cefotaxime in peds).
Chemical	- Chemical conjunctivitis is classically due to silver nitrate eyedrops that used to be given to neonates for gonococcal prophylaxis. However, they have been replaced by erythromycin ointment.
Viral	- Viral conjunctivitis will present as teary eye that is either unilateral or bilateral, and either itchy or non-itchy. I've seen variable presentation on USMLE. You just need to know this is the most likely diagnosis in both school-age kids and adults, and is caused by **adenovirus,** which is a DNA virus. Treatment is supportive with saline rinse.
Allergic	- USMLE wants you to know that "cobblestoning" of the tarsal conjunctiva is buzzy for allergic conjunctivitis. - Likewise, for allergic rhinitis, they can say cobblestoning of the nasal mucosa. - Some students have asked if allergic has to be bilateral, which it need not. And it doesn't have to be itchy either.

Herpesviridae infections	
Herpes keratitis	- Keratitis is inflammation of the cornea. - HSV1/2 can cause dendritic (tree-like) pattern in the eye on fluorescein instillation. 

	- May or may not cause vesicles around the eye as well. - Tx with topical acyclovir/valacylcovir.
Herpes zoster ophthalmicus	- VZV (HHV-3) can cause shingles of the eye. - Causes keratitis with a similar dendritic pattern to HSV 1/2. - May or may not cause vesicles around the eye as well. - Tx with topical acyclovir/valacylcovir.
CMV retinitis	- The answer for blurry vision in severely immunocompromised patient (usually AIDS). - You don't need to know the fundoscopy. - Treat with ganciclovir.

FM MSK

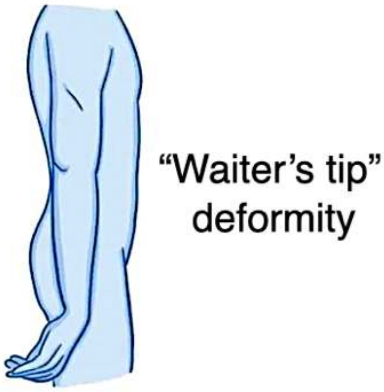
Rotator cuff injuries		
Rotator cuff muscles (SITS)	Function	HY Points
Supraspinatus	Abduction of arm **(first 15 degrees)**	- Answer if patient has difficulty abducting arm first 15 degrees. - Empty-can (thumb-down) / full-can (thumb-up) tests for diagnosis → patient abducts arm to 90 degrees with thumb up or down → downward pressure applied to arm → if elicits pain, answer = supraspinatus injury. - There is Q on new 2CK CMS IM form 7 where answer is supraspinatus tendonitis, and they say patient has reduced ability to abduct the first 60 degrees.
Infraspinatus	Lateral (external) rotation	- Notion of "pitcher injury" = infraspinatus does more harm than good for USMLE; vignette can by all means give a pitcher with (+) full-can test above and answer is supra-, not infra-, spinatus injury. - Q might say patient simply cannot externally rotate arm, nothing more, where teres minor isn't listed as another answer, so infraspinatus is only one that could be right.
Teres minor	Adduction; lateral rotation	- Same as infraspinatus, just know it externally rotates the arm. - Also adducts arm, so if patient has issues with both lateral rotation **and** adduction, answer is teres minor over infraspinatus. - I've never seen NBME material assess the diagnostic tests for infraspinatus or teres minor.
Subscapularis	Adduction; medial (internal) rotation	- Can medially rotate and adduct the arm. - I've had students get asked Gerber lift-off test on both Step 1 as well as 2CK Family Med shelf, where answer = subscapularis. - Gerber lift-off test = patient places dorsal aspect of hand on lower back, with palm facing posteriorly → examiner applies pressure into the patient's palm against his/her lower back → patient is then asked to move hand away under the pressure → if elicits pain / difficult to do, answer = subscapularis injury.

Upper limb nerve HY Points for FM	
Axillary	- Main innervation of the deltoid, allowing for abduction of arm 15-90 degrees. USMLE wants you to know deltoid has an origin on the lateral clavicle and axillary nerve innervating it is at C5/C6. - Palsy caused by **surgical neck of humerus fracture.** - USMLE vignette will often say "flattened deltoid" or "loss of sensation over lateral upper arm / deltoid."
Median	- Main innervation is lateral "3 and a half" fingers / thenar pad, and lateral forearm. - Does thumb abduction. (In contrast, ulnar nerve does thumb adduction) - NBME wants "palmar cutaneous branch of median nerve" as answer for sensation over thenar region. - Palsy caused by **supracondylar fracture of humerus**, or "**distal shaft fracture."** Former is buzzy; latter sounds non-specific, but I've seen it this way on NBME. - Entrapment of median nerve causes carpal tunnel syndrome; will present as paresthesia/numbness of lateral hand / thenar region; can be caused by hypothyroidism (GAG deposition), acromegaly (growth of tendons), and pregnancy (edema); can occur bilaterally in construction workers using jackhammer. - Tx for carpal tunnel ultra-HY on 2CK FM forms. "Use of wrist pad when using computer" is answer on new 2CK NBME. If not listed, "wrist splint" is answer on FM form. If vignette says wrist splint fails, NSAIDs are wrong answer and not proven.

	USMLE wants "triamcinolone injection into carpal tunnel" (not IV steroids) as next answer. Surgery is always wrong answer for carpal tunnel on USMLE. - 2CK wants "electrophysiological testing" and "electromyography and nerve conduction studies" as next best step in diagnosis for carpal tunnel.
Ulnar	- Main innervation of medial "1 and a half" fingers, and medial forearm. - Ulnar nerve also does **finger abduction and adduction** (i.e., interosseous muscles). - USMLE loves **Froment sign** for ulnar nerve injury, which is inability to pinch a piece of paper between the thumb and index finger (ulnar nerve needed for thumb adduction against index finger, despite thumb being most lateral digit). - Distal compression (i.e., of wrist and hand only, not forearm) is aka **Guyon canal syndrome** and is caused by **hook of hamate fracture;** this can sometimes be seen in cyclists due to handlebar compression; presents as paresthesias / numbness of 4th and 5th fingers + hypothenar eminence. - Proximal compression (i.e., medial forearm + wrist/hand) is aka **cubital tunnel syndrome** and is one of the most underrated diagnoses on USMLE, since its yieldness, especially on 2CK, is comparable to carpal tunnel syndrome, but students often haven't heard of it. Essentially, patient will get paresthesias of medial forearm + hand, where it "sounds like carpal tunnel but on the ulnar side instead" → answer = cubital tunnel syndrome. - Tx for cubital tunnel syndrome is "overnight elbow splint." Surgery is wrong answer on USMLE.
Radial	- Main innervation for finger, wrist, and elbow extension. - Innervates **BEST** → **B**rachioradialis, **E**xtensors, **S**upinator, **T**riceps. - Palsy occurs as a result of **midshaft fracture of the humerus,** or as a result of fracture at the **radial groove** (latter is obvious). - Retired Step 1 NBME Q says construction worker sustains "comminuted spiral fracture of humerus" (unusual, since spiral fracture classically = child abuse), and they ask for the resulting defect → answer = "loss of radial nerve function." - Highest yield point is that injury results in pronated forearm + wrist drop.
Musculocutaneous	- Main innervation of the biceps. - Just need to know injury results in loss of sensation over lateral forearm + decreased biceps function. - USMLE doesn't give a fuck about what kind of injury causes palsy.

Frequently confused shoulder conditions	
Subacromial bursitis vs rotator cuff tendonitis	- In NBME vignettes, I've seen both can give Hx of patient doing frequent overhead movement (i.e., painting a fence) + pain with palpation + pain that's worse when lying on one's shoulder in bed at night, making differentiating these difficult. - Subacromial bursitis will only present with above findings, which collectively are known as impingement syndrome. - Rotator cuff tendonitis will present with weakness when performing exam maneuvers (as described in prior table).
Biceps tendonitis	- Presents as anterior shoulder pain with focal tenderness over the biceps tendon (i.e., when pressing on anterior shoulder).
Adhesive capsulitis	- Aka "frozen shoulder," or arthrofibrosis. - Decreased passive and active motion of shoulder in all directions. - Idiopathic, but increased risk in diabetes. - Tx = range of motion exercises / physiotherapy. - This is a Dx that is LY on Step 1, but for whatever reason, becomes HY on 2CK.

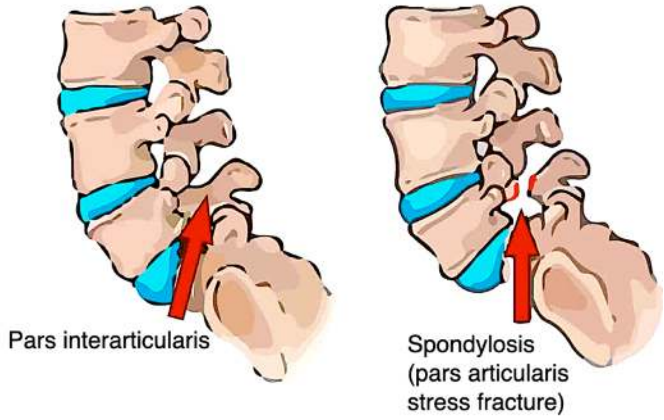
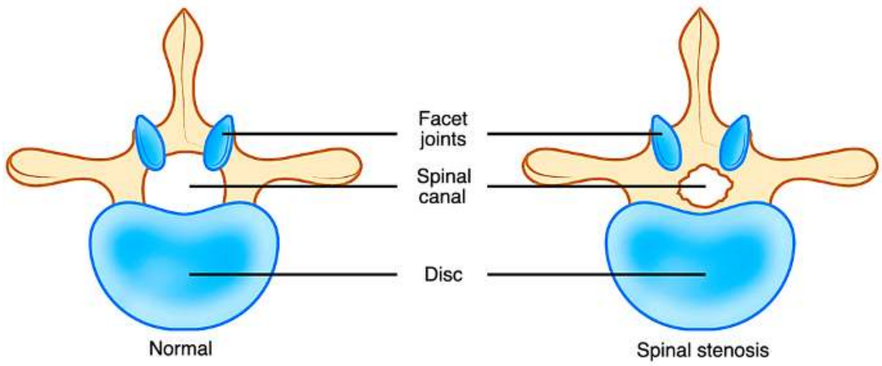
<table>
<tr><th>Winged scapula</th></tr>
<tr><td>
Winged scapula
- Long thoracic nerve injury, which innervates serratus anterior.
- Can occur post-mastectomy.</td></tr>
</table>

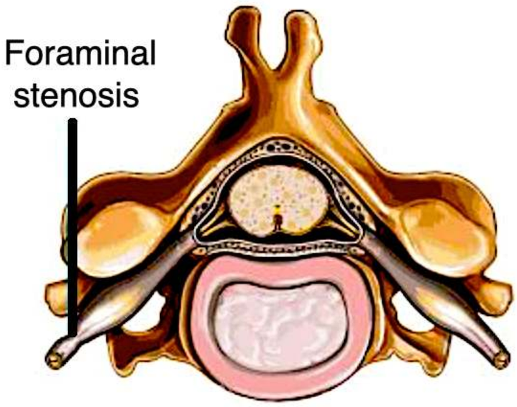
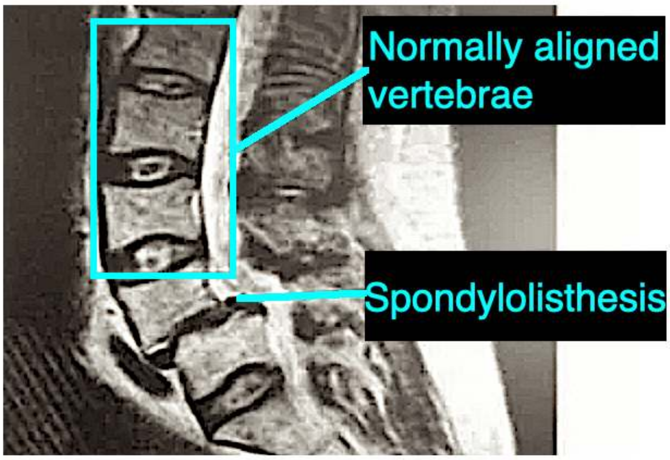
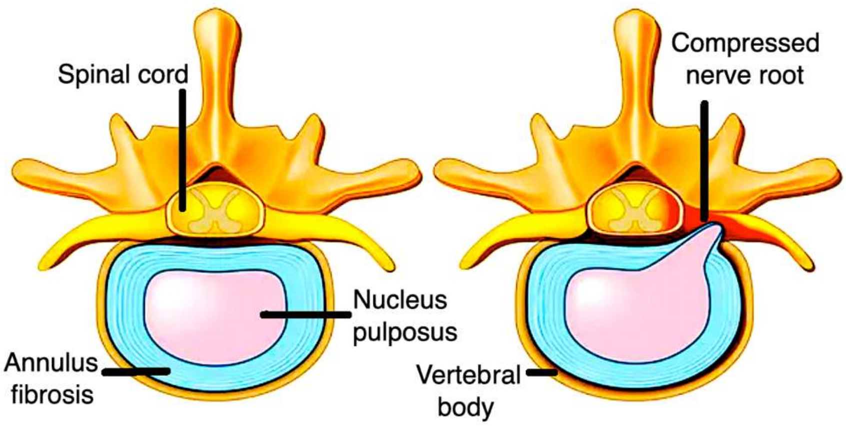
<table>
<tr><th colspan="2">Brachial plexus injuries</th></tr>
<tr><td colspan="2">- Both can be caused in neonates by breech / traumatic labor.
- Can also be caused by injuries such as grabbing onto tree branch while falling.</td></tr>
<tr><td>Erb-Duchenne</td><td>- Upper brachial plexus injury (C5-C6).
- Sometimes rather than Erb-Duchenne written as the diagnosis, the answer will just be “upper brachial plexus” when they ask for what’s injured.


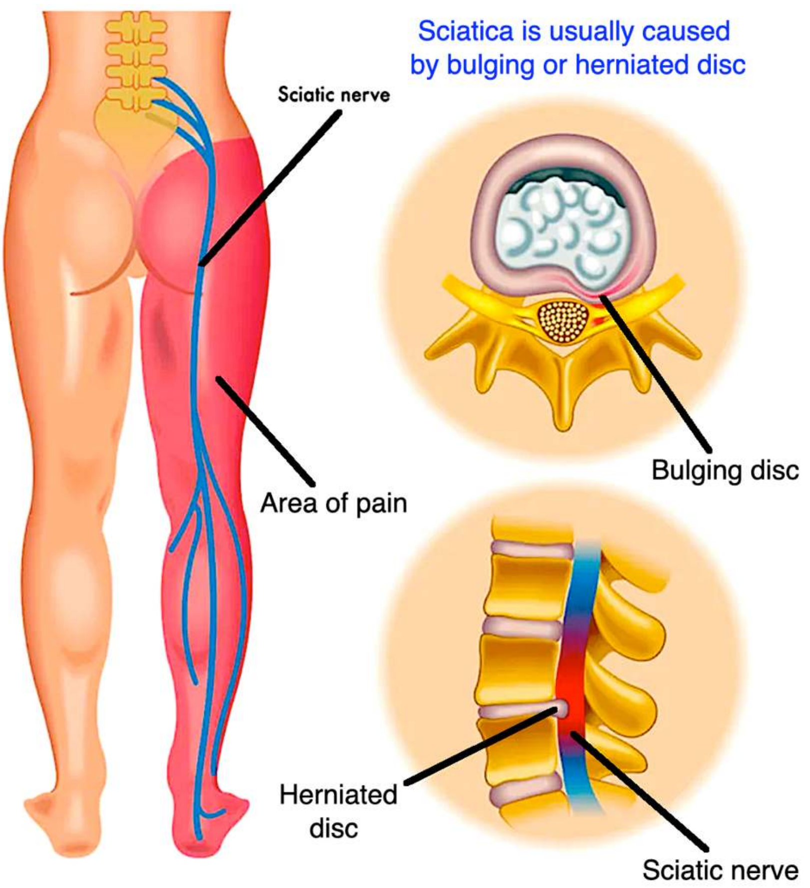
- Arm is adducted, pronated, and wrist flexed.</td></tr>
<tr><td>Klumpke</td><td>- Lower brachial plexus injury (C8-T1).
- Sometimes rather than Klumpke written as the diagnosis, the answer will just be “lower brachial plexus” when they ask for what’s injured.

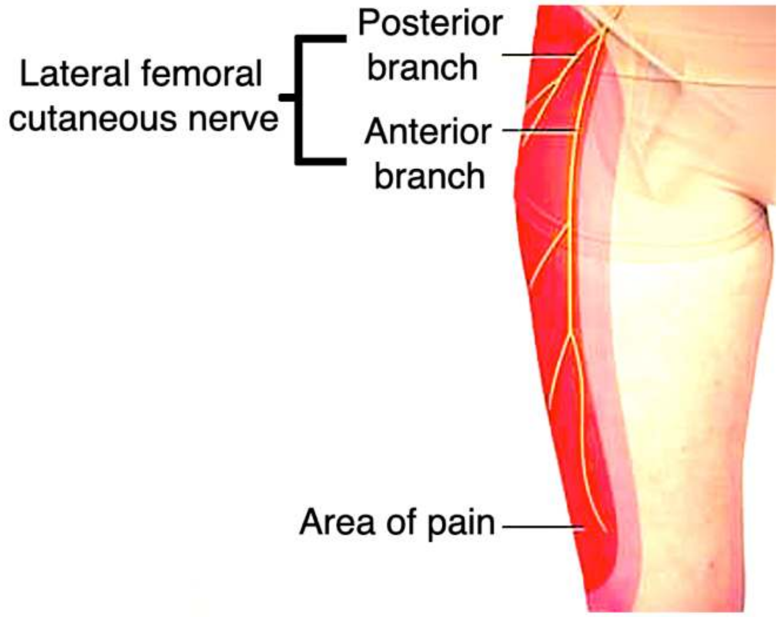
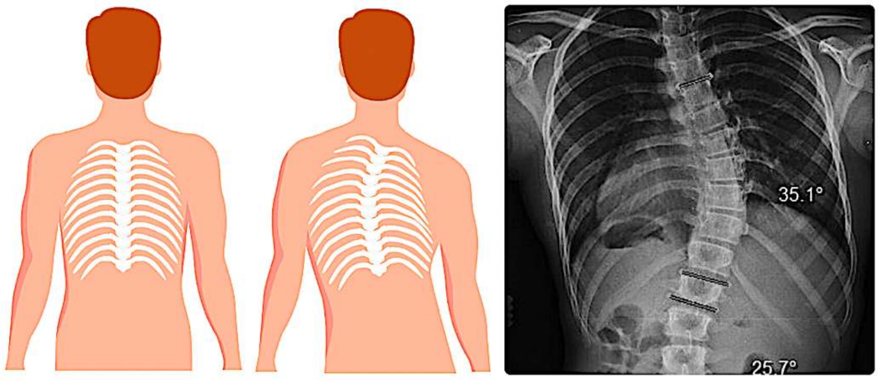
</td></tr>
</table>

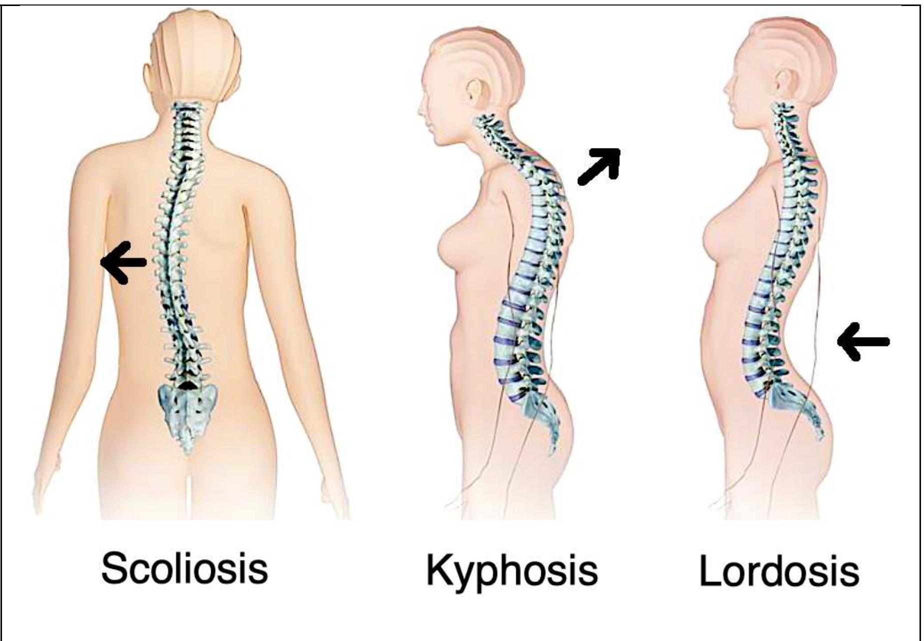
Other upper limb DDx for FM	
deQuervain tenosynovitis	- Tenosynovitis means inflammation of tendon sheaths. - deQuervain is classic in breastfeeding women and is worsened with Finkelstein test (shown below) → 1) thumb is placed in palm; 2) 2nd-5th fingers are wrapped over the palm; 3) patient ulnar deviates the wrist → this causes pain.  - Patient should avoid offending activity, but since this is often breastfeeding, steroid injection can provide immediate relief.
Ganglion cyst	- Gelatinous collection of joint fluid; can occur on ankles and flexor areas as well, but classic location is dorsum of hand/wrist.  - If Q asks you most likely trajectory if untreated, answer = spontaneous regression. - Tx is needle drainage if disturbing to the patient; recurrence common because the root opening to the tendon sheath is not removed.
Lateral epicondylitis	- Tennis elbow. - Lateral elbow pain worsened when patient extends wrist against resistance. - NBME asks "extensor carpi radialis brevis" as answer for site of inflammation. - Tx = "forearm strap" on 2CK FM forms.
Medial epicondylitis	- Golfer elbow. - Medial elbow pain worsened when patient flexes wrist against resistance. - Inflammation at flexor carpi radialis and pronator teres. - Tx = forearm strap.
Radial head subluxation	- "Nursemaid elbow"; HY for 2CK Peds. - Child stops using arm + arm pronated and partially flexed. - Hx of child having arm pulled/yanked, or child holding hands and running with older sibling + the child falls, resulting in elbow pull. - Tx = hyper-pronation, OR supination when arm partially flexed. Either is correct. Both will not be listed at the same time as answers.
Olecranon bursitis	- Elbow pain, usually following contact injury. - Tx = compression bandage + NSAIDs. Steroid injection is wrong answer on USMLE.

HY MSK spinal conditions for USMLE	
Cervical spondylosis	 - The answer on USMLE if they say patient over 50 has neck pain + MRI shows degenerative changes of cervical spine. - Can occur in lumbar spine, but USMLE likes cervical spine for this. - Technically defined as degeneration of pars interarticularis component of vertebral body.
Atlantoaxial subluxation	- Increased mobility between the first (atlas) and second (axis) vertebrae. - Really HY on 2CK Surg and Neuro forms in patients who have **rheumatoid arthritis.** - Must do CT or flexion/extension x-rays of cervical spine prior to surgery when a patient will be intubated; I've seen both of these as answers for different Surg Qs. - Q on one of the Neuro CMS forms gives patient with RA not undergoing surgery who has paresthesias of upper limbs → answer is just MRI of cervical spine (implying atlantoaxial subluxation has already occurred).
Lumbar spinal stenosis	- Narrowing of the spinal canal. - The answer on USMLE if they mention a patient over 50 who has lower back pain that's worse when walking down a hill (i.e., relieved when leaning forward), or when standing/walking for extended periods of time. - Can cause "neurogenic claudication," where the vignette sounds like the patient has intermittent claudication, but they'll make it clear the peripheral pulses are normal and that the patient doesn't have cardiovascular disease; this shows up in particular on 2CK Neuro CMS forms. - Technically an osteoarthritic change of the spine; therefore increased risk in obesity (but not mandatory for questions). 
Cervical foraminal stenosis	- The answer on USMLE if they say old woman has difficulty fastening buttons + weakness of hand muscles + loss of sensation of little finger → answer = "C7-T1 foraminal stenosis" (offline NBME 20). - Not stenosis of cervical spinal canal, but stenosis of foramen where nerve exits.

	
Spondylolisthesis	 - The answer on USMLE if they a "step-off" between infra-/suprajacent vertebrae. In other words, they'll say one vertebra "juts out" or has a "step-off" compared to those above/below it. - Can be due to trauma or idiopathic development.
Disc herniation	 - Herniation of nucleus pulposus through a tear in annulus fibrosis. - The answer on USMLE if they mention **radiculopathy** (i.e., shooting pain down a leg) after lifting a heavy weight or bending over (e.g., while gardening). They can write the answer as "herniated nucleus pulposus." - As I mentioned with the radiculopathies above, be aware of the L4, L5, and S1 differences. - Be aware that cervical disc herniation "is a thing," meaning it's possible and also assessed on USMLE. 2CK neuro forms ask this a couple times, where patient has shooting pain down an arm, and answer is "C8 disc herniation."

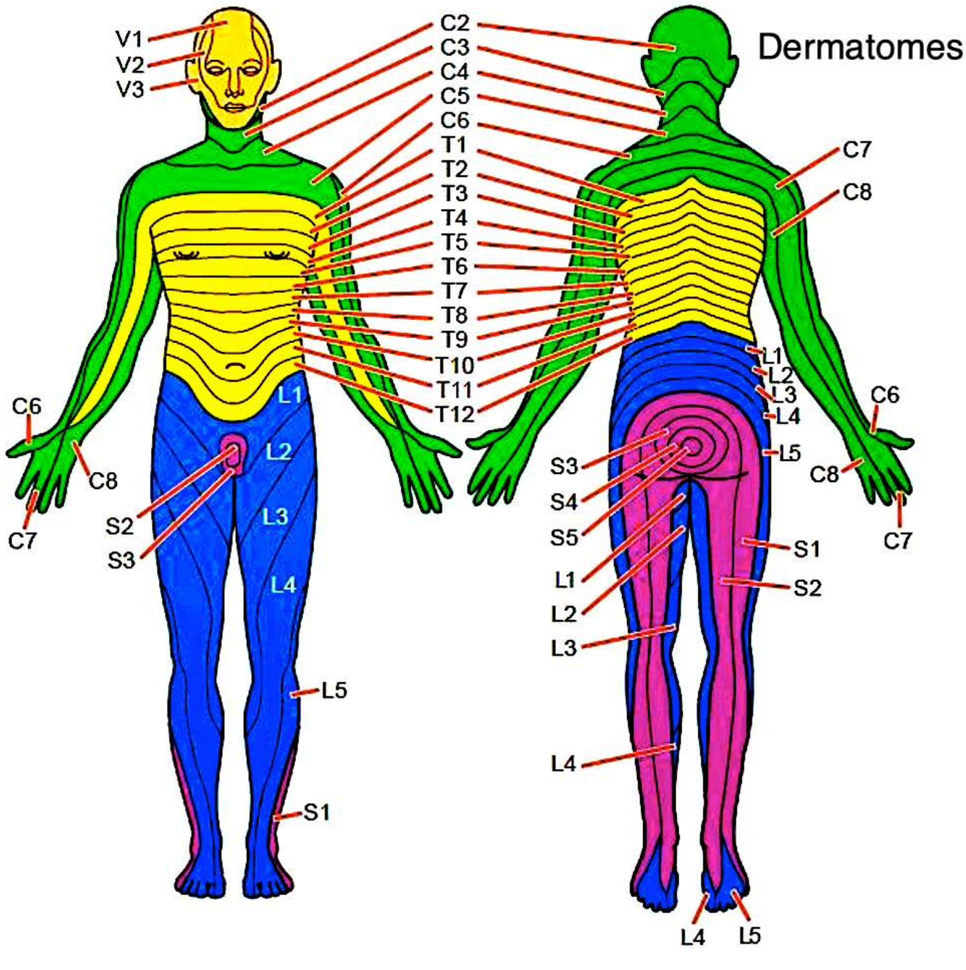
	- If suspected, newest NBMEs want "no diagnostic studies indicated." X-ray and MRI are not indicated unless there is motor/sensory abnormality (i.e., weakness or numbness). But for mere radiculopathy (i.e., radiating pain), no imaging necessary on new NBME content. - Straight-leg raise test is **not** reliable. Mere pain alone is a **negative** test. The test is only positive when they say it reproduces **radiculopathy/radiating pain.** There is a 2CK Q where they say straight-leg test causes pain (i.e., negative test) and answer is "no further management indicated" (i.e., Dx is only lumbosacral strain). - Tx is NSAIDs + light exercise as tolerated. Bed rest is wrong answer on USMLE.
Lumbosacral strain	- The answer on USMLE if they say patient has paraspinal muscle spasm following lifting of heavy box **without** radiculopathy. If they say radiculopathy, the answer is disc herniation instead. - Straight-leg test can cause pain (i.e., **negative** test). The test is only positive if they it reproduces radiating pain. **-** Do **not** x-ray. This is really HY for 2CK. Apparently lumbar spinal x-rays are one of the most frivolously ordered tests, and USMLE wants you to know that you do not order one for simple lumbosacral strain. - Tx is NSAIDs + light exercise as tolerated. Bed rest is wrong answer on USMLE.
Sciatica	- 90% of the time is due to disc herniation.  - Straight-leg test classically (+) – i.e., reproduces radiating pain. - Tx = Light exercise as tolerated + NSAIDs. Bed rest is wrong answer on USMLE. - On one of the 2CK CMS forms, ibuprofen straight-up is listed as the answer.
Meralgia paresthetica	- The answer on USMLE if they say patient has pain or paresthesias running down the lateral thigh. - Due to entrapment of lateral femoral cutaneous nerve. - Often seen as incorrect answer choice on Step, so at least be aware of it.

	
Scoliosis	- Sideways curvature of spine, creating an S- or C-shaped curve. - Usually idiopathic; affects 3% of population; girls 4:1.  Normal Spine Scoliosis Rib Rotation Scoliosis X-Ray with Rib Rotation - Can be associated with Marfan syndrome, Friedreich ataxia, NF1. - Adams forward bend test used to diagnose. - USMLE wants you to know most children do not need treatment, but that curvatures will remain throughout life. - Answer = bracing if curvature is >25 degrees and child is still growing. - I've never seen surgery as answer for scoliosis on NBME; literature says recommended only when curvature >40 degrees.
Kyphosis	- Abnormal convex curvature of thoracic spine. - Usually idiopathic due to old age; can be due to degenerative disc disease and compression fractures (osteoporosis). - If severe, can in theory cause restrictive lung disease due to impaired chest wall expansion.



Lower limb nerve HY Points for USMLE	
Common peroneal (fibular) nerve	- The answer on USMLE if patient loses both eversion **and** dorsiflexion of the foot. - Sensation to upper third of lateral leg (around and below lateral knee). - Splits into superficial and deep peroneal (fibular) nerves.
Superficial peroneal nerve	- The answer on USMLE if patient loses only eversion of the foot, but dorsiflexion stays intact. - Sensation to lower lateral leg and dorsum of foot.
Deep peroneal nerve	- The answer on USMLE if patient only loses dorsiflexion of the foot, but eversion stays intact. - **D**eep for **D**orsiflexion, which means superficial is the one that does eversion instead. - Loss of dorsiflexion causes a high-steppage gait (patient has to lift foot high into the air with each step). - Also does sensation to webbing between 1st and 2nd toes. I've never seen NBME Qs ask or give a fuck about this sensation detail, but students get fanatical about it as if it's supposed to be high-yield.
Tibial nerve	- The answer on USMLE if patient loses plantarflexion of the foot (can't stand on tippytoes). - Sensation to bottom of foot / heel.
Sciatic nerve	- The answer on USMLE if patient has motor dysfunction of tibial and common peroneal nerves at the same time, or has sciatica (shooting pain down leg). - Splits into the common peroneal nerve and tibial nerve. - Does not supply sensation to thigh; sensation encompasses that supplied by the combination of the common peroneal nerve and tibial nerves. - Supplies some motor function to muscles of thigh but USMLE doesn't care. - Sciatica = shooting pain from the lower back down the leg usually as the result of disc herniation; 2CK Neuro forms simply want NSAIDs as treatment; straight-leg test is classically used in part to diagnose, but I've seen this test show up on NBME material for simple lumbosacral strain (i.e., the test is non-specific and not reliable).
Obturator nerve	- The answer on USMLE if patient has inability to adduct the hip with loss of sensation to medial thigh.
Femoral nerve	- The answer on USMLE if patient cannot extend knee and/or has buckling at the knee.

	- Also does sensation to anterior thigh + medial leg (not thigh), although I haven't seen sensation specifically asked for femoral nerve.
Saphenous nerve	- The answer on USMLE if patient loses sensation to medial leg. - Pure sensory branch of the femoral nerve.
Sural nerve	- The answer on USMLE if patient loses sensation to lower lateral leg. In contrast, if sensation loss is upper lateral leg, that's common peroneal nerve instead. - Often confused with saphenous. Good way to remember is: sura**L** is **L**ateral, therefore saphenous must be the one that's medial.
Superior gluteal nerve	- The answer on USMLE if patient has Trendelenburg gait → opposite side of pelvis will fall while walking, so patient will tilt trunk toward side of lesion while walking to maintain level pelvis. - Innervates gluteus medius and minimus.
Inferior gluteal nerve	- The answer on USMLE if patient cannot squat, stand up from a chair, or go up/down stairs. - Innervates gluteus maximus.



Lower limb reflexes / radiculopathies	
L4 radiculopathy	- The answer on USMLE if patient loses knee (patellar) reflex + has weakened knee extension. - Pain / paresthesias / numbness in L4 distribution (anterior thigh + medial leg). - Disc herniation of L3-4. - Just remember that L4 is the one where the knee reflex is fucked up.
L5 radiculopathy	- The answer on USMLE if patient loses dorsiflexion. - Pain / paresthesias / numbness in L5 distribution (lateral + anterior leg).

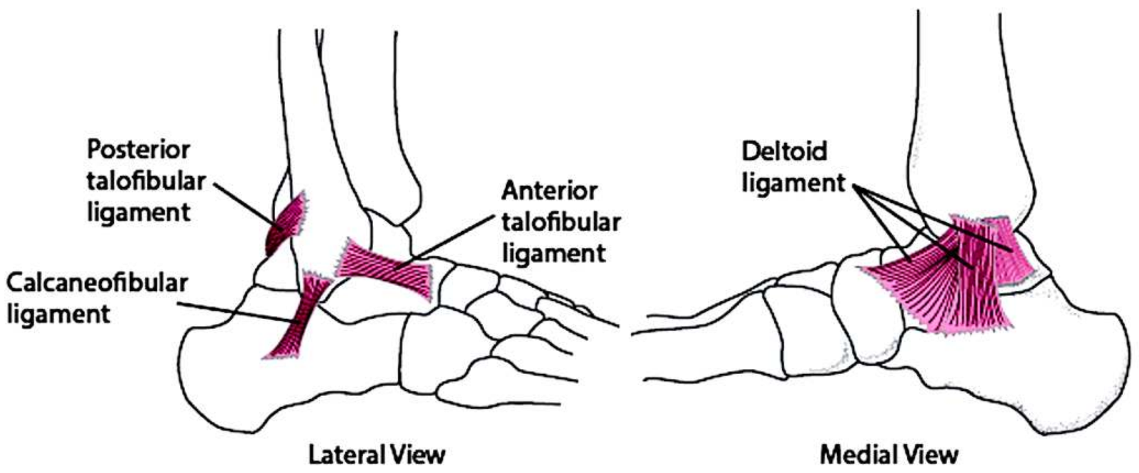
	- Disc herniation of L4-L5.
S1 radiculopathy	- The answer on USMLE if patient loses ankle (Achilles) reflex + has weakened plantar flexion. - Pain / paresthesias / numbness in S1 distribution (sole of foot + lower leg). - Disc herniation of L5-S1. - Just remember that S1 is the one where the ankle reflex is fucked up. - **SALT** → **S**1, **A**chilles, **L**ateral leg dermatome, **T**ibial motor issue (plantar flexion).

Lower limb DDx for FM	
Trochanteric bursitis	- Vignette will be lateral hip pain that is worsened with palpation + lying on one's side in bed. - Tx = NSAIDs.
Septic bursitis	- USMLE will give inflammation of knee joint that sounds like septic arthritis, but they will say there's **no** joint effusion. This is how it presents in a 2CK NBME vignette, where students constantly ask "why not septic arthritis?" And they say in the vignette there's **no** joint effusion. - Tx = antibiotics.
Prepatellar bursitis	- Aka housemaid's knee; presents as anterior knee pain in people who are frequently on their knees (painters, plumbers, etc.). - Tx = NSAIDs.
Patellar tendonitis ("Jumper's knee")	- Inflammation of patellar tendon. - The answer on USMLE if they describe anterior knee pain that initially occurs only after finishing sports (e.g., basketball game), then progresses to more chronic pain. - Tx on NBME = "quadricep strengthening exercises."
Patellofemoral instability	- Presentation is annoyingly similar to patellar tendonitis, but do not confuse. - Patellofemoral instability is misalignment of the patella at the trochlear groove of the femur. - Q can mention crepitus. - Shows up on 2CK Peds CMS form 6 as teenage girl who has knee pain worse after jumping or running + has crepitus → answer = "patellofemoral instability" (patellar tendonitis / "jumper's knee" not listed as answer).
Patellofemoral pain syndrome	- Aka chondromalacia patellae; name implies softening of cartilage in the knee. - The answer on NBME if they say pain that worsens when sitting for long periods of time, or when going up or down stairs. - Classic in obesity or those who squat heavy weight (knees think you're obese). - Tx = quadriceps strengthening exercises.
Anterior cruciate ligament injury	- ACL is answer if (+) anterior drawer test or Lachman test → excessive anterior displacement of tibia relative to femur. - Classically injured when knee is hyper-extended, or with a rotational force on a fixed, planted knee.
Posterior cruciate ligament injury	- PCL is answer if (+) posterior drawer test → excessive posterior displacement of tibia relative to femur. - Classically injured when knee hits the dashboard in car accident.
Lateral collateral ligament injury	- LCL is the answer if varus test induces excessive lateral motion of the knee compared to the unaffected side. - Varus test = hand placed on medial knee and pushing outward + other hand placed on lateral ankle and pushing inward.
Medial collateral ligament injury	- MCL is the answer if valgus test induces excessive medial motion of the knee compared to the unaffected side. - Valgus test = hand placed on lateral knee and pushing inward + other hand placed on medial ankle and pulling outward.
Lateral meniscal tear	- Lateral knee pain where patient experiences "locking" or "catching" of the knee in partial flexion.

	- Diagnosed with McMurray test → internal rotation of leg with concurrent knee extension causes lateral knee pain / "catching."
Medial meniscal tear	- Medial knee pain where patient experiences "locking" or "catching" of the knee in partial flexion. - Diagnosed with McMurray test → external rotation of leg with concurrent knee extension causes medial knee pain / "catching."
"Unhappy triad"	- Refers to a trio injury of the ACL, medial collateral ligament, and either the medial or lateral meniscus. - Students are sometimes fanatical about this triad as though it has yieldness. USMLE doesn't give a fuck. I cannot recall a single NBME question that has ever assessed this. This Dx primarily resides within the domain of Qbank, not the NBME.
Pes anserine bursitis	- The answer on USMLE if patient has inferomedial knee pain.
Iliotibial band syndrome	- The answer on USMLE if they say lateral knee pain, usually in a runner. - Iliotibial band runs from the hip to the knee. Pain may occur anywhere along the hip, lateral thigh, and lateral knee, but is worst in the latter. Tx = conservative with physiotherapy; NSAIDs for pain.
Osgood-Schlatter disease	- Buzzy vignette is knee pain in fast-growing teenage male who plays soccer. Don't pigeon-hole things, but that's classic vignette. - Inflammation of the patellar ligament at the tibial tuberosity. - Mechanism is repeated stress on the growth plate of the superior tibia.
Plantar fasciitis	- The answer on USMLE if they give severe heel pain that is worst when first getting out of bed in the morning.
Morton neuroma	- The answer on USMLE if patient has pain + abnormal growth occurring between the 2nd and 3rd metatarsals, usually worsened with high-heel shoes. - Benign growth/tumor of intertarsal nerve; cause is idiopathic. - First step in diagnosis is x-ray to rule out arthritis + fractures. Ultrasound is then done to confirm Dx, which will show thickening of interdigital/intertarsal nerve. - Tx = orthotics (comfortable shoes) + steroid injection.
Neurogenic joint	- Aka Charcot joint, where patient injures joint due to lack of joint sensation from peripheral neuropathy. - Usually seen in diabetes; can also be seen in neurosyphilis. - The answer on USMLE when they say diabetic patient has "disorganization of the tarsometatarsal joints" on foot x-ray.

Ankle sprains

- A sprain is an injury to ligaments (connective tissue holding two bones together).
- Might seem pedantic, but ankle injuries are HY for Family Med.



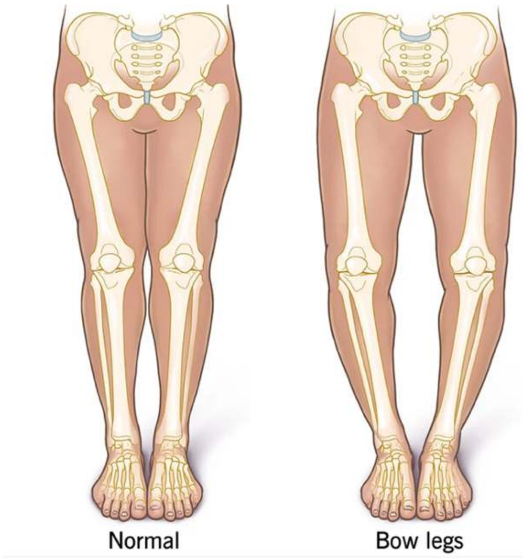
- Anterior talofibular ligament is on the lateral side of the ankle and will be injured if the foot inverts (rolls inward). This is the most commonly injured ankle ligament.
- The deltoid ligament is stronger and on the medial side of the ankle. This injury is more rare and occurs if the ankle forcibly everts (rolls outward).
- Below X-ray shows a ligamental ankle injury (left) followed by its repair (right). The Q wants to know which ligament was fucked up.

- Answer = deltoid ligament. Notice the large joint space on the medial aspect of the ankle in the left x-ray.

Ottawa criteria
- Exceedingly HY for FM. Used to assess probability of ankle fracture and tells us whether we need to order an X-ray or vs supportive care, e.g., RICE (rest, ice, compression, elevation). Before development of this criteria, x-rays for the ankle used to be ordered frivolously, with most showing no fracture. - Only order an x-ray for the ankle if: - Pain in the malleolar zone, AND any of the following: - Tenderness **posterior** to the lateral or medial malleolus; OR - Tenderness on the tip of the lateral or medial malleolus; OR - Patient cannot bear weight when walking four steps. - The above might seem nitpicky and pedantic, but this is HY for 2CK FM as I said. Examples: - 25M + twisted ankle yesterday + moderate edema of lateral side of ankle with ecchymoses + tenderness to palpation lateral and anterior to lateral malleolus + patient can weight-bear; Q asks, in addition to 2-day ice pack application, what is next best step? → answer on Family Med form = "use a soft protective brace and early range of motion exercises"; wrong answer = "x-ray of the ankle to rule out fracture." - 40M + playing basketball + rolls ankle + pain anterior to lateral malleolus + swelling of ankle + no pain posterior to lateral malleolus + patient can bear weight; Q wants next best step in management → answer = rest, ice, compression, elevation (RICE); wrong answer is x-ray; the patient doesn't fulfill the Ottawa criteria for x-raying the ankle; he has pain in the malleolar zone but does not have pain posterior to the malleolus or on the tip of the malleolus, and he can bear weight. - 26F + went running and rolled her ankle + pain in lateral ankle + tenderness posterior to malleolus + can bear weight; Q wants next best step → answer = x-ray of ankle; patient fulfills Ottawa criteria for x-ray → she has pain in malleolar region + tenderness posterior to the malleolus; although she can bear weight, the former two findings satisfy the Ottawa criteria.

HY Fracture points for FM	
Spiral fracture	- Pathognomonic for child abuse. - Caused by rotational/twisting force applied to a limb. - USMLE doesn't expect you to diagnose based on imaging.
Simple fracture	- Aka closed fracture – i.e., the skin is not broken and the underlying bone does not pierce the skin.
Compound fracture	- Aka open fracture – i.e., the skin is broken and the underlying bone pierces the skin.
Comminuted fracture	- Fracture where the bone is broken in at least two pieces.
Greenstick fracture	- Bone bends and cracks instead of breaking completely into two pieces. - More common in Peds than adults. - Children's bones are more flexible than adult bones adults' bones because they have ↑ collagen content and ↓ mineral content, allowing them to bend more. - When a force that would cause a complete fracture in an adult bone is applied to a child's bone, it might only cause a greenstick fracture, which is an incomplete fracture with the bone bending and breaking only on one side, resembling a green branch of a tree that bends and splinters on one side without breaking completely.
Linear skull fracture	- Most common type of skull fracture, where there is a break in the skull but the bone has not moved. - USMLE wants you to know this is classically associated with **epidural hematoma** (asked on 2CK CMS form).
Base of skull fracture	- Presents with tetrad of Battle sign (bruising over mastoid process), raccoon eyes (bruising around the eyes), rhinorrhea, and otorrhea.
Metatarsal stress fracture	- The answer on USMLE for pain in the metatarsal area of the foot in long-distance runners with low BMI. - Rather than asking diagnosis directly, USMLE will often give a vignette where a female long-distance runner with low BMI already has a metatarsal stress fracture, and then they'll ask what she's at greatest risk of developing → answer = osteoporosis.
Clavicular fracture	- Occurs with fall on outstretched hand (FOSH), or occasionally as a result of handlebar injury / impaction, with force transferred up to clavicle. - Most common site of break is the middle-third of the clavicle. - Tx = Figure-of-8 sling.
Scaphoid fracture	- As discussed earlier, will present as pain over anatomic snuffbox in patient with FOSH. - X-ray will usually be negative acutely. Must do thumb-spica cast to prevent avascular necrosis of scaphoid, followed by repeat x-ray in 2-3 weeks.
Lunate fracture	- The answer on USMLE if FOSH with pain in central palm + **no** pain over anatomic snuffbox.
Hook of hamate fracture	- Cause of distal ulnar nerve injury / Guyon canal syndrome. - Often from handlebar injury / impaction.
Surgical neck of humerus fracture	- Causes axillary nerve injury → loss of deltoid function + sensation over deltoid.
Midshaft fracture of humerus	- Causes radial nerve injury → wrist-drop + pronated arm.
Supracondylar fracture of humerus	- Aka "distal shaft fracture." - Causes median nerve injury → motor/sensory dysfunction of forearm muscles, first three fingers and thenar region.
Vertebral compression fracture	- Synonymous with osteoporosis on USMLE (i.e., post-menopausal, corticosteroid-use, Cushing syndrome). - Will often give point tenderness over a vertebra.
Pseudofracture	- Band of low-density bone that looks like fracture on x-ray but not actual fracture. - Synonymous with vitamin D deficiency (osteomalacia/rickets) on USMLE. - Can be seen in renal failure, since 1,25-D3 is low. Osteomalacia due to renal failure is called renal osteodystrophy.

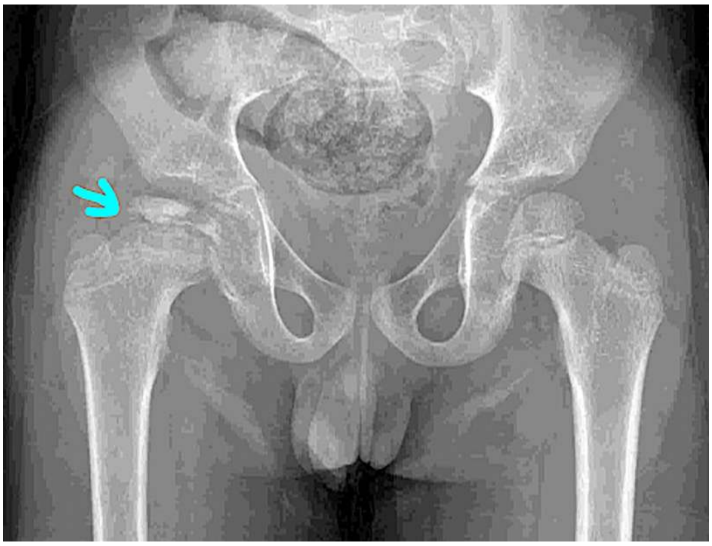
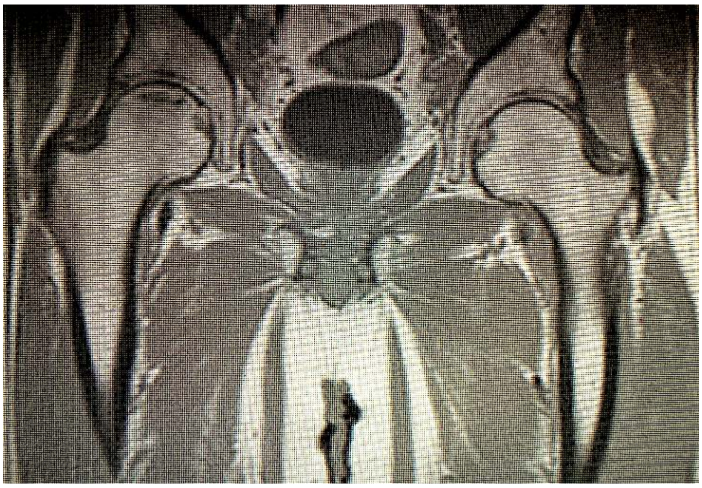
Orbital floor fracture	- USMLE wants you to know this can cause entrapment of **inferior rectus and inferior oblique** muscles. - Vignette will say guy got hit in eye by baseball + has impaired upward gaze. - I talk about extraocular muscles and lesions in my neuroanatomy document, but this is one notable point you should be aware of here.

Bone issues in kids	
Rickets	- Rickets = vitamin D deficiency in children. Osteomalacia = vitamin D deficiency in adults. - Rickets = craniotabes (soft skull), rachitic rosary (bony knobs at costochondral junctions), genu varum (bowing of tibias). - Activated vitamin D (1,25) is necessary to convert unmineralized osteoid into mineralized hydroxyapatite, therefore hardening bones. - In both rickets and osteomalacia, patient will have "increased unmineralized osteoid," or "deficient mineralization of osteoid." - Important cause of Vit D deficiency on USMLE is impaired intestinal malabsorption (i.e., CF, Crohn). For CF, answer can be written as "exocrine pancreas insufficiency." - "Pseudofracture" on x-ray is buzzy finding in osteomalacia/rickets. - Patients have ↓ Ca^{2+}, ↓ PO_4^{3-}, ↑ PTH (due to ↓ negative feedback at parathyroid glands). 
Tibia vara	- Aka Blount disease. 

<table>
<tr><td></td><td>- Bowing of the tibias after the age of 2 years in a patient whom rickets has been ruled out.
- Can be unilateral or bilateral.
- Bowing of one or both tibias is sometimes normal until age 2 years.
- Treatment is surgery (osteotomy).</td></tr>
<tr><td>Osteopetrosis</td><td>- Osteoclast dysfunction resulting in recurrent fractures in children due to bone density being too high. Sounds weird, but bone strength is based on balanced internal architecture of canals and networks, not just density alone.
- HY DDx against osteogenesis imperfecta and child abuse.
- Osteoclast dysfunction is due to deficiency of carbonic anhydrase II. This enzyme inside osteoclasts normally allows osteoclasts to form H^+ to resorb bone.</td></tr>
<tr><td>Growing pains</td><td>- No, this is not a joke. This is the answer straight-up on a 2CK NBME form.
- Vignette is healthy child age 3-12 who awakens from sleep with throbbing pain in the legs.
- No treatment necessary. You just need to know this Dx is exists and isn't a troll.</td></tr>
<tr><td>Talipes equinovarus</td><td>- Aka clubbed foot.

- USMLE just wants you to know that this is treated initially with serial casting.
- Usually idiopathic; can be seen in Potter sequence.
- Not the same as rocker-bottom foot (aka congenital vertical talus), seen in Edward syndrome.</td></tr>
<tr><td>Arthrogryposis</td><td>- You just need to know this is fancy name for a child born with multiple joint contractures.
- If they give you a child with not just a clubbed foot, but also knee and/or elbow contractures, etc., the answer is arthrogryposis.</td></tr>
</table>

HY Pediatric hip disorders for USMLE	
Primary hip dysplasia	- Aka developmental dysplasia of the hip. - Mechanism is "poorly developed acetabulum." - Initial diagnosis is with Ortolani and Barlow maneuvers, where a "clicking and clunking" is elicited on physical exam. - After the O&A maneuvers, next best step is ortho referral. Sounds wrong, but if it's listed, it's the answer before going to imaging. - Definitively diagnose with hip ultrasound if <6 months of age; hip x-ray if >6 months of age. - Treatment is "abduction harness," aka Pavlik harness, which positions the child's legs in a frog-leg-appearing manner.
Legg-Calve-Perthes	- Aka idiopathic avascular necrosis of the femoral head. - If the etiology for the avascular necrosis is known (i.e., Gaucher, sickle cell, corticosteroids), then the diagnosis is just "avascular necrosis," not LCP.

	- Vignette will be child 5-8 years old with hip pain. - First step in diagnosis is hip x-ray, which will show a "contracted" or flattened femoral head. The word "contracted" is HY and synonymous with avascular necrosis. - If x-ray is negative, diagnose with bone scan or MRI (on 2CK form). - Tx = hip replacement.  X-ray showing flattened/contracted femoral head (compare with the normal side that looks rounder).  On MRI, the right femoral head (left side of image) appears hypointense (more black). The necrotic / lack of bone in the black superior portion of the femoral head means the remaining white part of the femoral head is "flattened" or "contracted." The left femoral head (right side of image) shows a small area of necrosis as well (black medial portion).
Slipped capital femoral epiphysis (SCFE)	- Classic vignette is a 10-13-year-old (pre-adolescent) overweight boy with a painful limp. - NBME will write answer / mechanism as "displacement of the epiphysis of the femoral head." - Resources tend to emphasize obesity as the main risk factor, but maybe only ~1/2 of NBME Qs for SCFE give the child as overweight. This causes problems for students, where they rely on seeing high BMI to think SCFE. - This has led me to conclude that the **age** matters the most, since they will always give a kid who's about 10-13-ish. If they give you a kid who's younger, think LCP instead.

	- 2CK NBME Q gives 13M with painful gait + no mention of weight → answer is SCFE. - Another 2CK Q outright says BMI 20 in a 13-year-old who's 6 feet tall, and answer is SCFE. - X-ray shows "ice cream falling off the cone."  X-ray shows the "ice cream slipping off its cone." - Offline 2CK gives Q where they say x-ray in 5-year-old shows "contracted capital femoral epiphysis" → answer is LCP, not SCFE. As I said above, "contracted" is HY for LCP. In this case, the younger age + the word "contracted" win over the words "femoral capital epiphysis." Tx = surgical pinning.

HY Arthritides / joint pathologies for FM	
JIA/JRA	- Juvenile idiopathic arthritis (formerly juvenile rheumatoid arthritis; JRA). - 2CK forms love JIA. It will sound like regular RA but just in a kid. - USMLE will structure these Qs where they want you to pick between JRA and septic arthritis (SA) as answer choices. This can be confusing since SA can occur in patients with JRA. They might say a kid has a low-grade fever and a warm, red, painful knee (sounds like SA), but then they say he's had similar **episodes** in the past (i.e., they want JRA over SA). This is because SA is usually a one-off event; for JRA, however, the vignette will say "intermittent" or "episodic." Low-grade fever can occur in autoimmune flares (not limited to JRA; HY for sarcoidosis as well). - "Salmon-pink" maculopapular rash only in ~50% of JRA Qs. Often described as a buzzy finding, but I'd say about half of JRA vignettes don't even mention rash. - 2CK forms are obsessed with anemia of chronic disease in JRA. HY point is that MCV can absolutely be low. Resources push normal MCV for AoCD. This is absolute nonsense. Plenty of 2CK NBME Qs give MCV as 70s in AoCD. - Can be associated with serous pericarditis. - If the Q asks what arthrocentesis will show in JRA, the answer is leukocytes. - For antibodies, anti-cyclic citrullinated peptide (anti-CCP) **is more specific** than rheumatoid factor. Both should be ordered for JRA (and RA in adults). - Tx for JRA has arms of management: 1) symptoms; 2) disease progression. For symptoms, give NSAIDs first, followed by steroids. These do **not** slow disease progression. NSAIDs and steroids are for symptoms only. For disease progression, we use disease-modifying anti-rheumatic drugs (DMARDs), which slow disease progression. Methotrexate is given first, followed by adding an anti-TNF-α agent (i.e., infliximab, adalimumab, or etanercept).
Psoriasis	- Plaques are described as silvery and scaly and over extensor regions (elbows); plaques can also show up on the face (i.e., forehead and lip).

	- Auspitz sign is bleeding of the scales if removed. - 15% of patients with psoriasis will get arthritis before any skin findings. - HY point for USMLE is that psoriasis is part of the HLA-B27 constellation (**PAIR** → **P**soriasis, **A**nkylosing spondylitis, **I**BD, **R**eactive arthritis). - For example, if 17M has silvery plaque on elbow and forehead + bloody stool → the latter is most likely IBD due to HLA-B27 association. - Don't confuse psoriasis + IBD combo with dermatitis herpetiformis + Celiac disease combo. - Treatment for plaque psoriasis is topicals first → USMLE wants calcipotriene (vitamin D derivative), triamcinolone or hydrocortisone (both corticosteroids), and coal tar. Choose in that order if you are forced. Chronic application of topical steroids causes dermal collagen thinning, so they are not preferred prior to topical vitamin D. - Oral meds are given if patient fails topicals, OR if patient has systemic psoriasis (i.e., arthritis). **Oral methotrexate** is HY on new NBME material as the first-line oral agent used. - An old NBME has **oral acitretin** (a vitamin A derivative) as an answer, where methotrexate is not listed.
Sacroiliitis	- Sacroiliitis is broad term that refers to arthritis of lower back; ankylosing spondylitis (AS) is most severe form. - Vignette will almost always be male 20s-40s who has lower back pain worse in the morning that improves throughout the day. However, there is Q on 2CK Peds form where AS is diagnosis in an 8-year-old, so you need to know it's possible in peds. - Lower back pain in patient with IBD, psoriasis, or reactive arthritis points toward sacroiliitis (HLA-B27 PAIR). - High ESR and anemia of chronic disease high-yield. - Diagnose with x-rays of the lumbosacral spine and sacroiliac joints. - USMLE wants "slit-lamp examination" to look for anterior uveitis in ankylosing spondylitis. Any autoimmune disease can theoretically increase risk of this eye finding, but for whatever reason USMLE likes it for AS. - Treat same as JRA.
Lupus	- Arthritis is most common presenting feature of lupus (90%). USMLE vignette will pretty much always give arthritis in lupus Qs. - Anti-double-stranded-DNA (dsDNA) antibodies go up with acute flares and are most closely related to renal prognosis for lupus nephritis. - Anti-Smith (ribonucleoprotein) antibodies are **most specific** for SLE (more than dsDNA). - Anti-phospholipid syndrome due to lupus anticoagulant (antibodies against β2-microglobulin or cardiolipin in patient with SLE). - Anti-phospholipid syndrome can cause false-positive syphilis VDRL test (HY). - Malar rash is type III hypersensitivity. Harder Qs won't mention this finding because it's too buzzy. You need to know **thrombocytopenia** is frequently seen in lupus due to anti-hematologic cell line antibodies. Antibodies can also target WBC and RBC → looks like aplastic anemia, but it's not → answer = "increased peripheral destruction," not "decreased bone marrow production." - Can cause lupus cerebritis (confusion / delirium-like episodes) and transverse myelitis (presents as Brown-Sequard syndrome). Peds form gives lupus cerebritis. - Similar to RA, lupus can cause pericarditis. - Flares cause decreased serum complement protein C3. - Congenital complement protein C1q deficiency causes ↑ risk of developing SLE; sounds nitpicky but it's on new Step 1 NBME exam. - For lupus nephritis, **biopsy** as the first step in management; steroids first is wrong answer; biopsy first sounds wrong but it guides management. - Lupus nephritis = diffuse proliferative glomerulonephritis (DPGN) on USMLE. - Treat flares of lupus with steroids. USMLE doesn't care about other Txs.

Septic arthritis	- Shows up as young healthy patient with recent trauma (e.g., car accident) or high-intensity exercise causing microtrauma (e.g., kickboxing/soccer tournament, long hike), as well as in sickle cell disease. - Also occurs in patients with Hx of RA/JRA, OA, and prosthetic joints. - USMLE will give hot, red, painful joint in one of the above patient groups, almost always with a fever. - First step in management is arthrocentesis. A high-yield point is that joint aspirate can be **negative** for organisms. Do not rule-out SA if NBME says no organisms. - *Staph aureus* is most common pathogen overall. - Choose *Salmonella* in sickle cell. But you have to use your head. *Salmonella* is a gram-negative rod, so if they say gram-positive cocci is cultured in sickle cell, the answer is still *staph aureus*. - *Gonococcus* is the answer for sexually active patients; presents two ways on USMLE: 1) monoarthritis of the knee in sexually active young patient, with no other information provided; or 2) as a triad: mono- or polyarthritis, cutaneous papules, and tenosynovitis (inflammation of tendon sheaths). - Treatment is surgical drainage of the joint + antibiotics. USMLE doesn't care about the exact antibiotics.
Toxic synovitis	- Aka transient synovitis. - Presents as hip inflammation/pain in child after a viral infection. - Toxic synovitis is diagnosis of exclusion, meaning the vignette gives you various findings that make septic arthritis less likely. **Leukocytosis** and **inability to bear weight** make septic arthritis the likely answer over toxic synovitis. Hard questions will not mention warmth or redness as ways to differentiate. Fever can present in both. - Treatment is NSAIDs (ibuprofen) → asked on 2CK Peds.
Reactive arthritis	- Classically presents as triad of 1) urethritis or abdominal infection, 2) polyarthritis, and 3) "eye-itis" (i.e., conjunctivitis, episcleritis, or anterior uveitis). - *Chlamydia* is the classic organism that causes reactive arthritis. *Gonococcus* does **not** cause reactive arthritis on USMLE. - *Rubella*, Hep B+C, and *Yersinia* can also cause reactive arthritis. - Part of HLA-B27 constellation (PAIR), as mentioned above.

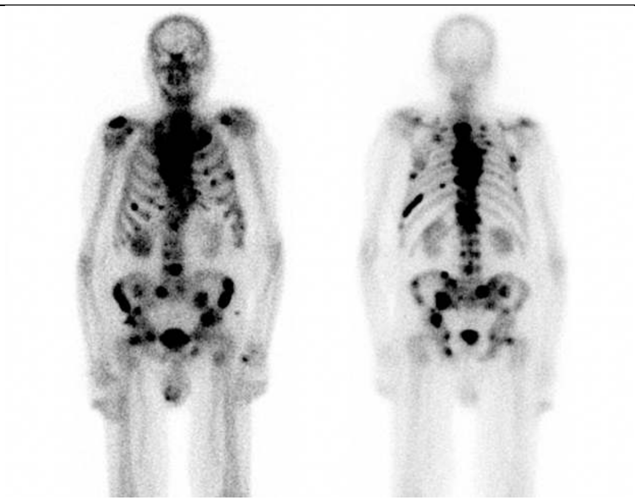
Paget disease
- Idiopathic disorder of increased bone turnover. Bone is described as having mixed osteoblastic and -clastic phases, where bone appears heterogenous on x-ray. - Buzzy vignette is male over 50 who's hat doesn't fit him anymore + has tinnitus (narrowing of acoustic foramina). - 19/20 questions will give isolated increase in serum ALP. You need to know Ca^{2+}, $PO4^{3-}$, and PTH are all normal in Paget disease. There is one Q on a 2CK CMS form where ALP is given as not elevated, but it's a one-off Q and rare. - High-output cardiac failure can occur due to intraosseous AV-fistulae, where patient has an S3 heart sound with high, rather than low, ejection fraction.

Osteoporosis
- Bone density >2.5 SD below mean compared to young adult woman. Osteopenia is 1.5-2.5 SD below mean. - Bone densitometry done at age 65 (2CK Family medicine). - If Q forces you to choose between female gender and age as most important risk factor, choose gender. - If Q gives you a female and forces you to choose between family history and age, choose family history. - Males are unlikely to develop osteoporosis, even with family history of females with the disorder. - If Q gives two women without family history and asks what is most protective against osteoporosis, answer is ethnicity. Black race is protective against osteoporosis.

- If Q gives old woman who has femoral fracture + no mention of osteoporosis in the question, answer = "activity level before fracture" as most important predictor of success in the rehabilitation of the patient → weight-bearing exercise during life is protective against osteoporosis later. - USMLE loves corticosteroids and Cushing as causes of osteoporosis. - Compression fractures = osteoporosis on USMLE; e.g., patient with RA on steroids who has compression fracture → easy Dx of osteoporosis. - Low/low-normal BMI causing osteoporosis is HY; 2CK NBMEs have a couple of nonsense Qs where they give BMI of 19 and 20 in young woman, where they ask what patient is at increased risk of; answer = osteoporosis. Student says, "Wait, but isn't low BMI under 18.5?" I agree. But it's on NBME. - Metatarsal stress fracture HY in low-BMI young female runners who have low bone density. - USMLE also is known to assess low vitamin D in the setting of intestinal malabsorption (i.e., CF, Crohn) as cause of osteoporosis, even though that makes no sense, since low Vit D causes osteomalacia. - Serum calcium, phosphate, PTH, and ALP are all normal in osteoporosis. - Tx = weight-bearing exercise first (NBME has "go for a long walk outside daily" as correct; wrong answer = "increase participation in swimming pool-based exercise classes to at least three times weekly"). - Calcium and vitamin D are the first medical / pharmacologic treatment. - Bisphosphonates can be used after Ca^{2+}/VitD. - Teriparatide is an *N*-terminus PTH analogue that stimulates bone development. - Denosumab is a RANK-L monoclonal antibody.

HY bone tumor points for FM	
Osteosarcoma	- Most common primary bone cancer; usually in patients age 10-30. - *Rb* mutations (congenital retinoblastoma) are associated with osteosarcoma (HY) – i.e., 1-year-old boy has enucleation of eye for retinoblastoma; what is he at risk of developing later in life? → answer = osteosarcoma. - Can also occur in Paget disease of bone patients (older age). - Buzzy findings are "Codman triangle" and "Sunburst appearance." - Codman triangle = periosteal reaction with lifting of periosteum off the bone. - Sunburst appearance = periosteal reaction described on NBME as "spiculated new born formation."  - The white arrow is the Codman triangle; the white star is the sunburst appearance. - I've seen osteosarcoma on Step 1 NBME written as "pleomorphic neoplastic cells producing new woven bone" as correct answer choice.
Chondrosarcoma	- Can occur in any long bone, as well as the **pelvis/hip.**

	- USMLE describes it as "glistening" or "shiny" in appearance. - The histo is exceedingly HY for Step 1 but not for Family Med. Just know this as DDx.
Ewing sarcoma	- The answer on USMLE if they give bone tumor in a child that presents similarly to osteomyelitis (i.e., fever and bone pain in a kid). - If bone scan is performed, it is most likely to show uptake in the diaphysis; in contrast, osteomyelitis will show uptake in the metaphysis. - Histo will show "small blue cells of neuroendocrine origin" and/or "onion-skinning." - Associated with t(11;22) translocation; don't confuse this with the 22q11 deletion in DiGeorge syndrome.
Osteoid osteoma	- Answer on USMLE for bone tumor that presents with pain relieved with NSAIDs.
Osteoma	- Answer on USMLE for bone tumor in Gardner syndrome (Familial adenomatous polyposis + soft tissue tumors [usually osteomas or lipomas]). - Benign bone tumors that usually grow from the skull.
Giant cell tumor	- Aka osteoclastoma. - Age of onset usually 20-40. - Has a "soap bubble" appearance.
Unicameral bone cyst	- Underrated diagnosis for USMLE. Asked on 2CK exam. - Benign bone tumor that looks similar to osteoclastoma but age of onset usually birth to age 20, rather than 20-40. - Unicameral means "one chamber"; it is fluid-filled.  2CK wants you to know this image for unicameral bone cyst.
Metastases	- USMLE loves mets to the vertebrae, particularly from breast, prostate, and lung. - The exam will not show images of spinal cancer mets, but they will give vignette of either lytic lesions of vertebrae in patient with background of cancer, or will give neurologic syndrome (i.e., of cauda equina). - Diffuse bone pain in patient with background of cancer = mets.

	 - Above image shows Technetium-99 bone scan of cancer mets. Similar imaging is on 2CK forms for prostate mets.

HY myopathies / muscular dystrophies for USMLE	
Duchenne / Becker muscular dystrophy	- XR disorder caused by mutation in dystrophin (*DMD*) gene. - Mutation results in disruption of α-/β-dystroglycan, which is required for proper internal **cytoskeletal** anchoring of the muscle cell to the extracellular matrix. - Presents with pseudohypertrophy, where muscles appear large but are replaced with fibroadipose tissue (connective tissue stromal cells). - Duchenne presents in a young boy who implements Gower maneuver to stand up (uses arms to walk up off the floor because leg muscles are weak). - Becker presents in adolescence or young adulthood (less severe form of Duchenne). - Duchenne is classically frameshift mutation; Becker is classically not frameshift.
Polymyalgia rheumatica (PMR)	- Usually patient over 50 with proximal muscle pain and stiffness. - No weakness on physical exam + creatine kinase (CK) levels are normal. If one or both of these findings is present, the answer is polymyositis, not PMR. - Can present with high ESR and low-grade fever (any autoimmune disease flare can present with low-grade fever). - PMR can present with or without temporal (giant cell) arteritis. Temporal arteritis can present bilaterally on NBME exams; do steroids first to prevent blindness, followed by biopsy second. - Temporal arteritis can cause jaw claudication (pain in the jaw during episodes). In contrast to temporomandibular joint dysfunction (a separate diagnosis), jaw claudication will not be precipitated by eating. - No specific diagnostic test; diagnosis is made clinically. - Tx = steroids. NSAIDs are wrong answer and are not proven to be effective.
Polymyositis / Dermatomyositis	- Usually patient over 50 with proximal muscle pain and stiffness. These findings are **not** unique to PMR. The USMLE will happily give pain and stiffness in polymyositis vignettes. - Key distinction between polymyositis and PMR is that polymyositis will present with **1) muscle weakness on physical exam, and/or 2) increased serum CK.** - The muscle weakness *must be on physical exam.* If the patient complains of "weakness" but there is no physical exam findings mentioned in vignette or physical exam shows 5/5 strength, there's no weakness. Patients will sometimes mention "weakness," even though they really just have pain and/or stiffness.

	- If polymyositis presents with skin findings, it is called dermatomyositis – i.e., Gottron papules (violaceous papules on the knuckles), mechanics' hands (rough-surfaced hands), shawl rash (body rash), heliotrope rash (violaceous eyelids / periorbital rash; don't confuse with malar rash of SLE). - Patients often positive for anti-Jo1 antibodies. - USMLE wants "electromyography and nerve conduction studies" as first step in management for polymyositis/dermatomyositis. This is what they ask on 2CK NBME forms. I have not seen them ask anti-Jo1 antibodies vs EMG+NCS as two separate answer choices. Usually anti-Jo1 antibodies are mentioned in the vignette rather than as the test you need to order. - Muscle biopsy is confirmatory, showing CD8+ T cell infiltration. The histo can be described as "CD8 + T cells and macrophages surrounding muscle fibers." - For whatever reason, dermatomyositis can be a paraneoplastic syndrome of ovarian cancer (shows up on NBME). Tx = steroids.  Gottron papules
Fibromyalgia	- This is a psych condition, not an actual muscle disorder, but is often confused with polymyositis and PMR. - Labs will be normal. ESR will not be elevated. Patient will not have fever. - Will be described as woman 20s-50s with multiple (and often symmetric) muscle tenderness points. - Treatment is SSRIs. USMLE can write this as "anti-depressant therapy." This confuses students ("But she doesn't have depression though.") → Right. But SSRIs are still anti-depressant medication.
Temporomandibular joint dysfunction	- The answer on USMLE if they give jaw pain that is **precipitated by eating.** - Often confused with jaw claudication seen in temporal arteritis. In the latter, however, the pain is not precipitated by eating.
Myotonic dystrophy	- Autosomal dominant, CTG trinucleotide repeat expansion disease. - Myotonia is inability to relax muscles. - The answer on USMLE if they say patient cannot relax grip on doorknob / handshake, or cannot let go of golf club. - Sometimes associated with early / frontal balding.
Hypothyroid myopathy	- Myopathy can occur in both hypo- and hyperthyroidism, yes, but this is especially HY for hypothyroidism on USMLE. - They will often sneak this in as proximal muscle weakness or increased CK in patient who has ongoing fatigue, dysthymia, menstrual irregularities, etc.
Drug-induced myopathy	- Classically seen when statins and fibrates are combined, but both drugs can cause myopathy independently. - Mild CK elevations are normal and expected in patients when commencing these agents. Dose does not need to be decreased for mild CK elevations. - USMLE wants "P450-mediated interaction" as the cause of the myopathy when statins and fibrates are combined.
Mitochondrial myopathy	- Broad term that can refer to numerous mitochondrial diseases.

	- USMLE wants you to know the patient has a mitochondrial disorder when he/she presents with **hypotonia, ear/eye problems, and lactic acidosis.** You want to memorize this tetrad as synonymous with mitochondrial disorders. - “Ragged red fibers” can be a buzzy descriptor in mitochondrial myopathy Qs. - Mitochondrial disorders are maternally inherited only. - Heteroplasmy refers to offspring having varying disease severity based on variation in allocation of diseased mitochondrial genes (I talk more about this stuff in my HY biochemistry PDF).
Inclusion body myositis	- The answer on USMLE if they tell you patient 50 or older has months to years of progressive muscle weakness + biopsy of muscle shows **basophilic rimmed vacuoles.**

Connective tissue disorders	
Marfan	- Autosomal dominant; chromosome 15, *FBN1/2*. - Codes for fibrillin, which is a glycoprotein that forms a sheath around elastin. - Tall, lanky body habitus with flat feet, chest wall abnormalities (i.e., pectus excavatum or carinatum), flat feet (pes planus), scoliosis, mitral valve prolapse (mid-systolic click), increased risk for aortic dissection (can retrograde propagate toward aortic root, causing root dilatation and aortic regurgitation [decrescendo diastolic murmur]). - Is not associated with berry/saccular aneurysms (unlike Ehlers-Danlos and ADPKD).
Ehlers-Danlos	- Defect in collagen III synthesis; can be written as “Defect in synthesis of fibrillar collagen.” - This can be confusing because fibrillin is completely unrelated and refers to Marfan syndrome. It is just coincidental that “fibrillar collagen” means type III collagen. - Wrong answer = “abnormal synthesis of extracellular glycoprotein” (refers to fibrillin for Marfan syndrome, which as mentioned above, is a glycoprotein that forms a sheath around elastin. - Presents as easy bruising + hyperextensible skin. 
Osteogenesis imperfecta	- Collagen I defect that results in recurrent fractures in a child; important DDx are child abuse and osteopetrosis. - Blue sclerae too buzzy and often not mentioned. - Conductive hearing loss due to defective ossicles (middle ear bones). - Many different types of OI, some resulting in miscarriage. Harder vignette can mention child with multiple fractures, where the mom has Hx of recurrent miscarriage.
Williams	- Autosomal dominant; chromosome 7. - Defect in elastin. - Elfin-like facies, hypercalcemia, well-developed verbal skills. - Can cause supravalvular aortic stenosis.

Mixed connective tissue disease	- Answer on USMLE if they give you a patient who has anti-U1-ribonucleoprotein (U1-RNP) antibodies. - Patient presents as having combined features/symptoms from three different disorders → LPS → Lupus, Polymyositis, Scleroderma.

Basic hernia points for FM	
Direct inguinal hernia	- Occurs usually in older men. - Small bowel herniates medial to inferior epigastric vessels. - Occurs due to weakness of abdominal wall musculature / transversalis fascia. - Q might say older patient has palpable mass in groin that reduces when he lies down, or worsens when he coughs. Tx = "operative management" or "elective hernia repair," since it is not an overt emergency but closure should be performed prior to any type of incarceration and strangulation (ischemia leading to pain, fever, and necrosis).
Indirect inguinal hernia	- Seen classically in male infants, but can occur any age. - Small bowel herniates lateral to inferior epigastric vessels, through deep inguinal ring. - Mechanism is patent processus vaginalis. This is also the mechanism for hydrocele (asked on NBME). - Since it passes through deep inguinal ring, it can be reduced with pressure / a finger placed over deep inguinal ring. Direct inguinal hernia, since it does not pass through deep inguinal ring, will not reduce with pressure applied here. - Tx = elective hernia repair. - For hydroceles, observation is the answer under the age of 1 (HY on Peds forms).

HY neck MSK masses for FM	
Thyroglossal duct cyst	- The answer on USMLE if they say there's a painless, midline neck lump in a child that moves upward with swallowing or protrusion of the tongue. This buzzy description is seen for maybe only about half of Qs. - Can also be described as painless mass inferior to the hyoid bone that demonstrates uptake with a Technetium-99 scan. - USMLE wants "endoderm of foramen cecum" as the embryology.
Sternocleidomastoid injury	- The answer on USMLE if they say nodular mass in the lateral neck in an infant who had been born via forceps delivery (risk factor for damage to the muscle).
Branchial cleft cyst	- The answer on USMLE if they give idiopathic lateral neck mass in infant that may or may not have an opening to the skin.
Lingual thyroid	- Neonates or infants. - Answer for hypothyroidism + a midline neck lump located high in the neck. - They say nothing about protrusion of the tongue or uptake into the mass (of course this is thyroglossal duct cyst instead). - Can sometimes cause dysphagia (trouble swallowing), dysphonia (voice changes), or dyspnea (difficulty breathing).

Highest yield "MSK pharm" for FM	
- I discussed some of these in the Neuro and Psych sections, but I reiterate a few due to their yieldness.	
Alendronate	- Bisphosphonate; inhibits osteoclasts. This MOA is HY. - Used for osteoporosis after Ca^{2+}/VitD. - I've seen pamidronate (not alendronate) show up on 2CK forms for Tx of hypercalcemia (after normal saline is given). - Students get fanatical about bisphosphonates causing osteonecrosis of the jaw. The yieldness of this adverse effect is basically non-existent on NBME exams.

	USMLE wants you to know bisphosphonates cause **pill-induced esophagitis.** This is very HY for 2CK FM forms in particular (K^+ supplements also cause esophagitis).
Teriparatide	- *N*-terminus PTH analogue that can induce bone formation. Even though PTH causes bone resorption, this agent stimulates osteoblast-mediated bone formation more than it induces RANK-L-mediated activation of osteoclasts. - Can be used for severe/advanced osteoporosis.
Denosumab	- Monoclonal antibody against RANK-L. - Can be used for severe/advanced osteoporosis.
Baclofen	- Agonizes GABAB. - Used for spasticity, classically in multiple sclerosis, but I've seen one NBME Q where it's used for random spasticity in an older dude. - Students frequently remember GABAB for this drug, but often say "antagonist" when I probe them further. So remember: it's an agonist, not an antagonist, at GABAB.
Cyclobenzaprine	- Muscle relaxant used for spasms. - Structurally similar to TCAs; helps modulate pain sensation at brain stem.
Benztropine	- Muscarinic (cholinergic) receptor antagonist. - Used to treat acute dystonia due to anti-psychotics. - If patient starts anti-psychotic and then gets stiff neck, oculogyric crisis (abnormal eye movements), or muscle rigidity **without** fever, the answer = benztropine.
Diphenhydramine / Chlorpheniramine	- First-generation histamine-1 (H1) antagonists. - Diphenhydramine is quite possibly the highest-yield drug on USMLE. - Used to treat acute dystonia, similar to benztropine, as well as motion sickness. - H1 blockers can treat allergies in theory, but they have **nasty anti-cholinergic (anti-muscarinic) side-effects.** - The anti-cholinergic side-effects are interestingly a *good* thing, however, when we want to treat acute dystonia. Psych Qs will either list benztropine or diphenhydramine (or chlorpheniramine) as the answer, but not both at the same time. - For whatever reason, anti-cholinergic effects treat motion sickness. Scopolamine is an anti-cholinergic used to treat motion sickness classically. But I've seen NBME ask diphenhydramine straight-up for this as well – i.e., the nasty anti-cholinergic side-effects are, once again, a good thing if the aim is Tx of motion sickness. - 1^{st}-gen H1 blockers can cause **cognitive dysfunction** (delirium, as well as worsening of dementia) and **drowsiness.** Therefore avoid in elderly and locomotive/machine operators if at all possible. - 1^{st}-gen H1 blockers can also cause anti-α1-adrenergic effects (orthostatic hypotension). - I talk about all of the pharm-related stuff in a lot more detail in my free pharm modules on the website.
Dantrolene	- Blocks ryanodine Ca^{2+} channel. - Tx for neuroleptic malignant syndrome (NMS) and malignant hyperthermia (MH). - If patient gets muscle rigidity and fever following commencement of anti-psychotic, or following administration of succinylcholine during surgery, answer = dantrolene. (Bromocriptine for NMS is low-yield and rarely seen on NBME). - NBME will sometimes give vignette of NMS or MH, and then the answer for Tx is "decreases sarcoplasmic calcium release." - In NMS and MH, the ryanodine channel, which allows calcium to move from the sarcoplasmic reticulum into the cytosol, gets stuck open, so high amounts of calcium moves into the cytoplasm. The cell then needs to use a lot of ATP to pump the calcium back into the sarcoplasmic reticulum. This generates heat → fever. Dantrolene closes this channel.

FM Immuno

Vaccine info for FM	
- The vaccine schedule continually gets updated, so rather than putting/writing out a schedule here where students constantly DM me and say, "How updated is your FM PDF vaccine stuff??" I advise you to Google the latest schedules to check for changes / to ensure you're looking at the newest info. - What I will do here is write for you some salient HY points about vaccines as per my observations on 2CK Family Med material.	
HepA	- Give if going to endemic areas (e.g., Mexico, India). - Give to MSM and IV drug users. Latter in particular sounds weird, considering HepA is fecal-oral, not parenteral, but it's a HY point.
HepB	- Birth, 2 months, 6 months. - We used to give it at 4 months as well, but apparently not anymore. - If mother's status is unknown, do not give HepB IVIG to neonate unless mother's status comes back (+). For the neonate in this setting, the answer is "give vaccine but only HepB IVIG if mother's status is positive." - I talked about HepB serology (i.e., surface/core Abs, window period, etc.) in detail in the Gastro section of this PDF.
Rotavirus	- 2, 4, 6 months. - Oral live-attenuated.
DTaP	- Diphtheria, Tetanus, acellular Pertussis. - 2, 4, 6 months, then again at 15-18 months, then again at 4-6 years. - School-age kids require Tdap booster at 11-12 years, followed by Td booster every 10 years thereafter. - Pregnant women should get Tdap at 27-36 weeks to protect neonate from pertussis. For cuts/wounds post-vaccine: - 0-5 years post-vaccine: no Tx is necessary. - 6-9 years post-vaccine: if clean wound: no Tx; if dirty wound: give Td booster. - 10+ years post-vaccine: if clean wound: give Td booster; if dirty wound: IVIG + vaccine. - In other words, only ever give IVIG if it's a dirty wound + has been 10+ years.
HIB	- *Haemophilus influenzae* type B. - 2, 4, 6 months, then again at 15-18 months. - Patients with **asplenia/splenectomy** require additional dose (along with *Neisseria meningitidis* and *Strep pneumo*).
Polio	- Killed IM at 2, 4, 6 months, then again at 4-6 years.
S. pneumo	- PCV15 at 2, 4, 6 months, then again 12-15 months. - Age ≥65: PCV20 once; OR, PCV15 then PPSV23 a year later. - For **asplenia/surgical splenectomy**, sickle cell, cochlear implant, CSF leak, chronic renal failure/nephrotic syndrome, HIV, and immunosuppressed patients: extra dose of PCV15, followed by PPSV23 8+ weeks later.
Influenza	- IM killed vaccine: start age 6 months, then give yearly through life; safe to give during pregnancy. - Intranasal live-attenuated: ages 2-45; immunocompetent, non-pregnant persons.
Coronavirus	- IM vaccine starting at 6 months; 2-3-doses.
MMR	- Mumps, measles, rubella. - First dose 12-15 months; second dose 4-6 years. - Not safe during pregnancy or in HIV patients with CD4 count <200.
HPV	- Human papillomavirus. - 2 doses at age 11-12, separated by 6 months.
Meningococcal	- First dose at 11-12; second dose age 16. - Extra dose for **asplenia/splenectomy** patients.

FM Endocrine

Acromegaly
- Caused by excess growth hormone secretion (usually by a tumor) following closure of the growth plates (↑ GH secretion prior to growth plate closure causes gigantism). - As mentioned earlier, GH goes to the liver, which causes it to secrete IGF-1. It is then IGF-1 that induces growth effects at tissues. An NBME Q gives easy vignette of acromegaly and then asks what you check; serum GH is wrong; answer is serum IGF-1. - Acromegaly causes prognathism (lantern jaw), large ears/nose, arthritis and carpal tunnel syndrome (due to ↑ joint and tendon growth), enlargement of hands/feet, hypertension, type II diabetes (GH causes insulin resistance), and ↑ risk of arrhythmia and cardiomyopathy. - Treatment is somatostatin analogue (octreotide). Somatostatin is aka growth hormone-inhibiting hormone (GHIH) and shuts off its secretion from the anterior pituitary. It also ↓ secretion of insulin and glucagon from the pancreas.

Vasopressin conditions	
- Vasopressin (aka anti-diuretic hormone; ADH). - ADH is produced by both the supraoptic nucleus and paraventricular nucleus of the hypothalamus → stored in posterior pituitary. - ADH ↑ free water reabsorption by the medullary collecting duct (MCD) of the kidney by causing aquaporin insertion. - If ADH is ↑, we get ↑ free water reabsorption and dilution of our serum → serum sodium and osmolality ↓. Likewise, our urine becomes more concentrated, meaning urine osmolality and specific gravity ↑. - If ADH is ↓, we get ↓ free water reabsorption and concentration of our serum → serum sodium and osmolality ↑. Likewise, our urine becomes more dilute, meaning urine osmolality and specific gravity ↓.	
SIADH	- Syndrome of Inappropriate Anti-Diuretic Hormone secretion → means too much ADH (vasopressin secretion). - Central SIADH → follows head trauma, meningitis, brain cancer, and pain (latter on 2CK Surg). - Ectopic SIADH → small cell lung cancer secreting ADH. - Drug-induced ADH → ultra-rare on USMLE, but carbamazepine can do it. - Patient will have dilute serum and concentrated urine: - ↓ serum sodium, ↓ serum osmolality, ↓ serum specific gravity. - ↑ urinary osmolality, ↑ urinary specific gravity. - You must know serum sodium is normally 135-145 mEq/L. So in SIADH, it's <135. - Serum vs urinary osmolalities will be all over the place and you do *not* need to memorize values. They might say serum osmolality is 250 and urinary is 750, and then you say, "Well I can tell serum is dilute compared to urine, which sounds like SIADH." - Specific gravity will be 1.000-1.030 on USMLE. Sounds obscure, but it shows up quite a bit, particularly on 2CK Qs. It's to my observation that values 1.000-1.006ish are "dilute"; 1.024-1.030 are "concentrated." For 9/10 Qs, values will be what you expect. - "Fluid restriction" is first answer in diagnosis on 2CK. We want to see how serum/urinary values change in response. - Demeclocycline is answer on 2CK offline NBME 8 for treatment of SIADH. Demeclocycline is technically a tetracycline antibiotic but isn't used because it can cause insensitivity to ADH at the kidney (i.e., nephrogenic diabetes insipidus). So we essentially cause a 2nd problem that cancels out the 1st problem.
DI	- Central diabetes insipidus → means not enough ADH secretion by hypothalamus, or the posterior pituitary is unable to release it properly. - Nephrogenic DI → insensitivity to ADH at the kidney (serum ADH is ↑).

	- Similar to central SIADH, central DI can be caused by head trauma, meningitis, and cancer. - Nephrogenic DI is caused by **lithium, demeclocycline, hypercalcemia,** and **NSAIDs.** - 2CK Q gives patient with primary hyperparathyroidism and ↑↑ serum calcium + nephrogenic DI; Q asks cause of the DI → answer = hypercalcemia. High calcium can cause renal insensitivity to vasopressin. - Serum vs urinary values are the opposite of SIADH: - ↑ serum sodium (>145 mEq/L), ↑ serum osmolality, ↑ serum specific gravity. - ↓ urinary osmolality, ↓ urinary specific gravity. - When we are trying to first diagnose DI, the first thing we do is fluid restriction, same as with SIADH. We want to see how serum/urinary values change first. - After we determine that the urine is staying dilute + the serum is staying concentrated, the next best step is desmopressin (analogue of vasopressin). If the urine gets more concentrated, (i.e., if the drug works), we know central DI is the diagnosis and we're merely deficient in ADH. - If desmopressin doesn't work, we know we have nephrogenic DI. I should point out that even in nephrogenic DI, desmopressin might work *but only very little,* whereas with central DI, administration will ↑↑ urinary osmolality robustly. It will be obvious on USMLE. But my point is, don't say, "Oh well desmopressin worked like 5% so we can't have nephrogenic DI here." - Treatment for central DI is therefore desmopressin. - Treatment for nephrogenic DI is NSAID + a thiazide. Sounds weird, but ↓ Na^+ reabsorption induced by thiazides in the early-DCT promote compensatory ↑ Na^+ reabsorption in the PCT, where water follows Na^+ and our net loss of fluid is less than without the thiazide. In healthy individuals, however, they *will* lose more net fluid with the thiazide. The NSAID presumably ↓ renal blood flow, which will ↓ GFR and ↓ net fluid loss. - There is difficult 2CK NBME Q where they say patient is on lithium + has ↑ urinary output, and fluid restriction is wrong answer to this question (I say hard because 9/10 times, fluid restriction is correct when it's listed); answer = NSAID + thiazide diuretic. The implication is, if it's obvious what the patient's diagnosis is already nephrogenic DI, going straight to Tx is acceptable.
PP	- Psychogenic polydipsia means the patient is simply drinking too much. - Both the urine and serum will be dilute. - Serum vs urinary values are the opposite of SIADH: - ↓ serum sodium (<135 mEq/L), ↓ serum osmolality, ↓ serum specific gravity. - ↓ urinary osmolality, ↓ urinary specific gravity. - I'd say 3/4 Qs on USMLE are obvious and will say some psych patient is drinking lots to "clear himself from evil spirits," etc. - Probably 1/4 Qs won't be an obvious psych vignette, but will just show you the lab values where you have to say, "The urine and serum are *both* dilute, so this is psychogenic polydipsia." - First step in diagnosis is fluid restriction in order to see how urinary/serum values change.

Thyroid endocrine terminology	
Primary hypothyroidism	- ↓ T3, ↓ T4, ↑ TSH. - The thyroid gland itself is under-producing thyroid hormone. TSH goes ↑ due to ↓ negative-feedback at hypothalamus + anterior pituitary. - HY examples are Hashimoto thyroiditis (autoimmune destruction of thyroid gland) and iodine deficiency (insufficient thyroid hormone precursor).
Secondary hypothyroidism	- ↓ T3, ↓ T4, ↓ TSH. - The thyroid gland is under-producing thyroid hormone due to inadequate stimulation (i.e., the anterior pituitary isn't producing sufficient TSH). - HY causes are Sheehan syndrome (ischemic infarction of anterior pituitary postpartum) and pituitary tumors (e.g., prolactinoma, which impinge on and necrose the thyrotropes in the anterior pituitary that produce TSH).

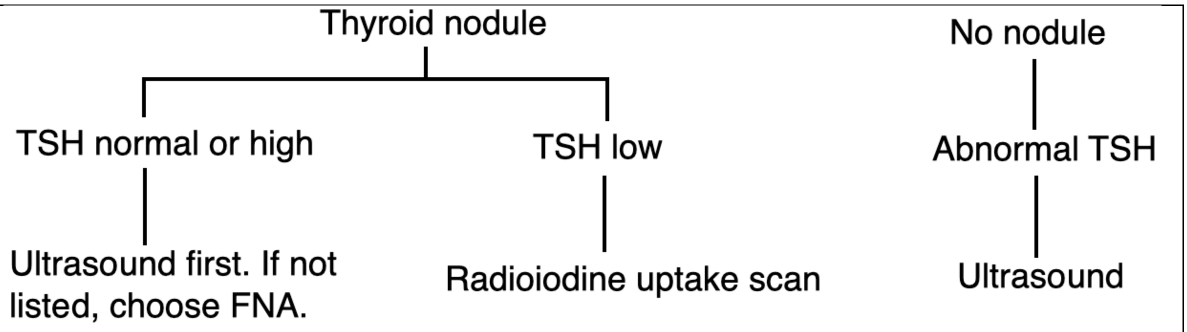
	- ↓ TRH from the hypothalamus causing ↓ TSH could be another theoretical cause, but I've never seen this assessed.
Primary hyperthyroidism	- ↑ T3, ↑ T4, ↓ TSH. - The thyroid gland itself is over-producing, causing suppression of TSH. - HY example is Graves disease (activating-antibody against TSH receptor), toxic adenoma (autonomously secreting nodule), and toxic multinodular goiter (multiple autonomously secreting nodules). - TSH is suppressed due to ↑ negative-feedback at the hypothalamus / anterior pituitary.
Secondary hyperthyroidism	- ↑ T3, ↑ T4, ↑ TSH. - TSH- or TRH-secreting tumor causing over-stimulation of thyroid gland. - Despite the ↑ thyroid hormone, TSH can't be suppressed because there is autonomous secretion from a tumor. - In theory, hypothalamic / anterior pituitary glandular hyperplasia, or inflammation (autoimmune hypophysitis) could also cause ↑ TSH leading to secondary hyperthyroidism.

HY Thyroid diagnoses	
Hashimoto thyroiditis	- Autoimmune destruction of thyroid gland. - Lymphocytic infiltrate seen on biopsy (asked on NBME). - Anti-microsomal (aka anti-thyroperoxidase) + anti-thyroglobulin antibodies. - ↓ T3, ↓ T4, ↑ TSH. - ^{131}I uptake scan is ↓ or patchy (i.e., the gland is not producing hormone). - Buzzy findings are weight gain, brittle hair, dry/doughy skin, and cold intolerance. The USMLE often will omit these findings because they're too easy. - HY additional findings I've seen show up in NBME Qs are: bradycardia (55-60), menstrual irregularity, dysthymia/depression/apathy, carpal tunnel syndrome (due to glycosaminoglycan [GAG] deposition), ↑ cholesterol, proximal muscle weakness with ↑ creatine kinase (hypothyroid myopathy); transaminitis (sounds weird, but LFTs can be ↑). - Associated with other autoimmune diseases (e.g., type I diabetes, pernicious anemia, vitiligo) and immunodeficiencies (e.g., IgA deficiency). For USMLE, a HY big-picture concept is that "Autoimmune diseases go together," where if you have one autoimmune disease, there's ↑ risk of other autoimmune diseases. The strict HLA associations aren't important (outside of HLA-B27). In addition, "Autoimmune diseases and immunodeficiencies to together," where having autoimmune disease in general ↑ risk of immunodeficiencies, and vice versa. - ↑ risk of non-Hodgkin lymphoma (e.g., primary CNS lymphoma). - Tx = levothyroxine or triiodothyronine.
Graves disease	- Activating autoantibody against TSH receptor. This antibody is called thyroid-stimulating immunoglobulin (TSI). - ↑ T3, ↑ T4, ↓ TSH. - Lymphocytic infiltrate + hyperplastic thyroid follicles seen on biopsy. - ^{131}I uptake scan shows diffusely ↑ uptake. - Exophthalmos (aka proptosis; protrusion of eyes) and pretibial myxedema (non-pitting edema of shins due to GAG deposition) are specific findings in Graves. The vignette need not mention either of these findings, but if the Q specifically tells you "there's no exophthalmos," they're saying it's not Graves. - Buzzy findings are tachycardia, heat intolerance, sweating, and palpitations. - Myopathy can also occur with ↑ CK, but USMLE tends to give this finding in hypothyroidism. - Can cause atrial fibrillation in younger patients (normally AF is elderly).

	- "Thyroid storm" is an acute exacerbation of Graves, often precipitated by a stress factor such as surgery, trauma, or infection, where thyroid hormone production goes ↑↑↑, causing dangerous physiologic responses, such as hyperthermia, tachycardia, and altered mental status. - Thyroid storm can also cause adrenal crisis (i.e., acute ↓ in blood pressure due to rapid consumption of cortisol), especially in those with concurrent adrenal insufficiency taking exogenous steroids. Tx for the low BP is IV glucocorticoid (i.e., hydrocortisone or methylprednisolone). - Tx for general Graves is thionamides (propylthiouracil [PTU] and methimazole). - Both PTU and methimazole can cause neutropenia/agranulocytosis. - Propranolol is used for tachycardia in hyperthyroidism. - Oral prednisone is specifically used for Tx of exophthalmos, not thionamides. - Tx of thyroid storm on USMLE is tetrad of: 1) PTU, 2) propranolol, 3) steroids, and 4) potassium iodide.
deQuervain thyroiditis	- Aka subacute granulomatous thyroiditis, or just simply subacute thyroiditis. - Mechanism is viral infection followed by a **painful/tender** thyroid. There is inflammation of the thyroid gland, which causes the spacing between the cells to increase slightly, allowing for the release of pre-formed thyroid hormone into the blood. Therefore we have ↓ TSH, ↑ T3, ↑ T4, ↓ ^{131}I uptake. - The gland is not over-producing thyroid hormone. This is why uptake is not increased. - Subacute granulomatous thyroiditis can be either hypo- or hyperthyroid. The key detail you need to know is that **uptake is always decreased *even if the patient is hyper-.*** **-** DeQuervain vignettes will almost always be given as hyperthyroidism because the USMLE wants to specifically assess that you know uptake is decreased. If they give you hypo-, of course you'll select decreased for uptake. - Decreased uptake applies to all thyroiditis conditions (i.e, deQuervain, drug-induced, and postpartum). - Viral infections are often asymptomatic, so most deQuervain vignettes will not mention the viral infection.
Drug-induced thyroiditis	- Caused by lithium or amiodarone on USMLE. - Can in theory be hypo-, eu-, or hyperthyroid. - Will usually present on USMLE as painless hypothyroidism in patient being treated for bipolar disorder (lithium) or started on new anti-arrhythmic (amiodarone). - Will present with ↓ ^{131}I uptake, where inflammation of the gland can sometimes cause leakage of thyroid hormone into the blood, but the gland itself is not demonstrating increased production of hormone.
Subclinical hypothyroidism	- ↑ TSH, ↔ T3, ↔ T4. **-** In subclinical hypothyroidism, the patient will be asymptomatic (hence subclinical) and will have normal T3 and T4, despite an elevated TSH. - Most patients with subclinical do not need to be treated. - Don't treat unless TSH >10, patient is pregger, or anti-Hashimoto Abs are +.
Euthyroid sick syndrome	- Vignette will give patient who's had recent major trauma or surgery, or for patients in hospital on ventilators. - ↔ TSH, ↔ T4, ↓ T3, ↑ rT3. - Etiologies are manifold, but one proposed mechanism is that in times of stress (usually acute), spikes in cortisol and other inflammatory mediators can inhibit peripheral conversion of T4 to T3. This causes T3 to go down. - T4, however, rather than being detectable in excess, is converted to an inactive form of thyroid hormone called reverse T3 (rT3), so rT3 is high. - T4 is therefore still in the normal range, as is TSH (presumably because T4 is normal).

	- Call it weird all you want, but USMLE asks this. This is one of the most "underrated" thyroid diagnoses, as students will often have heard of it but then disregard it, thinking it's minutiae and probably won't show up.
Toxic adenoma	- Autonomously secreting thyroid nodule (aka hot nodule). - ^{131}I scan shows uptake onto into the area of a single nodule. - ↑ T3, ↑ T4, ↓ TSH. - Not malignant (i.e., no metastatic potential).
Toxic multinodular goiter	- Autonomously secreting thyroid nodules (i.e., many hot nodules). - ^{131}I scan shows multinodular uptake. - ↑ T3, ↑ T4, ↓ TSH. - Not malignant (i.e., no metastatic potential). - I don't think I've ever seen this as a correct answer on USMLE, but it shows up quite a bit as a distractor, e.g., for deQuervain Qs. - Occurs in elderly. You might get a younger patient with hyperthyroidism due to deQuervain, and you can eliminate toxic multinodular goiter because 1) the patient is young, and 2) deQuervain is painful/tender, whereas TMG is not.
Riedel thyroiditis	- Obscure fibrosing disorder characterized by replacement of the thyroid gland with dense, fibrous tissue (sometimes called "woody thyroid" for this reason). - Can sometimes resemble anaplastic thyroid carcinoma, where it extends into adjacent structures such as the esophagus or trachea.
Factitious thyrotoxicosis	- Aka surreptitious thyrotoxicosis. - USMLE need not mention the patient is a pharmacist (implying access to the hormone). - Q will give hyperthyroid patient with small, non-palpable thyroid gland (atrophic due to suppressed TSH). - As mentioned earlier, T4 is converted to T3, but T3 isn't converted to T4. - So if thyroxine is administered, we will have ↑ T3, ↑ T4, ↓ TSH. - If triiodothyronine is administered, we will have ↑ T3, **↓ T4,** ↓ TSH.
Struma ovarii	- Abstruse ovarian tumor that for some magical reason secretes T3 and T4.

Thyroid nodule evaluation
- When evaluating for thyroid cancer, the first step is "palpation of thyroid gland" before "check serum TSH." There is an NBME Q for 2CK where both are listed and palpation is correct (i.e., do a history and exam before jumping into tests). - Once a nodule is palpated, *then* serum TSH is the next best step. - If patient with nodule has normal or high TSH (i.e., is euthyroid or hypothyroid), do ultrasound before fine-needle aspiration (FNA). This is assessed on the new 2CK Free 120, where ultrasound is correct over FNA for evaluation of thyroid nodule. It had long been pushed in resources that ultrasound is always wrong. But this is on Free 120. - 2CK IM CMS Q gives euthyroid patient with nodule where FNA is the answer, but ultrasound isn't listed. - So my conclusion (based on NBME/Free 120 content; not my opinion) is: for eu- or hypothyroid patient with thyroid nodule, ultrasound is correct over FNA if both are listed. If ultrasound isn't listed, choose FNA. - If patient has low TSH (i.e., is hyperthyroid), do ^{131}I uptake scan, not ultrasound. - Since carcinomas are non-secretory of thyroid hormone, if a patient is hyperthyroid, we're not concerned about carcinoma, which is why we don't go the ultrasound then FNA route. We just do uptake to better see if the patient's etiology for hyperthyroidism is Graves (diffuse), toxic adenoma (single nodular uptake), or toxic multinodular goiter (multifocal nodular uptake). - A caveat about the above algorithm for nodule evaluation is that if a patient has hyperthyroidism (i.e., low TSH) *and does not have a nodule*, then you will choose ultrasound over uptake scan. I know this sounds annoying, but USMLE assesses this. There is a new NBME Q where they give low TSH in young patient with no mention of nodule, and the answer is ultrasound over uptake scan. But in Qs where TSH is low and the patient has a nodule, the answer is uptake scan first over ultrasound. If you want a high score, then know the difference.



- TSH is screening for everyone not pregnant.
- T3+T4 can be ordered after TSH, but Q often omits this.
- Screening in pregnancy is Free-T4.
- Most diagnostic/accurate for everyone is Free-T4.
- Algorithms that are more detailed as far as size of nodule, etc., are basically nonsense for USMLE.

Thyroid in pregnancy

- For pregnancy on USMLE, choose the combo of **no change TSH, no change free T4, ↑ total T4** for women who have no thyroid symptoms.
- **Estrogen** causes ↑ thyroid-binding globulin (TBG) production by the liver. TBG is the protein carrier molecule for thyroid hormone in the blood. An NBME Q asks for which hormone causes the ↑ TBG in pregnancy → answer = estrogen, not progesterone.
- Free T4 is the physiologically active form of thyroid hormone. T4 protein-bound to TBG (99%) has minimal effect. Free T4 + TBG-bound T4 = total T4.
- TBG will mop up free T4, causing free T4 to transiently decrease and TSH to rise (less negative feedback). This rise in TSH will stimulate more production of T4 by the thyroid gland, making total T4 go up. The absolute amount of free T4 will increase back to normal, thereby suppressing TSH back to normal. But the total amount of T4 is now increased – i.e., free T4 is normal again, but TBG-bound T4 is higher.
- T3 is normal because free T4 is normal. Free T4 is peripherally converted to T3. I've never seen anything about "free T3" on NBME material and I wouldn't worry about it.
- A student might say, "Wait, but why are you giving the above bold arrows if you just gave me all sorts of transient changes in the arrows based on TBG?." It's because the bold arrows are what the USMLE wants. Pregnant women who are euthyroid will have normal free T4 and increased total T4, and their TSH will be normal. The changes due to TBG rising are likely synchronous and slow enough that the patient's TSH and free T4 stay within reference ranges.
- Postpartum (silent) thyroiditis can result in either hypo- or hyperthyroidism following parturition. These arrows are unrelated to the aforementioned ones. The highest yield point you need to know is that ^{131}I uptake into the thyroid gland is low, even if the patient is hyperthyroid. This is the same for deQuervain and drug-induced thyroiditis, where uptake is always low. This is because with thyroiditis conditions, there is merely increased spacing between the cells of the thyroid gland due to inflammation, allowing thyroid hormone to leak out into the blood. The gland itself is not excessively producing thyroid hormone. Then we have negative feedback causing low TSH, and in turn less stimulation of the thyroid gland, which is why uptake is low.
- If USMLE asks you about levothyroxine dosing during pregnancy, the answer is "increase dose by 50%."
- Avoid methimazole in first trimester (teratogenic).
- PTU is the answer for thyroid storm during pregnancy, even though longer-term use in 2nd and 3rd trimesters isn't considered ideal because of hepatic toxicity risk.

Thyroid in kids	
Thyroglossal duct cyst	- Benign, painless, midline neck mass in a school-age kid that arises from a persistent thyroglossal duct (i.e., the embryologic remnant of the thyroid gland's descent from the base of the tongue to its final position in the neck). - 2/3 of Qs will give a painless midline neck lump that moves upwards with swallowing or protrusion of the tongue (due to its attachment to the hyoid bone and/or the base of the tongue). - 1/3 of Qs won't mention the buzzy upward movement with swallowing or protrusion of the tongue; instead, it will just say a kid has a painless midline neck mass just inferior to the hyoid bone that demonstrates ^{99}Tc uptake.
Lingual thyroid	- Obscure cause of hypothyroidism that shows up on an offline NBME. - Refers to presence of ectopic thyroid tissue located at the base of the tongue due to failure of the thyroid gland to descend from the foramen cecum. - This ectopic tissue may be the only functioning thyroid tissue in the body, resulting in hypothyroidism. - The USMLE Q will say a kid has hypothyroidism + a midline neck lump located high in the neck. They say nothing about protrusion of the tongue or uptake into the mass. - Can sometimes cause dysphagia (trouble swallowing), dysphonia (voice changes), or dyspnea (difficulty breathing).
Cretinism	- Aka congenital hypothyroidism. - Most common causes are iodine deficiency in the mother during pregnancy (worldwide) and fetal thyroid dysgenesis (western countries). - Leads to mental retardation (poor myelin sheath development), impaired bone growth, hypotonia, macroglossia, and protuberant abdomen (due to umbilical hernia; can be confused with kwashiorkor, which is protein-calorie malnutrition causing ascites). - Screened for at birth using the heel-prick test to prevent exacerbation.
TBG deficiency	- The USMLE does not expect you to know about some obscure condition called thyroid-binding globulin deficiency. The reason they ask about this is because the role of TBG on thyroid hormones in pregnancy is exceedingly high-yield, so if you know the reasoning/mechanism behind the latter, you can easily infer what would be seen in TBG deficiency (i.e., the inverse). - ↔ TSH, ↔ T3, ↔ free T4, ↓ total T4. - In pregnancy, since TBG is high, this ultimately results in high total T4 despite a normal free T4. So the student can easily infer, "Well, if our TBG merely is low, rather than high, then total T4 must be low while free is same. Sort of like pregnancy but just the opposite direction."

Hyperparathyroidism conditions	
Primary	- ↑ Ca^{2+}, ↓ $PO4^{3-}$, ↑ PTH. - Primary hyperparathyroidism is usually due to a PTH-secreting adenoma, but can also be due to diffuse four-gland hyperplasia. - Can be part of MEN-1 or -2A (discussed later).
Secondary	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↑ PTH, ↓ 1,25-D3. - Secondary hyperparathyroidism is due to renal failure. - There are two reasons why calcium is low: 1) The kidney cannot reabsorb it in the late-DCT like it's supposed to. 2) The kidney cannot synthesize activated 1,25-D3 in the PCT like it's supposed to. Since 1,25-D3 absorbs calcium in the small bowel, and 1,25-D3 levels are lower, serum calcium will also be lower. - The low serum calcium will cause PTH to go up (deceased negative-feedback at the calcium-sensing receptors on the parathyroid glands).

	- Since the etiology for the increased PTH secretion is not the parathyroid glands themselves, we call this secondary hyperparathyroidism. If the etiology is due to the parathyroid glands themselves (i.e., adenoma or diffuse hyperplasia), we call that primary hyperparathyroidism. - The calcium and phosphate arrows for secondary hyperPTH are the opposite of primary hyperPTH. - Phosphate is high in renal failure, despite the high PTH, because the kidney is not able to downregulate the PCT phosphate transporters, so there is too much phosphate reabsorption.
Tertiary	- ↑ Ca^{2+}, ↑ $PO4^{3-}$, ↑ PTH. - In order to understand tertiary hyperparathyroidism, let's first review secondary hyperparathyroidism: - Patients with renal failure will initially develop secondary hyperparathyroidism, where calcium is low, phosphate is high, and PTH is high. Calcium is low due to failure of reabsorption at the late-DCT of the kidney and because of decreased synthesis of activated vitamin D3 (leading to decreased small bowel absorption). Phosphate is high due to failure to downregulate PCT reabsorption pumps (i.e., more pumps reabsorb phosphate). PTH is high because calcium is low (less negative feedback). - Tertiary hyperparathyroidism results from hyperplasia of the parathyroid glands in patients with long-standing secondary hyperparathyroidism, such that even if the renal failure is brought under control and serum calcium is brought back into the normal range, PTH continues being autonomously secreted at higher basal levels than prior to the renal disease, effectively resetting the body's setpoint for calcium homeostasis. - This causes a rise in serum calcium in a patient with renal failure. This should immediately raise a red flag for tertiary hyperPTH, since renal failure patients will almost always have low, not high, calcium. - Even though the parathyroid glands will be hyper-secreting PTH, this is not primary hyperparathyroidism, since the etiology for the high PTH was not idiopathic adenoma or hyperplasia. - Phosphate is high, not low, because the patient has renal failure. Phosphate is always high in renal failure. Even though PTH is high, remember that the kidney can't downregulate the PCT reabsorption pumps the way they're supposed to.

Hypoparathyroidism conditions	
DiGeorge	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↓ PTH, ↓ T cell levels, ↔ B cell levels. - Mechanism is most frequently 22q11 deletion, resulting in agenesis of the 3rd and 4th pharyngeal pouches. - Tetralogy of Fallot and truncus arteriosus are common heart defects. - Agenesis of the parathyroid glands → primary hypoparathyroidism → low calcium + high phosphate. - DiGeorge is characterized by T cell deficiency. Recurrent viral, fungal, and protozoal infections are characteristic. - Absent thymic shadow = T cell deficiency → DiGeorge. - Scanty lymph nodes/tonsils = B cell deficiency → Bruton agammaglobulinemia. - If Q says kid has both absent thymic shadow *and* scanty lymph nodes/tonsils → answer = SCID.
Post-thyroidectomy	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↓ PTH. - Thyroidectomy can result in removal of or damage to the parathyroid glands, resulting in decreased PTH secretion and, in turn, hypocalcemia. Phosphate goes up because there is increased renal reabsorption (PTH normally downregulates apical PCT phosphate transporters, thereby promoting excretion). - Twitching of the masseter with stimulation is called Chvostek sign of hypocalcemia. - Hypocalcemia causes "up" findings – i.e., muscle tetany and hyperreflexia.

	- Trousseau sign of hypocalcemia is carpopedal spasm with blood pressure cuff inflation (i.e., twitching of the hand/wrist). The USMLE can give you either Chvostek or Trousseau sign when Ca^{2+} is low. - Hypercalcemia, in contrast, causes "down" findings – i.e., muscle flaccidity and hyporeflexia. - Do not confuse Trousseau sign of hypocalcemia with Trousseau and Troisier signs of malignancy. - Troisier sign of malignancy is a palpable left supraclavicular lymph node sometimes seen in visceral malignancy (aka Virchow node). - Trousseau sign of malignancy is migratory thrombophlebitis classically seen in head of pancreas adenocarcinoma (but can be other adenocarcinomas, such as bronchogenic).
Hypomagnesemia	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↓ PTH, ↓ Mg^{2+}. **-** Alcoholics are susceptible to hypomagnesemia as a result of dietary deficiency (EtOH is 7kcal/g, so they fill up on alcohol). **-** Low magnesium can cause hypocalcemia and hypokalemia nonresponsive to supplementation (i.e., you give calcium or potassium for low serum levels, but the serum levels do not appreciably rise). **-** Basal levels of magnesium are required for proper functioning of the parathyroid gland, so low magnesium can cause low PTH secretion. **-** Patients with hypomagnesemia may indeed have completely normal serum levels for calcium, phosphate, and PTH, but if we are forced to choose arrows, we would choose a Ca^{2+}/$PO4^{3-}$/PTH combo that reflects low PTH secretion. **-** The Step 2 exam will often just say an alcoholic has low calcium or potassium nonresponsive to supplementation, and then the next best step in management is "check serum magnesium levels." This isn't hard at all. But some students will be like "Oh wow."

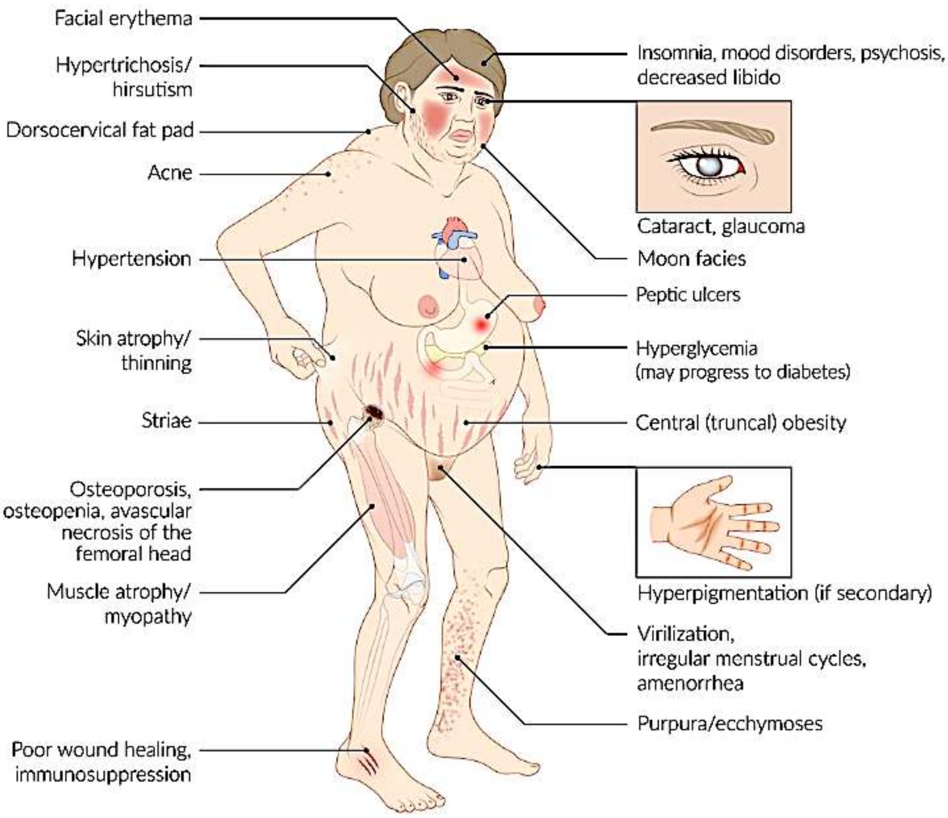
Other calcium-related conditions	
Paraneoplastic PTHrp	- Squamous cell carcinoma of the lung and renal cell carcinoma both secrete parathyroid hormone-related peptide (PTHrp). *This is not the same as endogenous PTH.* - USMLE Qs for renal cell carcinoma and squamous cell of the lung will have: ○ ↑ Serum Ca^{2+}, ↓ serum $PO4^{3-}$, ↓ PTH. - PTHrp exerts very similar physiologic effects as PTH, causing high serum calcium and low phosphate. - PTH secretion by the parathyroid glands will be suppressed in this setting due to the high serum calcium. - NBME will give you lung cancer + high serum calcium + low serum PTH, with the answer being squamous cell carcinoma of the lung. The low PTH will throw some people off. But this is not weird. PTH is not the same as PTHrp (which will be up-arrow).
Metastases	- ↑ Serum Ca^{2+}, ↓ PTH. - Once metastases seed at bony locations, cytokine activity causes lysis of the bone and release of calcium into the blood. Hypercalcemia is common in the setting of metastatic malignancies. - PTH is suppressed due to the high calcium.
Paget disease	- ↔ Serum Ca^{2+}, ↔ serum $PO4^{3-}$, ↔ PTH, ↑ ALP. - Idiopathic disorder of increased bone turnover. Bone is described as having mixed osteoblastic and -clastic phases, appearing heterogenous on x-ray. - Buzzy vignette is male over 50 who's hat doesn't fit him anymore + has tinnitus (narrowing of acoustic foramina).

	- 19/20 questions will give isolated increase in serum ALP. You need to know Ca^{2+}, PO_4^{3-}, and PTH are all normal in Paget disease. There is one Q on a 2CK CMS form where ALP is given as not elevated, but it's a one-off Q and rare. - High-output cardiac failure can occur due to intraosseous AV-fistulae, where patient has an S3 heart sound with high, rather than low, ejection fraction.
Milk-alkali syndrome	- Diagnosis of exclusion on USMLE, meaning we eliminate to get there. - Textbook scenario is **hypercalcemia and metabolic alkalosis** (high bicarb) in someone taking too many calcium carbonate antacid tablets for GERD or dyspepsia (indigestion/upset stomach). This can lead to **urolithiasis.** - What the USMLE vignette will do, however, is not mention the patient's Hx, but will simply give you a kidney stone + hypercalcemia + bicarb that is elevated (or toward the upper limit of normal). Then you eliminate to get there, where you say, "Even though they don't mention the textbook presentation, no other answers explain the high calcium + high bicarb." - Normal bicarb is 22-28 mEq/L, but an NBME Q for milk-alkali syndrome gives a bicarb of 27, where this throw students off. Acid-base disorders can occasionally have bicarb in the normal range, albeit in the direction you expect.

Primary adrenal DDx affecting aldosterone	
Conn syndrome	- Aldosterone-secreting tumor. - Causes ↑ Na^+, ↓ K^+, ↑ HCO_3^-, ↑ pH, ↓ renin. - In ~50% of aldosterone-derangement Qs, Na^+ will be normal. This confuses students, but it's because the body has strong ability to maintain Na^+ (i.e., also via ADH). - Tx = spironolactone (aldosterone receptor antagonist) prior to surgery.
Familial hyperaldosteronism	- There are gene mutations that can cause ↑ primary adrenal production of aldo. - If the USMLE gives you Q where various family members have ↑ BP + ↑ aldosterone and they don't tell you any other info, including the renin level, the answer is primary hyperaldosteronism, not renovascular hypertension (i.e., FMD or RAS). - There is a Q on a 2CK NBME form where they do this, and the student says, "But wait, how are we supposed to know it's primary and not FMD/RAS?" It's because if ↑ aldo is familial, it's likely due to gene mutations causing ↑ primary production from the zona glomerulosa.
Addison disease	- Autoimmune destruction of the adrenal cortex, leading to ↓ aldosterone *and* ↓ cortisol. - I haven't seen the USMLE ever care about "decreased androgens," so the effect is predominantly on the zona glomerulosa and fasciculata. - Called "primary adrenal insufficiency" because the adrenal gland itself is under-producing hormones. - Addison is idiopathic / can be seen in patients with Hx of other autoimmune diseases, either in themselves or family members (i.e., the vignette throws in the side-detail that the brother has RA, or the sister has IBD, or the patient has Hx of thyroiditis). - ↓ Na^+, ↑ K^+, ↓ pH, ↓ bicarb. - Blood pressure will either be low-normal or low, due to both ↓ aldosterone as well as ↓ cortisol. I explicate this because students tend to just mention the ↓ aldosterone as the reason BP is ↓, but the effect of ↓ cortisol is HY for USMLE. **-** ACTH and α-MSH are both ↑ due to ↓ cortisol → hyperpigmentation. Although I'd say they only mention this in ~50% of vignettes. - Patients can present with eosinophilia. This is because cortisol normally plays a role in helping to sequester eosinophils within the spleen + facilitates eosinophil apoptosis, so ↓ cortisol means ↑ eosinophils. Don't go chasing stool ova and parasites. Qs can give you eosinophils of 15-25%, where you're like wtf? They should normally be <5%.

	- Classic Addison vignette is patient (young or old, I've seen both) who has ongoing fatigue + hyperpigmentation, and then the answer is just "ACTH stimulation test." - Exogenous ACTH should normally cause a robust ↑↑ in cortisol from the adrenal fasciculata. In the setting of primary adrenal insufficiency, since the adrenal gland is fucked up, giving exogenous ACTH won't do jack shit. - ↓ cortisol causes chronic fatigue syndrome. This is one of the most over-looked details in NBME Qs and in real life. Any time a patient has ongoing chronic fatigue that is unexplained, cortisol levels should be checked. If they are ↓, hydrocortisone will immediately boost the patient. - Tx for Addison = fludrocortisone +/- hydrocortisone. Fludrocortisone is a mineralocorticoid similar to aldosterone. Hydrocortisone is a glucocorticoid similar to cortisol.
Waterhouse-Friderichsen	- Meningococcal septicemia causing bilateral hemorrhagic necrosis of the adrenal glands, resulting in ↓ aldosterone and ↓ cortisol. - Vignette will be ↓ BP + ↓ Na^+ + ↑ K^+ + ↓ bicarb in patient with meningococcal infection (e.g., recent meningitis + non-blanching rash), where answer = ACTH stimulation test.

Cushing syndrome vs disease
- Cushing syndrome = how the patient looks/presents; can refer to any cause of Cushingoid presentation (i.e., exogenous glucocorticoids, ACTH-secreting tumor of anterior pituitary, cortisol-secreting tumor of adrenal cortex, small cell bronchogenic carcinoma secreting ACTH). **- Cushing disease** = only ACTH-secreting tumor of anterior pituitary. - If patient walks through the door and appears Cushingoid due to an ACTH-secreting tumor of the anterior pituitary, we would say, "The patient has Cushing syndrome caused by Cushing disease." - Highest yield findings for Cushingoid appearance on USMLE, as per my observation, are: **purple striae** on the abdomen / stretch marks, **osteoporosis** (compression fractures of vertebrae)**,** and **avascular necrosis of the femoral head** (hip pain). Findings such as buffalo hump, moon facies, central obesity, peripheral muscle wasting, hypertension, and diabetes type II are important and pass-level, but you really need to know those 3 bold findings I write above. - In Peds, Cushings can present as precocious puberty without all of the classic findings. - Chronically elevated cortisol can cause **hypokalemia** similar to high aldosterone. Sometimes NBME will give you a vignette of Cushings + mention potassium is 3.0 mEq/L (NR 3.5-5.0), and you're like, "why the fuck is the potassium low. Is aldosterone high?" No. All glucocorticoids have some degree of mineralocorticoid effect at the kidney similar to aldosterone. In patients who have high cortisol, the effect longer term can occasionally cause hypokalemia. This detail is particularly important for 2CK vignettes.



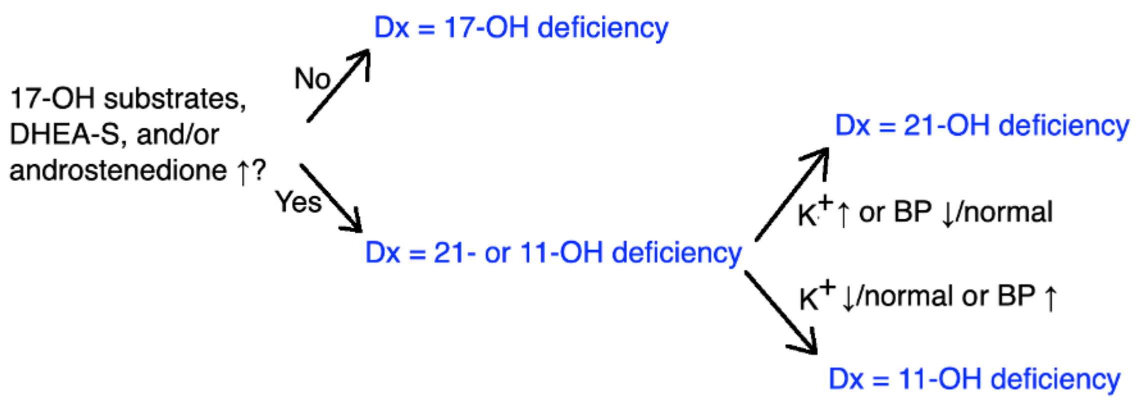
Dexamethasone suppression test	
- Dexamethasone is a cortisol analogue that is capable of exerting negative-feedback at the hypothalamus and anterior pituitary. Given to healthy individuals, the result is suppression of ACTH and cortisol levels. - The dexamethasone suppression test is done to determine the etiology of *endogenous* Cushing syndrome (i.e., the patient isn't taking exogenous steroids). This is because if the patient is taking steroids, clearly we know the cause of the Cushings already. - Low-dose dex will suppress ACTH and cortisol in healthy patients. It is effectively a "yes or no" test for Cushing syndrome, where if there's suppression of cortisol with low-dose dex, then we can say, "Patient doesn't have Cushings"; if cortisol doesn't suppress, then we know the patient has Cushings, but we just don't know the cause yet. - High-dose dex helps us differentiate Cushing disease from other causes of Cushing syndrome. This is because only Cushing disease will have suppressed cortisol levels. - Therefore, if cortisol goes ↓ with high-dose dex, we know the diagnosis is Cushing disease. - If cortisol does not go ↓ with high-dose dex, we look at ACTH level. - If ACTH is ↓, we say, the only way that possible to have ↑ cortisol with ↓ ACTH is if the diagnosis is an adrenal cortical adenoma of the zona fasciculata secreting cortisol. - If ACTH is ↑, we know the cause is ectopic ACTH secretion from small cell bronchogenic carcinoma. The reason we know the ACTH isn't from the pituitary is because high-dose Dex didn't suppress cortisol.	
Cushing disease	- ↑ ACTH + ↑ cortisol. - Suppresses with high-dose dex. - If high-dose dex is given and they ask for the arrows for how ACTH and cortisol *will change*, the answer is "↓ ACTH, ↓ cortisol."
Small cell lung cancer	- ↑ ACTH + ↑ cortisol. - Does not suppress with high-dose dex. - If high-dose dex is given and they ask for the arrows for how ACTH and cortisol *will change*, the answer is "no change ACTH, no change cortisol."

Adrenal adenoma	- ↓ ACTH + ↑ cortisol. - Does not suppress with high-dose dex. - Same as with small cell lung cancer, if high-dose dex is given and they ask for the arrows for how ACTH and cortisol *will change*, the answer is "no change ACTH, no change cortisol."
Exogenous steroids	- ↓ ACTH + **↓ cortisol (holy shit).** - Exogenous steroids (i.e., prednisone, hydrocortisone) are not the same thing as cortisol. The result is suppression of endogenous cortisol. - Students get this wrong all of the time.

Miscellaneous	
Glucocorticoid psychosis	- Glucocorticoids can cause **"glucocorticoid psychosis,"** which can present as an overt psychosis, delirium-like presentation, or even just as general mood disturbance, i.e., depression. - The vignette can give you Cushing syndrome + depression, or **recent IV methylprednisolone injection** + delirium-like presentation 24-48 hours later.
Licorice	- Licorice contains glycyrrhizic acid, which leads to ↑ mineralocorticoid effect, at the kidney (i.e., similar to aldosterone). - Can cause hypokalemia in those who drink excessive licorice tea.

Congenital adrenal hyperplasia (CAH)

- Autosomal recessive.
-21-, 11-, and 17-hydroxylase deficiencies. 21-hydroxylase deficiency is most common.
- ACTH is ↑ in all three types. This is the mechanism USMLE wants for the hyperplasia. That is, cortisol production is impaired in all three, so ACTH goes ↑ to compensate → stimulates hyperplasia of adrenals.
- The USMLE will give a vignette of CAH, followed by "which of the following is the most likely explanation for these findings?" → answer = "Increased ACTH secretion." Sounds weird, but it's on NBME.
- The relevance of these conditions is that they are a cause of ambiguous genitalia and/or virilization.
- Presentations are variable, with Qs giving vignettes in neonates, children, and adults. So the way to differentiate them comes down to electrolytes and BP.
- For both 21- **AND** 11-hydroxylase deficiencies:
  - **↑ 17-OH substrates (i.e., 17-OH-progesterone and 17-OH-pregnenolone).**
  - **↑ DHEA-S, ↑ androstenedione.**
  - If you see any of these up in the Q, you know right away answer is either 21- or 11-deficiency. This is because it's impossible to get 17-OH substrates in 17-hydroxylase deficiency.



21-hydroxylase deficiency	- **↑ Serum K^+, ↓ Na^+, ↓ pH, ↓ $HCO3^-$, ↓ glucose.** - **↓/↔ BP.**

	- Cortisol and aldosterone, as well as their precursors, are all low.
11-hydroxylase deficiency	- **↓/↔ Serum K^+, ↑/↔ Na^+, ↑/↔ pH, ↑/↔ $HCO3^-$, ↑/↔ glucose.** - **↑ BP.** - Cortisol and aldosterone are low, but their precursors, which have some effect, are high.
17-hydroxylase deficiency	- **↓ 17-OH substrates (i.e., 17-OH-progesterone and 17-OH-pregnenolone)** - **↓ DHEA-S, ↓ androstenedione** - **↓ Serum K^+, ↑ Na^+, ↑ pH, ↑ $HCO3^-$** (high aldosterone) - **↓ glucose** (cortisol and its precursor are low).

Catecholamine-secreting tumors	
Pheochromocytoma	- Catecholamine-secreting tumor of adrenal medulla. - Derived from neural crest (i.e., chromaffin cells of adrenal medulla). - Occurs in adults; can calcify. - Causes paroxysmal (i.e., comes and goes) ↑↑ BP, which presents as periodic palpitations and pounding headaches. - Because the HTN is paroxysmal, the patient can have BP measured in the office at 120/80, so do not exclude pheo in this case. - Stressors (e.g., sitting up on the examination table) can trigger ↑ NE + E release from the tumor, where BP shoots up to >180/100. This is different from whitecoat hypertension, where young patient has 140/90 measured due to being slightly nervous. - Diagnose with urinary/serum metanephrines. If positive, then do CT abdomen. - Tx = phenoxybenzamine (irreversible α1-blocker). If you give beta-blocker first, this will kill the patient. This is because if you block beta receptors, the NE and E have little to bind to now except α receptors (mainly α1), so we get ↑↑ arteriolar constriction and BP shoots up. This is called “unopposed alpha.” - Some students ask about giving propranolol after phenoxybenzamine. The answer is, yes, in theory this can be done. But USMLE doesn’t assess this. They want straight-up alpha-blockade. - Pheo can be idiopathic or part of condition such as MEN 2A/B or NF1. - If a patient has medullary thyroid cancer, you must check for pheochromocytoma before doing thyroidectomy. This is because if the patient has MEN 2 and you don’t check for pheo first, doing surgery will cause ↑↑ BP due to catecholamine release and kill the patient.
Neuroblastoma	- Essentially the “pediatric pheo”; *N-myc* gene. - Secretes catecholamines; can calcify (same as pheo in adults). - Grows from the median (midline) sympathetic chain. Usually intra-abdominal but can sometimes be in the posterior mediastinum. Sounds odd, but the latter shows up on an 2CK NBME form. - Causes ↑ BP and findings such as opsoclonus-myoclonus syndrome (dancing eyes) and violaceous eyelids (sounds like heliotrope rash of dermatomyositis but unrelated). - Diagnose with urinary homovanillic acid (HVA) and vanillylmandelic acid (VMA), which are metabolites of catecholamines. - Metaiodobenzylguanidine scan can be done after HVA and VMA detected.

Carcinoid tumor
- Neuroendocrine tumor of small bowel, appendix, or lung that is S-100 (+) and consists of small blue cells. - Secretes serotonin. - Presents as flushing, tachycardia, diaphoresis, and diarrhea. - Tricuspid regurg common (holosystolic murmur that increases with inspiration) due to tricuspid lesions. - Diagnose with urinary 5-hydroxyindole acetic acid (5-HIAA). - Don't confuse with serotonin syndrome, which is a drug interaction (e.g., starting a monoamine oxidase inhibitor too soon after discontinuing an SSRI, or if a patient takes St John wort with an SSRI).

Polyglandular syndromes	
- All MEN syndromes and Carney complex are autosomal dominant.	
MEN 1	- *MEN1* gene mutation. - Parathyroid adenoma (or hyperplasia), pituitary adenoma, pancreatic adenoma. ○ Zollinger-Ellison syndrome (gastrinoma) is highest yield pancreatic tumor in MEN 1. Will present as recurrent peptic ulcers. I discuss this stuff in detail in the HY Gastro PDF. ○ Parathyroid adenoma often presents with **calcium urolithiasis,** where they will mention flank or groin pain in patient with Hx of ulcers (Zollinger-Ellison). ○ Prolactinoma is highest yield pituitary tumor. ○ USMLE need not give the full constellation of findings for the MEN syndromes. There is an NBME Q floating around where the patient has a parathyroid adenoma – nothing more – and the answer is just *MEN1* as the gene that's mutated.
MEN 2A	- *RET* proto-oncogene mutation. - Parathyroid adenoma (or hyperplasia), pheochromocytoma, medullary thyroid carcinoma. ○ As mentioned earlier, *RET* mutations need not cause the full constellation of findings. NBME Qs can give isolated medullary thyroid carcinoma in a family, and the answer is just *RET.* ○ In theory, isolated parathyroid adenoma could also be *RET*, but with the NBME example I mentioned above they listed *MEN1* without also listing *RET*.
MEN 2B	- *RET* proto-oncogene mutation. - Marfanoid body habitus, mucosal neuromas, pheochromocytoma, medullary thyroid carcinoma. ○ -Oid means looks like but ain't. So the patient might look like he/she has Marfan's (i.e., tall, lanky) but doesn't, because this isn't *FBN1/2* mutation. ○ As mentioned earlier, before doing thyroidectomy for medullary thyroid carcinoma, you must check for pheochromocytoma (because stress of surgery will ↑↑ catecholamines from the tumor and kill the patient) by first doing urinary metanephrines. If positive, then do CT of the abdomen.
Carney complex	- Hyperpigmentation, atrial myxoma, endocrine hypersecretion. ○ The hyperpigmentation can present as perioral melanosis (i.e., hyperpigmentation around the lips/mouth) that sounds similar to Peutz-Jegher syndrome, but instead of the Q focusing on hamartomatous colonic polyps, they mention a cardiac tumor and another endocrine organ hyper-secreting. And you're like huh? ○ Atrial myxoma normally shows up in adults; cardiac rhabdomyoma normally shows up in children with tuberous sclerosis. In Carney, the Q can say atrial myxoma *in a kid*. So it's the only example on USMLE that breaks the rule. ○ Endocrine hypersecretion is classically Cushing syndrome (granular hyperpigmentation of the zona fasciculata) or hyperthyroidism. ○ The reason you likely haven't heard of Carney complex before is because students coming out of the USMLE didn't even realize this is what they saw.

<table>
<tr><th colspan="2">Diabetes mellitus (DM)</th></tr>
<tr><td colspan="2">- The word diabetes means polydipsia + polyuria (i.e., drinking a lot and peeing a lot).
- I see students frequently misconstrue diabetes mellitus with diabetes insipidus, since both vignettes will give patients who have increased thirst and urination.
- A key detail that distinguishes acute diabetes mellitus from diabetes insipidus is that DM vignettes can often mention weight loss due to ↑ catabolism. Weight loss is also seen in acute type II, even though the patient is usually overweight. Vomiting is another HY finding in acute DM.
- So if vignette says ↑ thirst + ↑ urination + weight loss +/- vomiting, this is DM, not DI.
- For extensive info on the effects of diabetes on different organ systems (i.e., cardio, renal, ophthal, GIT, pregnancy, etc.), use those corresponding HY PDFs I've made. I will keep the focus here on endocrine and electrolytes / acid-base.
- Diagnosis of DM is made by measuring two fasting glucoses >126 mg/dL, any one random fasting glucose >200 mg/dL, or an HbA1c of >6.5%.</td></tr>
<tr><td>Type I</td><td>- Absence of insulin; autoimmune-mediated; no one specific gene, although HLA-DR3/4 are associated.
- Often described as "genetic susceptibility with environmental trigger" (i.e., viral infection, usually Coxsackie B).
- Occurs almost always in children or teenagers of normal BMI.
- Pathogenesis is usually combo of an antibody- (anti-GAD65 or -ZnT8) and T-cell-mediated process that leads to destruction of β-islet cells in the tail of the pancreas → ↓ insulin secretion → inability to recruit GLUT4 to cell surface of skeletal muscle and adipose tissue → inability to take up glucose into cells → cells are starved of energy → ↑ gluconeogenesis + ketogenesis.
- Hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with ↓ uptake by skeletal muscle and adipose tissue.
- Biopsy of pancreas shows inflammatory infiltrate in acute type I DM; in late type I (i.e., years later), it shows atrophy and fibrosis.
- It is the absence of insulin in type I DM that enables ketogenesis. In type II DM, ketones are low, not high, because insulin is present.
- Ketone bodies (USMLE likes β-hydroxybutyrate) are highly acidic, so will drop the blood pH (high anion-gap metabolic acidosis).
- Vignette will give kid or teenager with polyuria, polydipsia, and sometimes vomiting, who has severe metabolic acidosis (bicarb ↓), which is due to diabetic ketoacidosis. Breath can be fruity in odor (smell of ketones). DKA will have:
- ↓ pH, ↓ HCO_3^-, ↓ CO_2.
    - Ketone bodies are acidic → causes metabolic acidosis (decreased bicarbonate).
    - CO_2 is acidic, so we blow that off to compensate. Kussmaul breathing is deep, labored breathing seen in DKA.
- ↑ β-hydroxybutyrate, ↑ serum osmolality, ↑ anion gap.
    - β-hydroxybutyrate is a ketone body. Since we have ketosis (ketone production), it will be increased. I have seen this specific ketone body as an arrow on an offline Step 1 NBME.
    - Serum osmolality is increased because serum glucose is markedly increased (insulin isn't present to drive glucose into cells). Even though serum sodium is low (so you might think osmolality is low similar to SIADH), the ↑ glucose makes serum osmolality high.
    - DKA is the D in MUDPILES, so anion gap is up.
- ↓ Serum Na^+, ↑ serum K^+, ↓ total body K^+, ↑ GFR.
    - Serum sodium is decreased mostly due to dilutional hyponatremia. As mentioned earlier, serum osmolality is high, which causes fluid retention intravascularly → dilutes out serum sodium. In addition, low insulin means less glucose enters cells → less ATP production → less cellular Na^+/K^+-ATPase activation → sodium not pumped out of cells.
    - We describe the state of K^+ as: hyperkalemia (high serum K^+) despite a low total body potassium.
    - Serum potassium is high for two main reasons:</td></tr>
</table>

	1) **Potassium-proton shift:** Acidosis (due to ketosis) causes protons in the blood to exchange with potassium in the cell via the H^+/K^+-antiporter (i.e., H^+ goes into cell; K^+ moves out). 2) **Insulin drives potassium into cells (and we don't have insulin in DKA):** Insulin normally leads to the upregulation of a Na^+/K^+-ATPase antiporter that moves $3Na^+$ out for every $2K^+$ in. If insulin is low, then less K^+ will be moved into the cell → hyperkalemia (high potassium in blood). - Total body potassium is low because of increased losses at the kidney. Since serum potassium is high, more potassium is filtered through the glomerulus, so more is excreted (i.e., the kidney thinks there's too much potassium in the body since more is being filtered, so it tries to get rid of more of it). However, this causes bodily depletion of K^+ because the processes that cause hyperkalemia are unmitigated, so the kidney continues to excrete potassium. - GFR is increased because of hyperfiltration secondary to hyperglycemia. That is, since serum glucose is high, more glucose is filtered through the glomerulus, which pulls water with it → hyperfiltration. - Tx for DKA is normal saline first (i.e., 0.9% NaCl), prior to giving insulin. - Serum potassium will drop as insulin is given. If the Q tells you patient has DKA + shows you normal potassium (even though it's usually high) + insulin is given + patient now has arrhythmia, the answer is potassium for which electrolyte is disturbed (i.e., potassium was normal prior to giving insulin, but now it's been driven into cells and the patient's developed hypokalemia). - Bicarb is always a wrong answer for low pH in DKA. - Don't confuse DKA with a hypoglycemic episode. The latter will be a type I diabetic who went out to do exercise or who was cleaning the roof of his house, who is now found unconscious → if insulin dose is not decreased prior to exercise, patient can get hypoglycemia. The Tx is intramuscular glucagon (2CK likes this) or dextrose in water. - Sometimes the USMLE can give a bit of a weird Q where they say a patient is given insulin for diabetes but has prolonged hypoglycemia + they ask why → answer = "deficiency of counter-regulatory glucagon." You need to know that in advanced diabetes, the glucagon-producing α-islet cells of the pancreas can also get fucked up. I've also seen this Q asked for chronic alcoholism and pancreatectomy; in both cases, if insulin is given, prolonged hypoglycemia can result from "deficiency of counter-regulatory glucagon."
Type II	- Insulin resistance; seen usually in high-BMI adults, but can rarely show up in Peds (2CK NBME form gives type II in a 17-yr-old). - High BMI causes insulin resistance because adipose tissue secretes adipokines that impair insulin signaling. - Insulin is *high* initially (hyperinsulinemia) due to β-cell hypertrophy in order to compensate for peripheral insulin resistance. - Ketones are absent/low because the presence of insulin inhibits ketogenesis. - The two above points are the highest yield pieces of info for USMLE. - Biopsy of the pancreas initially shows β-cell hypertrophy, but years later will show amyloid deposition and cell shrinkage. - Same as with type I, hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with ↓ uptake by skeletal muscle and adipose tissue. - Hyperosmolar hyperglycemic state (HHS) is seen in type II; DKA is strictly type I on USMLE. - In HHS, glucose is usually >600 mg/dL (in DKA, it need not be this high) and bicarb is usually not below 20 mEq/L (NR 22-28); in DKA, bicarb can be as low as it wants (I've seen 5 mEq/L on NBME). - The USMLE vignette will give massive paragraph Q where they mention glucose is 700 and then ask for the cause of mental status change → answer = "hyperosmolality," or simply "hyperglycemia," or "osmotic diuresis" (causes severe dehydration). I point this out because this is particularly prevalent on 2CK NBMEs, where they'll give a 15-line bullshit paragraph with tons of lab values that the student spends 9 minutes reading, but meanwhile you can

	see glucose is 650 and then the answer is just "osmotic diuresis" or "hyperosmolality"; should take two seconds to answer. - Mental status change is technically due to hyperosmolality in the ECF of the brain pulling water out of the ICF (i.e., the cells). - Treat HHS with normal saline first prior to insulin. - I discuss the various meds for type II diabetes later in this PDF.
Other causes	- **Hereditary hemochromatosis** ("bronze diabetes") → iron deposition in tail of the pancreas, leading to impaired fasting glucose / overt diabetes. Hyperpigmentation is due to hemosiderin deposition. Patients can also get cardiomyopathy, infertility, and pseudogout. - **Cushing syndrome** (as discussed earlier) can cause diabetes due to the insulin resistance caused by glucocorticoids. - **Acromegaly** (growth hormone causes insulin resistance). - **Alcoholism** can cause chronic pancreatitis that not only leads to exocrine pancreatic burnout and steatorrhea, but can also cause diabetes due to loss of islet cells. - **Pancreatectomy** is self-explanatory. If you remove some of the tail of the pancreas, this can lead to diabetes. - **Tacrolimus** is a HY drug that can cause diabetes. Other drugs can do this, but tacrolimus is the notable one for USMLE. Tacrolimus is an immunosuppressant that ↓ IL-2 synthesis and T cell stimulation.

Electrolytes	
Calcium	- 8.4-10.2 mg/dL. - Low Ca causes "up" presentation of tetany and hyperreflexia; high Ca causes "down" presentation of muscle flaccidity and hyporeflexia; high Ca also associated with delirium/confusion (hypercalcemic crisis). - High calcium HY causes are sarcoidosis (due to ↑ vitamin D), primary hyperparathyroidism (adenoma, hyperplasia, MEN 1/2A), metastatic malignancy, multiple myeloma, and thiazides (cause - Ca reabsorption in DCT). - Low calcium HY causes are rickets/osteomalacia (↓ vitamin D), secondary hyperparathyroidism (renal failure), post-thyroidectomy (concomitant loss of parathyroids), DiGeorge syndrome (agenesis of 3rd and 4th pharyngeal pouches), and loop diuretics (↓ paracellular reabsorption), hypomagnesemia (hypo-Mg can cause hypo-Ca and hypo-K non-responsive to supplementation; usually seen in alcoholics).
Sodium	- 135-145 mEq/L. - Sodium derangement (high and low) causes CNS dysfunction – i.e., confusion, stupor, or coma. - Na can often be normal in USMLE vignettes despite your expectation that it might be characteristically deranged – e.g., patient has Conn syndrome and yet the sodium is normal, and you're like wtf? (because you expect it to be elevated); this is typical for USMLE vignettes. - High Na HY causes are primary hyperaldosteronism, renal artery stenosis, fibromuscular dysplasia, dehydration, diabetes insipidus. - Low Na HY causes are Addison disease, psychogenic polydipsia, SIADH.
Potassium	- 3.5-5.0 mEq/L. - Potassium derangement (high and low) causes cardiac dysfunction → arrhythmia (contrast this with Na, which causes CNS dysfunction). - High K HY causes are Addison disease, renal failure, polypharmacy with agents such as potassium-sparing diuretics and digoxin. - Low K HY causes are primary hyperaldosteronism, renal artery stenosis, fibromuscular dysplasia, Cushing disease (chronically high glucocorticoid levels can cause hypo-K due to distal renal secretion similar to mineralocorticoids), vomiting, diarrhea, loop diuretics and thiazides, and hypomagnesemia (hypo-Mg can cause hypo-K and hypo-Ca non-responsive to supplementation; usually seen in alcoholics).
Phosphate	- 2.5-4.5 mg/dL.

	- High phosphorus HY causes are hypoparathyroidism, secondary hyperparathyroidism, renal failure, tumor lysis syndrome, and sarcoidosis (↑ vitamin D). - Low phosphorus HY causes are primary hyperparathyroidism, rickets / osteomalacia, and refeeding syndrome.
Magnesium	- 1.7-2.2 mg/dL. - Magnesium derangement causes effects similar to Ca – i.e., low Mg presents with an "up" state of hyperreflexia and increased muscle tone; high Mg presents with a "down" state of hyporeflexia and muscle flaccidity. - Low Mg HY cause is alcoholism (decreased dietary intake); leads to hypocalcemia and hypokalemia non-response to supplementation.

Acid/base	
(If you want practice with acid/base USMLE Qs and/or are confused by certain things I list in this table, go through my HY Arrows PDF. I discuss this stuff in comprehensive detail).	
- Normal blood pH: 7.35-7.45. **- Normal blood $HCO3^-$: 22-28 mEq/L.** **- Normal blood pCO2: 33-44 mmHg.**	
Metabolic acidosis	- Low pH caused by low HCO3; pCO2 will go down to compensate (CO2 is acidic).
High anion-gap metabolic acidoses	- MUDPILES is mnemonic for high-anion-gap metabolic acidoses → Methanol, Uremia (renal failure), DKA, Phenformin (weird drug you don't need to worry about), Iron tablets / Isoniazid, Lactic acidosis, Ethylene glycol (antifreeze), Salicylates (aspirin). - If the patient has metabolic acidosis and the anion gap is normal in the Q, you can eliminate the MUDPILES answers and choose an answer like renal tubular acidosis (RTA). - Anion gap is calculated as $Na^+ - (HCO_3^- + Cl^-)$. - Anion gap of 8-12 is normal. So 13 or greater is high. I specifically write "13 or greater" because there is an NBME Q where anion gap comes out to 12, and they want renal tubular acidosis. I talk in detail in the HY Renal PDF about RTA.
Normal anion-gap metabolic acidoses	- Diarrhea, RTA, adrenal insufficiency, and CAH (21-deficiency) are most important.
Metabolic alkalosis	- High pH caused by high HCO3; pCO2 will go up to compensate. - Conn syndrome, renal artery stenosis, fibromuscular dysplasia, vomiting, loops and thiazides (promote RAS activation due to volume depletion).
Respiratory acidosis	- Low pH caused by high CO2; HCO3 goes up *if chronic* to compensate (bicarb is basic). - Acute (**normal** bicarb): hypoventilation due to opioids, benzos, barbiturates. - Chronic (high bicarb): COPD, obstructive sleep apnea, obesity, ankylosing spondylitis, severe kyphoscoliosis.
Respiratory alkalosis	- High pH caused by low CO2; HCO3 goes down *if chronic* to compensate. - Acute (**normal** bicarb): asthma attack, panic attack, most pulmonary emboli, altitude sickness (first day at high altitude). - Chronic (**low** bicarb): living at high altitude, pregnancy (progesterone upregulates respiratory center).
- Should be noted that primary metabolic acid/base derangements are compensated for instantly by changes in respiration (i.e., it's easy to retain or breathe off CO2 merely by changing respiratory rate), whereas primary respiratory derangements are compensated for **slowly** because it takes time for the kidney to alter bicarb excretion. - Winter formula is used to calculate predicted pCO2 based on any change in bicarb, where: Predicted pCO2 = (1.5xHCO3) + 8 +/- 2. For example, if a patient has DKA and bicarb is 14 mEq/L, we expect pCO2 to be 27-31 mmHg. If pCO2 is lower than 27, then the patient has a concurrent respiratory alkalosis; if higher than 31, then the patient has a concurrent respiratory acidosis.	

Salicylate toxicity
- Diagnosis is salicylate (aspirin) toxicity. Tinnitus (ear-ringing) is the most common first symptom. - This patient has mixed metabolic acidosis-respiratory alkalosis (i.e., both a metabolic acidosis and respiratory alkalosis at the same time). - Aspirin initially causes an isolated respiratory alkalosis (within the first 20 minutes) due to direct upregulation of respiratory centers in the brain. - However, aspirin itself is an acid (salicylic acid), so it will ultimately cause a metabolic acidosis that wins over the respiratory alkalosis (i.e., pH will ultimately go low). - Anion gap is high. Salicylates are the S in MUDPILES. - Sodium and potassium are normal in salicylate toxicity. I have seen these arrows listed before, which causes students confusion, but they are unchanged. Sometimes the Q will give you numerical values where you need to calculate anion gap – i.e., Na^+ - ($HCO3^-$ + Cl^-). The value must be 13 or greater since it is one of the MUDPILES. Normal range is 8-12. - In summary, once again, for aspirin: ○ First 20 mins: acute respiratory alkalosis (**↑ pH, ↔ $HCO3^-$, ↓ CO2).** ○ After 20 mins: mixed metabolic acidosis-respiratory alkalosis (**↓ pH, ↓ $HCO3^-$, ↓ CO2, ↑ anion-gap).** - The low bicarbonate we see after 20 minutes **is not compensation.** It coincidentally goes low (as we'd expect for renal compensation in the setting of respiratory alkalosis). But remember that the kidney cannot induce this change until a minimum of 12-24 hours later. The reason bicarb is low after 20 minutes is because aspirin itself is an acid that is causing a metabolic acidosis and driving the bicarb low. - A 2CK-level point is that Na^+ and Cl^- can be abnormal insofar as the bicarb is low and anion gap is high. - Potassium must always be normal. - You might get an aspirin Q where you're forced to choose either an up or down arrow for chloride and you say, "I haven't heard of that before. I thought it would just be normal." But if you're forced to choose, clearly you'd go with the answer where chloride is low, since anion gap is calculated as Na^+ - (HCO_3^- + Cl^-), and a low chloride would make the anion gap higher than the answer choice with high chloride.

Renal failure key disturbances
- As discussed earlier, renal failure causes secondary hyperparathyroidism, where we have: ○ ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↑ PTH, ↓ 1,25-D3. - However, it is also pass-level you know that in renal failure we also have: ○ ↑ K^+, ↓ HCO_3^-, ↓ pH, ↑ anion gap (U in MUDPILES = uremia = renal failure), with Na^+ variable. ○ The kidney normally secretes potassium distally in the cortical collecting duct under the action of aldosterone. So if the kidney is fucked up, we can't do that, so K^+ is ↑. ○ Likewise, we normally have HCO_3^- reabsorbed in the PCT and H^+ secreted in the cortical collecting duct, so if we can't do either of those things, blood HCO_3^- and pH ↓.

HY Pulmonary acid/base findings
- Acute asthma attack and PE: **↓ pCO2, no change HCO_3^-, ↑ pH,** ○ The key here is that these conditions cause acute respiratory alkalosis, not acidosis, as I mentioned in above table. Students normally fuck this up and think pCO2 is high. pCO2 only goes ↑ if the patient becomes tired as an asthma attack prolongs. But initially, for both of these conditions, we have low pCO2 (holy shit), not high. Bicarb is unchanged because it takes a minimum of 12-24 hours for bicarb excretion to be sufficiently modulated by the kidney. ○ Tangentially, metabolic acid-base disturbances are instantly compensated for by changes in pCO2 by simply breathing more or less. - Patients with COPD and obstructive sleep apnea are **chronic CO2 retainers,** where we have: ○ **↑ pCO2, ↑ HCO_3^-, ↓ pH.** ○ Bicarb goes up to compensate.

Miscellaneous HY acid/base	
Vomiting	**- ↓ Serum K^+, ↓ Cl^-, ↑ $HCO3^-$, ↑ CO2, ↑ pH.** - Very high-yield you know that vomiting causes a hypokalemic, hypochloremic, metabolic alkalosis. - Vomitus contains lots of potassium, chloride, and protons. In addition, volume depletion can promote distal renal losses of K^+ and H^+ secondary to RAAS upregulation. - CO2 is acidic, so we want to retain it to compensate.
Loop/thiazide diuretic abuse	- **↓ Serum K^+, ↓ Cl^-, ↑ $HCO3^-$, ↑ CO2, ↑ pH.** - Furosemide is a loop diuretic. It will block the apical $2Cl^-/K^+/Na^+$ symporter in the thick ascending limb. Since water follows ions, and ion reabsorption is impaired, increased diuresis (urination) occurs. Volume loss causes upregulation of RAAS and distal secretion of protons under the control of aldosterone, leading to metabolic alkalosis. This is referred to as "contraction alkalosis," where volume contraction induced by diuretics promotes metabolic alkalosis. In addition, additional distal secretion of K^+ occurs. - Thiazide diuretics will also follow this arrow pattern. They block the Na^+/Cl^- symporter on the apical membrane of the early-DCT, leading to volume depletion and ↑ RAAS, thereby ↑ H^+ and K^+ secretion distally. - CO2 is acidic and retained because it compensates for our metabolic alkalosis. - It should be noted that these arrows are the same as for vomiting.
Diarrhea	**- ↓ Serum K^+, ↑ Cl^-, ↓ $HCO3^-$, ↓ pH, ↓ CO2, ↔ anion gap.** - Stool is rich in potassium and bicarb. Diarrhea and laxative abuse can cause a normal anion-gap metabolic acidosis. Since CO2 is acidic, we blow this off to compensate. - Serum chloride is high due to cellular shifts. - Anion gap is normal because diarrhea is not part of MUDPILES.
Laxative abuse	- **↓ Serum K^+, ↑ Cl^-, ↓ $HCO3^-$, ↓ CO2, ↓ pH, ↔ anion gap.** - Stool is rich in potassium and bicarb. Diarrhea and laxative abuse can cause a normal anion-gap metabolic acidosis. Since CO2 is acidic, we blow this off to compensate. - Serum chloride is high due to cellular shifts.

Endocrine pharm for FM (keeping things HY)	
Metformin	- MOA = "↑ Glycolysis + ↓ gluconeogenesis." - Only used for type II DM. - Causes lactic acidosis. - Discontinue if bicarb is low (means lactic acidosis). - Discontinue (or don't commence) if creatinine is 1.5 or greater, as reduced renal function ↑ risk of lactic acidosis. - Choose insulin over metformin for regulation of glycemic control in pregnancy / intrapartum, as it allows for optimal fine-tuning / control of glucose. - USMLE can give you patient on metformin who had CT with contrast + gets high creatinine as a result + they ask you what would have prevented the increase in creatinine prior to the CT → answer = "infusion of 0.9% saline" or "adequate intravenous hydration"; the wrong answer is "discontinuation of metformin." This one is a little tricky, where you need to know that adequate hydration saline hydration is the #1 way to prevent contrast nephropathy. There is ↑ risk of lactic acidosis due to metformin in patients who have renal insufficiency, but metformin itself doesn't cause renal insufficiency. In other words, ↑ creatinine is a reason to stop metformin, but ↑ creatinine doesn't result from metformin itself.
Insulin	- USMLE doesn't give a fuck that you know about all of the different insulin types and their durations of action. - What USMLE might do on 2CK is give you a patient with diabetes who wakes up with severe hyperglycemia + ask you what needs to be done. The answer = **decrease, not increase (holy shit),** dose of evening intermediate-acting insulin. This is called **Somogyi phenomenon,** where it appears as though because morning glucose is elevated, this means evening/night insulin is insufficient. But it's actually

	the opposite, where the evening/night dose is too high, which pushes the patient into overnight hypoglycemia, causing a compensatory sympathetic response that results in a robust rebound hyperglycemia by morning. - The other Q they will give is a patient with type II DM who is already on metformin + glyburide who has uncontrolled HbA1c at 12% + give you either low bicarb or high creatinine → they ask what needs to be done → answer = “discontinue metformin and glyburide and commence intermediate-acting insulin.” I’ve made YouTube clips on this Q and have talked about it many times, as it gets asked in our Telegram group probably once a month. The point is that 1) if a patient has low bicarb while on metformin, this means lactic acidosis due to metformin, so you must choose an answer choice where metformin is discontinued; and 2) a patient who has a creatinine of 1.5 or greater needs to be off metformin (I’ve seen max 1.4 on NBME where metformin is acceptable), since renal insufficiency is a risk factor for lactic acidosis due to metformin.
Sulfonylureas	- Glyburide, glipizide, glibenclamide. - Insulin secretagogues – i.e., ↑ insulin secretion by blocking the ATP-gated K^+ channel on β-islet cells without the need for ATP. - Glyburide and glipizide tend to be the ones that shows up repeatedly on NBME. I don’t think I’ve ever seen other sulfonylurea on NBME for that matter. - Glibenclamide was known for causing severe hypoglycemia. You could be aware of this in theory.
Meglitinides	- Repaglinide. - Similar to sulfonylureas, these ↑ insulin secretion by blocking the ATP-gated K^+ channel on β-islet cells without the need for ATP.
GLP-1 analogues	- Liraglutide, semaglutide (Ozempic). - Glucagon-like peptide-1 stimulates insulin release.
DPP-4 inhibitors	- Sitagliptin. - DPP-4 is an enzyme that breaks down GLP-1, so if we inhibit DPP-4, GLP-1 is ↑. - Therefore, ↑ insulin release from the pancreas.
Thiazolidinediones	- Pioglitazone. - ↑ insulin sensitivity at tissues by stimulating PPAR-γ. - Can cause liver toxicity.
α-glucosidase inhibitors	- Acarbose, miglitol - ↓ Glucose absorption in small bowel. - Can cause diarrhea.
SGLT-2 inhibitors	- Dapagliflozin. - ↓ Glucose reabsorption in the PCT of the kidney. - Can ↑ risk of candidal infections.
Amylin analogues	- Pramlintide. - Amylin is a hormone that is co-secreted with insulin from β-cells that suppresses glucagon secretion, slows gastric emptying, and promotes satiety, thereby preventing a rapid increase in postprandial blood glucose.
PTU/methimazole	- Propylthiouracil (PTU) and methimazole are both used for Graves. - Both inhibit thyroperoxidase; PTU has an additional MOA of inhibiting 5-deiodonise. - Both PTU and methimazole can cause **neutropenia/agranulocytosis.** You need to know this often presents as mouth ulcers on USMLE. For example, they’ll say patient was recently treated in hospital for thyroid storm + now has mouth ulcers and fever → answer = “drug adverse effect,” or “drug-induced neutropenia.” - Propranolol is used for tachycardia in hyperthyroidism (beta-blockade inhibits 5-deiodonase). Steroids are used for exophthalmos. - Methimazole is teratogenic in first-trimester.
Demeclocycline	- Shows up as Tx for SIADH on NBME. - Tetracycline antibiotic that causes nephrogenic DI as an adverse effect, but this is actually a “good thing” if we want to treat SIADH (i.e., let’s just treat one problem by causing another problem, and they somehow cancel out).

Desmopressin	- Vasopressin (ADH) analogue used as treatment for central DI. - Also used after fluid restriction in DI cases for differentiating central from nephrogenic.
Octreotide	- Somatostatin analogue used for acromegaly and esophageal varices.
Cinacalcet	- Increases sensitivity of calcium-sensing receptors on parathyroid glands. - Helps to suppress PTH in secondary hyperparathyroidism.

Fast FM bullet points

- Patient has pain in the shoulder when raising the arm to paint a fence; Dx? → subacromial bursitis (caused by repetitive overhead motion)
- Patient has pain in the shoulder when raising the arm above the head; subacromial bursitis isn't listed; Dx? → answer = rotator cuff injury
- Patient has pain in the shoulder lying on his or her side in bed; Dx? → rotator cuff injury
- Positive Gerber lift-off test → subscapularis injury → place dorsum of hand against lower back so palm faces posteriorly; examiner applies pressure into palm + asks patient to move hand → if pain, subscapularis injury
- Positive "empty can" or "full can" test; Dx? → supraspinatus injury → shoulder is abducted to 90 degrees; then downward pressure is applied; elicits pain when patient attempts to resist
- Resistance to lateral rotation of shoulder elicits pain; Dx? → infraspinatus or teres minor injury
- "Pitcher injury"? → infraspinatus injury; but if a pitcher has positive full or empty can test, use your head → still supraspinatus injury
- Patient has elbow pain after leaning on elbow for long periods; Dx? → olecranon bursitis
- Patient has pain in lateral forearm with extension of elbow against resistance; Dx + Tx? → answer = lateral epicondylitis (tennis elbow); Tx = forearm strap
- Patient has pain in medial forearm with flexion of elbow against resistance; Dx + Tx? → answer = medial epicondylitis (golfer elbow); Tx = forearm strap
- Pain on lateral wrist in breastfeeding woman → deQuervain tenosynovitis → answer = immediate steroid injection into the wrist. Dx with Finkelstein test (place thumb in palm of hand; then wrap four digits over thumb; then ulnar deviate; pain at lateral wrist with ulnar deviation is + test)
- Lump on the dorsum of hand alongside a tendon; painless; slightly mobile; Dx? → ganglion cyst; Tx = needle drainage
- Wrist fracture + posterior displacement of radius; Dx? → Colles fracture ("dinner fork deformity")
- Wrist fracture + anterior displacement of radius; Dx? → Smith fracture
- Proximal ulnar fracture + anterior displacement of radial head; Dx? → Monteggia fracture
- Radial shaft fracture + displacement of distal radioulnar joint; Dx? → Galeazzi fracture

- Fracture of forearm in child with “bending” of the bone? → greenstick fracture → bone is soft so part of bone remains intact
- Fracture in child abuse → spiral fracture (from rotational force/twisting of limb); also posterior rib fractures
- Patient has hip pain lying on his or her side in bed?; Dx? → trochanteric bursitis
- Patient has lateral hip pain when palpated, when abducting against resistance, or when standing on that foot; Dx? → greater trochanteric syndrome (gluteus medius or minimus tendonopathy)
- Patient has “locking” or “catching” of the knee; Dx? → meniscal tear
- Patient has pain in the medial knee; Dx? → anserine bursitis
- Patient has pain in the lateral knee; Dx? → iliotibial band syndrome
- 44F + pain worse in knee when going down/up stairs or when sitting for long periods of time + crepitus + BMI 39; Dx? → answer = patellofemoral syndrome (patellofemoral pain syndrome); next best step in Mx = “strengthening exercises for quadriceps muscles” + RICE (rest, ice, compression, elevation).
- 25F + pain in anterior knee on the inferior kneecap + plays basketball + pain initially worse while playing but past few weeks hurts when done playing as well; Dx? → answer = patellar tendonitis (Jumper’s knee); next best step in Mx = “strengthening exercises for quadriceps muscles” + RICE.
- 15M + 5’11” + plays soccer + knee pain; Dx? → Osgood-Schlatter → inflammation of patellar ligament at the tibial tuberosity; occurs in fast-growing, active teenagers; USMLE wants “repeated avulsion microfractures” as an answer
- Patient has knee pain after spending long periods of time on her knees painting; Dx? → prepatellar bursitis
- 32M + pain in anterior knee + fever 100.5F + joint effusion **not** present; Dx? → septic bursitis
- Any patient with red, warm, tender knee; next best step in Mx + Dx? → joint aspiration (arthrocentesis); septic arthritis till proven otherwise
- 6M + viral infection + now has hip pain +/- fever; Dx? → answer = **toxic synovitis (aka transient synovitis), not septic arthritis** → inflammation of the synovial lining of hip joint; Tx is supportive.

- 6M + suspected JRA + red, hot, painful knee → **must do arthrocentesis** to rule out septic arthritis. If the vignette sounds like classic transient synovitis (affects *hip, not knee*), you do not need to do an arthrocentesis.
- 5F + 2-day Hx of limp and left hip pain + a week ago had watery stools and a temp of 100F + pain with weight-bearing and movement + no swelling or erythema; Tx? → answer = ibuprofen (toxic synovitis).
- Biggest risk factor for septic arthritis → abnormal joint architecture
- Pt groups most likely to get SA → prosthetic joints, RA/OA, recent intense exercise/joint trauma; peds (JRA)
- Pt group most likely to get SA → those with prosthetic joints (can't be more abnormal than fake joint)
- Pt with OA or RA has red, warm, tender knee → do arthrocentesis (septic arthritis)
- 17F had kickboxing tournament last weekend + knee is red, warm, tender → arthrocentesis (SA)
- Kid + recurrent knee redness, warmth, pain + fever → Juvenile rheumatoid arthritis (JRA; Still disease)
- Kid + recurrent joint pain +/- high ESR +/- rash → JRA
- Kid + recurrent joint pain + anemia → JRA (anemia of chronic disease)
- Kid with suspected JRA has sore knee → must do arthrocentesis to rule out septic arthritis
- Patient has "knock-knees" (i.e., knees touch); Dx? → genu valgum
- Child has bow legs; Dx? → genu varum → can be seen in rickets
- 9F + both legs bowed + parents noticed bowing since she started to walk + recently bowing worse in right leg + x-ray while standing shows collapse of the medial aspect of the metaphysis of proximal tibia + rest of vignette describes healthy, thriving patient; Dx? → answer = tibia vara (Blount disease); wrong answer is rickets; should be noted that bowing is physiologic age < 2 years; tibia vara.
- Patient has lateral thigh pain; Dx? → meralgia paresthetica → due to lateral femoral cutaneous nerve entrapment
- 18F + anorexia + runs long distances + has foot pain; Dx? → metatarsal stress fracture
- 25M + wakes up with heel pain + gradually improves throughout the morning; Dx? → plantar fasciitis
- 29M + pain in ball of the foot; Dx? → metatarsalgia → overuse injury / from jumping or sports
- 44F + frequently wears high-heel shoes + painful lump on the underside of her foot between her third and fourth toes; Dx? → answer = Morton neuroma → benign growth of nerve tissue between the 2nd

and 3rd, or 3rd and 4th, metatarsal heads; usually from chronic irritation from high-heel shoes; Mulder sign is replication of Sx when the metatarsal heads are compressed together.

- 42M + diabetes + decreased range of motion of the shoulder in all directions; Dx? → adhesive capsulitis, aka "frozen shoulder" → Tx = range-of-motion exercises / physiotherapy
- 4-month-old + "clicking/clunking" on physical exam → (+) Ortolani and Barlow maneuvers → primary hip dysplasia (congenital hip dysplasia) → once these are positive, **the next best step is ORTHO REFERRAL if it is listed** → referral always sounds wrong, but this is the correct answer if it's listed; if it's not listed, do ultrasound if under 6 months, or x-ray if over 6 months. Tx is with abduction harness (Pavlik harness; looks frog-leg-like)
- 5-8-year-old boy with painful limp; no other risk factors; x-ray shows contracted capital epiphysis; Dx? → Legg-Calve-Perthes (idiopathic avascular necrosis); the word "contracted" **wins** over "capital epiphysis" → this is a Q on one of the NBME forms where everyone selects slipped capital femoral epiphysis (SCFE), but it's Legg-Calve-Perthes; Tx = hip replacement
- 5-8-year-old boy with painful limp + sickle cell disease; Dx? → avascular necrosis (but **not** Legg-Calve-Perthes, because LCP is idiopathic)
- 5-8-year-old boy + painful limp + x-ray is negative + bone scan confirms diagnosis; answer? → USMLE wants you to know that x-ray can be negative initially in avascular necrosis, **but bone scan or MRI can also pick it up**
- 11-13-year-old overweight boy with a painful limp; Dx? → SCFE; Tx = surgical pinning
- Tissue mass in palm of hand + bent fingers; Dx? → Dupuytren contracture → seen in alcoholism, diabetes, Norwegian descent, and epilepsy. Tx is surgical
- 2-year-old boy running + playing with 8-year-old sister + they were holding hands and he fell + now he holds arm pronated by his side; Dx? → **nursemaid's elbow** → radial head subluxation
- Tx for nursemaid's elbow → hyperpronation OR gentle supination (both are correct answers; only one will be listed)
- Kid falls on outstretched arm + pain over anatomical snuffbox; Dx? → scaphoid fracture
- Kid falls on outstretched arm + pain over anatomical snuffbox; next best step in Mx? → x-ray

- Kid falls on outstretched arm + pain over anatomical snuffbox + x-ray is negative; next best step in Mx? → **thumb-spica cast** → x-ray is often negative in scaphoid fracture; must cast to prevent scaphoid avascular necrosis → re-x-ray in 2-3 weeks
- When is figure-of-8-strap the answer? → clavicular fracture
- What part of the clavicle gets fractured easiest? → middle-third
- First Tx for carpal tunnel syndrome in patient who can't stop offending activity (e.g., office worker) → **wrist splint first;** then **triamcinolone (steroid) injection into the carpal tunnel**; do not select anything surgical as it's always wrong on the USMLE; NSAIDs are a wrong answer and not proven to help
- 32F + paresthesias in thenar region of hand +/- hand weakness + sensation intact over dorsum of hand; next best step in Dx? → NBME answer = "Electrophysiological testing"; call it weird, but it's what they want. Examination findings such as Tinel sign, Phalen maneuver, Flick test are insufficient for diagnosis.
- Tx for cubital tunnel syndrome → elbow splint
- What is cubital tunnel syndrome → ulnar nerve entrapment at elbow → presents similarly to carpal tunnel syndrome but just in an ulnar distribution and involves the forearm.
- What is Guyon canal syndrome → ulnar nerve entrapment at the wrist → **hook of hamate fracture** or chronic handle bar impaction in avid cyclists
- Vegan + they ask for nutrient deficiency + B12 is not listed; what's the answer? → FM shelf wants **calcium** as the answer (normally get from dairy + fish); B9 (folate) is wrong because we get that from dark green leafy vegetables
- 82F + tea and toast diet for past 6 months; MCV is elevated; is the nutrient deficiency B9 or B12? → FM shelf answer = B9 (folate) deficiency → stores deplete within six months → "tea and toast" used to be classic for vitamin C deficiency, but the shelf uses this colloquialism for folate deficiency. If scurvy is the answer, they will say the patient "appears ill" and has bleeding from gums or around hair follicles (perifollicular hemorrhages)
- 16M + painful testes + fever + positive cremasteric reflex; Dx? → answer = epididymitis
- Most common organism causing epididymitis? → Chlamydia in sexually active younger males; E. coli in elderly males. This is also the same for prostatitis. If the vignette tells you no organisms grow on culture, choose Chlamydia

- 16M + acutely painful testes + negative cremasteric reflex; Dx? → answer = testicular torsion; do Doppler ultrasound followed by surgical detorsion
- 6M + painful testis + superior pole shows blue/black discoloration + bowel sounds are decreased + abdomen is rigid; Dx? → answer = **strangulated hernia, not testicular torsion** → answer = "operative management"
- 6M + painful testis + superior pole shows blue dot + cremasteric reflex is intact; Dx? → answer = torsion of **appendix testis** → this is on the new peds form but is fair game for FM → torsion of appendix testis is different from testicular torsion; the latter presents with negative cremasteric reflex; but torsion of appendix testis causes "blue dot sign"
- 8-month-old boy + unilateral testicular enlargement + light shone to it causes transillumination; answer = "persistent processus vaginalis" (hydrocele)
- Tx for hydrocele? → **observe** until the age of one as most spontaneously resolve; this is almost always the answer; after the age of one, surgical management can be considered
- 3M + hard nodule on testis; Dx? → yolk sac tumor (endodermal sinus tumor) → serum AFP may be elevated
- 3M + hard nodule on testis + serum AFP + beta-hCG are elevated; Dx? → answer = mixed germ cell tumor (embryonal cancer), **not** yolk sac tumor (yolk sac tumor is *only* high AFP; in mixed germ cell, beta-hCG is also elevated)
- 22M + heaviness and/or bogginess of testes; Dx? → varicocele → one FM shelf Q literally says "bag of worms" (normally this is so buzzy that we'd say this wouldn't show up on an actual form, but it does, so it must be mentioned here) → Dx with Doppler ultrasound → elective surgical intervention may be performed to reduce risk of sub-fertility due to increased testicular temperature
- 8-month-old boy has undescended testis; Tx? → answer = observation until 6 months; do not do orchiopexy before 6 months for USMLE purposes. For hydrocele, observe till one year.
- Tx for scabies? → answer = topical permethrin
- Tx for pediculosis (lice)? → answer = topical permethrin
- Tx for acne:
    - Topical retinoids first (i.e., topical tretinoin; NOT oral isotretinoin); **cause photosensitivity (rash);** also used for photoaging; mechanism is decreasing sebum production; topical

tretinoin (not oral isotretinoin) is not a teratogen and does ***not*** have any effect on pregnancy or male sperm

  - Benzoyl peroxide used second; often coadministered with topic retinoids; mechanism is the killing of bacteria
  - Topical clindamycin
  - Oral tetracycline; **causes photosensitivity (blistering)**
  - Oral isotretinoin; must do beta-hCG in women; recommend barrier contraception even if on OCP; can cause elevations in LFTs; can cause dyslipidemia; main complaint is dry skin + peeling; takes several weeks to really start working but ultra-effective according to most patients; can be commenced earlier in patients with severe nodulocystic acne; works by diffusely shutting of sebum production

- 22F + pain radiating down one arm; Dx? → answer = cervical disc herniation.
- 68M + pain in the neck + MRI shows degenerative changes; Dx? → cervical spondylosis.
- 58F + long-standing Hx of RA + bilateral paresthesias in the upper limbs; Dx? → atlantoaxial subluxation secondary to RA → answer = cervical spine MRI.
- When to give a statin on the USMLE? → Guidelines from 2019 ACC/AHA:
  - Anyone age 20-39 with an LDL > 190 mg/dL
  - Anyone age 40-75 with an LDL > 70 mg/dL
  - Age >75 → assessment of risk status + clinician-patient discussion are recommended before commencing or discontinuing a statin
  - Diabetics <40 if **LDL > 100 mg/dL; hypertension, smoking Hx, CKD, albuminuria, FHx of CVD in first degree relative (sibling or parent), HDL <40 mg/dL.**
  - Anyone age <19 with familial hypercholesterolemia
  - Balance of risk factors contributes to an ASCVD risk score that determines *intensity* of statin administered to the patient (not assessed on USMLE); what the USMLE cares about is you knowing whether or not to give a statin period.
- First drug to start in T2DM → metformin
- Side-effect of metformin → lactic acidosis

- 40F + T2DM + on glyburide + HbA1c = 9.6%; what do we do? → answer = switch to metformin (beta-islet cells are burned out if sulfonylurea isn't working).
- 40F + T2DM + on glyburide + metformin + HbA1c = 9.6^; what do we do? → answer = switch to intermediate-acting insulin (needs insulin if oral hypoglycemics aren't working).
- 40F + T2DM + on glyburide + HbA1c 5.9% + BP 138/82 + creatinine of 1.0; what do we do? → answer = start ACEi (e.g., enalapril) → start an ACEi if BP > 130/80 (either #) or evidence of protein in the urine.
- 40F + T2DM + on glyburide + HbA1c 6.7% + LDL is 112 mg/dL; what do we do? → answer = commence statin to bring LDL under 70 mg/dL (old guidelines said <100).
- How to Dx TB?
    - PPD skin test is performed first diagnostically. If history of BCG vaccine, do interferon-gamma release assay (IGRA) instead. **Do not do IGRA in addition to PPD.**
    - If PPD is negative, repeat after one week. If negative again, no further studies indicated. Repeats performed within 1 week may cause a false (+) secondary to a "booster reaction."
    - If IGRA is negative, no further studies indicated.
    - If PPD or IGRA is positive, do CXR. **Do not repeat positive PPD tests.**
    - If CXR is negative, treat for latent TB / give TB prophylaxis. On the USMLE, "treatment for latent TB" and "administer TB prophylaxis" mean the same thing.
    - If CXR is positive, treat for active TB.
- How to manage a positive PPD test?
    - Measure induration only. **Erythema does not count.**
- 5+ mm
    - Recent contact with people with active TB
    - HIV + status
    - Organ transplant recipients
    - Chronic prednisone use (>15mg/day for >1 month); anti-TNF-α agent use
    - Findings consistent with TB on CXR
- 10+ mm
    - **Immigrant status** (Western countries not included)

  - IV drug users
  - Healthcare workers; prison workers; homeless shelter personnel
  - TB laboratory personnel
  - Children under 4 years of age
- 15+ mm
  - Everyone
- How to Tx latent TB?
  - **9 months INH + pyridoxine (vitamin B6) - The USMLE Steps 1 and 2CK assess this as the answer.**
  - 4 months rifampin
  - 3 months INH + rifapentine + pyridoxine
  - Vitamin B6 must be given with INH to prevent vitamin B6 deficiency.
- Tx of active TB
  - **Rifampin, INH, pyrazinamide, ethambutol (RIPE) for 2 months, followed by RI alone for 4 more months (6 months total)**
  - And of course add pyridoxine (annoying that it sounds similar to pyrazinamide)
- Travel + self-limiting watery or brown/green diarrhea → Traveler diarrhea = ETEC HL or HS toxin
- Bloody diarrhea + poultry consumption → Campylobacter jejuni or Salmonella spp.
- Abx (clindamycin, beta-lactam, cephalosporin) + diarrhea → C. difficile
- Dx of C. diff → stool AB toxin test, **not** stool culture
- Fever of 104 + abdo distension in C diff → toxic megacolon → laparotomy
- Tx of C. diff → **vancomycin, not metronidazole (updated guidelines as of Feb 2018)**
- Bloody diarrhea + travel → Entamoeba histolytica
- Tx of E. histolytica → metronidazole + iodoquinol; can give paromomycin
- Close quarters or military barracks or cruise ship + watery diarrhea → Norwalk virus
- Child <5 years + watery diarrhea → rotavirus
- Few organisms causing bloody diarrhea → Shigella
- Bloody diarrhea + appendicitis-like pain (pseudoappendicitis) in a child → Yersinia enterocolitica

- Bloody diarrhea + reactive arthritis in an adult → Y. enterocolitica, Campylobacter, Shigella, Salmonella
- Diarrhea + Guillain-Barre syndrome → Campylobacter
- GBSyndrome CSF? → albuminocytologic dissociation (high protein + normal cells)
- GBSyndrome Dx? → electromyography + nerve conduction studies (on NBME)
- Cardiac ischemia + need to evaluate → ECG stress test first-line
- Cardiac ischemia + abnormal baseline ECG (e.g., BBB) → Echo stress test (need normal ECG to do ECG stress test)
- Cardiac ischemia + patient can't exercise → dobutamine + ECG/echo
- ECG shows diffuse ST-segment elevations → pericarditis
- Pericarditis Tx → NSAID, or steroid, or colchicine
- Central chest pain worse when supine; better when leaning forward → pericarditis
- Lateral chest pain after viral infection + increased CK → pleurodynia (intercostal muscle spasm)
- ST-segment depressions in the anterior ECG leads → posterior MI
- Electrical alternans on ECG → pericardial tamponade / pericardial effusion
- Pulsus paradoxus (drop in systolic BP >10 mm with inspiration) → cardiac tamponade or severe asthma
- Beck triad → hypotension + muffled heart sounds + JVD → pericardial tamponade
- Tx of tamponade → pericardiocentesis or pericardial window
- Tamponade → do echo before pericardiocentesis if both listed (even though sounds wrong, on 2CK NBME)
- Community-acquired pneumonia (CPP) + bilateral CXR infiltrates → Mycoplasma
- CPP + lobar pattern (right-lower lobe consolidation + dullness to percussion) → Strep pneumo
- CPP + lobar pattern, but they say "interstitial" in the vignette description → Mycoplasma, not S. pneumo
- Empiric Tx for CPP → azithromycin

- Tx for CPP is pt on Abx past three months → fluoroquinolone over azithro
- Pneumonia in CF patient <10 years → S. aureus exceeds Pseudomonas
- Pneumonia in CF patient >10 years → Pseudomonas exceeds S. aureus.
- Pneumonia after influenza infection → USMLE wants S. aureus
- Pneumonia + rabbits → F. tularensis
- Pneumonia + cattle → Coxiella (Q fever)
- Pneumonia + birds → Chlamydia psittaci
- Tx for aspirin toxicity → bicarb (increased excretion through urinary alkalinization)
- Tx for diabetic neuropathic pain? → answer = TCA (i.e., amitriptyline). Second-line is gabapentin
- Tx for herpetic / post-herpetic neuralgia (i.e., from shingles)? → gabapentin
- 82M diabetic + neuropathic pain + already taking carbamazepine + gabapentin to no avail; next best step? → switch the meds to nortriptyline (a TCA) → student then asks, "Wait, I thought you said TCAs are first-line. Why does this Q have the guy on those two meds then?" → two points: 1) we don't like giving TCAs to elderly because of their anticholinergic and anti-alpha-1 side-effects, so this vignette happen to try other agents first, but if you're asked first-line, always choose TCA; and 2) if we do give a TCA to an elderly patient, we choose nortriptyline because it carries fewer adverse effects.
- How to differentiate cluster headache from trigeminal neuralgia? → cluster will be a male 20s-40s with 11/10 lancinating pain behind the eye waking him up at night (he may pace around the room until it goes away); details such as lacrimation and rhinorrhea are too easy and will likely not show up on the shelf. In contrast, trigeminal neuralgia will be 11/10 lancinating pain behind the eye (or along the cheek / jaw if V2 or V3 branches affected; it's when V1 is affected that this diagnoses are more readily confused) → TN is brought on by a minor stimulus such as brushing one's hair or teeth, or a gust of wind.
- Tx and prophylaxis for cluster headache? → Tx = 100% oxygen; prophylaxis = **verapamil.**
- Tx and prophylaxis for trigeminal neuralgia? → Tx = goes away on its own because it lasts only seconds; prophylaxis = **carbamazepine.**
- Tx and prophylaxis for migraine? → Tx = NSAID, followed by triptan (triptans are NOT prophylaxis; they are for abortive therapy only after NSAIDs); prophylaxis = **propranolol.**

- 32M + diffuse headache relieved by acetaminophen + sleep; Dx? → answer = tension-type headache; Tx = rest + taper caffeine (if taking it).
- Other HY uses for propranolol?
    - Migraine prophylaxis (FM form gives patient with HTN + migraine; answer = propranolol)
    - Akathisia (with antipsychotic use)
    - Thyroid storm (decreases peripheral conversion of T4 to T3)
    - Essential tremor (bilateral resting tremor in young adult; autosomal dominant; patient will self-medicate with EtOH, which decreases tremor).
    - Hypertension + idiopathic tremor (i.e., tremor need not be essential if patient has HTN → answer on FM form is "beta-adrenergic blockade" for the HTN Tx).
    - Esophageal varices prophylaxis (patients at risk of bleeds)
    - Hypertrophic obstructive cardiomyopathy (increases preload → decreases murmur)
    - Social phobia
- Important point about Graves? → proptosis/exophthalmos + pretibial myxedema are specific; in other words, there are numerous causes of hyperthyroidism (e.g., toxic multinodular goiter, toxic adenoma, etc.), but only Graves will cause the eye findings
- Tx for ophthalmopathy in Graves → **steroids**
- Why do the eye findings occur in Graves? → glycosaminoglycan deposition in/around extra-ocular muscles
- High BP + fever + increased CK → thyroid storm (Tx with PTU + propranolol)
- What is the role of potassium iodide (KI) in hyperthyroid Tx? → shuts off gland production (Wolff-Chaikoff effect) → answer in person exposed to nuclear fallout or radioiodine vapors in laboratory
- Hashimoto parameters → high TSH, low T3, low T4, decreased iodine uptake
- Mechanism for Hashimoto → antibodies against thyroperoxidase + thyroglobulin; anti-microsomal
- Histo of Hashimoto → lymphocytic infiltrate (easy to remember bc the non-eponymous name for Hashimoto is chronic lymphocytic thyroiditis)
- Tx for Hashimoto = levothyroxine (synthetic T4)
- 50M + low mood + BMI 26 → Hashimoto
- Proximal muscle weakness in Hashimoto + increased serum CK → hypothyroid myopathy

- 45M + decreased ability to get up from chair unassisted + HR of 60 → Hashimoto
- 45M + high cholesterol + high hepatic AST + HR of 55 → Hashimoto (hypothyroidism can cause bradycardia, high cholesterol, and high AST [the latter is weird, correct])
- Thyroid cancer in Hashimoto → thyroid lymphoma (autoimmune diseases → increased risk of lymphoma)
- 22M + viral infection + very tender thyroid → subacute granulomatous thyroiditis (de Quervain)
- De Quervain parameters → triphasic → causes hyper-, then hypo-, then rebounds to euthyroid state → most important point is that **iodine uptake is DECREASED in de Quervain + drug-induced thyroiditis, even if patient is in a hyper-phase.**
- 22M + very tender thyroid + HR of 88 + tremulousness + heat intolerance → low TSH, high T3, high T4, **decreased iodine uptake** (in contrast to Graves, which is painless and uptake is high)
- Tx for subacute thyroiditis → aspirin first, not steroids; steroids may be used later
- Drugs causing thyroiditis → lithium + amiodarone
- Surreptitious thyrotoxicosis →self- injection of thyroxine → low TSH, high T3, high T4, small thyroid gland with decreased uptake
- Injection of triiodothyroinine (T3) → TSH will go down, T3 goes up (clearly), **T4 does not go up because T3 isn't converted to T4; only T4 is converted to T3**
- Injection of thyroxine → TSH will go down (negative feedback), T4 goes up (clearly), T3 goes up (due to peripheral conversion), reverse T3 also goes up
- What is reverse T3? → an inactive form of T3; T4 is converted peripherally into T3 (active) and reverse T3 (inactive)
- Anything else I need to know about reverse T3? → it's increased in euthyroid sick syndrome → times of stress/surgery/illness → cortisol increases → cortisol decreases conversion of T4 to active T3, so more T4 is converted to reverse T3 → parameters in euthyroid sick syndrome: normal TSH, normal T4, low T3, high reverse T3
- What is subclinical hypothyroidism → high TSH but normal T3 + T4 (don't confuse with ESS)
- Subclinical hypothyroidism Tx → don't treat unless TSH >10 (normal is 0.5-5), Hashimoto Abs are present, or patient is pregnant → they ask this info on 2CK + Step 3
- Want to check thyroid function, what's the first thing to order → TSH

- Want to check thyroid function, what's the first thing to order → free T4
- What is free T4 → most thyroid hormone is protein-bound and inactive; free T4 tells you definitively whether the patient has thyroid derangement or not
- Pregnancy and thyroid → estrogen causes increased thyroid-binding globulin production by the liver → mops of T4 → less free T4 → less negative feedback at hypothalamus + anterior pituitary → TSH goes up transiently to compensate → more T4 made → free T4 rebounds to normal but now total T4 is high → parameters you need to know for pregnancy: normal TSH + high total T4 + normal free T4 + normal free T3 + high thyroid-binding globulin
- Hyperthyroidism in pregnancy → LH, FSH, TSH, hCG all share same alpha-subunit; their beta-subunits differ; some women have increases sensitivity of TSH receptor to alpha-subunit, so high hCG in early pregnancy can stimulate thyroid gland and cause transient hyperthyroidism
- Hashimoto in pregnancy → increase pregnant woman's dose of levothyroxine by 50%
- Graves in pregnancy → avoid methimazole in first trimester (teratogenic; causes aplasia cutis congenita) → give PTU in first-trimester → in second + third trimesters, switch from PTU to methimazole (methimazole no longer teratogenic later in pregnancy + PTU is heavily hepatotoxic)
- Pt being treated for Graves + mouth ulcers → agranulocytosis (neutropenia) caused by methimazole or PTU.
- Young child with normal free T4 and low total T4 → thyroid-binding globulin deficiency (opposite of pregnancy)
- Young child + large belly + large tongue + hypotonia → cretinism (congenital hypothyroidism)
- Most common cause of cretinism → iodine deficiency
- Most common cause of hyperthyroidism in elderly → toxic multinodular goiter
- Evaluation of thyroid cancer, first step? → palpation of thyroid gland (on FM 2CK form as answer)
- If thyroid nodule present, then check TSH; if TSH normal or high → answer = ultrasound first if listed; if not listed, answer = FNA; if TSH low, do radioiodine uptake scan; thyroid cancer is cold, not hot, which is why no FNA with low TSH
- Tx for shingles (not the pain; the actual shingles)? → answer = oral acyclovir (one of the forms writes "oral acyclovir" rather than oral valacyclovir, but both are fine) → however if patient is on chemotherapy / is immunocompromised, give IV acyclovir

- Patient older than 50 + temporal headache + high ESR; Dx + Tx? → temporal arteritis → give immediate IV methylprednisolone (steroids) *then* do biopsy temporal artery (never biopsy first)
- Patient with temporal arteritis + jaw pain; why the jaw pain? → can cause temporomandibular joint claudication
- Patient with temporal arteritis has muscle pain; Dx? → polymyalgia rheumatica (PMR)
- Tx of PMR? → oral steroids
- How to differentiate PMR from polymyositis? → both can have high ESR; PMR tends to have "muscle pain + stiffness" **with preserved muscle strength**; polymyositis will present with proximal muscle weakness +/- pain and stiffness (the idea is: PMR more likely to have pain + stiffness, but NO muscle weakness; polymyositis less likely to have pain + stiffness, but WILL have proximal muscle weakness)
- Difference between polymyositis and dermatomyositis? → dermatomyositis has heliotrope rash (around the eyes; don't confuse with malar rash) +/- shawl rash +/- Gottron papules (violaceous papules on the knuckles) +/- mechanics' hands (rough/scaly palms)
- Dx of polymyositis/dermatomyositis? → anti-Jo1 and anti-Mi2 first; if positive, do confirmatory muscle biopsy
- Mechanism of hyponatremia in heart failure? → answer = "baroreceptor-mediated ADH release" → heart doesn't pump as well → decreased systolic impulse → stretch of carotid sinus baroreceptors → increased ADH release. This is the same autoregulation mechanism that will cause decreased parasympathetic activity via vagus nerve.
- Nocturnal enuresis; when is it pathologic? → after age 5
- 6M + nocturnal enuresis; next best step? USMLE / NBME / shelf wants the following order:
    - Behavioral answer first; e.g., spend more time with child; decrease overt stressors as much as possible
    - If the above not an answer, do **star chart** (positive reinforcement therapy; i.e., don't wet the bed and get a star; get 5 stars for extra dessert; 100 and we go to Disneyland)
    - If star chart not listed or already attempted, next answer is **enuresis alarm**
    - Medications like imipramine and desmopressin are always wrong; water deprivation after 5pm is also always wrong.

  - Students mess these Qs up because they'll see enuresis alarm as correct on one form, but on a different form it's star chart, so know the above order
- 72M + intermittent claudication + absent distal pulses + Hx of coronary artery bypass grafting + high BP that's been gradually increasing past two years; Dx? → renal artery stenosis
- 32F + high BP + high aldosterone/renin → fibromuscular dysplasia (tunica media proliferation in renal arteries) → this is **not** renal artery stenosis → if you say "renal artery stenosis," that means atherosclerosis
- Increased creatinine following medication administered to someone with renal artery stenosis; what was the drug? → ACEi or ARB
- Tx for RAS + FMD → initially medical therapy with cautious use of ACEi or ARB; definitive is renal angioplasty + stenting; FMD is not curable
- How to differentiate viral from bacterial upper respiratory tract infection (URTI)? → CENTOR criteria
  - If 0 or 1 point, the URTI is unlikely to be bacterial (i.e., it's likely to be viral). If 2-4 points, chance is much greater that URTI is bacterial.
  - **1) <u>Absence of</u>** cough (i.e., no cough = 1 point; if patient has cough = 0 points).
  - **2)** Fever.
  - **3)** Tonsillar exudates.
  - **4)** Lymphadenopathy (cervical, submandibular, etc.).
- There is a version of the criteria that includes age, but on the USMLE it can cause you to get questions wrong. So just use the simplified above four points.
  - If 0-1 point, answer = "supportive care"; or "no treatment necessary"; or "warm saline gargle" (same as supportive care)
  - If 2-4 points, next best step = "rapid Strep test." If rapid Strep test is negative, answer = throat culture, **NOT** sputum culture
  - While waiting on the throat culture results, we send the patient home with amoxicillin or penicillin for presumptive Strep pharyngitis
  - If child is, e.g., 12 years old, and develops a rash with the beta-lactam, answer = beta-lactam allergy

  - If the vignette is of a 16-17 year-old who has been going on dates recently (there will be no confusion; the USMLE will make it clear), the answer = EBV mononucleosis; therefore do a heterophile antibody test (Monospot test)
  - **EBV is the odd virus out that usually presents with all four (+) CENTOR criteria**
  - This is why it's frequently misdiagnosed as Strep pharyngitis. It is HY to know that beta-lactams given to patients with EBV may cause rash via a hypersensitivity response to the Abx in the setting of antibody production to the virus. EBV, in a patient who does not receive Abx, can cause a mild maculopapular rash. But the rash with beta-lactam + EBV causes a more intense pruritic response generally 7-10 days following Abx administration on the extensor surfaces + pressure points

- Tx for TCA toxicity → sodium bicarb → dissociates drug from myocardial sodium channels
- Ataxia, confusion, ophthalmoplegia → Wernicke encephalopathy (B1 deficiency)
- Retrograde amnesia + confabulation in alcoholic → Korsakoff psychosis
- Wernicke-Korsakoff syndrome → mammillary bodies
- Hx of many pregnancies + downward movement of vesicourethral junction → stress incontinence
- Tx of stress incontinence → pelvic floor exercises (Kegel); if insufficient → mid-urethral sling
- Hyperactive detrusor or detrusor instability → urge incontinence
- Need to run to bathroom when sticking key in a door → urge incontinence
- Incontinence in multiple sclerosis patient or perimenopausal → urge incontinence
- Tx of urge incontinence → oxybutynin (muscarinic cholinergic antagonist) or mirabegron (beta-3 agonist)
- Incontinence + high post-void volume (usually 3-400 in question; normal is <50 mL) → overflow incontinence
- Incontinence in diabetes → overflow incontinence due to neurogenic bladder
- Tx for overflow incontinence in diabetes → bethanechol (muscarinic cholinergic agonist)
- Incontinence in BPH → overflow incontinence due to outlet obstruction → eventual neurogenic bladder

- Tx for overflow incontinence in BPH → insert catheter first; then give alpha-1 blocker of 5-alpha-reductase inhibitor; then TURP if necessary
- Exquisitely tender prostate on digital rectal exam → prostatitis
- Prostatitis Tx → ciprofloxacin (fluoroquinolone → DNA gyrase / topoisomerase II inhibitor)
- Costovertebral angle tenderness + fever → pyelonephritis
- Costovertebral angle tenderness + granular casts → pyelonephritis (correct, super-weird; NOT acute tubular necrosis; this is on 2CK NBME)
- Tx for pyelonephritis → ciprofloxacin, OR ampicillin + gentamicin
- Saddle anesthesia + urinary retention → cauda equina syndrome
- Perianal anesthesia + urinary retention or incontinence → conus medullaris syndrome
- Gradual-onset dementia + no sensory or motor dysfunction → Alzheimer
- Mini-mental state exam score low + patient tries to do well → Alzheimer
- Mini-mental state exam score low + patient is apathetic / takes long to perform tasks / does poorly on reverse serial 7s → depression (pseudodementia)
- Patient complains about memory loss → normal aging, not Alzheimer
- First-line Tx for Alzheimer → donepezil (cholinesterase inhibitor → increases cholinergic transmission); can also give galantamine or rivastigmine
- NMDA receptor (glutamate receptor) antagonist used in Alzheimer Tx → memantine
- Step-wise dementia/decline and/or sensory/motor disturbance → vascular dementia
- Hx of hypertension + dementia + sensory/motor disturbance → vascular dementia
- Visual hallucinations + Parkinsonism + dementia → Lewy body dementia
- Apathy + disinhibition + personality change + dementia → frontotemporal dementia (Pick disease)
- Urinary incontinence + ataxia + CNS dysfunction (wet, wobbly, wacky) → Normal-pressure hydrocephalus
- Wet, wobbly, wacky + Parkinsonism → still NPH
- Parkinsonism in young patient → Wilson disease till proven otherwise

- Parkinsonism in older patient → Parkinson disease
- Parkinsonism + axial dystonia → progressive supranuclear palsy
- Tx of UTI → TMP/SMX or nitrofurantoin
- Tx of cystitis → nitrofurantoin (need not be pregnant)
- Waiter tip position in kid → upper brachial plexus injury → C5/6 → Erb-Duchenne palsy
- Claw hand → lower brachial plexus → C8/T1 → Klumpke palsy
- Pronated arm + wrist-drop → radial nerve injury
- Midshaft fracture of humerus → Radial nerve injury
- Supracondylar fracture of humerus → median nerve injury
- Surgical neck of humerus fracture → axillary nerve injury
- Medial epicondylar injury → ulnar nerve injury
- Weakened biceps + loss of sensation of lateral forearm → musculocutaneous nerve injury
- Paresthesias + pain following burn or casting → compartment syndrome
- Compartment syndrome Dx → measure compartment pressure
- Compartment syndrome Tx → fasciotomy of uncast
- Guy lifts heavy box → severe lower back pain + muscle spasm + no radiculopathy → lumbosacral strain only
- Lumbosacral strain diagnosis → DO NOT x-ray
- Lumbosacral strain Tx → NSAIDs + exercise as tolerated; bedrest is the WRONG answer
- Guy lifts heavy box → severe lower back pain + radiculopathy → herniated disc; yes, x-ray.
- Point tenderness over a vertebra in older woman → osteoporosis (compression fracture)
- Point tenderness over a vertebra in younger patient on steroids → osteoporosis (compression fracture)
- Point tenderness over a vertebra in patient with autoimmune disease → recognize patient is on steroids → osteoporosis (compression fracture)
- Point tenderness over a vertebra in IV drug user → epidural abscess

- "Step-off" of one vertebra relative to another → spondylolisthesis
- Back pain worse in the morning and gets better throughout day in male 20s-40s → ankylosing spondylitis
- Bamboo spine → ankylosing spondylitis
- Dx of AS → x-ray of sacroiliac joints
- Back pain worse when standing or walking for long periods of time → lumbar spinal stenosis
- Radiculopathy down an arm → cervical disc herniation
- Bilateral paresthesias in the arms in rheumatoid arthritis patient → atlantoaxial subluxation
- Bilateral paresthesias in the arms in rheumatoid arthritis patient → MRI of spine to Dx atlantoaxial subluxation
- Prior to surgery in rheumatoid arthritis patient → cervical CT or flexion/extension x-rays of cervical spine to check for atlantoaxial subluxation
- Back pain in elderly patient with hypercalcemia → multiple myeloma or metastases
- Back in pain in patient with history of other type of cancer → metastases
- Suspected spinal mets → MRI
- Metastases to long bones in prostate cancer → osteoblastic (Dx with bone scan); spine do MRI
- High hemoglobin +/- pruritis after shower +/- plethora +/- splenomegaly → polycythemia vera
- High hemoglobin + low EPO → polycythemia vera
- Pruritis after shower → basophilia
- High hemoglobin + lung disease / low pO2 → secondary polycythemia (high EPO)
- Polycythemia + hypercalcemia + smoker + red urine → RCC (paraneoplastic EPO + PTH-rp)
- Blurry vision or Raynaud or pain in fingers or headache → hyperviscosity syndrome
- Hereditary spherocytosis → AD, ankyrin or spectrin or band protein deficiency; Tx = splenectomy
- Treatment for ITP → steroids, then IVIG, then splenectomy
- Tx for hereditary hemochromatosis → serial phlebotomy
- Tx for secondary hemochromatosis (transfusional siderosis) → chelation therapy (deferoxamine)

- Viral infection + tinnitus + vertigo +/- neurosensory hearing loss → labyrinthitis
- Viral infection + vertigo → vestibular neuritis
- Tx for multiple sclerosis flares → IV steroids (IV methylprednisolone)
- Given to MS patients between flares → interferon-beta
- Tx for spasticity in MS → baclofen (GABA-B receptor agonist)
- Incontinence in MS → urge (hyperactive detrusor, as mentioned earlier)
- New-onset murmur + fever → endocarditis till proven otherwise
- Empiric Tx for endocarditis → vancomycin or ampicillin/sulbactam, PLUS gentamicin
- Beta-lactam (e.g., nafcillin) or cephalosporin + rash + renal issue → interstitial nephritis
- Interstitial nephritis → WBCs in the urine (eosinophils)
- Fixed splitting of S2 → atrial septal defect
- Holosystolic murmur at left sternal border PLUS either parasternal heave or palpable thrill → VSD
- Holosystolic murmur at left sternal border PLUS diastolic rumble → also VSD
- To-and-fro murmur → PDA (on 2CK NBME)
- Pan-systolic-pan-diastolic murmur → PDA
- Continuous, machinery-like murmur → PDA
- Congenital rubella syndrome → PDA
- Heart problem in neonate of mom with SLE → congenital third-degree heartblock
- Heart problem in William syndrome → supravalvular aortic stenosis
- Bicuspid aortic valve → aortic stenosis
- Mid-systolic (crescendo-decrescendo) murmur + gets worse with Valsalva → HOCM
- Mid-systolic (crescendo-decrescendo) murmur + no change or softens with Valsalva → aortic stenosis
- Myxomatous degeneration of mitral valve → mitral valve prolapse
- Marfan or Ehlers-Danlos syndrome → MVP or aortic regurg
- Rheumatic heart disease acutely (onset of Group A Strep infection) → mitral regurg
- Rheumatic heart disease later on (years after infection) → mitral stenosis

- Mid-systolic click → MVP
- Late-peaking systolic murmur with ejection click → another way they describe aortic stenosis
- Bounding pulses + massively wide pulse pressure → aortic regurg
- Brisk upstroke + precipitous downstroke of pulse → aortic regurg
- Syncope + angina + dyspnea (SAD) → aortic stenosis
- Dyspnea in second trimester of pregnancy → mitral stenosis
- Dyspnea late in pregnancy → peripartum cardiomyopathy
- Screening at age 50 → mammogram (every two years) + colonoscopy (every ten years)
- Colon cancer in first-degree relative (sibling or parent) → start at age 40 or ten years before diagnosis in relative, whichever is earlier, and do every 5 years.
- Breast imaging (if performed) → ultrasound only under age 30; over age 30 do mammogram +/- ultrasound
- Dysphagia to solids and liquids at the same time to start → says neurogenic cause → achalasia
- Dysphagia to solids that **progresses** to solids and liquids → esophageal cancer
- Halitosis +/- gurgling sound when swallowing +/- regurgitation of undigested food → Zenker
- Zenker + achalasia initial Dx modality → barium swallow
- After barium swallow is done and shows bird's beak appearance → monometry to confirm Dx of achalasia
- Pt with Hx of GERD + dysphagia → straight to endoscopy to rule out cancer
- Diabetic pt with new-onset GERD → diabetic gastroparesis
- Diabetic pt with new-onset GERD → give metoclopramide, not PPI
- Diabetic gastroparesis before giving med → endoscopy first to rule out physical obstruction
- Endoscopy negative for diabetic gastroparesis → do gastric-emptying scintigraphy
- Bulimia nervosa or anorexia → never give buproprion (seizure risk)
- Electrolyte abnormality in anorexia → hypokalemia
- Most common cause of death in anorexia → arrhythmia from hypokalemia

- Refeeding syndrome → worry about hypophosphatemia
- Tx of anorexia + depression → mirtazapine (alpha-2 antagonist); stimulates appetite
- Amenorrhea in anorexia → low FSH + low estrogen (hypogonadotropic)
- Premature ovarian failure + Turner syndrome + menopause → high FSH (low inhibin) + low estrogen
- Trichotillomania (eating one's hair) + GI symptoms → gastric bezoar (hair ball)
- Tx for trichotillomania? → SSRI.
- Kid with high lymphocytes → ALL or pertussis (weird bc bacterial, but lymphocytes often >30k)
- Dry cough in winter → cough-variant asthma (1/3 of asthmatics only have cough)
- Young African American woman + dry cough + normal CXR → asthma (activation of mast cells), not sarcoidosis
- Young African American woman + dry cough + nodularity on CXR → sarcoidosis (noncaseating granulomas)
- Electrolyte disturbance in sarcoid → hypercalcemia → LOW PTH + high calcium
- Hypercalcemia in sarcoid, why? → epithelioid (activated) macrophages produce 1-alpha hydroxylase, thereby activating vitamin D3
- Increased calcium in sarcoid → means decreased calcium in feces (bc D3 increased small bowel absorption)
- Tx for sarcoid → steroids
- Outpatient Tx of asthma → SABA, then low-dose ICS, then maximize dose of ICS, then LABA, then use any number of drugs (e.g., mast cell stabilizers, anti-leukotriene, etc.), then oral steroids last resort
- Kid with asthma on SABA inhaler + not effective + next best step? → ICS (fluticasone)
- Kid with asthma on SABA inhaler + most effective way to decrease recurrence? → oral steroids (not next best step, but certainly most effective)
- 40s male + hematuria + hemoptysis → Goodpasture syndrome
- Antibodies in Goodpasture → Anti-GBM (anti-collagen IV)
- Dx of Goodpasture → antibodies first, but confirmatory is renal biopsy showing linear immunofluorescence

- Hematuria + hemoptysis + “head-itis” (mastoiditis, sinusitis, otitis, nasal septal perforation) → Wegener
- New name for Wegener → granulomatosis with polyangiitis
- Dx of Wegener → c-ANCA (anti-PR3; anti-proteinase 3)
- Asthma + eosinophilia → Churg-Strauss
- New name for CS → eosinophilic granulomatosis with polyangiitis
- Dx of CS → p-ANCA (anti-MPO; anti-myeloperoxidase)
- Hematuria in isolation + p-ANCA in serum → microscopic polyangiitis (MP)
- Teenage girl with Hx of cutaneous candida infections since childhood → chronic mucocutaneous candidiasis
- MCC → answer = T cell dysfunction = impaired cell-mediated immunity on the USMLE
- Bacterial + fungal + protozoal + viral infections since birth → SCID
- Bacterial infections since age 6 months → Bruton
- Bacterial infections only since birth → Bruton (rare as hell to say from birth, but it’s on new 2CK NBME)
- SCID XR variant → common gamma-chain mutation (IL-2 receptor deficiency)
- SCID AR variant → adenosine deaminase deficiency
- Bruton mechanism → tyrosine kinase mutation
- Hyper IgM syndrome → deficiency of CD40 ligand on T cell (can’t activate CD40 on B cell to induce isotype class switching)
- Greasy, scaly scalp + itchy + papules + adult → seborrheic dermatitis
- Tx for SD → azole or selenium shampoo
- Tx for tinea capitis → oral griseofulvin for patient only
- How to decrease risk of tinea capitis → avoidance of sharing of hats
- Tx of onychomycosis (nailbed fungus) → oral terbinafine
- Tx of tinea pedis → topical terbinafine or topical azole

- Tx of tinea corporis (ring worm) → topical azole (clotrimazole or miconazole)
- Tx of cutaneous candida → oral azole
- Tx of oropharyngeal candida → nystatin mouthwash
- Tx of esophageal candidiasis → oral azole, not nystatin mouthwash
- Tx of vaginal candidiasis → topical nystatin before trying oral azole
- Odynophagia (painful swallowing) in immunocompromised pt → esophageal candidiasis till proven otherwise
- CNS fungal infection or fungemia (rigors/chills) → amphotericin B
- Cryptococcal meningitis → amphotericin B + flucytosine, then do fluconazole taper
- Simple fungal pneumonia → fluconazole
- Sporothrix schenckii (rose thorn + finger papule) → itraconazole
- Hypopigmentation on upper back / trunk → tinea versicolor (Malassezia furfur)
- Tx of tinea versicolor → topical selenium
- Most common cause of impetigo → S. aureus now exceeds S. pyogenes
- Tx of impetigo → topical mupirocin
- Beefy red, well-demarcated skin plaque → erysipelas
- Most common cause of erysipelas → Group A Strep (S. pyogenes) >>> S. aureus
- More diffuse pink skin lesion + tenderness + fever → cellulitis
- Most common cause of cellulitis → S. aureus exceeds S. pyogenes
- Tx of erysipelas + cellulitis → oral dicloxacillin or oral cephalexin
- Give killed IM influenza vaccine when? → Every year in fall/winter only; start from 6 months of age
- Killed IM Influenza vaccine safe in pregnancy? → Yes, give anytime to pregnant women
- Live-attenuated intranasal influenza vaccine guidelines? → Only give age 2-49 to non-pregnant, non-immunocompromised persons
- Vaccines at age 2, 4, 6 months: HepB, Polio Salk, Pneumo PCV13, DPT, HiB, rotavirus (also give HepB at birth)

- HPV vaccine when → age 9-45
- Mom's HepB status unknown→ give neonate HepB vaccine; only give immunoglobulin if mom comes back +
- MMR → first dose at 12-15 months; second dose age 4-6 years
- Varicella → one dose between 12-18 months
- Age 65 or older → give Pneumo PCV13 followed by PPSV23 6-12 months later
- Asplenia or sickle cell → PCV13 + PPSV23 + HiB + Meningococcal
- Young adult + non-smoker + has emphysema + relative died of hepatic cirrhosis → alpha-1 anti-trypsin deficiency
- CREST syndrome lung pathology? → can cause pulmonary fibrosis → pulmonary hypertension
- Restrictive lung disease → normal or increased FEV1/FVC
- Obstructive lung disease → decreased FEV1/FVC
- Why is FEV1/FVC normal or high in restrictive? → radial traction on outside of airways is sticky (keeps airways from closing)
- Apex to base lung changes when sitting/standing → **both** ventilation + perfusion increase apex to base
- Most common cause of otitis media → Strep pneumo
- Tx of otitis media → oral amoxicillin only
- Tx of recurrent OM → amoxicillin/clavulanate
- When to do tympanostomy tube → three or more OM in 6 months, or 4 or more in a year
- Most common cause of otitis externa → Pseudomonas
- Tx of otitis externa → topical ciprofloxacin + hydrocortisone drops
- Prevention of OE in someone with constant water exposure (e.g., crew team) → alcohol-acetic acid drops
- Tx of earwax buildup → carbamide peroxide drops
- Low hematocrit + low MCV + low transferrin + low TIBC + transferrin saturation normal or low → anemia of chronic disease

- Low hematocrit + low MCV + high transferrin + high TIBC + transferrin saturation super-low → iron deficiency anemia
- Low hematocrit + low MCV + increased red cell distribution width (RDW) → iron deficiency anemia
- Low hematocrit + low MCV + decreased RDW → thalassemia
- Low hematocrit + low MCV + low iron + low ferritin → iron deficiency
- Low hematocrit + low MCV + low iron + normal or high ferritin → anemia of chronic disease.
- Low hematocrit + low MCV + low iron + normal ferritin in pregnant woman on iron supplements → thalassemia
- Microcytic anemia that doesn't improve with iron supplementation → thalassemia
- Dx of thalassemia → hemoglobin electrophoresis
- Low hematocrit + normal MCV + low iron + normal or high ferritin → anemia of chronic disease
- Tx of anemia of chronic disease if renal failure is cause → answer = EPO
- Tx of anemia of chronic disease if renal failure not cause (IBD, RA, SLE, etc.) → CANNOT give EPO; Tx underlying condition.
- High BP + smoker + TIA or stroke or retinal artery occlusion. How to best decrease stroke risk → Answer = lisinopril, not smoking cessation
- Normotensive old pt + TIA or stroke or retinal artery occlusion → atrial fibrillation
- Hypertensive pt + stroke → do carotid duplex ultrasound
- Normotensive pt + stroke → do ECG; if ECG normal → Holter monitor
- High BMI female + irregular menstrual cycles → anovulation
- Anovulation + hirsutism → PCOS
- Anovulation. Cause USMLE wants? → insulin resistance → causes abnormal GnRH pulsation
- Why hirsutism in anovulation → abnormal GnRH pulsation causes high LH/FSH ratio
- Why high LH/FSH ratio important in anovulation/PCOS → ovulation stimulated when follicle not ready → no ovulation (anovulation) → follicle retained as cyst
- What's LH do? → Stimulates theca interna cells (females) and Leydig cells (males) to make androgens

- What's FSH do? → Stimulates granulosa cells (females) and Sertoli cells (males) to make aromatase; also primes follicles
- Tx for PCOS → if high BMI, weight loss first always on USMLE
- Tx for PCOS if they ask for meds and/or weight loss already tried → OCPs (if not wanting pregnancy); clomiphene (if wanting pregnancy)
- PCOS increases risk of what → endometrial cancer (unopposed estrogen)
- Tx of prostate cancer → flutamide + leuprolide together (if they force a sequence, choose F then L).
- Tx of acute gout → indomethacin (NSAID) first on USMLE; then steroids, then colchicine
- Tx of acute gout if indomethacin + steroids not listed → colchicine
- Tx of acute gout in pt with renal insufficiency or Hx of renal transplant → steroids
- Tx of chronic gout (decrease recurrence) → allopurinol or febuxostat (xanthine oxidase inhibitors)
- Never give which drug to pt with Hx of uric acid stones or over-producer → probenecid (uricosuric)
- What are rasburicase / pegloticase → urate oxidase analogues → cleave uric acid directly
- Young kid + self-mutilation + red-orange crystals in diaper → Lesch-Nyhan syndrome (X-linked)
- Lesch-Nyhan enzyme → HGPRT deficiency
- Crystal type in pseudogout → calcium pyrophosphate deposition disease
- Two main causes of pseudogout → primary hyperparathyroidism + **hemochromatosis**
- Two ways pseudogout presents → monoarthritis of large joint (i.e., knee) or **osteoarthritis-like presentation in someone with primary hyperparathyroidism or hemochromatosis**
- 32M + dark skin on forearms + increased fasting glucose; Dx? → hemochromatosis (bronze diabetes)
- Same male + painful hands + x-ray shows DIP involvement. Joint pain Dx? → **pseudogout**
- Tx of pseudogout → same as gout acutely; Tx underlying condition for chronic
- Biggest risk factor for osteoarthritis → obesity
- Tx of osteoarthritis → weight loss; if normal BMI → acetaminophen before NSAIDs
- Patient with OA taking naproxen (NSAID) + peripheral edema → increased renal retention of sodium

- Patient taking NSAID + edema; why? → NSAID decreases renal blood flow → PCT increases Na reabsorption to compensate for perceived low volume status → water follow sodium
- Tx of rheumatoid arthritis → Two-armed: symptom-relief + disease-modifying anti-rheumatic drugs (DMARDs)
- Symptom-relief for RA → NSAID first, then steroids (these do symptoms only; do not slow disease progression)
- DMARDs for early RA → always methotrexate first; if insufficient, add another DMARD (sulfasalazine or leflunomide); if insufficient add anti-TNF-alpha agent
- Methotrexate MOA → dihydrofolate reductase inhibitor
- Methotrexate side-effects → pulmonary fibrosis + hepatotoxicity + mouth ulcers (neutropenia)
- Sulfasalazine MOA → metabolized into sulfapyridine + mesalamine in the gut by bacteria
- Mesalamine is 5-ASA absorbed as the Tx for RA; only NSAID considered to be DMARD
- Leflunomide MOA → dihydroorotate dehydrogenase inhibitor (pyrimidine synthesis)
- Most specific Abs in RA → anti-CCP (cyclic citrullinated peptide), not RF (rheumatoid factor)
- X-ray of hands in RA vs OA → Only OA has DIPs involved; RA is PIPs + MCPs
- Symmetry in RA vs OA → RA is symmetrical; OA is not
- Most common presentation finding in SLE → arthritis (>90%)
- Woman 20s-40s + arthritis + thrombocytopenia → SLE
- Woman 20s-40s + arthritis + mouth ulcer + circular skin lesions → SLE
- Malar rash + low RBCs + low WBCs + low platelets; mechanism for low cell lines? → increased peripheral destruction (antibodies against hematologic cells lines seen in SLE; isolated thrombocytopenia most common)
- Tx of SLE flare → steroids
- Tx of lupus nephritis → mycophenolate mofetil
- Tx of discoid lupus → hydroxychloroquine
- Most specific Abs for SLE → anti-Smith (RNP), not anti-dsDNA
- Which Abs go up in acute SLE flares → anti-dsDNA (and C3 goes down)

- Drug-induced lupus Abs → anti-histone
- Drugs that cause DIL → Mom is HIPP → Minocycline, Hydralazine, INH, Procainamide, Penicillamine
- Viral infection + all three cell-lines are down → viral-induced aplastic anemia
- Viral-induced aplastic anemia; next best step in Dx? → bone marrow aspiration
- Viral-induced aplastic anemia; mechanism? → defective bone marrow production (contrast with SLE)
- Viral infection + low platelets → ITP (immune thrombocytopenic purpura)
- Woman 30s-40s with random bruising at different stages of healing → (also ITP; first rule out abuse)
- Mechanism of ITP → Abs against GpIIb/IIIa on platelets
- Dx of ITP → answer = low platelet count; don't choose increased bleeding time
- ITP Tx → steroids first, then IVIG, then splenectomy
- ITP episode → next best step in management → steroids
- ITP episode → most effective way to decrease recurrence → splenectomy (not first-line, but most effective)
- Family Hx of heme condition treated with splenectomy → hereditary spherocytosis (autosomal dominant)
- Bleeding time meaning? → platelet problem
- PT and aPTT meaning? → clotting factor problem
- Heme findings in ITP → increased BT, normal PT, normal aPTT
- Heme findings in hemophilia → increased aPTT; bleeding time and PT are normal
- Cause of hemophilia → X-linked recessive; hemophilia A (factor VIII def); hemophilia B (factor IX def)
- Tx of hemophilia A → desmopressin for hemophilia A (increases VIII release); then give factor VIII
- Tx of hemophilia B → give factor IX
- Classic hemophilia presentation → hemarthrosis in school-age boy; bleeding after circumcision in neonate
- Inheritance pattern of vWD → AD
- Heme findings in vWD → bleeding time always high; PT always normal; aPTT elevated half the time

- What is main function of vWF? → bridges platelet GpIb to underlying collagen (adhesion, not aggregation)
- What is secondary function of vWF → stabilizes factor VIII in plasma (that's why aPTT only half time increased)
- vWD presentation → always one platelet problem + one clotting factor problem
- Platelet problem? → epistaxis, bruising, petechiae → generally mild and cutaneous
- Clotting factor problem → menorrhagia, excessive bleeding with tooth extraction, hemarthrosis (but hemarthrosis very very rare in vWD; it is seen in hemophilia)
- vWD treatment → desmopressin → increases release of vWF
- Vitamin K deficiency heme parameters? → Increased PT + aPTT; bleeding time normal
- Cause of vitamin K deficiency in adults → chronic Abx knock out colonic flora
- Cause of sickle cell → glutamic acid to valine mutation on beta-chain
- HY drugs that cause agranulocytosis → clozapine, ganciclovir, propylthiouracil, methimazole, methotrexate, ticlopidine
- How will agranulocytosis (neutropenia) present on USMLE? → mouth ulcers + fever
- Tx for febrile neutropenia / neutropenic fever → immediate broad-spectrum IV Abx
- Broad-spectrum Abx example? → Pipericillin/tazobactam; cefepime + vancomycin
- What is ganciclovir used for? → Tx of CMV (DNA polymerase inhibitor)
- PTU and methimazole are used for what? → Tx of Graves
- Familial thyroid cancer → medullary (even if they mention nothing else related to MEN 2A/2B); apple-green birefringence on Congo red stain due to amyloid deposition; serum calcitonin high
- Calcitonin mechanism of action → inhibits osteoclast activity (**not** the opposite of PTH; in other words, doesn't put calcium back into bone; it merely caps the Ca that can resorb out of the bone)
- Most common thyroid cancer → papillary; extends lymphatogenously; has papillary structure and psammoma bodies on LM; don't worry about buzzywordy things like Orphan Annie nuclei

- Follicular carcinoma → literally just thyroid follicles on biopsy; will be a cold nodule, like any other type of thyroid cancer (for instance, if you see follicles but it's a hot nodule w/ increased uptake, that's a toxic adenoma, rather than follicular thyroid cancer); spreads hematogenously
- Hashimoto + thyroid cancer + no other histo description → thyroid lymphoma → autoimmune diseases increase the risk of non-Hodgkin lymphoma
- MEN1 → pituitary, pancreas, parathyroid (MEN1 gene; chromosome 11)
- MEN2A → parathyroid, medullary thyroid carcinoma, pheochromocytoma
- MEN2B → medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas, Marfanoid body habitus ("oid" means looks like but ain't)
- Riedel thyroiditis → fibrosis of thyroid → can extend into adjacent structures, e.g., the esophagus, and resemble anaplastic carcinoma
- 17F + painless lateral neck mass + mediastinal mass; Dx? → Hodgkin lymphoma
- 42M + painless lateral neck mass + hepatomegaly; Dx? → Hodgkin lymphoma
- Electrolyte abnormality in Cushing syndrome → hypokalemia (chronically high glucocorticoid can cause potassium wasting distally in the kidney similar to aldosterone)
- Pt has tachy + diaphoresis + diarrhea after drug → serotonin syndrome (tramadol; MOA too soon after stopping SSRI)
- Pt has tachy + diaphoresis + diarrhea + tricuspid valve lesion → carcinoid syndrome
- Cause of carcinoid syndrome → usually small bowel or appendiceal tumor that has metastasized to liver (if not metastasized, liver can process serotonin derivatives it receives); can also be due to bronchogenic carcinoid; tumors are S-100 positive and of neural crest origin
- Dx of serotonin + carcinoid syndromes → urinary 5-hydroxyindole acetic acid (5-HIAA)
- Tx of serotonin syndrome → remove offending agents + administer cyproheptadine (serotonin receptor antagonist)
- Tx of carcinoid syndrome → Tx underlying condition
- Asthma (outpatient) → albuterol (short-acting beta-2 agonist; SABA) inhaler for immediate Mx → if insufficient, start low-dose ICS (inhaled corticosteroid) preventer → if insufficient, maximize dose of

ICS preventer → if insufficient, add salmeterol inhaler (long-acting beta-2 agonist; LABA); in other words:

- **1) SABA; then**
- **2) low-dose ICS; then**
- **3) maximize dose ICS; then**
- **4) LABA.**
- That initial order is universal. Then you need to know last resort is **oral corticosteroids, however they are most effective.** In other words:
- 12M has ongoing wheezing episodes + is on albuterol inhaler; next best step? → add low-dose ICS
- 12M has ongoing wheezing episodes + is on albuterol inhaler; what's most likely to decrease recurrence → oral corticosteroids (student says "wtf? I thought you said ICS was what we do next and that oral steroids are last resort" Yeah, you're right, but they're still **most effective** at decreasing recurrence. This isn't something I'm romanticizing; this distinction is assessed on the FM NBME forms.
- After the LABA and before the oral steroids, any number of agents can be given in any order – i.e., nedocromil or cromolyn sodium, zileuton, montelukast, zafirlukast.
- MOA of nedocromil and cromolyn sodium → mast cell stabilizers
- MOA of zileuton → lipoxygenase inhibitor (enzyme that makes leukotrienes from arachidonic acid)
- MOA of the -lukasts → leukotriene LTC, D, and E4 inhibitors. LTB4 receptor agonism is unrelated and induces neutrophilic chemotaxis (LTB4, IL-8, kallikrein, platelet-activating factor, C5a, bacterial proteins)
- 16M goes snowboarding all day + takes pain reliever for sore muscles afterward + next day develops wheezing out on the slopes again; what's going on? → took **aspirin + this is Samter triad** (now cumbersomely known as aspirin-exacerbated respiratory disease [AERD]) → triad of aspirin-induced asthma + aspirin hypersensitivity + **nasal polyps**). Just to be clear, other NSAIDs can precipitate Samter triad, but the literature + USMLE will make it explicitly about aspirin.
- 16M takes aspirin + gets wheezing; what are we likely to see on physical exam? → answer on USMLE = nasal polyps.

- "Wait I don't understand. Why would aspirin cause asthma?" → arachidonic acid can be shunted down either the cyclooxygenase or lipoxygenase pathways; if you knock out COX irreversibly by giving aspirin (or reversibly with another NSAID), more arachidonic acid will be shunted down the lipoxygenase pathway → more leukotrienes → more bronchoconstriction
- Kid has Hx of AERD; physician considers agent to decrease his recurrence of Sx → zileuton, or -lukasts (both are correct; and only one will be listed).
- Kid has Hx of AERD; what agent is most likely to decrease his recurrence of Sx → **oral steroids** (sounds wrong, but once again, you need to know oral steroids are most effective for preventing asthma, period; this is exceedingly HY, especially on family medicine forms). We simply don't want to give them because of their nasty side-effects (Cushing syndrome).
- Any weird asthma Txs? → **omalizumab** → monoclonal antibody against IgE → used for intractable, severe asthma unresponsive to oral steroids + in patients who have eosinophilia + high IgE levels (I asked a pulmonologist about this drug years ago when I was in MS3 and he said he was managing 1000 patients with asthma and just three were on omalizumab).
- Acute asthma Mx (emergencies) → most important piece of info straight-up is: USMLE wants you to know that inhaled corticosteroids (ICS) have **no role in acute asthma management.** First thing we do is give oxygen (any USMLE Q that shows depressed O2 sats, answer is always O2) + nebulized albuterol (face mask with mist); **IV steroids** are then administered. The Mx algorithm is more complicated, but that is what you need for the USMLE.
- Acid-base disturbance in asthma? → respiratory alkalosis → **low O2, low CO2, high pH, normal bicarb**
- "Wait, why the low CO2? Aren't you not able to breathe?" → low CO2 is due to high respiratory rate; even if your bronchioles are constricted + filled with secretions, CO2 can diffuse really quickly; in contrast, O2 diffuses slowly and requires healthy airways; that's why with a high RR, O2 and CO2 are both low (O2 can't get in, but CO2 can still get out); 19 times out of 20 on the USMLE, if your respiratory rate is high, CO2 is low.
- "19 times out of 20? Then what's the exception." → I've seen COPD questions where the patient will have a RR of 28 but a super-high CO2, and the answer is chronic respiratory acidosis + acute respiratory acidosis (acute on chronic) → in the event of emphysema, where you literally have reduced surface area for gas exchange, even if your RR is high, CO2 has no way of diffusing out.

- "Wait, why is bicarb normal in acute asthma attack? Shouldn't it go low to compensate if CO2 is low?" → not enough time for bicarb to change; takes a minimum of 12-24 hours for renal elimination to have an effect on serum levels; this is why in altitude sickness, where CO2 is low (due to high RR bc of lower atmospheric O2), azetazolamide (carbonic anhydrase inhibitor) can be given to increase bicarb loss in the PCT of the kidney to essentially force a metabolic acidosis to compensate.
- 12M + acute asthma episode + given O2 + nebulized albuterol + IV steroids + his acid-base disturbance is as we talked about above → after 30 minutes, new values are: low O2, normal CO2, normal pH, normal bicarb; why? → he's getting tired → low O2 means he should still be hyperventilating, so for CO2 and pH to have normalized means his RR is decreasing → **answer on USMLE = intubate.** When O2 and CO2 are both initially down, that's called a type I respiratory failure; then eventually it will invert, where this patient will have a respiratory acidosis with low O2, high CO2, low pH, normal bicarb (type II respiratory failure when O2 and CO2 are the opposite).
- 12M + red urine 1-3 days after upper respiratory tract infection (URTI) → IgA nephropathy, not PSGN; can also get IgA nephropathy from GI infections
- 12M + red urine 1-2 weeks after URTI or skin infection → PSGN → can get it from Group A Strep skin infections
- 6F + red urine + abdo pain + arthralgias + violaceous lesions on buttocks + thighs; Dx? → Henoch-Schonlein purpura; red urine = IgA nephropathy → HSP is tetrad of 1) IgA nephropathy, 2) palpable purpura, 3) arthralgias, 4) abdo pain
- 13F has never had a period + has suprapubic mass + nausea + vomiting; next best step in Mx? → answer = **do beta-hCG** → she's pregnant; this is HY. Correct, girls can get pregnant without ever having had a period → must rule out
- 14F has massive unilateral breast mass + mom is freaking out bc her sister died of breast cancer → answer = follow-up in six months → **virginal breast hypertrophy** is normal during puberty
- 15M has unilateral mass behind his nipple +/- tenderness of it → answer = reassurance → physiologic gynecomastia of puberty (higher androgens are aromatized to estrogens)
- Girl is Tanner stage 3; which of the following is true? → **answer = menarche is imminent** → USMLE asks this Q straight up and it's exceedingly HY and frequent

- 17F + really pad period pain + physical exam is normal → answer = primary dysmenorrhea = prostaglandin hypersecretion (PGF2alpha) → give NSAIDs
- Tx for hypertension? → if patient has pre-diabetes, diabetes, or any cardiovascular/cerebrovascular disease of any kind → **answer = ACEi or ARB first.** These agents decrease morbidity and mortality in these patient groups. If patient has none of the above (i.e., your typical fat American middle-age male who's a little overweight but otherwise just has essential hypertension), the answer = **HCTZ or dihydropyridine CCB.** You might think that's really weird (i.e., "why not just give an ACEi or ARB anyway to anyone if they're good for morbidity/mortality?"), but the basis is: you're not going to live to 120 just because you start taking a statin when it's not indicated; well the same is true here: there's no evidence of further improvement or morbidity/mortality in pts without the above risk factors if started on ACEi or ARB). This knowledge about how to Tx HTN is HY for FM shelves in particular
- 32F + pedal + forearm edema after commencing anti-hypertensive agent; Dx? → answer = fluid retention / edema caused by dihydropyridine CCB (e.g., nifedipine) → really HY side-effect of d-CCBs!
- Side-effects of thiazides → hyperGLUC → hyperglycemia, -lipidemia, -uricemia, calcemia
- Whom should you never give thiazides to? → prediabetics or diabetics → will push people into type II DM and make current DMs worse. One of the worst/frequent pharmacologic mistreatments. Also don't give to pts with Hx of gout (bc of hyperuricemia risk)
- Diabetic pt on HCTZ for HTN → take them the fuck off the thiazide and put them on an ACEi or ARB.
- Important use of thiazide apart from HTN management in select patients → decreased risk of nephro- / ureterolithiasis (stones) because they cause hypocalciuria (and hence hypercalcemia)
- 72F + 6-month Hx of small painless papule from a chickenpox scar on her chin; Dx? → answer = Marjolin ulcer (squamous cell carcinoma) → SCC growing from previous scar or burn site

Risk factor bullet points

- 45M + BMI of 40 + Hx of hypertension (HTN) + 10-pack-yr Hx of smoking; Q asks what is most likely to be beneficial for this patient → answer = smoking cessation.
    - When in doubt, "smoking cessation" is the most common overall answer on NBME/USMLE for the #1 way to improve general health. The stem might be a long, rambling vignette where the dude is overweight, smokes, has high blood pressure, etc., and they ask broadly/vaguely what is most likely to improve mortality/morbidity → answer = smoking cessation.
- 67F + diabetic + smoker + HTN; Q asks which of the following is biggest risk factor for developing an MI in this patient → answer = diabetes.
    - Diabetes (I and II), followed by smoking, followed by HTN, in that order, are the most acceleratory (i.e., worst) risk factors for atherosclerosis. Patients with diabetes are managed as a cardiovascular disease equivalent.
    - HTN is *most common* risk factor in the population for atherosclerosis, but diabetes, followed by smoking, are the two *most acceleratory* / worst. HTN doesn't cause plaque development as fast as diabetes or smoking.
    - HTN is only most acceleratory for atherosclerosis of the *carotid arteries*, which I will discuss in detail below. *If not the carotids*, we have diabetes → smoking → HTN.
    - If Q asks about how to decrease peri- or post-operative MI risk, answer will be smoking cessation on NBME. This is presumably because smoking cessation acutely improves blood flow + oxygen utilization at myocardium.
- 67F + smoker + HTN + high BMI; Q asks biggest risk factor for developing an MI in this patient → answer = smoking. As we said above, the order of importance for atherosclerosis leading to MI is diabetes → smoking → HTN. This is asked on one of the NBME exams, where it's a 1-2-liner Q followed by smoking being correct over HTN.
- 60F + diabetic + HTN + smoker + about to undergo hip surgery; Q asks for the best way to reduce perioperative MI risk → answer = smoking cessation. As mentioned above, smoking cessation is #1 way to reduce peri- and post-operative MI risk, since acute smoking cessation improves myocardial oxygenation and coronary autoregulation.

- 65F + 80-pack-year Hx of smoking + CXR shows hyperinflation + loud P2 on auscultation + JVD + ECG shows right-axis deviation + 2-year Hx of type II diabetes; Q asks what is most likely to reduce MI risk in this patient → answer = smoking cessation. Inflated lung fields mean COPD. Loud P2 means pulmonary hypertension (pulmonic valve slams shut due to high distal pressure). JVD means impaired right-heart filling. Right-axis deviation means right ventricular hypertrophy. You need to avoid being intransigently rigid when approaching questions. If the vignette overwhelmingly emphasizes one presentation over another (i.e., obvious cor pulmonale versus mere peripheral mention of recent diabetes), you need to be able to reason that the Q wants a particular answer (i.e., smoking here over diabetes).
- 60F + Hx of MI 5 years ago + diabetic + smoker + HTN; Q asks number-one risk factor for an MI occurring in this patient → answer = "Hx of myocardial infarction." Not complicated. Akin to psych questions that want you to know biggest risk factor for suicide is Hx of previous suicide attempt.
- Question gives 40s male with duodenal ulcers caused by *H. pylori*. Following antibiotic treatment, they ask what lifestyle variable is most likely to promote healing of ulcers → answer = smoking cessation. Reducing alcohol intake can also help, but I've seen smoking cessation as answer on NBME, where they don't have abstinence from alcohol as an answer.
- Patient has autoimmune disease (i.e., SLE, RA, etc.); question wants to know how to decrease recurrence of flares → answer = smoking cessation.
- 4F + recurrent otitis media + parents smoke but only outside; Q asks #1 way to decrease recurrence of otitis media in this patient (answers are all lifestyle/household variables); answer = parental smoking cessation; even though "only outside," second-hand smoke is important cause of recurrent upper respiratory tract/ear infections and sudden infant death syndrome on USMLE.
- 11F + rhinoconjunctivitis past month; Q asks what to ask parents about → answer = "recent pets in household"; if recent allergy-like presentation in a patient with no prior Hx, inquire about pets.
- 55M + gangrene of the fingers + HTN + drinks alcohol and smokes; Q wants to know #1 way to improve this patient's condition → answer = smoking cessation; diagnosis is Buerger disease (thromboangiitis obliterans), which is idiopathic digital gangrene, generally in middle-age men who are heavy smokers.

- 66M + experiences stroke + has Hx of hypertension and smoking; Q asks #1 way to decrease risk of recurrent stroke → **answer = lisinopril; smoking cessation is wrong answer.** This vignette is exceedingly HY, particularly for 2CK. You need to know hypertension eclipses smoking as the bigger risk factor for stroke. The order of risk factors for stroke on NBME/USMLE is: atrial fibrillation (AF) → HTN → smoking.
  - The answer on USMLE for reducing stroke risk in a hypertensive patient without AF will often just be "lisinopril," rather than "management of hypertension." Lisinopril, to my observation, is a favorite drug on NBME forms. Patients who have stroke risk need to be on an ACEi or ARB.
  - High blood pressure leads to a strong systolic impulse pounding the carotids → endothelial damage → atheromata development (carotid stenosis due to atherosclerosis) → plaque launches off to the brain/eye causing stroke, TIA, and retinal artery occlusion.
  - Should be noted that of course smoking cessation should be implemented in patients at risk for stroke, but HTN eclipses smoking for risk factor importance.
  - Patients who have diabetes and/or are smokers who don't have HTN will absolutely develop diffuse atherosclerosis, but vessels such as the abdominal aorta, coronaries, and popliteals classically develop plaques first. It is specifically hypertension that affects the *carotids*.
- 66M + experiences retinal artery occlusion + has Hx of hypertension and smoking; last line of the Q says "carotid duplex ultrasonography shows 90% occlusion"; Q asks what's the best way to decrease recurrence of stroke → answer = carotid endarterectomy; lisinopril is wrong answer (offline NBME 8 for 2CK). Student says, "Wait, didn't you just say lisinopril is the answer they want for reducing risk of stroke → You're right. But if they tell you specifically that the patient has high degree of occlusion on carotid ultrasound, then answer = carotid endarterectomy over lisinopril.
  - When to do carotid endarterectomy: symptomatic carotid stenosis >70% (i.e., Hx of stroke, TIA, or retinal artery occlusion) or asymptomatic carotid stenosis >80% (i.e., no Hx of stroke, TIA, or retinal artery occlusion; mere presence of a carotid bruit does ***not*** refer to symptoms).
  - Students get pedantic about the above %s, but in reality, the NBME won't make the degree of occlusion borderline. They'll either say something like 90% or 30%.

- If patient is under endarterectomy % thresholds, the treatment = the triad of 1) antiplatelet therapy (usually aspirin alone is answer on NBME, but can be combo of aspirin + dipyridamole, OR clopidogrel alone); 2) statin; 3) ACEi or ARB.

- 65M + S4 heart sound + paradoxical splitting of S2 + left bundle branch block + lateralized apical impulse + ECG shows left-axis deviation + BP is 160/95 + smoker past 20 years; Q asks what is most likely to have prevented this patient's findings? → answer = "management of hypertension"; smoking cessation is wrong answer. **Before you freak out,** I'll unpack the above findings:
  - S4 heart sounds on USMLE = stiff left ventricle from high afterload (usually systemic hypertension, but can be aortic stenosis). When you see S4, you should be thinking "ok that's diastolic dysfunction and a stiff LV from some form of afterload, most likely AS or HTN."
  - Paradoxical splitting of S2 (when A2 occurs after P2), LBBB, and left-axis deviation on ECG all = left ventricular hypertrophy. That's it. You don't need to worry about mechanisms or anything more.
  - So the above vignette, if we summarize it, basically screams hypertensive heart disease. The patient has LVH due to HTN. The smoking is bad, yes, but the focus of the vignette is clearly the HTN. This is a 2CK-level Q. For those of you studying for Step 1 confused about cardio stuff, my HY Cardio PDF discusses these things in more detail.
- 79M + irregularly irregular rhythm on ECG + history of HTN; Q asks most important risk factor for stroke in this patient → answer = atrial fibrillation, not hypertension. Once again, the order for stroke risk on NBME/USMLE is atrial fibrillation → hypertension → smoking. If they mention any of these in the same patient, AF eclipses HTN, which eclipses smoking. This is on NBMEs, not my opinion.
  - AF can cause left atrial mural thrombi that launch off to the brain/eye; HTN causes carotid atheromatous plaques that launch off.
  - AF is most common in patients over 75, or occasionally in select patient groups such as those with hyperthyroidism.
- 79M + atrial fibrillation + HTN; Q wants to know best way to decrease stroke risk in this patient → answer = warfarin; aspirin is wrong answer; USMLE wants you to know CHADS2 score (there are variations, but this simple one suffices).
  - Each component is one point → Congestive heart failure; Hypertension, Age >75, Diabetes.

  - Stroke or TIA is 2 points.
  - If patient has Afib and 0 or 1 points from CHADS2 score, give aspirin; if 2+ points, give warfarin.
  - NBME doesn't really assess the non-warfarin NOACs. The exam-writers seem to be pretty old-school and just stick with warfarin, likely because it avoids ambiguity for NOAC use cases.
- 45M + HTN + smoker + BMI of 30; Q wants to know best way to decrease BP in this patient; answer = weight loss; smoking cessation is wrong answer; HY you are aware that weight loss is more important than smoking cessation to decrease BP. Yes, smoking cessation helps, but weight loss is the most effective lifestyle modification.
  - Since HTN is biggest risk factor for stroke (in patients without Afib), and the best way to decrease BP is weight loss, this means weight reduction is crucial for decreasing stroke risk.
- 45M + big paragraph vignette that mentions Hx of prosthetic valve and venous disease; Q asks most important indication for anticoagulation in this patient (in other words, what's biggest risk factor for a clot) → answer = prosthetic valve. USMLE wants prosthetic valve as most important indication for using warfarin. The other two HY indications are post-heparin for DVT and elevated CHADS2 for AF. "Venous disease" does not refer to DVT; venous disease simply means valvular insufficiency causing brawny edema/pigmentation of lower extremities, or sometimes just varicose veins; venous disease is of course an important risk factor for an eventual DVT. USMLE will not give prosthetic valve and DVT for the same Q (some students proceed to ask annoying what-ifs like that).
- 53F + cirrhosis + lung cancer + severe abdominal pain requiring surgery + laparotomy shows purple, necrosed bowel; diagnosis + risk factors? → Dx is mesenteric venous thrombosis; risk factors are both the cirrhosis (portal vein entering liver is formed from superior mesenteric vein and splenic vein, so cirrhosis means increased pressure/stasis in any of these veins), as well as hypercoagulable state due to malignancy. You should know that cancer/malignancy is important risk factor for clots/DVT, and even marantic endocarditis for that matter.
- 53M + cirrhosis + blood in stool + no pain; Dx + risk factor? → internal hemorrhoids (painless; external are painful); risk factor is increased venous pressure in superior rectal veins from cirrhosis (for external hemorrhoids choose inferior rectal veins); other important risk factor is third trimester of pregnancy.

- 53M + cirrhosis + vomiting copious blood; Dx + risk factor? → ruptured esophageal varices; risk factor is increased pressure in left gastric vein due to cirrhosis (esophageal veins feed into left gastric, which feeds into portal). Mallory-Weiss tears due to alcoholism with retching/vomiting often present with "just a little blood"; classically, varices = lots of blood. This isn't a 100% rule, but it's usually the case that varices = lots of blood; MW tear = little blood.
- 50M + undergoing dental procedure; Q asks about risk for endocarditis and if prophylaxis is indicated → answer on USMLE is almost always do **not** give prophylaxis. Only indications are the presence of prosthetic material in the heart or incompletely/unrepaired cyanotic congenital heart disease. The mere presence of a mitral valve prolapse or small ASD/VSD that never became cyanotic is **not** an indication for prophylaxis. Hx of prior endocarditis is other obvious indication.
- 45F + HTN + high LDL and TGAs + low HDL + BMI of 23; Q asks most important modification in this patient → answer = "exercise program"; weight loss program is wrong answer (makes sense, since BMI is normal); smoking not listed as answer (if it were, it would be correct answer); this Q shows up on 2CK FM form.
- 60M + HTN + LV ejection fraction of 35%; Q asks which drug would improve mortality the most → answer = lisinopril; ACEi or ARB is first-line for heart failure, as well as for HTN in patients who have diabetes, pre-diabetes, atherosclerotic disease, proteinuria, and/or elevated creatinine or renin. This is not some arbitrary list; this is exceedingly HY for 2CK. If patient has HTN but does not bear any of the aforementioned scenarios, first-line for HTN is a dihydropyridine CCB (e.g., nifedipine) or HCTZ. It should also be noted that HTN in diabetes is >130/80, whereas everyone else it's >140/90.
- 48M + BP 150/90 on multiple visits + normal BMI + non-diabetic; Q wants to know the best therapy for this patient → answer = hydrochlorothiazide; answer can also be a dCCB. Vignette doesn't mention diabetes, pre-diabetes, atherosclerotic disease, or proteinuria (otherwise answer would be ACEi or ARB); if patient is non-diabetic with HTN and has mere elevation in renin or creatinine, I have not seen the NBME force a specific drug here, but I would choose the ACEi or ARB unless the patient has renal artery stenosis or fibromuscular dysplasia (don't give ACEi or ARB in the latter patients).
- 38M + diabetic + no protein in the urine + LDL 95 mg/dL + BP 135/80 + HbA1c of 7%; Q asks what is most likely to decrease morbidity in this patient → answer = lisinopril; as mentioned above, HTN in

diabetes is >130/80; first-line is ACEi or ARB. LDL should be <70-100 mg/dL (but on NBME, if <100 a statin isn't indicated); HbA1c should be in the 7s or below.

- 38M + diabetic + mild proteinuria + LDL 95 mg/dL + BP 120/80 + HbA1c of 7%; Q asks what is most likely to decrease morbidity in this patient? → answer = lisinopril; once again, start an ACEi or ARB in a diabetic if there is 1) any proteinuria; 2) any elevation in creatinine or renin; or 3) BP >130/80.
- 38M + diabetic + no proteinuria + LDL 110 mg/dL + BP 120/80 + HbA1c of 7%; Q asks what is most likely to decrease morbidity in this patient? → answer = statin. LDL should be under 100 mg/dL in diabetics on NBME (the literature says <70-100 depending on the source, but I've seen NBME Qs where 95 mg/dL LDL is written in vignette and statin isn't indicated).
- 38M + no proteinuria + LDL 95 mg/dL + BP 120/80 + HbA1c of 10%; Q asks what is most likely to decrease morbidity in this patient? → answer = commence metformin; patient clearly has diabetes with an HbA1c >7s.
    - Diagnosis of diabetes is any of the following: HbA1c >6.5%; two fasting glucose measurements 126 mg/dL or greater; any one random glucose 200 mg/dL or greater.
    - Patients can have normal glucose levels but high HbA1c and therefore be diabetic.
    - It's to my observation on NBMEs that if the Q wants glycemic control (e.g., "commence metformin") as the answer, they will give HbA1c as >9%, even though technically any elevation in HbA1c could warrant its commencement.
    - The USMLE Q will often give HbA1c in the 7s as tolerable in a diabetic, where they will make the focus of the Q something else, e.g., LDL >100 (answer = statin) or proteinuria (answer = lisinopril).
- 50M + diabetic + high LDL + low HDL + high BMI; Q asks best way to decrease long-term complications in this patient → answer = "good glycemic control"; generic answer, similar to smoking cessation, but this is HY.
- 50M + diabetic on metformin and glyburide + HbA1c is 12% + bicarb of 20; Q asks #1 way to improve glycemic control → answer = "switch from metformin and glyburide to intermediate-acting insulin."
    - This above Q is on an FM form and is one of the most frequently asked-about Qs I know of. Students lose the forest for the trees, thinking the USMLE gives a fuck about hyper-specific Tx regimens. They don't. Notice the patient has low bicarb (NR 22-28). So you know right

away metformin needs to be discontinued (the Q only gives two answers where metformin is discontinued).

- If patient is already on glyburide and HbA1c is very high, the assumption is pancreatic burnout has already occurred (patient needs functioning β-cells for sulfonylureas to work), so increasing dose of glyburide won't work. The remaining answer is just switching to insulin.
- Since metformin causes lactic acidosis, it is important you know to stop / don't commence it if patient has low bicarb or creatinine 1.5 or greater (renal insufficiency can increase risk for lactic acidosis due to metformin); there is an NBME Q where patient has creatinine of 1.4 mg/dL and HbA1c of 12.5% and answer is commence metformin. Although normal creatinine is 0.7-1.2 mg/dL, I've observed across NBME Qs that a max of 1.3-1.4 can be considered "normal" if very old (mid-80s+) or urgently requiring glycemic regulation.

- 50M + on metformin + received IV contrast + now has creatinine of 1.5 mg/dL; Q wants to know what would have prevented this condition → answer = intravenous hydration; Dx is contrast nephropathy; metformin does not cause high creatinine; it is merely the case that if a patient has high creatinine, then we discontinue metformin due to lactic acidosis risk.
- 50M + heavy smoker + Hx of HTN + BMI of 30; Q wants to know best way to decrease diabetes risk in this patient → answer = weight loss; smoking cessation is wrong answer. For development of type II diabetes, maintaining a normal BMI is most crucial. The answer can also sometimes be "low calorie diet" for these Qs; "low carbohydrate diet" is wrong answer.
- 65M + long history of alcoholism and smoking + pulsatile mass in epigastrium; Q wants to know #1 way to have prevented this patient's condition → answer = smoking cessation; the demographic that classically gets AAA is: elderly males who are ever-smokers; alcohol is not salient risk factor; diabetes is actually slightly *protective* of AAA due to the stiffening effect of glycosylation on endothelium.
  - Guidelines vary on screening for AAA, but updated literature says both men and women 65-75 who have ever-smoked should get a one-off abdominal ultrasound to screen for AAA. Lower thresholds can be used in patients with family history.
- 50F + subarachnoid hemorrhage; Q asks biggest risk factor → answer = HTN most common overall in general population; otherwise, if patient has ADPKS or Ehlers-Danlos, those would be more specific.

- 82F + wobbly gait for 4 months + head CT shows crescent-shaped bleed; what is biggest risk factor? → diagnosis is subdural hematoma; often slow-accumulating; elderly have reduced cerebral volume, making superior cerebral (bridging) veins more susceptible to friability; other notable risk factors are alcoholism, dementia (further reduction in brain mass), deceleration injury (car accident or shaken-baby syndrome). There is no lucid interval.
- 14M + hits head while skateboarding + loses consciousness briefly + head CT shows biconvex-shaped bleed; what is major risk factor → answer = head temporal trauma resulting in rupture of middle meningeal artery.
- 81M + Alzheimer + recurrent hemorrhagic stroke; what is major risk factor for the bleeds? → Alzheimer → can lead to amyloid angiopathy, causing intracerebral/intraparenchymal hemorrhage that presents as recurrent hemorrhagic stroke.
- 50F + has diabetic foot ulcer; Q asks how this condition could have best been prevented → answer = wearing comfortable-fitting shoes; wrong answer is nightly foot checks. This is asked on 2CK NBME. Both are important to prevent foot ulcers, but if both choices are listed together, choose wearing comfortable-fitting shoes.
- 62F + smoker + HTN + pain in legs with walking + diminished peripheral pulses + punched-out ulcer on bottom of foot + ankle-brachial indices are reduced; Q asks, in addition to smoking cessation and BP control, which of the following is indicated for this patient → answer = recommend an exercise/walking program; diagnosis is peripheral arterial disease (PAD).
    - Since PAD is caused by atherosclerosis, we'd have the same risk factors as before – i.e., diabetes > smoking > HTN, with the caveat being that if the vignette hyper-emphasizes / pushes one risk factor onto you (e.g., they give you a 12-line paragraph where 10 lines are just about HTN), then obviously choose the emphasized factor.
    - HY first step in diagnosis is measuring ABIs (<0.9 is diagnostic).
    - After low ABIs are measured, the next step, if listed, is doing an exercise stress test to determine exercise tolerance.
    - If exercise stress test not listed, answer = recommend walking/exercise program. Do not choose cilostazol first. This is a wrong answer almost always.

- 51M + brawny edema and hyperpigmentation of lower legs + large, sloughy ulcer at left medial malleolus + peripheral pulses strong + ultrasound confirms diagnosis; Q asks next step in management → answer = compression stockings; diagnosis is venous disease.
    - Biggest risk factor for venous disease is valvular insufficiency (often idiopathic/familial).
    - USMLE wants duplex ultrasonography of the legs to diagnose, followed by compression stockings as next best step.
    - If patient has active superficial thrombophlebitis (i.e., painful, palpable cord at the ankle), choose subcutaneous enoxaparin, not compression stockings as next best step. This must be treated with heparin, same as DVT. This is easy to get wrong, since compression stockings are the answer almost always for venous disease.
    - Strong peripheral pulses tells you it's not arterial disease (hence in this case, venous disease).
- 50M + being treated for Hodgkin + develops cardiomegaly + bilateral crackles; what is most likely risk factor for this acute presentation → doxorubicin/Adriamycin causing dilated cardiomyopathy.
- 50M + being treated for Hodgkin + develops progressive dry cough + CXR shows reticulonodular pattern; what is most likely risk factor for this acute presentation → bleomycin causing pulmonary fibrosis; reticular, or reticulonodular, or bilateral granular patterning = "honeycombing" = HY for pulmonary fibrosis.
- 50F + being treated for breast cancer + develops cardiomegaly; Q asks biggest risk factor → answer = trastuzumab causing cardiotoxicity.
- 44M + worked in shipyard 20 years ago part-time + 6-month history of shortness of breath; CXR shows supra-diaphragmatic soft tissue densities; Q wants diagnosis → answer = asbestosis; risk factor is working in construction involving insulation, or shipyards; pleural and supra-diaphragmatic plaques are HY for asbestosis; mesothelioma will present as grossly white cancer that circumferentially envelopes the lungs.
- 66M + worked in aerospace industry for 30 years + 1-year Hx of declining pulmonary function; biopsy of lung parenchyma shows non-caseating granulomas; Q asks diagnosis → answer = berylliosis; risk factor is working in aeronautical industry; this is one of the three HY lung conditions causing granulomas (sarcoidosis + berylliosis are both non-caseating; TB is caseating).

- 32F + smoked for 5 years + loud P2 heart sound + JVD + bullous changes seen on CXR + father died of alcoholic cirrhosis; Q wants risk factor for this patient's condition → answer = "deficiency of neutrophil elastase"; diagnosis is cor pulmonale due to COPD from α1-antitrypsin deficiency; latter leads to pan-acinar emphysema (bullous changes on CXR are buzzy/HY for emphysema); can also cause cirrhosis; vignettes can absolutely mention smoking or alcohol Hx, where patients are at increased/accelerated risk for emphysema and cirrhosis (i.e., 5 years is too fast typically for COPD to develop). The loud P2 means pulmonary hypertension. JVD means we have right heart failure. Since the etiology of the right heart failure is pulmonary (i.e., not left heart), we call it cor pulmonale.
- 42F + long-standing history of rheumatoid arthritis managed with prednisone + undergoes appendectomy + experiences drop of BP to 80/40 during surgery; risk factor for this precipitous drop in BP? → chronic prednisone use causing adrenal suppression → can't mount stress response in setting of stressor (i.e., surgery, trauma, infection) → can cause BP to fall precipitously → after IV fluids, give hydrocortisone. Normally, cortisol (i.e., glucocorticoid) from the adrenals helps maintain BP by upregulating alpha-1 receptors on arterioles; NE and E (catecholamines) agonize alpha-1 to maintain BP; this is called permissive effects of glucocorticoids on catecholamines. Patients with suppressed adrenals have insufficient cortisol after exogenous prednisone is consumed in setting of stressor → insufficient alpha-1 expression on arterioles → drop in BP; hydrocortisone is Tx because without the glucocorticoid, NE and E can't do their job.
- 42F + long-standing rheumatoid arthritis managed with multiple medications + 6-month Hx of dry cough + CXR shows reticular pattern; what's the risk factor for this condition? → answer = methotrexate-induced pulmonary fibrosis + rheumatoid lung (the autoimmune disease itself, independent of methotrexate, can cause pulmonary fibrosis, so patients will have restrictive lung disease due to both methotrexate and the RA).
- 39F + rheumatoid arthritis + low RBCs, WBCs and platelets + splenomegaly; Q wants risk factor for this condition → answer = rheumatoid arthritis; diagnosis is Felty syndrome (RA + neutropenia + splenomegaly); sometimes RBCs and platelets can be down as well.
- 50F + BMI 40 + smoker + HTN; Q asks #1 way to decrease risk of osteoarthritis in this patient → answer = weight loss (exceedingly HY).

- 62F + has peripheral edema + using high doses of naproxen for past 6 weeks to manage her osteoarthritis; Q wants to know #1 way to minimize risk for this patient's condition → answer = avoidance of NSAIDs; patient has fluid retention due to NSAIDs (decreased renal blood flow causes kidney to reabsorb fluid to compensate).
- 50M + 6-month history of declining mental function + microcytic anemia + drinks home-distilled liquor + smoker; Q wants to know biggest risk factor for this patient's condition → answer = home-distilled alcohol; diagnosis is lead poisoning; in this case, you need to be aware old-school home distillation equipment is a cause of lead poisoning, which can cause microcytic anemia.
- 50M + swollen big toe + smoker + works on farm in California + drinks home-distilled liquor; Q wants to know biggest risk factor for this patient's condition → answer = home-distilled liquor; patient has podagra (gout of big toe) from alcohol; the fact that it's home-distilled doesn't mean lead poisoning in this context, but this is on NBME. Smoking doesn't cause gout; farming can be risk factor for organophosphate poisoning.
- 70M + high creatinine + taking amitriptyline; Q wants to know best way to prevent this patient's condition → answer = avoidance of anticholinergic medications; patient has high creatinine due to post-renal obstruction → BPH in the setting of a TCA, which is anticholinergic (urinary retention).
    - In elderly, be very cautious administering TCAs, anti-psychotics, or 1st-generation H1-blockers (i.e., diphenhydramine). They can cause delirium, worsening of dementia, and in old men in particular, exacerbation of BPH.
- 23F + works on fruit farm + doesn't wear mask + doesn't wear gloves + develops pinpoint pupils and salivation; Q wants to know what would most likely have prevented this patient's condition → answer = wearing gloves; wrong answer = face mask; organophosphate poisoning occurs through the skin, not inhalation (on NBME).
- 16F + consumed pills + now has tinnitus + low bicarb, low CO2, and high anion-gap; risk factor? → aspirin (salicylate) causing high anion-gap mixed metabolic acidosis / respiratory alkalosis.
- 25M + uses a variety of illicit drugs + develops Parkinsonism; risk factor? → MPTP use (synthetic heroin) can cause Parkinsonism.
- 16F + consumed bottle of pills in suicide attempt + now has severe elevation in ALT and AST; what is risk factor for this presentation → acetaminophen toxicity; antidote = *N*-acetylcysteine.

- 55M + high BMI + high blood lipids + does not drink or smoke + ALT and AST very slightly elevated; Q asks #1 way to have prevented this condition → answer = weight loss; non-alcoholic steatohepatitis (NASH) occurs in patients with metabolic syndrome / high BMI → it's assumed most chronically overweight patients will have some degree of fatty liver.
    - Patients will start out with **no change** in their hepatic enzymes; this is very important.
    - There are a couple of NBME Qs for 2CK where they give patient with metabolic syndrome and normal liver enzymes, and then the answer is just NASH (you can eliminate other answers as obviously wrong); student will ask how this makes sense if labs are all normal; my response is, once again, that labs start out normal in NASH, but it assumed that most people with obesity have some degree of ensuing steatosis.
- 55M + drinks 2 beers a day + smokes a pack a day + all laboratory findings normal except slight elevation in both AST and ALT (AST slightly higher than ALT); Q asks what is most likely to have prevented the findings in this patient → answer = cessation of alcohol; students should know smoking isn't a significant risk factor for hepatic disease; alcoholic liver disease need not present with significant alcohol intake or with prominent elevations in transaminases where AST/ALT is 2:1.
- 55M + obese + gurgling sounds when swallowing + sometimes regurgitates undigested food + halitosis; Dx + risk factor? → Zenker; risk factor is obesity; can also be seen in setting of dysphagia (e.g., from stroke Hx) due to increased oropharyngeal pressure → increased risk of outpouching.
- 28F + spoon-shaped nails + chapped lips + trouble swallowing; Dx + risk factor? → Plummer-Vinson syndrome; risk factor is iron deficiency anemia. PV syndrome is triad of 1) iron deficiency anemia (can present as koilonychia), 2) angular cheilitis, and 3) esophageal webs (dysphagia).
- 28F + eating clay, starch, and ice; Dx + risk factor? → pica due to iron deficiency.
- 40M + dysphagia to solids and liquids + barium swallow shows birds beak appearance; Dx + risk factor? → achalasia; loss of myenteric plexus; often idiopathic; can be due to *Trypanosoma cruzi* (Chagas disease) as risk factor; latter can also cause dilated cardiomyopathy.
- 40M + Hx of heavy smoking/alcohol + new-onset dysphagia; Dx + risk factor? → squamous cell carcinoma of esophagus; heavy smoking/alcohol are major risk factors; do immediate endoscopy.
- 40M + Hx of GERD + new-onset dysphagia; Dx + risk factor? → adenocarcinoma of esophagus; GERD / Barrett is major risk factor; do immediate endoscopy.

  - In short, do immediate endoscopy on USMLE to look for esophageal cancer in any patient with new-onset dysphagia who has Hx of GERD or heavy smoking/alcohol.
  - Progression of dysphagia from solids only to solids + liquids is HY for cancer.
  - Dysphagia to solids + liquids from the start suggests neurogenic etiology (achalasia).
- 45M + abdo pain an hour after meals + no other PMHx; Dx + risk factor? → duodenal ulcer due to *H. pylori*; can also be gastrinoma. If they give you no information apart from duodenal ulcer presentation, *H. pylori* is answer. Gastrinoma is patient who has recurrent duodenal ulcers despite *H. pylori* Tx. Gastrinoma can also be part of MEN 1.
- 67M + abdo pain an hour after meals + Hx of CABG four years ago + intermittent claudication; Dx + risk factor? → chronic mesenteric ischemia due to atherosclerosis of SMA/IMA; don't confuse with duodenal ulcers; will sound like duodenal ulcer except will be an older patient with extensive cardiovascular history. The abdo pain is angina of the bowel (↑ oxygen demand during digestion).
- 65M + long-standing diabetes + diarrhea; Dx + risk factor? → diabetic neuropathy to bowel affecting hypogastric nerves (loss of anti-peristalsis sympathetic nerves → causes too much peristalsis).
- 65M + long-standing diabetes + constipation; Dx + risk factor? → diabetic neuropathy to bowel affecting pelvic splanchnic nerves (loss of pro-peristalsis parasympathetic nerves).
- 65M + long-standing diabetes + new-onset GERD; Dx + risk factor? → diabetic gastroparesis; do endoscopy first to rule out physical obstruction, followed by gastric-emptying scintigraphy to diagnose; can give metoclopramide and/or erythromycin (motilin receptor agonist) to Tx.
- 27F + exophthalmos + tachycardia + diarrhea; Dx + risk factor for GIT symptoms? → "motility disorder" due to hyperthyroid state (in this case, Graves); "motility disorder" is also answer for hypothyroidism causing constipation, as well as for the diabetes conditions above.
- 44M + alcoholic + low calcium; following administration of calcium, serum levels do not rise appreciably; what is risk factor/cause → alcoholism causing hypomagnesemia; latter can cause hypocalcemia and/or hypokalemia refractory to supplementation.
- 23M + lost in woods for 3 weeks without food + given big meal following rescue + develops arrhythmia; Q wants Dx + risk factor? → refeeding syndrome; presents as hypophosphatemia causing arrhythmia; patient can have normal BMI; being given food after a long time without can cause a surge production of glycolytic intermediates that trap phosphate within the liver.

- 27F + painful thyroid + low uptake of radioiodine on scan; Dx + risk factor? → subacute granulomatous thyroiditis (aka deQuervain); risk factor is viral infection; answer on USMLE for tender/painful thyroid.
- 27F + gave birth 4 weeks ago + non-tender thyroid + hypothyroid state; Dx + risk factor? → post-partum thyroiditis; risk factor is recent pregnancy (makes sense).
- 55F + started on new medication regimen + now has hypothyroidism; Dx + risk factor? → drug-induced thyroiditis; lithium and amiodarone are HY causes.
- 55F + started on new medication regimen + now has mouth ulcers; Dx + risk factor? → drug-induced neutropenia (agranulocytosis); HY drugs are clozapine, methimazole, PTU, ganciclovir, methotrexate.
- 44M + ataxia + confusion + ophthalmoplegia + confabulations; what is risk factor for this condition → alcoholism causing thiamine deficiency, resulting in Wernicke-Korsakoff syndrome (WKS). ACOW → Ataxia, Confusion, Ophthalmoplegia = Wernicke. If we add confabulations (Korsakoff psychosis) on top of ACOW, we now call it WKS.
- 60M + had gastrectomy performed 6 months ago + now has neuropathy; Dx + risk factor? → B12 deficiency due to gastrectomy (loss of parietal cells which produce intrinsic factor).
- 60M + had gastrectomy performed 6 months ago + now has neuropathy and confusion; Dx + risk factor → thiamine (B1) deficiency due to gastrectomy; if confusion is present, this is suggestive of Wernicke; the neuropathy can be dry beri beri. Absolutely asinine question, but asked twice on 2CK NBME material. In other words, just be vigilant for B1 deficiency after gastrectomy, not just B12.
- 24F + vegan + high MCV + hyper-segmented neutrophil on smear; Dx + risk factor? → B12 deficiency due to veganism (can also be strict vegetarianism).
- 40F + moving around legs in bed aimlessly at night; Dx + risk factor? → restless leg syndrome; risk factor is iron deficiency anemia (always check iron and ferritin first); otherwise, patient is at risk for developing Parkinson disease later; Tx with gabapentin or D2 agonist (e.g., pramipexole / ropinirole).
- 44M + fever 103F + diffuse abdominal pain with fluid wave; Dx + risk factor? → spontaneous bacterial peritonitis (SBP); major risk factors are cirrhosis, recent peritoneal dialysis, or nephrotic syndrome (any cause of ascites, but those are the ones I've seen on NBME forms); next best step is paracentesis; in terms of what we look for, choose "white cell count and differential" before "gram stain and culture of the fluid" if both are listed. That sequence is assessed on 2CK NBMEs.

- 36F + new-onset shortness of breath and tachycardia; patient has 5-pack-year Hx of smoking + recently prescribed combined oral contraceptive pill; Q asks which of the following would have prevented this patient's acute presentation → answer = avoidance of combined contraceptives; diagnosis is pulmonary embolism; combined contraceptives (i.e., containing estrogen in addition to progesterone in order to minimize breakthrough bleeding) are contraindicated in women 35+ who are smokers. They are also contraindicated in women who have migraines with aura, active breast cancer, or Hx of thromboembolic disease. This is because estrogen-containing contraception (and HRT) increases risk of thromboembolic events (estrogen upregulates fibrinogen and factors V and VIII).
- 55F + perimenopausal + family Hx of osteoporosis + severe vasomotor symptoms; Q wants to know which aspect of patient's history makes HRT a consideration; answer = severe vasomotor symptoms (i.e., hot flashes, vaginal dryness, urge incontinence); this is the only approved indication for HRT; preservation of bone density is not an indication; this is because the increased risk of breast cancer and thromboemboli (i.e., DVT, stroke, MI) negatively outweigh the bone-preserving effects in most circumstances.
- 55F + perimenopausal + severe hot flashes + started on HRT a few months ago + stopped taking the progesterone component because of effects on her moods; Q asks what is most likely to be seen in this patient? → answer = endometrial hyperplasia; unopposed estrogen is risk factor for endometrial hyperplasia, which in turn can lead to endometrial cancer (↑ cell proliferation = ↑ risk of mutation).
- 55F + started taking HRT three months ago + has appearance of new breast cyst + mammogram prior to HRT showed no such cyst; Q asks next best step in management; answer = biopsy of the cyst; HRT is risk factor for breast cancer, even when it is combined with progesterone (the only role progesterone has as part of HRT is to prevent endometrial cancer); 2CK CMS form has vignette of new-onset simple (i.e., clear; hypoechoic) cyst following HRT where biopsy is answer.
- 50F + BMI of 30 + irregular menses for many years + new-onset heavier vaginal bleeding; Q asks what is most likely risk factor for this patient's condition → answer = anovulation; diagnosis is endometrial cancer due to history of anovulation/PCOS. Mechanism: high BMI → insulin resistance → abnormal GnRH pulsation → high LH/FSH ratio → high LH over-stimulates theca interna cells (hirsutism); low FSH leads to poor follicular development (anovulation / retention of follicles as cysts) → lack of

ovulation means no formation of corpus luteum → since corpus luteum normally secretes progesterone, there is no progesterone production → unopposed estrogen → endometrial hyperplasia and increased risk of endometrial cancer (progesterone normally limits growth of endometrium; estrogen stimulates it).

- 65F + started on tamoxifen as part of her breast cancer chemotherapy regimen; Q asks what aspect of her history makes her best candidate for this treatment → answer = history of hysterectomy; tamoxifen is risk factor for endometrial cancer due to its partial agonist effects at endometrium. Raloxifene is the classic SERM used in most patients who still have a uterus, as it is not a partial agonist at endometrium; both agents are antagonists at breast and partial agonists at bone.
- 16F + concerned about risk for ovarian cancer because her great aunt was diagnosed with it; there is no family Hx of BRCA mutations; Q asks next best step in management for this patient → answer = oral contraceptive pills (any type); random factoid you need to know is that OCPs decrease risk of ovarian cancer by ~50%, presumably due to synchronization of cycles. In summary regarding OCPs in relation to cancer risk: ↓↓ ovarian cancer (~50%); ↓ endometrial cancer; ↑ cervical cancer (not direct effect; cause is ↑ HPV exposure due to ↓ condom use); no change breast cancer (some studies have suggested a possible increased risk, but there is no significance across meta-analyses).
- 28F + pregnant at 35 weeks' gestation + cocaine use + painful contractions of abdomen + vaginal bleeding; Dx + risk factors? → abruptio placentae; risk factors are cocaine use and deceleration injury (i.e., fall or car accident). Presents as painful third-trimester bleeding.
- 28F + pregnant at 35 weeks' gestation + painless vaginal bleeding; Dx + risk factors? → placenta previa; notable risk factor is previous C-section (on Obgyn CMS form); implication is, if lining of uterus has been disturbed, it makes sense that there would be future risk of abnormal implantation. Presents as painless third-trimester bleeding.
- 28F + post-partum hemorrhage + placenta is seen attached to myometrial layer; Dx + risk factor? → placenta accreta (placental attachment to surface of myometrium); risk factor is previous C-section, as well as concurrent placenta previa (latter sounds weird but is asked on Obgyn form) – i.e., placental attachment over the internal os increases risk of deeper attachment. Same deal holds true for placenta increta (attachment into myometrium) and percreta (attachment through myometrium). Should be noted in general that the most common cause of post-partum bleeding is uterine atony.

- 34F + giving birth + fetal anterior shoulder caught behind mother's pubic symphysis; Dx + risk factor? → shoulder dystocia; risk factor is fetal macrosomia due to maternal diabetes (causes big baby). This can cause clavicular fracture in the newborn.
- Above 34F + experiences excessive vaginal bleeding despite McRobert's maneuver to deliver in setting of shoulder dystocia; Dx + risk factor? → vaginal laceration due to macrosomia from maternal diabetes.
    - In short: maternal diabetes causes fetal macrosomia → results in shoulder dystocia → results in fetal clavicular fracture + maternal vaginal laceration during delivery.
- 32F + one-year Hx of amenorrhea + had dilation & curettage performed to remove miscarriage last year; Dx + risk factor? → Asherman syndrome; uterine synechiae (scarring) due to instrumentation within the uterus; NBME likes amenorrhea as the major symptom.
- 30F + 38 weeks' gestation + fever 103F + CVA tenderness; Dx + risk factor? → pyelonephritis; uterus can compress ureters in third trimester, increasing risk for pyelo; in addition, progesterone slows ureteral peristalsis.
- 30F + pregnant + severe epigastric pain after meals starting during third-trimester; Dx + risk factor? → cholelithiasis; progesterone slows biliary peristalsis (causing biliary sludge) + estrogen upregulates HMG-CoA reductase (increasing cholesterol concentration within bile).
- 4-month-old girl + small amount of blood in stool + exclusively breastfed until age 2 months + on formula since; Q wants to know biggest risk factor for this patient's condition → answer = "not being exclusively breastfed until 6 months of life"; diagnosis is milk-protein allergy; for whatever reason it can cause blood in the stool, but this is a HY part of the vignette and confuses students; Tx is switching to a hydrolyzed casein formula; switching to soy formula is wrong answer as there is an allergy crossover in up to 25% of cases (literature varies on exact %). Vignette can also mention kid is started right away on soy formula (not cow-milk-based); answer is still switch to hydrolyzed casein formula. I talk about this stuff in high detail in my HY Pediatrics PDF.
- 2M + iron deficiency anemia + drinks lots of cows milk; what is risk factor for anemia? → answer = drinking greater than 24 oz of cows milk daily; sounds weird, but 2CK Peds component wants you to know that is a risk factor for iron deficiency in kids under age 2.

- 2M + recent viral infection + now has intermittent vomiting and crying with blood in the stool; Dx + risk factor? → intussusception; risk factor is recent viral infection; can cause adenitis of bowel leading to telescoping; almost all intussusceptions occur under age 2; if elderly, can be colorectal cancer.
- 67 + chest pain when arguing with wife + blood in stool; Dx + risk factor? → angiodysplasia; risk factor is aortic stenosis (combo of angiodysplasia + aortic stenosis + acquired vWD = Heyde syndrome); angiodysplasia is tortuosity of superficial vessels within the colon that are prone to painlessly bleed; third most common cause of GI bleeding in elderly after diverticulosis and cancer; Q on offline Step 1 NBME gives old guy arguing with his wife + gets chest pain + blood in stool; answer = angiodysplasia.
- 35F + Hx of recurrent miscarriages + positive VDRL test; Dx + risk factor? → SLE causing antiphospholipid syndrome, resulting in recurrent miscarriages due to uteroplacental microthromboses; antiphospholipid syndrome can cause a false (+) VDRL syphilis test.
- 35F + pregnant + fetus shows intrauterine growth restriction; risk factor? → often uteroplacental insufficiency, e.g., from smoking, cocaine use, HTN, SLE.
- Family Med Q shows glowing red rash on face in someone being treated for acne; Q asks best way to minimize this condition → answer = avoidance of sun exposure; topic tretinoin (vitamin A) can cause photosensitivity (usually red rash); oral tetracycline sometimes used for acne tends to cause more of a blistering photosensitivity. Oral isotretinoin (high-dose vitamin A used for severe acne) is big risk teratogen; must do pregnancy test + recommend contraception before commencing.
- Neonate born with small right ventricle; risk factor? → lithium during pregnancy is risk factor for Ebstein anomaly (atrialization of right ventricle).
- Neonate born with spina bifida in mother who takes various medications; risk factor? → anti-epileptics (especially valproic acid) cause B9 (folate) malabsorption and neural tube defects.
- 35F + pregnant + abnormal AFP measurement; risk factor? → dating error is most common cause of abnormal AFP measurement; next best step is fetal ultrasound to get correct dates (i.e., crown-rump length); re-measure AFP is wrong answer (the lab result wasn't wrong, just the ultrasound date estimate was).
- 35F + pregnant + high AFP measurement despite correct dates; Dx + risk factor? → likely neural tube defect (i.e., spina bifida) due to folate (B9) deficiency; B9 required during first 3-4 weeks of gestation

for neural tube development; high AFP can also be due to anencephaly, multiple-gestation pregnancy, omphalocele, or gastroschisis.

- 40F + pregnant + low AFP + low estriol + high β-hCG + high inhibin A; Dx + risk factor? → Down syndrome due to increased maternal age.
- Neonate + duodenal atresia and/or Hirschsprung; risk factor? –> Down syndrome is HY cause of these two findings.
- Young child born to mother age 41 + long, smooth philtrum + widely spaced eyes + single palmar crease + slanted palpebral fissures; risk factor? → fetal alcohol syndrome; Down syndrome is wrong answer → if Q mentions any change whatsoever regarding the philtrum, the answer is fetal alcohol syndrome, not Down; there is a hard Q on the 2CK Psych CMS form that sounds like Down syndrome, but answer is FAS due to the philtrum changes.
- 8F + long Hx of recurrent pulmonary infections + 6-month history of peripheral neuropathy; Dx + risk factor? → cystic fibrosis resulting in vitamin E deficiency (can present as neuropathy). This sounds low-yield and weird, but it's all over the NBME exams for Step 1 (i.e., vitamin E deficiency due to CF).
- 4M + low Hb + Hx of diarrhea after meals; Dx + risk factor for the anemia? → Celiac disease; major risk factor for anemia in peds (flattening of villi causes impaired iron absorption in duodenum.
- 4M + recent gastroenteritis + now has diarrhea after meals; Dx + risk factor? → lactose intolerance; risk factor is recent GI viral infection; the latter can cause transient lactose intolerance due to sloughing of brush border (hence losing disaccharidase).
- 4M + small, shrunken kidney + tubular atrophy seen on biopsy; Dx + risk factor? → chronic pyelonephritis; major risk factor is recurrent acute pyelo, either from vesicoureteral reflux or posterior urethral valves (can be written on NBME as "congenital urethral obstruction").
- 8F + puffy eyes and ankles + ascites + protein in urine; Dx + risk factor? → minimal change disease; risk factor is viral infection; if presents in adult, can be Hodgkin lymphoma.
- 27M + sickle cell + nephrotic syndrome; Dx + risk factor? → focal segmental glomerulosclerosis caused by sickle cell; in short, FSGS is the answer if patient has nephrotic syndrome in sickle cell. Can also rarely be caused by HIV / heroin use.
- 27M + sickle cell + dark urine; Dx + risk factor? → renal papillary necrosis; this is answer on USMLE if patient has blood in the urine in sickle cell.

- 26F + IV drug user + recently treated with 6 weeks of nafcillin for MSSA endocarditis + now has rash and eosinophils in the urine; Dx + risk factor for renal condition? → interstitial nephropathy (aka tubulointerstitial nephropathy/-nephritis); risk factor is beta-lactam, cephalosporin, or NSAID; answer on USMLE if patient gets WBCs (eosinophils) in urine +/- maculopapular rash following any of the aforementioned drugs; these agents do **not** classically cause acute tubular necrosis.
- 26F + IV drug user + develops oliguria following gentamicin and vancomycin empiric Tx for endocarditis; Dx + risk factor? → acute tubular necrosis due to gentamicin (aminoglycoside).
- 38F + rheumatoid arthritis + took gold salts for treatment + develops proteinuria; Dx + risk factor? → gold salts causing membranous glomerulonephritis; dapsone can also cause this.
- 40F + history of hepatitis B + proteinuria; Dx + risk factor? → membranous glomerulonephritis due to hepatitis B; this is renal Dx if patient has nephrotic syndrome in setting of hepatitis B or C.
- 40F + history of hepatitis C + blood in urine; Dx + risk factor? → membranoproliferative glomerulonephritis; this is answer on USMLE if patient has nephritic syndrome in hepatitis B or C.
- 45M + various medium-sized arteries with fibrinoid necrosis and segmental, beaded, “rosary sign”; diagnosis + risk factor? → polyarteritis nodosa; major risk factor is hepatitis B (30% of patients with PA are seropositive for HepB).
- 40F + SLE + blood in urine; Dx + risk factor? → diffuse proliferative glomerulonephritis due to SLE; answer on USMLE if patient has nephritic syndrome in SLE; if nephrotic syndrome in SLE, patient just has regular membranous glomerulonephritis. But DPGN is >>> higher yield for SLE.
- 12F + blood in urine 1-3 **days** after sore throat; Dx + risk factor? → IgA nephropathy; answer on USMLE when patient has blood in urine 1-3 **days** following sore throat (viral infection).
- 12F + blood in urine 1-3 **weeks** after sore throat; Dx + risk factor? → PSGN is answer on USMLE when patient has blood in urine 1-3 **weeks** following sore throat (group A strep infection).
- 50M + flank pain + blood in urine + high serum calcium + high hemoglobin; Dx + risk factor? → renal cell carcinoma; major risk factor is smoking; can produce ectopic PTHrp (similar to squamous cell of lung) and EPO.
- 50M + blood in urine + Hx of smoking; Dx + risk factor? → bladder cancer (transitional cell carcinoma); major risk factor is smoking. Aniline dyes (industrial dyes) and naphthylamine (moth balls) are also assessed for causing transitional cell carcinoma of bladder.

- 23M + swam in lake in Africa + now has blood in urine; Dx + risk factor? → squamous cell carcinoma of the bladder due to *Schistosoma hematobium*; risk factor is swimming in fresh water in Africa.
- 40M + comes back from Africa with hemolytic condition + image shows schizonts of malaria; Q asks what patient is at greatest risk for → answer = hypoglycemia; you need to know malaria (particularly *P. falciparum*) can lead to hypoglycemia. If you think it's weird, don't take it up with me. It's on one of the Step 1 NBME exams.
- 7F + running around barefoot in rural Louisiana + gets microcytic anemia + high eosinophils; Dx + risk factor? → Dx = hookworms; risk factor is running around barefoot (USMLE wants you to know you get them through your feet, not from food); they cause microcytic anemia.
- 60F + low serum sodium + concentrated urine + history of weight loss; Dx + risk factor? → SIADH due to small cell lung cancer; smoking is risk factor.
- 60F + high calcium + lung cancer; Dx + risk factor? → squamous cell carcinoma secreting PTHrp.
- 60F + lung cancer + difficulty getting up from chair a few times; Dx + risk factor? → Lambert-Eaton due to small cell (anti-presynaptic voltage-gated Ca channel antibodies).
- 60F + lung cancer + ataxia; Dx + risk factor? → small cell cerebellar dysfunction (anti-Hu/-Yo) Abs.
- 35F + worsening diplopia, dysphagia, and eyelid ptosis over the course of a day; Dx and risk factor? → myasthenia gravis (Abs against post-synaptic nicotinic acetylcholine receptors); often idiopathic, but can be due to thymoma as risk factor.
- 35F + ovarian mass + purple discoloration of knuckles + violaceous eyelids + proximal muscle weakness; Dx + risk factor? → dermatomyositis due to ovarian cancer. Dermatomyositis usually idiopathic, but a risk factor for it that's assessed is ovarian cancer.
- 64F + back pain + nephrotic syndrome + S4 heart sound; Dx + risk factor? → renal amyloidosis and cardiac amyloidosis; major risk factor for amyloidosis on USMLE is multiple myeloma.
- 2M + enucleation of eye for retinoblastoma; Q asks about this being risk factor for which of the following in the future? → answer = osteosarcoma; hereditary retinoblastoma can also lead to osteosarcoma.

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